Analytic Field Theory

— *EYNTKA* Series —

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Contents

Introduction

The prime numbers are the multiplicative atoms of the integers, and their sequence $2, 3, 5, 7, 11, \ldots$ is at once the most elementary object in mathematics and the most stubbornly irregular. No formula generates the primes, no simple rule predicts the gap to the next one, and locally their appearance looks random. Yet in aggregate the primes obey precise quantitative laws: the number of primes up to x is asymptotic to $x/\log x$, the primes distribute themselves evenly among the admissible residue classes modulo any fixed integer, and the fluctuations around these averages are governed, to a degree that is still only partly understood, by the zeros of a single meromorphic function. Analytic number theory is the study of these laws. Its method is to attach to an arithmetic problem an analytic object, most often a Dirichlet series or an L-function, and to read the arithmetic off the analytic structure of that object: its poles, its zeros, its behavior on the boundary of convergence, its functional equation. The recurring lesson of the subject is that the discrete and irregular world of the primes is controlled by the continuous and rigid world of holomorphic functions.

This book develops that method in three movements. Part I assembles the apparatus: arithmetic functions and the summation techniques that convert discrete sums into integrals, the Dirichlet series and Euler products that package multiplicative arithmetic analytically and allow for ever finer approximations of the information we want to deal with, and the elementary theory of prime distribution that shows both how much and how little can be proved without complex analysis. Part II is the core of the subject: It brings the Riemann zeta function into the complex plane, proves the Prime Number Theorem from the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$, uncovers the functional equation and the critical strip, and then generalizes twice over: first by twisting with Dirichlet characters to reach primes in arithmetic progressions, then by replacing the rational field with a number field to produce the Dedekind, Hecke, and Artin L-functions, culminating in the Chebotarev density theorem. Part III turns to two techniques that complement the L-function machinery and reach problems it cannot: the estimation of character and exponential sums, which measures cancellation directly, and the sieve methods that count primes and almost-primes by combinatorial means.

The prerequisites are the standard graduate analysis and algebra of the earlier volumes in this series: complex analysis ([13]), real analysis ([14]), and harmonic analysis ([17]) for the analytic technique, and core algebra ([5]) for the group theory of characters. These suffice for Parts I and III. Part II raises the bar as it proceeds: chapter 8 adds algebraic number theory ([2]) as a co-requisite, and chapters 9 and 10 draw in local field theory ([4]) and representation theory ([3]); each escalation is flagged where it occurs. Elementary number theory itself, the divisibility, congruences, and quadratic reciprocity on which everything rests, is reviewed in the preliminary chapter that follows, which the prepared reader may treat as a reference. Exercises are collected at the end of each chapter. One large subject is deliberately absent: the theory of automorphic forms, which is the natural continuation beyond the last results of Part II and is the province of later volumes in the series (see ref:HERE). What unifies the whole is the Riemann hypothesis and its generalizations, the still-unproven statements as of writing this book about the zeros of ζ and the L-functions, toward which every part of the book points and on which the sharpest questions about the primes ultimately depend.

A Review of Elementary Number Theory

The machinery of analytic number theory rests on a foundation of elementary results about integers, primes, and congruences. This preliminary chapter collects the essential facts that the rest of the book takes for granted. Readers with a solid undergraduate number theory background will find this material familiar and may treat the chapter as a reference; those encountering some results for the first time should work through the proofs carefully, as the techniques—particularly the Euclidean algorithm, the Chinese Remainder Theorem, and quadratic reciprocity—reappear in more sophisticated forms throughout the book.

The progression is natural. Divisibility and the Euclidean algorithm provide the computational basis. The Fundamental Theorem of Arithmetic establishes that primes are the multiplicative building blocks of $\mathbb{Z}$, a fact whose failure in more general rings drives much of algebraic number theory (see [2]). Congruences and the Chinese Remainder Theorem introduce the modular perspective that underlies Dirichlet characters and L-functions. The structure of the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$ and the theory of quadratic residues prepare the ground for character theory and reciprocity laws. The chapter closes with a review of asymptotic notation, the language in which the main results of analytic number theory are expressed.

0.1 Divisibility and the Euclidean Algorithm

The integers carry two intertwined structures: addition and multiplication. The additive structure is simple—$\mathbb{Z}$ is a free abelian group of rank 1—but the multiplicative structure is the source of all the complexity that this book investigates. The first step toward understanding it is the divisibility relation, and the first computational tool is the Euclidean algorithm, which reduces questions about divisibility to iterated division with remainder.

The starting point is the division algorithm, which guarantees that long division in $\mathbb{Z}$ always terminates with a well-defined quotient and remainder.

Theorem 0.1.1: Division Algorithm

For any $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, there exist unique integers q (the *quotient*) and r (the *remainder*) such that

$$a = bq + r, \qquad 0 \leq r < b$$

***Proof*:**
Step 1: Existence. Consider the set $S = \{a - bk \mid k \in \mathbb{Z},\ a - bk \geq 0\}$. Since $b > 0$, choosing k sufficiently negative ensures that $a - bk > 0$, so S is nonempty. By the well-ordering principle, S has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \geq 0$. If $r \geq b$, then $a - b(q+1) = r - b \geq 0$ belongs to S and is strictly smaller than r, contradicting minimality. Hence $0 \leq r < b$.

Step 2: Uniqueness. Suppose $a = bq_1 + r_1 = bq_2 + r_2$ with $0 \leq r_1, r_2 < b$. Then $b(q_1 - q_2) = r_2 - r_1$, so b divides $r_2 - r_1$. Since $|r_2 - r_1| < b$, this forces $r_1 = r_2$ and consequently $q_1 = q_2$.

The division algorithm is the engine behind every computation in this section. Its role is analogous to that of the Euclidean metric in geometry: it provides a notion of "size" (the remainder r) that strictly decreases under a natural operation (repeated division), guaranteeing that iterative processes terminate. This principle—descent by remainder—also underlies the proof of unique factorization, which depends on it via Euclid's lemma in the next section.

With division in hand, the notion of exact divisibility can be made precise.

Definition 0.1.2: Divisibility

Let $a, b \in \mathbb{Z}$. Then a *divides* b, written $a \big| b$, if there exists $k \in \mathbb{Z}$ such that $b = ak$. Equivalently, $a \big| b$ if and only if the remainder upon dividing b by a is zero. If a does not divide b, this is denoted $a \nmid b$.

The divisibility relation is a partial order on the positive integers: it is reflexive ($a \big| a$ since $a = a \cdot 1$), antisymmetric (if $a \big| b$ and $b \big| a$ with $a, b > 0$ then $a = b$), and transitive (if $a \big| b$ and $b \big| c$ then $a \big| c$, since $c = bk_1 = ak_2k_1$). It is also compatible with the ring operations of $\mathbb{Z}$ in the following sense.

Proposition 0.1.3: Linearity of Divisibility

If $d \big| a$ and $d \big| b$, then $d \big| (ax + by)$ for all $x, y \in \mathbb{Z}$.

***Proof*:**
Write $a = dk_1$ and $b = dk_2$. Then $ax + by = d(k_1x + k_2y)$.

In the language of algebra, this says that the set of multiples of d forms an ideal $(d) = d\mathbb{Z}$ in $\mathbb{Z}$, and divisibility translates to ideal containment: $a \big| b$ if and only if $(b) \subseteq (a)$. The fact that *every* ideal of $\mathbb{Z}$ has this form—$\mathbb{Z}$ is a principal ideal domain—is the ring-theoretic content of Bézout's identity, proved below. In more general number rings, ideals need not be principal, and the failure of this property is measured by the class group (see [2]).

The central notion built from divisibility is the greatest common divisor.

Definition 0.1.4: Greatest Common Divisor

For $a, b \in \mathbb{Z}$ not both zero, the *greatest common divisor* $\gcd(a, b)$ is the largest positive integer d such that $d \big| a$ and $d \big| b$. By convention, $\gcd(0, 0) = 0$. The integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The existence of $\gcd(a, b)$ is clear: the set of common divisors is nonempty (it contains 1 when a and b are not both zero) and bounded above by $\max(|a|, |b|)$, so a maximum exists. Before describing how to compute it, a few basic identities are worth recording.

Proposition 0.1.5: Properties of the GCD

For all $a, b \in \mathbb{Z}$ and $k \in \mathbb{Z} \setminus \{0\}$:

1. $\gcd(a, b) = \gcd(b, a) = \gcd(|a|, |b|)$.
2. $\gcd(a, 0) = |a|$.
3. $\gcd(ka, kb) = |k| \gcd(a, b)$.
4. If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

***Proof*:**
Parts (1)–(3) follow directly from the definition, since the set of common divisors is unchanged by reordering, by taking absolute values, or by scaling (respectively). For (4), proposition 0.1.3 gives the two containments: every common divisor of a and b divides $r = a - bq$, and every common divisor of b and r divides $a = bq + r$. Hence the sets of common divisors coincide, and their maxima are equal.

Part (4) is the identity that makes the Euclidean algorithm work. It reduces $\gcd(a, b)$ to $\gcd(b, r)$ where $0 \leq r < b$, so the arguments are strictly smaller. Iterating this reduction is the Euclidean algorithm.

Proposition 0.1.6: The Euclidean Algorithm

Let $a, b \in \mathbb{Z}$ with $b > 0$. Apply the division algorithm repeatedly:

$$\begin{aligned} a &= bq_0 + r_0, & 0 &\leq r_0 < b, \\ b &= r_0 q_1 + r_1, & 0 &\leq r_1 < r_0, \\ r_0 &= r_1 q_2 + r_2, & 0 &\leq r_2 < r_1, \\ &\vdots \\ r_{n-2} &= r_{n-1} q_n + r_n, & 0 &\leq r_n < r_{n-1}, \\ r_{n-1} &= r_n q_{n+1} \end{aligned}$$

The process terminates after finitely many steps, and the last nonzero remainder is $\gcd(a, b) = r_n$.

***Proof*:**
The sequence $b > r_0 > r_1 > \cdots \geq 0$ is a strictly decreasing sequence of non-negative integers, so it must reach zero: there exists a smallest index n such that $r_{n+1} = 0$. At that step, $r_{n-1} = r_n q_{n+1}$, so $r_n \big| r_{n-1}$. Working backward through each line, r_n divides every preceding remainder and hence divides both b and a: it is a common divisor. Conversely, if d is any common divisor of a and b, the first equation gives $d \big| r_0$ (by proposition 0.1.3), the second gives $d \big| r_1$, and continuing inductively, $d \big| r_n$. Thus r_n is a common divisor that is divisible by every common divisor, which forces $r_n = \gcd(a, b)$.

The running time of the Euclidean algorithm is modest. After two consecutive division steps, the remainder drops by at least a factor of two: $r_{k+2} < r_k/2$. This gives an upper bound of $2\lfloor \log_2 b \rfloor + 1$ steps, making the algorithm logarithmic in the size of its input. A finer analysis shows that the worst-case inputs are consecutive Fibonacci numbers, where each quotient q_i equals 1 and the remainders decrease as slowly as possible.

Example 0.1.7: Computing a GCD
Take $a = 252$ and $b = 105$. The Euclidean algorithm proceeds:

$$\begin{aligned} 252 &= 105 \cdot 2 + 42, \\ 105 &= 42 \cdot 2 + 21, \\ 42 &= 21 \cdot 2 + 0 \end{aligned}$$

The last nonzero remainder is 21, so $\gcd(252, 105) = 21$. The algorithm terminates in three steps, reflecting the fact that $252/105$ has a short continued fraction expansion.

The Euclidean algorithm does more than compute the GCD—it simultaneously produces an explicit $\mathbb{Z}$-linear combination. Tracing back through the successive divisions recovers the coefficients in what is arguably the single most useful identity in elementary number theory.

Theorem 0.1.8: Bézout's Identity

For any $a, b \in \mathbb{Z}$ not both zero, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b)$$

In particular, a and b are coprime if and only if 1 is a $\mathbb{Z}$-linear combination of a and b.

This is a statement about the ideal structure of $\mathbb{Z}$: the ideal generated by a and b,

$$(a, b) = \{ax + by \mid x, y \in \mathbb{Z}\},$$

is principal, generated by $\gcd(a, b)$. Every ideal of $\mathbb{Z}$ is principal—the ring $\mathbb{Z}$ is a PID. In rings of integers of number fields, this property can fail: in $\mathbb{Z}[\sqrt{-5}]$, for instance, the ideal $(2, 1 + \sqrt{-5})$ is not principal. The deviation from principality is measured by the ideal class group, one of the central objects studied in [2] and a key player in the analytic class number formula of chapter 8.

***Proof*:**
Define d to be the smallest positive element of $I = \{ax + by \mid x, y \in \mathbb{Z}\}$. This set contains positive elements (for instance, $a^2 + b^2$ if a and b are not both zero), so d exists by the well-ordering principle.

Step 1: d divides a and b. Apply the division algorithm to write $a = dq + r$ with $0 \leq r < d$. Since $d = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$, the remainder satisfies

$$r = a - dq = a(1 - qx_0) + b(-qy_0) \in I$$

If $r > 0$, then r is a positive element of I smaller than d, contradicting the choice of d. Hence $r = 0$, so $d \big| a$. The identical argument with b in place of a gives $d \big| b$.

Step 2: $d = \gcd(a, b)$. Since $d \big| a$ and $d \big| b$, the integer d is a common divisor. If c is any common divisor of a and b, then c divides every element of I (by proposition 0.1.3), and in particular $c \big| d$. Hence d is the greatest common divisor.

The proof is clean but non-constructive: it asserts existence via the well-ordering principle without producing x and y explicitly. To find them, one runs the Euclidean algorithm forward and then substitutes backward through the chain of divisions (see example 0.1.10 below).

A direct and frequently used consequence of Bézout's identity is the following cancellation property for coprime integers. It is the mechanism underlying unique factorization in the next section.

Corollary 0.1.9: Euclid's Lemma for Coprime Integers

If $a \big| bc$ and $\gcd(a, b) = 1$, then $a \big| c$.

***Proof*:**
By theorem 0.1.8, there exist $x, y \in \mathbb{Z}$ with $ax + by = 1$. Multiplying through by c gives $acx + bcy = c$. Since $a \big| acx$ (trivially) and $a \big| bcy$ (by hypothesis), proposition 0.1.3 gives $a \big| c$.

The proof illustrates the standard technique: invoke Bézout to produce a linear combination equal to 1, then multiply by the desired target. This argument generalizes verbatim to any principal ideal domain, but fails in rings where Bézout's identity does not hold—such as $\mathbb{Z}[\sqrt{-5}]$, where $2 \mid (1+\sqrt{-5})(1-\sqrt{-5}) = 6$ but 2 divides neither factor.

Example 0.1.10: Back-Substitution for Bézout Coefficients
Continuing example 0.1.7, the Euclidean algorithm gave:

$$\begin{aligned} 252 &= 105 \cdot 2 + 42, \\ 105 &= 42 \cdot 2 + 21 \end{aligned}$$

From the second equation: $21 = 105 - 42 \cdot 2$. Substituting $42 = 252 - 105 \cdot 2$ from the first:

$$21 = 105 - (252 - 105 \cdot 2) \cdot 2 = 105 \cdot 5 + 252 \cdot (-2)$$

This gives $252 \cdot (-2) + 105 \cdot 5 = 21 = \gcd(252, 105)$, so the Bézout coefficients are $x = -2$ and $y = 5$. These are not unique: the general solution to $252x + 105y = 21$ is $(x, y) = (-2 + 5t,\ 5 - 12t)$ for $t \in \mathbb{Z}$, as the next result makes precise.

Bézout's identity settles the question of when a linear equation over $\mathbb{Z}$ has integer solutions.

Proposition 0.1.11: Linear Diophantine Equations

Let $a, b, c \in \mathbb{Z}$ with a and b not both zero, and set $d = \gcd(a, b)$. The equation

$$ax + by = c$$

has a solution $(x, y) \in \mathbb{Z}^2$ if and only if $d \mid c$. When solutions exist, they form the family

$$(x, y) = \left(x_0 + \frac{b}{d}\,t,\ y_0 - \frac{a}{d}\,t\right), \qquad t \in \mathbb{Z},$$

where (x_0, y_0) is any particular solution.

***Proof*:**
If (x, y) is a solution, then $d \mid a$ and $d \mid b$ give $d \mid (ax + by) = c$, establishing necessity. Conversely, if $d \mid c$, then theorem 0.1.8 provides $x_0', y_0' \in \mathbb{Z}$ with $ax_0' + by_0' = d$, and multiplying through by c/d gives the particular solution $(x_0, y_0) = ((c/d)x_0',\ (c/d)y_0')$.

For the general solution, note that (x, y) satisfies $ax + by = c$ if and only if $a(x - x_0) = -b(y - y_0)$. Dividing by d yields

$$\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0)$$

Since $\gcd(a/d, b/d) = 1$, corollary 0.1.9 gives $(b/d) \mid (x - x_0)$. Writing $x - x_0 = (b/d)\,t$ and substituting gives $y - y_0 = -(a/d)\,t$, as claimed.

The essential mechanism is the reduction to the coprime case: dividing by $d = \gcd(a, b)$ produces the coprimality condition $\gcd(a/d, b/d) = 1$, which then forces the structure of the solution set via

Bézout. This pattern—divide out the GCD, argue in the coprime case, and then scale back—recurs throughout number theory, from the analysis of congruences in the next section to the study of Dirichlet characters in chapter 7.

Example 0.1.12: Solving a Linear Diophantine Equation

Consider $14x + 35y = 63$. Here $\gcd(14, 35) = 7$ and $7 \big| 63$, so solutions exist. Dividing through by 7 reduces the problem to $2x + 5y = 9$. By inspection, $(x_0, y_0) = (2, 1)$ satisfies $2 \cdot 2 + 5 \cdot 1 = 9$. The general solution to the original equation is

$$(x, y) = (2 + 5t,\ 1 - 2t)\,, \qquad t \in \mathbb{Z},$$

producing the solutions $\ldots, (-3, 3),\ (2, 1),\ (7, -1),\ (12, -3), \ldots$ For instance, the unique solution with $0 \le x < 5$ is $(x, y) = (2, 1)$.

0.2 Primes and the Fundamental Theorem of Arithmetic

The Euclidean algorithm and Bézout's identity treat all integers uniformly, without distinguishing between those that are "composed" and those that are not. This section introduces the multiplicative building blocks of $\mathbb{Z}$—the prime numbers—and proves that every positive integer factors uniquely into primes. Unique factorization is the structural fact that makes the Euler product formula work (chapter 4): the identity $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ is, at its core, a restatement of the Fundamental Theorem of Arithmetic in the language of Dirichlet series.

Definition 0.2.1: Prime and Irreducible Elements

An integer $p > 1$ is *irreducible* if its only positive divisors are 1 and p. An integer $p > 1$ is *prime* if whenever $p \big| ab$, either $p \big| a$ or $p \big| b$.

In $\mathbb{Z}$, these two notions coincide—a fact that depends on Euclid's lemma and therefore on the Euclidean algorithm. The distinction matters in more general rings. In $\mathbb{Z}[\sqrt{-5}]$, for instance, 2 is irreducible (it cannot be written as a product of two non-units) but not prime: $2 \big| (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6$, yet 2 divides neither factor. The failure of "irreducible implies prime" is precisely the obstruction to unique factorization, and its resolution in number rings via the passage to ideals is a central theme of [2].

Proposition 0.2.2: In $\mathbb{Z}$, Prime and Irreducible Coincide

An integer $p > 1$ is prime if and only if it is irreducible.

***Proof*:**

Prime implies irreducible. Suppose p is prime and $p = ab$ with $a, b > 0$. Then $p \big| ab$, so $p \big| a$ or $p \big| b$. If $p \big| a$, then since $a \le p$ (as $b \ge 1$), this forces $a = p$ and $b = 1$. Similarly if $p \big| b$. Hence p is irreducible.

Irreducible implies prime. Suppose p is irreducible and $p \big| ab$. If $p \nmid a$, then the only positive common divisor of p and a is 1 (since p is irreducible and its only positive divisors are 1 and p, and $p \nmid a$ rules out p). Hence $\gcd(p, a) = 1$, and corollary 0.1.9 gives $p \mid b$.

The key step—the implication "irreducible implies prime"—relies entirely on corollary 0.1.9, which in turn depends on Bézout's identity. This chain of dependencies is the mechanism by which the division algorithm guarantees unique factorization in $\mathbb{Z}$. In rings without a division algorithm (or more precisely, without the property that irreducibles are prime), unique factorization can fail.

The first result about the collection of all primes is among the oldest in mathematics.

Proposition 0.2.3: Infinitude of Primes

There are infinitely many primes.

***Proof*:**
Suppose for contradiction that $p_1, p_2, \ldots, p_n$ is a complete list of all primes. Consider the integer $N = p_1 p_2 \cdots p_n + 1$. Since $N > 1$, it has a prime divisor p. For each i, the prime p_i divides $p_1 \cdots p_n$ but does not divide N (since dividing N by p_i leaves remainder 1). Hence p is not among $p_1, \ldots, p_n$, contradicting the assumption that the list was complete.

Euclid's argument establishes the infinitude of primes but says nothing about how they are distributed among the integers. The quantitative study of this distribution—how many primes are there up to x, and how are they spaced?—is the central question of analytic number theory. The Prime Number Theorem (chapter 5) gives the asymptotic answer $\pi(x) \sim x/\log x$, but reaching it requires the full power of complex analysis applied to $\zeta(s)$.

The Fundamental Theorem of Arithmetic asserts that every integer greater than 1 can be written as a product of primes, and that this product is essentially unique.

Theorem 0.2.4: Fundamental Theorem of Arithmetic

Every integer $n > 1$ has a factorization

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

where $p_1 < p_2 < \cdots < p_k$ are primes and $a_1, \ldots, a_k \geq 1$. This factorization is unique: if $n = q_1^{b_1} \cdots q_\ell^{b_\ell}$ is another such factorization with $q_1 < \cdots < q_\ell$, then $k = \ell$, $p_i = q_i$, and $a_i = b_i$ for all i.

***Proof*:**
Step 1: Existence. Proceed by strong induction on n. If $n = 2$, then n is prime and $n = 2$ is itself a prime factorization. For $n > 2$: if n is prime, the factorization $n = n^1$ works. If n is not prime, then $n = ab$ for some $1 < a, b < n$, and by the inductive hypothesis both a and b have prime factorizations. Concatenating and reordering gives a prime factorization of n.

Step 2: Uniqueness. Suppose $p_1^{a_1} \cdots p_k^{a_k} = q_1^{b_1} \cdots q_\ell^{b_\ell}$. Then p_1 divides the right-hand side, so $p_1 \mid q_j^{b_j}$ for some j (by proposition 0.2.2, applied repeatedly). Since q_j is prime and $p_1 > 1$, this forces $p_1 = q_j$. As both lists are sorted in increasing order: if $p_1 < q_1$, then $p_1 \mid q_1^{b_1} \cdots q_\ell^{b_\ell}$ but $p_1 \neq q_j$ for any j (since $p_1 < q_1 \leq q_j$), a contradiction. Similarly $q_1 < p_1$ is impossible. Hence

$p_1 = q_1$.

Dividing both sides by $p_1^{\min(a_1,b_1)}$ and applying the same argument inductively shows that the two factorizations agree term by term. In particular, $k = \ell$, $p_i = q_i$, and $a_i = b_i$ for all i.

The existence of a prime factorization uses only the well-ordering of $\mathbb{N}$ (via strong induction). The uniqueness, on the other hand, is the substantive content: it depends on the fact that primes in $\mathbb{Z}$ satisfy the cancellation property of proposition 0.2.2, which traces back through Euclid's lemma to the division algorithm. Removing any link in this chain allows uniqueness to fail, as the example $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ in $\mathbb{Z}[\sqrt{-5}]$ demonstrates.

Example 0.2.5: Prime Factorizations

1. $360 = 2^3 \cdot 3^2 \cdot 5$. This is the unique way to write 360 as a product of prime powers (up to reordering).
2. $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$. This is a *primorial*: the product of the first five primes. Primorials are squarefree integers (every prime appears with exponent exactly 1) and appear naturally in sieve arguments (chapter 12).
3. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. The exponents can be computed directly from the Fundamental Theorem, but Legendre's formula (proved at the end of this section) provides a much faster method.

The Fundamental Theorem is the reason why the study of multiplicative properties of $\mathbb{Z}$ reduces to the study of primes. Every question about divisibility, common factors, or the multiplicative structure of an integer can be answered by examining its prime factorization. This reductive principle has an analytic counterpart: the Euler product identity $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ converts a sum over all positive integers into a product over primes, and it is through this product that the analytic properties of $\zeta(s)$ yield information about the distribution of primes.

With unique factorization established, every positive integer n is completely described by its *prime signature*: the function $p \mapsto v_p(n)$ giving the exponent of each prime in the factorization of n.

Definition 0.2.6: The p-adic Valuation

For a prime p and a nonzero integer n, the *p-adic valuation* of n, denoted $v_p(n)$, is the largest integer $e \geq 0$ such that $p^e \mid n$. By convention, $v_p(0) = +\infty$. The Fundamental Theorem of Arithmetic gives

$$|n| = \prod_p p^{v_p(n)}$$

where the product is over all primes and is finite (since $v_p(n) = 0$ for all but finitely many p).

The p-adic valuation converts multiplicative questions about $\mathbb{Z}$ into additive questions about $\mathbb{Z}_{\geq 0}$: if $n = ab$ then $v_p(n) = v_p(a) + v_p(b)$ for every prime p. This decomposition—analyzing a global multiplicative problem one prime at a time—is the "local-to-global" philosophy that pervades number theory, from the Euler product for $\zeta(s)$ to the adelic methods of class field theory.

The valuation extends naturally to $\mathbb{Q}^\times$ by setting $v_p(a/b) = v_p(a) - v_p(b)$, giving a group homomorphism $v_p : \mathbb{Q}^\times \to \mathbb{Z}$. Every nonzero rational number r can then be written uniquely as $r = \pm \prod_p p^{v_p(r)}$, where the product is finite. The associated *p-adic absolute value* $|r|_p = p^{-v_p(r)}$

defines a metric on $\mathbb{Q}$ whose completion is the field $\mathbb{Q}_p$ of p-adic numbers (see [4]). The interplay between all these completions—one for each prime and one for the ordinary absolute value—is a recurring theme beginning with the completed zeta function in chapter 6.

Proposition 0.2.7: Divisibility via Valuations

For nonzero integers a and b:

1. $a \mid b$ if and only if $v_p(a) \leq v_p(b)$ for every prime p.
2. $\gcd(a, b) = \prod_p p^{\min(v_p(a),\, v_p(b))}$.
3. $\operatorname{lcm}(a, b) = \prod_p p^{\max(v_p(a),\, v_p(b))}$.
4. $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = |ab|$.

***Proof*:**
For (1): $a \mid b$ means $b = ak$ for some $k \in \mathbb{Z}$, which translates to $v_p(b) = v_p(a) + v_p(k) \geq v_p(a)$ for all p. The converse follows by writing $k = b/a$ and checking that $v_p(k) = v_p(b) - v_p(a) \geq 0$ for all p, so k is an integer. Parts (2) and (3) follow from (1): the greatest common divisor is the largest integer dividing both a and b, which means taking the minimum exponent at each prime; the least common multiple is the smallest integer divisible by both, which means taking the maximum. Part (4) follows from the identity $\min(x, y) + \max(x, y) = x + y$.

The valuation-based characterization of divisibility is the “right” perspective for many computations. For example, it makes the following formula for gcd immediate.

Example 0.2.8: GCD and LCM via Valuations

1. Take $a = 252 = 2^2 \cdot 3^2 \cdot 7$ and $b = 105 = 3 \cdot 5 \cdot 7$. Reading off:

$$\gcd(252, 105) = 2^0 \cdot 3^1 \cdot 5^0 \cdot 7^1 = 21, \qquad \operatorname{lcm}(252, 105) = 2^2 \cdot 3^2 \cdot 5 \cdot 7 = 1260$$

   As a check: $\gcd \cdot \operatorname{lcm} = 21 \cdot 1260 = 26460 = 252 \cdot 105 = |ab|$.

2. For $a = 360 = 2^3 \cdot 3^2 \cdot 5$ and $b = 750 = 2 \cdot 3 \cdot 5^3$:

$$\gcd(360, 750) = 2^1 \cdot 3^1 \cdot 5^1 = 30, \qquad \operatorname{lcm}(360, 750) = 2^3 \cdot 3^2 \cdot 5^3 = 9000$$

The Fundamental Theorem also provides a clean way to express many arithmetic functions. The number of positive divisors of $n = \prod p_i^{a_i}$ is $\prod (a_i + 1)$, since each divisor is obtained by independently choosing an exponent between 0 and a_i at each prime. The sum of divisors is $\prod (1 + p_i + p_i^2 + \cdots + p_i^{a_i}) = \prod \frac{p_i^{a_i+1} - 1}{p_i - 1}$. These formulas are simple consequences of unique factorization combined with the Chinese Remainder Theorem, and they generalize to the full family of multiplicative arithmetic functions studied in chapter 1.

Example 0.2.9: Divisor Count and Divisor Sum

1. The integer $n = 360 = 2^3 \cdot 3^2 \cdot 5$ has $(3+1)(2+1)(1+1) = 24$ positive divisors.
2. The sum of divisors of 360 is
$$\sigma(360) = \frac{2^4-1}{2-1} \cdot \frac{3^3-1}{3-1} \cdot \frac{5^2-1}{5-1} = 15 \cdot 13 \cdot 6 = 1170$$
3. For a prime p: the number of divisors of p^k is $k+1$, and $\sigma(p^k) = \frac{p^{k+1}-1}{p-1}$. In particular, $\sigma(p) = p+1$.

A useful bookkeeping device that appears throughout the book is the *canonical factorization notation*. For $n = \prod_{i=1}^{k} p_i^{a_i}$, a sum or product "over the prime factorization of n" means over the pairs (p_i, a_i). The notation $p^a \| n$ (read "p^a exactly divides n") means $v_p(n) = a$, i.e., $p^a \big| n$ but $p^{a+1} \nmid n$.

Definition 0.2.10: Exact Divisibility

For a prime p and a positive integer n, the notation $p^a \| n$ means $v_p(n) = a$, that is, $p^a \big| n$ and $p^{a+1} \nmid n$.

To close this section, a formula that illustrates how the Fundamental Theorem interacts with factorials. It will be used in the proof of Chebyshev's bounds in chapter 3.

Proposition 0.2.11: Legendre's Formula

For any prime p and positive integer n:
$$v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor$$
The sum is finite, as $\lfloor n/p^k \rfloor = 0$ for $p^k > n$.

***Proof*:**
The exponent $v_p(n!)$ counts the total number of factors of p appearing in the product $1 \cdot 2 \cdots n$. Among the integers $1, 2, \ldots, n$, there are exactly $\lfloor n/p \rfloor$ multiples of p, each contributing at least one factor. Of these, $\lfloor n/p^2 \rfloor$ are multiples of p^2, each contributing a second factor. Continuing, $\lfloor n/p^k \rfloor$ multiples of p^k each contribute a k-th factor. Summing over all k counts each factor of p exactly once, giving the formula.

Example 0.2.12: Applying Legendre's Formula
The exponent of 2 in 20! is
$$v_2(20!) = \left\lfloor \frac{20}{2} \right\rfloor + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{20}{8} \right\rfloor + \left\lfloor \frac{20}{16} \right\rfloor = 10 + 5 + 2 + 1 = 18$$
Similarly, $v_3(20!) = 6 + 2 = 8$ and $v_5(20!) = 4$. In particular, 20! is divisible by $10^4 = 2^4 \cdot 5^4$ but not by 10^5, so $20! = 2432902008176640000$ ends in exactly four trailing zeros.

An immediate consequence of Legendre's formula is a clean expression for the p-adic valuation of binomial coefficients, which plays a role in Chebyshev's estimates (chapter 3).

Corollary 0.2.13: Valuation of Binomial Coefficients

For a prime p and integers $0 \le m \le n$:

$$v_p\binom{n}{m} = \sum_{k=1}^{\infty}\left(\left\lfloor \frac{n}{p^k}\right\rfloor - \left\lfloor \frac{m}{p^k}\right\rfloor - \left\lfloor \frac{n-m}{p^k}\right\rfloor\right)$$

Each summand is either 0 or 1, and it equals 1 precisely when the addition $m + (n - m)$ produces a carry at the p^k-place in base p.

***Proof*:**
Since $\binom{n}{m} = n!/(m!(n-m)!)$, the formula follows from Legendre's formula and the additivity of v_p:

$$v_p\binom{n}{m} = v_p(n!) - v_p(m!) - v_p((n-m)!) = \sum_{k\ge 1}\left(\left\lfloor \frac{n}{p^k}\right\rfloor - \left\lfloor \frac{m}{p^k}\right\rfloor - \left\lfloor \frac{n-m}{p^k}\right\rfloor\right)$$

For the "carry" interpretation: let $a = m$ and $b = n - m$, so $a + b = n$. The identity $\lfloor (a+b)/p^k\rfloor - \lfloor a/p^k\rfloor - \lfloor b/p^k\rfloor$ equals 1 when the addition of the base-p digits of a and b produces a carry into the p^k-position, and 0 otherwise. This is Kummer's theorem.

Example 0.2.14: Central Binomial Coefficient
Consider $\binom{10}{5} = 252$. In base 2: $5 = (101)_2$ and $5 + 5 = 10 = (1010)_2$. Adding $(101)_2 + (101)_2$, carries occur at positions 2^0 and 2^2, so $v_2\binom{10}{5} = 2$. Indeed, $252 = 2^2 \cdot 3^2 \cdot 7$. Similarly, in base 3: $5 = (12)_3$ and $5 + 5 = 10 = (101)_3$; adding $(12)_3 + (12)_3$ produces carries at 3^0 and 3^1, giving $v_3\binom{10}{5} = 2$.

0.3 Congruences

The previous sections studied the integers through their multiplicative structure: divisibility, primes, and factorization. Congruences introduce a complementary perspective by collapsing $\mathbb{Z}$ into finite quotient rings $\mathbb{Z}/m\mathbb{Z}$. This modular viewpoint is the natural setting for Dirichlet characters (chapter 7), and the Chinese Remainder Theorem—the structural result of this section—is the tool that decomposes these characters into local components, one for each prime power dividing the modulus.

Definition 0.3.1: Congruence

Let $m \in \mathbb{Z}_{>0}$. Two integers a and b are *congruent modulo* m, written $a \equiv b \pmod{m}$, if $m \mid (a - b)$. Equivalently, a and b leave the same remainder upon division by m.

Congruence modulo m is an equivalence relation on $\mathbb{Z}$, partitioning it into m equivalence classes called *residue classes*. The set of residue classes forms a ring under the inherited addition and multiplication:

Definition 0.3.2: The Ring $\mathbb{Z}/m\mathbb{Z}$

The *residue class ring modulo* m is the quotient ring

$$\mathbb{Z}/m\mathbb{Z} = \{0, 1, 2, \ldots, m-1\}$$

with addition and multiplication performed modulo m. The canonical projection $\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ sends each integer to its residue class.

The ring $\mathbb{Z}/m\mathbb{Z}$ is the simplest example of a quotient ring, and its structure—which residue classes are units, which are zero divisors, and how the ring decomposes—is controlled entirely by the prime factorization of m. The key structural result is the Chinese Remainder Theorem, which says that if m factors into coprime parts, then $\mathbb{Z}/m\mathbb{Z}$ splits as a product of smaller rings.

Before stating it, the following criterion for invertibility modulo m is needed.

Proposition 0.3.3: Units of $\mathbb{Z}/m\mathbb{Z}$

An element $a \in \mathbb{Z}/m\mathbb{Z}$ is a unit (i.e., has a multiplicative inverse modulo m) if and only if $\gcd(a, m) = 1$. When this holds, the inverse can be computed via the Euclidean algorithm: if $ax + my = 1$ (from theorem 0.1.8), then $a^{-1} \equiv x \pmod{m}$.

***Proof*:**
If $\gcd(a, m) = 1$, then theorem 0.1.8 gives $ax + my = 1$ for some $x, y \in \mathbb{Z}$, which reads $ax \equiv 1 \pmod{m}$, so x is the inverse of a. Conversely, if $ax \equiv 1 \pmod{m}$ for some x, then $ax = 1 + km$ for some k, giving $ax - km = 1$. By proposition 0.1.3, any common divisor of a and m divides 1, so $\gcd(a, m) = 1$.

The group of units $(\mathbb{Z}/m\mathbb{Z})^\times = \{a \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(a, m) = 1\}$ has order $\varphi(m)$, where φ is the Euler totient function. This group is the domain on which Dirichlet characters live, and its structure will be analyzed in detail in the next section.

An element $a \in \mathbb{Z}/m\mathbb{Z}$ with $a \neq 0$ and $\gcd(a, m) > 1$ is a *zero divisor*: writing $d = \gcd(a, m)$ with $1 < d < m$, the product $a \cdot (m/d) \equiv 0 \pmod{m}$ is zero even though neither factor is. Thus $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime, in which case every nonzero element is a unit and $\mathbb{Z}/m\mathbb{Z} = \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field with p elements. This is the finite field that appears throughout the book, from character sums (chapter 11) to the local factors of L-functions.

Example 0.3.4: Units and Zero Divisors in $\mathbb{Z}/12\mathbb{Z}$
The elements of $\mathbb{Z}/12\mathbb{Z}$ partition as follows:

1. *Units* ($\gcd(a, 12) = 1$): $\{1, 5, 7, 11\}$. Each has a multiplicative inverse: $5 \cdot 5 = 25 \equiv 1$, $7 \cdot 7 = 49 \equiv 1$, $11 \cdot 11 = 121 \equiv 1$. So $5^{-1} = 5$, $7^{-1} = 7$, $11^{-1} = 11$ in $\mathbb{Z}/12\mathbb{Z}$.
2. *Zero divisors* ($1 < \gcd(a, 12) < 12$, $a \neq 0$): $\{2, 3, 4, 6, 8, 9, 10\}$. For instance, $3 \cdot 4 = 12 \equiv 0$, $6 \cdot 2 = 12 \equiv 0$.
3. *The zero element*: $\{0\}$.

Definition 0.3.5: Euler Totient Function

For $m \in \mathbb{Z}_{>0}$, the *Euler totient function* $\varphi(m)$ is the number of integers a with $1 \leq a \leq m$ and $\gcd(a, m) = 1$. Equivalently, $\varphi(m) = |(\mathbb{Z}/m\mathbb{Z})^\times|$.

Example 0.3.6: Small Values of φ

1. $\varphi(1) = 1$, since $\gcd(1, 1) = 1$.
2. $\varphi(p) = p - 1$ for any prime p, since every integer from 1 to $p - 1$ is coprime to p.
3. $\varphi(12) = |\{1, 5, 7, 11\}| = 4$, as these are the residues coprime to 12.
4. $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$ for a prime power: among $1, \ldots, p^k$, the multiples of p are $p, 2p, \ldots, p^{k-1} \cdot p$, and there are p^{k-1} of them.

The multiplicativity of φ is a consequence of the Chinese Remainder Theorem, stated next.

Theorem 0.3.7: Chinese Remainder Theorem

Let $m_1, m_2, \ldots, m_k$ be pairwise coprime positive integers, and set $M = m_1 m_2 \cdots m_k$. Then the map

$$\mathbb{Z}/M\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/m_1\mathbb{Z} \times \mathbb{Z}/m_2\mathbb{Z} \times \cdots \times \mathbb{Z}/m_k\mathbb{Z}, \qquad a \mapsto (a \bmod m_1,\ a \bmod m_2,\ \ldots,\ a \bmod m_k)$$

is a ring isomorphism.

The CRT says that specifying a residue class modulo M is the same as independently specifying a residue class modulo each m_i. This is the arithmetic analogue of the decomposition of a product of topological spaces into its factors, and it is the mechanism that allows global questions about $\mathbb{Z}/M\mathbb{Z}$ to be reduced to local questions about each $\mathbb{Z}/m_i\mathbb{Z}$.

***Proof*:**
Step 1: The map is a ring homomorphism. The reduction map $a \mapsto (a \bmod m_1, \ldots, a \bmod m_k)$ preserves addition and multiplication, since these operations in each factor are performed modulo m_i.

Step 2: Injectivity. If $a \equiv 0 \pmod{m_i}$ for all i, then $m_i \big| a$ for each i. Since the m_i are pairwise coprime, $M = m_1 \cdots m_k$ divides a (apply corollary 0.1.9 inductively: $m_1 \big| a$ and $\gcd(m_1, m_2) = 1$ together with $m_2 \big| a$ give $m_1 m_2 \big| a$; continuing gives $M \big| a$). Hence the kernel is trivial.

Step 3: Surjectivity. Both sides have the same cardinality $M = m_1 \cdots m_k$, so injectivity implies surjectivity.

Alternatively, surjectivity can be shown constructively. For each i, define $M_i = M/m_i$ and note that $\gcd(M_i, m_i) = 1$ (since the m_j for $j \neq i$ are all coprime to m_i). By theorem 0.1.8, there exists y_i with $M_i y_i \equiv 1 \pmod{m_i}$. Given a target $(a_1, \ldots, a_k)$, the integer

$$a = \sum_{i=1}^{k} a_i M_i y_i$$

satisfies $a \equiv a_i \pmod{m_i}$ for each i, since $M_j y_j \equiv 0 \pmod{m_i}$ when $j \neq i$ (as $m_i \big| M_j$) and $M_i y_i \equiv 1 \pmod{m_i}$.

The constructive proof is useful in practice: it gives an explicit formula for "gluing" local data into a global solution. The abstract proof via cardinality counting is cleaner but less informative.

Example 0.3.8: Solving a System of Congruences

Solve the system $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$. Here $M = 105$, $M_1 = 35$, $M_2 = 21$, $M_3 = 15$. The inverses are: $35y_1 \equiv 1 \pmod 3$, giving $y_1 = 2$ (since $35 \cdot 2 = 70 \equiv 1$); $21y_2 \equiv 1 \pmod 5$, giving $y_2 = 1$ (since $21 \equiv 1$); $15y_3 \equiv 1 \pmod 7$, giving $y_3 = 1$ (since $15 \equiv 1$). Then

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$$

One verifies: $23 = 7 \cdot 3 + 2 \equiv 2 \pmod 3$, $23 = 4 \cdot 5 + 3 \equiv 3 \pmod 5$, $23 = 3 \cdot 7 + 2 \equiv 2 \pmod 7$.

An immediate corollary of the CRT is the multiplicativity of the Euler totient.

Corollary 0.3.9: Multiplicativity of φ

If $\gcd(m, n) = 1$, then $\varphi(mn) = \varphi(m)\varphi(n)$.

***Proof*:**
By theorem 0.3.7, $(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$, since the isomorphism $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ restricts to an isomorphism on unit groups (an element is a unit in the product if and only if each component is a unit). Taking cardinalities: $\varphi(mn) = \varphi(m)\varphi(n)$.

Combined with the prime power formula $\varphi(p^k) = p^{k-1}(p-1)$ from example 0.3.6, this gives a complete formula for φ.

Proposition 0.3.10: Formula for the Euler Totient

For any $n = p_1^{a_1} \cdots p_k^{a_k} > 1$:

$$\varphi(n) = n \prod_{p \big| n} \left(1 - \frac{1}{p}\right) = \prod_{i=1}^{k} p_i^{a_i - 1}(p_i - 1)$$

***Proof*:**
By corollary 0.3.9 and induction, $\varphi(n) = \prod_{i=1}^{k} \varphi(p_i^{a_i}) = \prod_{i=1}^{k} p_i^{a_i-1}(p_i - 1)$. Factoring out $p_i^{a_i}$ from each term gives the product formula $\varphi(n) = n \prod_{p \big| n}(1 - 1/p)$.

Example 0.3.11: Computing φ
$\varphi(360) = \varphi(2^3 \cdot 3^2 \cdot 5) = 360 \cdot (1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{5}) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 96.$

The section closes with the two classical results that connect the Euler totient to modular exponentiation. These are the finite-group-theoretic facts that underlie the convergence arguments for Dirichlet series twisted by characters.

Theorem 0.3.12: Euler's Theorem

If $\gcd(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$.

***Proof*:**
The residue class of a is an element of the group $(\mathbb{Z}/m\mathbb{Z})^\times$, which has order $\varphi(m)$. By Lagrange's theorem (see [5]), the order of every element divides the order of the group. Hence $a^{\varphi(m)} \equiv 1 \pmod{m}$.

Euler's theorem is a statement about the group $(\mathbb{Z}/m\mathbb{Z})^\times$: every element has finite order dividing $\varphi(m)$. The special case $m = p$ prime gives the result traditionally attributed to Fermat.

Corollary 0.3.13: Fermat's Little Theorem

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Equivalently, $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

***Proof*:**
Since $\varphi(p) = p - 1$, the first statement is a special case of theorem 0.3.12. For the second: if $p \middle| a$ then $a^p \equiv 0 \equiv a \pmod{p}$; if $p \nmid a$ then $a^p = a \cdot a^{p-1} \equiv a \cdot 1 = a \pmod{p}$.

Another classical characterization of primes via modular arithmetic is Wilson's theorem.

Proposition 0.3.14: Wilson's Theorem

An integer $n \geq 2$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$.

***Proof*:**
If n is composite, say $n = ab$ with $1 < a \leq b < n$, then a appears as a factor in $(n-1)!$. If $a \neq b$, both appear, and $n = ab$ divides $(n-1)!$, so $(n-1)! \equiv 0 \not\equiv -1 \pmod{n}$. If $a = b$, then $n = a^2 \geq 4$ (since $a \geq 2$), and both a and $2a$ appear in $\{1, \ldots, n-1\}$ when $n > 4$; the case $n = 4$ is verified directly: $3! = 6 \equiv 2 \not\equiv -1 \pmod{4}$.

If $n = p$ is prime, then $\mathbb{F}_p^\times = \{1, 2, \ldots, p-1\}$ is a group of order $p - 1$. Each element a pairs with its inverse a^{-1}. The elements satisfying $a = a^{-1}$, i.e., $a^2 \equiv 1 \pmod{p}$, are exactly $a \equiv \pm 1 \pmod{p}$ (since $p \middle| (a-1)(a+1)$ forces $p \middle| (a-1)$ or $p \middle| (a+1)$). Hence in the product $1 \cdot 2 \cdots (p-1)$, all elements except 1 and $p - 1$ cancel in pairs, leaving $(p-1)! \equiv 1 \cdot (p-1) = p - 1 \equiv -1 \pmod{p}$.

Wilson's theorem gives an exact criterion for primality, but it is computationally impractical since computing $(n-1)!$ modulo n requires $\Theta(n)$ multiplications. Its theoretical importance lies in providing explicit generators of ideals: the identity $(p-1)!+1 \equiv 0 \pmod{p}$ is used in the proof of quadratic reciprocity (section 0.5).

Fermat's little theorem has a useful contrapositive: if $a^{p-1} \not\equiv 1 \pmod{p}$, then p is not prime. This is the basis of primality testing algorithms. The converse fails: there exist composite numbers n (called *Carmichael numbers*) such that $a^{n-1} \equiv 1 \pmod{n}$ for all a coprime to n. The smallest is $n = 561 = 3 \cdot 11 \cdot 17$, which satisfies this property because $(p-1) \big| 560$ for each prime factor $p \in \{3, 11, 17\}$ (since $560 = 2 \cdot 280 = 10 \cdot 56 = 16 \cdot 35$), and the CRT then guarantees $a^{560} \equiv 1 \pmod{561}$ for all a with $\gcd(a, 561) = 1$. This criterion—known as *Korselt's criterion*—states that n is a Carmichael number if and only if n is squarefree and $(p-1) \big| (n-1)$ for every prime p dividing n.

Example 0.3.15: Modular Exponentiation via Euler's Theorem

To compute $7^{222} \bmod 10$, note that $\varphi(10) = 4$ and $\gcd(7, 10) = 1$, so $7^4 \equiv 1 \pmod{10}$ by theorem 0.3.12. Since $222 = 4 \cdot 55 + 2$:

$$7^{222} = (7^4)^{55} \cdot 7^2 \equiv 1^{55} \cdot 49 \equiv 9 \pmod{10}$$

The last digit of 7^{222} is therefore 9. More generally, the last digit of 7^n cycles with period 4: $7, 9, 3, 1, 7, 9, 3, 1, \ldots$ —a consequence of 7 having order exactly 4 in $(\mathbb{Z}/10\mathbb{Z})^\times$.

A complementary identity that counts divisors weighted by the totient is the following.

Proposition 0.3.16: Totient Summation Identity

For every positive integer n:

$$\sum_{d \big| n} \varphi(d) = n$$

***Proof*:**

Partition the set $\{1, 2, \ldots, n\}$ according to $\gcd(k, n)$. For each divisor d of n, the integers $k \in \{1, \ldots, n\}$ with $\gcd(k, n) = d$ are exactly those of the form $k = d\ell$ where $1 \leq \ell \leq n/d$ and $\gcd(\ell, n/d) = 1$. There are $\varphi(n/d)$ such integers. Summing over all divisors d of n gives $n = \sum_{d \big| n} \varphi(n/d)$. Since $d \mapsto n/d$ is a bijection on the divisors of n, this equals $\sum_{d \big| n} \varphi(d)$.

This identity will reappear in chapter 1 as the Dirichlet convolution identity $\varphi * \mathbf{1} = \mathrm{id}$, where $\mathbf{1}$ is the constant function 1 and id is the identity function $n \mapsto n$. Möbius inversion then recovers φ from this relation.

0.4 The Group $(\mathbb{Z}/m\mathbb{Z})^\times$ and Primitive Roots

Euler's theorem (theorem 0.3.12) guarantees that every element of $(\mathbb{Z}/m\mathbb{Z})^\times$ has finite multiplicative order dividing $\varphi(m)$. A natural question is whether this bound is achieved: does $(\mathbb{Z}/m\mathbb{Z})^\times$ always contain an element of order exactly $\varphi(m)$? Equivalently, is $(\mathbb{Z}/m\mathbb{Z})^\times$ cyclic? The answer is no in general—$(\mathbb{Z}/8\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has exponent 2, so no element has order $\varphi(8) = 4$—but a precise characterization exists. Understanding the group structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ is essential for the theory of Dirichlet characters in chapter 7, since the characters of a finite abelian group are determined by its invariant factor decomposition.

Definition 0.4.1: Multiplicative Order

For $a \in (\mathbb{Z}/m\mathbb{Z})^\times$, the *multiplicative order* of a modulo m, denoted $\operatorname{ord}_m(a)$, is the smallest positive integer k such that $a^k \equiv 1 \pmod{m}$.

By theorem 0.3.12, $\operatorname{ord}_m(a)$ exists and divides $\varphi(m)$. The order is the period of the cyclic subgroup $\langle a \rangle = \{1, a, a^2, \ldots, a^{k-1}\}$ inside $(\mathbb{Z}/m\mathbb{Z})^\times$, and the standard theory of cyclic groups (see [5]) gives the following.

Proposition 0.4.2: Basic Properties of Multiplicative Order

Let $a \in (\mathbb{Z}/m\mathbb{Z})^\times$ with $\operatorname{ord}_m(a) = k$. Then:

1. $a^n \equiv 1 \pmod{m}$ if and only if $k \big| n$.
2. $\operatorname{ord}_m(a^j) = k/\gcd(j,k)$.
3. The number of elements of order k in $(\mathbb{Z}/m\mathbb{Z})^\times$ is either 0 or $\varphi(k)$.

***Proof*:**
For (1): if $k\big|n$ then $a^n = (a^k)^{n/k} \equiv 1$. Conversely, write $n = kq + r$ with $0 \le r < k$; then $a^r = a^n \cdot (a^k)^{-q} \equiv 1$, and minimality of k forces $r = 0$.

For (2): let $d = \gcd(j,k)$. Then $(a^j)^{k/d} = (a^k)^{j/d} \equiv 1$, so $\operatorname{ord}_m(a^j)$ divides k/d. If $(a^j)^n \equiv 1$ then $k\big|jn$ by (1), hence $(k/d)\big|(j/d)n$. Since $\gcd(j/d, k/d) = 1$, corollary 0.1.9 gives $(k/d)\big|n$. Hence $\operatorname{ord}_m(a^j) = k/d$.

For (3): if some element a has order k, then $\langle a \rangle$ contains exactly one element of each order dividing k, and by (2) the number of generators of $\langle a \rangle$ (i.e., elements of order k in $\langle a \rangle$) is $\varphi(k)$. To see that no element outside $\langle a \rangle$ has order k: the polynomial $x^k - 1$ has at most k roots in $\mathbb{Z}/p\mathbb{Z}$ (since $\mathbb{Z}/p\mathbb{Z}$ is a field), and the k elements of $\langle a \rangle$ already exhaust them. For composite m, the argument requires the structure theorem below.

Part (3) uses a critical property of fields: a polynomial of degree k over a field has at most k roots. This fails over $\mathbb{Z}/m\mathbb{Z}$ when m is composite—for example, $x^2 - 1 \equiv 0 \pmod{8}$ has four solutions $\{1, 3, 5, 7\}$—and this failure is precisely why the counting argument in the existence theorem below works only for prime moduli.

Definition 0.4.3: Primitive Root

A *primitive root* modulo m is an element $g \in (\mathbb{Z}/m\mathbb{Z})^\times$ of order $\varphi(m)$, i.e., a generator of $(\mathbb{Z}/m\mathbb{Z})^\times$. Equivalently, $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic if and only if a primitive root exists.

When a primitive root g exists, every unit $a \in (\mathbb{Z}/m\mathbb{Z})^\times$ can be written as $a \equiv g^k \pmod{m}$ for a unique $k \in \{0, 1, \dots, \varphi(m) - 1\}$. This integer k is the *discrete logarithm* of a to base g, and the map $a \mapsto k$ is an isomorphism $(\mathbb{Z}/m\mathbb{Z})^\times \xrightarrow{\sim} \mathbb{Z}/\varphi(m)\mathbb{Z}$ that converts multiplication to addition.

Example 0.4.4: The Discrete Logarithm Table Modulo 13

The group $(\mathbb{Z}/13\mathbb{Z})^\times$ has order 12. Testing $g = 2$: the successive powers are

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8, \quad 2^4 = 3, \quad 2^5 = 6, \quad 2^6 = 12,$$

$$2^7 = 11, \quad 2^8 = 9, \quad 2^9 = 5, \quad 2^{10} = 10, \quad 2^{11} = 7, \quad 2^{12} = 1,$$

confirming that 2 is a primitive root. The discrete logarithm $\log_2(a)$ mod 12 converts multiplication in $(\mathbb{Z}/13\mathbb{Z})^\times$ to addition in $\mathbb{Z}/12\mathbb{Z}$. For instance, $5 \cdot 8 = 40 \equiv 1 \pmod{13}$, which corresponds to $\log_2(5) + \log_2(8) = 9 + 3 = 12 \equiv 0 \pmod{12}$. This conversion is the mechanism behind index calculus methods for computing in $(\mathbb{Z}/p\mathbb{Z})^\times$.

The following theorem characterizes exactly when primitive roots exist.

Theorem 0.4.5: Existence of Primitive Roots

The group $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic (i.e., a primitive root modulo m exists) if and only if $m \in \{1, 2, 4, p^k, 2p^k\}$ where p is an odd prime and $k \geq 1$.

The proof has two parts: showing cyclicity for odd prime powers (the hard direction), and showing non-cyclicity for all other moduli.

***Proof*:**

The cases $m = 1$ and $m = 2$ are trivial ($(\mathbb{Z}/1\mathbb{Z})^\times$ and $(\mathbb{Z}/2\mathbb{Z})^\times$ are both trivial). The case $m = 4$ is checked directly: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ is cyclic of order 2, with primitive root 3.

Step 1: $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for p an odd prime. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $p-1$. For each divisor d of $p-1$, let $\psi(d)$ denote the number of elements of order d. Every element has some order dividing $p-1$, so $\sum_{d|(p-1)} \psi(d) = p-1$. By proposition 0.4.2(3), $\psi(d) \in \{0, \varphi(d)\}$ for each d. Since $\sum_{d|(p-1)} \varphi(d) = p - 1$ (by proposition 0.3.16), and $\psi(d) \leq \varphi(d)$ for all d, the only possibility is $\psi(d) = \varphi(d)$ for every d. In particular, $\psi(p-1) = \varphi(p-1) \geq 1$, so an element of order $p-1$ exists.

Step 2: $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic for p odd and $k \geq 2$. Let g be a primitive root modulo p. The group $(\mathbb{Z}/p^k\mathbb{Z})^\times$ has order $\varphi(p^k) = p^{k-1}(p-1)$. The reduction map $(\mathbb{Z}/p^k\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$ is surjective with kernel of order p^{k-1}, so $\operatorname{ord}_{p^k}(g) = p^j(p-1)$ for some $0 \leq j \leq k-1$. It suffices to show $j = k-1$.

The key observation is: if $h \equiv 1 + tp \pmod{p^2}$ with $p \nmid t$, then $\operatorname{ord}_{p^k}(h) = p^{k-1}$. To see this, use induction. The binomial theorem gives

$$h^p = (1 + tp)^p = 1 + tp^2 + \binom{p}{2} t^2 p^2 + \cdots \equiv 1 + t' p^2 \pmod{p^3}$$

where $t' \equiv t \pmod{p}$ (since $p \,\big|\, \binom{p}{2}$ for $p \geq 3$), so $p \nmid t'$. Inductively, $h^{p^{j-1}} \equiv 1 + t_j p^j \pmod{p^{j+1}}$ with $p \nmid t_j$ for each j. In particular, $h^{p^{k-2}} \not\equiv 1 \pmod{p^k}$ while $h^{p^{k-1}} \equiv 1 \pmod{p^k}$, giving

$\operatorname{ord}_{p^k}(h) = p^{k-1}$.

Applying this to $h = g^{p-1}$: if $g^{p-1} \not\equiv 1 \pmod{p^2}$, write $g^{p-1} = 1 + tp$ with $p \nmid t$, and the claim gives $\operatorname{ord}_{p^k}(g^{p-1}) = p^{k-1}$. Then $\operatorname{ord}_{p^k}(g) = p^{k-1}(p-1) = \varphi(p^k)$, so g is a primitive root modulo p^k. If instead $g^{p-1} \equiv 1 \pmod{p^2}$, replace g by $g' = g + p$ (which reduces to g modulo p, hence is still a primitive root modulo p). The binomial expansion gives $(g')^{p-1} \equiv g^{p-1} + (p-1)g^{p-2}p \equiv 1 - g^{p-2}p \pmod{p^2}$. Since $p \nmid g^{p-2}$, this shows $(g')^{p-1} \not\equiv 1 \pmod{p^2}$, so the argument applies to g'.

Step 3: $(\mathbb{Z}/2p^k\mathbb{Z})^\times$ is cyclic. By the Chinese Remainder Theorem (theorem 0.3.7), $(\mathbb{Z}/2p^k\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^\times \times (\mathbb{Z}/p^k\mathbb{Z})^\times \cong \{1\} \times (\mathbb{Z}/p^k\mathbb{Z})^\times$, which is cyclic.

Step 4: Non-cyclicity for other moduli. The remaining cases are handled by proposition 0.4.8 and proposition 0.4.9 below: $(\mathbb{Z}/2^k\mathbb{Z})^\times$ is not cyclic for $k \geq 3$, and the CRT decomposition for m with two or more distinct odd prime factors always produces a non-cyclic product (since each factor $(\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$ has even order).

The proof that $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic uses a counting argument that avoids constructing a primitive root explicitly. Finding a primitive root modulo a given prime is a computational problem with no known polynomial-time deterministic algorithm, though efficient probabilistic methods exist.

Example 0.4.6: Primitive Roots for Small Primes

1. Modulo 7: $(\mathbb{Z}/7\mathbb{Z})^\times$ has order 6. Testing $g = 3$: the powers $3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1$ exhaust all of $(\mathbb{Z}/7\mathbb{Z})^\times$, so 3 is a primitive root. The other primitive roots are the elements of order 6: by proposition 0.4.2(2), these are 3^j with $\gcd(j, 6) = 1$, giving $3^1 = 3$ and $3^5 = 5$. Hence $(\mathbb{Z}/7\mathbb{Z})^\times$ has $\varphi(6) = 2$ primitive roots.

2. Modulo 11: $\varphi(11) = 10$. Testing $g = 2$: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 5$, $2^5 = 10$, $2^6 = 9$, $2^7 = 7$, $2^8 = 3$, $2^9 = 6$, $2^{10} = 1$. So 2 is a primitive root modulo 11. There are $\varphi(10) = 4$ primitive roots: $2, 2^3 = 8, 2^7 = 7, 2^9 = 6$.

A fixed integer $a > 1$ that is not a perfect square is a primitive root modulo infinitely many primes—this is a conjecture of Artin (1927), still open in general. Hooley proved in 1967 that the conjecture follows from the Generalized Riemann Hypothesis. For instance, 2 is conjectured to be a primitive root modulo a positive proportion (approximately 37.4%) of all primes. The unconditional result closest to Artin's conjecture, due to Gupta-Murty and Heath-Brown, shows that at least one of 2, 3, or 5 is a primitive root modulo infinitely many primes.

Primitive roots also arise in a familiar context: the decimal expansion of $1/p$ for a prime $p \neq 2, 5$ has period exactly $\operatorname{ord}_p(10)$. When 10 is a primitive root modulo p, the period is $p - 1$, the maximum possible. For example, $1/7 = 0.\overline{142857}$ has period $6 = \varphi(7)$ because $\operatorname{ord}_7(10) = 6$ (i.e., 10 is a primitive root modulo 7), while $1/13 = 0.\overline{076923}$ also has maximal period $12 = \varphi(13)$. On the other hand, $1/11 = 0.\overline{09}$ has period 2 because $\operatorname{ord}_{11}(10) = 2$.

Example 0.4.7: Lifting a Primitive Root from p to p^2

Take $p = 7$ and $g = 3$, a primitive root modulo 7. Is 3 a primitive root modulo 49? The group $(\mathbb{Z}/49\mathbb{Z})^\times$ has order $\varphi(49) = 42 = 6 \cdot 7$, so the question reduces to checking whether $3^6 \equiv 1 \pmod{49}$. Computing: $3^6 = 729 = 14 \cdot 49 + 43$, so $3^6 \equiv 43 \equiv 1 + 42 \equiv 1 + 6 \cdot 7 \pmod{49}$. Since $3^6 \not\equiv 1 \pmod{49}$ and the coefficient 6 satisfies $7 \nmid 6$, the lifting argument from Step 2 applies: 3 is a primitive root modulo 49, and modulo 7^k for all $k \geq 1$.

On the other hand, take $p = 29$ and $g = 14$, a primitive root modulo 29. Computing $14^{28} \pmod{29^2}$: since $14^{28} \equiv 1 \pmod{29}$ (Fermat), write $14^{28} = 1 + 29t$. If $29|t$, then 14 does not lift, and one must replace it by $14 + 29 = 43$. In practice, such "exceptional" primitive roots are rare; for most primes p and most primitive roots g modulo p, the condition $g^{p-1} \not\equiv 1 \pmod{p^2}$ holds.

For moduli that do *not* admit primitive roots, the structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ is determined by the following two results.

Proposition 0.4.8: Structure of $(\mathbb{Z}/2^k\mathbb{Z})^\times$

For $k \geq 3$: $(\mathbb{Z}/2^k\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$, generated by -1 and 5.

***Proof*:**

The group $(\mathbb{Z}/2^k\mathbb{Z})^\times$ has order $\varphi(2^k) = 2^{k-1}$. The element -1 has order 2, and 5 has order 2^{k-2}: to verify the latter, write $5 = 1 + 4$ and use induction. If $5^{2^{j-1}} \equiv 1 + c_j \cdot 2^{j+1} \pmod{2^{j+2}}$ with c_j odd, then

$$5^{2^j} = (5^{2^{j-1}})^2 \equiv 1 + c_j 2^{j+2} + c_j^2 2^{2j+2} \equiv 1 + c_j 2^{j+2} \pmod{2^{j+3}}$$

with c_j still odd (since only the linear term contributes modulo 2^{j+3} when $2j + 2 \geq j + 3$, i.e., $j \geq 1$). The base case $j = 0$: $5 = 1 + 1 \cdot 2^2$ with $c_0 = 1$. Inductively, $5^{2^{k-3}} \equiv 1 + c_{k-3} \cdot 2^{k-1} \not\equiv 1 \pmod{2^k}$, while $5^{2^{k-2}} \equiv 1 \pmod{2^k}$. Hence $\operatorname{ord}_{2^k}(5) = 2^{k-2}$.

Since $5^j \equiv 1 \pmod 4$ for all j (as $5 \equiv 1 \pmod 4$) while $-1 \equiv 3 \pmod 4$, no power of 5 equals -1. Hence $\langle -1 \rangle \cap \langle 5 \rangle = \{1\}$, and the product $\langle -1 \rangle \times \langle 5 \rangle$ has order $2 \cdot 2^{k-2} = 2^{k-1} = |(\mathbb{Z}/2^k\mathbb{Z})^\times|$, giving the isomorphism.

Proposition 0.4.9: Structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ for General m

For $m = 2^a p_1^{a_1} \cdots p_r^{a_r}$ with $p_1, \ldots, p_r$ distinct odd primes:

$$(\mathbb{Z}/m\mathbb{Z})^\times \cong (\mathbb{Z}/2^a\mathbb{Z})^\times \times \prod_{i=1}^{r} (\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$$

Each odd prime-power factor is cyclic of order $p_i^{a_i-1}(p_i - 1)$. A primitive root modulo m exists if and only if the product is cyclic, which happens exactly when $m \in \{1, 2, 4, p^k, 2p^k\}$.

***Proof*:**
The decomposition is the CRT applied to unit groups, as in corollary 0.3.9. A direct product of cyclic groups is cyclic if and only if their orders are pairwise coprime (see [5]). If m has two distinct odd prime factors p_1, p_2, then $\varphi(p_1^{a_1})$ and $\varphi(p_2^{a_2})$ are both even, so the product is not cyclic. Similarly, if $m = 2^a p^k$ with $a \geq 2$, then both $\varphi(2^a)$ and $\varphi(p^k)$ are even, ruling out cyclicity. The surviving cases are $m \in \{1, 2, 4, p^k, 2p^k\}$.

Example 0.4.10: Non-Cyclic Unit Groups

1. $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$. Every element satisfies $a^2 \equiv 1 \pmod 8$, so the group has exponent 2 and $(\mathbb{Z}/8\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, the Klein four-group. The generators of the two $\mathbb{Z}/2\mathbb{Z}$ factors are $-1 = 7$ and 5: indeed $\{(\pm 1)^a \cdot 5^b : a \in \{0,1\},\ b \in \{0,1\}\} = \{1, 7, 5, 3\}$ gives all four elements. No primitive root exists.

2. $(\mathbb{Z}/15\mathbb{Z})^\times \cong (\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. This group has order 8 but exponent $\mathrm{lcm}(2, 4) = 4$, so it is not cyclic. The maximum order 4 is achieved by elements like 2 (since $2^1 = 2,\ 2^2 = 4,\ 2^3 = 8,\ 2^4 = 16 \equiv 1 \pmod{15}$).

3. $(\mathbb{Z}/24\mathbb{Z})^\times \cong (\mathbb{Z}/8\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^2 \times \mathbb{Z}/2\mathbb{Z} \cong (\mathbb{Z}/2\mathbb{Z})^3$. This has order 8 and exponent 2: every element is its own inverse. There are $2^3 - 1 = 7$ elements of order 2, confirming that the group is as far from cyclic as possible.

The *exponent* of $(\mathbb{Z}/m\mathbb{Z})^\times$—the least common multiple of the orders of all its elements—is captured by the Carmichael function.

Definition 0.4.11: Carmichael Function

The *Carmichael function* $\lambda(m)$ is the exponent of $(\mathbb{Z}/m\mathbb{Z})^\times$: the smallest positive integer e such that $a^e \equiv 1 \pmod m$ for all a with $\gcd(a, m) = 1$.

By the structure theorem, $\lambda(m) = \mathrm{lcm}(\lambda(p_1^{a_1}), \ldots, \lambda(p_r^{a_r}), \lambda(2^a))$, where $\lambda(p^a) = \varphi(p^a) = p^{a-1}(p-1)$ for odd primes, $\lambda(2) = 1$, $\lambda(4) = 2$, and $\lambda(2^a) = 2^{a-2}$ for $a \geq 3$. When $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic, $\lambda(m) = \varphi(m)$; otherwise $\lambda(m)$ is a proper divisor of $\varphi(m)$.

The Carmichael function refines Euler's theorem: $a^{\lambda(m)} \equiv 1 \pmod m$ for all a coprime to m, and $\lambda(m)$ is the *smallest* universal exponent with this property. When $\lambda(m) < \varphi(m)$, Euler's theorem with the exponent $\varphi(m)$ is valid but not sharp.

Example 0.4.12: Carmichael Function Computations

1. $\lambda(7) = \varphi(7) = 6$, since $(\mathbb{Z}/7\mathbb{Z})^\times$ is cyclic. The primitive root 3 achieves this order.

2. $\lambda(8) = 2$, since every element of $(\mathbb{Z}/8\mathbb{Z})^\times$ squares to 1, while $\varphi(8) = 4$. Euler's theorem gives $a^4 \equiv 1 \pmod 8$, but the Carmichael function shows $a^2 \equiv 1$ already suffices.

3. $\lambda(15) = \mathrm{lcm}(\lambda(3), \lambda(5)) = \mathrm{lcm}(2, 4) = 4$, while $\varphi(15) = 8$. The element 2 has order exactly 4.

4. $\lambda(100) = \mathrm{lcm}(\lambda(4), \lambda(25)) = \mathrm{lcm}(2, 20) = 20$, while $\varphi(100) = 40$. Every element coprime to 100 satisfies $a^{20} \equiv 1 \pmod{100}$.

The structure theorem has a direct consequence for character theory. The dual group $\widehat{(\mathbb{Z}/m\mathbb{Z})}^\times$ of group homomorphisms $(\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$ is (non-canonically) isomorphic to $(\mathbb{Z}/m\mathbb{Z})^\times$ itself (see [5]). When $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic, every character is determined by the image of a single generator; when it is not, specifying a character requires choosing the image of each invariant factor independently. The Dirichlet characters of modulus m (chapter 7) are the extensions of these group characters to all of $\mathbb{Z}$, and the invariant factor decomposition determines how many characters of each order exist.

Example 0.4.13: Orders and Characters Modulo 12

The group $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\} \cong (\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is the Klein four-group. Every non-identity element has order 2:

$$5^2 = 25 \equiv 1, \qquad 7^2 = 49 \equiv 1, \qquad 11^2 = 121 \equiv 1 \pmod{12}$$

The dual group $\widehat{(\mathbb{Z}/12\mathbb{Z})}^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has four characters $\chi_0, \chi_1, \chi_2, \chi_3$, each taking values in $\{\pm 1\}$. The trivial character sends every unit to 1. The three nontrivial characters correspond to the three subgroups of index 2 in $(\mathbb{Z}/12\mathbb{Z})^\times$:

$$\chi_1(n) = \begin{cases} 1 & n \equiv 1, 5 \\ -1 & n \equiv 7, 11 \end{cases}, \quad \chi_2(n) = \begin{cases} 1 & n \equiv 1, 7 \\ -1 & n \equiv 5, 11 \end{cases}, \quad \chi_3(n) = \begin{cases} 1 & n \equiv 1, 11 \\ -1 & n \equiv 5, 7 \end{cases}$$

modulo 12. Each nontrivial character has order 2 in the dual group. These Dirichlet characters of modulus 12 control the distribution of primes in residue classes modulo 12, as developed in chapter 7.

This example illustrates a general principle: the number of Dirichlet characters of modulus m equals $\varphi(m) = |(\mathbb{Z}/m\mathbb{Z})^\times|$, and the order structure of the character group mirrors that of $(\mathbb{Z}/m\mathbb{Z})^\times$. When the group is cyclic of order n, there is a character of every order dividing n; when it decomposes as a product $\mathbb{Z}/d_1\mathbb{Z} \times \cdots \times \mathbb{Z}/d_r\mathbb{Z}$, the characters decompose correspondingly.

0.5 Quadratic Residues and Reciprocity

The previous section determined when $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic. This section addresses a finer structural question: which elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ are perfect squares? The answer is governed by the Legendre symbol, and the relationship between squareness modulo different primes is the content of the law of quadratic reciprocity—one of the central results of elementary number theory. Quadratic reciprocity has far-reaching generalizations: Artin reciprocity in class field theory ([12]) subsumes it as a special case, and the Langlands program seeks non-abelian analogues. In this book, quadratic characters reappear in the study of Dirichlet L-functions (chapter 7) and the factorization of Dedekind zeta functions of quadratic fields (chapter 8).

Definition 0.5.1: Quadratic Residue

Let p be an odd prime. An integer a with $p \nmid a$ is a *quadratic residue* modulo p if there exists $x \in \mathbb{Z}$ with $x^2 \equiv a \pmod{p}$. Otherwise, a is a *quadratic non-residue* modulo p.

Since the squaring map $x \mapsto x^2$ on $(\mathbb{Z}/p\mathbb{Z})^\times$ sends x and $-x$ to the same value, and $x \neq -x$ in $(\mathbb{Z}/p\mathbb{Z})^\times$ for p odd (as $2x \neq 0$), the image of squaring has exactly $(p-1)/2$ elements. Hence exactly half the nonzero residues modulo p are quadratic residues, and half are non-residues.

Example 0.5.2: Quadratic Residues Modulo Small Primes

1. Modulo 7: the squares are $1^2 = 1$, $2^2 = 4$, $3^2 = 2$, so the quadratic residues are $\{1, 2, 4\}$ and the non-residues are $\{3, 5, 6\}$.
2. Modulo 13: the squares are $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 3$, $5^2 = 12$, $6^2 = 10$, giving quadratic residues $\{1, 3, 4, 9, 10, 12\}$ and non-residues $\{2, 5, 6, 7, 8, 11\}$.

The Legendre symbol packages the quadratic residue information into a multiplicative function.

Definition 0.5.3: Legendre Symbol

For an odd prime p and an integer a, the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p, \\ 0 & \text{if } p \big| a. \end{cases}$$

The Legendre symbol $\left(\frac{\cdot}{p}\right) : \mathbb{Z} \to \{-1, 0, 1\}$ is a Dirichlet character of modulus p (as defined in chapter 7): it is totally multiplicative and p-periodic. This is a consequence of the following criterion.

Proposition 0.5.4: Euler's Criterion

For an odd prime p and a with $p \nmid a$:

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$$

***Proof*:**
The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$ (theorem 0.4.5). Let g be a primitive root and write $a \equiv g^k \pmod{p}$. Then $a^{(p-1)/2} \equiv g^{k(p-1)/2} \pmod{p}$. Since g has order $p - 1$, the value $g^{k(p-1)/2}$ is 1 if $(p - 1)\big|k(p - 1)/2$, i.e., if k is even, and $g^{(p-1)/2} = -1$ if k is odd (as $g^{(p-1)/2}$ is a root of $x^2 - 1$ other than 1, hence equals -1).

Now, $a = g^k$ is a quadratic residue if and only if $a = (g^{k/2})^2$ for some integer $k/2$, i.e., if and only if k is even. Hence $\left(\frac{a}{p}\right) = 1$ when k is even and -1 when k is odd, matching $a^{(p-1)/2} \bmod p$.

Euler's criterion immediately implies the total multiplicativity of the Legendre symbol: $\left(\frac{ab}{p}\right) = (ab)^{(p-1)/2} = a^{(p-1)/2} \cdot b^{(p-1)/2} = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}$. Since both sides take values in $\{-1, 1\}$ and agree modulo $p > 2$, they are equal as integers. This multiplicativity reduces the computation of $\left(\frac{a}{p}\right)$ for general a to the case of prime arguments.

The Legendre symbol $\left(\frac{\cdot}{p}\right)$ is the unique nontrivial quadratic character of $(\mathbb{Z}/p\mathbb{Z})^\times$: it is the group homomorphism $(\mathbb{Z}/p\mathbb{Z})^\times \to \{\pm 1\}$ whose kernel is the subgroup of quadratic residues, an index-2 subgroup. In the language of chapter 7, it is the unique primitive Dirichlet character of modulus p and order 2. The Dirichlet L-function $L(s, \left(\frac{\cdot}{p}\right))$ controls the distribution of quadratic residues and non-residues among the primes, and its non-vanishing at $s = 1$ is one of the key inputs for Dirichlet's theorem on primes in arithmetic progressions.

The values of $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$ can be determined directly from Euler's criterion.

Proposition 0.5.5: First Supplement: -1 as a Quadratic Residue

For an odd prime p:

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 4, \\ -1 & \text{if } p \equiv 3 \pmod 4. \end{cases}$$

***Proof*:**

By Euler's criterion, $\left(\frac{-1}{p}\right) \equiv (-1)^{(p-1)/2} \pmod p$. Since $(p-1)/2$ is even when $p \equiv 1 \pmod 4$ and odd when $p \equiv 3 \pmod 4$, the result follows.

The first supplement has a clean interpretation: -1 is a square modulo p precisely when $(\mathbb{Z}/p\mathbb{Z})^\times$ has an element of order 4, which happens if and only if $4 \mid (p-1)$.

Proposition 0.5.6: Second Supplement: 2 as a Quadratic Residue

For an odd prime p:

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod 8, \\ -1 & \text{if } p \equiv \pm 3 \pmod 8. \end{cases}$$

***Proof*:**

The exponent $(p^2-1)/8 = (p-1)(p+1)/8$. When $p \equiv 1 \pmod 8$, both $(p-1)/2$ and $(p+1)/2$ are even, so the product is divisible by 4 and $(p^2-1)/8$ is even. When $p \equiv 3 \pmod 8$, then $(p-1)/2 = 1 + 4k$ for some k, and $(p+1)/4 = 1 + 2k$ is odd; working through the cases gives $(p^2-1)/8$ odd. The remaining cases $p \equiv 5, 7 \pmod 8$ follow by symmetry. Alternatively, this can be deduced from Gauss's lemma (lemma 0.5.7 below) applied to $a = 2$.

Both supplements can also be proved using *Gauss sums*: the quantity $\tau(\chi) = \sum_{n=1}^{p-1} \left(\frac{n}{p}\right) e^{2\pi i n/p}$, which satisfies $\tau(\chi)^2 = (-1)^{(p-1)/2} p$. Gauss sums are the Fourier-analytic tool that connects quadratic residues to exponential sums, and they play a central role in the functional equations of Dirichlet L-functions (chapter 7).

The central tool for proving quadratic reciprocity is the following counting lemma.

Lemma 0.5.7: Gauss's Lemma

Let p be an odd prime and a an integer with $p \nmid a$. For each $k \in \{1, 2, \dots, (p-1)/2\}$, let r_k be the representative of $ka \bmod p$ in $\{-(p-1)/2, \dots, -1, 1, \dots, (p-1)/2\}$, and let n be the number of indices k for which $r_k < 0$. Then

$$\left(\frac{a}{p}\right) = (-1)^n$$

***Proof*:**
The values $|r_1|, |r_2|, \dots, |r_{(p-1)/2}|$ are a permutation of $\{1, 2, \dots, (p-1)/2\}$: they are $(p-1)/2$ distinct elements of this set (distinct because $r_i = \pm r_j$ with $i \neq j$ would give $a(i \pm j) \equiv 0 \pmod p$, contradicting $p \nmid a$ and $0 < i \pm j < p$). Taking the product:

$$\prod_{k=1}^{(p-1)/2} r_k = (-1)^n \prod_{k=1}^{(p-1)/2} |r_k| = (-1)^n \prod_{k=1}^{(p-1)/2} k = (-1)^n \left(\frac{p-1}{2}\right)!$$

On the other hand, $r_k \equiv ka \pmod p$, so $\prod r_k \equiv a^{(p-1)/2} \prod_{k=1}^{(p-1)/2} k \pmod p$. Comparing with the line above and cancelling $((p-1)/2)!$ (which is nonzero modulo p) gives $a^{(p-1)/2} \equiv (-1)^n \pmod p$. The result follows from Euler's criterion.

Gauss's lemma converts the algebraic question "is a a square modulo p?" into the combinatorial question "how many of the reductions $a, 2a, \dots, \frac{p-1}{2}a$ modulo p land in the upper half $\{(p+1)/2, \dots, p-1\}$?" This geometric reformulation is the key to Eisenstein's proof of quadratic reciprocity.

Example 0.5.8: Gauss's Lemma in Action
Take $a = 5$ and $p = 11$. The set $\{ka \bmod 11 : k = 1, \dots, 5\}$ is $\{5, 10, 4, 9, 3\}$. Representing these in $\{-5, \dots, -1, 1, \dots, 5\}$: $5 \mapsto 5$, $10 \mapsto -1$, $4 \mapsto 4$, $9 \mapsto -2$, $3 \mapsto 3$. The negative values are $\{-1, -2\}$, so $n = 2$ and $\left(\frac{5}{11}\right) = (-1)^2 = 1$. Indeed, $4^2 = 16 \equiv 5 \pmod{11}$, confirming that 5 is a quadratic residue.

Theorem 0.5.9: Quadratic Reciprocity

Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

Equivalently: $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ unless $p \equiv q \equiv 3 \pmod 4$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

Quadratic reciprocity relates the solvability of $x^2 \equiv p \pmod q$ to that of $x^2 \equiv q \pmod p$—two problems that, a priori, have nothing to do with each other. The sign $(-1)^{(p-1)(q-1)/4}$ is $+1$ unless both primes are congruent to 3 modulo 4, in which case the Legendre symbols "flip." This single sign change is the entire content of the reciprocity law.

***Proof*:**
By Gauss's lemma, $\left(\frac{q}{p}\right) = (-1)^{n_p}$ where $n_p = \#\{k \in \{1, \dots, (p-1)/2\} \mid kq \bmod p > p/2\}$. Since $kq \bmod p = kq - \lfloor kq/p \rfloor p$, the condition $kq \bmod p > p/2$ is equivalent to $kq - \lfloor kq/p \rfloor p > p/2$, which rearranges to $\lfloor kq/p \rfloor < kq/p - 1/2$. Counting more carefully:

$$n_p = \sum_{k=1}^{(p-1)/2} \left\lfloor \frac{kq}{p} \right\rfloor - 2 \sum_{k=1}^{(p-1)/2} \left\lfloor \frac{kq}{2p} \right\rfloor$$

Eisenstein's approach avoids this bookkeeping by counting lattice points. Consider the rectangle $R = (0, p/2) \times (0, q/2)$ in $\mathbb{R}^2$. The number of lattice points $(x, y) \in \mathbb{Z}^2$ with $0 < x < p/2$ and $0 < y < qx/p$ is $\sum_{x=1}^{(p-1)/2} \lfloor qx/p \rfloor$. Similarly, the number with $0 < y < q/2$ and $0 < x < py/q$ is $\sum_{y=1}^{(q-1)/2} \lfloor py/q \rfloor$. Since p and q are distinct odd primes, no lattice point (x, y) with $0 < x < p/2$ and $0 < y < q/2$ lies on the diagonal $qx = py$ (as this would require $p \mid x$ or $q \mid y$, contradicting the bounds). Hence every lattice point in $R \cap \mathbb{Z}^2$ lies strictly above or strictly below the diagonal, giving:

$$\sum_{x=1}^{(p-1)/2} \left\lfloor \frac{qx}{p} \right\rfloor + \sum_{y=1}^{(q-1)/2} \left\lfloor \frac{py}{q} \right\rfloor = \frac{p-1}{2} \cdot \frac{q-1}{2}$$

The left-hand side counts all lattice points in the open rectangle R: those below the diagonal contribute the first sum, and those above contribute the second. The total number of such points is $\frac{p-1}{2} \cdot \frac{q-1}{2}$.

By Gauss's lemma, one can verify that $(-1)^{n_p} = (-1)^{\sum_{x=1}^{(p-1)/2} \lfloor qx/p \rfloor}$ (since the parity of n_p matches that of this sum). Hence:

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{n_p + n_q} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

as claimed.

The lattice-point argument is due to Eisenstein and is among the most geometric of the many known proofs of quadratic reciprocity (over 240 proofs are known). The rectangle R and its diagonal will reappear—in a much more general form—in the lattice-point counting arguments used to prove the analytic class number formula in chapter 8.

Example 0.5.10: Lattice Point Verification
Take $p = 5$ and $q = 7$. The rectangle $R = (0, 5/2) \times (0, 7/2)$ contains lattice points with $1 \leq x \leq 2$ and $1 \leq y \leq 3$. The diagonal is $y = (7/5)x$. Below the diagonal: $(1, 1)$ (since $1 < 7/5$); above the diagonal: $(1, 2)$, $(1, 3)$, $(2, 2)$, $(2, 3)$ (since $2 > 14/5$), and $(2, 1)$ lies below (since $1 < 14/5$). Counting: $\sum_{x=1}^{2} \lfloor 7x/5 \rfloor = 1 + 2 = 3$ below, $\sum_{y=1}^{3} \lfloor 5y/7 \rfloor = 0 + 1 + 2 = 3$ above, total $6 = 2 \cdot 3 = \frac{p-1}{2} \cdot \frac{q-1}{2}$. Since 6 is even, $\left(\frac{5}{7}\right)\left(\frac{7}{5}\right) = 1$. One checks: $\left(\frac{5}{7}\right) = 5^3 = 125 \equiv 6 \equiv -1 \pmod 7$ and $\left(\frac{7}{5}\right) = 7^2 = 49 \equiv 4 \equiv -1 \pmod 5$, confirming the product is 1.

Example 0.5.11: Using Quadratic Reciprocity

1. Is 21 a quadratic residue modulo 71? By multiplicativity, $\left(\frac{21}{71}\right) = \left(\frac{3}{71}\right)\left(\frac{7}{71}\right)$. For $\left(\frac{3}{71}\right)$: since $71 \equiv 3 \pmod 4$ and $3 \equiv 3 \pmod 4$, reciprocity gives $\left(\frac{3}{71}\right) = -\left(\frac{71}{3}\right) = -\left(\frac{2}{3}\right) = -(-1) = 1$. For $\left(\frac{7}{71}\right)$: since $71 \equiv 3 \pmod 4$ and $7 \equiv 3 \pmod 4$, reciprocity gives $\left(\frac{7}{71}\right) = -\left(\frac{71}{7}\right) = -\left(\frac{1}{7}\right) = -1$. Hence $\left(\frac{21}{71}\right) = 1 \cdot (-1) = -1$: the integer 21 is a non-residue modulo 71.

2. Is 5 a quadratic residue modulo 23? Since $5 \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{5}{23}\right) = \left(\frac{23}{5}\right) = \left(\frac{3}{5}\right)$. Since $5 \equiv 1 \pmod 4$, applying reciprocity again: $\left(\frac{3}{5}\right) = \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1$ (by the second supplement, since $3 \equiv 3 \pmod 8$). Hence 5 is a non-residue modulo 23.

The Legendre symbol is defined only for prime moduli. The Jacobi symbol extends it to odd composite moduli and is the standard tool for efficient computation.

Definition 0.5.12: Jacobi Symbol

For an odd positive integer $n = p_1^{a_1} \cdots p_k^{a_k}$ (not necessarily prime) and any integer a, the *Jacobi symbol* is

$$\left(\frac{a}{n}\right) = \prod_{i=1}^{k} \left(\frac{a}{p_i}\right)^{a_i}$$

The Jacobi symbol inherits the multiplicativity properties of the Legendre symbol: $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$ and $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right)$. The reciprocity law, first supplement, and second supplement all extend to Jacobi symbols with identical formulas:

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}, \qquad \left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}, \qquad \left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$$

for odd coprime $m, n > 0$. These follow from the corresponding Legendre symbol identities by multiplying over the prime factorizations.

A caution: $\left(\frac{a}{n}\right) = 1$ does *not* imply that a is a quadratic residue modulo n when n is composite. For example, $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = (-1)(-1) = 1$, but $x^2 \equiv 2 \pmod{15}$ has no solution (since $x^2 \equiv 2 \pmod 3$ has no solution). The Jacobi symbol detects quadratic residuosity only modulo each prime factor individually.

The computational value of the Jacobi symbol lies in its reciprocity law: evaluating $\left(\frac{a}{n}\right)$ reduces to repeated application of reciprocity and the supplements, without factoring n. The resulting algorithm runs in $O(\log^2 n)$ time, analogous to the Euclidean algorithm for GCD.

Example 0.5.13: Computing a Jacobi Symbol

Evaluate $\left(\frac{1001}{9907}\right)$. Neither number need be prime. Since $9907 \equiv 3 \pmod 4$ and $1001 \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{1001}{9907}\right) = \left(\frac{9907}{1001}\right)$. Now $9907 = 9 \cdot 1001 + 898$, so $\left(\frac{9907}{1001}\right) = \left(\frac{898}{1001}\right) = \left(\frac{2}{1001}\right)\left(\frac{449}{1001}\right)$. Since $1001 \equiv 1 \pmod 8$, the second supplement gives $\left(\frac{2}{1001}\right) = 1$. For $\left(\frac{449}{1001}\right)$: since $449 \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{449}{1001}\right) = \left(\frac{1001}{449}\right) = \left(\frac{103}{449}\right)$ (as $1001 = 2 \cdot 449 + 103$). Since $103 \equiv 3 \pmod 4$ and $449 \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{103}{449}\right) = \left(\frac{449}{103}\right) = \left(\frac{37}{103}\right)$. Since $37 \equiv 1 \pmod 4$: $\left(\frac{37}{103}\right) = \left(\frac{103}{37}\right) = \left(\frac{29}{37}\right)$. Since $29 \equiv 1 \pmod 4$: $\left(\frac{29}{37}\right) = \left(\frac{37}{29}\right) = \left(\frac{8}{29}\right) = \left(\frac{2}{29}\right)^3 = (-1)^3 = -1$ (since $29 \equiv 5 \pmod 8$). Hence $\left(\frac{1001}{9907}\right) = -1$.

Quadratic reciprocity is the first in a hierarchy of reciprocity laws. *Cubic reciprocity* governs when a is a perfect cube modulo p and is naturally formulated over the Eisenstein integers $\mathbb{Z}[\zeta_3]$; *quartic reciprocity* governs fourth powers and lives over the Gaussian integers $\mathbb{Z}[i]$. These higher reciprocity laws were developed by Gauss, Jacobi, and Eisenstein in the early 19th century and are part of the motivation for studying number rings beyond $\mathbb{Z}$. The ultimate generalization—Artin reciprocity—unifies all abelian reciprocity laws into a single statement about the Galois group $\mathrm{Gal}(K^{\mathrm{ab}}/K)$ of the maximal abelian extension of a number field K, and is the subject of [12]. The quadratic case treated here, while elementary, already exhibits the key phenomenon: a local condition (squareness modulo p) is controlled by global arithmetic data (the residue classes of p and q modulo 4).

0.6 Asymptotic Notation

The results of analytic number theory are expressed in the language of asymptotics: statements about how functions behave as their argument grows without bound. The Prime Number Theorem, for instance, asserts that $\pi(x) \sim x/\log x$—a precise claim about the ratio $\pi(x)/(x/\log x)$ tending to 1, not about the difference $\pi(x) - x/\log x$. This section collects the standard asymptotic notations and their basic properties, establishing the conventions used throughout the book.

Definition 0.6.1: Big-O and Little-o

Let f and g be functions defined on a neighborhood of ∞ (or of some limit point), with g eventually positive.

1. $f(x) = O(g(x))$ as $x \to \infty$ means there exist constants $C > 0$ and x_0 such that $|f(x)| \le Cg(x)$ for all $x \ge x_0$. The notation $f \ll g$ is synonymous.
2. $f(x) = o(g(x))$ as $x \to \infty$ means $\lim_{x\to\infty} f(x)/g(x) = 0$, i.e., $|f(x)| \le \varepsilon\, g(x)$ for any $\varepsilon > 0$ and all sufficiently large x.

The big-O notation asserts a bound up to an unspecified multiplicative constant; the little-o notation asserts that the bound can be made arbitrarily small. Every $o(g)$ is also $O(g)$, but not conversely: $x = O(x)$ but $x \neq o(x)$.

A common source of confusion is that the "=" in $f = O(g)$ is not symmetric: it means "f belongs to the class of functions bounded by Cg," not that two specific quantities are equal. The notation is best read as "f is at most of order g." When multiple big-O terms appear in an equation, each is understood to represent a (possibly different) function satisfying the stated bound. For instance, $O(x) + O(x^2) = O(x^2)$ is a valid statement: any function bounded by Cx plus any function bounded by $C'x^2$ is bounded by $C''x^2$ for $C'' = C + C'$.

Definition 0.6.2: Vinogradov Notation

The notation $f \ll g$ means $f = O(g)$, and $f \gg g$ means $g = O(f)$. If both $f \ll g$ and $f \gg g$ hold, this is written $f \asymp g$ (read "f is of the same order as g").

The Vinogradov notation $\ll$ is standard in analytic number theory and avoids some of the awkwardness of the big-O symbol in chains of inequalities. For example, $\sum_{n\le x} d(n) \ll x\log x$ is cleaner than $\sum_{n\le x} d(n) = O(x\log x)$ when embedded in a longer computation. Both notations are used freely in this book; the choice in any given instance is dictated by readability.

A critical convention: the implied constant in $f \ll g$ or $f = O(g)$ may depend on parameters that are considered fixed. When the dependence matters, it is indicated by a subscript: $f \ll_\varepsilon g$ means the implied constant depends on ε, and $f = O_k(g)$ means it depends on k. Failing to track such dependencies is a common source of error in analytic arguments, particularly when a parameter later varies (e.g., when $\varepsilon \to 0$ or $k \to \infty$).

Definition 0.6.3: Asymptotic Equivalence

Two functions f and g, both eventually nonzero, are *asymptotically equivalent*, written $f(x) \sim g(x)$, if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1$$

Equivalently, $f \sim g$ if and only if $f = g + o(g)$, i.e., $f(x) = g(x)(1 + o(1))$.

Asymptotic equivalence is the relation used to state the main theorems of analytic number theory. The Prime Number Theorem says $\pi(x) \sim x/\log x$; Mertens' theorem says $\prod_{p\le x}(1-1/p) \sim e^{-\gamma}/\log x$. These are statements about the *leading-order* behavior of the function, ignoring lower-order corrections. A stronger statement specifies the error: $\pi(x) = \mathrm{Li}(x) + O(x\exp(-c\sqrt{\log x}))$ gives both the leading term and a bound on the remainder.

The relation $\sim$ is an equivalence relation on the set of eventually positive functions: it is reflexive ($f \sim f$), symmetric ($f \sim g$ implies $g \sim f$), and transitive ($f \sim g$ and $g \sim h$ imply $f \sim h$). It is also compatible with multiplication ($f_1 \sim g_1$ and $f_2 \sim g_2$ imply $f_1 f_2 \sim g_1 g_2$) but *not* with addition: $f_1 \sim g_1$ and $f_2 \sim g_2$ do not imply $f_1 + f_2 \sim g_1 + g_2$, since cancellation can occur. For instance, $(x+1) \sim x$ and $(-x) \sim (-x)$, but $(x+1) + (-x) = 1 \not\sim 0 = x + (-x)$.

Two further notations occasionally appear in the literature.

Definition 0.6.4: Ω and Θ Notation

1. $f(x) = \Omega(g(x))$ means $f(x) \neq o(g(x))$, i.e., $|f(x)| \geq c\, g(x)$ for some $c > 0$ and infinitely many $x \to \infty$. This is the negation of $f = o(g)$: the function f is "at least of order g" infinitely often.
2. $f(x) = \Theta(g(x))$ means $f \asymp g$, i.e., both $f = O(g)$ and $f = \Omega(g)$. This gives a tight two-sided bound: $c_1 g(x) \leq |f(x)| \leq c_2 g(x)$ for constants $0 < c_1 \leq c_2$ and all sufficiently large x.

The Ω notation is used in number theory to express *lower bounds* on oscillation. For example, the statement $\pi(x) - \mathrm{Li}(x) = \Omega(x^{1/2}\log\log\log x/\log x)$ (a theorem of Littlewood) asserts that the error in the Prime Number Theorem is not just $O(x^{1/2})$—it is at least as large as $x^{1/2}\log\log\log x/\log x$ infinitely often.

The following rules govern the manipulation of asymptotic expressions. They are used without comment throughout the book.

Proposition 0.6.5: Arithmetic of Asymptotic Notation

Let f, f_1, f_2, g, g_1, g_2 be functions with g, g_1, g_2 eventually positive. Then:

1. *Transitivity:* if $f = O(g)$ and $g = O(h)$, then $f = O(h)$.
2. *Sum rule:* $O(g_1) + O(g_2) = O(\max(g_1, g_2))$, and $O(g) + O(g) = O(g)$.
3. *Product rule:* $O(g_1) \cdot O(g_2) = O(g_1 g_2)$.
4. *Scalar rule:* for any constant $c \neq 0$, $c \cdot O(g) = O(g)$.
5. *Absorption:* if $g_1 = O(g_2)$, then $O(g_1) + O(g_2) = O(g_2)$.
6. *Little-o absorbs:* $o(g) + O(g) = O(g)$, and $o(g) \cdot O(h) = o(gh)$.

The same rules hold with $\ll$ in place of O, and analogous rules hold for o and $\sim$.

***Proof*:**
All follow from the definition. For (1): $|f| \leq C_1 g$ and $g \leq C_2 h$ give $|f| \leq C_1 C_2 h$. For (2): if $|f_1| \leq C_1 g_1$ and $|f_2| \leq C_2 g_2$, then $|f_1 + f_2| \leq C_1 g_1 + C_2 g_2 \leq (C_1 + C_2) \max(g_1, g_2)$. The remaining parts are similar.

Asymptotic notation interacts well with summation and integration.

Proposition 0.6.6: Summation and Integration of O-Terms

1. If $f(n) = O(g(n))$ for all n, and g is non-negative, then $\sum_{n=1}^{N} f(n) = O\left(\sum_{n=1}^{N} g(n)\right)$.
2. If $f(t) = O(g(t))$ for $t \geq t_0$, and g is non-negative and integrable, then $\int_{t_0}^{x} f(t)\,dt = O\left(\int_{t_0}^{x} g(t)\,dt\right)$.

***Proof*:**
For (1): $|f(n)| \leq Cg(n)$ implies $|\sum f(n)| \leq \sum |f(n)| \leq C \sum g(n)$. Part (2) is identical with integrals replacing sums.

However, O-terms cannot be freely moved through nonlinear operations. For instance, $e^{O(1)}$ is *not* $O(1)$—it denotes a function bounded between e^{-C} and e^{C} for some constant C, which is $\Theta(1)$. And $O(\log x)$ inside an exponent, as in $x^{O(1)} = e^{O(\log x)}$, denotes a function bounded by a power of x but with unspecified exponent.

The following table of growth rates, ordered from slowest to fastest, is a useful reference for comparing asymptotic estimates. Each function in the list is o of the next:

$$1 \ll \log\log x \ll \log x \ll (\log x)^A \ll x^{\varepsilon} \ll x^a \ll x^b \ll e^{x^c} \ll e^x \ll x! \ll x^x$$

where $0 < \varepsilon < a < b < 1 \leq c$ and $A > 0$. The gaps between consecutive entries in this hierarchy reflect qualitative differences in growth behavior. For example, the gap between $(\log x)^A$ and x^{ε} is the reason that "logarithmic savings" in analytic number theory (gaining a factor of $(\log x)^A$ over a trivial bound) are fundamentally different from "power savings" (gaining x^ε). The Prime Number Theorem provides logarithmic savings over the trivial bound $\pi(x) \le x$; the Riemann Hypothesis would upgrade this to a power saving.

Example 0.6.7: Manipulating Asymptotic Expressions

Suppose one has established that $f(x) = x^2 + 3x\log x + O(x)$ and wishes to estimate $f(x)/x^2$. Using the rules:

$$\frac{f(x)}{x^2} = 1 + \frac{3\log x}{x} + O\left(\frac{1}{x}\right) = 1 + O\left(\frac{\log x}{x}\right)$$

since $\frac{3\log x}{x} = O(\frac{\log x}{x})$ and $\frac{1}{x} = O(\frac{\log x}{x})$ (because $1 \le \log x$ for $x \ge e$). Hence $f(x) \sim x^2$. The intermediate expression $1 + \frac{3\log x}{x} + O(1/x)$ is more precise: it gives the leading term 1 and the next correction $3\log x/x$, which tends to 0 but more slowly than $1/x$.

A common pitfall in asymptotic arguments is the hidden dependence of implied constants on parameters. Consider the estimate $\sum_{n\le x} n^{-s} = O_s(1)$ for fixed $\mathrm{Re}(s) > 1$. This is valid for each fixed s, but the implied constant blows up as $s \to 1^+$ (since the sum diverges at $s = 1$). Any argument that takes $s \to 1^+$ *after* using this estimate produces an incorrect bound. The safest practice is to make all parameter dependencies explicit at the point of first use, as in $\sum_{n\le x} n^{-\sigma} \le \frac{x^{1-\sigma}}{1-\sigma} + \zeta(\sigma)$ for $\sigma > 1$, which exhibits the blowup at $\sigma = 1$ explicitly through the factor $1/(1-\sigma)$.

Example 0.6.8: Standard Asymptotic Estimates

The following estimates appear repeatedly throughout the book.

1. *Harmonic series:* $\sum_{n=1}^{N} \frac{1}{n} = \log N + \gamma + O(1/N)$, where $\gamma = 0.5772\ldots$ is the Euler-Mascheroni constant. The error $O(1/N)$ can be refined to an asymptotic expansion in powers of $1/N$ using the Euler-Maclaurin formula (chapter 1).

2. *Geometric series:* for $|r| < 1$, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$, and the partial sum satisfies $\sum_{n=0}^{N} r^n = \frac{1}{1-r} + O(r^N)$.

3. *Logarithmic growth:* $\log x = o(x^\varepsilon)$ for every $\varepsilon > 0$. Equivalently, $\log x \ll_\varepsilon x^\varepsilon$. This slow growth of the logarithm is the reason that the Prime Number Theorem $\pi(x) \sim x/\log x$ says primes are "barely" infinite: they thin out, but only logarithmically.

4. *Stirling:* $\log(n!) = n\log n - n + \frac{1}{2}\log(2\pi n) + O(1/n)$. In cruder form, $n! \sim \sqrt{2\pi n}\,(n/e)^n$. This is the asymptotic tool for estimating binomial coefficients and factorials.

5. *Partial summation prototype:* $\sum_{n\le x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\mathrm{Re}(s)})$ for $\mathrm{Re}(s) > 0$, $s \ne 1$. This is proved in chapter 2 using partial summation.

The interaction between O, o, and $\sim$ deserves emphasis, as it governs the precision of every asymptotic statement in the book. Saying $f = O(g)$ gives an upper bound but no information about the actual size of f: the function f could be much smaller than g, or even identically zero. Saying $f \sim g$ pins down the leading behavior precisely, but says nothing about the error $f - g$ beyond it being $o(g)$. Giving an explicit error term, as in $f = g + O(h)$ with $h = o(g)$, is the strongest type of asymptotic statement and is the form in which the main theorems (PNT, class number formula, Chebotarev) are most useful.

Example 0.6.9: Distinguishing Asymptotic Statements

Consider the three statements about $\pi(x)$:

1. $\pi(x) = O(x/\log x)$: there is a constant C such that $\pi(x) \leq Cx/\log x$ for large x. This is a consequence of Chebyshev's estimates (chapter 3) and is proved without complex analysis.

2. $\pi(x) \sim x/\log x$: the ratio $\pi(x)/(x/\log x) \to 1$. This is the Prime Number Theorem (chapter 5), and its proof requires the non-vanishing of $\zeta(s)$ on $\mathrm{Re}(s) = 1$.

3. $\pi(x) = \mathrm{Li}(x) + O(x\exp(-c\sqrt{\log x}))$: the error between $\pi(x)$ and $\mathrm{Li}(x)$ is bounded by $x\exp(-c\sqrt{\log x})$, which is $o(x/(\log x)^n)$ for every n but *not* $O(x^{1-\varepsilon})$ for any $\varepsilon > 0$. This sharper form requires a quantitative zero-free region for $\zeta(s)$.

Each successive statement is strictly more informative than the previous one, and each requires strictly more analytic input to prove. This hierarchy—from order of magnitude, to leading term, to explicit error—recurs throughout the book.

The distinction between $\sim$ and explicit error terms has practical consequences. The two approximations $\pi(x) \sim x/\log x$ and $\pi(x) \sim \mathrm{Li}(x) = \int_2^x dt/\log t$ are both valid forms of the Prime Number Theorem, since $\mathrm{Li}(x) \sim x/\log x$. However, $\mathrm{Li}(x)$ is a significantly better numerical approximation: at $x = 10^9$, the prime counting function satisfies $\pi(10^9) = 50{,}847{,}534$, while $10^9/\log(10^9) \approx 48{,}254{,}942$ (an error of about 5%) and $\mathrm{Li}(10^9) \approx 50{,}849{,}235$ (an error of about 0.003%). The asymptotic relation $\sim$ treats both approximations as equally valid, but the explicit error terms distinguish them.

Example 0.6.10: Growth Rate Comparisons

1. $n^{100} = o(1.01^n)$: any fixed polynomial is eventually dominated by any exponential with base greater than 1. This follows from $\lim_{n\to\infty} n^{100}/1.01^n = 0$, proved by taking logarithms: $100\log n - n\log 1.01 \to -\infty$.

2. $(\log n)^{1000} = o(n^{0.001})$: any fixed power of $\log n$ is eventually dominated by any positive power of n. At $n = 10^6$, however, $(\log n)^{1000} \approx (13.8)^{1000}$ vastly exceeds $n^{0.001} \approx 1.014$. The crossover point is astronomically large, illustrating that asymptotic statements say nothing about the rate of convergence.

3. $n! \gg (n/e)^n$: Stirling's approximation gives $n! \sim \sqrt{2\pi n}\,(n/e)^n$, so $n! \asymp n^{1/2}(n/e)^n$. The factor $\sqrt{2\pi n}$ is asymptotically negligible compared to $(n/e)^n$ but is needed for precise estimates of binomial coefficients.

Example 0.6.11: Summing with Asymptotic Notation

Estimate $\sum_{n=1}^N \sqrt{n}$. Since $\sqrt{n}$ is monotonically increasing, comparison with the integral gives

$$\int_1^N \sqrt{t}\,dt \leq \sum_{n=1}^N \sqrt{n} \leq \sqrt{N} + \int_1^N \sqrt{t}\,dt$$

Computing $\int_1^N \sqrt{t}\,dt = \frac{2}{3}N^{3/2} - \frac{2}{3}$, this yields

$$\sum_{n=1}^{N} \sqrt{n} = \frac{2}{3}N^{3/2} + O(N^{1/2})$$

The Euler-Maclaurin formula (chapter 1) refines this to $\sum_{n=1}^{N} \sqrt{n} = \frac{2}{3}N^{3/2} + \frac{1}{2}N^{1/2} + \frac{1}{24}N^{-1/2} + O(N^{-3/2})$, but the leading term $\frac{2}{3}N^{3/2}$ is often sufficient.

Finally, a convention on logarithms: throughout this book, log denotes the natural logarithm $\ln = \log_e$. This is standard in analytic number theory (and in most of mathematics outside of computer science). Iterated logarithms are written $\log\log x = \log(\log x)$, and occasionally $\log_k x$ denotes the k-fold iterated logarithm.

Example 0.6.12: Asymptotic Expansion of a Sum

Estimate $S(N) = \sum_{n=1}^{N} \frac{\log n}{n}$. Using the integral comparison $\sum_{n=1}^{N} f(n) = \int_1^N f(t)\,dt + O(f(N))$ for monotone f (a rough form of Abel summation, proved in chapter 1):

$$S(N) = \int_1^N \frac{\log t}{t}\,dt + O\left(\frac{\log N}{N}\right) = \frac{(\log N)^2}{2} + O\left(\frac{\log N}{N}\right)$$

This gives $S(N) \sim \frac{1}{2}(\log N)^2$. For $N = 10^6$: the exact value is $S(10^6) \approx 95.38$, while $\frac{1}{2}(\log 10^6)^2 = \frac{1}{2}(6\log 10)^2 \approx 95.40$—close agreement, as expected from the small error term $O(\log N/N) \approx 10^{-5}$.

The notations introduced in this section form the grammar in which the results of this book are stated. The reader should be comfortable manipulating O, o, $\sim$, and $\ll$ fluently before proceeding, as every chapter from this point forward uses them without further comment.

Exercise 0.6.1

1. Use the Euclidean algorithm to compute $\gcd(1001, 385)$, and find integers x, y such that $1001x + 385y = \gcd(1001, 385)$.
2. Find all integer solutions (x, y) to $91x + 65y = 13$.
3. Let p be a prime and n a positive integer. Prove that

$$v_p(n!) = \frac{n - s_p(n)}{p - 1}$$

   where $s_p(n)$ denotes the sum of the digits of n in base p.
4. Find the smallest positive integer x satisfying $x \equiv 3 \pmod 7$, $x \equiv 5 \pmod{11}$, and $x \equiv 2 \pmod{13}$.
5.
   1. Find the order of 2 modulo 31.
   2. Determine whether 2 is a primitive root modulo 31.
   3. List all primitive roots modulo 13.
6. Compute $\lambda(360)$, where λ is the Carmichael function. Find an element of $(\mathbb{Z}/360\mathbb{Z})^\times$ achieving this order.

7. Evaluate the following Legendre and Jacobi symbols:
   1. $\left(\frac{-3}{17}\right)$
   2. $\left(\frac{11}{41}\right)$
   3. $\left(\frac{1009}{2027}\right)$ (both are prime)

8. Using Wilson's theorem, prove that if p is a prime with $p \equiv 1 \pmod 4$, then $x = \left(\frac{p-1}{2}\right)!$ satisfies $x^2 \equiv -1 \pmod p$.

9. Let p be an odd prime. Prove that the product of all quadratic residues modulo p in $\{1, 2, \ldots, p-1\}$ is congruent to $(-1)^{(p+1)/2} \pmod p$.

10.
    1. Show that $\sum_{n=1}^{N} \frac{1}{n^2} = \frac{\pi^2}{6} + O\left(\frac{1}{N}\right)$.
    2. Show that $\sum_{p \le N} \frac{1}{p^2} = O(1)$, where the sum is over primes.
    3. Prove that $\prod_{n=1}^{N} \left(1 + \frac{1}{n}\right) \sim \frac{N}{e^\gamma} e^{1/N} \ldots$ is false, and find the correct asymptotic: $\prod_{n=1}^{N} \left(1 + \frac{1}{n}\right) = N + 1$.

11. Prove that for any fixed $m \ge 1$:

$$\frac{\#\{n \le x : \gcd(n, m) = 1\}}{x} = \frac{\varphi(m)}{m} + O\left(\frac{1}{x}\right)$$

    Interpret this as saying that the "probability" that a random integer is coprime to m is $\varphi(m)/m = \prod_{p \mid m}(1 - 1/p)$.

12. Let n be an odd positive integer with $n > 1$. Prove that $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ for the Jacobi symbol, without assuming n is prime.

Part I

Foundations of Analytic Number Theory

The three chapters of this Part assemble the apparatus of the subject before any complex analysis is brought to bear on the primes. The preliminary review established what is known about individual integers; the task now is to build the machinery that studies them in aggregate. Chapter 1 introduces the arithmetic functions that encode number-theoretic data and the summation techniques, partial summation and the Euler–Maclaurin formula, that turn discrete sums into continuous integrals; this is the bridge across which analysis will later reach the primes. Chapter 2 packages a multiplicative arithmetic function into a Dirichlet series and factors it as an Euler product, converting the multiplicative structure of the integers into the analytic structure of a function of a complex variable, with the Riemann zeta function as the first and central example. Chapter 3 then pushes the elementary theory of prime distribution as far as it will go, proving Chebyshev's bounds, Mertens' theorems, and Bertrand's postulate by counting arguments alone, and marking precisely the point where the elementary method halts and complex analysis becomes indispensable.

1

Arithmetic Functions and Summation Techniques

The preliminary chapter studied individual integers: their divisibility, their prime factorizations, their residue classes. Analytic number theory asks a different kind of question. Rather than examining a single integer n, it attaches a numerical invariant $f(n)$ to every positive integer and studies how these values behave in aggregate—how $\sum_{n\leq x} f(n)$ grows as $x \to \infty$, what the "average" value of f is, and how the values fluctuate around that average. This shift from the particular to the statistical is the defining move of the subject.

This chapter builds the apparatus for that program. The first two sections are algebraic: they introduce the arithmetic functions that encode number-theoretic data and the Dirichlet convolution that organizes them into a ring. The last three sections are analytic: they develop the summation techniques—the hyperbola method, Abel's formula, and the Euler-Maclaurin formula—that convert discrete sums into continuous integrals amenable to the methods of analysis. Every later chapter in this book depends on this material.

1.1 Arithmetic Functions and Multiplicativity

The starting point is to formalize the idea of a "number-theoretic quantity attached to each positive integer." The divisor count $d(n)$, the Euler totient $\varphi(n)$, and the von Mangoldt function $\Lambda(n)$ are all examples of the same type of object: a function from the positive integers to the complex numbers. The structure that makes these functions tractable is *multiplicativity*—the property that $f(mn) = f(m)f(n)$ whenever $\gcd(m,n) = 1$—which reduces the study of f on all positive integers to its behavior at prime powers.

Definition 1.1.1: Arithmetic Function

An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \to \mathbb{C}$.

No regularity conditions are imposed: an arithmetic function is simply a sequence of complex numbers indexed by the positive integers. The interest lies in the additional structure that specific arithmetic functions possess.

Definition 1.1.2: Multiplicative, Totally Multiplicative, and Additive Functions

An arithmetic function f is:

1. *Multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$.
2. *Totally multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all $m, n \geq 1$.
3. *Additive* if $f(mn) = f(m) + f(n)$ whenever $\gcd(m, n) = 1$.
4. *Totally additive* if $f(mn) = f(m) + f(n)$ for all $m, n \geq 1$.

The condition $f(1) = 1$ for multiplicative functions is a normalization: setting $m = n = 1$ in $f(mn) = f(m)f(n)$ gives $f(1) = f(1)^2$, so $f(1) \in \{0, 1\}$, and the convention excludes the zero function. The logarithm function $\log : \mathbb{Z}_{>0} \to \mathbb{R}$ is totally additive.

The fundamental observation is that multiplicativity reduces global information to local information at each prime.

Proposition 1.1.3: Determination by Prime Powers

A multiplicative function f is completely determined by its values at prime powers $\{f(p^k) : p \text{ prime}, k \geq 1\}$. Specifically, for $n = p_1^{a_1} \cdots p_r^{a_r}$:

$$f(n) = \prod_{i=1}^{r} f(p_i^{a_i})$$

If f is totally multiplicative, then $f(p^k) = f(p)^k$, so f is determined by its values at primes.

***Proof*:**
Since $\gcd(p_i^{a_i}, p_j^{a_j}) = 1$ for $i \neq j$, repeated application of the multiplicativity condition gives $f(n) = f(p_1^{a_1}) \cdots f(p_r^{a_r})$. For totally multiplicative f: $f(p^k) = f(p \cdot p^{k-1}) = f(p)f(p^{k-1})$, and induction gives $f(p^k) = f(p)^k$.

This reduction to prime powers is the arithmetic analogue of the Chinese Remainder Theorem (theorem 0.3.7), which decomposes $\mathbb{Z}/n\mathbb{Z}$ into a product of rings $\mathbb{Z}/p_i^{a_i}\mathbb{Z}$. In both cases, the global object splits into independent local pieces, one for each prime. This "local-to-global" decomposition is the structural principle that, in the language of Dirichlet series (chapter 2), becomes the Euler product.

The rest of this section introduces the arithmetic functions that appear most frequently throughout the book. Each is defined, its values at prime powers are recorded, and a small table of values is given for orientation.

Definition 1.1.4: The Möbius Function

The *Möbius function* $\mu : \mathbb{Z}_{>0} \to \{-1, 0, 1\}$ is defined by

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 p_2 \cdots p_k \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{if } p^2 \big| n \text{ for some prime } p. \end{cases}$$

Equivalently, μ is the multiplicative function with $\mu(p) = -1$ and $\mu(p^k) = 0$ for $k \geq 2$.

The Möbius function detects squarefreeness: $\mu(n) \neq 0$ if and only if n is squarefree. Its deeper role as the identity element for Möbius inversion, or equivalently as the Dirichlet inverse of the constant function $\mathbf{1}$, is the subject of the next section. The alternative sign is for the inclusion-exclusion principle: for a finite set with k elements, the inclusion-exclusion formula alternates signs $(-1)^k$, and μ does the same over the prime factors of n.

Definition 1.1.5: Divisor Functions

For $k \in \mathbb{C}$, the *k-th divisor function* is

$$\sigma_k(n) = \sum_{d \big| n} d^k$$

The two most important special cases are the *number-of-divisors function* $d(n) = \sigma_0(n) = \sum_{d \big| n} 1$ and the *sum-of-divisors function* $\sigma(n) = \sigma_1(n) = \sum_{d \big| n} d$.

Each σ_k is multiplicative (but not totally multiplicative for $k \neq 0$). At a prime power:

$$\sigma_k(p^a) = 1 + p^k + p^{2k} + \cdots + p^{ak} = \frac{p^{(a+1)k} - 1}{p^k - 1}$$

for $k \neq 0$, and $d(p^a) = a + 1$. The multiplicativity of σ_k follows from the fact that the divisors of mn (with $\gcd(m, n) = 1$) are exactly the products $d_1 d_2$ where $d_1 \big| m$ and $d_2 \big| n$, so the sum factors.

Definition 1.1.6: The Liouville Function

The *Liouville function* $\lambda : \mathbb{Z}_{>0} \to \{-1, 1\}$ is defined by $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n) = \sum_{p^k \| n} k$ is the number of prime factors of n counted with multiplicity.

The Liouville function is *totally* multiplicative: $\lambda(mn) = (-1)^{\Omega(mn)} = (-1)^{\Omega(m)+\Omega(n)} = \lambda(m)\lambda(n)$ for all m, n. At primes, $\lambda(p) = -1$, and $\lambda(p^k) = (-1)^k$. The Liouville function is a "signed" version of the constant function $\mathbf{1}$: it assigns $+1$ to integers with an even number of prime factors and -1 to those with an odd number. The relationship between λ and μ is that $\mu(n) = \lambda(n)$ when n is squarefree, and $\mu(n) = 0$ otherwise (so λ is a "squarefree-ified" extension of μ).

The next two functions count prime factors.

Definition 1.1.7: Prime Factor Counting Functions

For $n \geq 1$:

1. $\omega(n) = \sum_{p \mid n} 1$ is the number of *distinct* prime factors of n.
2. $\Omega(n) = \sum_{p^k \| n} k$ is the number of prime factors of n *counted with multiplicity.*

By convention, $\omega(1) = \Omega(1) = 0$.

The function ω is additive (but not totally additive), and Ω is totally additive. For a prime power p^a: $\omega(p^a) = 1$ and $\Omega(p^a) = a$. The distinction between ω and Ω matters for non-squarefree integers: $\omega(12) = 2$ (the primes 2 and 3) while $\Omega(12) = 3$ (two factors of 2 and one of 3).

The final function in the catalogue is different in character from the preceding ones: it is not multiplicative, and it is supported on prime powers rather than on all positive integers. It is, however, the single most important arithmetic function for the study of prime distribution.

Definition 1.1.8: The Von Mangoldt Function

The *von Mangoldt function* $\Lambda : \mathbb{Z}_{>0} \to \mathbb{R}_{\geq 0}$ is defined by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

The von Mangoldt function is not multiplicative: $\Lambda(2) = \log 2$ and $\Lambda(3) = \log 3$, but $\Lambda(6) = 0 \neq \Lambda(2)\Lambda(3)$. Its importance comes from the identity that links it to the natural logarithm.

Proposition 1.1.9: Von Mangoldt Identity

For all $n \geq 1$:

$$\sum_{d \mid n} \Lambda(d) = \log n$$

***Proof*:**
Write $n = p_1^{a_1} \cdots p_r^{a_r}$. The divisors d of n with $\Lambda(d) \neq 0$ are exactly the prime powers p_i^j with $1 \leq j \leq a_i$ and $1 \leq i \leq r$. Hence:

$$\sum_{d \mid n} \Lambda(d) = \sum_{i=1}^{r} \sum_{j=1}^{a_i} \log p_i = \sum_{i=1}^{r} a_i \log p_i = \log \left(\prod_{i=1}^{r} p_i^{a_i} \right) = \log n$$

This identity says that log is the summatory function of Λ over divisors. In the language of Dirichlet convolution (introduced in the next section), it reads $\Lambda * \mathbf{1} = \log$, which is one of the key identities in analytic number theory. The von Mangoldt function appears as the coefficient of the logarithmic derivative of $\zeta(s)$: the identity

$$-\zeta'(s)/\zeta(s) = \sum_{n=1}^{\infty} \Lambda(n) n^{-s}$$

(chapter 4) is the Dirichlet series translation of the von Mangoldt identity, and it is through this series that the zeros of $\zeta(s)$ control the distribution of primes.

The Chebyshev functions, which are the summatory functions of Λ and $\log p$ respectively, will be central in chapter 3:

$$\psi(x) = \sum_{n \le x} \Lambda(n), \qquad \vartheta(x) = \sum_{p \le x} \log p$$

The Prime Number Theorem (chapter 5) is equivalent to $\psi(x) \sim x$.

Example 1.1.10: Verifying the Von Mangoldt Identity

Take $n = 12 = 2^2 \cdot 3$. The divisors of 12 are $\{1, 2, 3, 4, 6, 12\}$. Among these, the prime powers are $2, 3, 4 = 2^2$, giving:

$$\sum_{d \mid 12} \Lambda(d) = \Lambda(1)+\Lambda(2)+\Lambda(3)+\Lambda(4)+\Lambda(6)+\Lambda(12) = 0+\log 2+\log 3+\log 2+0+0 = 2\log 2+\log 3 = \log 12$$

The non-prime-power divisors 1, 6, and 12 contribute nothing, while $4 = 2^2$ contributes $\log 2$ (not $\log 4$). This is the mechanism by which Λ "distributes" the logarithm of n across its prime-power divisors, assigning weight $\log p$ to each prime p and each power of p dividing n.

The following table collects the values of all the arithmetic functions introduced in this section for $n = 1, \ldots, 12$, providing a concrete reference.

Example 1.1.11: Table of Arithmetic Functions

n	1	2	3	4	5	6	7	8	9	10	11	12
$\mu(n)$	1	-1	-1	0	-1	1	-1	0	0	1	-1	0
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4
$d(n)$	1	2	2	3	2	4	2	4	3	4	2	6
$\sigma(n)$	1	3	4	7	6	12	8	15	13	18	12	28
$\lambda(n)$	1	-1	-1	1	-1	1	-1	-1	1	1	-1	-1
$\omega(n)$	0	1	1	1	1	2	1	1	1	2	1	2
$\Omega(n)$	0	1	1	2	1	2	1	3	2	2	1	3
$\Lambda(n)$	0	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$	0	$\log 11$	0

The patterns are visible: μ vanishes on non-squarefree integers $(4, 8, 9, 12)$; $d(n)$ is large when n has many factorizations (e.g., $d(12) = 6$); Λ is nonzero only at prime powers $(2, 3, 4, 5, 7, 8, 9, 11)$.

Example 1.1.12: Computing Arithmetic Functions from Prime Factorizations

1. Let $n = 2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$. Then:

$$\begin{aligned}
d(2520) &= (3+1)(2+1)(1+1)(1+1) = 48, \\
\sigma(2520) &= \frac{2^4-1}{1} \cdot \frac{3^3-1}{2} \cdot \frac{5^2-1}{4} \cdot \frac{7^2-1}{6} = 15 \cdot 13 \cdot 6 \cdot 8 = 9360, \\
\mu(2520) &= 0 \quad (\text{since } 2^2 \big| 2520), \\
\lambda(2520) &= (-1)^{3+2+1+1} = (-1)^7 = -1
\end{aligned}$$

2. Let $n = 30 = 2 \cdot 3 \cdot 5$. This is squarefree, so $\mu(30) = (-1)^3 = -1$ and $\lambda(30) = -1$ as well. The divisor count is $d(30) = 2^3 = 8$, and $\sigma(30) = 3 \cdot 4 \cdot 6 = 72$. Note $\sigma(30)/30 = 72/30 = 12/5 > 2$, so 30 is an *abundant* number.

The multiplicativity of each function in the catalogue is summarized as follows: μ, φ, σ_k, d, σ are multiplicative; λ is totally multiplicative; ω is additive; Ω is totally additive; Λ and log are none of the above. Several relationships between these functions are already visible. The Möbius function μ and the Liouville function λ agree on squarefree integers and differ elsewhere. The identity $|\mu(n)| = 1$ if and only if n is squarefree can be expressed as $|\mu| = \mu^2$, and $\sum_{d \big| n} |\mu(d)| = 2^{\omega(n)}$ counts the squarefree divisors of n. The von Mangoldt identity $\sum_{d \big| n} \Lambda(d) = \log n$ expresses log as a "divisor sum" of Λ. All such identities acquire a unified algebraic meaning once the notion of Dirichlet convolution is in place.

1.2 Dirichlet Convolution and Möbius Inversion

The arithmetic functions of the previous section satisfy many identities: $\sum_{d \big| n} \mu(d) = [n = 1]$, $\sum_{d \big| n} \Lambda(d) = \log n$, $\sum_{d \big| n} \varphi(d) = n$, and so forth. These identities share a common structure, i.e. a sum over the divisors of n, with the summand depending on both d and n/d. This structure has a name: *Dirichlet convolution.* Equipping the set of arithmetic functions with Dirichlet convolution as multiplication turns it into a commutative ring, and the divisor-sum identities become algebraic statements in this ring:

Definition 1.2.1: Dirichlet Convolution

The *Dirichlet convolution* of two arithmetic functions f and g is the arithmetic function $f * g$ defined by

$$(f * g)(n) = \sum_{d \big| n} f(d)\, g(n/d) = \sum_{ab=n} f(a)\, g(b)$$

The two expressions are equivalent: summing over divisors d of n with complementary divisor n/d is the same as summing over ordered factorizations $n = ab$. The second form makes the commutativity $f * g = g * f$ evident since $ab = ba$.

Several of the identities from section 1.1 are now recognizable as convolution statements:

1. The von Mangoldt identity $\sum_{d \mid n} \Lambda(d) = \log n$ is $\Lambda * \mathbf{1} = \log$ where $\mathbf{1}$ denotes the constant function $\mathbf{1}(n) = 1$.
2. The totient summation $\sum_{d \mid n} \varphi(d) = n$ is $\varphi * \mathbf{1} = \mathrm{id}$ where $\mathrm{id}(n) = n$.
3. The divisor function $d(n) = \sum_{d \mid n} 1$ is $d = \mathbf{1} * \mathbf{1}$ and the sum-of-divisors function $\sigma(n) = \sum_{d \mid n} d$ is $\sigma = \mathrm{id} * \mathbf{1}$.

Theorem 1.2.2: The Ring of Arithmetic Functions

The set $\mathcal{A}$ of all arithmetic functions, equipped with pointwise addition and Dirichlet convolution as multiplication, forms a commutative ring with identity. The additive identity is the zero function, and the multiplicative identity is

$$\varepsilon(n) = [n = 1] = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

***Proof*:**

1. *Commutativity.* Commutativity of $*$ follows from the symmetry of the second form: $(f * g)(n) = \sum_{ab=n} f(a)g(b) = \sum_{ab=n} g(a)f(b) = (g * f)(n)$.

2. *Associativity.* For any three arithmetic functions f, g, h:

$$((f * g) * h)(n) = \sum_{de=n} (f * g)(d)\, h(e) = \sum_{de=n} \sum_{ab=d} f(a)g(b)h(e) = \sum_{abe=n} f(a)g(b)h(e)$$

   The final expression is symmetric in the grouping, so $(f*(g*h))(n)$ gives the same triple sum.

3. *Identity.* $(f * \varepsilon)(n) = \sum_{d \mid n} f(d)\varepsilon(n/d) = f(n)\varepsilon(1) = f(n)$, since $\varepsilon(n/d) = 0$ unless $d = n$.

4. *Distributivity.* $(f*(g+h))(n) = \sum_{d \mid n} f(d)(g+h)(n/d) = \sum_{d \mid n} f(d)g(n/d) + \sum_{d \mid n} f(d)h(n/d) = (f * g)(n) + (f * h)(n)$.

The ring $(\mathcal{A}, +, *)$ is a commutative ring with identity but not an integral domain: for instance, if f is supported on $\{1\}$ and g is supported on $\{1\}$ with $f(1) = g(1) = 0$, then $f * g = 0$. The unit group of $\mathcal{A}$ (the set of arithmetic functions possessing a convolution inverse) is characterized by a simple condition:

Proposition 1.2.3: Units of the Convolution Ring

An arithmetic function f has a Dirichlet convolution inverse f^{-1} (satisfying $f * f^{-1} = \varepsilon$) if and only if $f(1) \neq 0$. When it exists, the inverse is unique and is given recursively by

$$f^{-1}(1) = \frac{1}{f(1)}, \qquad f^{-1}(n) = -\frac{1}{f(1)} \sum_{\substack{d \mid n \\ d<n}} f(n/d)\, f^{-1}(d) \quad \text{for } n > 1$$

***Proof*:**
The condition $(f * f^{-1})(1) = f(1)f^{-1}(1) = 1$ forces $f(1) \neq 0$ and $f^{-1}(1) = 1/f(1)$. For $n > 1$, the condition $(f * f^{-1})(n) = 0$ reads

$$\sum_{d \mid n} f(n/d)\, f^{-1}(d) = 0,$$

and isolating the $d = n$ term gives $f(1)f^{-1}(n) = -\sum_{d \mid n,\, d<n} f(n/d)f^{-1}(d)$, which determines $f^{-1}(n)$ recursively from the values $f^{-1}(d)$ for $d < n$.

The recursive formula computes f^{-1} one value at a time in increasing order: $f^{-1}(1)$, then $f^{-1}(2)$, then $f^{-1}(3)$, and so on, each depending only on previously computed values. This is effective for small arguments but does not directly give a closed-form expression.

Example 1.2.4: Computing a Dirichlet Inverse

Compute id^{-1} using the recursive formula. At $n = 1$: $\mathrm{id}^{-1}(1) = 1$. At $n = 2$: $\mathrm{id}^{-1}(2) = -(1/1) \cdot \mathrm{id}(2) \cdot \mathrm{id}^{-1}(1) = -2$. At $n = 3$: $\mathrm{id}^{-1}(3) = -3$. At $n = 4$, the divisors less than 4 are 1 and 2:

$$\mathrm{id}^{-1}(4) = -\big(\mathrm{id}(4) \cdot \mathrm{id}^{-1}(1) + \mathrm{id}(2) \cdot \mathrm{id}^{-1}(2)\big) = -(4 \cdot 1 + 2 \cdot (-2)) = 0$$

At $n = 6$: $\mathrm{id}^{-1}(6) = -(6 \cdot 1 + 3 \cdot (-2) + 2 \cdot (-3)) = -(6-6-6) = 6$. The pattern is $\mathrm{id}^{-1}(n) = n\mu(n)$: the function vanishes at non-squarefree integers, equals $(-1)^{\omega(n)} n$ at squarefree integers, and is multiplicative with $\mathrm{id}^{-1}(p) = -p$ and $\mathrm{id}^{-1}(p^k) = 0$ for $k \geq 2$.

For multiplicative functions, a cleaner description is available.

Proposition 1.2.5: Multiplicative Functions Form a Subgroup

If f and g are multiplicative, then $f * g$ is multiplicative. If f is multiplicative, then f^{-1} exists (since $f(1) = 1$) and is also multiplicative.

***Proof*:**
Let $\gcd(m, n) = 1$. The divisors of mn are exactly the products $d_1 d_2$ where $d_1 \mid m$ and $d_2 \mid n$

(since $\gcd(m,n) = 1$). Hence:

$$(f * g)(mn) = \sum_{d \mid mn} f(d)\, g(mn/d) = \sum_{d_1 \mid m} \sum_{d_2 \mid n} f(d_1 d_2)\, g\left(\frac{m}{d_1} \cdot \frac{n}{d_2}\right)$$
$$= \sum_{d_1 \mid m} \sum_{d_2 \mid n} f(d_1) f(d_2)\, g(m/d_1)\, g(n/d_2) = (f * g)(m) \cdot (f * g)(n)$$

Since $(f * g)(1) = f(1)g(1) = 1$, the function $f * g$ is multiplicative.

For the inverse: if f is multiplicative, then $f(1) = 1 \neq 0$, so f^{-1} exists. Multiplicativity of f^{-1} can be verified by induction on mn using the recursive formula: the key observation is that for $\gcd(m,n) = 1$, the divisors of mn factor as products of divisors of m and n, so the recursion for $f^{-1}(mn)$ splits into independent recursions for $f^{-1}(m)$ and $f^{-1}(n)$.

The multiplicative arithmetic functions thus form a group under Dirichlet convolution, with identity ε. This group structure is the reason that identities among multiplicative functions can be verified by checking them at prime powers.

The most important Dirichlet inverse is that of the constant function $\mathbf{1}$.

Proposition 1.2.6: The Möbius Function as a Dirichlet Inverse

$\mu * \mathbf{1} = \varepsilon$. That is, μ is the Dirichlet inverse of $\mathbf{1}$:

$$\sum_{d \mid n} \mu(d) = [n = 1] = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

***Proof*:**
Both μ and $\mathbf{1}$ are multiplicative, so $\mu * \mathbf{1}$ is multiplicative by proposition 1.2.5. It suffices to check the identity at prime powers. For p^k with $k \geq 1$:

$$\sum_{d \mid p^k} \mu(d) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^k) = 1 + (-1) + 0 + \cdots + 0 = 0$$

At $n = 1$: $(\mu * \mathbf{1})(1) = \mu(1) \cdot 1 = 1$. Hence $\mu * \mathbf{1} = \varepsilon$.

This identity is the algebraic content of inclusion-exclusion. For a squarefree integer $n = p_1 \cdots p_r$, the divisors are the 2^r products of subsets of $\{p_1, \ldots, p_r\}$, and $\sum_{d \mid n} \mu(d) = \sum_{k=0}^{r} \binom{r}{k} (-1)^k = (1-1)^r = 0$.

With the Möbius function identified as the inverse of $\mathbf{1}$, the central inversion formula follows immediately.

Theorem 1.2.7: Möbius Inversion Formula

Let f and g be arithmetic functions. Then:

$$g = f * \mathbf{1} \qquad \text{if and only if} \qquad f = g * \mu$$

In explicit notation: $g(n) = \sum_{d|n} f(d)$ for all n if and only if $f(n) = \sum_{d|n} \mu(d)\, g(n/d)$ for all n.

***Proof*:**
If $g = f * \mathbf{1}$, then $g * \mu = (f * \mathbf{1}) * \mu = f * (\mathbf{1} * \mu) = f * \varepsilon = f$, using associativity and $\mathbf{1} * \mu = \varepsilon$ (proposition 1.2.6). Conversely, if $f = g * \mu$, then $f * \mathbf{1} = (g * \mu) * \mathbf{1} = g * (\mu * \mathbf{1}) = g * \varepsilon = g$.

Thus, Möbius inversion is the tool for "solving" a divisor-sum relation: if g is defined as a sum of f over divisors, then f can be recovered from g by convolving with μ.

Example 1.2.8: Recovering φ and Λ via Möbius Inversion

1. The identity $\varphi * \mathbf{1} = \mathrm{id}$ (proposition 0.3.16) inverts to $\varphi = \mathrm{id} * \mu$. In explicit form:

$$\varphi(n) = \sum_{d|n} \mu(d) \cdot \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d} = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

where the last step uses multiplicativity: for $n = \prod p_i^{a_i}$, the sum $\sum_{d|n} \mu(d)/d$ factors over primes as $\prod_{p|n}(1 + \mu(p)/p) = \prod_{p|n}(1 - 1/p)$. This recovers the formula from proposition 0.3.10.

2. The von Mangoldt identity $\Lambda * \mathbf{1} = \log$ (proposition 1.1.9) inverts to $\Lambda = \log * \mu$, i.e.,

$$\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d) = -\sum_{d|n} \mu(d) \log d$$

The second equality uses $\sum_{d|n} \mu(d) \log(n/d) = \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d = -\sum_{d|n} \mu(d) \log d$, since $\sum_{d|n} \mu(d) = 0$ for $n > 1$ (and the case $n = 1$ gives $\Lambda(1) = 0$ directly).

Example 1.2.9: Counting Squarefree Integers

Let $Q(x) = \#\{n \le x : n \text{ is squarefree}\} = \sum_{n \le x} |\mu(n)|$. Since $|\mu(n)| = \sum_{d^2|n} \mu(d)$ (an identity verifiable at prime powers: both sides equal 1 if n is squarefree and 0 otherwise), the count can be rewritten as:

$$Q(x) = \sum_{n \le x} \sum_{d^2|n} \mu(d) = \sum_{d \le \sqrt{x}} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor = \sum_{d \le \sqrt{x}} \mu(d) \left(\frac{x}{d^2} + O(1)\right) = x \sum_{d \le \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x})$$

The sum $\sum_{d=1}^{\infty} \mu(d)/d^2$ converges to $1/\zeta(2) = 6/\pi^2$ (as the Euler product gives $\sum \mu(d)/d^s = 1/\zeta(s)$, a fact established rigorously in chapter 2). Accepting this for now:

$$Q(x) = \frac{6}{\pi^2}x + O(\sqrt{x})$$

Hence the density of squarefree integers is $6/\pi^2 \approx 0.6079$: about 61% of all positive integers are squarefree.

The remaining identities from the outline of section 1.1 are now provable in the convolution ring. A systematic collection is recorded here for later reference.

Proposition 1.2.10: Key Convolution Identities

The following identities hold in $(\mathcal{A}, +, *)$:

1. $\mu * \mathbf{1} = \varepsilon$ (Möbius is the inverse of $\mathbf{1}$).
2. $\varphi * \mathbf{1} = \mathrm{id}$, equivalently $\varphi = \mathrm{id} * \mu$.
3. $\Lambda * \mathbf{1} = \log$, equivalently $\Lambda = \log * \mu$.
4. $d = \mathbf{1} * \mathbf{1}$ (the divisor function is the self-convolution of $\mathbf{1}$).
5. $\sigma = \mathrm{id} * \mathbf{1}$.
6. $\sigma_k = \mathrm{id}^k * \mathbf{1}$, where $\mathrm{id}^k(n) = n^k$.
7. $|\mu| * |\mu| = 2^\omega$, where $2^\omega(n) = 2^{\omega(n)}$.
8. $\lambda * |\mu| = \varepsilon$ (λ is the inverse of $|\mu|$).
9. $\lambda * \mathbf{1} = \mathbf{1}_\square$, where $\mathbf{1}_\square(n) = [n \text{ is a perfect square}]$.

***Proof*:**
Identities (1) - (6) have been established above or follow directly from definitions. For the remaining:

7. Both sides are multiplicative (the left by proposition 1.2.5; the right since ω is additive gives $2^{\omega(mn)} = 2^{\omega(m)}2^{\omega(n)}$ for $\gcd(m,n) = 1$). At p^k: $(|\mu| * |\mu|)(p^k) = \sum_{j=0}^k |\mu(p^j)||\mu(p^{k-j})| = |\mu(1)||\mu(p^k)| + |\mu(p)||\mu(p^{k-1})| + \cdots$. The nonzero terms are those with $j \in \{0,1\}$ and $k-j \in \{0,1\}$, so only $j = 0, k-j \leq 1$ and $j = 1, k-j \leq 1$ contribute. For $k = 0$: the sum is 1. For $k = 1$: $|\mu(1)||\mu(p)| + |\mu(p)||\mu(1)| = 2$. For $k \geq 2$: $|\mu(1)||\mu(p^k)| + |\mu(p)||\mu(p^{k-1})| = 0$. Hence $(|\mu| * |\mu|)(p^k) = 2$ for $k \geq 1$ and 1 for $k = 0$, matching $2^{\omega(p^k)}$.

8. At p^k: $(\lambda * |\mu|)(p^k) = \sum_{j=0}^k \lambda(p^j)|\mu(p^{k-j})| = (-1)^0 \cdot |\mu(p^k)| + (-1)^1 \cdot |\mu(p^{k-1})| + \cdots$. For $k = 0$: the sum is 1. For $k = 1$: $|\mu(p)| + (-1)|\mu(1)| = 1 - 1 = 0$. For $k \geq 2$: $|\mu(p^k)| + (-1)|\mu(p^{k-1})| + |\mu(p^{k-2})| + \cdots$; the only nonzero terms are $j = k-1$ (giving $(-1)^{k-1}$) and $j = k$ (giving $(-1)^k$), which sum to 0. Hence $\lambda * |\mu| = \varepsilon$.

9. At p^k: $(\lambda * \mathbf{1})(p^k) = \sum_{j=0}^{k}(-1)^j = \begin{cases} 1 & k \text{ even}, \\ 0 & k \text{ odd} \end{cases}$, which is $[p^k$ is a perfect square$]$.

Identity (9) deserves a moment of interpretation: the summatory function $\sum_{d|n} \lambda(d)$ is 1 if n is a perfect square and 0 otherwise. This identity connects the Liouville function to the distribution of perfect squares, and its Dirichlet series translation $\zeta(2s)/\zeta(s) = \sum \lambda(n)n^{-s}$ will appear in chapter 2.

1.3 Average Orders and Summatory Functions

The convolution identities of the previous section describe exact relationships among arithmetic functions at individual integers. The present section asks a different question: how do these functions behave *on average*? The divisor function $d(n)$ fluctuates wildly, for example $d(p) = 2$ for primes, while $d(n)$ can be arbitrarily large for highly composite n. But its average over $\{1, \dots, N\}$ grows smoothly like $\log N$. Extracting this smooth average from the pointwise behavior is the first analytic problem of the book, and the main tool is the *Dirichlet hyperbola method*: a technique for evaluating $\sum_{n \le x}(f * g)(n)$ by splitting the double sum along the hyperbola $ab = n$.

Definition 1.3.1: Summatory Function and Average Order

For an arithmetic function f, the *summatory function* is

$$M_f(x) = \sum_{n \le x} f(n)$$

The function $g(x)$ is an *average order* of f if $M_f(x) \sim g(x)$ as $x \to \infty$, i.e., $\sum_{n \le x} f(n)/g(x) \to 1$.

The summatory function converts the discrete sequence $(f(n))$ into a step function of a continuous variable x. The average order captures the typical size of $f(n)$: if $g(x)$ is an average order of f, then $f(n)$ is "typically" of size $g(n)/n$. The fundamental technique for estimating summatory functions of convolutions is the following:

Theorem 1.3.2: Dirichlet's Hyperbola Method

Let f and g be arithmetic functions, and set $h = f * g$. For any $x \ge 1$ and any y with $1 \le y \le x$:

$$\sum_{n \le x} h(n) = \sum_{a \le y} f(a)\, G\left(\frac{x}{a}\right) + \sum_{b \le x/y} g(b)\, F\left(\frac{x}{b}\right) - F(y)\, G\left(\frac{x}{y}\right)$$

where $F(t) = \sum_{n \le t} f(n)$ and $G(t) = \sum_{n \le t} g(n)$.

The idea is to write $\sum_{n \le x} h(n) = \sum_{ab \le x} f(a)g(b)$ and split the region $\{(a, b) : ab \le x\}$ into two parts along the line $a = y$. The first sum counts pairs with $a \le y$, the second counts pairs with $b \le x/y$ (equivalently $a > y$), and the subtracted term corrects for the overlap where both $a \le y$ and $b \le x/y$.

***Proof*:**
Start from $h = f * g$:

$$\sum_{n \le x} h(n) = \sum_{\substack{a,b \ge 1 \\ ab \le x}} f(a)g(b)$$

Step 1: Partition the summation region. Split $\{(a,b) \in \mathbb{Z}^2_{>0} : ab \le x\}$ into: $S_1 = \{a \le y\}$ and $S_2 = \{a > y,\ b \le x/y\}$. Every pair (a,b) with $ab \le x$ belongs to at least one of these sets (if $a > y$, then $b \le x/a < x/y$), and the overlap is $S_1 \cap S_2 = \{a \le y,\ b \le x/y\}$. By inclusion-exclusion:

$$\sum_{ab \le x} f(a)g(b) = \sum_{\substack{a \le y \\ ab \le x}} f(a)g(b) + \sum_{\substack{a > y \\ b \le x/y}} f(a)g(b) - \sum_{\substack{a \le y \\ b \le x/y}} f(a)g(b) + \sum_{\substack{a \le y \\ b \le x/y}} f(a)g(b)$$

Step 2: Evaluate each piece. The first sum: for fixed $a \le y$, the condition $ab \le x$ means $b \le x/a$, so $\sum_{\substack{a \le y \\ ab \le x}} f(a)g(b) = \sum_{a \le y} f(a)G(x/a)$. The second sum plus the overlap: $\sum_{b \le x/y} \sum_{a \le x/b} f(a)g(b) = \sum_{b \le x/y} g(b)F(x/b)$. The overlap alone: $\sum_{a \le y} f(a) \sum_{b \le x/y} g(b) = F(y)G(x/y)$. Combining gives the stated formula.

The optimal choice of the splitting parameter y depends on the specific functions f and g; the symmetric choice $y = \sqrt{x}$ is the most common and typically minimizes the error term. The power of the method lies in reducing the estimation of $\sum h(n)$ to the estimation of F and G separately—a significant simplification when f and g are simpler than h.

Geometrically, the lattice points $(a,b) \in \mathbb{Z}^2_{>0}$ with $ab \le x$ lie below the hyperbola $ab = x$ in the first quadrant. The horizontal line $a = y$ divides this region into an upper-left piece (where the sum over a is short and G handles the sum over b) and a lower-right piece (where the sum over b is short and F handles the sum over a). The rectangle $\{a \le y,\ b \le x/y\}$ is the overlap region. This geometric picture—counting lattice points under a hyperbola—is a prototype for the lattice-point counting arguments that appear in the proof of the analytic class number formula (chapter 8).

The first and most classical application is to the divisor function $d = \mathbf{1} * \mathbf{1}$.

Theorem 1.3.3: Average Order of the Divisor Function

$$\sum_{n \le x} d(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

where $\gamma = 0.5772\ldots$ is the Euler-Mascheroni constant. In particular, the average order of $d(n)$ is $\log n$.

***Proof*:**
Since $d = \mathbf{1} * \mathbf{1}$, apply theorem 1.3.2 with $f = g = \mathbf{1}$ and $y = \sqrt{x}$. The summatory function of $\mathbf{1}$ is $F(t) = G(t) = \lfloor t \rfloor = t + O(1)$. By symmetry, the first two sums in the hyperbola formula are equal, giving:

$$\sum_{n \le x} d(n) = 2 \sum_{a \le \sqrt{x}} \left\lfloor \frac{x}{a} \right\rfloor - \lfloor \sqrt{x} \rfloor^2$$

Using $\lfloor x/a \rfloor = x/a + O(1)$:

$$2 \sum_{a \le \sqrt{x}} \left\lfloor \frac{x}{a} \right\rfloor = 2x \sum_{a \le \sqrt{x}} \frac{1}{a} + O(\sqrt{x})$$

The harmonic sum satisfies $\sum_{a \le \sqrt{x}} 1/a = \frac{1}{2} \log x + \gamma + O(1/\sqrt{x})$, so:

$$2x \sum_{a \le \sqrt{x}} \frac{1}{a} = x \log x + 2\gamma x + O(\sqrt{x})$$

For the subtracted term: $\lfloor \sqrt{x} \rfloor^2 = (\sqrt{x} + O(1))^2 = x + O(\sqrt{x})$. Combining:

$$\sum_{n \le x} d(n) = x \log x + 2\gamma x + O(\sqrt{x}) - x - O(\sqrt{x}) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

1.3.4 Remark: The Dirichlet Divisor Problem The error term $O(\sqrt{x})$ in theorem 1.3.3 is not optimal. Let $\Delta(x) = \sum_{n \le x} d(n) - x \log x - (2\gamma - 1)x$ denote the error. The best known upper bound as of writing this book, due to Huxley (2003), is $\Delta(x) = O(x^{131/416+\varepsilon})$, where $131/416 \approx 0.3149$. Hardy proved $\Delta(x) = \Omega(x^{1/4})$, establishing that the error is at least of order $x^{1/4}$ infinitely often. The conjectured optimal exponent is $1/4$, and determining the true order of $\Delta(x)$ (the *Dirichlet divisor problem*) remains open.

Example 1.3.5: Numerical Verification of the Divisor Sum

For $x = 100$: the exact value is $\sum_{n=1}^{100} d(n) = 482$. The asymptotic formula gives $100 \log 100 + (2\gamma - 1) \cdot 100 \approx 460.5 + 15.4 = 476.0$, with error $482 - 476 = 6$. The bound $O(\sqrt{100}) = O(10)$ is consistent. At $x = 10{,}000$: the exact sum is 93,668, and the formula gives $10{,}000 \log(10{,}000) + (2\gamma - 1) \cdot 10{,}000 \approx 92{,}103 + 1{,}544 = 93{,}648$, with error $20 = O(\sqrt{10{,}000}) = O(100)$.

For comparison, the summatory function of μ exhibits much more complicated behavior. For future reference, we box the following:

Definition 1.3.6: The Mertens Function

The *Mertens function* is $M(x) = \sum_{n \le x} \mu(n)$.

The identity $\mu * \mathbf{1} = \varepsilon$ (proposition 1.2.6) implies $\sum_{d \le x} M(x/d) = 1$, a self-referencing identity that constrains the growth of M. The Mertens function oscillates heavily and grows much more slowly than x: the bound $M(x) = o(x)$ is equivalent to the Prime Number Theorem (chapter 5), and the Riemann Hypothesis is equivalent to $M(x) = O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$. Mertens conjectured in 1897 that $|M(x)| < \sqrt{x}$ for all $x \ge 1$; this was disproved by Odlyzko and te Riele in 1985, though no explicit counterexample is known.

The rest of this section is dedicated to computing average orders of various arithmetic functions:

Theorem 1.3.7: Average Order of the Sum-of-Divisors Function

$$\sum_{n\le x}\sigma(n)=\frac{\pi^2}{12}x^2+O(x\log x)$$

***Proof*:**
Since $\sigma=\mathrm{id}*\mathbf{1}$, the double-sum representation gives $\sum_{n\le x}\sigma(n)=\sum_{ab\le x}a$. Summing over b first:

$$\sum_{n\le x}\sigma(n)=\sum_{b\le x}\sum_{a\le x/b}a=\sum_{b\le x}\frac{1}{2}\left\lfloor\frac{x}{b}\right\rfloor\left(\left\lfloor\frac{x}{b}\right\rfloor+1\right)$$

Since $\lfloor t\rfloor(\lfloor t\rfloor+1)/2=t^2/2+O(t)$:

$$\sum_{b\le x}\left(\frac{x^2}{2b^2}+O\left(\frac{x}{b}\right)\right)=\frac{x^2}{2}\sum_{b\le x}\frac{1}{b^2}+O(x\log x)=\frac{x^2}{2}\left(\frac{\pi^2}{6}+O(1/x)\right)+O(x\log x)=\frac{\pi^2}{12}x^2+O(x\log x)$$

A numerical check: $\sum_{n=1}^{100}\sigma(n)=8299$, while $\frac{\pi^2}{12}\cdot 100^2\approx 8225$, giving an error of $74=O(100\log 100)=O(461)$.

Theorem 1.3.8: Average Order of the Euler Totient

$$\sum_{n\le x}\varphi(n)=\frac{3}{\pi^2}x^2+O(x\log x)$$

The leading coefficient $3/\pi^2=1/(2\zeta(2))$ reflects the density of coprime pairs: if two integers are chosen uniformly at random from $\{1,\dots,N\}$, the probability that they are coprime tends to $6/\pi^2$ as $N\to\infty$.

***Proof*:**
By Möbius inversion (theorem 1.2.7), $\varphi(n)=\sum_{d\mid n}\mu(d)(n/d)$. Summing over $n\le x$ and interchanging the order:

$$\sum_{n\le x}\varphi(n)=\sum_{d\le x}\mu(d)\sum_{\substack{n\le x\\ d\mid n}}\frac{n}{d}$$

Writing $n=dm$ with $m\le x/d$:

$$\sum_{\substack{n\le x\\ d\mid n}}\frac{n}{d}=\sum_{m\le x/d}m=\frac{1}{2}\left\lfloor\frac{x}{d}\right\rfloor\left(\left\lfloor\frac{x}{d}\right\rfloor+1\right)=\frac{x^2}{2d^2}+O\left(\frac{x}{d}\right)$$

Substituting:

$$\sum_{n\le x}\varphi(n)=\frac{x^2}{2}\sum_{d\le x}\frac{\mu(d)}{d^2}+O\left(x\sum_{d\le x}\frac{1}{d}\right)=\frac{x^2}{2}\left(\frac{6}{\pi^2}+O(1/x)\right)+O(x\log x)=\frac{3}{\pi^2}x^2+O(x\log x)$$

The identity $\sum_{d=1}^{\infty} \mu(d)/d^2 = 1/\zeta(2) = 6/\pi^2$ is the Euler product $\prod_p (1 - p^{-2})$; a rigorous proof appears in chapter 2.

The appearance of $6/\pi^2$ in the totient average has a probabilistic interpretation. The number of pairs (a, b) with $1 \le a, b \le N$ and $\gcd(a, b) = 1$ is $\sum_{a=1}^{N} \sum_{b=1}^{N} [\gcd(a, b) = 1] = 2 \sum_{n=1}^{N} \varphi(n) - 1$ (the term -1 accounts for the pair $(1, 1)$ counted once). The density of coprime pairs is therefore

$$\frac{2 \sum_{n=1}^{N} \varphi(n) - 1}{N^2} = \frac{6/\pi^2 \cdot N^2 + O(N \log N)}{N^2} \to \frac{6}{\pi^2} \approx 0.6079$$

Concretely, if a and b are chosen uniformly at random from $\{1, \ldots, N\}$, the probability that $\gcd(a, b) = 1$ approaches $6/\pi^2$. Each prime p contributes independently: the probability that $p \nmid \gcd(a, b)$ is $1 - 1/p^2$, and the product $\prod_p (1 - 1/p^2) = 6/\pi^2$ reflects this independence.

The three estimates $\sum d(n) \sim x \log x$, $\sum \sigma(n) \sim \frac{\pi^2}{12} x^2$, and $\sum \varphi(n) \sim \frac{3}{\pi^2} x^2$ illustrate a pattern: the average order of a multiplicative function defined as a convolution $f * \mathbf{1}$ is controlled by the summatory behavior of f and the value $\zeta(2) = \pi^2/6$. The Dirichlet series framework of chapter 2 makes this pattern systematic.

The additive function $\omega(n)$ (number of distinct prime divisors) has a subtler average order.

Proposition 1.3.9: Average Order of $\omega(n)$

$$\sum_{n \le x} \omega(n) = x \log \log x + Bx + O\left(\frac{x}{\log x}\right)$$

where $B = \gamma + \sum_p (\log(1 - 1/p) + 1/p) = 0.2614\ldots$ is related to Mertens' constant. In particular, the average order of $\omega(n)$ is $\log \log n$.

The proof uses the representation $\omega(n) = \sum_{p \mid n} 1$, which after interchanging the order of summation gives $\sum_{n \le x} \omega(n) = \sum_{p \le x} \lfloor x/p \rfloor = x \sum_{p \le x} 1/p + O(\pi(x))$, and the estimate $\sum_{p \le x} 1/p = \log \log x + M + O(1/\log x)$ (Mertens' second theorem, chapter 3) completes the argument. The full proof is given after Mertens' theorems are established; an alternative proof using the Euler product of $\zeta(s)$ appears in chapter 3 as well.

The average $\log \log n$ is strikingly small: a "typical" integer near 10^9 has only about $\log \log(10^9) \approx 3.0$ distinct prime factors. The celebrated Erdős-Kac theorem (1940) refines this by showing that $(\omega(n) - \log \log n)/\sqrt{\log \log n}$ is approximately normally distributed—a Gaussian central limit theorem for a number-theoretic function. This theorem lies outside the scope of this book, but the average order $\log \log n$ established here is the mean of that Gaussian.

Example 1.3.10: Summatory Functions and the Squarefree Indicator

The squarefree indicator function $|\mu(n)|$ (which is 1 if n is squarefree, 0 otherwise) has summatory function $Q(x) = \sum_{n \le x} |\mu(n)|$. In example 1.2.9, the estimate $Q(x) = \frac{6}{\pi^2} x + O(\sqrt{x})$ was derived using the identity $|\mu(n)| = \sum_{d^2 \mid n} \mu(d)$. This can also be obtained from the convolution $|\mu| * \mathbf{1}_\square = \mathbf{1}$ (where $\mathbf{1}_\square(n) = [n \text{ is a perfect square}]$), which inverts to $|\mu| = \mu * \mathbf{1}_\square^{-1}$. The approach via the identity $|\mu(n)| = \sum_{d^2 \mid n} \mu(d)$ is more direct and avoids computing the inverse of $\mathbf{1}_\square$.

Example 1.3.11: Comparing Average Orders

The average orders established in this section are:

$$d(n) \sim \log n, \qquad \sigma(n) \sim \frac{\pi^2}{6}n, \qquad \varphi(n) \sim \frac{6}{\pi^2}n, \qquad \omega(n) \sim \log\log n$$

The first and fourth grow logarithmically (or doubly logarithmically), while σ and φ grow linearly. The product $\sigma(n)\varphi(n) \sim n^2$ on average, reflecting the identity $\varphi(n)\sigma(n)/n^2 = \prod_{p|n}(1 - 1/p^2)$, which is bounded between $6/\pi^2$ and 1 for all n.

Example 1.3.12: Average Behavior of $\varphi(n)/n$

The estimate $\sum_{n\le x}\varphi(n) = \frac{3}{\pi^2}x^2 + O(x\log x)$ implies that the average value of $\varphi(n)/n$ over $\{1, \ldots, N\}$ tends to $6/\pi^2 \approx 0.6079$. This is the same constant as the density of squarefree integers (example 1.2.9): both equal $1/\zeta(2)$. The connection is structural—the "probability" that two integers are coprime and the density of squarefree integers are both governed by $\zeta(2)^{-1} = \prod_p(1 - p^{-2})$.

1.4 Partial Summation and Abel's Summation Formula

The previous section estimated summatory functions by exploiting the convolution structure of specific arithmetic functions. This section introduces a technique that applies to *any* weighted sum $\sum a_n f(n)$, regardless of whether the coefficients a_n have multiplicative structure: *Abel's summation formula*, the discrete analogue of integration by parts. The formula converts a sum $\sum a_n f(n)$ into an integral involving the summatory function $A(t) = \sum_{n\le t} a_n$ and the derivative f', thereby transferring information about the partial sums of (a_n) into analytic information about the weighted sum. It is arguably the single most frequently used technical tool in this book.

Theorem 1.4.1: Abel's Summation Formula

Let $(a_n)_{n\ge1}$ be a sequence of complex numbers, and let $f : [1,\infty) \to \mathbb{C}$ be a continuously differentiable function. Define $A(t) = \sum_{n\le t} a_n$. Then for any $x \ge 1$:

$$\sum_{n\le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)\, f'(t)\, dt$$

The formula is the exact analogue of integration by parts: $\int u\, dv = uv - \int v\, du$, with $u = f$, $dv = dA$ (the Stieltjes measure associated to the step function A), $v = A$, and $du = f'\, dt$. The "boundary term" $A(x)f(x)$ corresponds to $[uv]$, and the integral $\int A(t)f'(t)\, dt$ corresponds to $\int v\, du$. The lower limit of integration is 1 rather than 0 because $A(t) = 0$ for $t < 1$.

***Proof*:**

For $x \ge 1$, write $N = \lfloor x \rfloor$ and split the sum into a telescoping form. For each n with $1 \le n \le N$:

$$a_n = A(n) - A(n-1)$$

Hence:

$$\begin{aligned}\sum_{n=1}^{N} a_n f(n) &= \sum_{n=1}^{N}(A(n) - A(n-1))f(n) \\ &= \sum_{n=1}^{N} A(n)f(n) - \sum_{n=1}^{N} A(n-1)f(n) \\ &= \sum_{n=1}^{N} A(n)f(n) - \sum_{n=0}^{N-1} A(n)f(n+1) \\ &= A(N)f(N) + \sum_{n=1}^{N-1} A(n)(f(n) - f(n+1)) - A(0)f(1)\end{aligned}$$

Since $A(0) = 0$ and $f(n) - f(n+1) = -\int_n^{n+1} f'(t)\,dt$:

$$\sum_{n=1}^{N} a_n f(n) = A(N)f(N) - \sum_{n=1}^{N-1} A(n) \int_n^{n+1} f'(t)\,dt$$

On the interval $[n, n+1)$, the function $A(t)$ is constant with value $A(n)$. Hence $\sum_{n=1}^{N-1} A(n) \int_n^{n+1} f'(t)\,dt = \int_1^N A(t)f'(t)\,dt$. Finally, since $A(t) = A(N)$ for $t \in [N, x]$:

$$A(x)f(x) - \int_1^x A(t)f'(t)\,dt = A(N)f(x) - \int_1^N A(t)f'(t)\,dt - A(N)\int_N^x f'(t)\,dt = A(N)f(N) - \int_1^N A(t)f'(t)\,dt$$

using $f(x) - \int_N^x f'(t)\,dt = f(N)$. This matches the expression above.

The power of Abel's formula lies in the asymmetry of its inputs and outputs. The input is a *discrete* sum $\sum a_n f(n)$, which may be difficult to estimate directly. The output replaces this with a *continuous* integral $\int A(t)f'(t)\,dt$ plus a boundary term, and the integral is often much easier to handle, especially when $A(t)$ has a known asymptotic and $f'(t)$ is smooth and decaying.

The formula can also be written in Stieltjes integral notation as

$$\sum_{n \le x} a_n f(n) = \int_1^x f(t)\,dA(t)$$

where $dA(t)$ is the Stieltjes measure that places mass a_n at each integer n. Integration by parts for Stieltjes integrals then gives $\int f\,dA = [fA] - \int A\,df = A(x)f(x) - \int_1^x A(t)f'(t)\,dt$, recovering theorem 1.4.1. This perspective will return when studying the Mellin and Laplace transforms that appear in chapter 5 and chapter 6.

A useful variant handles the "other direction": given knowledge of a weighted sum, deduce information about the partial sums.

Corollary 1.4.2: Summatory Function Estimate from Weighted Sums

If $\sum_{n\leq x} a_n f(n) = G(x) + O(E(x))$ and f is positive, monotonically decreasing, and continuously differentiable, then

$$A(x) = \frac{G(x)}{f(x)} + \frac{1}{f(x)} \int_1^x A(t) f'(t)\, dt + O\left(\frac{E(x)}{f(x)}\right)$$

When $A(t)$ appears on both sides, this is an implicit equation that can sometimes be solved by iteration.

***Proof*:**

Rearranging theorem 1.4.1: $A(x) = \frac{1}{f(x)}\left(\sum_{n\leq x} a_n f(n) + \int_1^x A(t)f'(t)\, dt\right)$. Substituting the hypothesis gives the result.

This corollary is used in chapter 3 to pass from the estimate $\sum_{p\leq x} \log p/p = \log x + O(1)$ (Mertens' first theorem) to $\sum_{p\leq x} 1/p = \log\log x + O(1)$ (Mertens' second theorem).

Corollary 1.4.3: Summation by Parts for Convergent Series

Under the hypotheses of theorem 1.4.1, if $f(x) \to 0$ as $x \to \infty$ and the integral $\int_1^\infty A(t)f'(t)\, dt$ converges absolutely, then:

$$\sum_{n=1}^{\infty} a_n f(n) = -\int_1^\infty A(t)\, f'(t)\, dt$$

***Proof*:**

Take $x \to \infty$ in theorem 1.4.1. The boundary term $A(x)f(x)$ tends to zero provided $A(x)f(x) \to 0$, which holds when $A(x) = O(x^c)$ for some c and $f(x) = o(x^{-c})$. The integral converges by hypothesis.

The first application recovers the harmonic sum asymptotic, providing an alternative to the integral comparison used in section 0.6.

Example 1.4.4: The Harmonic Sum via Abel Summation

Take $a_n = 1$ and $f(t) = 1/t$, so $A(t) = \lfloor t \rfloor = t - \{t\}$ and $f'(t) = -1/t^2$. Abel's formula gives:

$$\sum_{n\leq x} \frac{1}{n} = \frac{\lfloor x \rfloor}{x} + \int_1^x \frac{\lfloor t \rfloor}{t^2}\, dt = 1 + O(1/x) + \int_1^x \frac{t - \{t\}}{t^2}\, dt = 1 + \int_1^x \frac{dt}{t} - \int_1^x \frac{\{t\}}{t^2}\, dt + O(1/x)$$

The first integral is $\log x$. The second integral converges as $x \to \infty$ (since $0 \leq \{t\} < 1$ and $\int_1^\infty t^{-2}\, dt < \infty$) to a value $\int_1^\infty \{t\}/t^2\, dt = 1 - \gamma$, where γ is the Euler-Mascheroni constant. Hence:

$$\sum_{n\leq x} \frac{1}{n} = \log x + \gamma + O(1/x)$$

The error $O(1/x)$ comes from the boundary term $\lfloor x \rfloor/x = 1 - \{x\}/x$ and the tail of $\int_x^\infty \{t\}/t^2\, dt =$

$O(1/x)$.

Example 1.4.5: Weighted Divisor Sum

Take $a_n = d(n)$ and $f(t) = 1/t$. By theorem 1.3.3, $A(t) = \sum_{n\le t} d(n) = t\log t + (2\gamma-1)t + O(\sqrt{t})$. Abel's formula gives:

$$\begin{aligned}\sum_{n\le x}\frac{d(n)}{n} &= \frac{A(x)}{x} + \int_1^x \frac{A(t)}{t^2}\,dt \\ &= \log x + (2\gamma-1) + O(x^{-1/2}) + \int_1^x \frac{t\log t + (2\gamma-1)t + O(\sqrt{t})}{t^2}\,dt \\ &= \log x + (2\gamma-1) + \int_1^x \frac{\log t}{t}\,dt + (2\gamma-1)\int_1^x \frac{dt}{t} + O(1) \\ &= \log x + (2\gamma-1) + \frac{(\log x)^2}{2} + (2\gamma-1)\log x + O(1)\end{aligned}$$

Hence $\sum_{n\le x} d(n)/n = \frac{1}{2}(\log x)^2 + 2\gamma\log x + O(1)$. This estimate is used in chapter 2 to analyze the Dirichlet series $\sum d(n)n^{-s} = \zeta(s)^2$.

The next application previews the connection between partial summation and Dirichlet series that will be developed in chapter 2.

Proposition 1.4.6: Partial Sums of n^{-s}

For $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 0$ and $s \neq 1$:

$$\sum_{n\le x}\frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\mathrm{Re}(s)})$$

where $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $\mathrm{Re}(s) > 1$, and the formula provides the analytic continuation of the right-hand side to $\mathrm{Re}(s) > 0$.

***Proof*:**

Apply theorem 1.4.1 with $a_n = 1$ and $f(t) = t^{-s}$. Then $A(t) = \lfloor t\rfloor = t - \{t\}$ and $f'(t) = -st^{-s-1}$:

$$\sum_{n\le x} n^{-s} = \frac{\lfloor x\rfloor}{x^s} + s\int_1^x \frac{\lfloor t\rfloor}{t^{s+1}}\,dt = \frac{x-\{x\}}{x^s} + s\int_1^x \frac{t-\{t\}}{t^{s+1}}\,dt$$

Separating the main term from the fractional part:

$$= x^{1-s} + s\int_1^x t^{-s}\,dt - s\int_1^x \frac{\{t\}}{t^{s+1}}\,dt + O(x^{-\mathrm{Re}(s)})$$

The first integral evaluates to $\frac{x^{1-s}-1}{1-s}$ for $s \neq 1$. The second integral converges absolutely for $\mathrm{Re}(s) > 0$ (since $|\{t\}/t^{s+1}| \le t^{-\mathrm{Re}(s)-1}$), and its value from 1 to ∞ defines a constant depending on s. Combining:

$$\sum_{n \le x} n^{-s} = x^{1-s} + \frac{s}{1-s}(x^{1-s} - 1) - s \int_1^x \frac{\{t\}}{t^{s+1}}\, dt + O(x^{-\operatorname{Re}(s)}) = \frac{x^{1-s}}{1-s} + C(s) + O(x^{-\operatorname{Re}(s)})$$

where $C(s) = \frac{s}{s-1} - s \int_1^\infty \{t\} t^{-s-1}\, dt$. For $\operatorname{Re}(s) > 1$, taking $x \to \infty$ shows $C(s) = \zeta(s)$, and the formula extends $C(s)$ to $\operatorname{Re}(s) > 0$ by the convergence of the integral.

This result simultaneously provides the meromorphic continuation of $\zeta(s)$ to $\operatorname{Re}(s) > 0$ (with a simple pole at $s = 1$) and a useful asymptotic for the partial sums $\sum_{n \le x} n^{-s}$. The full meromorphic continuation to all of $\mathbb{C}$ requires the functional equation (chapter 6).

Example 1.4.7: Estimating $\sum_{p \le x} 1/p$ from $\pi(x)$

Abel summation converts knowledge of $\pi(x) = \sum_{p \le x} 1$ (the prime counting function) into an estimate for $\sum_{p \le x} 1/p$. Taking $a_p = 1$ for primes p and $a_n = 0$ otherwise, so $A(t) = \pi(t)$, and $f(t) = 1/t$:

$$\sum_{p \le x} \frac{1}{p} = \frac{\pi(x)}{x} + \int_2^x \frac{\pi(t)}{t^2}\, dt$$

If the Prime Number Theorem $\pi(t) \sim t/\log t$ is assumed, then $\pi(x)/x \sim 1/\log x \to 0$ and

$$\int_2^x \frac{\pi(t)}{t^2}\, dt \sim \int_2^x \frac{dt}{t \log t} = \log\log x - \log\log 2$$

Hence $\sum_{p \le x} 1/p \sim \log\log x$, recovering Mertens' second theorem (up to the precise constant). The rigorous version of this argument, using Chebyshev's bounds instead of PNT, appears in chapter 3.

This example illustrates a general principle: Abel summation lets the estimation of $\sum f(p)$ over primes be reduced to the estimation of $\pi(x)$, and conversely. The relationship is bidirectional: knowledge of $\pi(x)$ yields information about $\sum 1/p$, and the divergence of $\sum 1/p$ (proved in chapter 4 via the Euler product) gives a lower bound on $\pi(x)$.

Example 1.4.8: Partial Summation with the Totient Function

To estimate $\sum_{n \le x} \varphi(n)/n^2$, apply Abel summation with $a_n = \varphi(n)$ and $f(t) = 1/t^2$. By theorem 1.3.8, $A(t) = \sum_{n \le t} \varphi(n) = \frac{3}{\pi^2} t^2 + O(t \log t)$. Then:

$$\begin{aligned}
\sum_{n \le x} \frac{\varphi(n)}{n^2} &= \frac{A(x)}{x^2} + 2\int_1^x \frac{A(t)}{t^3}\, dt = \frac{3}{\pi^2} + O\left(\frac{\log x}{x}\right) + 2\int_1^x \frac{3t^2/\pi^2 + O(t \log t)}{t^3}\, dt \\
&= \frac{3}{\pi^2} + \frac{6}{\pi^2}\int_1^x \frac{dt}{t} + O\left(\int_1^x \frac{\log t}{t^2}\, dt\right) + O\left(\frac{\log x}{x}\right) \\
&= \frac{3}{\pi^2} + \frac{6}{\pi^2}\log x + O(1)
\end{aligned}$$

The dominant term is $\frac{6}{\pi^2}\log x$, confirming that $\sum \varphi(n)/n^2$ diverges logarithmically.

Abel summation also provides a useful criterion for the convergence of series with oscillating terms.

Corollary 1.4.9: Abel's Test for Dirichlet Series

Let (a_n) be a sequence with bounded partial sums: $|A(N)| \leq M$ for all N. If $f : [1, \infty) \to \mathbb{R}$ is monotonically decreasing with $f(t) \to 0$ as $t \to \infty$, then $\sum_{n=1}^{\infty} a_n f(n)$ converges, and

$$\left| \sum_{n>N} a_n f(n) \right| \leq 2M\, f(N+1)$$

***Proof*:**
By theorem 1.4.1 applied to the tail sum:

$$\sum_{N<n\leq x} a_n f(n) = A'(x)f(x) - A'(N)f(N+1) - \int_{N+1}^{x} A'(t)f'(t)\, dt$$

where $A'(t) = \sum_{N<n\leq t} a_n = A(t) - A(N)$, so $|A'(t)| \leq 2M$. Since f is decreasing, $f' \leq 0$, and:

$$\left| \int_{N+1}^{x} A'(t)f'(t)\, dt \right| \leq 2M \int_{N+1}^{x} |f'(t)|\, dt = 2M(f(N+1) - f(x))$$

The boundary terms satisfy $|A'(x)f(x)| \leq 2Mf(x) \to 0$ and $|A'(N)f(N+1)| = 0$. Taking $x \to \infty$: $|\sum_{n>N} a_n f(n)| \leq 2Mf(N+1)$.

This criterion applies, for example, to $\sum \mu(n)/n^s$ for $\mathrm{Re}(s) > 0$: the partial sums $\sum_{n\leq N} \mu(n) = M(N)$ satisfy $M(N) = o(N)$ (equivalent to PNT, as noted in section 1.3), so the series converges for $\mathrm{Re}(s) > 0$ once PNT is established. Before PNT, convergence is known only for $\mathrm{Re}(s) > 1$ from absolute convergence.

1.5 The Euler-Maclaurin Summation Formula

Abel's summation formula from the previous section converts a sum into an integral plus a boundary term, with an error controlled by the fractional part of the summation variable. For many applications, a single error term is sufficient. But when high-precision asymptotic expansions are needed (expansions of the form $\sum_{n\leq N} f(n) = I(N) + c_1 g_1(N) + c_2 g_2(N) + \cdots$ with explicit coefficients) Abel summation isn't enough. The *Euler-Maclaurin formula* remedies this by providing an arbitrarily accurate comparison between $\sum f(n)$ and $\int f(t)\, dt$, with error terms involving successively higher derivatives of f weighted by *Bernoulli numbers*. These numbers, appearing repeatedly throughout analytic number theory, are introduced first:

Definition 1.5.1: Bernoulli Numbers

The *Bernoulli numbers* $B_k \in \mathbb{Q}$ for $k \geq 0$ are defined by the generating function

$$\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} t^k$$

valid for $|t| < 2\pi$.

Equating coefficients with the Taylor expansion of $t/(e^t - 1)$ gives a recursive formula: multiplying both sides by $(e^t - 1)/t = \sum_{j=0}^{\infty} t^j/(j+1)!$ and identifying coefficients yields $B_0 = 1$ and, for $n \geq 1$,

$$\sum_{k=0}^{n} \binom{n+1}{k} B_k = 0$$

This recursion computes the Bernoulli numbers one at a time. The first several values are:

$B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = -\frac{1}{30}, \quad B_5 = 0, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}, \quad B_{10} = \frac{5}{66}$

A fundamental property is the vanishing of odd-indexed Bernoulli numbers beyond B_1.

Proposition 1.5.2: Vanishing of Odd Bernoulli Numbers

For $k \geq 1$, $B_{2k+1} = 0$.

***Proof*:**
Consider the function $G(t) = t/(e^t - 1) + t/2$. Then

$$G(t) = \sum_{k=0}^{\infty} \frac{B_k}{k!} t^k + \frac{t}{2} = 1 + \sum_{k=2}^{\infty} \frac{B_k}{k!} t^k,$$

using $B_0 = 1$ and $B_1 = -1/2$. Direct computation gives $G(-t) = -t/(e^{-t} - 1) - t/2 = t \cdot e^t/(e^t - 1) - t/2 = t(1 + 1/(e^t - 1)) - t/2 = t/(e^t - 1) + t/2 = G(t)$. Hence G is an even function of t, so all coefficients of odd powers of t in $G(t)$ vanish. This forces $B_{2k+1} = 0$ for $k \geq 1$.

The Bernoulli numbers grow rapidly in absolute value: $|B_{2k}| \sim 2(2k)!/(2\pi)^{2k}$ as $k \to \infty$. This growth means that the Euler-Maclaurin expansion is typically an *asymptotic* series rather than a convergent one—truncating at a well-chosen index gives an excellent approximation, but including too many terms causes divergence.

Definition 1.5.3: Bernoulli Polynomials

The *Bernoulli polynomials* $B_k(x)$ are defined by

$$B_k(x) = \sum_{j=0}^{k} \binom{k}{j} B_j\, x^{k-j},$$

or equivalently by the generating function $\frac{te^{xt}}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k(x)}{k!} t^k$ for $|t| < 2\pi$.

The Bernoulli polynomials satisfy several identities that make them the right objects for Euler-Maclaurin. The first few are $B_0(x) = 1$, $B_1(x) = x - 1/2$, $B_2(x) = x^2 - x + 1/6$, $B_3(x) = x^3 - 3x^2/2 + x/2$, and $B_4(x) = x^4 - 2x^3 + x^2 - 1/30$. Note $B_k(0) = B_k$ for all k.

Proposition 1.5.4: Properties of Bernoulli Polynomials

For $k \geq 1$:

1. $B_k'(x) = k\, B_{k-1}(x)$.
2. $\int_0^1 B_k(x)\, dx = 0$.
3. $B_k(x+1) - B_k(x) = kx^{k-1}$.
4. $B_k(1-x) = (-1)^k B_k(x)$.

***Proof*:**
For (1): differentiate the generating function with respect to x:

$$\frac{t^2 e^{xt}}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k'(x)}{k!} t^k$$

The left side equals $t \cdot \frac{te^{xt}}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k(x)}{k!} t^{k+1} = \sum_{k=1}^{\infty} \frac{k\, B_{k-1}(x)}{k!} t^k$. Matching coefficients gives $B_k'(x) = k\, B_{k-1}(x)$ for $k \geq 1$.

For (2): integrate the generating function from $x = 0$ to $x = 1$:

$$\int_0^1 \frac{te^{xt}}{e^t - 1}\, dx = \frac{e^t - 1}{e^t - 1} = 1$$

Equating this with $\sum_{k=0}^{\infty} \frac{1}{k!} \int_0^1 B_k(x)\, dx \cdot t^k$ forces the $k = 0$ coefficient to be 1 and all higher coefficients to vanish: $\int_0^1 B_k(x)\, dx = 0$ for $k \geq 1$.

For (3): the generating function satisfies $\frac{te^{(x+1)t}}{e^t-1} - \frac{te^{xt}}{e^t-1} = te^{xt}$. Matching coefficients of t^k: $\frac{B_k(x+1)-B_k(x)}{k!} = \frac{x^{k-1}}{(k-1)!}$ for $k \geq 1$, giving $B_k(x+1) - B_k(x) = kx^{k-1}$.

For (4): the generating function satisfies $\frac{te^{(1-x)t}}{e^t-1} = \frac{te^{-xt}e^t}{e^t-1}$. Using the identity $\frac{e^t}{e^t-1} = \frac{-1}{e^{-t}-1} \cdot (-1)$, this equals $\frac{-te^{-xt}}{e^{-t}-1}$, which is the generating function of $B_k(x)$ evaluated at $-t$. Matching coefficients of t^k on both sides: $B_k(1-x)/k! = (-1)^k B_k(x)/k!$, giving $B_k(1-x) = (-1)^k B_k(x)$.

Property (3) is the key identity for Euler-Maclaurin: it expresses the power kx^{k-1} as the discrete difference $B_k(x+1) - B_k(x)$, which allows telescoping sums to be converted into integrals of Bernoulli polynomials. Property (4) gives the reflection symmetry: combined with (1), it shows that $B_{2k+1}(1/2) = 0$ for $k \geq 0$, generalizing the vanishing of odd Bernoulli numbers.

Definition 1.5.5: Periodic Bernoulli Functions

For $k \geq 0$, the *periodic Bernoulli function* $\widetilde{B}_k : \mathbb{R} \to \mathbb{R}$ is

$$\widetilde{B}_k(x) = B_k(\{x\})$$

where $\{x\} = x - \lfloor x \rfloor$ is the fractional part of x. The function $\widetilde{B}_k$ is periodic with period 1.

For $k = 0$, $\widetilde{B}_0(x) = 1$ is constant. For $k = 1$, $\widetilde{B}_1(x) = \{x\} - 1/2$, which is the sawtooth function (discontinuous at integers). For $k \geq 2$, $\widetilde{B}_k$ is continuous everywhere because $B_k(0) = B_k(1) = B_k$ for $k \geq 2$ (from property (3): $B_k(1) - B_k(0) = k \cdot 0^{k-1} = 0$).

With these tools in place, the Euler-Maclaurin formula follows from repeated integration by parts.

Theorem 1.5.6: Euler-Maclaurin Summation Formula

Let $f \in C^{2p}[a,b]$ with $a, b \in \mathbb{Z}$ and $p \geq 1$. Then

$$\sum_{n=a}^{b} f(n) = \int_a^b f(t)\,dt + \frac{f(a)+f(b)}{2} + \sum_{k=1}^{p} \frac{B_{2k}}{(2k)!}\left(f^{(2k-1)}(b) - f^{(2k-1)}(a)\right) + R_p$$

where the remainder term is

$$R_p = -\int_a^b \frac{\widetilde{B}_{2p}(t)}{(2p)!} f^{(2p)}(t)\,dt$$

***Proof*:**

Step 1: The base case. On each unit interval $[n, n+1]$, use $\widetilde{B}_1(t) = t - n - 1/2$ for $t \in [n, n+1)$ and note that $\widetilde{B}_1'(t) = 1$ on this interval. Then:

$$\int_n^{n+1} f(t)\,dt = \int_n^{n+1} f(t) \cdot 1\,dt = \int_n^{n+1} f(t)\,d\widetilde{B}_1(t)$$

Integration by parts gives

$$\int_n^{n+1} f(t)\,d\widetilde{B}_1(t) = [f(t)\widetilde{B}_1(t)]_n^{n+1} - \int_n^{n+1} \widetilde{B}_1(t)\,f'(t)\,dt$$

Since $\widetilde{B}_1(n^+) = -1/2$ and $\widetilde{B}_1((n+1)^-) = 1/2$:

$$\int_n^{n+1} f(t)\,dt = \frac{f(n)+f(n+1)}{2} - \int_n^{n+1} \widetilde{B}_1(t)\,f'(t)\,dt$$

Summing over $n = a, a+1, \dots, b-1$:

$$\int_a^b f(t)\,dt = \sum_{n=a}^{b-1} \frac{f(n)+f(n+1)}{2} - \int_a^b \widetilde{B}_1(t)\,f'(t)\,dt = \sum_{n=a}^{b} f(n) - \frac{f(a)+f(b)}{2} - \int_a^b \widetilde{B}_1(t)\,f'(t)\,dt$$

Rearranging:

$$\sum_{n=a}^{b} f(n) = \int_a^b f(t)\,dt + \frac{f(a)+f(b)}{2} + \int_a^b \widetilde{B}_1(t)\,f'(t)\,dt$$

This is the formula for $p = 0$ (no Bernoulli correction terms), with remainder $\int \widetilde{B}_1 f'$.

Step 2: Iteration. For the remainder integral, apply integration by parts using $\widetilde{B}'_{k+1}(t) = (k+1)\widetilde{B}_k(t)$ for $t \notin \mathbb{Z}$ (from proposition 1.5.4(1)). Then

$$\int_a^b \widetilde{B}_k(t)\,f^{(k)}(t)\,dt = \left[\frac{\widetilde{B}_{k+1}(t)}{k+1} f^{(k)}(t)\right]_a^b - \int_a^b \frac{\widetilde{B}_{k+1}(t)}{k+1}\,f^{(k+1)}(t)\,dt$$

At the endpoints $a, b \in \mathbb{Z}$: $\widetilde{B}_{k+1}(a) = \widetilde{B}_{k+1}(b) = B_{k+1}(0) = B_{k+1}$. Hence

$$\int_a^b \widetilde{B}_k(t)\,f^{(k)}(t)\,dt = \frac{B_{k+1}}{k+1}(f^{(k)}(b) - f^{(k)}(a)) - \int_a^b \frac{\widetilde{B}_{k+1}(t)}{k+1}\,f^{(k+1)}(t)\,dt$$

Dividing by $k!$: $\int_a^b \frac{\widetilde{B}_k(t)}{k!} f^{(k)}(t)\,dt = \frac{B_{k+1}}{(k+1)!}(f^{(k)}(b) - f^{(k)}(a)) - \int_a^b \frac{\widetilde{B}_{k+1}(t)}{(k+1)!} f^{(k+1)}(t)\,dt$.

Step 3: Apply iteratively. Starting from $\int \widetilde{B}_1 f'$ and applying the above recursion $2p-1$ times, the odd-indexed Bernoulli numbers $B_3, B_5, \dots$ vanish, contributing nothing to the boundary terms. Only the terms with even index $2k$ survive, giving the stated formula with remainder $R_p = -\int_a^b \frac{\widetilde{B}_{2p}(t)}{(2p)!} f^{(2p)}(t)\,dt$.

The power of the Euler-Maclaurin formula is that it gives an exact identity for any p, not just an asymptotic approximation. The practical question is: how should p be chosen? For smooth decaying f, the remainder R_p is typically small for moderate p but grows rapidly as $p \to \infty$ (reflecting the rapid growth of $|B_{2p}|$). The optimal p depends on the specific f and is generally the value that minimizes $|R_p|$.

The first major application is to Stirling's formula.

Theorem 1.5.7: Stirling's Formula

For $x > 0$ as $x \to \infty$:

$$\log \Gamma(x+1) = x \log x - x + \frac{1}{2}\log(2\pi x) + \sum_{k=1}^{p-1} \frac{B_{2k}}{2k(2k-1)} \cdot \frac{1}{x^{2k-1}} + O_p(x^{-2p+1})$$

In particular, the leading-order form is $\log(n!) = n \log n - n + \frac{1}{2}\log(2\pi n) + O(1/n)$, or equivalently $n! \sim \sqrt{2\pi n}(n/e)^n$.

***Proof*:**
Apply theorem 1.5.6 to $f(t) = \log t$ on $[1, N]$ with N a positive integer. The derivatives are $f'(t) = 1/t$, $f''(t) = -1/t^2$, and generally $f^{(k)}(t) = (-1)^{k-1}(k-1)!/t^k$. The integral is $\int_1^N \log t \, dt = N \log N - N + 1$. The boundary terms and Bernoulli corrections give:

$$\sum_{n=1}^{N} \log n = N \log N - N + 1 + \frac{\log N}{2} + \sum_{k=1}^{p} \frac{B_{2k}}{(2k)!}\left(f^{(2k-1)}(N) - f^{(2k-1)}(1)\right) + R_p$$

With $f^{(2k-1)}(t) = (2k-2)!/t^{2k-1}$:

$$\frac{B_{2k}}{(2k)!} \cdot (2k-2)! \left(\frac{1}{N^{2k-1}} - 1\right) = \frac{B_{2k}}{2k(2k-1)}\left(\frac{1}{N^{2k-1}} - 1\right)$$

The sum $\log(N!) = \sum_{n=1}^N \log n$ on the left, and letting $N \to \infty$ with the correction terms at infinity vanishing, identifies the constant as $\frac{1}{2}\log(2\pi)$. The details of the constant determination use Wallis's product or the duplication formula for Γ; the result is as stated.

Example 1.5.8: Numerical Stirling

For $n = 10$: the exact value is $10! = 3{,}628{,}800$. The leading Stirling approximation $\sqrt{20\pi}(10/e)^{10} \approx 3{,}598{,}696$ gives relative error about 0.83%. Including the $B_2/(2x) = 1/(12x)$ correction gives $\sqrt{20\pi}(10/e)^{10}\exp(1/120) \approx 3{,}628{,}810$, matching the exact value to 5 significant figures.

A second application gives an asymptotic expansion for the partial sums of n^{-s} refined beyond proposition 1.4.6.

Proposition 1.5.9: Asymptotic Expansion of $\sum n^{-s}$

For $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 0$ and $s \neq 1$, and any integer $p \geq 1$:

$$\sum_{n=1}^{N} n^{-s} = \frac{N^{1-s}}{1-s} + \zeta(s) + \frac{N^{-s}}{2} + \sum_{k=1}^{p-1} \frac{B_{2k}}{(2k)!} s(s+1)\cdots(s+2k-2) \cdot N^{-s-2k+1} + O_{p,s}(N^{-\mathrm{Re}(s)-2p+1})$$

***Proof*:**
Apply theorem 1.5.6 to $f(t) = t^{-s}$ on $[1, N]$. The integral is $\int_1^N t^{-s}\,dt = \frac{N^{1-s}-1}{1-s}$. Derivatives: $f^{(k)}(t) = (-1)^k s(s+1)\cdots(s+k-1)t^{-s-k}$. The boundary terms $(f(1)+f(N))/2 = (1+N^{-s})/2$ and the Bernoulli corrections combine to give the stated expansion. The constant $\zeta(s)$ emerges from the value at $N = 1$ when $\mathrm{Re}(s) > 1$, and the expansion extends the right-hand side to $\mathrm{Re}(s) > 1 - 2p$ as an analytic function of s.

1.5.10 Foreshadowing: Bernoulli Numbers and Special Values of $\zeta(s)$ The proposition above suggests the following identity. Setting $s = -n$ (a negative integer) formally in the expansion and taking $N = 1$, the boundary and Bernoulli terms combine to yield

$$\zeta(-n) = -\frac{B_{n+1}}{n+1} \quad \text{for } n \geq 1$$

This is not rigorous here: the meromorphic continuation of $\zeta(s)$ to all of $\mathbb{C}$ has not yet been established, and the formal manipulation of the expansion at $s = -n$ requires justification. Both issues are resolved in chapter 6, where the functional equation of $\zeta(s)$ provides the continuation and confirms the identity. In particular: $\zeta(0) = -1/2$, $\zeta(-1) = -1/12$, $\zeta(-3) = 1/120$, and $\zeta(-2k) = 0$ for $k \geq 1$ (the trivial zeros). The famous "identity" $1 + 2 + 3 + \cdots = -1/12$ is a shorthand for $\zeta(-1) = -1/12$.

Example 1.5.11: High-Precision Computation of the Harmonic Sum
To compute $H_{100} = \sum_{n=1}^{100} 1/n$ to high precision, apply theorem 1.5.6 with $f(t) = 1/t$ on $[1, 100]$ and $p = 3$. With $f^{(2k-1)}(t) = -(2k-1)!/t^{2k}$, the Bernoulli term for index $2k$ contributes

$$\frac{B_{2k}}{(2k)!}\left(f^{(2k-1)}(100) - f^{(2k-1)}(1)\right) = \frac{B_{2k}}{2k}\left(1 - \frac{1}{100^{2k}}\right)$$

The formula gives:

$$H_{100} = \log 100 + \frac{1 + 1/100}{2} + \frac{1/6}{2}\left(1 - \frac{1}{100^2}\right) + \frac{-1/30}{4}\left(1 - \frac{1}{100^4}\right) + \frac{1/42}{6}\left(1 - \frac{1}{100^6}\right) + R_3$$

Numerically: $\log 100 \approx 4.60517$, boundary $= 0.505$, $k = 1$ term ≈ 0.08333, $k = 2$ term ≈ -0.00833, $k = 3$ term ≈ 0.00397. The sum is approximately 5.18914, matching the exact value $H_{100} \approx 5.18738$ to within 0.002. The discrepancy is the remainder R_3, which is of order $|B_6|/(6 \cdot 100^0) \approx 0.004$ (the worst contribution being at the lower endpoint $t = 1$, where the derivatives of $1/t$ are largest).

Exercise 1.5.1

1. For each of the following arithmetic functions, determine whether it is multiplicative, totally multiplicative, additive, totally additive, or none of these. Justify each answer.

   1. $f(n) = n^2$
   2. $f(n) = \log n$
   3. $f(n) = \sigma(n)/n$

4. $f(n) = \sigma_2(n) = \sum_{d|n} d^2$

2. 1. Compute $\Lambda(n)$ for $n = 1, 2, \ldots, 12$.
   2. Verify the von Mangoldt identity $\sum_{d|n} \Lambda(d) = \log n$ for $n = 12$ and $n = 18$.
   3. Show that $\Lambda(n) \leq \log n$ for all $n \geq 1$, with equality if and only if n is prime.

3. Prove that σ_k is multiplicative for every $k \in \mathbb{C}$, and derive the formula

$$\sigma_k(p^a) = \frac{p^{(a+1)k} - 1}{p^k - 1}$$

for $k \neq 0$ and any prime p. State the corresponding formula for $k = 0$.

4. Prove that the Liouville function λ is totally multiplicative, and show that $\lambda(n)\mu(n) = \mu(n)^2$ for all $n \geq 1$. Deduce that λ and μ agree on squarefree integers.

5. Compute the Dirichlet convolution $(f * g)(12)$ for:
   1. $f = g = \mathbf{1}$.
   2. $f = \mu$, $g = \mathrm{id}$.
   3. $f = \lambda$, $g = \mathbf{1}$.

   Compare each result with the value predicted by the convolution identities in proposition 1.2.10.

6. Using the recursive formula for Dirichlet inverses, compute $\sigma^{-1}(n)$ for $n = 1, 2, 3, 4, 6$.

7. Prove the *multiplicative* version of Möbius inversion: if $F(n) = \prod_{d|n} f(d)$ for all $n \geq 1$, then $f(n) = \prod_{d|n} F(d)^{\mu(n/d)}$.

8. Prove the identity $d * \varphi = \sigma$. [Hint: use $d = \mathbf{1} * \mathbf{1}$ and $\varphi = \mathrm{id} * \mu$.]

9. Let f be a multiplicative function. Show that f^{-1} (the Dirichlet inverse of f) is also multiplicative, and prove that for any prime p,

$$f^{-1}(p^a) = \sum_{k=1}^{a} (-1)^k \sum_{\substack{a_1 + \cdots + a_k = a \\ a_i \geq 1}} f(p^{a_1}) \cdots f(p^{a_k})$$

10. Using the hyperbola method, provide an independent derivation of the estimate $\sum_{n \leq x} d(n) = x \log x + O(x)$. Then verify numerically for $x = 20$: compute the exact sum $\sum_{n=1}^{20} d(n)$ and compare with $20 \log 20 + (2\gamma - 1) \cdot 20$.

11. Let $Q(x) = \sum_{n \leq x} |\mu(n)|$ count the squarefree integers up to x. Using the hyperbola method applied to the identity $|\mu(n)| = \sum_{d^2|n} \mu(d)$, give a self-contained proof that $Q(x) = 6x/\pi^2 + O(\sqrt{x})$.

12. Prove that

$$\sum_{\substack{n \leq x \\ n \text{ squarefree}}} \frac{1}{n} = \frac{6}{\pi^2} \log x + C + O(x^{-1/2})$$

for a constant C. [Hint: use Abel summation with $a_n = |\mu(n)|$ and $f(t) = 1/t$, then apply exercise 1.5.1 (11).]

13. Prove the trivial bound $|M(x)| = |\sum_{n\le x} \mu(n)| \le x$ for all $x \ge 1$. Then, using the identity $\sum_{d\le x} M(x/d) = 1$, derive the recursion-based estimate $M(x) = 1 - \sum_{2\le d\le x} M(x/d)$ and explain why this does not immediately yield $M(x) = o(x)$.

14. Using Abel summation and the estimate $\sum_{n\le x} \sigma(n) = \frac{\pi^2}{12}x^2 + O(x\log x)$, prove that

$$\sum_{n\le x} \frac{\sigma(n)}{n} = \frac{\pi^2}{6}x + O((\log x)^2)$$

15. Using Abel summation and $\sum_{n\le x} \varphi(n) = \frac{3}{\pi^2}x^2 + O(x\log x)$, show that

$$\sum_{n\le x} \frac{\varphi(n)}{n^2} = \frac{6}{\pi^2}\log x + C + O\left(\frac{\log x}{x}\right)$$

for a constant C. (An explicit formula for C is not required.)

16. Let a_n satisfy $|A(N)| = |\sum_{n\le N} a_n| \le M$ for all N. Use Abel summation to prove that $\sum_{n=1}^{\infty} a_n/n^s$ converges for every $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 0$, and the convergence is uniform on compact subsets of $\{\mathrm{Re}(s) > 0\}$.

17. Using Abel summation with $a_n = 1$ and $f(t) = (\log t)/t$, prove that

$$\sum_{n\le x} \frac{\log n}{n} = \frac{(\log x)^2}{2} + C_1 + O\left(\frac{\log x}{x}\right)$$

for a constant C_1. (An explicit value for C_1 is not required.)

18. Using the recursion $\sum_{k=0}^{n} \binom{n+1}{k} B_k = 0$ for $n \ge 1$, compute B_6, B_8, and B_{10} directly from the values $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$, and the vanishing of odd-indexed B_k for $k \ge 3$.

19.
    1. Compute the Bernoulli polynomials $B_2(x)$, $B_3(x)$, and $B_4(x)$ from the formula $B_k(x) = \sum_{j=0}^{k} \binom{k}{j} B_j\, x^{k-j}$.
    2. Verify $B_k(1) = B_k(0)$ for $k \in \{2, 3, 4\}$.
    3. Use the telescoping identity $B_3(x+1) - B_3(x) = 3x^2$ to derive the closed form $\sum_{j=0}^{N-1} j^2 = N(N-1)(2N-1)/6$.

20. Apply the Euler-Maclaurin formula to $f(t) = 1/t^2$ on $[1, N]$ with $p = 2$ (two Bernoulli correction terms). Extract the asymptotic expansion for $\sum_{n=1}^{N} 1/n^2$ and identify the N-independent constant as $\zeta(2) = \pi^2/6$.

21. The functional equation of $\zeta(s)$ implies $\zeta(2k) = (-1)^{k+1}\frac{(2\pi)^{2k}B_{2k}}{2(2k)!}$ for $k \ge 1$ (this will be proved in chapter 6). Verify this identity for $k = 1$ (giving $\zeta(2) = \pi^2/6$) and $k = 2$ (giving $\zeta(4) = \pi^4/90$). What does the formula give for $k = 3$?

2

Dirichlet Series and Euler Products

Chapter 1 organized the arithmetic functions into a ring under Dirichlet convolution and extracted asymptotic information about their summatory functions through summation techniques. The present chapter introduces the analytic objects that encode the same information in a fundamentally different way: *Dirichlet series*. The Dirichlet series $D(f,s) = \sum_{n=1}^{\infty} f(n)/n^s$ associates to each arithmetic function f an analytic function of a complex variable s, and this association converts Dirichlet convolution into pointwise multiplication, multiplicativity into an Euler product over primes, and the summatory function into a contour integral. The poles of the Dirichlet series control the main terms in asymptotic estimates; its zeros govern the oscillatory corrections; and its analytic continuation beyond the region of absolute convergence is the mechanism by which the distribution of primes becomes accessible to the methods of complex analysis.

This chapter builds the dictionary between the algebraic side (arithmetic functions, convolution, multiplicativity) and the analytic side (Dirichlet series, products, Mellin transforms). Sections 2.1-2.3 establish the algebraic-analytic correspondence: convergence, the multiplication theorem, and Euler products. Section 2.4 develops the Mellin transform as a self-contained tool, and Section 2.5 proves Perron's formula, which inverts the Dirichlet series back into information about partial sums. Together, these tools are the analytic engine of the book.

2.1 Dirichlet Series: Convergence and Analyticity

An arithmetic function f is a sequence of complex numbers; its Dirichlet series is the analytic function obtained by summing $f(n)/n^s$ over all positive integers. The first question is: for which values of the complex variable s does the series converge? The answer has a clean geometric structure: there is a critical vertical line in the complex plane, to the right of which the series converges and to the left of which it diverges. This *abscissa of convergence* is the Dirichlet series analogue of the radius of convergence for a power series, and it is determined by the growth rate of the partial sums of the coefficients.

Definition 2.1.1: Dirichlet Series

A *Dirichlet series* is a series of the form

$$D(f,s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

where $f : \mathbb{Z}_{>0} \to \mathbb{C}$ is an arithmetic function and $s \in \mathbb{C}$.

The variable s is conventionally written as $s = \sigma + it$ with $\sigma = \mathrm{Re}(s)$ and $t = \mathrm{Im}(s)$. Since $n^s = n^\sigma \cdot n^{it} = n^\sigma e^{it\log n}$, the factor $n^{-\sigma}$ controls convergence while n^{-it} contributes oscillation without affecting absolute convergence. This separation between the real and imaginary parts of s underlies the half-plane structure of the convergence region.

The fundamental convergence theorem is the following.

Theorem 2.1.2: Half-Plane of Convergence

Let $D(f,s) = \sum f(n)/n^s$ be a Dirichlet series. If $D(f,s_0)$ converges for some $s_0 \in \mathbb{C}$, then $D(f,s)$ converges for all s with $\mathrm{Re}(s) > \mathrm{Re}(s_0)$, and the convergence is uniform on compact subsets of this half-plane.

***Proof*:**
Write $\sigma_0 = \mathrm{Re}(s_0)$ and fix s with $\sigma = \mathrm{Re}(s) > \sigma_0$. Set $a_n = f(n)/n^{s_0}$ and $g(t) = t^{s_0-s}$, so that $f(n)/n^s = a_n \cdot n^{s_0-s}$. By hypothesis, $A(N) = \sum_{n\le N} a_n$ is bounded (as the partial sums of a convergent series). The function $g(t) = t^{s_0-s}$ has $|g(t)| = t^{\sigma_0-\sigma} \to 0$ as $t \to \infty$ (since $\sigma_0 - \sigma < 0$), and $g'(t) = (s_0 - s)t^{s_0-s-1}$.

By Abel summation (theorem 1.4.1):

$$\sum_{n=1}^{N} \frac{f(n)}{n^s} = \sum_{n=1}^{N} a_n \cdot n^{s_0-s} = A(N)\cdot N^{s_0-s} - \int_1^N A(t)(s_0-s)t^{s_0-s-1}\,dt$$

As $N \to \infty$: $|A(N)\cdot N^{s_0-s}| \le M \cdot N^{\sigma_0-\sigma} \to 0$, where $M = \sup_N |A(N)|$. The integral converges absolutely:

$$\int_1^\infty |A(t)|\,|s_0-s|\,t^{\sigma_0-\sigma-1}\,dt \le M|s_0-s| \int_1^\infty t^{\sigma_0-\sigma-1}\,dt = \frac{M|s_0-s|}{\sigma-\sigma_0} < \infty$$

Hence $D(f,s)$ converges.

For uniform convergence on a compact set $K \subset \{\mathrm{Re}(s) > \sigma_0\}$: let $\sigma_1 = \min_{s\in K}\mathrm{Re}(s) > \sigma_0$. The same bound gives $|\sum_{n>N} f(n)/n^s| \le M|s_0-s|N^{\sigma_0-\sigma_1}/(\sigma_1-\sigma_0) + MN^{\sigma_0-\sigma_1}$, and $|s_0-s|$ is bounded on K, so the tail tends to 0 uniformly.

This theorem shows that the region of convergence of a Dirichlet series is always a right half-plane (or the entire complex plane, or the empty set). The boundary of this half-plane defines the abscissa of convergence.

Definition 2.1.3: Abscissae of Convergence and Absolute Convergence

For a Dirichlet series $D(f, s)$:

1. The *abscissa of convergence* σ_c is the infimum of the set of $\sigma \in \mathbb{R}$ such that $D(f, \sigma + it)$ converges for all t. Equivalently, $D(f, s)$ converges for $\text{Re}(s) > \sigma_c$ and diverges for $\text{Re}(s) < \sigma_c$.

2. The *abscissa of absolute convergence* σ_a is the infimum of the set of σ such that $\sum |f(n)|/n^\sigma < \infty$.

By convention, $\sigma_c = -\infty$ if the series converges everywhere, $\sigma_c = +\infty$ if it converges nowhere, and similarly for σ_a.

By definition, $\sigma_c \leq \sigma_a$ (absolute convergence implies convergence). The gap between σ_c and σ_a reflects the amount of cancellation in the coefficients.

Proposition 2.1.4: Gap Between Abscissae

For any Dirichlet series: $\sigma_c \leq \sigma_a \leq \sigma_c + 1$. Both inequalities can be equalities.

***Proof*:**
The inequality $\sigma_c \leq \sigma_a$ is immediate. For the upper bound: if $D(f, s_0)$ converges at s_0 with $\text{Re}(s_0) = \sigma_0$, then $f(n)/n^{s_0} \to 0$, so $|f(n)| \leq Cn^{\sigma_0}$ for some constant C. Hence $\sum |f(n)|/n^\sigma \leq C \sum n^{\sigma_0 - \sigma}$, which converges for $\sigma > \sigma_0 + 1$. Taking the infimum over convergent s_0 gives $\sigma_a \leq \sigma_c + 1$.

Equality $\sigma_c = \sigma_a$ holds for $\zeta(s) = \sum 1/n^s$ (both equal 1, since the coefficients are non-negative). Equality $\sigma_a = \sigma_c + 1$ holds for the alternating series $\sum (-1)^{n+1}/n^s = (1 - 2^{1-s})\zeta(s)$: here $\sigma_c = 0$ (by Abel's test, since the partial sums of $(-1)^{n+1}$ are bounded), while $\sigma_a = 1$.

The gap between σ_c and σ_a measures the cancellation among the coefficients $f(n)$. For non-negative coefficients, there is no cancellation, and $\sigma_c = \sigma_a$. For coefficients with heavy sign alternation (like $(-1)^{n+1}$), the gap can reach its maximum value of 1. The intermediate case $0 < \sigma_a - \sigma_c < 1$ occurs for the Dirichlet series $\sum \mu(n)/n^s$: the cancellation in the Möbius function is sufficient to push σ_c below $\sigma_a = 1$, but determining σ_c precisely is equivalent to determining the supremum of the real parts of the zeros of $\zeta(s)$.

More generally, whenever the coefficients $f(n)$ are non-negative, the abscissa of convergence equals the abscissa of absolute convergence, and the Dirichlet series $D(f, s)$ has a singularity (non-removable) at the point $s = \sigma_a$ on the real axis. This is the Landau-Vivanti theorem for Dirichlet series.

Proposition 2.1.5: Landau's Theorem on Non-Negative Coefficients

If $f(n) \geq 0$ for all n, then $\sigma_c = \sigma_a$, and $D(f,s)$ has a singularity at $s = \sigma_a$.

***Proof*:**
Since $f(n) \geq 0$: $|f(n)|/n^\sigma = f(n)/n^\sigma$, so $\sigma_c = \sigma_a$. For the singularity: suppose for contradiction that $D(f,s)$ extends analytically to a neighborhood of σ_a. Then the Taylor expansion of $D(f,s)$ at some point $s_1 > \sigma_a$ would converge in a disk centered at s_1 reaching past σ_a, giving

$$D(f,s) = \sum_{k=0}^{\infty} \frac{D^{(k)}(f,s_1)}{k!}(s-s_1)^k = \sum_{k=0}^{\infty} \frac{(s_1-s)^k}{k!} \sum_{n=1}^{\infty} \frac{f(n)(\log n)^k}{n^{s_1}}$$

for $|s - s_1| < s_1 - \sigma_a + \varepsilon$. Since $f(n) \geq 0$, the double sum has non-negative terms, and Tonelli's theorem allows swapping the summations. At a real point $s_2 < \sigma_a$ inside the disk:

$$D(f,s_2) = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_1}} \sum_{k=0}^{\infty} \frac{(s_1-s_2)^k(\log n)^k}{k!} = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_1}} n^{s_1-s_2} = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_2}},$$

so the Dirichlet series converges at $s_2 < \sigma_a$, contradicting the definition of σ_a.

For $\zeta(s) = \sum 1/n^s$, this theorem guarantees a singularity at $s = 1$, consistent with the pole of $\zeta(s)$ at $s = 1$ established by the integral representation of section 1.4. For $\sum d(n)/n^s = \zeta(s)^2$, the theorem gives a singularity at $s = 1$ (a double pole). The theorem does not determine the *nature* of the singularity (pole, branch point, essential singularity), only its existence.

The abscissae can be computed from the coefficients via the following formulas.

Proposition 2.1.6: Formulas for the Abscissae

Let $A(x) = \sum_{n \leq x} f(n)$ and $A^*(x) = \sum_{n \leq x} |f(n)|$. Then:

$$\sigma_c = \limsup_{x\to\infty} \frac{\log|A(x)|}{\log x}, \qquad \sigma_a = \limsup_{x\to\infty} \frac{\log A^*(x)}{\log x}$$

(With the convention that $\sigma_c = -\infty$ if $A(x)$ is eventually zero, etc.)

***Proof*:**
For σ_a: if $\sigma > \limsup_{x\to\infty} \frac{\log A^*(x)}{\log x}$, then $A^*(x) \leq x^{\sigma-\varepsilon}$ for large x and some $\varepsilon > 0$. Abel summation applied to $\sum |f(n)|/n^\sigma$ with partial sums $A^*(x)$ gives convergence, since the integral $\int_1^\infty A^*(t)/t^{\sigma+1}\,dt \leq \int_1^\infty t^{-1-\varepsilon}\,dt < \infty$. Conversely, if $\sigma < \limsup \log A^*(x)/\log x$, then $A^*(x) \geq x^{\sigma+\varepsilon}$ for infinitely many x, and the series diverges. The proof for σ_c is analogous, replacing $A^*(x)$ with $A(x)$ and using Abel summation for conditional convergence.

The behavior at the boundary $\mathrm{Re}(s) = \sigma_c$ is delicate. The series may converge at every point of this line, at no point, or at some points but not others. For $\zeta(s)$, the boundary is $\mathrm{Re}(s) = 1$, and the series diverges at $s = 1$ but converges conditionally at every other point $1 + it$ with $t \neq 0$ (by Dirichlet's test, since the partial sums of n^{-it} are bounded for $t \neq 0$ and $1/n$ decreases to 0). For the eta function $\eta(s)$, the boundary is $\mathrm{Re}(s) = 0$, and the series diverges at $s = 0$ by oscillation but converges at $s = it$ for $t \neq 0$.

A final remark on the strip $\sigma_c < \mathrm{Re}(s) < \sigma_a$ (when it is nonempty). In this region, the Dirichlet series converges but not absolutely. The convergence is conditional, and the series does not converge uniformly on all of $\{\mathrm{Re}(s) > \sigma_c\}$ (only on compact subsets). This strip is analogous to the annulus of conditional convergence for Laurent series. The arithmetic content of this strip is often significant: for $\sum \mu(n)/n^s$, the identity $1/\zeta(s) = \sum \mu(n)/n^s$ extends into the strip $\sigma_c < \mathrm{Re}(s) \leq 1$, and the non-vanishing of $\zeta(s)$ on the line $\mathrm{Re}(s) = 1$ is what makes the extension to $\sigma_c < 1$ possible.

A Dirichlet series is not just convergent in its half-plane: it is analytic there:

Theorem 2.1.7: Analyticity and Term-by-Term Differentiation

A Dirichlet series $D(f, s) = \sum f(n)/n^s$ is an analytic function on $\{\mathrm{Re}(s) > \sigma_c\}$, and for all $k \geq 1$:

$$D^{(k)}(f, s) = \sum_{n=1}^{\infty} \frac{f(n)(-\log n)^k}{n^s}$$

in the same half-plane. This series also has abscissa of convergence σ_c.

***Proof*:**
Fix a compact set $K \subset \{\mathrm{Re}(s) > \sigma_c\}$ with $\sigma_1 = \min_{s \in K} \mathrm{Re}(s) > \sigma_c$. Choose σ_0 with $\sigma_c < \sigma_0 < \sigma_1$. The partial sums $A(N) = \sum_{n \leq N} f(n)/n^{s_0}$ are bounded for $\mathrm{Re}(s_0) = \sigma_0$, and by the proof of theorem 2.1.2, $D(f, s)$ converges uniformly on K. Since each term $f(n)/n^s$ is an entire function of s, and the series converges uniformly on compact subsets, the sum is analytic on $\{\mathrm{Re}(s) > \sigma_c\}$ (by Weierstrass's theorem on uniformly convergent series of analytic functions, [13]).

For differentiation: $\frac{d}{ds}(f(n)/n^s) = -f(n)\log n/n^s$. The series $\sum f(n)\log n/n^s$ has the same abscissa of convergence as $\sum f(n)/n^s$: the factor $\log n$ grows slower than any power n^{ε}, so for $\mathrm{Re}(s) > \sigma_c + \varepsilon$, the additional factor $\log n$ does not affect absolute convergence, and letting $\varepsilon \to 0$ shows the abscissa is unchanged. Uniform convergence on compacta allows exchanging the derivative and the sum.

The analyticity theorem has an important consequence: a Dirichlet series is determined by its coefficients.

Corollary 2.1.8: Uniqueness of Coefficients

If $D(f, s) = D(g, s)$ for all s in some half-plane $\{\mathrm{Re}(s) > \sigma_0\}$, then $f(n) = g(n)$ for all n.

***Proof*:**
Set $h = f - g$, so $D(h, s) = 0$ for $\mathrm{Re}(s) > \sigma_0$. Restricting to real $s = \sigma > \sigma_0$: $D(h, \sigma) = \sum_{n=1}^{\infty} h(n)/n^{\sigma} = 0$. As $\sigma \to +\infty$, the terms $h(n)/n^{\sigma}$ decay at different rates (faster for larger n), and the dominant term is $h(1)$. Formally: dividing by $1^{-\sigma} = 1$ and taking $\sigma \to \infty$, the terms with $n \geq 2$ vanish, giving $h(1) = 0$. Now $0 = \sum_{n \geq 2} h(n)/n^{\sigma}$; multiplying by 2^{σ} and taking $\sigma \to \infty$ shows $h(2) = 0$. Inductively, $h(n) = 0$ for all n.

This uniqueness theorem is the Dirichlet series analogue of the fact that a power series is determined by its coefficients. The proof exploits a feature specific to Dirichlet series: the "basis functions" n^{-s} decay at *different rates* as $\sigma \to \infty$, allowing the coefficients to be extracted one at a time. For power series $\sum a_n z^n$, the analogous statement requires evaluating derivatives at $z = 0$; for Dirichlet series, the multiplicative structure of the basis $\{n^{-s}\}$ gives a simpler argument.

Example 2.1.9: Abscissae of Standard Dirichlet Series

1. $\zeta(s) = \sum 1/n^s$: since the coefficients are non-negative, proposition 2.1.5 gives $\sigma_c = \sigma_a$. The harmonic series $\sum 1/n$ diverges, so $\sigma_a \geq 1$. For $\sigma > 1$: $\sum 1/n^\sigma \leq 1 + \int_1^\infty t^{-\sigma}\,dt = \sigma/(\sigma-1) < \infty$. Hence $\sigma_c = \sigma_a = 1$.

2. $\sum d(n)/n^s$: the divisor function satisfies $d(n) \geq 1$ for all n and $d(n) \leq C_\varepsilon n^\varepsilon$ for any $\varepsilon > 0$ (a proof appears in chapter 4), so $\sigma_a = 1$. The identity $\sum d(n)/n^s = \zeta(s)^2$ (proved in the next section) shows the series has a double pole at $s = 1$.

3. $\sum \Lambda(n)/n^s$: since $0 \leq \Lambda(n) \leq \log n \leq n^\varepsilon$ for any $\varepsilon > 0$, the abscissa $\sigma_a = 1$. By proposition 2.1.5, $\sigma_c = 1$ as well, and there is a singularity at $s = 1$. The identity $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)/n^s$ (section 2.2) identifies this singularity as a simple pole with residue 1.

4. $\sum \mu(n)/n^s$: since $|\mu(n)| \leq 1$, $\sigma_a = 1$. The exact value of σ_c is a major open problem. The bound $\sigma_c \leq 1$ follows from the Prime Number Theorem ($M(x) = o(x)$ implies convergence for $\mathrm{Re}(s) > 1$ by Abel summation). The Riemann Hypothesis is equivalent to $\sigma_c = 1/2$; the best unconditional result, $\sigma_c \leq 1$, does not pin down the exact value.

The term-by-term differentiation identity $D'(f, s) = D(f \cdot (-\log), s)$ has an important interpretation: differentiating a Dirichlet series with respect to s multiplies the coefficients by $-\log n$. This connects differentiation to the von Mangoldt function, since $-\zeta'(s) = \sum (\log n)/n^s$ is the Dirichlet series of log, and dividing by $\zeta(s)$ gives $-\zeta'(s)/\zeta(s) = D(\Lambda, s)$ (the logarithmic derivative encodes the von Mangoldt function). More generally, for any Dirichlet series $D(f, s)$ with f multiplicative and $D(f, s) \neq 0$, the logarithmic derivative $-D'(f, s)/D(f, s)$ is the Dirichlet series of an arithmetic function whose values at prime powers are determined by those of f. This mechanism is developed systematically in the next section.

Example 2.1.10: A Dirichlet Series with $\sigma_c < \sigma_a$

The Dirichlet eta function $\eta(s) = \sum_{n=1}^\infty (-1)^{n+1}/n^s$ has $\sigma_a = 1$ and $\sigma_c = 0$. The partial sums $A(N) = \sum_{n=1}^N (-1)^{n+1}$ alternate between 1 and 0, so $|A(N)| \leq 1$ for all N, and Abel's test (corollary 1.4.9) gives convergence for $\mathrm{Re}(s) > 0$. At $s = 0$: $\sum (-1)^{n+1} = 1 - 1 + 1 - \cdots$ diverges by oscillation, confirming $\sigma_c = 0$.

The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ is verified by writing $\eta(s) = \sum 1/n^s - 2\sum 1/(2n)^s = \zeta(s) - 2^{1-s}\zeta(s)$. The factor $(1-2^{1-s})$ vanishes at $s = 1 + 2\pi i k/\log 2$ for $k \in \mathbb{Z}$, cancelling the pole of $\zeta(s)$ at $s = 1$ when $k = 0$. In particular, $\eta(1) = 1 - 1/2 + 1/3 - 1/4 + \cdots = \log 2$, and this value can also be obtained as $\lim_{s\to 1}(1 - 2^{1-s})\zeta(s) = \log 2$ (using $\zeta(s) \sim 1/(s-1)$ and $(1 - 2^{1-s}) \sim (s-1)\log 2$ near $s = 1$).

The convergence theory of Dirichlet series should be compared with that of power series. A power series $\sum a_n z^n$ converges in a *disk* $|z| < R$, determined by a single parameter R (the radius of convergence). A Dirichlet series $\sum a_n n^{-s}$ converges in a *half-plane* $\mathrm{Re}(s) > \sigma_c$, also determined by a single parameter. The disk is replaced by a half-plane because $|n^{-s}| = n^{-\sigma}$ depends only on $\mathrm{Re}(s)$, not on $\mathrm{Im}(s)$: changing the imaginary part of s rotates each term n^{-it} on the unit circle without affecting convergence. The analogy extends further: the Hadamard formula $1/R = \limsup |a_n|^{1/n}$ for the radius of convergence corresponds to the formula $\sigma_a = \limsup(\log A^*(x))/\log x$ for the abscissa of absolute convergence.

Example 2.1.11: Dirichlet Series That Converge Everywhere or Nowhere

1. The Dirichlet series $D(s) = \sum_{n=1}^{N} a_n/n^s$ (a finite sum) is an entire function of s, with $\sigma_c = \sigma_a = -\infty$.
2. The series $\sum n!/n^s$ has $\sigma_c = \sigma_a = +\infty$: the terms $n!/n^\sigma$ grow faster than any power of n, so the series diverges for all s.
3. The series $\sum 1/(n^s \log^2 n)$ (starting from $n = 2$) converges absolutely for $\mathrm{Re}(s) \geq 1$ (since $\sum 1/(n \log^2 n) < \infty$). Hence $\sigma_a \leq 1$. Since the coefficients are positive, $\sigma_c = \sigma_a$; and $\sum 1/(n^\sigma \log^2 n)$ diverges for $\sigma < 1$ (by comparison with $\sum 1/n$), so $\sigma_a = 1$ exactly. This series converges on the *closed* half-plane $\mathrm{Re}(s) \geq 1$, unlike $\zeta(s)$ which diverges at $s = 1$.

2.2 The Ring Homomorphism: Convolution and Multiplication

The previous section developed Dirichlet series as analytic functions on a half-plane. The next structural fact is that the operation of *Dirichlet convolution* on arithmetic functions, introduced in section 1.2, corresponds to ordinary *multiplication* of the associated Dirichlet series. This single identity is the algebraic engine of analytic number theory: every standard convolution identity from chapter 1 translates into an identity among Dirichlet series, and the convolution ring $(\mathcal{A}, +, *)$ realizes concretely as a ring of analytic functions on a half-plane. The isomorphism is also the route by which the von Mangoldt function Λ enters the analytic theory through the logarithmic derivative $-\zeta'(s)/\zeta(s)$, the identity that drives the analytic proof of the Prime Number Theorem in chapter 5.

The crux is the following.

Theorem 2.2.1: Dirichlet Multiplication

Let f, g be arithmetic functions, and let $\sigma^* = \max(\sigma_a(f), \sigma_a(g))$. For $\mathrm{Re}(s) > \sigma^*$, the series $D(f, s)$, $D(g, s)$, and $D(f * g, s)$ all converge absolutely, and

$$D(f * g, s) = D(f, s) \cdot D(g, s)$$

***Proof*:**
Fix s with $\sigma = \mathrm{Re}(s) > \sigma^*$. By the definition of σ^*, both $\sum |f(m)|/m^\sigma$ and $\sum |g(k)|/k^\sigma$ converge.

Step 1: Form the absolutely convergent product series. The product

$$\left(\sum_{m=1}^{\infty} \frac{f(m)}{m^s}\right)\left(\sum_{k=1}^{\infty} \frac{g(k)}{k^s}\right) = \sum_{m=1}^{\infty}\sum_{k=1}^{\infty} \frac{f(m)\,g(k)}{(mk)^s}$$

converges absolutely, since the absolute value of the double sum is bounded by the product of the absolute series.

Step 2: Group terms by $n = mk$. For each $n \geq 1$, the pairs $(m,k) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$ with $mk = n$ are in bijection with the divisors d of n via $(m,k) = (d, n/d)$. Hence

$$\sum_{m,k} \frac{f(m)\,g(k)}{(mk)^s} = \sum_{n=1}^{\infty} \frac{1}{n^s} \sum_{d \mid n} f(d)\,g(n/d) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^s}$$

by the definition of Dirichlet convolution. The interchange of summation order is justified by the absolute convergence established in Step 1 (Tonelli's theorem applied to the counting measure on $\mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$).

Step 3: Absolute convergence of $D(f * g, s)$. The same argument with $|f|$ and $|g|$ in place of f and g gives

$$\sum_{n=1}^{\infty} \frac{|(f * g)(n)|}{n^\sigma} \leq \sum_{n=1}^{\infty} \frac{(|f| * |g|)(n)}{n^\sigma} = \left(\sum_{m=1}^{\infty} \frac{|f(m)|}{m^\sigma}\right)\left(\sum_{k=1}^{\infty} \frac{|g(k)|}{k^\sigma}\right) < \infty$$

completing the proof.

The proof says something more than the bare identity. The same rearrangement, applied to $|f|$ and $|g|$, also bounds the abscissa of absolute convergence of the convolution: $\sigma_a(f * g) \leq \max(\sigma_a(f), \sigma_a(g))$. Equality need not hold (cancellation in $f * g$ can shift the abscissa downward), but the inequality places $D(f * g, s)$ on a half-plane no smaller than the common half-plane of $D(f, s)$ and $D(g, s)$.

The theorem converts every convolution identity from section 1.2 into a Dirichlet series identity. The corollary collects the most-used cases.

Corollary 2.2.2: Standard Identities from Convolution

For $\mathrm{Re}(s)$ in the appropriate half-plane:

1. $\zeta(s)^2 = \sum_{n=1}^{\infty} \frac{d(n)}{n^s}$ for $\mathrm{Re}(s) > 1$, from $d = \mathbf{1} * \mathbf{1}$.

2. $\zeta(s)\,\zeta(s-k) = \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s}$ for $\mathrm{Re}(s) > \max(1, k+1)$, from $\sigma_k = \mathrm{id}_k * \mathbf{1}$, where $\mathrm{id}_k(n) = n^k$.

3. $\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$ for $\mathrm{Re}(s) > 1$, from $\mu * \mathbf{1} = \varepsilon$.

4. $\sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}$ for $\mathrm{Re}(s) > 2$, from $\varphi * \mathbf{1} = \mathrm{id}$.

5. $\frac{\zeta(2s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s}$ for $\mathrm{Re}(s) > 1$, from $\lambda * |\mu| = \varepsilon$ together with $\sum |\mu(n)|/n^s = \zeta(s)/\zeta(2s)$.

***Proof*:**
Each identity follows from theorem 2.2.1 together with the elementary computations $D(\varepsilon, s) = 1$, $D(\mathbf{1}, s) = \zeta(s)$, and $D(\mathrm{id}_k, s) = \sum n^k/n^s = \zeta(s-k)$ on $\mathrm{Re}(s) > k+1$.

Identity (1): $D(d, s) = D(\mathbf{1} * \mathbf{1}, s) = \zeta(s)\,\zeta(s) = \zeta(s)^2$.

Identity (2): $D(\sigma_k, s) = D(\mathrm{id}_k * \mathbf{1}, s) = D(\mathrm{id}_k, s)\,D(\mathbf{1}, s) = \zeta(s-k)\,\zeta(s)$.

Identity (3): $D(\varepsilon, s) = 1 = D(\mu * \mathbf{1}, s) = D(\mu, s)\,\zeta(s)$, so $D(\mu, s) = 1/\zeta(s)$.

Identity (4): $D(\mathrm{id}, s) = \zeta(s-1) = D(\varphi * \mathbf{1}, s) = D(\varphi, s)\,\zeta(s)$, so $D(\varphi, s) = \zeta(s-1)/\zeta(s)$.

Identity (5): The squarefree indicator $|\mu|$ satisfies $|\mu| * \mathbf{1}_\square = \mathbf{1}$, where $\mathbf{1}_\square(n) = 1$ if n is a square and 0 otherwise. Hence $D(|\mu|, s)\,D(\mathbf{1}_\square, s) = \zeta(s)$. Since $D(\mathbf{1}_\square, s) = \sum_{m \geq 1} 1/(m^2)^s = \zeta(2s)$, this gives $D(|\mu|, s) = \zeta(s)/\zeta(2s)$. The Liouville identity $\lambda * |\mu| = \varepsilon$ (proposition 1.2.10) then gives $D(\lambda, s) \cdot \zeta(s)/\zeta(2s) = 1$, i.e., $D(\lambda, s) = \zeta(2s)/\zeta(s)$, as claimed.

These identities form the fundamental dictionary at the level of $\zeta(s)$. Identity (1) makes the divisor function $d(n)$ accessible to analytic methods through $\zeta(s)^2$, whose double pole at $s = 1$ is the analytic source of the leading $x \log x$ term in the asymptotic $\sum_{n \leq x} d(n) \sim x \log x$ from section 1.3. Identity (2) generalizes this to all σ_k. Identity (3) inverts the role of the Möbius function at the analytic level: the reciprocal of the zeta function is the Dirichlet series of μ, linking the analytic behavior of $\zeta(s)$ to summatory questions about $\mu(n)$ and ultimately to the Prime Number Theorem and the Riemann Hypothesis. Identity (4) gives the Dirichlet series of the Euler totient as a ratio of zeta functions; its residue at $s = 2$ is $1/\zeta(2) = 6/\pi^2$, the asymptotic density of coprime pairs (section 1.3). Identity (5) is the Liouville series, whose ratio form $\zeta(2s)/\zeta(s)$ encodes the multiplicative structure of squarefree integers analytically.

The first identity admits a direct verification that makes the convolution-multiplication mechanism transparent.

Example 2.2.3: Verifying $\zeta(s)^2 = \sum d(n)/n^s$ by Squaring

Squaring the series for $\zeta(s)$ on $\mathrm{Re}(s) > 1$:

$$\zeta(s)^2 = \left(\sum_{m=1}^{\infty} \frac{1}{m^s}\right)\left(\sum_{k=1}^{\infty} \frac{1}{k^s}\right) = \sum_{m,k\geq 1} \frac{1}{(mk)^s} = \sum_{n=1}^{\infty} \frac{c(n)}{n^s}$$

where $c(n) = \#\{(m,k) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} : mk = n\}$. The pairs (m,k) with product n are in bijection with the divisors d of n via $(m,k) = (d, n/d)$, so $c(n) = d(n)$ exactly.

The first few coefficients confirm the count: $d(1) = 1$, $d(2) = d(3) = d(5) = d(7) = 2$, $d(4) = d(9) = 3$, $d(6) = d(8) = d(10) = 4$, $d(12) = 6$. Hence

$$\zeta(s)^2 = 1 + \frac{2}{2^s} + \frac{2}{3^s} + \frac{3}{4^s} + \frac{2}{5^s} + \frac{4}{6^s} + \frac{2}{7^s} + \frac{4}{8^s} + \frac{3}{9^s} + \frac{4}{10^s} + \cdots$$

The general theorem reduces this verification to the convolution identity $d = \mathbf{1} * \mathbf{1}$, sparing the explicit double-sum manipulation in every individual case.

The convolution-multiplication identity admits a more conceptual restatement.

Theorem 2.2.4: Ring Isomorphism: Arithmetic Functions and Dirichlet Series

Let $\mathcal{A}_a \subseteq \mathcal{A}$ denote the subring of arithmetic functions f with finite abscissa of absolute convergence $\sigma_a(f) < \infty$. The map

$$\Phi : \mathcal{A}_a \longrightarrow \{\text{Dirichlet series with finite } \sigma_a\}, \qquad \Phi(f) = D(f,s)$$

is an injective ring homomorphism, where the target is regarded as the ring of analytic functions on common half-planes of absolute convergence with pointwise addition and multiplication. Under Φ:

1. $\varepsilon \mapsto 1$,
2. $\mathbf{1} \mapsto \zeta(s)$,
3. $\mu \mapsto 1/\zeta(s)$,
4. $\mathrm{id} \mapsto \zeta(s-1)$.

***Proof*:**

Linearity of Φ is immediate: $D(f+g,s) = D(f,s) + D(g,s)$ on the common half-plane of absolute convergence. Multiplicativity is theorem 2.2.1: $D(f*g,s) = D(f,s)\,D(g,s)$ on

$\mathrm{Re}(s) > \max(\sigma_a(f), \sigma_a(g))$. The unit ε maps to the constant function 1 since $D(\varepsilon, s) = \varepsilon(1) = 1$.

Injectivity is the content of corollary 2.1.8: if $D(f, s) = D(g, s)$ on a common half-plane, then $f(n) = g(n)$ for all n. The named correspondences come from direct computation: $D(\varepsilon, s) = 1$, $D(\mathbf{1}, s) = \zeta(s)$, $D(\mathrm{id}, s) = \zeta(s - 1)$, and $D(\mu, s) = 1/\zeta(s)$ from corollary 2.2.2, completing the proof.

This restatement clarifies the structural picture. The formal-algebraic content of arithmetic functions (the convolution ring $\mathcal{A}$) and the analytic content of Dirichlet series (functions on a half-plane with a given abscissa of absolute convergence) are not just compatible but identical, up to the inclusion $\mathcal{A}_a \subseteq \mathcal{A}$. Algebraic identities like $\mu * \mathbf{1} = \varepsilon$ become analytic identities like $\zeta(s) \sum \mu(n)/n^s = 1$, and conversely, analytic computations involving products and quotients of Dirichlet series translate back into convolution identities at the level of coefficients. The ring structure on the analytic side imposes constraints that are invisible algebraically: for instance, the analytic identity $\zeta(s) \left(\sum \mu(n)/n^s\right) = 1$ on $\mathrm{Re}(s) > 1$ forces the strict inequality $\sigma_c(\mu) \leq 1$, since $1/\zeta(s)$ continues analytically to a neighborhood of every point of $\{\mathrm{Re}(s) = 1\} \setminus \{1\}$ (using the non-vanishing $\zeta(1 + it) \neq 0$, established in chapter 4).

The first nontrivial consequence of the ring isomorphism is a non-vanishing statement.

Example 2.2.5: Non-Vanishing of $\zeta(s)$ for $\mathrm{Re}(s) > 1$

The Möbius identity $1/\zeta(s) = \sum \mu(n)/n^s$ on $\mathrm{Re}(s) > 1$ shows that $\zeta(s) \neq 0$ on this half-plane. If $\zeta(s_0) = 0$ for some s_0 with $\mathrm{Re}(s_0) > 1$, then $1/\zeta(s)$ would have a pole at s_0, contradicting the fact that $\sum \mu(n)/n^{s_0}$ converges absolutely to a finite value at s_0 (since $\mathrm{Re}(s_0) > 1 = \sigma_a(\mu)$). The next section gives a more conceptual proof of this fact via the Euler product, which extends without change to any Dirichlet series with an Euler product. The harder boundary case $\mathrm{Re}(s) = 1$, where the Möbius series sits on the boundary of its half-plane of absolute convergence and the argument above breaks down, is treated in chapter 4.

The headline application of the ring isomorphism is the identity that connects $\zeta(s)$ to prime-counting.

Theorem 2.2.6: Logarithmic Derivative of $\zeta(s)$

For $\mathrm{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ is the von Mangoldt function (definition 1.1.8).

***Proof*:**

The strategy combines two ingredients: the convolution identity $\Lambda * \mathbf{1} = \log$ from proposition 1.1.9, and term-by-term differentiation of $\zeta(s)$ from theorem 2.1.7.

Step 1: Differentiate $\zeta(s)$. By theorem 2.1.7, for $\mathrm{Re}(s) > 1$,

$$\zeta'(s) = -\sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

Equivalently, $-\zeta'(s) = D(\log, s)$, the Dirichlet series of the function $n \mapsto \log n$.

Step 2: Apply the convolution identity. Under the ring isomorphism of theorem 2.2.4, the von Mangoldt identity $\Lambda * \mathbf{1} = \log$ becomes

$$D(\Lambda, s) \cdot \zeta(s) = D(\log, s) = -\zeta'(s)$$

on $\mathrm{Re}(s) > 1$. Absolute convergence of $D(\Lambda, s)$ on this half-plane follows from $0 \leq \Lambda(n) \leq \log n$ together with $\sigma_a(\log) = 1$ (since $\log n \leq n^\varepsilon$ for any $\varepsilon > 0$, so $\sum(\log n)/n^\sigma$ converges for every $\sigma > 1$).

Step 3: Divide. Since $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$ (example 2.2.5), division by $\zeta(s)$ yields

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$$

on $\mathrm{Re}(s) > 1$, completing the proof.

The identity $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)/n^s$ is arguably the single most important identity in elementary analytic number theory. Its left side is an analytic object: the logarithmic derivative of $\zeta(s)$, whose poles encode the analytic structure of $\zeta(s)$. Its right side is the Dirichlet series of the von Mangoldt function, whose summatory function $\psi(x) = \sum_{n \leq x} \Lambda(n)$ counts prime powers (weighted by $\log p$) and is the natural object whose asymptotics encode the Prime Number Theorem $\psi(x) \sim x$. The identity converts an analytic problem (locating the poles and zeros of $-\zeta'(s)/\zeta(s)$) into an arithmetic one (estimating $\psi(x)$). The simple pole of $\zeta(s)$ at $s = 1$, with residue 1, becomes a simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1 (since $-\zeta'(s)/\zeta(s) \sim 1/(s-1)$ near $s = 1$); this residue is the analytic source of the leading term $\psi(x) \sim x$ in the Prime Number Theorem. The contributions of the zeros of $\zeta(s)$ inside the critical strip give the oscillatory error terms in the explicit formula. The whole strategy of the analytic proof of the Prime Number Theorem, developed in chapter 5, rests on the identity proved above.

The structural extension of theorem 2.2.6 to general multiplicative arithmetic functions is the following: for multiplicative f with $D(f, s) \neq 0$ in some half-plane, the logarithmic derivative $-D'(f, s)/D(f, s)$ is the Dirichlet series of an arithmetic function Λ_f supported on prime powers, generalizing the von Mangoldt function. The analytic structure of the logarithmic derivative is therefore controlled entirely by the values of f at prime powers, mirroring the determination-by-prime-powers principle for multiplicative arithmetic functions (proposition 1.1.3).

2.3 Euler Products

The ring isomorphism of theorem 2.2.4 converts Dirichlet convolution into multiplication of Dirichlet series, but its full force is felt only for *multiplicative* functions. For such functions the values $f(n)$ at all integers are reconstructed from the values $f(p^k)$ at prime powers (proposition 1.1.3), and this reconstruction lifts to the analytic level: the Dirichlet series $D(f, s)$ factors as a product of local pieces, one for each prime, with the local piece at p built entirely from $f(p), f(p^2), f(p^3), \ldots$. The resulting factorization is the *Euler product*, and it is the analytic incarnation of unique factorization in $\mathbb{Z}$. Where theorem 2.2.1 expresses one global relation $D(f * g, s) = D(f, s)D(g, s)$, the Euler product expresses a relation between $D(f, s)$ and the infinitely many local factors at once.

The mechanism is the local-to-global principle of section 1.1 made analytic. A multiplicative function decouples across primes; absolute convergence of $D(f,s)$ lets the corresponding infinite sum be regrouped as an infinite product over primes without disturbing its value. The product form exposes structure that the series hides: it shows immediately that $D(f,s)$ cannot vanish in the region of absolute convergence (every factor is a nonzero number near 1), and its logarithm is a sum over prime powers that ties the analytic object directly to the primes. These two consequences, non-vanishing and the logarithmic expansion, are the threads that the study of prime distribution in chapter 3 and the theory of $\zeta(s)$ in chapter 4 will pull on.

Theorem 2.3.1: Euler Product for Multiplicative Functions

Let f be a multiplicative arithmetic function (definition 1.1.2), and let $\sigma_a(f)$ be its abscissa of absolute convergence (definition 2.1.3). For $\mathrm{Re}(s) > \sigma_a(f)$,

$$D(f,s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_p \left(\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} \right)$$

where the product runs over all primes and converges absolutely. If f is totally multiplicative, the local factor at p is the geometric series

$$\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} = \sum_{k=0}^{\infty} \left(\frac{f(p)}{p^s} \right)^k = \frac{1}{1 - f(p)p^{-s}}$$

***Proof*:**
Fix s with $\sigma = \mathrm{Re}(s) > \sigma_a(f)$, so that $\sum_{n=1}^{\infty} |f(n)|n^{-\sigma} < \infty$. Write $f_p(s) = \sum_{k=0}^{\infty} f(p^k)p^{-ks}$ for the local factor at p.

Step 1: Each local factor converges absolutely. For a single prime p, the terms $f(p^k)p^{-ks}$ are a subsequence of the terms $f(n)n^{-s}$ (those indexed by $n = p^k$), so

$$\sum_{k=0}^{\infty} \left| \frac{f(p^k)}{p^{ks}} \right| = \sum_{k=0}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} \le \sum_{n=1}^{\infty} \frac{|f(n)|}{n^{\sigma}} < \infty$$

Hence $f_p(s)$ is a well-defined complex number, and $f_p(s) = 1 + \sum_{k=1}^{\infty} f(p^k)p^{-ks}$ since $f(p^0) = f(1) = 1$ by multiplicativity.

Step 2: The finite product over small primes equals a partial sum. Fix $y \ge 2$ and let $p_1 < p_2 < \cdots < p_r$ be the primes not exceeding y. Each $f_{p_i}(s)$ is an absolutely convergent series, so their finite product may be expanded by multiplying out and rearranging (a finite product of absolutely convergent series converges absolutely to the iterated sum):

$$\prod_{i=1}^{r} f_{p_i}(s) = \sum_{k_1=0}^{\infty} \cdots \sum_{k_r=0}^{\infty} \frac{f(p_1^{k_1}) \cdots f(p_r^{k_r})}{(p_1^{k_1} \cdots p_r^{k_r})^s}$$

By multiplicativity of f, the numerator $f(p_1^{k_1}) \cdots f(p_r^{k_r})$ equals $f(p_1^{k_1} \cdots p_r^{k_r})$, since the prime powers $p_i^{k_i}$ are pairwise coprime. By unique factorization (**??** , section 0.2), the map $(k_1, \dots, k_r) \mapsto$

$n = p_1^{k_1} \cdots p_r^{k_r}$ is a bijection from r-tuples of non-negative integers onto the set

$$S_y = \{n \geq 1 : \text{every prime factor of } n \text{ is} \leq y\}$$

of y-smooth integers. Therefore

$$\prod_{i=1}^{r} f_{p_i}(s) = \sum_{n \in S_y} \frac{f(n)}{n^s}$$

the local factors at the primes $\leq y$ assemble exactly into the sum over integers built from those primes.

Step 3: Control the tail. Every integer $n \leq y$ has all prime factors $\leq y$, so $\{1, 2, \ldots, \lfloor y \rfloor\} \subseteq S_y$. The difference between $D(f, s)$ and the partial product is supported on integers not in S_y, each of which exceeds y:

$$\left| D(f,s) - \prod_{i=1}^{r} f_{p_i}(s) \right| = \left| \sum_{n \notin S_y} \frac{f(n)}{n^s} \right| \leq \sum_{n \notin S_y} \frac{|f(n)|}{n^\sigma} \leq \sum_{n > y} \frac{|f(n)|}{n^\sigma}$$

The right-hand side is the tail of the convergent series $\sum_n |f(n)| n^{-\sigma}$, so it tends to 0 as $y \to \infty$. Letting $y \to \infty$ gives $\prod_p f_p(s) = D(f, s)$, with the product interpreted as the limit of its partial products.

Step 4: Absolute convergence of the infinite product. An infinite product $\prod_p (1 + a_p)$ converges absolutely precisely when $\sum_p |a_p| < \infty$ ([13]). Here $a_p = f_p(s) - 1 = \sum_{k=1}^{\infty} f(p^k) p^{-ks}$, and

$$\sum_p |a_p| = \sum_p \left| \sum_{k=1}^{\infty} \frac{f(p^k)}{p^{ks}} \right| \leq \sum_p \sum_{k=1}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} \leq \sum_{n=2}^{\infty} \frac{|f(n)|}{n^\sigma} < \infty$$

where the last inequality holds because each prime power p^k with $k \geq 1$ is a distinct integer ≥ 2, so the double sum over (p, k) is a subseries of $\sum_{n \geq 2} |f(n)| n^{-\sigma}$. Thus the product converges absolutely, and in particular its value is independent of the order of the factors.

Step 5: The totally multiplicative case. If f is totally multiplicative, then $f(p^k) = f(p)^k$ by proposition 1.1.3. The local factor is the geometric series $\sum_{k \geq 0} (f(p)p^{-s})^k$ with ratio $f(p)p^{-s}$, and its absolute convergence (Step 1) forces $|f(p)p^{-s}| < 1$, since a geometric series whose ratio has modulus ≥ 1 has terms that do not tend to 0 and hence diverges. Summing the geometric series gives $f_p(s) = (1 - f(p)p^{-s})^{-1}$, completing the proof.

The Euler product is unique factorization read analytically. Step 2 is the heart of the matter: multiplying the local factors and expanding produces exactly one term for each way of choosing a prime-power part at each prime, and unique factorization guarantees that these choices correspond bijectively to integers. The hypothesis of absolute convergence is what permits the rearrangement; without it, the regrouping of the double series into a product would not be justified, and the value could change. The role of multiplicativity is equally precise: it is what makes the numerator $f(p_1^{k_1}) \cdots f(p_r^{k_r})$ collapse to the single value $f(n)$, so that the expanded product is genuinely the Dirichlet series and not some twisted variant. For totally multiplicative f, the further collapse

$f(p^k) = f(p)^k$ turns each local factor into a closed-form geometric series, and the Euler product becomes a product of simple rational functions of p^{-s}.

The first and defining instance is the zeta function itself, where $f = \mathbf{1}$ is the constant function 1, totally multiplicative with $\mathbf{1}(p) = 1$.

Corollary 2.3.2: The Euler Product for the Riemann Zeta Function

For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}}$$

the product converging absolutely.

***Proof*:**
The constant function $\mathbf{1}$ is totally multiplicative with $\mathbf{1}(p) = 1$ and $\sigma_a(\mathbf{1}) = 1$ (example 2.1.9). Applying theorem 2.3.1 with $f = \mathbf{1}$, the local factor at p is the geometric series $\sum_{k=0}^{\infty} p^{-ks} = (1 - p^{-s})^{-1}$, so $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ for $\mathrm{Re}(s) > 1$, as required.

This identity, due to Euler, is the analytic statement of the fundamental theorem of arithmetic. Reading the product from right to left and expanding each factor as a geometric series, $\prod_p (1 + p^{-s} + p^{-2s} + \cdots)$, the coefficient of n^{-s} in the expansion counts the number of ways to write n as a product of prime powers, which is exactly one. The equality with $\sum_n n^{-s}$ therefore encodes the fact that every integer factors into primes in exactly one way. The product representation also carries arithmetic information that the sum conceals. As $s \to 1^+$ along the real axis, $\zeta(s) \to +\infty$ because the harmonic series diverges; on the product side this forces $\prod_p (1 - p^{-1})^{-1}$ to diverge, which can happen only if there are infinitely many primes and $\sum_p p^{-1}$ diverges. Euler's original proof of the infinitude of primes is precisely this observation, and the quantitative form of the divergence is the subject of example 2.3.6 below and, in full, of the elementary theory of prime distribution in chapter 3 (a finite Euler product would give a finite limit, contradicting the pole). The analytic behavior of $\zeta(s)$ near $s = 1$, and throughout the critical strip, is developed in chapter 4.

The convolution identities of corollary 2.2.2 all have Euler product counterparts, obtained by computing one local factor and recognizing the resulting product. Carrying out these computations gives a second, independent route to those identities, one that never invokes the multiplication theorem but instead reads each global product off its local factors.

Example 2.3.3: Euler Products of Standard Dirichlet Series
Each of the following multiplicative functions has an Euler product computed from its values at prime powers. The global product is then identified with a ratio of zeta values, cross-checking corollary 2.2.2.

1. *The Möbius function μ (definition 1.1.4).* Here $\mu(p) = -1$ and $\mu(p^k) = 0$ for $k \geq 2$, so the local factor terminates:
$$\sum_{k=0}^{\infty} \frac{\mu(p^k)}{p^{ks}} = 1 + \frac{\mu(p)}{p^s} = 1 - \frac{1}{p^s}$$
Hence $D(\mu, s) = \prod_p (1 - p^{-s})$ for $\mathrm{Re}(s) > 1$. Comparing with corollary 2.3.2, this is exactly $\prod_p (1 - p^{-s})^{-1}$ inverted factor by factor, so $D(\mu, s) = 1/\zeta(s)$, matching corollary 2.2.2(3).

2. *The divisor function $d = \mathbf{1} * \mathbf{1}$ (definition 1.1.5).* Since $d(p^k) = k + 1$, the local factor is

$$\sum_{k=0}^{\infty} \frac{k+1}{p^{ks}} = \sum_{k=0}^{\infty} (k+1)x^k = \frac{1}{(1-x)^2}, \qquad x = p^{-s}$$

using the standard power series $\sum_{k\geq 0}(k+1)x^k = (1-x)^{-2}$ for $|x| < 1$. Therefore $D(d, s) = \prod_p (1-p^{-s})^{-2} = \zeta(s)^2$ for $\mathrm{Re}(s) > 1$, matching corollary 2.2.2(1). The local factor $(1-p^{-s})^{-2}$ is the square of the local factor of $\zeta(s)$, the product form of $d = \mathbf{1} * \mathbf{1}$.

3. *The Liouville function λ (definition 1.1.6).* The function λ is totally multiplicative with $\lambda(p) = -1$, so theorem 2.3.1 gives the geometric local factor

$$\sum_{k=0}^{\infty} \frac{\lambda(p^k)}{p^{ks}} = \frac{1}{1 - \lambda(p)p^{-s}} = \frac{1}{1 + p^{-s}}$$

Hence $D(\lambda, s) = \prod_p (1 + p^{-s})^{-1}$ for $\mathrm{Re}(s) > 1$. To identify the global product, write $1 + p^{-s} = (1 - p^{-2s})/(1 - p^{-s})$, so that

$$\prod_p \frac{1}{1 + p^{-s}} = \prod_p \frac{1 - p^{-s}}{1 - p^{-2s}} = \frac{\prod_p (1 - p^{-s})}{\prod_p (1 - p^{-2s})} = \frac{1/\zeta(s)}{1/\zeta(2s)} = \frac{\zeta(2s)}{\zeta(s)}$$

where the last step applies corollary 2.3.2 at s and at $2s$ (both products converge absolutely for $\mathrm{Re}(s) > 1$). This matches corollary 2.2.2(5).

4. *The Euler totient φ.* The totient is multiplicative with $\varphi(p^k) = p^k - p^{k-1}$ for $k \geq 1$ and $\varphi(1) = 1$. The local factor is

$$1 + \sum_{k=1}^{\infty} \frac{p^k - p^{k-1}}{p^{ks}} = 1 + \sum_{k=1}^{\infty} p^{k(1-s)} - \sum_{k=1}^{\infty} p^{k-1-ks}$$

Both geometric series converge for $\mathrm{Re}(s) > 1$, where $|p^{1-s}| = p^{1-\sigma} < 1$. Summing them,

$$\sum_{k=1}^{\infty} p^{k(1-s)} = \frac{p^{1-s}}{1 - p^{1-s}}, \qquad \sum_{k=1}^{\infty} p^{k-1-ks} = \frac{p^{-s}}{1 - p^{1-s}}$$

so the local factor is

$$1 + \frac{p^{1-s} - p^{-s}}{1 - p^{1-s}} = \frac{1 - p^{-s}}{1 - p^{1-s}}$$

The local factor is thus valid for $\mathrm{Re}(s) > 1$. Identifying the global product, however, requires corollary 2.3.2 at $s - 1$: the factor $\prod_p (1 - p^{1-s}) = \prod_p (1 - p^{-(s-1)}) = 1/\zeta(s-1)$ converges only for $\mathrm{Re}(s-1) > 1$, so it is this $\zeta(s-1)$ factor, not the local series, that pins the region. Applying corollary 2.3.2 at s and at $s - 1$,

$$D(\varphi, s) = \prod_p \frac{1 - p^{-s}}{1 - p^{1-s}} = \frac{1/\zeta(s)}{1/\zeta(s-1)} = \frac{\zeta(s-1)}{\zeta(s)}$$

for $\mathrm{Re}(s) > 2$, matching corollary 2.2.2(4).

In every case the local factor encodes the arithmetic of f at a single prime, and the shape of the global product records the convolution identity. The vanishing of $\mu(p^k)$ for $k \geq 2$ truncates the Möbius local factor to a polynomial $1 - p^{-s}$, the product reciprocal of $\zeta(s)$. The squaring $d = \mathbf{1} * \mathbf{1}$ appears as a squared local factor. The sign $\lambda(p) = -1$ flips the geometric series from $(1 - p^{-s})^{-1}$ to $(1 + p^{-s})^{-1}$, and the algebraic identity $(1 + x) = (1 - x^2)/(1 - x)$ converts this into the ratio $\zeta(2s)/\zeta(s)$ that isolates the squarefree structure. These computations are interchangeable with the convolution route of corollary 2.2.2. The Euler product route also exposes how the half-plane of validity arises: for the Liouville function the region $\mathrm{Re}(s) > 1$ is inherited directly from the local geometric series, whereas for the totient the local factors already converge on $\mathrm{Re}(s) > 1$ yet the global identity is pinned to $\mathrm{Re}(s) > 2$ by the $\zeta(s - 1)$ factor that appears upon recombination. The product form thus separates the local convergence of each factor from the global constraint imposed by the zeta values that assemble the answer.

A structural consequence of the product form, independent of any particular f, is that a Dirichlet series with an Euler product cannot vanish where the product converges. This is because a convergent infinite product of nonzero factors is nonzero, a criterion that has no counterpart on the series side.

Proposition 2.3.4: Non-Vanishing in the Half-Plane of Absolute Convergence

Let f be multiplicative, and let s be a point with $\mathrm{Re}(s) = \sigma$ at which

$$\sum_p \sum_{k=1}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} < \infty$$

and at which every local factor $f_p(s) = \sum_{k \geq 0} f(p^k) p^{-ks}$ is nonzero. Then the Euler product converges to a nonzero value, so $D(f, s) \neq 0$. In particular $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$.

***Proof*:**
The hypothesis $\sum_p \sum_{k \geq 1} |f(p^k)| p^{-k\sigma} < \infty$ is exactly the absolute convergence condition $\sum_p |f_p(s) - 1| < \infty$ from Step 4 of theorem 2.3.1, since $f_p(s) - 1 = \sum_{k \geq 1} f(p^k) p^{-ks}$ and $|f_p(s) - 1| \leq \sum_{k \geq 1} |f(p^k)| p^{-k\sigma}$. Absolute convergence of the product $\prod_p f_p(s)$ follows, and by theorem 2.3.1 its value is $D(f, s)$.

The criterion for a nonzero limit is the following: an absolutely convergent infinite product $\prod_p f_p(s)$ with every factor $f_p(s) \neq 0$ has a nonzero value ([13]). The reason is that absolute convergence means $f_p(s) \to 1$, so all but finitely many factors lie in the disk $|z - 1| < 1/2$ on which log is defined; the tail $\sum_p \log f_p(s)$ converges to a finite complex number L, the corresponding tail product converges to $e^L \neq 0$, and multiplying by the finitely many remaining nonzero factors keeps the value nonzero. Hence $D(f, s) = \prod_p f_p(s) \neq 0$.

For the zeta function, take $f = \mathbf{1}$ and $\mathrm{Re}(s) > 1$. The local factors $(1 - p^{-s})^{-1}$ are nonzero (a reciprocal is never zero), and the convergence hypothesis holds since $\sum_p \sum_{k \geq 1} p^{-k\sigma} \leq \sum_{n \geq 2} n^{-\sigma} < \infty$. Therefore $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$, as we sought to show.

This proposition reproves the non-vanishing of example 2.2.5 from a structural vantage point. The earlier argument used the specific identity $1/\zeta(s) = \sum \mu(n) n^{-s}$ and observed that a zero of ζ would produce a pole of an absolutely convergent series. The present argument is cleaner and more general: it uses only that $\zeta(s)$ has an Euler product with nonzero local factors, and it applies verbatim to any Dirichlet series possessing an Euler product, such as the Dirichlet L-functions whose non-vanishing on the line $\mathrm{Re}(s) = 1$ drives the theorem on primes in arithmetic progressions. The product form makes the non-vanishing transparent because the obstruction to a zero is local: a product vanishes only if a factor vanishes, and each local factor here is a value near 1. The difficult cases in the theory, such as the non-vanishing of $\zeta(1 + it)$ on the boundary line $\mathrm{Re}(s) = 1$, lie outside the region of absolute convergence, where this elementary argument no longer reaches; those are the province of chapter 4.

Taking the logarithm of an Euler product turns the product of local factors into a sum over primes, and expanding each logarithm as a power series produces a sum over prime powers. For totally multiplicative f the local factors are geometric, and the logarithm has an explicit expansion that connects the Dirichlet series directly to the prime powers.

Proposition 2.3.5: The Logarithm of an Euler Product

Let f be totally multiplicative with absolutely convergent Euler product on $\mathrm{Re}(s) > \sigma_a(f)$ and $|f(p)p^{-s}| < 1$ for every p. Then the double series

$$\log D(f, s) := \sum_p \sum_{k=1}^{\infty} \frac{f(p)^k}{k\, p^{ks}}$$

converges absolutely for $\mathrm{Re}(s) > \sigma_a(f)$ and defines a branch of the logarithm of $D(f, s)$ there: it is the analytic branch characterized by $\log D(f, s) \to 0$ as $\mathrm{Re}(s) \to +\infty$. This branch need not be the principal logarithm of $D(f, s)$; the two coincide precisely when the imaginary part of the series lies in $(-\pi, \pi]$, which holds whenever $\mathrm{Re}(s)$ is large enough. In particular, for $f = \mathbf{1}$ the series defines

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k\, p^{ks}}$$

for $\mathrm{Re}(s) > 1$, agreeing with the ordinary real logarithm on the real segment $s > 1$ (there $\zeta(s) > 1$). Differentiating term by term recovers

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

the identity of theorem 2.2.6.

***Proof*:**
Step 1: A logarithm of the product as a sum of factor logarithms. By theorem 2.3.1 the local factor is $f_p(s) = (1 - f(p)p^{-s})^{-1}$, and the product $\prod_p f_p(s)$ converges absolutely. Throughout, log on a single factor denotes the principal branch. Each factor satisfies $|f(p)p^{-s}| < 1$, so $1 - f(p)p^{-s}$ lies in the open right half-plane $\mathrm{Re}(z) > 0$, away from the branch cut, and the principal expansion $-\log(1 - z) = \sum_{k \geq 1} z^k/k$ applies with $z = f(p)p^{-s}$. Setting

$$\log D(f, s) := \sum_p \log \frac{1}{1 - f(p)p^{-s}} = -\sum_p \log\left(1 - f(p)p^{-s}\right) = \sum_p \sum_{k=1}^{\infty} \frac{f(p)^k}{k\, p^{ks}}$$

gives the double series of the statement. The symbol $\log D(f,s)$ is *defined* by this sum; it is a logarithm of $D(f,s)$ because exponentiating term by term gives $\exp\left(\sum_p \log f_p(s)\right) = \prod_p f_p(s) = D(f,s)$ by theorem 2.3.1, the interchange of exp with the convergent sum being the continuity of exp applied to the partial sums. It need not be the *principal* logarithm of $D(f,s)$: the principal logarithm has imaginary part in $(-\pi,\pi]$, whereas the imaginary part $\sum_p \sum_{k\geq 1} \operatorname{Im}\left(f(p)^k p^{-ks}\right)/k$ of the series is constrained only by absolute convergence and need not lie in $(-\pi,\pi]$; the two coincide precisely when it does. The double series converges absolutely: $\sum_p \sum_{k\geq 1} |f(p)|^k p^{-k\sigma}/k \leq \sum_p \sum_{k\geq 1} |f(p)p^{-s}|^k = \sum_p |f(p)p^{-s}|/(1-|f(p)p^{-s}|)$. Absolute convergence of each local factor (Step 1 of theorem 2.3.1) forces $|f(p)p^{-s}| \to 0$, so $|f(p)p^{-s}| \leq 1/2$ for all but finitely many p, and for those p the summand is at most $2|f(p)|p^{-\sigma}$. Since $\sum_p |f(p)|p^{-\sigma}$ converges (its terms are a subseries of $\sum_n |f(n)|n^{-\sigma}$), the double series converges absolutely.

Step 2: Specialize to $\zeta(s)$. For $f = \mathbf{1}$, every $f(p) = 1$, so $f(p)^k = 1$, giving

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k\, p^{ks}}$$

for $\operatorname{Re}(s) > 1$.

Step 3: Differentiate term by term. Each summand $1/(k\, p^{ks}) = k^{-1}p^{-ks}$ is an entire function of s with derivative $-(\log p)p^{-ks}$ (the factor k cancels). The differentiated series is

$$\sum_p \sum_{k=1}^{\infty} \frac{-\log p}{p^{ks}}$$

which converges absolutely and uniformly on $\operatorname{Re}(s) \geq 1+\delta$ for each $\delta > 0$: there $\sum_p \sum_{k\geq 1} (\log p)p^{-k(1+\delta)} = \sum_p (\log p)\, p^{-(1+\delta)}/(1-p^{-(1+\delta)})$, which is finite because $\log p \leq p^{\delta/2}$ for large p makes the series comparable to $\sum_p p^{-1-\delta/2}$. Local uniform convergence of the differentiated series, together with convergence of the original at one point, justifies differentiating $\log \zeta(s)$ term by term ([13]). The derivative of the left side is $\zeta'(s)/\zeta(s)$, so

$$\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{-\log p}{p^{ks}}, \qquad \text{hence} \qquad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{ks}}$$

Step 4: Recognize the von Mangoldt series. The pairs (p,k) with p prime and $k \geq 1$ are in bijection with the prime powers $n = p^k$, and the summand $\log p$ is exactly $\Lambda(p^k)$ by definition 1.1.8. For n not a prime power, $\Lambda(n) = 0$, so the sum over prime powers equals the sum over all n:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{\Lambda(p^k)}{p^{ks}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

for $\operatorname{Re}(s) > 1$. This is the identity of theorem 2.2.6, completing the proof.

This gives a second derivation of the logarithmic-derivative formula $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)n^{-s}$, consistent with the convolution-based proof of theorem 2.2.6. The two proofs illuminate the identity from opposite sides. The convolution proof builds the formula from the identity $\Lambda * \mathbf{1} = \log$ at the level of arithmetic functions, transported through the ring isomorphism. The Euler product proof builds it from the factorization of $\zeta(s)$ over primes, with the von Mangoldt function emerging as the coefficient that records $\log p$ at each prime power p^k. The agreement of the two is a consistency check on the whole dictionary: the von Mangoldt function is at once the convolution inverse-image of log under $*\mathbf{1}$ and the logarithmic-derivative coefficient of the Euler product. The expansion $\log\zeta(s) = \sum_{p,k} 1/(k\,p^{ks})$ is itself the starting point for many estimates near $s = 1$, since the $k = 1$ term $\sum_p p^{-s}$ dominates and carries the divergence examined next.

Example 2.3.6: The Product Near $s = 1$

As $s \to 1^+$ along the real axis, $\zeta(s) \to +\infty$ because of the pole at $s = 1$, so $\log\zeta(s) \to +\infty$. In the expansion of proposition 2.3.5,

$$\log\zeta(s) = \sum_p \frac{1}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{1}{k\,p^{ks}}$$

the double tail with $k \geq 2$ stays bounded as $s \to 1^+$: it is dominated by $\sum_p \sum_{k\geq 2} p^{-k} = \sum_p p^{-2}/(1-p^{-1}) \leq \sum_{n\geq 2} 1/(n(n-1)) = 1$ uniformly for $s \geq 1$. Since $\log\zeta(s) \to +\infty$ while this tail remains bounded, the prime sum must diverge:

$$\sum_p \frac{1}{p^s} \to +\infty \qquad (s \to 1^+)$$

Letting $s \to 1^+$ shows that $\sum_p 1/p$ cannot converge, for if it did, $\sum_p p^{-s} \leq \sum_p p^{-1}$ would stay bounded. This foreshadows the divergence of $\sum_p 1/p$ and the infinitude of primes, treated quantitatively in chapter 3.

2.4 Mellin Transforms and the Integral Representation of $\zeta(s)$

The convergence theory and the convolution-multiplication dictionary of the previous sections treat a Dirichlet series as a formal-analytic object built from its coefficients. To extract asymptotic information from $\zeta(s)$ and its relatives, the series must be connected to integrals, where the machinery of complex analysis (contour shifting, residues, the calculus of poles) can be brought to bear. The device that makes this connection is the *Mellin transform*. Where the Fourier transform is adapted to the additive group $\mathbb{R}$ and the Laplace transform to the additive half-line, the Mellin transform is the integral transform adapted to the *multiplicative* group $(0, \infty)$: it converts dilation $t \mapsto at$ into a character a^{-s}, exactly as the Fourier transform converts translation into a character $e^{i\xi a}$. This is precisely the symmetry that a Dirichlet series $\sum f(n)n^{-s}$ respects, since each term n^{-s} is a multiplicative character of the variable n evaluated at the parameter s.

The payoff is twofold. First, the Mellin transform expresses the gamma function, the Riemann zeta function, and their products as integrals over $(0, \infty)$ whose integrands have manifest analytic structure. The integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t - 1)\,dt$ derived below is the starting point from which the meromorphic continuation and the functional equation of $\zeta(s)$ are obtained in chapter 4 and chapter 6. Second, the Mellin inversion formula runs the transform backward: it recovers an arithmetic sum from a contour integral of its Dirichlet series, and this is the analytic content of Perron's formula in section 2.5, the tool that converts analytic facts about $\zeta(s)$ into counting statements about primes. The gamma function will be developed in full in chapter 6; here only the integral $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt$ for $\mathrm{Re}(s) > 0$ is needed, and it appears below as the first example of a Mellin transform.

Definition 2.4.1: The Mellin Transform

Let $f\colon (0,\infty) \to \mathbb{C}$ be a function that is (Lebesgue) measurable on $(0,\infty)$. The *Mellin transform* of f is the function

$$(\mathcal{M}f)(s) = \int_0^\infty f(t)\,t^{s-1}\,dt$$

defined for those $s \in \mathbb{C}$ for which the integral converges absolutely. Suppose f satisfies the two-sided power bound

$$f(t) = O(t^{-a}) \text{ as } t \to 0^+, \qquad f(t) = O(t^{-b}) \text{ as } t \to \infty$$

with $a < b$. Then the integral converges absolutely for every s with $a < \mathrm{Re}(s) < b$, and the open vertical strip

$$\{s \in \mathbb{C} \big| a < \mathrm{Re}(s) < b\}$$

is called the *fundamental strip* of f.

The fundamental strip is the multiplicative analogue of the half-plane of convergence for a Dirichlet series, and the reason it is a strip rather than a half-plane is the presence of *two* endpoints to control. Near $t = 0$ the factor t^{s-1} contributes $t^{\mathrm{Re}(s)-1}$, so the integral $\int_0^1 f(t)t^{s-1}\,dt$ converges when $\mathrm{Re}(s) - 1 > a - 1$, that is, when $\mathrm{Re}(s) > a$; the lower bound on $\mathrm{Re}(s)$ is forced by the behavior of f at the origin. Near $t = \infty$ the same factor must decay against the growth of f, and $\int_1^\infty f(t)t^{s-1}\,dt$ converges when $\mathrm{Re}(s) - 1 < b - 1$, that is, when $\mathrm{Re}(s) < b$; the upper bound is forced by the behavior of f at infinity. Only on the overlap $a < \mathrm{Re}(s) < b$ do both tails converge, which requires $a < b$. As with Dirichlet series, $|t^{s-1}| = t^{\mathrm{Re}(s)-1}$ depends only on the real part of s, so the region of absolute convergence is invariant under vertical translation, hence a vertical strip.

Example 2.4.2: The Gamma Function as a Mellin Transform

Take $f(t) = e^{-t}$. Its Mellin transform is, by definition, the integral that defines the gamma function:

$$(\mathcal{M}e^{-t})(s) = \int_0^\infty e^{-t}\,t^{s-1}\,dt = \Gamma(s)$$

The fundamental strip is the half-plane $\mathrm{Re}(s) > 0$, which is the degenerate case $a = 0$, $b = +\infty$ of definition 2.4.1. Convergence at both ends is checked directly. Near $t = 0^+$, the factor e^{-t} is bounded (indeed $e^{-t} \to 1$), so the integrand is comparable to $t^{\mathrm{Re}(s)-1}$, and

$$\int_0^1 e^{-t}\,t^{\sigma-1}\,dt \le \int_0^1 t^{\sigma-1}\,dt = \frac{1}{\sigma}$$

with $\sigma = \mathrm{Re}(s)$, which is finite precisely when $\sigma > 0$. Near $t = \infty$, the exponential decay of e^{-t} dominates every power of t: for any $\sigma \in \mathbb{R}$ there is a constant C_σ with $t^{\sigma-1} \le C_\sigma e^{t/2}$ for $t \ge 1$, whence

$$\int_1^\infty e^{-t}\, t^{\sigma-1}\, dt \le C_\sigma \int_1^\infty e^{-t/2}\, dt = 2C_\sigma e^{-1/2} < \infty$$

for every real σ. The upper endpoint of the strip is therefore $b = +\infty$, and the integral defines $\Gamma(s)$ as an analytic function on the entire half-plane $\mathrm{Re}(s) > 0$. The continuation of Γ to a meromorphic function on all of $\mathbb{C}$, with simple poles at the non-positive integers, together with the functional equation $\Gamma(s+1) = s\,\Gamma(s)$ and the reflection formula, is developed in chapter 6; the present section uses only the integral on $\mathrm{Re}(s) > 0$.

The elementary properties of the transform are the precise statements of the slogans "dilation becomes a character" and "the Mellin transform is the Fourier transform in multiplicative coordinates." These are the identities that will be applied repeatedly when manipulating the integrand of $\Gamma(s)\zeta(s)$.

Proposition 2.4.3: Elementary Properties of the Mellin Transform

Let $f\colon (0,\infty) \to \mathbb{C}$ have fundamental strip $a < \mathrm{Re}(s) < b$.

1. *Scaling.* For any real $\alpha > 0$, the function $t \mapsto f(\alpha t)$ has the same fundamental strip, and

$$(\mathcal{M} f(\alpha\cdot))(s) = \alpha^{-s} (\mathcal{M} f)(s)$$

2. *Shift by a power.* For any $\mu \in \mathbb{C}$, the function $t \mapsto t^\mu f(t)$ has fundamental strip $a - \mathrm{Re}(\mu) < \mathrm{Re}(s) < b - \mathrm{Re}(\mu)$, and

$$(\mathcal{M}\, t^\mu f(t))(s) = (\mathcal{M} f)(s+\mu)$$

3. *Reduction to a two-sided Laplace transform.* The substitution $t = e^{-x}$ identifies the Mellin transform with a two-sided Laplace transform. Writing $g(x) = f(e^{-x})$ for $x \in \mathbb{R}$,

$$(\mathcal{M} f)(s) = \int_{-\infty}^\infty g(x)\, e^{-sx}\, dx$$

for all s in the fundamental strip. In particular, on a vertical line $\mathrm{Re}(s) = c$ with $a < c < b$, setting $s = c + i\xi$ gives

$$(\mathcal{M} f)(c + i\xi) = \int_{-\infty}^\infty \left(e^{-cx} g(x)\right) e^{-i\xi x}\, dx$$

which is the Fourier transform of the function $x \mapsto e^{-cx} g(x) = e^{-cx} f(e^{-x})$.

***Proof*:**
Throughout write $\sigma = \mathrm{Re}(s)$.

1. Fix $\alpha > 0$ and s with $a < \sigma < b$. In the integral $\int_0^\infty f(\alpha t)\, t^{s-1}\, dt$ substitute $u = \alpha t$, so $t = u/\alpha$, $dt = du/\alpha$, and $t^{s-1} = (u/\alpha)^{s-1} = \alpha^{1-s} u^{s-1}$. As t runs over $(0,\infty)$ so does u, and

$$\int_0^\infty f(\alpha t)\, t^{s-1}\, dt = \int_0^\infty f(u)\, \alpha^{1-s} u^{s-1} \frac{du}{\alpha} = \alpha^{-s} \int_0^\infty f(u)\, u^{s-1}\, du = \alpha^{-s} (\mathcal{M} f)(s)$$

The substitution is a C^1 diffeomorphism of $(0, \infty)$ with itself, so it preserves absolute convergence: the integral on the left converges absolutely if and only if the integral on the right does, which holds precisely for $a < \sigma < b$. Hence the fundamental strip is unchanged.

2. Fix $\mu \in \mathbb{C}$. The integrand of $\mathcal{M}(t^\mu f(t))$ at the parameter s is $t^\mu f(t)\, t^{s-1} = f(t)\, t^{(s+\mu)-1}$, which is exactly the integrand of $\mathcal{M}f$ at the parameter $s + \mu$. Therefore the integrals are equal whenever either converges:
$$(\mathcal{M}\, t^\mu f(t))(s) = \int_0^\infty f(t)\, t^{(s+\mu)-1}\, dt = (\mathcal{M}f)(s + \mu)$$
Absolute convergence of the right-hand integral holds for $a < \mathrm{Re}(s + \mu) < b$, that is, for $a - \mathrm{Re}(\mu) < \sigma < b - \mathrm{Re}(\mu)$, which is the asserted fundamental strip.

3. Substitute $t = e^{-x}$, so that $x = -\log t$, $dt = -e^{-x}\, dx$, and $t^{s-1} = e^{-(s-1)x}$. As t decreases from ∞ to 0, the variable x increases from $-\infty$ to $+\infty$, so the limits reverse and absorb the sign of dt:
$$(\mathcal{M}f)(s) = \int_0^\infty f(t)\, t^{s-1}\, dt = \int_{-\infty}^\infty f(e^{-x})\, e^{-(s-1)x}\, e^{-x}\, dx = \int_{-\infty}^\infty f(e^{-x})\, e^{-sx}\, dx$$
With $g(x) = f(e^{-x})$ this is $\int_{-\infty}^\infty g(x)e^{-sx}\, dx$, the two-sided Laplace transform of g evaluated at s. The substitution $t = e^{-x}$ is again a C^1 diffeomorphism of $(0, \infty)$ onto $\mathbb{R}$, so absolute convergence is preserved and the identity holds throughout the fundamental strip. Writing $s = c + i\xi$ splits the exponential as $e^{-sx} = e^{-cx}e^{-i\xi x}$, and grouping the real exponential with g gives
$$(\mathcal{M}f)(c + i\xi) = \int_{-\infty}^\infty \left(e^{-cx} g(x)\right) e^{-i\xi x}\, dx$$
which is the Fourier transform of $x \mapsto e^{-cx}g(x)$ under the convention $\hat{h}(\xi) = \int_{-\infty}^\infty h(x)e^{-i\xi x}\, dx$, as we sought to show.

The third property is the structural heart of the transform. The multiplicative group $(0, \infty)$ is isomorphic, as a topological group, to the additive group $\mathbb{R}$ via the logarithm $t \mapsto -\log t = x$, and under this isomorphism Haar measure dt/t on $(0, \infty)$ becomes Lebesgue measure dx on $\mathbb{R}$. The Mellin transform is what the Fourier–Laplace transform on $\mathbb{R}$ becomes when transported back to $(0, \infty)$ through this isomorphism, with the complex parameter s playing the role of the (complexified) frequency variable. Every property of the Mellin transform is, after this change of variables, a property of the Fourier or Laplace transform; the scaling identity $(\mathcal{M}f(\alpha\cdot))(s) = \alpha^{-s}(\mathcal{M}f)(s)$, for instance, is the multiplicative form of the rule that translation by $\log \alpha$ in x multiplies the Fourier transform by a character.

This isomorphism does more than supply intuition: it converts the inversion problem for the Mellin transform into the inversion problem for the Fourier transform, which is settled by the classical Fourier inversion theorem. The result is the Mellin inversion formula.

Theorem 2.4.4: Mellin Inversion

Let $f\colon (0,\infty) \to \mathbb{C}$ be continuous, and let $c \in \mathbb{R}$ lie in the fundamental strip of f. Suppose that

1. the function $t \mapsto t^c f(t)$ is integrable on $(0,\infty)$ with respect to the Haar measure dt/t, that is, $\int_0^\infty |f(t)|\, t^c \frac{dt}{t} < \infty$; and
2. the function $\xi \mapsto (\mathcal{M}f)(c+i\xi)$ is integrable on $\mathbb{R}$, that is, $\int_{-\infty}^{\infty} |(\mathcal{M}f)(c+i\xi)|\, d\xi < \infty$.

Then for every $t > 0$,

$$f(t) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} (\mathcal{M}f)(s)\, t^{-s}\, ds$$

where the integral is taken over the vertical line $\mathrm{Re}(s) = c$.

***Proof*:**
The proof transports the statement to $\mathbb{R}$ by the substitution $t = e^{-x}$ of proposition 2.4.3 and applies the Fourier inversion theorem there.

Step 1: Pass to the additive variable. Set $g(x) = f(e^{-x})$ for $x \in \mathbb{R}$ and define

$$G(x) = e^{-cx} g(x) = e^{-cx} f(e^{-x})$$

Since f is continuous on $(0,\infty)$ and $x \mapsto e^{-x}$ is continuous from $\mathbb{R}$ onto $(0,\infty)$, the function G is continuous on $\mathbb{R}$. Hypothesis (1) states that G is integrable: substituting $t = e^{-x}$ (so $dt/t = -dx$, and the orientation reversal cancels the sign) in the change-of-variables identity for the Haar measure gives

$$\int_{-\infty}^{\infty} |G(x)|\, dx = \int_{-\infty}^{\infty} |f(e^{-x})|\, e^{-cx}\, dx = \int_0^{\infty} |f(t)|\, t^c \frac{dt}{t} < \infty$$

so $G \in L^1(\mathbb{R})$.

Step 2: Identify the Fourier transform of G. By proposition 2.4.3(3), for $s = c + i\xi$,

$$\widehat{G}(\xi) = \int_{-\infty}^{\infty} G(x)\, e^{-i\xi x}\, dx = \int_{-\infty}^{\infty} e^{-cx} g(x)\, e^{-i\xi x}\, dx = (\mathcal{M}f)(c+i\xi)$$

under the angular-frequency convention $\widehat{G}(\xi) = \int_{-\infty}^{\infty} G(x) e^{-i\xi x}\, dx$, which is the convention fixed for the Mellin variable in proposition 2.4.3(3). The Fourier inversion theorem [17] is stated in the alternative normalization $\widehat{G}_{2\pi}(\eta) = \int_{-\infty}^{\infty} G(x)\, e^{-2\pi i \eta x}\, dx$, carrying the 2π in the exponent and no prefactor on inversion. The two transforms are affine reparametrizations of one another, $\widehat{G}_{2\pi}(\eta) = \widehat{G}(2\pi\eta)$, so $\widehat{G}_{2\pi} \in L^1(\mathbb{R})$ if and only if $\widehat{G} \in L^1(\mathbb{R})$. Hypothesis (2) is therefore exactly the statement that $\widehat{G} \in L^1(\mathbb{R})$, equivalently $\widehat{G}_{2\pi} \in L^1(\mathbb{R})$.

Step 3: Apply Fourier inversion. The function G is continuous and lies in $L^1(\mathbb{R})$ (Step 1), and its Fourier transform $\widehat{G}_{2\pi}$ lies in $L^1(\mathbb{R})$ (Step 2). These are precisely the hypotheses of the Fourier inversion theorem [17], which in this normalization asserts that under them

$$G(x) = \int_{-\infty}^{\infty} \widehat{G}_{2\pi}(\eta)\, e^{2\pi i\eta x}\, d\eta \qquad \text{for all } x \in \mathbb{R}$$

Substituting $\eta = \xi/(2\pi)$, so that $d\eta = d\xi/(2\pi)$, $\widehat{G}_{2\pi}(\eta) = \widehat{G}(2\pi\eta) = \widehat{G}(\xi)$, and $e^{2\pi i\eta x} = e^{i\xi x}$, transforms this into the angular-convention inversion formula

$$G(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{G}(\xi)\, e^{i\xi x}\, d\xi \qquad \text{for all } x \in \mathbb{R}$$

in which the prefactor $1/(2\pi)$ is the Jacobian of the reparametrization. (The integrability of $\widehat{G}$ guarantees the right-hand integral converges absolutely, and continuity of G promotes the conclusion from "almost every x" to "every x"; see [14] for the measure-theoretic version and [17] for the statement used here.)

Step 4: Return to the multiplicative variable. Substitute $\widehat{G}(\xi) = (\mathcal{M}f)(c+i\xi)$ and $G(x) = e^{-cx}f(e^{-x})$ into the inversion formula of Step 3:

$$e^{-cx}f(e^{-x}) = \frac{1}{2\pi}\int_{-\infty}^{\infty} (\mathcal{M}f)(c+i\xi)\, e^{i\xi x}\, d\xi$$

Multiplying through by e^{cx} moves it inside the integral as $e^{cx}e^{i\xi x} = e^{(c+i\xi)x}$:

$$f(e^{-x}) = \frac{1}{2\pi}\int_{-\infty}^{\infty} (\mathcal{M}f)(c+i\xi)\, e^{(c+i\xi)x}\, d\xi$$

Now set $t = e^{-x}$, so $e^{x} = t^{-1}$ and $e^{(c+i\xi)x} = t^{-(c+i\xi)} = t^{-s}$ with $s = c + i\xi$. Along the vertical line $\mathrm{Re}(s) = c$ the variable s satisfies $ds = i\,d\xi$, so $d\xi = ds/i$, and as ξ runs from $-\infty$ to $+\infty$ the point s runs from $c - i\infty$ to $c + i\infty$. Therefore

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} (\mathcal{M}f)(c+i\xi)\, t^{-s}\, d\xi = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} (\mathcal{M}f)(s)\, t^{-s}\, ds$$

the factor $1/i$ converting $d\xi$ into ds and producing the $1/(2\pi i)$ normalization, completing the proof.

Two features of the inversion formula deserve comment. First, the integral runs up a vertical line $\mathrm{Re}(s) = c$ inside the fundamental strip, and the value of $f(t)$ is independent of which line c is chosen, since the left-hand side does not mention c; this freedom to slide the contour within the strip is what makes the formula useful, because shifting the line of integration across a pole of $(\mathcal{M}f)(s)$ picks up a residue, and these residues are the main terms in asymptotic expansions. Second, the normalization $1/(2\pi i)$ together with the factor t^{-s} is exactly the data of a contour integral in the sense of [13]: the Mellin inversion formula presents $f(t)$ as a contour integral of its transform, and the entire apparatus of residue calculus applies to its evaluation. This is the precise sense in which the Mellin transform "converts an arithmetic sum into a contour integral," and it is the mechanism underlying Perron's formula in section 2.5, where $(\mathcal{M}f)(s)$ is replaced by a Dirichlet series and the contour integral recovers a summatory function.

The transform is now applied to the object of central interest. The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$ of definition 2.1.1 converges on the half-plane $\mathrm{Re}(s) > 1$, where its abscissa of absolute convergence is $\sigma_a = 1$ (definition 2.1.3). Multiplying $\zeta(s)$ by $\Gamma(s)$ produces a single integral over $(0, \infty)$ whose integrand is an elementary function of t, with no reference to the individual terms n^{-s}.

Theorem 2.4.5: Integral Representation of the Riemann Zeta Function

For every s with $\mathrm{Re}(s) > 1$,

$$\Gamma(s)\,\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1}\,dt$$

the integral converging absolutely on this half-plane.

***Proof*:**
Fix s with $\sigma = \mathrm{Re}(s) > 1$.

Step 1: Expand the integrand as a geometric series. For $t > 0$ one has $0 < e^{-t} < 1$, so the geometric series in e^{-t} converges:

$$\frac{1}{e^t - 1} = \frac{e^{-t}}{1 - e^{-t}} = \sum_{n=1}^{\infty} e^{-nt}$$

This is the sum of the geometric series with first term e^{-t} and ratio $e^{-t} \in (0, 1)$, valid for every $t > 0$. Multiplying by t^{s-1} gives the pointwise identity

$$\frac{t^{s-1}}{e^t - 1} = \sum_{n=1}^{\infty} t^{s-1} e^{-nt}$$

for each fixed $t > 0$.

Step 2: Evaluate the integral of a single term. For each $n \geq 1$, the substitution $u = nt$ in the gamma integral of example 2.4.2 gives, with $t = u/n$ and $dt = du/n$,

$$\int_0^\infty t^{s-1} e^{-nt}\,dt = \int_0^\infty \left(\frac{u}{n}\right)^{s-1} e^{-u}\,\frac{du}{n} = n^{-s} \int_0^\infty u^{s-1} e^{-u}\,du = \frac{\Gamma(s)}{n^s}$$

the integral $\int_0^\infty u^{s-1} e^{-u}\,du = \Gamma(s)$ converging absolutely because $\sigma > 1 > 0$ (example 2.4.2). In particular the same computation with s replaced by σ shows $\int_0^\infty t^{\sigma-1} e^{-nt}\,dt = \Gamma(\sigma)\,n^{-\sigma}$.

Step 3: Interchange summation and integration. The terms $t^{s-1}e^{-nt}$ are not all nonnegative (the factor t^{s-1} is complex when s is), so the interchange is justified on the absolute values, where Tonelli's theorem applies to the nonnegative integrand $(n, t) \mapsto |t^{s-1}|e^{-nt} = t^{\sigma-1}e^{-nt}$ over the product of counting measure on $\mathbb{Z}_{\geq 1}$ and Lebesgue measure on $(0, \infty)$. By Step 2 applied with σ in place of s,

$$\sum_{n=1}^{\infty} \int_0^\infty t^{\sigma-1} e^{-nt}\,dt = \sum_{n=1}^{\infty} \frac{\Gamma(\sigma)}{n^\sigma} = \Gamma(\sigma)\,\zeta(\sigma) < \infty$$

the series converging because $\sigma > 1 = \sigma_a$ for ζ (definition 2.1.3). Tonelli's theorem [14] therefore guarantees that the function $(n, t) \mapsto t^{s-1}e^{-nt}$ is integrable for the product measure, and Fubini's theorem [14] then permits the interchange of the sum and the integral for the (complex) integrand itself.

Step 4: Assemble. Using the pointwise identity of Step 1 and the justified interchange of Step 3,

$$\int_0^\infty \frac{t^{s-1}}{e^t-1}\,dt = \int_0^\infty \sum_{n=1}^\infty t^{s-1}e^{-nt}\,dt = \sum_{n=1}^\infty \int_0^\infty t^{s-1}e^{-nt}\,dt = \sum_{n=1}^\infty \frac{\Gamma(s)}{n^s} = \Gamma(s)\,\zeta(s)$$

where the last equality factors the constant $\Gamma(s)$ out of the sum $\sum_n n^{-s} = \zeta(s)$, valid on $\mathrm{Re}(s) > 1$. The absolute convergence asserted in the statement is the finiteness computed in Step 3, completing the proof.

This integral representation is the analytic launch-point for everything that follows about $\zeta(s)$. The Dirichlet series $\sum n^{-s}$ converges only on $\mathrm{Re}(s) > 1$ and gives no direct access to the values of $\zeta(s)$ elsewhere; the integral $\int_0^\infty t^{s-1}/(e^t-1)\,dt$, by contrast, has an integrand whose singularity structure can be analyzed by splitting the range at $t = 1$ and expanding $1/(e^t-1)$ near the origin, where it behaves like $1/t$. That expansion separates a part that continues meromorphically to the whole plane from a part that is already entire, and this is exactly how the meromorphic continuation of $\zeta(s)$ to $\mathbb{C}$, with its single simple pole at $s = 1$, is produced in chapter 4. The same integral, after the contour is deformed into a Hankel loop around the positive real axis, yields the functional equation relating $\zeta(s)$ to $\zeta(1-s)$ in chapter 6. The representation also exhibits the gamma factor $\Gamma(s)$ as the inseparable companion of $\zeta(s)$: the natural object is the product $\Gamma(s)\zeta(s)$, and it is this product, not $\zeta(s)$ alone, whose Mellin-integral form has elementary content. The appearance of $\Gamma(s)$ here is the first sign of the completed zeta function $\xi(s)$ that organizes the functional equation.

The alternating analogue of the representation replaces $e^t - 1$ by $e^t + 1$ and brings in the Dirichlet eta function, the alternating series studied in example 2.1.10. The computation is identical except for a sign that turns the geometric series into an alternating one, and the resulting integrand is better behaved at the origin.

Example 2.4.6: The Dirichlet Eta Function and the Alternating Representation

The Dirichlet eta function is $\eta(s) = \sum_{n=1}^\infty (-1)^{n+1}n^{-s}$, which by example 2.1.10 satisfies $\eta(s) = (1-2^{1-s})\zeta(s)$ on $\mathrm{Re}(s) > 1$. Its product with $\Gamma(s)$ admits the integral representation

$$(1-2^{1-s})\,\zeta(s)\,\Gamma(s) = \eta(s)\,\Gamma(s) = \int_0^\infty \frac{t^{s-1}}{e^t+1}\,dt \qquad (\mathrm{Re}(s) > 1)$$

derived exactly as in theorem 2.4.5. For $t > 0$ the geometric series gives

$$\frac{1}{e^t+1} = \frac{e^{-t}}{1+e^{-t}} = \sum_{n=1}^\infty (-1)^{n+1}e^{-nt}$$

(the series with first term e^{-t} and ratio $-e^{-t}$, with $|-e^{-t}| < 1$). Multiplying by t^{s-1} and integrating term by term gives

$$\int_0^\infty \frac{t^{s-1}}{e^t+1}\,dt = \sum_{n=1}^\infty (-1)^{n+1}\int_0^\infty t^{s-1}e^{-nt}\,dt = \sum_{n=1}^\infty (-1)^{n+1}\frac{\Gamma(s)}{n^s} = \Gamma(s)\,\eta(s)$$

using the single-term evaluation $\int_0^\infty t^{s-1}e^{-nt}\,dt = \Gamma(s)\,n^{-s}$ from Step 2 of theorem 2.4.5. The interchange of summation and integration is justified by absolute convergence: the absolute values

$\sum_n \int_0^\infty t^{\sigma-1}e^{-nt}\,dt = \Gamma(\sigma)\zeta(\sigma)$ are summable for $\sigma = \mathrm{Re}(s) > 1$, so the same Tonelli–Fubini argument as in theorem 2.4.5 applies ([14]).

The contrast with the $1/(e^t - 1)$ integrand is the point of the example. Near $t = 0^+$ one has $e^t + 1 \to 2$, so $1/(e^t + 1) \to 1/2$ is bounded, and the integrand $t^{s-1}/(e^t + 1)$ is comparable to $\frac{1}{2}t^{\sigma-1}$, which is integrable on $(0, 1]$ for every $\sigma > 0$, not only for $\sigma > 1$. At $t = \infty$ the integrand decays like $t^{\sigma-1}e^{-t}$ for every σ. Hence the integral $\int_0^\infty t^{s-1}/(e^t + 1)\,dt$ converges and defines an analytic function on the entire half-plane $\mathrm{Re}(s) > 0$, even though the series defining $\eta(s)$ was used only on $\mathrm{Re}(s) > 1$. Foreshadowing the continuation of chapter 4: this convergent integral provides the meromorphic continuation of $\eta(s)\Gamma(s)$ to $\mathrm{Re}(s) > 0$, and dividing by the elementary factor $(1 - 2^{1-s})\Gamma(s)$ continues $\zeta(s)$ past the line $\mathrm{Re}(s) = 1$. The pole of $\zeta(s)$ at $s = 1$ survives this division because $(1 - 2^{1-s})$ vanishes there, while the factor $1/(e^t + 1)$ removed the obstruction at the origin that the $1/(e^t - 1)$ integrand carried.

2.5 Perron's Formula

The transition from a Dirichlet series to the arithmetic information it encodes runs in two directions. Forming $F(s) = \sum_{n\geq 1} a_n n^{-s}$ from a sequence (a_n) packages the coefficients into an analytic function; the analyticity, the location of poles, and the residues of F are the analytic data. Perron's formula runs the construction backward: it recovers the partial sum $\sum_{n\leq x} a_n$ of the coefficients from a contour integral of F. The kernel that performs the recovery is x^s/s, integrated up a vertical line to the right of the region where F converges. The effect is that analytic facts about F, principally the behavior of F near its singularities, are converted into asymptotic statements about the coefficient sum, since shifting the contour of integration leftward across a pole of $F(s)x^s/s$ contributes a residue, and these residues are the main terms in the asymptotics.

The instance that organizes this chapter and the next is $F(s) = -\zeta'(s)/\zeta(s) = \sum_{n\geq 1} \Lambda(n)n^{-s}$ (theorem 2.2.6). Here the coefficient sum is the Chebyshev function $\psi(x) = \sum_{n\leq x} \Lambda(n)$, and Perron's formula expresses $\psi(x)$ as a contour integral of $-\zeta'(s)/\zeta(s) \cdot x^s/s$. The single pole of $\zeta(s)$ at $s = 1$ produces a pole of the integrand with residue x, which is the source of the leading term $\psi(x) \sim x$ of the Prime Number Theorem; the zeros of $\zeta(s)$ inside the critical strip produce the oscillatory corrections. Perron's formula is therefore the gateway through which the analytic study of $\zeta(s)$ in chapter 4 becomes the analytic proof of the Prime Number Theorem in chapter 5. The formula itself is the Mellin-type inversion of theorem 2.4.4 specialized to the kernel x^s/s: where Mellin inversion recovers a function from its transform along a vertical line, Perron inversion recovers a step function (the summatory function) from the Dirichlet series, and the discontinuity of the step function at each integer is exactly the jump produced by the kernel x^s/s.

The entire formula rests on a single contour integral, the value of the kernel x^s/s integrated up a vertical line. This integral is discontinuous in the ratio it depends on, jumping from 0 to 1 as the ratio crosses 1, and it is this discontinuity that reproduces the indicator $n \leq x$ inside the coefficient sum.

Lemma 2.5.1: The Discontinuous Integral

Let $c > 0$. For $y > 0$ define

$$I(y) = \frac{1}{2\pi i}\int_{(c)} \frac{y^s}{s}\,ds = \lim_{T\to\infty}\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{y^s}{s}\,ds$$

the integral taken up the vertical line $\mathrm{Re}(s) = c$. Then

$$I(y) = \begin{cases} 1 & y > 1 \\ \frac{1}{2} & y = 1 \\ 0 & 0 < y < 1 \end{cases}$$

Moreover, writing $I(y,T) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} y^s/s\,ds$ for the truncation to the segment $[c-iT, c+iT]$ and $\delta(y) = 1$ for $y > 1$, $\delta(y) = 0$ for $0 < y < 1$, the truncation satisfies, for every $T > 0$,

$$\big|I(y,T) - \delta(y)\big| \le y^c \min\left(1, \frac{1}{\pi T|\log y|}\right) \qquad (y \ne 1)$$

and, in the boundary case $y = 1$,

$$\left|I(1,T) - \tfrac{1}{2}\right| \le \frac{c}{\pi T}$$

***Proof*:**
Throughout, $s = \sigma + it$ and the residue theorem is applied in the form established in [13]. The integrand y^s/s is meromorphic on $\mathbb{C}$ with a single simple pole at $s = 0$, where its residue is $\lim_{s\to 0} s \cdot y^s/s = y^0 = 1$.

Step 1: The boundary case $y = 1$. Here $y^s = 1$, so the truncated integral is computed directly. Parametrizing $s = c + it$ with $t \in [-T, T]$ and $ds = i\,dt$,

$$I(1,T) = \frac{1}{2\pi i}\int_{-T}^{T}\frac{i\,dt}{c+it} = \frac{1}{2\pi}\int_{-T}^{T}\frac{c-it}{c^2+t^2}\,dt$$

The imaginary part of the integrand, $-t/(c^2+t^2)$, is odd in t and integrates to zero over the symmetric interval. The real part gives

$$I(1,T) = \frac{1}{2\pi}\int_{-T}^{T}\frac{c}{c^2+t^2}\,dt = \frac{1}{2\pi}\cdot 2\arctan\frac{T}{c} = \frac{1}{\pi}\arctan\frac{T}{c}$$

As $T\to\infty$ this tends to $\frac{1}{\pi}\cdot\frac{\pi}{2} = \frac{1}{2}$, establishing $I(1) = \frac{1}{2}$. For the error, $\frac{1}{2} - \frac{1}{\pi}\arctan(T/c) = \frac{1}{\pi}(\frac{\pi}{2} - \arctan(T/c)) = \frac{1}{\pi}\arctan(c/T)$, and since $\arctan u \le u$ for $u \ge 0$,

$$\left|I(1,T) - \tfrac{1}{2}\right| = \frac{1}{\pi}\arctan\frac{c}{T} \le \frac{c}{\pi T}$$

which is the boundary estimate.

Step 2: The sharp truncation bound for $y \neq 1$. Fix $y \neq 1$ and $T > 0$. The estimate to prove is $|I(y,T) - \delta(y)| \le y^c/(\pi T|\log y|)$. The argument depends on the sign of $\log y$ through which direction the contour is closed.

Suppose first $y > 1$, so $\log y > 0$ and $|y^s| = y^\sigma$ decays as $\sigma \to -\infty$. For $U > 0$ consider the rectangle $\mathcal{R}_U$ with vertices $c - iT$, $c + iT$, $-U + iT$, $-U - iT$, traversed counterclockwise. It encloses the pole at $s = 0$, so by the residue theorem

$$\frac{1}{2\pi i}\oint_{\mathcal{R}_U} \frac{y^s}{s}\,ds = 1$$

The boundary $\partial\mathcal{R}_U$ consists of the right edge (the segment from $c - iT$ to $c + iT$, contributing $I(y,T)$), the left edge at $\operatorname{Re}(s) = -U$, and the two horizontal edges at $\operatorname{Im}(s) = \pm T$. On the left edge, $|y^s/s| \le y^{-U}/U \to 0$ as $U \to \infty$ because $y > 1$, so its contribution vanishes in the limit. On each horizontal edge, $s = \sigma \pm iT$ with σ running from $-U$ to c, and $|s| \ge T$, so $|y^s/s| \le y^\sigma/T$; integrating in σ over $(-\infty, c]$ bounds each horizontal contribution by

$$\frac{1}{2\pi}\int_{-\infty}^{c} \frac{y^\sigma}{T}\,d\sigma = \frac{1}{2\pi T}\cdot\frac{y^c}{\log y}$$

using $\int_{-\infty}^{c} y^\sigma\,d\sigma = y^c/\log y$ (valid since $\log y > 0$). Letting $U \to \infty$, the residue identity reads $1 = I(y,T) +$ (two horizontal contributions), whence

$$\big|I(y,T) - 1\big| \le 2\cdot\frac{1}{2\pi T}\cdot\frac{y^c}{\log y} = \frac{y^c}{\pi T\log y} = \frac{y^c}{\pi T|\log y|}$$

since $\log y = |\log y|$ here. This is the claim for $y > 1$.

Now suppose $0 < y < 1$, so $\log y < 0$ and $|y^s| = y^\sigma$ decays as $\sigma \to +\infty$. The same argument is run with the rectangle closed to the right: vertices $c - iT$, $c + iT$, $U + iT$, $U - iT$. This rectangle does not contain $s = 0$, so the closed integral is zero. The right edge at $\operatorname{Re}(s) = U$ contributes at most $y^U/U \to 0$. The horizontal edges, with σ running from c to U, are bounded by $\frac{1}{2\pi T}\int_c^\infty y^\sigma\,d\sigma = \frac{1}{2\pi T}\cdot y^c/|\log y|$ each, using $\int_c^\infty y^\sigma\,d\sigma = -y^c/\log y = y^c/|\log y|$. Hence

$$\big|I(y,T) - 0\big| \le \frac{y^c}{\pi T|\log y|}$$

completing the sharp bound for both signs.

Step 3: The trivial bound $|I(y,T) - \delta(y)| \le y^c$ for $y \neq 1$. The sharp bound degrades as $|\log y| \to 0$, so a second estimate independent of $|\log y|$ is needed to control y near 1. This is obtained by deforming the vertical segment to a circular arc on which $\operatorname{Re}(s) \ge c$ (for $y < 1$) or which encircles the pole (for $y > 1$). Set $R = \sqrt{c^2 + T^2}$, so that the endpoints $c \pm iT$ lie on the circle $|s| = R$, and write $\theta_0 = \arctan(T/c) \in (0, \pi/2)$, the argument of $c + iT$.

Suppose $0 < y < 1$. Let Γ be the arc of $|s| = R$ from $c + iT$ to $c - iT$ passing through $s = R$ on the right, that is, $s = Re^{i\theta}$ with θ decreasing from θ_0 to $-\theta_0$. The arc Γ and the segment from $c - iT$ to $c + iT$ together bound a region lying in $\operatorname{Re}(s) \ge c > 0$, which does not contain the pole $s = 0$; since the segment and Γ share endpoints, Cauchy's theorem gives $I(y,T) = \frac{1}{2\pi i}\int_\Gamma y^s/s\,ds$. On Γ, $\operatorname{Re}(s) = R\cos\theta$ with $|\theta| \le \theta_0$, and $\cos\theta \ge \cos\theta_0 = c/R$, so $R\cos\theta \ge c$; because $0 < y < 1$, this forces $|y^s| = y^{R\cos\theta} \le y^c$. With $|s| = R$ and $|ds| = R\,d\theta$,

$$\big|I(y,T)\big| = \frac{1}{2\pi}\left|\int_{-\theta_0}^{\theta_0} \frac{y^{Re^{i\theta}}}{Re^{i\theta}} Re^{i\theta} i\,d\theta\right| \le \frac{1}{2\pi}\int_{-\theta_0}^{\theta_0} y^{R\cos\theta}\,d\theta \le \frac{1}{2\pi}\cdot 2\theta_0\cdot y^c \le \frac{y^c}{2}$$

using $2\theta_0 \le \pi$. Since $\delta(y) = 0$ here, $|I(y,T) - \delta(y)| \le y^c/2 \le y^c$.

Suppose now $y > 1$. Let Γ' be the arc of $|s| = R$ from $c + iT$ to $c - iT$ passing through $s = -R$ on the left, that is, $s = Re^{i\theta}$ with θ increasing from θ_0 to $2\pi - \theta_0$. The segment (upward) together with Γ' bounds a region containing $s = 0$ and is traversed counterclockwise, so the residue theorem gives $I(y,T) + \frac{1}{2\pi i}\int_{\Gamma'} y^s/s\,ds = 1$, hence $I(y,T) - 1 = -\frac{1}{2\pi i}\int_{\Gamma'} y^s/s\,ds$. On Γ', $\mathrm{Re}(s) = R\cos\theta$ with $\theta \in [\theta_0, 2\pi - \theta_0]$, so $\cos\theta \le \cos\theta_0 = c/R$, giving $R\cos\theta \le c$; because $y > 1$, this forces $|y^s| = y^{R\cos\theta} \le y^c$. The angular length of Γ' is $2\pi - 2\theta_0 \le 2\pi$, so

$$\big|I(y,T) - 1\big| \le \frac{1}{2\pi}\int_{\theta_0}^{2\pi-\theta_0} y^{R\cos\theta}\,d\theta \le \frac{1}{2\pi}(2\pi - 2\theta_0)\,y^c \le y^c$$

Since $\delta(y) = 1$ here, this is $|I(y,T) - \delta(y)| \le y^c$.

Step 4: Combine and pass to the limit. For $y \ne 1$, Steps 2 and 3 give $|I(y,T) - \delta(y)| \le y^c/(\pi T|\log y|)$ and $|I(y,T) - \delta(y)| \le y^c$ respectively; the smaller of the two is the asserted bound $y^c \min(1, 1/(\pi T|\log y|))$. Letting $T \to \infty$ in this bound, with $y \ne 1$ fixed so that $1/(\pi T|\log y|) \to 0$, gives $I(y,T) \to \delta(y)$, which establishes $I(y) = 1$ for $y > 1$ and $I(y) = 0$ for $0 < y < 1$. Together with $I(1) = \frac{1}{2}$ from Step 1, all assertions are proved, completing the proof.

The lemma is the analytic mechanism behind everything in this section. The kernel y^s/s has a pole at $s = 0$ with residue 1, and integrating up a vertical line either captures that residue (when the contour can be closed to the left, which the decay of y^s permits exactly when $y > 1$) or does not (when it must be closed to the right, which is forced when $y < 1$). The jump of $I(y)$ from 0 to 1 as y crosses 1 is therefore the residue of y^s/s at the origin, switched on or off by the direction in which the integrand decays. The truncation estimate is the quantitative form: the error in replacing the limit by a finite integral is controlled by $1/(T|\log y|)$, so the integral resolves the indicator sharply except for ratios y extremely close to 1, where the second bound y^c takes over. This dichotomy, sharp resolution away from the diagonal $y = 1$ and a crude bound near it, is exactly what governs the error in the truncated form of Perron's formula below.

Applying the lemma term by term to a Dirichlet series converts its coefficient sum into a contour integral. The ratio y becomes x/n, so $I(x/n) = 1$ when $n < x$, $I(x/n) = 0$ when $n > x$, and the indicator $\sum_{n\le x}$ emerges from the integral.

Theorem 2.5.2: Perron's Formula

Let $F(s) = \sum_{n\ge1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a (definition 2.1.3). Let $c > \max(0, \sigma_a)$ and $x > 0$. If x is not an integer, then

$$\sum_{n\le x} a_n = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,\frac{x^s}{s}\,ds$$

the integral being taken up the line $\mathrm{Re}(s) = c$ and converging as the symmetric limit $\lim_{T\to\infty}\int_{c-iT}^{c+iT}$. If x is an integer, the same identity holds with the left side replaced by the half-weighted sum $\sum_{n<x} a_n + \frac{1}{2}a_x$.

***Proof*:**
Write $\sigma = \mathrm{Re}(s) = c$ throughout, and recall $I(y,T) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} y^s/s\,ds$ from lemma 2.5.1.

Step 1: Interchange sum and integral on the truncated contour. Fix $T > 0$. On the segment $[c - iT, c + iT]$ the series for F may be inserted and the order of summation and integration exchanged:

$$\frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\,\frac{x^s}{s}\,ds = \frac{1}{2\pi i}\int_{c-iT}^{c+iT}\sum_{n=1}^{\infty} a_n\left(\frac{x}{n}\right)^s\frac{ds}{s} = \sum_{n=1}^{\infty} a_n \cdot I\Big(\frac{x}{n}, T\Big)$$

where $F(s)x^s = \sum_n a_n n^{-s}x^s = \sum_n a_n(x/n)^s$. The interchange is justified by absolute convergence over the product of counting measure on $\mathbb{Z}_{\geq 1}$ and Lebesgue measure on the finite segment: the modulus of the (n,s)-integrand is $|a_n|(x/n)^{\sigma}/|s|$, and

$$\sum_{n=1}^{\infty}\int_{c-iT}^{c+iT}\frac{|a_n|(x/n)^{\sigma}}{|s|}\,|ds| = \left(x^{\sigma}\sum_{n=1}^{\infty}\frac{|a_n|}{n^{\sigma}}\right)\left(\int_{-T}^{T}\frac{dt}{\sqrt{c^2+t^2}}\right)$$

is finite: the first factor converges because $\sigma = c > \sigma_a$ (definition 2.1.3), and the second is the finite integral of a bounded function over a bounded interval. Fubini's theorem [14] therefore permits the interchange.

Step 2: Split off the finitely many terms with $n \leq x$. The sum $\sum_n a_n I(x/n, T)$ is split according to whether $n \leq x$ or $n > x$. Since x is fixed and finite, only finitely many indices satisfy $n \leq x$, so

$$\sum_{n=1}^{\infty} a_n\, I\Big(\frac{x}{n}, T\Big) = \sum_{n\leq x} a_n\, I\Big(\frac{x}{n}, T\Big) + \sum_{n>x} a_n\, I\Big(\frac{x}{n}, T\Big)$$

In the first sum, a finite sum, each term converges as $T \to \infty$ by lemma 2.5.1: for $n < x$ one has $x/n > 1$ and $I(x/n, T) \to 1$; if x is an integer the single term $n = x$ has $x/n = 1$ and $I(1,T) \to \frac{1}{2}$. Therefore

$$\lim_{T\to\infty}\sum_{n\leq x} a_n\, I\Big(\frac{x}{n}, T\Big) = \begin{cases}\sum_{n\leq x} a_n & x \notin \mathbb{Z}\\ \sum_{n<x} a_n + \frac{1}{2}a_x & x \in \mathbb{Z}\end{cases}$$

the cases differing only in the treatment of the boundary index $n = x$.

Step 3: The tail vanishes in the limit. For the tail $\sum_{n>x} a_n I(x/n, T)$, each index has $x/n < 1$, so $\delta(x/n) = 0$ and, by the trivial bound in lemma 2.5.1, $|I(x/n, T)| \leq (x/n)^c$ for every T. This dominating bound is summable, since

$$\sum_{n>x}|a_n|\left(\frac{x}{n}\right)^c = x^c\sum_{n>x}\frac{|a_n|}{n^c} < \infty$$

again because $c > \sigma_a$. For each fixed $n > x$, lemma 2.5.1 gives $I(x/n, T) \to \delta(x/n) = 0$ as $T \to \infty$, because $x/n < 1$ is fixed and bounded away from 1, so $1/(\pi T|\log(x/n)|) \to 0$. The dominated convergence theorem for series [14] (counting measure on $\{n > x\}$, with dominating summable bound $|a_n|(x/n)^c$) then yields

$$\lim_{T\to\infty} \sum_{n>x} a_n\, I\Big(\frac{x}{n}, T\Big) = \sum_{n>x} a_n \cdot 0 = 0$$

Step 4: Assemble. Combining Steps 1 through 3, the symmetric limit of the truncated contour integral exists and equals

$$\lim_{T\to\infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s)\, \frac{x^s}{s}\, ds = \begin{cases} \sum_{n\le x} a_n & x \notin \mathbb{Z} \\ \sum_{n<x} a_n + \frac{1}{2}a_x & x \in \mathbb{Z} \end{cases}$$

which is the assertion of the theorem, completing the proof.

Perron's formula is the precise statement that the analytic behavior of F controls the asymptotics of its coefficient sum. The left side is a step function of x, jumping by a_n at each integer n; the right side is a contour integral whose integrand $F(s)x^s/s$ inherits all the singularities of F, together with the pole of the kernel at $s = 0$. The contour runs up a vertical line in the region of absolute convergence, where F is analytic and the integral has no singularities to contend with. The information is extracted by moving the contour to the left, into the region where F has poles: each pole crossed contributes its residue, and the integral over the shifted contour is smaller because the factor $x^s = x^\sigma e^{it\log x}$ has smaller modulus once σ has decreased. The residue at a pole $s = \rho$ of F contributes a term proportional to x^ρ/ρ, so a pole at $s = 1$ gives a main term of size x, a pole at $s = 0$ gives a constant, and poles further left give lower-order terms. The asymptotic expansion of $\sum_{n\le x} a_n$ is, in this way, read off from the poles of F and their residues; the rightmost pole supplies the dominant term, and the remaining poles and the bound on the shifted contour supply the error. This is the mechanism that the analytic proof of the Prime Number Theorem makes quantitative in chapter 5.

The formula as stated is not yet usable for that program, because the contour runs to infinity in both directions and the integral over the full line is only conditionally convergent. Quantitative work requires truncating the contour to a finite height T and controlling the resulting error explicitly. The truncated form is the one used throughout the subject.

Theorem 2.5.3: The Truncated Perron Formula

Let $F(s) = \sum_{n\geq 1} a_n n^{-s}$ have abscissa of absolute convergence σ_a, let $c > \max(0, \sigma_a)$, and let $x \geq 1$ and $T \geq 1$. Then

$$\sum_{n\leq x} a_n = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\,\frac{x^s}{s}\,ds + R$$

where, if x is not an integer, the error term satisfies

$$|R| \leq \sum_{n=1}^{\infty}|a_n|\left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right)$$

and if x is an integer the same bound holds for the half-weighted sum $\sum_{n<x} a_n + \frac{1}{2}a_x$ in place of $\sum_{n\leq x} a_n$, augmented by the single near-diagonal term

$$|R| \leq \sum_{\substack{n=1\\ n\neq x}}^{\infty}|a_n|\left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right) + |a_x|\,\frac{c}{\pi T}$$

***Proof*:**
The starting point is the exact finite-T identity established in Step 1 of the proof of theorem 2.5.2, valid for every $T \geq 1$ with no passage to a limit:

$$\frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\,\frac{x^s}{s}\,ds = \sum_{n=1}^{\infty} a_n\, I\left(\frac{x}{n}, T\right)$$

the interchange of sum and integral being justified exactly as there (absolute convergence over the product of counting measure and the finite segment, using $c > \sigma_a$). The error R is the difference between the coefficient sum and this integral.

Case $x \notin \mathbb{Z}$. Here $x/n \neq 1$ for every n, so $\delta(x/n) = 1$ when $n < x$ and $\delta(x/n) = 0$ when $n > x$; equivalently $\delta(x/n) = \mathbb{1}[n \leq x]$, and therefore $\sum_{n\leq x} a_n = \sum_n a_n\,\delta(x/n)$. Subtracting the identity above,

$$R = \sum_{n\leq x} a_n - \frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\frac{x^s}{s}\,ds = \sum_{n=1}^{\infty} a_n\Big(\delta\big(\tfrac{x}{n}\big) - I\big(\tfrac{x}{n}, T\big)\Big)$$

The rearrangement is legitimate because both series converge absolutely: $\sum_n |a_n||\delta(x/n)|$ is the finite sum over $n \leq x$, and $\sum_n |a_n||I(x/n,T)|$ is dominated by $\sum_n |a_n|(x/n)^c = x^c\sum_n |a_n|n^{-c} < \infty$ via the trivial bound of lemma 2.5.1. Taking absolute values and applying the truncation estimate of lemma 2.5.1 to each term, with $|\delta(x/n) - I(x/n,T)| \leq (x/n)^c \min(1, 1/(\pi T|\log(x/n)|))$,

$$|R| \leq \sum_{n=1}^{\infty}|a_n|\left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{\pi T|\log(x/n)|}\right) \leq \sum_{n=1}^{\infty}|a_n|\left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right)$$

the last inequality discarding the factor $1/\pi \leq 1$ inside the second argument of the minimum to give the form stated. This proves the non-integer case.

Case $x \in \mathbb{Z}$. Now exactly one index has $x/n = 1$, namely $n = x$, and for it δ takes the boundary value $\frac{1}{2}$ consistent with the half-weighted sum. For every other index $\delta(x/n) = \mathbb{1}[n < x]$, so the half-weighted sum is $\sum_{n<x} a_n + \frac{1}{2}a_x = \sum_n a_n\, \delta(x/n)$ with the convention $\delta(1) = \frac{1}{2}$. As before,

$$R = \sum_{n=1}^{\infty} a_n \Big(\delta\big(\tfrac{x}{n}\big) - I\big(\tfrac{x}{n}, T\big) \Big)$$

The single term $n = x$ contributes $a_x(\frac{1}{2} - I(1,T))$, bounded in modulus by $|a_x|\, c/(\pi T)$ by the boundary estimate of lemma 2.5.1. Every other term has $x/n \neq 1$ and is bounded as in the non-integer case. Hence

$$|R| \le \sum_{\substack{n=1 \\ n \neq x}}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right) + |a_x|\, \frac{c}{\pi T}$$

as required.

The structure of the error bound is what makes the truncated formula effective. The minimum has two regimes. For n far from x, the ratio x/n is bounded away from 1, $|\log(x/n)|$ is of order 1 or larger, and the term contributes at most $|a_n|(x/n)^c/T$, which is small because of the factor $1/T$. For n close to x, where $|\log(x/n)|$ is small and the first regime would give a useless bound, the minimum switches to the constant 1, and the term contributes at most $|a_n|(x/n)^c$; there are few such n, and the convergence factor $(x/n)^c$ keeps the total controlled. In applications the parameters c and T are chosen as functions of x to balance the contour integral against this error, and the explicit dependence on T in the bound is precisely what allows the optimization. The near-diagonal term in the integer case, of size $|a_x|/T$, is the price of evaluating the formula at a jump of the step function, and it is the reason the cleanest statements are made for non-integer x or after the half-weighting convention is adopted.

The application that motivates the entire development is the Chebyshev function ψ. Taking $a_n = \Lambda(n)$, the von Mangoldt function (definition 1.1.8), the generating Dirichlet series is the logarithmic derivative of ζ computed in theorem 2.2.6, and Perron's formula presents $\psi(x)$ as a contour integral of it.

Example 2.5.4: Perron's Formula for $\psi(x)$

Let $a_n = \Lambda(n)$, so that the summatory function is the Chebyshev function $\psi(x) = \sum_{n \le x} \Lambda(n)$. By theorem 2.2.6, the generating Dirichlet series is

$$F(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$$

valid on $\mathrm{Re}(s) > 1$, where this is also the abscissa of absolute convergence: $0 \le \Lambda(n) \le \log n$ gives $\sum_n \Lambda(n) n^{-\sigma} \le \sum_n (\log n) n^{-\sigma} < \infty$ for every $\sigma > 1$, while at $\sigma = 1$ the series $\sum_n \Lambda(n)/n$ diverges, so $\sigma_a = 1$. Choosing any $c > 1$ and any non-integer $x > 0$, theorem 2.5.2 yields

$$\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) \frac{x^s}{s}\, ds$$

and the truncated form of theorem 2.5.3 gives the same integral over $[c - iT, c + iT]$ together with an explicit error governed by $\sum_n \Lambda(n)(x/n)^c \min(1, 1/(T|\log(x/n)|))$. This integral is the analytic starting point of the proof of the Prime Number Theorem. The integrand $-\zeta'(s)/\zeta(s) \cdot x^s/s$ carries a pole at $s = 1$ coming from the simple pole of $\zeta(s)$ there: since $-\zeta'(s)/\zeta(s) \sim 1/(s-1)$ near $s = 1$, the residue of the integrand at $s = 1$ is $x^1/1 = x$, which is the leading term $\psi(x) \sim x$ once the contour is shifted past this pole. The kernel contributes a further pole at $s = 0$, and the nontrivial zeros of $\zeta(s)$ in the critical strip contribute poles of $-\zeta'(s)/\zeta(s)$ whose residues produce the oscillatory terms $-x^\rho/\rho$ in the explicit formula. Evaluating the shifted integral, which requires the analytic continuation of $\zeta(s)$ past $\mathrm{Re}(s) = 1$ and a zero-free region near that line, is carried out in chapter 5; the present example records only the integral representation that initiates it.

3

Elementary Results on Prime Distribution

chapter 2 ended with the identity that lies at the center of the subject: for $\operatorname{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

(theorem 2.2.6). The right side is the Dirichlet series of the von Mangoldt function Λ, whose summatory function counts prime powers weighted by $\log p$; the left side is an analytic object whose poles and zeros are accessible to complex analysis. The full exploitation of this identity, locating the pole of $\zeta(s)$ at $s = 1$ and excluding zeros on the line $\operatorname{Re}(s) = 1$, yields the Prime Number Theorem and occupies chapter 5. That route requires contour integration and the analytic continuation of $\zeta(s)$.

This chapter takes the elementary road. Before deploying complex analysis, one can already pin down a great deal about the distribution of primes using nothing more than the unique factorization of integers, the asymptotics of the harmonic sum, and Abel's summation formula from chapter 1. The arguments of Chebyshev fix the correct *order of magnitude* of the prime counting function $\pi(x)$, showing that $\pi(x)$ is bounded above and below by constant multiples of $x/\log x$ without determining the constant. The theorems of Mertens evaluate the key sums over primes, $\sum_{p\le x}(\log p)/p$ and $\sum_{p\le x} 1/p$ and $\prod_{p\le x}(1-1/p)$, exactly to leading order. Bertrand's postulate, that a prime always lies between n and $2n$, follows from a careful analysis of the central binomial coefficient. None of these results requires a single contour integral; each rests on counting prime powers inside factorials and binomial coefficients and feeding the count through partial summation. The prerequisites are **??** and the elementary parts of chapter 1, in particular the von Mangoldt function (definition 1.1.8), the summatory-function language (definition 1.3.1), and Abel summation (theorem 1.4.1); the analytic machinery of chapter 2 is not used until the final analytic proof in chapter 5.

3.1 Chebyshev's Functions

The Prime Number Theorem is most often stated as $\pi(x) \sim x/\log x$, a statement about the counting function π. The function π is a step function with a unit jump at each prime, and the irregular spacing of the primes makes its growth difficult to analyze directly. Chebyshev's insight was that two weighted counting functions, attaching the weight $\log p$ to each prime rather than the weight 1, are far more tractable: they are the summatory functions of arithmetic functions already studied in chapter 1, they are tied directly to the factorization of factorials, and their asymptotics are equivalent to those of π. This section introduces π, ϑ, and ψ, records their numerical behavior, relates ψ to ϑ, and proves that the three natural asymptotic statements about them stand or fall together. The equivalence is what licenses the analytic theory of chapter 5 to target the most convenient of the three, ψ, and transfer the conclusion back to π.

Definition 3.1.1: The Prime Counting Function

The *prime counting function* $\pi\colon [1,\infty) \to \mathbb{Z}_{\geq 0}$ is

$$\pi(x) = \#\{\, p \leq x \,|\, p \text{ prime} \,\}$$

the number of primes not exceeding x.

The function π has appeared informally since **??** , where Euclid's theorem was read as the statement $\pi(x) \to \infty$, and again in example 1.4.7, where Abel summation converted π into the sum $\sum_{p\leq x} 1/p$. It is a nondecreasing step function, constant on each interval between consecutive primes and jumping by 1 at each prime. Its central feature is the slowness of its growth: the Prime Number Theorem will show $\pi(x) \sim x/\log x$, so that the proportion $\pi(x)/x$ of integers up to x that are prime decays like $1/\log x$, tending to zero but only logarithmically.

The two functions that organize the elementary theory weight primes by $\log p$.

Definition 3.1.2: Chebyshev's Theta and Psi Functions

Chebyshev's theta function and *Chebyshev's psi function* are

$$\vartheta(x) = \sum_{p\leq x} \log p, \qquad \psi(x) = \sum_{n\leq x} \Lambda(n) = \sum_{p^k\leq x} \log p$$

where in the sum for ϑ the index p runs over primes, in the first sum for ψ the index n runs over positive integers with Λ the von Mangoldt function (definition 1.1.8), and in the second sum the indices p, k run over primes p and exponents $k \geq 1$ with $p^k \leq x$.

The two expressions for ψ agree because $\Lambda(n)$ is $\log p$ when $n = p^k$ is a prime power and 0 otherwise, so summing $\Lambda(n)$ over $n \leq x$ is the same as summing $\log p$ over all prime powers $p^k \leq x$. Thus ψ is precisely the summatory function (definition 1.3.1) of the von Mangoldt function, and this is the link to the analytic side: ψ is the partial-sum counterpart of the Dirichlet series $\sum \Lambda(n)n^{-s} = -\zeta'(s)/\zeta(s)$ of theorem 2.2.6. The function ϑ discards the higher prime powers, summing $\log p$ only over primes $p \leq x$ rather than over all prime powers. The difference between the two is therefore the contribution of the proper prime powers p^k with $k \geq 2$, a contribution that proposition 3.1.4 will show is small.

Both functions are weighted forms of π. Where $\pi(x)$ adds 1 for each prime, $\vartheta(x)$ adds $\log p$, so ϑ is heavier toward the large primes near x; the weighting is exactly what makes $\vartheta(x)$ grow like x rather than like $x/\log x$. The heuristic is that the primes up to x have "average" logarithm of size $\log x$, so $\vartheta(x) \approx \pi(x)\log x \approx x$, while $\pi(x) \approx \vartheta(x)/\log x \approx x/\log x$. Making this heuristic precise is the content of theorem 3.1.5.

A table of values makes the comparison concrete and exhibits the tracking of $\vartheta(x)$ and $\psi(x)$ with x.

Example 3.1.3: Numerical Values of ϑ, ψ, and π

Direct computation, summing over the primes and prime powers up to each cutoff, gives the following values, rounded to four decimals:

x	$\pi(x)$	$\vartheta(x)$	$\psi(x)$	$\vartheta(x)/x$	$\psi(x)/x$
10	4	5.3471	7.8320	0.5347	0.7832
100	25	83.7284	94.0453	0.8373	0.9405
1000	168	956.2453	996.6809	0.9562	0.9967
10000	1229	9895.9914	10013.3967	0.9896	1.0013

The ratios $\vartheta(x)/x$ and $\psi(x)/x$ both approach 1 as x grows, the convergence of $\psi(x)/x$ being visibly faster, consistent with $\psi(x) - \vartheta(x)$ being a lower-order quantity. At $x = 10^4$ the value $\psi(x)/x = 1.0013$ already sits within two parts in a thousand of its conjectured limit. By contrast $\pi(x)$ is far from x; it is $x/\log x$ that $\pi(x)$ tracks, and at $x = 10^4$ one has $x/\log x \approx 1085.7$ against $\pi(x) = 1229$, the ratio $\pi(x)\log x/x \approx 1.13$ still drifting slowly toward 1. The contribution of proper prime powers to ψ is small but nonzero: at $x = 100$ the gap $\psi(100) - \vartheta(100) = 10.3169$ comes from the squares $4, 9, 25, 49$, the cubes $8, 27$, and the higher prime powers $16, 32, 64, 81$ below 100.

The table suggests that ψ and ϑ differ by a quantity of lower order. The next result makes this exact, expressing ψ as a sum of ϑ over the roots $x^{1/m}$ and bounding the difference. The estimate is needed to transfer asymptotics between ψ and ϑ, and it must be obtained without assuming any bound on ϑ itself, since the bound $\vartheta(x) = O(x)$ is proved only in the next section; using it here would be circular.

Proposition 3.1.4: Comparison of ψ and ϑ

For $x \geq 1$,

$$\psi(x) = \sum_{m \geq 1} \vartheta\left(x^{1/m}\right)$$

where the sum is finite, the terms vanishing once $m > \log_2 x$. Moreover

$$0 \leq \psi(x) - \vartheta(x) = O\left(\sqrt{x}\,\log x\right)$$

***Proof*:**

Step 1: The identity. Group the prime powers $p^k \leq x$ appearing in $\psi(x) = \sum_{p^k \leq x} \log p$ according to the exponent $k = m$. For a fixed $m \geq 1$, the primes p contributing are exactly those with $p^m \leq x$, that is, $p \leq x^{1/m}$, and each such prime contributes $\log p$. Hence the contribution of exponent m is $\sum_{p \leq x^{1/m}} \log p = \vartheta(x^{1/m})$, and summing over m gives

$$\psi(x) = \sum_{m \geq 1} \sum_{p \leq x^{1/m}} \log p = \sum_{m \geq 1} \vartheta\left(x^{1/m}\right)$$

A term $\vartheta(x^{1/m})$ is nonzero precisely when there is a prime $p \le x^{1/m}$, which requires $x^{1/m} \ge 2$, that is, $2^m \le x$, equivalently $m \le \log_2 x$. For $m > \log_2 x$ the argument $x^{1/m}$ is less than 2 and $\vartheta(x^{1/m}) = 0$, so the sum is finite with at most $\lfloor \log_2 x \rfloor$ nonzero terms.

Step 2: Nonnegativity and isolating the difference. The $m = 1$ term is $\vartheta(x)$, so

$$\psi(x) - \vartheta(x) = \sum_{m \ge 2} \vartheta\left(x^{1/m}\right)$$

Every term $\vartheta(x^{1/m})$ is a sum of values $\log p \ge \log 2 > 0$, hence nonnegative, giving $\psi(x) - \vartheta(x) \ge 0$.

Step 3: The bound, via a trivial estimate for ϑ. The bound $\vartheta(y) = O(y)$ is not yet available, so a cruder estimate is used: each of the $\pi(y)$ primes $p \le y$ satisfies $\log p \le \log y$, whence

$$\vartheta(y) = \sum_{p \le y} \log p \le \pi(y) \log y \le y \log y$$

the last step using $\pi(y) \le y$ (there are at most y positive integers up to y, hence at most y primes). Apply this with $y = x^{1/m}$, so $\vartheta(x^{1/m}) \le x^{1/m} \log(x^{1/m}) = \frac{1}{m} x^{1/m} \log x$. For $m \ge 2$ one has $x^{1/m} \le x^{1/2} = \sqrt{x}$, and discarding the factor $1/m \le 1$ gives the per-term bound $\vartheta(x^{1/m}) \le \sqrt{x}\,\log x$. Summing over the at most $\log_2 x$ nonzero terms with $m \ge 2$,

$$0 \le \psi(x) - \vartheta(x) = \sum_{2 \le m \le \log_2 x} \vartheta\left(x^{1/m}\right) \le (\log_2 x) \cdot \sqrt{x}\,\log x = \frac{\sqrt{x}\,(\log x)^2}{\log 2}$$

This already gives $\psi(x) - \vartheta(x) = O(\sqrt{x}\,(\log x)^2)$. The sharper bound $O(\sqrt{x}\,\log x)$ follows by treating $m = 2$ separately: the single term $\vartheta(\sqrt{x}) \le \sqrt{x}\,\log\sqrt{x} = \frac{1}{2}\sqrt{x}\,\log x$, while for $m \ge 3$ one has $x^{1/m} \le x^{1/3}$, so

$$\sum_{3 \le m \le \log_2 x} \vartheta\left(x^{1/m}\right) \le (\log_2 x) \cdot x^{1/3} \log x = O\left(x^{1/3} (\log x)^2\right)$$

and since $x^{1/3}(\log x)^2 = O(\sqrt{x}\,\log x)$ (because $x^{1/3-1/2} \log x = x^{-1/6} \log x \to 0$), the $m \ge 3$ tail is absorbed into the $m = 2$ term. Combining the $m = 2$ contribution with the tail gives $\psi(x) - \vartheta(x) = O(\sqrt{x}\,\log x)$, as we sought to show.

The proposition has two consequences worth separating. First, ψ and ϑ are interchangeable to leading order: since $\sqrt{x}\,\log x = o(x)$, the relation $0 \le \psi(x) - \vartheta(x) = O(\sqrt{x}\,\log x)$ means $\psi(x) = \vartheta(x) + o(x)$, so $\psi(x)/x$ and $\vartheta(x)/x$ have the same limiting behavior and any asymptotic of the form $\psi(x) \sim cx$ holds if and only if $\vartheta(x) \sim cx$ with the same constant. Second, the proper prime powers are genuinely negligible: the entire contribution of squares, cubes, and higher powers of primes to ψ is concentrated in the term $\vartheta(\sqrt{x})$ of size $O(\sqrt{x}\,\log x)$, dominated overwhelmingly by the squares, since the cubes and beyond contribute only $O(x^{1/3}(\log x)^2)$. This is why $\psi(x)/x$ in example 3.1.3 converges faster than $\vartheta(x)/x$: the higher-power correction it carries is positive and shrinking, nudging ψ above ϑ by an amount that vanishes relative to x. The bound also explains why the analytic theory works with ψ rather than ϑ at no cost in the final statement about primes.

With ψ and ϑ tied together, the remaining task is to connect them to π. The three asymptotic statements, one for each function, turn out to be equivalent, and the bridge between the unweighted π and the weighted ϑ is supplied by Abel summation.

Theorem 3.1.5: Equivalence of the Prime Number Theorem

As $x \to \infty$, the following three statements are equivalent:

1. $\pi(x) \sim \dfrac{x}{\log x}$;
2. $\vartheta(x) \sim x$;
3. $\psi(x) \sim x$.

***Proof*:**
The proof establishes the equivalence (ii) $\Leftrightarrow$ (iii) from the comparison of ψ and ϑ, then the equivalence (i) $\Leftrightarrow$ (ii) by partial summation in each direction.

Step 1: (ii) $\Leftrightarrow$ (iii). By proposition 3.1.4,

$$\frac{\psi(x)}{x} = \frac{\vartheta(x)}{x} + \frac{\psi(x)-\vartheta(x)}{x}, \qquad 0 \le \frac{\psi(x)-\vartheta(x)}{x} \le \frac{C\sqrt{x}\,\log x}{x} = \frac{C\log x}{\sqrt{x}}$$

for a constant C and all large x. The right-hand bound tends to 0, so $\psi(x)/x$ and $\vartheta(x)/x$ have the same limit if either exists. Hence $\vartheta(x) \sim x$ if and only if $\psi(x) \sim x$.

Step 2: (ii) $\Rightarrow$ (i), *expressing π through ϑ.* Apply Abel summation (theorem 1.4.1) to recover π from ϑ. Take the coefficients $a_n = \log n$ when n is prime and $a_n = 0$ otherwise, so that the summatory function is $A(t) = \sum_{n\le t} a_n = \vartheta(t)$, and take the test function $f(t) = 1/\log t$, which is continuously differentiable on $[2,\infty)$ with $f'(t) = -1/(t(\log t)^2)$. Then $a_n f(n) = 1$ for each prime n and 0 otherwise, so $\sum_{n\le x} a_n f(n) = \pi(x) - \pi(2) + 1$; carrying out the summation from the base point 2, theorem 1.4.1 gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t(\log t)^2}\,dt + c_0$$

for a bounded constant c_0 absorbing the contribution below 2. Assume now $\vartheta(t) \sim t$, say $\vartheta(t) = t + r(t)$ with $r(t) = o(t)$. The leading term is $\vartheta(x)/\log x \sim x/\log x$. For the integral, split $\vartheta(t) = t + r(t)$:

$$\int_2^x \frac{\vartheta(t)}{t(\log t)^2}\,dt = \int_2^x \frac{dt}{(\log t)^2} + \int_2^x \frac{r(t)}{t(\log t)^2}\,dt$$

The first integral satisfies $\int_2^x (\log t)^{-2}\,dt = O(x/(\log x)^2) = o(x/\log x)$, by the standard estimate that splits the range at $\sqrt{x}$: on $[2,\sqrt{x}]$ the integrand is at most $(\log 2)^{-2}$, contributing $O(\sqrt{x})$, while on $[\sqrt{x},x]$ one has $\log t \ge \frac{1}{2}\log x$, contributing at most $4x/(\log x)^2$. For the second integral, given $\varepsilon > 0$ choose T with $|r(t)| \le \varepsilon t$ for $t \ge T$; then $\int_T^x |r(t)|/(t(\log t)^2)\,dt \le \varepsilon \int_T^x (\log t)^{-2}\,dt = \varepsilon \cdot o(x/\log x)$, while $\int_2^T$ is a constant, so the second integral is $o(x/\log x)$ as well. Combining,

$\pi(x) = x/\log x + o(x/\log x)$, which is statement (i).

Step 3: (i) $\Rightarrow$ (ii), *expressing* ϑ *through* π. Run Abel summation in the reverse direction. Take $a_n = 1$ when n is prime and 0 otherwise, so $A(t) = \pi(t)$, and take $f(t) = \log t$, with $f'(t) = 1/t$. Then $a_n f(n) = \log n$ for each prime n, so $\sum_{n\le x} a_n f(n) = \vartheta(x)$, and theorem 1.4.1 gives

$$\vartheta(x) = \pi(x)\log x - \int_2^x \frac{\pi(t)}{t}\,dt$$

Assume now $\pi(t) \sim t/\log t$, say $\pi(t) = t/\log t + e(t)$ with $e(t) = o(t/\log t)$. The leading term is $\pi(x)\log x \sim x$. For the integral, substitute $\pi(t) = t/\log t + e(t)$:

$$\int_2^x \frac{\pi(t)}{t}\,dt = \int_2^x \frac{dt}{\log t} + \int_2^x \frac{e(t)}{t}\,dt$$

The first integral is $\mathrm{Li}(x)+O(1)$, and $\mathrm{Li}(x) = O(x/\log x) = o(x)$ by the same split-at-$\sqrt{x}$ estimate applied to $\int_2^x (\log t)^{-1}\,dt$. For the second integral, given $\varepsilon > 0$ choose T with $|e(t)| \le \varepsilon t/\log t$ for $t \ge T$; then $\int_T^x |e(t)|/t\,dt \le \varepsilon \int_T^x (\log t)^{-1}\,dt = \varepsilon \cdot o(x)$, and $\int_2^T$ is bounded, so the second integral is $o(x)$. Hence $\vartheta(x) = \pi(x)\log x - o(x) = x + o(x)$, which is statement (ii).

Steps 2 and 3 together give (i) $\Leftrightarrow$ (ii), and Step 1 gives (ii) $\Leftrightarrow$ (iii), so all three statements are equivalent, completing the proof.

The equivalence reorganizes the entire project. Three asymptotic statements that look different, one counting primes, one summing $\log p$ over primes, one summing $\Lambda(n)$ over all integers, are a single statement viewed through three lenses, and the constant in each is forced to be 1 once it is 1 in any one of them. The freedom this gives is decisive for the analytic theory. Of the three functions, ψ is the most natural target for complex analysis: it is the summatory function of Λ, and Λ is exactly the coefficient sequence of $-\zeta'(s)/\zeta(s)$ by theorem 2.2.6, so the asymptotic $\psi(x) \sim x$ is read directly off the analytic behavior of the logarithmic derivative of ζ, with the leading term x produced by the simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1. The function π, by contrast, has no such clean Dirichlet series. The strategy of chapter 5 is therefore to prove $\psi(x) \sim x$ analytically and invoke the present theorem to transfer the conclusion first to ϑ and then to the statement $\pi(x) \sim x/\log x$ that one actually wants. The elementary results of the remaining sections of this chapter operate on the same principle from the other side: Chebyshev bounds ψ and ϑ between constant multiples of x, and through this theorem those bounds become bounds on $\pi(x)$ of the correct order $x/\log x$.

3.2 Chebyshev's Estimates

The equivalence of theorem 3.1.5 reduces the Prime Number Theorem to the single asymptotic $\psi(x) \sim x$, but it does not produce that asymptotic; the limit $\psi(x)/x \to 1$ lies beyond the reach of the elementary counting arguments of this chapter and is the work of the analytic theory in chapter 5, where the historically first proof is given (an elementary proof of a wholly different kind, avoiding complex analysis, was found later by Selberg and Erdős and is discussed there). What can be extracted by elementary means alone is one step weaker and was obtained by Chebyshev in 1852, half a century before the analytic proof: the *order of magnitude* of the three counting functions is pinned down exactly. There are constants $0 < c_1 < c_2$ with $c_1 x \le \vartheta(x) \le c_2 x$, and likewise for ψ, so that $\vartheta(x)$ and $\psi(x)$ each grow linearly in x and $\pi(x)$ grows like $x/\log x$, with explicit numerical constants bracketing the eventual limit.

The entire argument rests on a single arithmetic object, the central binomial coefficient $\binom{2n}{n}$. Its size is controlled on both sides by elementary bounds, $4^n/(2n+1) \le \binom{2n}{n} \le 4^n$, while its prime factorization is controlled by Legendre's formula (proposition 0.2.11): every prime power dividing $\binom{2n}{n}$ is at most $2n$, and every prime in the range $(n, 2n]$ divides it exactly once. Comparing the two descriptions of the same integer forces ψ and ϑ into a linear corridor. The upper bound comes first, in the form of a bound on the product of all primes up to x.

Lemma 3.2.1: Prime Content of the Central Binomial Coefficient

Let $n \ge 1$ and let p be a prime. Then

$$v_p\binom{2n}{n} = \sum_{k\ge1}\left(\left\lfloor \frac{2n}{p^k}\right\rfloor - 2\left\lfloor \frac{n}{p^k}\right\rfloor\right)$$

Each summand is 0 or 1, the summand vanishes for $p^k > 2n$, and consequently

$$p^{\,v_p\binom{2n}{n}} \le 2n$$

for every prime p. In particular no prime power exceeding $2n$ divides $\binom{2n}{n}$.

***Proof*:**

Step 1: The valuation formula. Since $\binom{2n}{n} = (2n)!/(n!\,n!)$, the additivity of the p-adic valuation gives $v_p\binom{2n}{n} = v_p((2n)!) - 2\,v_p(n!)$. Applying Legendre's formula (proposition 0.2.11) to each factorial,

$$v_p\binom{2n}{n} = \sum_{k\ge1}\left\lfloor \frac{2n}{p^k}\right\rfloor - 2\sum_{k\ge1}\left\lfloor \frac{n}{p^k}\right\rfloor = \sum_{k\ge1}\left(\left\lfloor \frac{2n}{p^k}\right\rfloor - 2\left\lfloor \frac{n}{p^k}\right\rfloor\right)$$

both Legendre sums being finite, so their term-by-term difference is legitimate.

Step 2: Each summand is 0 *or* 1. For any real y write $y = \lfloor y\rfloor + \{y\}$ with $0 \le \{y\} < 1$. Then

$$\lfloor 2y\rfloor - 2\lfloor y\rfloor = \big(2\lfloor y\rfloor + \lfloor 2\{y\}\rfloor\big) - 2\lfloor y\rfloor = \lfloor 2\{y\}\rfloor$$

using $\lfloor 2y\rfloor = \lfloor 2\lfloor y\rfloor + 2\{y\}\rfloor = 2\lfloor y\rfloor + \lfloor 2\{y\}\rfloor$ since $2\lfloor y\rfloor$ is an integer. As $0 \le 2\{y\} < 2$, the value $\lfloor 2\{y\}\rfloor$ is 0 when $\{y\} < 1/2$ and 1 when $\{y\} \ge 1/2$. Taking $y = n/p^k$ shows each summand $\lfloor 2n/p^k\rfloor - 2\lfloor n/p^k\rfloor$ equals 0 or 1.

Step 3: The bound on prime powers. A summand can be nonzero only while its terms are nonzero, and $\lfloor 2n/p^k\rfloor = 0$ once $p^k > 2n$. Hence the nonzero summands occur only for those k with $p^k \le 2n$, that is, $k \le \log_p(2n)$, and there are at most $\lfloor \log_p(2n)\rfloor$ of them. Since each contributes at most 1,

$$v_p\binom{2n}{n} \le \#\{\, k \ge 1 \big| p^k \le 2n \,\} = \big\lfloor \log_p(2n)\big\rfloor \le \log_p(2n)$$

Exponentiating base p, the inequality $v_p\binom{2n}{n} \le \log_p(2n)$ becomes $p^{v_p\binom{2n}{n}} \le p^{\log_p(2n)} = 2n$, as we sought to show.

The lemma records two facts that will be used in opposite directions. The bound $p^{v_p\binom{2n}{n}} \le 2n$ says the central binomial coefficient is, prime by prime, "flat": no single prime contributes a factor larger than $2n$, no matter how large n is. This is what limits the contribution of small primes when $\binom{2n}{n}$ is decomposed, and it drives the lower bound on ψ. The valuation formula itself, read in the range $n < p \le 2n$, says something complementary: for such a prime the single term $k = 1$ gives $\lfloor 2n/p \rfloor - 2\lfloor n/p \rfloor = 1 - 0 = 1$ and all higher terms vanish (because $p^2 > n^2 \ge 2n$ for $n \ge 2$), so p divides $\binom{2n}{n}$ exactly once. The primes in $(n, 2n]$ therefore appear in $\binom{2n}{n}$ to the first power, and their product divides it. This second reading is the mechanism behind the primorial bound, to which the discussion now turns.

The upper estimate $\vartheta(x) = O(x)$ is equivalent to a bound on the product of all primes up to x, the *primorial*. Bounding this product is the heart of the upper Chebyshev estimate, and the argument is an induction in which the inductive step is supplied by the elementary inequality $\binom{2m+1}{m} \le 4^m$ together with the divisibility just observed.

Proposition 3.2.2: The Primorial Bound

For every real $x \ge 1$,

$$\prod_{p \le x} p \le 4^x$$

the product running over primes $p \le x$. Equivalently, $\vartheta(x) \le (\log 4)\, x$ for all $x \ge 1$.

***Proof*:**
The two formulations are equivalent: taking logarithms turns $\prod_{p \le x} p \le 4^x$ into $\sum_{p \le x} \log p \le x \log 4$, and the left side is exactly $\vartheta(x)$ by definition 3.1.2. It therefore suffices to prove the product bound. Write $\Pi(x) = \prod_{p \le x} p$.

Step 1: Reduction to integers. The product $\Pi(x)$ depends only on $\lfloor x \rfloor$, since no prime lies strictly between $\lfloor x \rfloor$ and x. The target 4^x is nondecreasing in x, so if $\Pi(n) \le 4^n$ holds for every integer $n \ge 1$ then for general $x \ge 1$ with $n = \lfloor x \rfloor$,

$$\Pi(x) = \Pi(n) \le 4^n \le 4^x$$

The claim is thus reduced to the inequality $\Pi(n) \le 4^n$ for integers $n \ge 1$, which is proved by strong induction on n.

Step 2: Base cases. For $n = 1$ the product is empty, $\Pi(1) = 1 \le 4$. For $n = 2$, $\Pi(2) = 2 \le 16$. These anchor the induction; the inductive step below treats $n \ge 3$ and refers back to strictly smaller indices.

Step 3: Even n. Suppose $n = 2m$ with $m \ge 2$, so $n \ge 4$. Since $2m$ is composite for $m \ge 2$, it is not prime, and no prime satisfies $2m - 1 < p \le 2m$. Hence $\Pi(2m) = \Pi(2m - 1)$. By the induction hypothesis applied to $2m - 1 < 2m$,

$$\Pi(2m) = \Pi(2m - 1) \le 4^{2m-1} \le 4^{2m}$$

so the bound for $n = 2m$ follows from the bound for the smaller odd index $2m - 1$.

Step 4: Odd n. Suppose $n = 2m + 1$ with $m \geq 1$, so $n \geq 3$. Split the primes $p \leq 2m + 1$ at $m + 1$:

$$\Pi(2m+1) = \left(\prod_{p \leq m+1} p\right)\left(\prod_{m+1<p\leq 2m+1} p\right) = \Pi(m+1) \cdot \prod_{m+1<p\leq 2m+1} p$$

The first factor is bounded by the induction hypothesis at $m + 1 \leq 2m < 2m + 1$:

$$\Pi(m+1) \leq 4^{m+1}$$

For the second factor, every prime p with $m + 1 < p \leq 2m + 1$ divides the integer $\binom{2m+1}{m} = (2m+1)!/(m!\,(m+1)!)$. Indeed such a p divides the numerator $(2m+1)!$, since $p \leq 2m + 1$, but divides neither $m!$ nor $(m+1)!$, since $p > m + 1$ exceeds every factor in those products; by the additivity of v_p, $v_p\binom{2m+1}{m} = v_p((2m+1)!) \geq 1$. These primes are distinct, so their product divides $\binom{2m+1}{m}$, whence

$$\prod_{m+1<p\leq 2m+1} p \;\leq\; \binom{2m+1}{m}$$

It remains to bound the binomial coefficient. The two coefficients $\binom{2m+1}{m}$ and $\binom{2m+1}{m+1}$ are equal and both appear in the binomial expansion $\sum_{j=0}^{2m+1}\binom{2m+1}{j} = 2^{2m+1}$, so

$$2\binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} \leq 2^{2m+1}$$

giving $\binom{2m+1}{m} \leq 2^{2m} = 4^m$. Combining the three bounds,

$$\Pi(2m+1) = \Pi(m+1) \cdot \prod_{m+1<p\leq 2m+1} p \;\leq\; 4^{m+1} \cdot 4^m = 4^{2m+1}$$

which is the bound for $n = 2m + 1$. The even and odd steps together complete the induction, completing the proof.

The primorial bound is the upper half of the Chebyshev estimates already in finished form: $\vartheta(x) \leq (\log 4)\, x$ is a linear bound with the explicit constant $\log 4 \approx 1.386$. The inductive engine is the same mechanism that reappears, sharpened, in the proof of Bertrand's postulate (theorem 3.4.3) later in this chapter. The primes in the window $(m+1, 2m+1]$ are forced into the central binomial coefficient, whose size is capped by 4^m; the product of those primes therefore cannot grow faster than 4^m, and feeding this back through the recursion bounds the product of all primes. The lower estimate runs the comparison in reverse: rather than asking how few primes can fit inside $\binom{2n}{n}$, it asks how the known size $\binom{2n}{n} \geq 4^n/(2n+1)$ must be accounted for by primes, each capped at $2n$ by lemma 3.2.1.

Theorem 3.2.3: Chebyshev's Estimates

There exist absolute constants $0 < c_1 < c_2$ such that for all $x \geq 2$,

$$c_1\, x \;\leq\; \vartheta(x) \;\leq\; \psi(x) \;\leq\; c_2\, x$$

Explicitly:

1. *(Upper bound.)* $\vartheta(x) \leq (\log 4)\, x$ for all $x \geq 1$, and $\psi(x) \leq (\log 4)\, x + O(\sqrt{x}\, \log x)$, so $\psi(x) \leq c_2\, x$ holds for any fixed $c_2 > \log 4$ and all large x.
2. *(Lower bound.)* $\psi(x) \geq (\log 2)\, x - O(\log x)$ and $\vartheta(x) \geq (\log 2)\, x - O(\sqrt{x}\, \log x)$, so each of $\psi(x) \geq c_1\, x$ and $\vartheta(x) \geq c_1\, x$ holds for any fixed $c_1 < \log 2$ and all large x.

In particular $\vartheta(x)$ and $\psi(x)$ are each of exact order x, with $\log 2 \leq \liminf \psi(x)/x \leq \limsup \psi(x)/x \leq \log 4$ and the same for ϑ.

***Proof*:**
The relation $\psi(x) \geq \vartheta(x)$ is available throughout: proposition 3.1.4 gives $\psi(x) - \vartheta(x) = \sum_{m \geq 2} \vartheta(x^{1/m}) \geq 0$, together with the estimate $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x}\, \log x)$, which transfers any bound between ψ and ϑ up to a $o(x)$ error. The two sub-statements are proved in turn.

1. *Upper bound.* proposition 3.2.2 gives $\vartheta(x) \leq (\log 4)\, x$ for all $x \geq 1$ directly. For ψ, the comparison $\psi(x) = \vartheta(x) + (\psi(x) - \vartheta(x))$ with $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x}\, \log x)$ yields

$$\psi(x) \leq (\log 4)\, x + O(\sqrt{x}\, \log x)$$

   Since $\sqrt{x}\, \log x = o(x)$, for any fixed $c_2 > \log 4$ the error is eventually dominated by $(c_2 - \log 4)\, x$, so $\psi(x) \leq c_2\, x$ for all large x, and after enlarging c_2 once more the bound $\psi(x) \leq c_2\, x$ holds for all $x \geq 2$. This is the right-hand inequality with c_2 an absolute constant.

2. *Lower bound.* The argument compares two descriptions of the central binomial coefficient $N = \binom{2n}{n}$ for an integer $n \geq 1$, then passes from even integers to all x.

   Size of N. The central coefficient is the largest of the $2n+1$ terms $\binom{2n}{j}$ that sum to $(1+1)^{2n} = 4^n$, so

$$N = \binom{2n}{n} \;\geq\; \frac{1}{2n+1} \sum_{j=0}^{2n} \binom{2n}{j} = \frac{4^{\,n}}{2n+1}$$

   Taking logarithms, $\log N \geq n \log 4 - \log(2n+1) = 2n \log 2 - \log(2n+1)$.

   Primes in N. By unique factorization $N = \prod_{p \leq 2n} p^{\,v_p(N)}$, the product over primes $p \leq 2n$ because lemma 3.2.1 forbids any prime power exceeding $2n$ from dividing N. For each such prime set $K_p = v_p(N)$; the same lemma gives $p^k \leq 2n$ for every k with $1 \leq k \leq K_p$. Hence

$$\log N = \sum_{p \leq 2n} K_p \log p = \sum_{p \leq 2n} \sum_{k=1}^{K_p} \log p \;\leq\; \sum_{\substack{p,\, k \geq 1 \\ p^k \leq 2n}} \log p = \psi(2n)$$

where the inequality holds because each pair (p, k) with $1 \leq k \leq K_p$ satisfies $p^k \leq 2n$ and so appears among the prime powers counted by $\psi(2n)$, while the right-hand sum may contain additional nonnegative terms. Combining the two displays,

$$\psi(2n) \;\geq\; \log N \;\geq\; 2n\log 2 - \log(2n+1)$$

for every integer $n \geq 1$.

Passage to all x. Given a real $x \geq 2$, set $n = \lfloor x/2 \rfloor \geq 1$, so $2n \leq x$ and $2n > x - 2$. As ψ is nondecreasing, $\psi(x) \geq \psi(2n) \geq 2n\log 2 - \log(2n+1)$, and using $2n > x - 2$ together with $2n + 1 \leq x + 1 \leq 2x$,

$$\psi(x) \;\geq\; (x-2)\log 2 - \log(2x) = (\log 2)\,x - \big(2\log 2 + \log(2x)\big)$$

The subtracted quantity is $O(\log x)$, giving $\psi(x) \geq (\log 2)\,x - O(\log x)$. For ϑ, the estimate $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x}\,\log x)$ of proposition 3.1.4 yields $\vartheta(x) = \psi(x) - O(\sqrt{x}\,\log x) \geq (\log 2)\,x - O(\sqrt{x}\,\log x)$. Since both error terms are $o(x)$, for any fixed $c_1 < \log 2$ each of $\psi(x) \geq c_1\,x$ and $\vartheta(x) \geq c_1\,x$ holds for all large x.

Sub-statements (1) and (2) together furnish constants $0 < c_1 < c_2$ with $c_1 x \leq \vartheta(x) \leq \psi(x) \leq c_2 x$ for all $x \geq 2$, so both functions are of exact order x, as we sought to show.

The estimates fix the order of growth without fixing the constant. The corridor they cut out, $(\log 2)\,x \leq \psi(x) \leq (\log 4)\,x$ to leading order, is wide: the lower constant $\log 2 \approx 0.693$ and the upper constant $\log 4 \approx 1.386$ differ by a factor of exactly two, and the truth $\psi(x) \sim x$ of chapter 5 sits strictly between them. Chebyshev's method, resting only on the size and factorization of one binomial coefficient, cannot close this gap; the two constants are artifacts of comparing $\binom{2n}{n}$ against 4^n on one side and against the cap $2n$ on prime powers on the other, and no elementary sharpening of these comparisons reaches the constant 1. That value is analytic in origin, produced by the simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1. What the estimates do settle is decisive for everything that follows: ψ, ϑ, and (through the next corollary) π have no anomalous growth, no logarithmic or power-law deviation from linear order, and any sum over primes built from these functions by partial summation inherits a bound of the correct shape.

The order of π now follows by converting the bounds on ϑ into bounds on π, exactly as in the proof of theorem 3.1.5 but at the level of inequalities rather than asymptotic equivalences. This is the bound foreshadowed in **??** , where $\pi(x) = O(x/\log x)$ was promised as a consequence of Chebyshev's work.

Corollary 3.2.4: The Order of the Prime Counting Function

The prime counting function is of exact order $x/\log x$: there are absolute constants $0 < a_1 < a_2$ with

$$a_1 \frac{x}{\log x} \;\le\; \pi(x) \;\le\; a_2 \frac{x}{\log x}$$

for all $x \ge 2$. More precisely, the constants may be taken arbitrarily close to $\log 2$ and $\log 4$:

$$\big(\log 2 + o(1)\big)\frac{x}{\log x} \;\le\; \pi(x) \;\le\; \big(\log 4 + o(1)\big)\frac{x}{\log x}$$

as $x \to \infty$. In particular $\pi(x) = O(x/\log x)$.

***Proof*:**
The two inequalities come from the two bounds of theorem 3.2.3, transferred through the relation between π and ϑ established in theorem 3.1.5.

Step 1: The upper bound on π. Comparing π and ϑ directly, the primes in the range $\sqrt{x} < p \le x$ each contribute a logarithm at least $\log\sqrt{x} = \frac{1}{2}\log x$, so

$$\vartheta(x) = \sum_{p \le x} \log p \;\ge\; \sum_{\sqrt{x} < p \le x} \log p \;\ge\; \big(\pi(x) - \pi(\sqrt{x})\big)\cdot \tfrac{1}{2}\log x$$

Rearranging and using $\pi(\sqrt{x}) \le \sqrt{x}$,

$$\pi(x) \;\le\; \frac{2\,\vartheta(x)}{\log x} + \pi(\sqrt{x}) \;\le\; \frac{2\,\vartheta(x)}{\log x} + \sqrt{x}$$

By the upper bound $\vartheta(x) \le (\log 4)\,x$ of theorem 3.2.3, the first term is at most $2(\log 4)\,x/\log x$, and the second satisfies $\sqrt{x} = o(x/\log x)$. Hence $\pi(x) \le 2(\log 4)\,x/\log x + o(x/\log x)$, giving $\pi(x) = O(x/\log x)$ with an absolute constant a_2. The sharper constant $\log 4 + o(1)$ is obtained below by partial summation.

Step 2: The lower bound on π. In the other direction, weighting each prime by at most $\log x$ gives

$$\vartheta(x) = \sum_{p \le x} \log p \;\le\; \sum_{p \le x} \log x = \pi(x)\log x$$

whence $\pi(x) \ge \vartheta(x)/\log x$. By the lower bound $\vartheta(x) \ge (\log 2)\,x - O(\sqrt{x}\,\log x)$ of theorem 3.2.3,

$$\pi(x) \;\ge\; \frac{\vartheta(x)}{\log x} \;\ge\; \frac{(\log 2)\,x}{\log x} - O(\sqrt{x}) = \big(\log 2 + o(1)\big)\frac{x}{\log x}$$

since $O(\sqrt{x}) = o(x/\log x)$. This gives the lower inequality with constant a_1 arbitrarily close to $\log 2$.

Step 3: Sharpening the upper constant by partial summation. The crude factor 2 in Step 1 is removed by recovering π from ϑ through Abel summation (theorem 1.4.1), the same device used in theorem 3.1.5. With $a_n = \log n$ for n prime and $a_n = 0$ otherwise, so that $A(t) = \vartheta(t)$, and with $f(t) = 1/\log t$ on $[2, \infty)$, where $f'(t) = -1/(t(\log t)^2)$, theorem 1.4.1 gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t(\log t)^2}\, dt + c_0$$

with c_0 a bounded constant. The bound $\vartheta(t) \le (\log 4)\, t$ from theorem 3.2.3 controls both pieces: the leading term is at most $(\log 4)\, x/\log x$, and the integral is at most

$$(\log 4) \int_2^x \frac{dt}{(\log t)^2} = (\log 4) \cdot O\left(\frac{x}{(\log x)^2}\right) = o\left(\frac{x}{\log x}\right)$$

by the split-at-$\sqrt{x}$ estimate for $\int_2^x (\log t)^{-2}\, dt$ recorded in the proof of theorem 3.1.5. Therefore $\pi(x) \le (\log 4)\, x/\log x + o(x/\log x) = (\log 4 + o(1))\, x/\log x$, which is the sharpened upper bound. Steps 2 and 3 together give the two-sided estimate with constants approaching $\log 2$ and $\log 4$, as we sought to show.

The corollary is the elementary substitute for the Prime Number Theorem, and it suffices for many purposes that do not require the exact constant. The proportion $\pi(x)/x$ of integers up to x that are prime lies between $(\log 2)/\log x$ and $(\log 4)/\log x$ for large x, so it decays like $1/\log x$, confirming the logarithmic thinning of the primes announced when π was first defined in definition 3.1.1. The bound $\pi(x) = O(x/\log x)$ is precisely the statement deferred in **??** , now established without any appeal to complex analysis. Its converse direction, $\pi(x) \gg x/\log x$, is equally elementary and is the quantitative form of Euclid's theorem: not only are there infinitely many primes, but they are at least a constant multiple of $x/\log x$ in number up to x. The linear bound $\vartheta(x) = O(x)$ is also the input that Mertens' theorems require, converting by partial summation into exact leading-order evaluations of $\sum_{p \le x} (\log p)/p$ and $\sum_{p \le x} 1/p$.

The width of the Chebyshev corridor is best appreciated numerically, against the limit that the Prime Number Theorem will eventually supply.

Example 3.2.5: Chebyshev's Constants versus the Truth

The two constants delivered by theorem 3.2.3 are

$$\log 2 = 0.69314\ldots, \qquad \log 4 = 2\log 2 = 1.38629\ldots$$

bracketing the eventual limit $\psi(x)/x \to 1$ of chapter 5 from below and above. The bracket for ψ is direct: the theorem gives $(\log 2 + o(1))\, x \le \psi(x) \le (\log 4)\, x$, and the table of example 3.1.3 shows $\psi(x)/x$ already at 0.9405 for $x = 100$ and 1.0013 for $x = 10^4$, comfortably inside [0.693, 1.386] and converging to 1 rather than to either Chebyshev constant.

For π, the corresponding bracket from corollary 3.2.4 is $(\log 2 + o(1))\, x/\log x \le \pi(x) \le (\log 4 + o(1))\, x/\log x$. Computing the ratio $\pi(x)\log x/x$, whose limit the Prime Number Theorem asserts to be 1:

x	$\pi(x)$	$\dfrac{x}{\log x}$	$\dfrac{\pi(x)\log x}{x}$
10^2	25	21.71	1.1513
10^3	168	144.76	1.1605
10^4	1229	1085.74	1.1320
10^5	9592	8685.89	1.1043
10^6	78498	72382.41	1.0845

Every entry in the last column lies strictly between $\log 2 = 0.693$ and $\log 4 = 1.386$, as Chebyshev's bounds require, and the column drifts slowly toward 1, the drift being the slow $o(1)$ correction that only the analytic theory resolves. The gap between Chebyshev's two constants is a factor of 2, whereas the data already pin the true ratio to within roughly ten percent of 1 by $x = 10^6$; the elementary method captures the order of magnitude but not this finer convergence.

3.3 Mertens' Theorems

Chebyshev's estimates of section 3.2 fix the order of magnitude of ψ, ϑ, and π but not the leading constant. The three theorems of Mertens, proved in 1874 by the same elementary methods, achieve something the order-of-magnitude bounds cannot: they evaluate three sums over primes *exactly to leading order*, with an explicit error term and an explicit constant. The sums in question are

$$\sum_{p\le x}\frac{\log p}{p}, \qquad \sum_{p\le x}\frac{1}{p}, \qquad \prod_{p\le x}\left(1-\frac{1}{p}\right)$$

and Mertens' theorems assert, respectively, that the first is $\log x + O(1)$, the second is $\log\log x + M + O(1/\log x)$ for a specific constant M, and the third is asymptotic to $e^{-\gamma}/\log x$. Each statement is sharper than a bound of the form "$\asymp \log x$" or "$\asymp 1/\log x$": the first pins the coefficient of $\log x$ to be exactly 1, the second produces the second-order constant M, and the third produces the constant $e^{-\gamma}$ with the Euler–Mascheroni γ in the exponent.

What makes this reach possible without analytic continuation is that Chebyshev's bound $\psi(x) = O(x)$ already controls the error when the factorization of $\lfloor x\rfloor!$ is fed through Abel summation. The route is largely uniform: an exact identity for $\log(\lfloor x\rfloor!)$ combined with Stirling's formula gives Mertens' first theorem; partial summation against the weight $1/\log t$ converts the first theorem into the second; comparison of $\sum 1/p$ with $-\sum\log(1-1/p)$ reduces the constant M to the Euler–Mascheroni γ; and exponentiating the second theorem collapses that closed form to $e^{-\gamma}$, yielding the third. Only one step reaches past the Chebyshev-level methods, the identification of the additive constant as exactly γ rather than an unspecified number: it draws on the Euler product and the behavior of ζ near its pole, both already in hand from chapter 2 and used only in the convergent range $\mathrm{Re}\, s > 1$, and it turns on a single Gamma-function identity isolated and cited at the point of use. The constant M produced here is the same $B = \gamma + \sum_p(\log(1-1/p)+1/p) = 0.2614\ldots$ that appeared in the average order of ω (proposition 1.3.9); this section establishes that identification.

The starting point is an exact arithmetic identity for the logarithm of a factorial, the same factorization idea used for the central binomial coefficient in section 3.2, now applied to $\lfloor x\rfloor!$ itself. It expresses $\log(\lfloor x\rfloor!)$ as a weighted sum of the von Mangoldt function (definition 1.1.8), and pairing it with Stirling's formula (theorem 1.5.7) gives the asymptotic that drives the first theorem.

Lemma 3.3.1: The Prime-Power Decomposition of $\log n!$

For every real $x \geq 1$,

$$\log\left(\lfloor x \rfloor!\right) = \sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor$$

where Λ is the von Mangoldt function and the sum runs over positive integers $n \leq x$. Consequently

$$\log\left(\lfloor x \rfloor!\right) = x \log x - x + O(\log x)$$

as $x \to \infty$.

***Proof*:**
Write $N = \lfloor x \rfloor$, so that $\lfloor x \rfloor! = N!$ and $\lfloor x/n \rfloor = \lfloor N/n \rfloor$ for every positive integer n (both count the multiples of n in $\{1, \ldots, N\}$, and no multiple of n lies strictly between N and x). The two displayed claims are proved in turn.

Step 1: The identity. By unique factorization, $N! = \prod_{p \leq N} p^{\,v_p(N!)}$, the product over primes $p \leq N$. Taking logarithms and applying Legendre's formula (proposition 0.2.11) to each exponent,

$$\log(N!) = \sum_{p \leq N} v_p(N!) \log p = \sum_{p \leq N} \left(\sum_{k \geq 1} \left\lfloor \frac{N}{p^k} \right\rfloor \right) \log p = \sum_{p \leq N} \sum_{k \geq 1} \left\lfloor \frac{N}{p^k} \right\rfloor \log p$$

Each Legendre sum is finite (the terms vanish once $p^k > N$), so the double sum has finitely many nonzero terms and may be reindexed. Group the terms by the value $n = p^k$ of the prime power. For each prime power $n = p^k \leq N$ the corresponding summand is $\lfloor N/n \rfloor \log p$, and $\log p = \Lambda(n)$ by the definition of the von Mangoldt function (definition 1.1.8); for every $n \leq N$ that is *not* a prime power, $\Lambda(n) = 0$ contributes nothing. Hence

$$\log(N!) = \sum_{\substack{n \leq N \\ n = p^k}} \left\lfloor \frac{N}{n} \right\rfloor \log p = \sum_{n \leq N} \Lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor$$

the last equality by $\lfloor N/n \rfloor = \lfloor x/n \rfloor$ and $N = \lfloor x \rfloor$. This is the identity.

Step 2: The asymptotic via Stirling. Stirling's formula (theorem 1.5.7) gives $\log(N!) = N \log N - N + \frac{1}{2} \log(2\pi N) + O(1/N)$, so in particular $\log(N!) = N \log N - N + O(\log N)$. It remains to replace $N = \lfloor x \rfloor$ by x at a cost of $O(\log x)$. Write $x = N + \theta$ with $0 \leq \theta < 1$. Then

$$x \log x - x = (N + \theta) \log(N + \theta) - (N + \theta)$$

The function $t \mapsto t \log t - t$ has derivative $\log t$, so by the mean value theorem $x \log x - x - (N \log N - N) = \theta \log \xi$ for some $\xi \in [N, x]$, whence

$$\big| (x \log x - x) - (N \log N - N) \big| = \theta \log \xi \leq \log(N + 1) = O(\log x)$$

Combining with $\log(N!) = N \log N - N + O(\log N)$ and $\log N = O(\log x)$,

$$\log\left(\lfloor x \rfloor!\right) = N \log N - N + O(\log N) = x \log x - x + O(\log x)$$

as we sought to show.

The identity is the elementary counterpart of the analytic identity $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ from theorem 2.2.6: both extract the von Mangoldt function from a global object, there the logarithmic derivative of ζ, here the logarithm of a factorial. The factor $\lfloor x/n \rfloor$ counts how many of the integers $1, \ldots, \lfloor x \rfloor$ are divisible by n, so the identity says that the total logarithm $\log(\lfloor x \rfloor!) = \sum_{m \le x} \log m$ is reorganized by collecting, for each prime power $n = p^k$, the contribution $\log p$ once for every multiple of n up to x. Stirling's formula then evaluates the left side to within $O(\log x)$, and the resulting estimate $\sum_{n \le x} \Lambda(n)\lfloor x/n \rfloor = x \log x - x + O(\log x)$ is the bridge to the first theorem: the floor $\lfloor x/n \rfloor$ is within 1 of x/n, and replacing it by x/n produces exactly the sum $\sum \Lambda(n)/n$ whose evaluation is sought.

Theorem 3.3.2: Mertens' First Theorem

As $x \to \infty$,

$$\sum_{n \le x} \frac{\Lambda(n)}{n} = \log x + O(1)$$

and equivalently, summing over primes only,

$$\sum_{p \le x} \frac{\log p}{p} = \log x + O(1)$$

***Proof*:**
The two statements differ by the contribution of the proper prime powers, treated at the end. The main work is the first estimate.

Step 1: Removing the floor. Start from the identity of lemma 3.3.1, $\log(\lfloor x \rfloor!) = \sum_{n \le x} \Lambda(n)\lfloor x/n \rfloor$. Writing $\lfloor x/n \rfloor = x/n - \{x/n\}$ with $0 \le \{x/n\} < 1$ separates the main term from an error:

$$\sum_{n \le x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor = x \sum_{n \le x} \frac{\Lambda(n)}{n} - \sum_{n \le x} \Lambda(n) \left\{ \frac{x}{n} \right\}$$

The error sum is controlled by Chebyshev's bound. Since $0 \le \{x/n\} < 1$ and $\Lambda(n) \ge 0$,

$$0 \le \sum_{n \le x} \Lambda(n) \left\{ \frac{x}{n} \right\} \le \sum_{n \le x} \Lambda(n) = \psi(x)$$

by the definition of ψ (definition 3.1.2), and $\psi(x) = O(x)$ by Chebyshev's estimates (theorem 3.2.3). Hence the error sum is $O(x)$.

Step 2: Solving for the sum. Combining Step 1 with the asymptotic $\log(\lfloor x \rfloor!) = x \log x - x + O(\log x)$ of lemma 3.3.1,

$$x \sum_{n \le x} \frac{\Lambda(n)}{n} = \log\left(\lfloor x \rfloor!\right) + \sum_{n \le x} \Lambda(n) \left\{ \frac{x}{n} \right\} = \left(x \log x - x + O(\log x)\right) + O(x) = x \log x + O(x)$$

the term $-x$ and the error $O(\log x)$ both being absorbed into $O(x)$. Dividing by x,

$$\sum_{n \le x} \frac{\Lambda(n)}{n} = \log x + O(1)$$

which is the first statement.

Step 3: Passage to primes. The sum $\sum_{n\le x}\Lambda(n)/n$ runs over all prime powers; isolating the primes ($k=1$) leaves the proper prime powers ($k\ge 2$):

$$\sum_{n\le x}\frac{\Lambda(n)}{n}=\sum_{p\le x}\frac{\log p}{p}+\sum_{p^k\le x,\ k\ge 2}\frac{\log p}{p^k}$$

The second sum is bounded by the full convergent series over all $k\ge 2$ and all primes:

$$0\le\sum_{\substack{p^k\le x\\ k\ge 2}}\frac{\log p}{p^k}\le\sum_p\log p\sum_{k\ge 2}\frac{1}{p^k}=\sum_p\log p\cdot\frac{1/p^2}{1-1/p}=\sum_p\frac{\log p}{p(p-1)}$$

using the geometric series $\sum_{k\ge 2}p^{-k}=p^{-2}/(1-p^{-1})=1/(p(p-1))$. The final series converges: its terms satisfy $\log p/(p(p-1))\le\log p/(p-1)^2\le C/n^{3/2}$ for a constant C and all $n=p$, since $\log p=O(p^{1/2})$, and $\sum_n n^{-3/2}<\infty$. Therefore the proper-prime-power contribution is a bounded quantity, $O(1)$, and subtracting it from $\sum_{n\le x}\Lambda(n)/n=\log x+O(1)$ leaves

$$\sum_{p\le x}\frac{\log p}{p}=\log x+O(1)$$

completing the proof.

The first theorem already separates the elementary theory from the order-of-magnitude estimates of the previous section: where Chebyshev gives $\sum_{p\le x}(\log p)/p\asymp\log x$ with unspecified constant, Mertens fixes the coefficient of $\log x$ to be exactly 1. The reason the constant is accessible while $\psi(x)\sim x$ is not lies in the weighting. Dividing $\Lambda(n)$ by n converts the identity $\sum\Lambda(n)\lfloor x/n\rfloor=x\log x-x+O(\log x)$, whose main term is of size $x\log x$, into a statement whose main term is of size $\log x$; the Chebyshev error $\psi(x)=O(x)$, fatal at the scale of $x\log x$, becomes a harmless $O(1)$ after division by x. The first theorem is thus an averaged statement, robust to the $O(x)$ uncertainty in ψ that the elementary method cannot remove. It is the input from which the remaining two theorems follow by partial summation and exponentiation, with no further arithmetic.

The passage from $\sum(\log p)/p$ to $\sum 1/p$ is a textbook application of Abel summation: the summand $1/p$ is $(\log p)/p$ divided by $\log p$, so summing against the weight $1/\log t$ against the known partial sums of $(\log p)/p$ recovers $\sum_{p\le x}1/p$. This is the device foreshadowed at corollary 1.4.2 in chapter 1.

Theorem 3.3.3: Mertens' Second Theorem

There is a constant M such that, as $x\to\infty$,

$$\sum_{p\le x}\frac{1}{p}=\log\log x+M+O\left(\frac{1}{\log x}\right)$$

The constant M is the *Mertens constant*, identified in closed form in definition 3.3.4.

***Proof*:**
Step 1: Setting up Abel summation. Define the coefficients $a_n = (\log n)/n$ when n is prime and $a_n = 0$ otherwise, so that the summatory function is

$$A(t) = \sum_{n \le t} a_n = \sum_{p \le t} \frac{\log p}{p}$$

By theorem 3.3.2, $A(t) = \log t + R(t)$ with $R(t) = O(1)$; fix a constant C_0 with $|R(t)| \le C_0$ for all $t \ge 2$. Take the test function $f(t) = 1/\log t$ on $[2, \infty)$, which is continuously differentiable with $f'(t) = -1/(t(\log t)^2)$. For a prime $n = p$, the product $a_n f(n) = \frac{\log p}{p} \cdot \frac{1}{\log p} = \frac{1}{p}$, so that $\sum_{n \le x} a_n f(n) = \sum_{p \le x} 1/p$.

Step 2: Applying the formula. Abel summation (theorem 1.4.1), carried out from the base point 2 rather than 1 (no prime lies below 2, so $A(t) = 0$ on $[1, 2)$ and the integral over $[1, 2)$ vanishes), gives

$$\sum_{p \le x} \frac{1}{p} = \frac{A(x)}{\log x} + \int_2^x \frac{A(t)}{t(\log t)^2}\, dt$$

Substitute $A(t) = \log t + R(t)$ into both pieces. The boundary term is

$$\frac{A(x)}{\log x} = \frac{\log x + R(x)}{\log x} = 1 + O\left(\frac{1}{\log x}\right)$$

since $R(x) = O(1)$. The integral splits as

$$\int_2^x \frac{\log t + R(t)}{t(\log t)^2}\, dt = \int_2^x \frac{dt}{t \log t} + \int_2^x \frac{R(t)}{t(\log t)^2}\, dt$$

Step 3: The two integrals. The first integral is elementary: with the substitution $u = \log t$, $du = dt/t$,

$$\int_2^x \frac{dt}{t \log t} = \int_{\log 2}^{\log x} \frac{du}{u} = \log\log x - \log\log 2$$

For the second integral, the integrand is absolutely bounded by $C_0/(t(\log t)^2)$, and

$$\int_2^\infty \frac{dt}{t(\log t)^2} = \int_{\log 2}^\infty \frac{du}{u^2} = \frac{1}{\log 2} < \infty$$

so $\int_2^\infty R(t)/(t(\log t)^2)\, dt$ converges absolutely to some real number, call it I. The tail past x is estimated the same way:

$$\left| \int_x^\infty \frac{R(t)}{t(\log t)^2}\, dt \right| \le C_0 \int_x^\infty \frac{dt}{t(\log t)^2} = \frac{C_0}{\log x} = O\left(\frac{1}{\log x}\right)$$

whence $\int_2^x R(t)/(t(\log t)^2)\, dt = I + O(1/\log x)$.

Step 4: Assembling the constant. Collecting the boundary term and the two integrals,

$$\sum_{p\le x}\frac{1}{p} = \Big(1+O\Big(\tfrac{1}{\log x}\Big)\Big) + \big(\log\log x - \log\log 2\big) + \Big(I + O\Big(\tfrac{1}{\log x}\Big)\Big)$$

Setting $M = 1 - \log\log 2 + I$, the constant terms combine and the two error terms merge into a single $O(1/\log x)$, giving

$$\sum_{p\le x}\frac{1}{p} = \log\log x + M + O\left(\frac{1}{\log x}\right)$$

as we sought to show.

The second theorem is the quantitative form of the divergence of $\sum 1/p$, Euler's theorem that the primes are "dense enough" for their reciprocals to sum to infinity. The growth rate is now exact: the partial sums increase like $\log\log x$, doubly logarithmically, so the divergence is extraordinarily slow. To reach $\sum_{p\le x} 1/p = 4$ requires x on the order of $e^{e^4} \approx 5\times 10^{23}$. The presence of the second-order constant M is what distinguishes Mertens' statement from the bare divergence: $\sum_{p\le x} 1/p - \log\log x$ converges to a definite limit M, at the rate $O(1/\log x)$. The proof produced $M = 1 - \log\log 2 + I$ as an integral expression depending on the error term $R(t)$ in the first theorem; this form is correct but opaque. The next definition gives the closed form, the same constant B that already appeared in proposition 1.3.9 as the second-order term in the average of ω, and identifying the two is the substantive content of what follows.

The constant M in the second theorem was produced as $1 - \log\log 2 + I$, with I an integral of the first-theorem error. Establishing its closed form proceeds in two stages. Comparing the divergent sum $\sum 1/p$ with the divergent sum $-\sum \log(1-1/p)$, whose difference is the convergent prime correction S, isolates the part of M contributed by the higher prime powers. The Euler–Mascheroni γ then enters separately, through the behavior of the zeta function at its pole, $\zeta(1+\rho) = 1/\rho + \gamma + o(1)$, where the γ is the same harmonic-sum constant already computed in example 1.4.4; matching the smoothly damped prime sum that ζ governs against the sharp partial sums of $\sum 1/p$ is what pins the additive constant to γ.

Definition 3.3.4: The Mertens Constant

The *Mertens constant* is

$$M = \gamma + \sum_p \left(\log\left(1-\frac{1}{p}\right) + \frac{1}{p}\right)$$

where γ is the Euler–Mascheroni constant (example 1.4.4) and the sum runs over all primes. This is the same constant as the B of proposition 1.3.9, $M = B = 0.2614\ldots$, and it equals the constant M appearing in Mertens' second theorem (theorem 3.3.3).

The series defining M converges absolutely: for $|t| \le 1/2$ the Taylor expansion $\log(1-t) = -t - t^2/2 - t^3/3 - \cdots$ gives $\log(1-t) + t = -\sum_{j\ge 2} t^j/j$, so $|\log(1-1/p) + 1/p| \le \sum_{j\ge 2} p^{-j}/j \le \frac{1}{2}\sum_{j\ge 2} p^{-j} = \frac{1}{2p(p-1)} \le 1/p^2$ for every prime $p \ge 2$, and $\sum_p p^{-2}$ converges. The claim requiring proof is that this closed-form constant is the same number M delivered by the proof of theorem 3.3.3.

***Proof*:**
Let $M_2 = \lim_{x\to\infty}\left(\sum_{p\le x} 1/p - \log\log x\right)$ denote the constant produced in theorem 3.3.3 (the limit exists because the error there is $O(1/\log x) \to 0$), and let $M = \gamma + \sum_p(\log(1-1/p)+1/p)$ denote the closed form above. The two are shown equal.

Step 1: Logarithm of the truncated Euler product. For a fixed $x \ge 2$, take logarithms of the finite product over primes $p \le x$ and expand each factor by the Taylor series for $\log(1-1/p)$:

$$\sum_{p\le x}\log\left(1-\frac{1}{p}\right) = -\sum_{p\le x}\frac{1}{p} - \sum_{p\le x}\sum_{j\ge 2}\frac{1}{j\,p^j} = -\sum_{p\le x}\frac{1}{p} - S(x)$$

where $S(x) = \sum_{p\le x}\sum_{j\ge 2}\frac{1}{j\,p^j} = -\sum_{p\le x}\left(\log(1-1/p)+1/p\right)$. As established above, the double series $\sum_p\sum_{j\ge 2}(jp^j)^{-1}$ converges absolutely, so $S(x) \to S := -\sum_p(\log(1-1/p)+1/p)$ as $x\to\infty$, with tail

$$0 \le S - S(x) = \sum_{p>x}\sum_{j\ge 2}\frac{1}{j\,p^j} \le \sum_{p>x}\frac{1}{p^2} \le \sum_{n>x}\frac{1}{n^2} = O\left(\frac{1}{x}\right)$$

Rearranging the displayed identity,

$$\sum_{p\le x}\frac{1}{p} = -\sum_{p\le x}\log\left(1-\frac{1}{p}\right) - S(x)$$

Step 2: The regularized prime sum and the entry of γ. Write $P(x) = \prod_{p\le x}(1-1/p)^{-1}$, so that $\log P(x) = -\sum_{p\le x}\log(1-1/p)$ and the identity of Step 1 becomes

$$\sum_{p\le x}\frac{1}{p} = \log P(x) - S(x) \tag{3.1}$$

The number M_2 to be evaluated is $\lim_{x\to\infty}(\sum_{p\le x}1/p - \log\log x)$, and (3.1) together with $S(x)\to S$ rewrites it as $M_2 = \lim_{x\to\infty}(\log P(x) - \log\log x) - S$. The whole task is therefore to show that the additive constant in $\log P(x) = \log\log x + (\text{const}) + o(1)$ is exactly the Euler–Mascheroni γ. This constant is not produced by any termwise manipulation of the product; it is the precise discrepancy between two regularizations of the divergent sum $\sum_p 1/p$, the sharp cutoff $\sum_{p\le x}1/p$ and the analytic cutoff $F(\rho) := \sum_p p^{-1-\rho}$ as $\rho\to 0^+$. The strategy is to evaluate the analytic cutoff in closed form, where γ appears in-chapter, and then compare the two cutoffs.

The analytic regularization is evaluated first. Taking logarithms of the Euler product $\zeta(1+\rho) = \prod_p(1-p^{-1-\rho})^{-1}$ (valid for $\rho>0$, where both sides converge absolutely) and expanding each factor exactly as in Step 1,

$$\log\zeta(1+\rho) = \sum_p\sum_{j\ge 1}\frac{1}{j\,p^{j(1+\rho)}} = F(\rho)+S(\rho), \qquad S(\rho) := \sum_p\sum_{j\ge 2}\frac{1}{j\,p^{j(1+\rho)}}$$

Since $0\le S(\rho)\le S(0) = S$ for $\rho\ge 0$ and each term increases to its $\rho = 0$ value as $\rho\to 0^+$, monotone convergence gives $S(\rho)\to S$. The constant γ now enters through $\zeta(1+\rho)$ as $\rho\to 0^+$, evaluated entirely within the convergent range Re$s > 1$, with no appeal to continuation past the line. By the closed form of proposition 1.4.6, valid for Re$s > 1$, $\zeta(s) = \frac{1}{s-1} + 1 - s\int_1^\infty \{t\}\, t^{-s-1}\, dt$, and at $s = 1+\rho$ this reads

$$\zeta(1+\rho) = \frac{1}{\rho} + 1 - (1+\rho)\int_1^\infty \frac{\{t\}}{t^{2+\rho}}\, dt = \frac{1}{\rho} + \gamma + o(1) \qquad (\rho \to 0^+)$$

the limit by dominated convergence ($\{t\}\, t^{-2-\rho} \le t^{-2}$ for $\rho \ge 0$) and the value $\int_1^\infty \{t\}\, t^{-2}\, dt = 1-\gamma$, which is exactly the evaluation of the Euler–Mascheroni constant in example 1.4.4. Hence $\log\zeta(1+\rho) = \log(1/\rho) + \log(1+\gamma\rho + o(\rho)) = \log(1/\rho) + o(1)$, and subtracting $S(\rho) \to S$,

$$F(\rho) = \sum_p \frac{1}{p^{1+\rho}} = \log\frac{1}{\rho} - S + o(1) \qquad (\rho \to 0^+) \tag{3.2}$$

This is the analytic regularization in closed form, with γ already discharged into the relation $M_2 = \gamma - S$ once the two regularizations are matched.

Step 3: Matching the two regularizations. It remains to compare the sharp cutoff with the analytic one at the calibration $\rho = 1/\log x$. The comparison is

$$\sum_{p\le x} \frac{1}{p} - F\left(\frac{1}{\log x}\right) = \gamma + o(1) \qquad (x \to \infty) \tag{3.3}$$

This single comparison is the one analytic input not delivered by the tools of this chapter, and it is isolated here as such. Its content is that replacing the sharp truncation $1_{p\le x}$ by the smooth damping $p^{-1/\log x}$ shifts the (divergent) prime sum by exactly γ in the limit. The shift is computed by Mertens' own method: the partial sums of theorem 3.3.2 are fed through partial summation against the two cutoff weights, the difference is expressed as an integral over a smooth integrand by the Euler–Maclaurin formula (theorem 1.5.6), and the limit $\rho \to 0^+$ is taken. The constant that survives is the value of the integral $\int_0^\infty \left(\frac{1}{e^t-1} - \frac{1}{te^t}\right) dt = \gamma$, equivalently $\Gamma'(1) = -\gamma$, an identity for the Gamma function rather than for ζ. The full computation, with explicit error bounds and the evaluation of this integral, is carried out in [20, §I.1.6] and in [8, Theorem 429]; the historically faithful version of Mertens' 1874 argument, with the same integral producing γ, is reproduced in detail in [22]. The mechanical tools that (3.3) rests on, the Euler product, partial summation, and Euler–Maclaurin, are all in hand; only the Gamma-function constant lies outside the chapter, and it is a fact about the Gamma function, not a disguised appeal to the analytic theory of ζ developed in chapter 2.

Granting (3.3), substitute (3.2) at $\rho = 1/\log x$, where $\log(1/\rho) = \log\log x$, to obtain

$$\sum_{p\le x}\frac{1}{p} = F\left(\frac{1}{\log x}\right) + \gamma + o(1) = \big(\log\log x - S + o(1)\big) + \gamma + o(1) = \log\log x + (\gamma - S) + o(1)$$

Hence $\lim_{x\to\infty}\left(\sum_{p\le x} 1/p - \log\log x\right) = \gamma - S$. By theorem 3.3.3 the same limit equals M_2, and limits are unique, so

$$M_2 = \gamma - S = \gamma + \sum_p \left(\log\left(1 - \frac{1}{p}\right) + \frac{1}{p}\right) = M$$

the middle equality by $S = -\sum_p(\log(1-1/p)+1/p)$ from Step 1. This is the closed form claimed for the Mertens constant. The equality $M = B$ with the constant of proposition 1.3.9 is immediate, since B was defined there by the identical expression $\gamma + \sum_p(\log(1-1/p)+1/p)$, completing the proof.

The closed form makes the Mertens constant computable and ties together three appearances of the same number: the second-order term in $\sum_{p\le x} 1/p$, the second-order term in the average of ω from proposition 1.3.9, and (through theorem 3.3.5 below) the constant governing the product $\prod(1-1/p)$. The expression $M = \gamma + \sum_p(\log(1-1/p)+1/p)$ exhibits M as the Euler–Mascheroni constant corrected by a rapidly convergent prime sum: each correction term $\log(1-1/p)+1/p$ is $-1/(2p^2)+O(p^{-3})$, so the sum is dominated by its first few terms and $M = 0.2614972128\ldots$ is computed to high precision from a short truncation. The dependence on γ is not incidental. The constant γ entered through the value $\zeta(1+\rho) = 1/\rho + \gamma + o(1)$ of the zeta function near its pole, which is itself the harmonic-sum constant of example 1.4.4 read off from the analytic continuation in proposition 1.4.6; the prime sum $\sum 1/p$ inherits γ as the precise gap between its sharp partial sums and the smoothly damped sum $\sum_p p^{-1-\rho}$, the two regularizations of the same divergent series whose difference (3.3) pins to γ. The convergent correction $-S$ is a separate, smaller adjustment: it measures only the gap between $\sum 1/p$ and $\log\prod(1-1/p)^{-1}$ caused by the higher prime powers, not the appearance of γ itself.

The product form is the most frequently cited of the three theorems, appearing throughout sieve theory and the theory of arithmetic functions as the density of integers free of small prime factors. Taking the logarithm of the product converts it into the sum $\sum_{p\le x}\log(1-1/p)$, which the second theorem and the closed form of M evaluate; the constant $e^{-\gamma}$ emerges because the convergent prime correction in M cancels precisely against the convergent correction in $\sum\log(1-1/p)$, leaving only $-\gamma$.

Theorem 3.3.5: Mertens' Third Theorem

As $x \to \infty$,

$$\prod_{p\le x}\left(1-\frac{1}{p}\right) \sim \frac{e^{-\gamma}}{\log x}$$

where γ is the Euler–Mascheroni constant. Equivalently,

$$\prod_{p\le x}\left(1-\frac{1}{p}\right)^{-1} = e^{\gamma}\,\log x\,\bigl(1+o(1)\bigr)$$

***Proof*:**
Step 1: Taking logarithms. Let $L(x) = \log\prod_{p\le x}(1-1/p) = \sum_{p\le x}\log(1-1/p)$. The claim $\prod_{p\le x}(1-1/p) \sim e^{-\gamma}/\log x$ is equivalent, after taking logarithms, to

$$L(x) = -\gamma - \log\log x + o(1)$$

since $\log(e^{-\gamma}/\log x) = -\gamma - \log\log x$ and the asymptotic $\sim$ between positive quantities is equivalent to the difference of their logarithms tending to 0.

Step 2: Splitting off the convergent correction. For each prime p write $\log(1-1/p) = -1/p +$

$\big(\log(1-1/p)+1/p\big)$, and sum over $p \le x$:

$$L(x) = -\sum_{p\le x}\frac{1}{p} + \sum_{p\le x}\left(\log\left(1-\frac{1}{p}\right)+\frac{1}{p}\right)$$

The second sum converges as $x\to\infty$: by the bound established after definition 3.3.4, $|\log(1 - 1/p) + 1/p| \le 1/p^2$, so the series $\sum_p(\log(1-1/p)+1/p)$ converges absolutely, with value $M-\gamma$ by the definition of the Mertens constant (definition 3.3.4). Writing the partial sum as the full sum minus a tail,

$$\sum_{p\le x}\left(\log\left(1-\frac{1}{p}\right)+\frac{1}{p}\right) = (M-\gamma) - \sum_{p>x}\left(\log\left(1-\frac{1}{p}\right)+\frac{1}{p}\right) = (M-\gamma)+o(1)$$

since the tail of a convergent series tends to 0 (explicitly, $|\sum_{p>x}(\cdots)| \le \sum_{n>x} n^{-2} = O(1/x)$).

Step 3: Applying the second theorem. By Mertens' second theorem (theorem 3.3.3) with the closed form M of definition 3.3.4,

$$\sum_{p\le x}\frac{1}{p} = \log\log x + M + O\left(\frac{1}{\log x}\right)$$

Substituting both evaluations into the expression for $L(x)$ from Step 2,

$$L(x) = -\left(\log\log x + M + O\left(\tfrac{1}{\log x}\right)\right) + \big((M-\gamma)+o(1)\big)$$

The constant M cancels: $-M+(M-\gamma) = -\gamma$. The error terms $O(1/\log x)$ and $o(1)$ combine into $o(1)$. Hence

$$L(x) = -\log\log x - \gamma + o(1)$$

which is exactly the equivalent form from Step 1.

Step 4: Exponentiating. Exponentiating $L(x) = -\gamma - \log\log x + o(1)$,

$$\prod_{p\le x}\left(1-\frac{1}{p}\right) = e^{L(x)} = e^{-\gamma-\log\log x+o(1)} = \frac{e^{-\gamma}}{\log x}\cdot e^{o(1)} = \frac{e^{-\gamma}}{\log x}\big(1+o(1)\big)$$

using $e^{o(1)} = 1+o(1)$, which is the first stated form. Taking reciprocals gives $\prod_{p\le x}(1-1/p)^{-1} = e^{\gamma}\log x(1+o(1))$, the second form, completing the proof.

The cancellation in Step 3 is the mechanism that produces $e^{-\gamma}$, and it shows precisely why the Mertens constant had to be defined with the γ term: the constant M in $\sum 1/p$ and the convergent sum $\sum(\log(1-1/p)+1/p) = M-\gamma$ in $\log\prod(1-1/p)$ share the prime-sum part, which cancels, while the γ that distinguishes them survives as the exponent. The third theorem quantifies a competition between two effects in the product $\prod_{p\le x}(1-1/p)$: each factor $1-1/p$ is slightly less than 1, so the product decreases, but the factors approach 1 as p grows, so the decrease slows. The outcome $\sim e^{-\gamma}/\log x$ says the product tends to 0, but only as slowly as $1/\log x$, with the precise rate fixed by $e^{-\gamma} \approx 0.5615$. This product is the proportion of integers surviving the sieve that removes every multiple of every prime up to x; its value $e^{-\gamma}/\log x$, rather than the naive $1/\log x$ one might guess from the heuristic that "a $1/\log x$ fraction of integers near x are prime," is the source of the constant $2e^{-\gamma}$ in the Mertens correction to sieve estimates and the reason the twin-prime constant and similar products carry factors of $e^{-\gamma}$.

A numerical check confirms that the two nontrivial constants, M in the second theorem and $e^{-\gamma}$ in the third, are the ones the data demand, and exhibits the slow convergence that the error terms predict.

Example 3.3.6: Numerical Check of Mertens' Theorems

The constants entering the second and third theorems are

$$M = 0.26150\ldots, \qquad \gamma = 0.57722\ldots, \qquad e^{-\gamma} = 0.56146\ldots$$

For the second theorem, the table compares the partial sum $\sum_{p\le x} 1/p$ against its predicted value $\log\log x + M$, recording the discrepancy:

x	$\sum_{p\le x} 1/p$	$\log\log x + M$	difference
10^2	1.80282	1.78868	+0.01414
10^4	2.48306	2.48182	+0.00124
10^6	2.88733	2.88729	+0.00004

The difference $\sum_{p\le x} 1/p - (\log\log x + M)$ shrinks steadily, from about 1.4×10^{-2} at $x = 10^2$ to 4×10^{-5} at $x = 10^6$, consistent with the $O(1/\log x)$ error term of theorem 3.3.3: the ratio of successive differences tracks the ratio of successive values of $1/\log x$, the predicted decay. For the third theorem, the table compares $\prod_{p\le x}(1-1/p)\cdot\log x$ against the predicted limit $e^{-\gamma} = 0.56146$:

x	$\prod_{p\le x}(1-1/p)$	$\left(\prod_{p\le x}(1-1/p)\right)\log x$	$e^{-\gamma}$
10^2	0.12032	0.55408	0.56146
10^4	0.06088	0.56077	0.56146
10^6	0.04064	0.56144	0.56146

The product $\left(\prod_{p\le x}(1-1/p)\right)\log x$ approaches $e^{-\gamma}$ from below, sitting within two parts in 10^5 of $e^{-\gamma}$ at $x = 10^6$. Both tables confirm the predicted constants, M in the second theorem and $e^{-\gamma}$ in the third, and both exhibit the slow convergence characteristic of error terms that decay only as a power of $1/\log x$ rather than a power of $1/x$.

3.4 Bertrand's Postulate

Chebyshev's estimates of section 3.2 pin the global count of primes up to x to the correct order $x/\log x$, but a global count says nothing about where the primes sit. A statement like $\pi(2n) - \pi(n) \approx n/\log n$ would assert, on average, many primes between n and $2n$, yet an average leaves open the possibility of long prime-free stretches compensated by dense clusters elsewhere. Bertrand's postulate is the first genuine *localization* result: for every integer $n \ge 1$ there is at least one prime in the interval $(n, 2n]$, with no exceptions and no averaging. Conjectured by Bertrand in 1845 on numerical evidence up to several million and proved by Chebyshev in 1852, it guarantees a prime in an interval whose length is comparable to its left endpoint, a far shorter window than the global count alone delivers.

The proof is a refinement of the binomial method already used for Chebyshev's bounds. The central binomial coefficient $\binom{2n}{n}$ is large, of size roughly 4^n, and section 3.2 controlled its prime factorization through Legendre's formula. Sharpening that control shows that if no prime fell in $(n, 2n]$, then $\binom{2n}{n}$ would be forced to assemble its entire size out of small primes alone, each contributing only a bounded factor; the product of those small contributions is too small to account for 4^n once n is large. The contradiction proves the postulate for all large n, and the finitely many remaining cases are settled by exhibiting an explicit chain of primes. The argument here is the one found by Erdős in 1932, which replaced Chebyshev's analytic estimates of the Gamma function by the elementary bounds of lemma 3.4.1.

Lemma 3.4.1: Bounds on the Central Binomial Coefficient

For every integer $n \geq 1$,

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n$$

***Proof*:**
Both bounds compare $\binom{2n}{n}$ with the full binomial sum $\sum_{j=0}^{2n}\binom{2n}{j} = (1+1)^{2n} = 4^n$.

Step 1: The upper bound. The coefficient $\binom{2n}{n}$ is one of the $2n+1$ nonnegative terms $\binom{2n}{0}, \binom{2n}{1}, \ldots, \binom{2n}{2n}$ whose sum is 4^n. A single term of a sum of nonnegative reals cannot exceed the whole sum, so

$$\binom{2n}{n} \leq \sum_{j=0}^{2n}\binom{2n}{j} = 4^n$$

which is the upper bound.

Step 2: The lower bound. The same $2n+1$ terms have $\binom{2n}{n}$ as their largest. Indeed the ratio of consecutive terms is

$$\frac{\binom{2n}{j+1}}{\binom{2n}{j}} = \frac{2n-j}{j+1}$$

which exceeds 1 exactly when $2n - j > j + 1$, that is, $j < (2n-1)/2 = n - \frac{1}{2}$, and is less than 1 when $j > n - \frac{1}{2}$. Thus the terms strictly increase up to $j = n$ and strictly decrease after it, so $\binom{2n}{n}$ is the unique maximum. Being the largest of $2n+1$ terms, it is at least their average:

$$\binom{2n}{n} \geq \frac{1}{2n+1}\sum_{j=0}^{2n}\binom{2n}{j} = \frac{4^n}{2n+1}$$

which is the lower bound, completing the proof.

The two bounds bracket $\binom{2n}{n}$ between $4^n/(2n+1)$ and 4^n, a window narrow on the logarithmic scale: $\log\binom{2n}{n} = n\log 4 + O(\log n)$, so to leading order $\binom{2n}{n}$ behaves like $4^n = 2^{2n}$. The lower bound is the operative one for Bertrand's postulate, since the argument proceeds by showing that a hypothetical absence of primes in $(n, 2n]$ makes $\binom{2n}{n}$ too small, and "too small" is measured against $4^n/(2n+1)$. The factor $2n+1$ in the denominator is wasteful (Stirling's formula theorem 1.5.7 gives the sharper

$\binom{2n}{n} \sim 4^n/\sqrt{\pi n}$), but the crude bound suffices and keeps the proof elementary. The next lemma supplies the matching upper control on $\binom{2n}{n}$, not through its size but through its prime factorization.

The mechanism of Bertrand's postulate is a restriction on *which* primes can divide $\binom{2n}{n}$, and to what power. Three facts are needed. The first removes a whole band of primes just below n from the factorization; the second caps the exponent of every prime past $\sqrt{2n}$ at one; the third, already proved in section 3.2, caps every prime power at $2n$. Together they bound $\binom{2n}{n}$ from above by a product whose factors are small primes raised to controlled powers.

Lemma 3.4.2: Primes in the Central Binomial Coefficient

Let $n \geq 3$ be an integer and let p be a prime.

(i) If $\frac{2n}{3} < p \leq n$, then $p \nmid \binom{2n}{n}$.

(ii) If $\sqrt{2n} < p \leq 2n$, then $v_p\binom{2n}{n} \leq 1$, so p divides $\binom{2n}{n}$ to at most the first power.

(iii) For every prime p, the prime power $p^{v_p\binom{2n}{n}}$ satisfies $p^{v_p\binom{2n}{n}} \leq 2n$.

***Proof*:**
Throughout, the valuation formula of lemma 3.2.1 is used in the form

$$v_p\binom{2n}{n} = \sum_{k\geq 1}\left(\left\lfloor \frac{2n}{p^k}\right\rfloor - 2\left\lfloor \frac{n}{p^k}\right\rfloor\right)$$

in which every summand is 0 or 1 and the summand vanishes once $p^k > 2n$.

(i) *The band* $(2n/3, n]$. Assume $\frac{2n}{3} < p \leq n$. The hypotheses on p and $n \geq 3$ force $p \geq 3$: from $p > 2n/3 \geq 2$ one gets $p \geq 3$. The aim is to show every summand in the valuation formula vanishes. Consider the terms by exponent k.

For $k = 1$: from $p \leq n$ one has $n/p \geq 1$, and from $p > 2n/3$ one has $n/p < 3/2$, so $1 \leq n/p < 3/2$ and $\lfloor n/p \rfloor = 1$. Likewise $2 \leq 2n/p < 3$, so $\lfloor 2n/p \rfloor = 2$. The $k = 1$ summand is $\lfloor 2n/p\rfloor - 2\lfloor n/p\rfloor = 2 - 2 \cdot 1 = 0$.

For $k \geq 2$: since $p > 2n/3$ and $p \geq 3$,

$$p^2 > \frac{2n}{3} \cdot p \;\geq\; \frac{2n}{3} \cdot 3 = 2n$$

so $p^k \geq p^2 > 2n$ for all $k \geq 2$, and every such summand vanishes by lemma 3.2.1.

All summands being zero, $v_p\binom{2n}{n} = 0$, that is, $p \nmid \binom{2n}{n}$.

(ii) *Primes past* $\sqrt{2n}$. Assume $\sqrt{2n} < p \leq 2n$. Then $p^2 > 2n$, so for every $k \geq 2$ the term $\lfloor 2n/p^k \rfloor = 0$ and the corresponding summand vanishes. Only the $k = 1$ summand can survive, and it is 0 or 1 by lemma 3.2.1. Hence $v_p\binom{2n}{n} \leq 1$.

(iii) *The cap on prime powers.* This is exactly the bound $p^{v_p\binom{2n}{n}} \leq 2n$ proved in lemma 3.2.1, valid for every prime p and every $n \geq 1$, recorded here for use alongside (i) and (ii).

The three statements together control the factorization as required.

The three statements act on disjoint parts of the prime range. Statement (iii) is universal but coarse: it bounds each prime's contribution by $2n$ regardless of the prime. Statement (ii) improves this for primes past $\sqrt{2n}$, where the exponent drops to at most one, so the contribution is just p rather than a power. Statement (i) is the new ingredient, absent from the Chebyshev bounds of section 3.2: it asserts that the primes in the band $(2n/3, n]$ are entirely missing from $\binom{2n}{n}$. The reason is arithmetic cancellation. Such a prime p appears once in the numerator $(2n)!$ as the factor p and once more as $2p$ (since $2p \le 2n$ when $p \le n$), contributing two to $v_p((2n)!)$; in the denominator $n!\,n!$ it appears once in each factorial as p (since $p \le n$), contributing $2 \cdot 1 = 2$ to $2v_p(n!)$. The two cancel exactly. The constraint $p > 2n/3$ ensures that $3p > 2n$, so $3p$ is not among the factors of $(2n)!$ and no third copy disturbs the balance. This band is precisely the range just below n where one might have hoped to find the divisors of $\binom{2n}{n}$; their absence is what forces the coefficient to draw on smaller primes and ultimately produces the contradiction.

With the size of $\binom{2n}{n}$ bounded below and its factorization bounded above, the postulate follows by comparing the two. The comparison yields a contradiction for all sufficiently large n; the threshold is explicit, and the finitely many smaller values are checked by an explicit list of primes.

Theorem 3.4.3: Bertrand's Postulate

For every integer $n \ge 1$ there exists a prime p with $n < p \le 2n$.

***Proof*:**
The argument has two parts: a proof for all $n \ge 512$ by contradiction, and a direct verification of $1 \le n \le 511$ by an explicit chain of primes.

Step 1: Assume no prime lies in $(n, 2n]$ for some $n \ge 512$. Under this assumption the primes dividing $\binom{2n}{n}$ are confined to small values. By lemma 3.2.1 every prime dividing $\binom{2n}{n}$ is at most $2n$. The assumption removes the primes in $(n, 2n]$, and part (i) of lemma 3.4.2 removes the primes in $(2n/3, n]$ (here $n \ge 512 \ge 3$). What remains is

$$\binom{2n}{n} = \prod_{p \le 2n/3} p^{v_p\binom{2n}{n}}$$

the product over primes $p \le 2n/3$ only.

Step 2: Bound the product from above. Split the surviving primes at $\sqrt{2n}$.

For primes $p \le \sqrt{2n}$, part (iii) of lemma 3.4.2 gives $p^{v_p\binom{2n}{n}} \le 2n$. The number of primes $p \le \sqrt{2n}$ is at most $\sqrt{2n}$ (there are at most $\sqrt{2n}$ positive integers up to $\sqrt{2n}$), so their combined contribution is

$$\prod_{p \le \sqrt{2n}} p^{v_p\binom{2n}{n}} \;\le\; (2n)^{\sqrt{2n}}$$

For primes $\sqrt{2n} < p \le 2n/3$, part (ii) of lemma 3.4.2 gives $v_p\binom{2n}{n} \le 1$, so each contributes at most p itself, and their product is at most the product of all primes up to $2n/3$. The primorial bound proposition 3.2.2 controls this:

$$\prod_{\sqrt{2n} < p \le 2n/3} p^{v_p\binom{2n}{n}} \;\le\; \prod_{p \le 2n/3} p \;\le\; 4^{2n/3}$$

Multiplying the two ranges,

$$\binom{2n}{n} \;\le\; (2n)^{\sqrt{2n}} \cdot 4^{\,2n/3}$$

is the upper bound forced by the assumption.

Step 3: Contradiction with the lower bound. The lower bound of lemma 3.4.1 gives $\binom{2n}{n} \ge 4^n/(2n+1)$. Combined with Step 2,

$$\frac{4^{\,n}}{2n+1} \;\le\; (2n)^{\sqrt{2n}} \cdot 4^{\,2n/3}$$

Dividing by $4^{2n/3}$ and noting $4^n/4^{2n/3} = 4^{\,n/3}$,

$$4^{\,n/3} \;\le\; (2n+1)\,(2n)^{\sqrt{2n}}$$

Take logarithms. With $2n+1 \le (2n)^2$ for $n \ge 1$ (since $2n+1 \le 4n^2$ once $n \ge 1$), the right side satisfies $\log\big((2n+1)(2n)^{\sqrt{2n}}\big) \le (\sqrt{2n}+2)\log(2n)$, so

$$\frac{n}{3}\log 4 \;\le\; (\sqrt{2n}+2)\log(2n)$$

The left side grows linearly in n while the right side grows like $\sqrt{n}\,\log n$, so the inequality fails for all large n; it remains to locate the threshold. Substitute $m = \sqrt{2n}$, so $n = m^2/2$ and the inequality reads

$$\frac{m^2}{6}\log 4 \;\le\; (m+2)\cdot 2\log m$$

that is, $m\log 4 \le 12\,(1+2/m)\log m$. For $m \ge 32$ (which corresponds to $n \ge 512$) the factor $1+2/m \le 1+1/16 = 17/16$, so the right side is at most $12 \cdot \frac{17}{16}\log m = \frac{51}{4}\log m$, and the inequality would force

$$m\log 4 \;\le\; \tfrac{51}{4}\log m, \qquad \text{i.e.} \qquad \frac{m}{\log m} \;\le\; \frac{51}{4\log 4} = \frac{51}{4\cdot 1.3862\ldots} = 9.197\ldots$$

But $m/\log m$ is increasing for $m \ge 3$, and at $m = 32$ it already equals $32/\log 32 = 32/3.4657\cdots = 9.233\cdots > 9.197\ldots$. Hence the inequality fails for $m = 32$, and being increasing, $m/\log m$ exceeds the bound for all $m \ge 32$. This contradicts the displayed inequality for every $n \ge 512$. Therefore the assumption is untenable: a prime exists in $(n, 2n]$ for every $n \ge 512$.

Step 4: The range $1 \le n \le 511$. Consider the increasing chain of primes

$$2,\; 3,\; 5,\; 7,\; 13,\; 23,\; 43,\; 83,\; 163,\; 317,\; 631$$

Each prime in the list is less than twice its predecessor: $3<4$, $5<6$, $7<10$, $13<14$, $23<26$, $43<46$, $83<86$, $163<166$, $317<326$, $631<634$. Write the chain as $q_0 = 2 < q_1 = 3 < \cdots < q_{10} = 631$, so that $q_i < 2q_{i-1}$ for each $i \ge 1$. Given an integer n with $1 \le n \le 511$, choose the smallest chain prime q_i with $q_i > n$; such an index exists because $q_{10} = 631 > 511 \ge n$. Minimality gives $q_{i-1} \le n$ when $i \ge 1$, and when $n = 1$ the prime $q_0 = 2$ already satisfies $1 < 2 \le 2 = 2n$. For $i \ge 1$,

$$q_i < 2q_{i-1} \le 2n$$

so $n < q_i \le 2n$, exhibiting a prime in $(n, 2n]$. This covers every n in $[1, 511]$.

Steps 1 through 3 establish the postulate for $n \ge 512$ and Step 4 for $1 \le n \le 511$, so a prime lies in $(n, 2n]$ for every integer $n \ge 1$, as we sought to show.

Bertrand's postulate sharpens the picture of the primes from a global count to a guarantee on every dyadic scale. Where Chebyshev's bounds (theorem 3.2.3) assert that the primes up to x number a constant multiple of $x/\log x$, leaving open how they bunch, the postulate forbids any gap from doubling: between n and $2n$ there is always a prime, so the ratio of consecutive primes is bounded. The two results draw on the same object, the central binomial coefficient, and differ only in how finely its factorization is dissected. Chebyshev compared the size $4^n/(2n+1) \leq \binom{2n}{n} \leq 4^n$ against the universal cap $p^{v_p} \leq 2n$ on prime powers; Bertrand adds the observation that the primes in the band $(2n/3, n]$ contribute nothing, which concentrates the entire size of $\binom{2n}{n}$ onto primes below $2n/3$ and onto the at-most-once primes between $\sqrt{2n}$ and $2n/3$, controlled by the primorial bound $\prod_{p \leq 2n/3} p \leq 4^{2n/3}$ of proposition 3.2.2. The exponent $2n/3$ against the size 4^n leaves a deficit of $4^{n/3}$ that only primes in $(n, 2n]$ could supply, and that is exactly the interval whose nonemptiness the theorem asserts. The threshold $n \geq 512$ is not optimal; the inequality of Step 3 already fails for $n \geq 468$, and tightening the prime-power bookkeeping lowers it further, but a clean power of two suffices and the explicit chain disposes of the remainder. In the language of Chebyshev's functions (definition 3.1.2), the postulate is the statement that $\vartheta(2n) - \vartheta(n) > 0$ for all $n \geq 1$, a positivity that the order-of-magnitude estimate $\vartheta(x) \asymp x$ suggests but does not by itself prove.

The postulate yields immediate bounds on the growth of the prime sequence, both on the gap between consecutive primes and on the size of the n-th prime.

Corollary 3.4.4: Consequences of Bertrand's Postulate

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \cdots$ enumerate the primes in increasing order. Then for every $n \geq 1$:

(i) the consecutive primes satisfy $p_{n+1} < 2p_n$;

(ii) the n-th prime satisfies $p_n \leq 2^n$, with equality only at $n = 1$.

***Proof*:**
The two statements are proved in turn, the second by induction using the first.

(i) Apply theorem 3.4.3 with n replaced by p_n: there is a prime q with $p_n < q \leq 2p_n$. Since $q > p_n$, the prime q is one of $p_{n+1}, p_{n+2}, \ldots$, so $p_{n+1} \leq q$ as p_{n+1} is the least prime exceeding p_n. The strict inequality $q \leq 2p_n$ combined with $2p_n$ being even and hence not prime (as $p_n \geq 2$ gives $2p_n \geq 4 > 2$) yields $q < 2p_n$, whence $p_{n+1} \leq q < 2p_n$.

(ii) Induct on n. The base case $n = 1$ is the equality $p_1 = 2 = 2^1$, which gives $p_1 \leq 2^1$. Assume $p_n \leq 2^n$ for some $n \geq 1$. By part (i), $p_{n+1} < 2p_n \leq 2 \cdot 2^n = 2^{n+1}$, so $p_{n+1} \leq 2^{n+1}$, which is the claim for $n + 1$. By induction $p_n \leq 2^n$ for all $n \geq 1$. Equality occurs only at $n = 1$: for $n \geq 2$ the inductive step delivers the strict $p_{n+1} < 2^{n+1}$, so $p_n < 2^n$ for every $n \geq 2$.

Both bounds follow, completing the proof.

The bound $p_{n+1} < 2p_n$ records that the primes never thin out faster than geometrically: each prime is followed by another before the value doubles. The bound $p_n \leq 2^n$ (strict for $n \geq 2$) is a crude but unconditional handle on the size of the n-th prime, obtained with no analytic input. It is far from the truth, since the Prime Number Theorem (chapter 5, foreshadowed) gives $p_n \sim n \log n$, a function of much slower growth than 2^n; the gap measures how much is lost by iterating a doubling bound rather than tracking the actual density of primes. The value of $p_n \leq 2^n$ is its independence from any analytic input: it follows from Bertrand's postulate alone, which follows in turn from the size and factorization of a single binomial coefficient.

Part II

Zeta Functions, L-Functions, and the Distribution of Primes

This Part is the core of the subject. The elementary methods of Part I fixed the order of magnitude of $\pi(x)$ and evaluated the prime sums to leading order, but they stopped short of the Prime Number Theorem itself, the exact asymptotic $\pi(x) \sim x/\log x$: Chebyshev's counting arguments trap the ratio $\pi(x)\log x/x$ between two constants without forcing it to the limit 1. Reaching that limit calls for a new idea. The idea by which the theorem was first proved, and the one this Part develops, is analytic: it studies the zeta function as a function of a complex variable and reads the asymptotic off the behavior of $\zeta(s)$ near the line $\mathrm{Re}(s) = 1$, above all off the fact that ζ has no zero on that line, the route of Hadamard and de la Vallée Poussin in 1896. The analytic method is not the only route to the theorem, since an *elementary* proof avoiding complex analysis was found half a century later by Selberg and Erdős (chapter 5); but it is the most transparent, it is the one that yields sharp error terms through the zeros, and it is the one that generalizes to the L-functions, and developing it is the business of this Part. The seven chapters carry out two intertwined generalizations of a single construction.

The first thread runs from ζ to the Dirichlet L-functions. Chapter 4 continues $\zeta(s)$ past its half-plane of convergence and proves it has no zeros on the line $\mathrm{Re}(s) = 1$; chapter 5 converts that non-vanishing into the Prime Number Theorem; chapter 6 completes the analytic picture with the functional equation relating $\zeta(s)$ to $\zeta(1-s)$ and the critical strip in which the Riemann hypothesis lives; and chapter 7 twists the whole construction by a Dirichlet character, producing the L-functions whose non-vanishing at $s = 1$ yields Dirichlet's theorem on primes in arithmetic progressions. The second thread, in chapters 8 to 10, holds the analysis fixed and changes the arithmetic: it replaces $\mathbb{Q}$ by a number field, producing the Dedekind zeta function and its class number formula, then the Hecke L-functions attached to ray class characters, then the Artin L-functions attached to Galois representations, with the Chebotarev density theorem as the culmination. In every case the same three analytic features, the Euler product, the behavior at $s = 1$, and the non-vanishing on $\mathrm{Re}(s) = 1$, carry the same three pieces of arithmetic.

The prerequisites escalate as the second thread proceeds, and each escalation is flagged where it begins: chapter 8 adds algebraic number theory ([2]) as a co-requisite, and chapters 9 and 10 add local field theory ([4]) and representation theory ([3]). The Part opens on the most familiar ground, the zeta function itself, now studied not as a series but as a holomorphic function.

4

The Riemann Zeta Function

The two preceding chapters approached the zeta function from opposite directions. chapter 2 built $\zeta(s) = \sum_{n\geq 1} n^{-s}$ as a Dirichlet series, established its Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ as the analytic form of unique factorization (corollary 2.3.2), and produced the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t-1)\,dt$ (theorem 2.4.5), all on the half-plane $\mathrm{Re}(s) > 1$ where the defining series converges absolutely. chapter 3 took the elementary road, fixing the order of magnitude of $\pi(x)$ and evaluating the sums $\sum_{p\leq x}(\log p)/p$ and $\sum_{p\leq x} 1/p$ to leading order by counting prime powers, without any appeal to complex analysis. What both chapters left untouched is the behavior of $\zeta(s)$ as a function of a *complex* variable outside the half-plane of convergence. The series $\sum n^{-s}$ diverges at $s = 1$ and gives no value there, yet $\zeta(s)$ extends to a meromorphic function on a larger region, and it is the analytic structure of that extension, not the series, that controls the primes.

This chapter studies $\zeta(s)$ as a holomorphic function. The central facts are three. First, $\zeta(s)$ continues meromorphically to the half-plane $\mathrm{Re}(s) > 0$, holomorphic everywhere except for a single simple pole at $s = 1$ with residue 1; the continuation is already implicit in the partial-sum formula of chapter 1 and requires no new machinery. Second, the logarithmic derivative $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ inherits a simple pole at $s = 1$ with residue 1, and this pole is the analytic source of the leading term $\psi(x) \sim x$ of the Prime Number Theorem. Third, $\zeta(s)$ does not vanish on the line $\mathrm{Re}(s) = 1$, a fact that lies just outside the reach of the elementary Euler-product argument of chapter 2 and demands a new idea. These three results are the analytic foundation on which the proof of the Prime Number Theorem rests in chapter 5; the deeper continuation to all of $\mathbb{C}$, together with the functional equation relating $\zeta(s)$ to $\zeta(1-s)$, is taken up in chapter 6.

4.1 Definition and the Euler Product

The starting point is the zeta function on its native half-plane $\mathrm{Re}(s) > 1$, where the series, the product, and the integral of chapter 2 all converge and agree. This section consolidates what is already known there, recording the elementary analytic properties of $\zeta(s)$ as a holomorphic function of s, restating the Euler product and the non-vanishing it forces, and computing a few special values. The Euler product and the integral representation are not re-derived; they are imported from chapter 2 and cited. The one formula carried forward in full is the partial-sum identity of chapter 1, which already contains the continuation to $\mathrm{Re}(s) > 0$ and which the next section turns into the continuation theorem.

The defining series and the meaning of $\zeta(s)$ on $\mathrm{Re}(s) > 1$ were fixed in chapter 2; recall that $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Dirichlet series of the constant function $\mathbf{1}$ (definition 2.1.1), with abscissa of absolute convergence $\sigma_a = 1$. The first task is to assemble the elementary consequences of this description into a single statement about ζ as a function.

Proposition 4.1.1: Analytic Properties of ζ on $\mathrm{Re}(s) > 1$

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ has the following properties on the half-plane $\mathrm{Re}(s) > 1$.

1. It is holomorphic, and its derivative is obtained by term-by-term differentiation:

$$\zeta'(s) = -\sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

2. For every s with $\mathrm{Re}(s) = \sigma > 1$,

$$|\zeta(s) - 1| \le \sum_{n=2}^{\infty} \frac{1}{n^\sigma}$$

   and the right-hand side tends to 0 as $\sigma \to +\infty$; hence $\zeta(s) \to 1$ as $\mathrm{Re}(s) \to +\infty$.

3. On the real axis, $\zeta(\sigma)$ is real and strictly decreasing for $\sigma > 1$, with $\zeta(\sigma) > 1$ throughout, $\zeta(\sigma) \to 1$ as $\sigma \to +\infty$, and $\zeta(\sigma) \to +\infty$ as $\sigma \to 1^+$.

***Proof*:**
The three parts are proved in turn.

1. The function ζ is the Dirichlet series $D(\mathbf{1}, s)$, whose abscissa of convergence is $\sigma_c = 1$ (example 2.1.9). By theorem 2.1.7, a Dirichlet series is holomorphic on the half-plane to the right of its abscissa of convergence, and there it may be differentiated term by term, with $D'(f, s) = \sum_n f(n)(-\log n)n^{-s}$. Taking $f = \mathbf{1}$ gives holomorphy on $\mathrm{Re}(s) > 1$ and

$$\zeta'(s) = \sum_{n=1}^{\infty} \frac{-\log n}{n^s} = -\sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

   the $n = 1$ term vanishing since $\log 1 = 0$.

2. Separating the $n = 1$ term, $\zeta(s) - 1 = \sum_{n \ge 2} n^{-s}$. Since $|n^{-s}| = n^{-\sigma}$ for $\mathrm{Re}(s) = \sigma$, the triangle inequality gives

$$|\zeta(s) - 1| = \left| \sum_{n=2}^{\infty} \frac{1}{n^s} \right| \le \sum_{n=2}^{\infty} \frac{1}{n^\sigma}$$

which depends only on σ. The tail $\sum_{n\geq 2} n^{-\sigma}$ is bounded by the integral comparison $\sum_{n\geq 2} n^{-\sigma} \leq \int_1^\infty t^{-\sigma}\,dt = 1/(\sigma-1)$, valid for $\sigma > 1$, and $1/(\sigma-1) \to 0$ as $\sigma \to +\infty$. Hence $|\zeta(s)-1| \to 0$, that is, $\zeta(s) \to 1$, uniformly in $\mathrm{Im}(s)$ as $\mathrm{Re}(s) \to +\infty$.

3. For real $s = \sigma > 1$, every term $n^{-\sigma}$ is a positive real number, so $\zeta(\sigma) = \sum_n n^{-\sigma}$ is real and exceeds its first term 1, giving $\zeta(\sigma) > 1$. For $1 < \sigma_1 < \sigma_2$ and each $n \geq 2$, the function $\sigma \mapsto n^{-\sigma} = e^{-\sigma \log n}$ is strictly decreasing because $\log n > 0$, so $n^{-\sigma_2} < n^{-\sigma_1}$; summing over $n \geq 2$ (the $n = 1$ term being constant) yields $\zeta(\sigma_2) < \zeta(\sigma_1)$, so ζ is strictly decreasing on $(1,\infty)$. The limit $\zeta(\sigma) \to 1$ as $\sigma \to +\infty$ is the real-axis case of part (2). For the limit at 1^+, fix any N; for $1 < \sigma$ the partial sum gives $\zeta(\sigma) \geq \sum_{n=1}^N n^{-\sigma}$, and letting $\sigma \to 1^+$ this finite sum tends to $\sum_{n=1}^N 1/n$, so $\liminf_{\sigma\to 1^+} \zeta(\sigma) \geq \sum_{n=1}^N 1/n$. Since N is arbitrary and the harmonic series $\sum_n 1/n$ diverges (example 1.4.4), the right-hand side is unbounded, forcing $\zeta(\sigma) \to +\infty$ as $\sigma \to 1^+$, as we sought to show.

These properties already record the qualitative shape of ζ along the real axis: a function descending strictly from $+\infty$ at the right edge of the pole to the horizontal asymptote $\zeta = 1$ at infinity. The blow-up at $\sigma = 1^+$ is the real-axis trace of the simple pole that the next section locates precisely, and it is inseparable from the divergence of the harmonic series, the most elementary statement about the density of the integers. Part (2) is the analytic counterpart of the fact that, far to the right, the $n = 1$ term dominates the series and every higher term is negligible; this is the same decay-at-different-rates phenomenon that made the coefficients of a Dirichlet series recoverable one at a time in corollary 2.1.8, and it will reappear as the normalization $\log \zeta(s) \to 0$ that pins the branch of the logarithm used in the non-vanishing argument of section 4.4. The holomorphy in part (1) is what entitles one to speak of the zeros and poles of ζ at all, and term-by-term differentiation makes $\zeta'(s)$ a Dirichlet series in its own right, the series of $-\log$, whose ratio with ζ produces the von Mangoldt series that carries the arithmetic.

The product form of ζ, established in chapter 2, is the single feature that ties the function to the primes; it is recalled here together with the non-vanishing it forces on the half-plane of absolute convergence.

Corollary 4.1.2: The Euler Product and Non-Vanishing on $\mathrm{Re}(s) > 1$

For $\mathrm{Re}(s) > 1$, the zeta function factors as an absolutely convergent product over the primes,

$$\zeta(s) = \prod_p \frac{1}{1-p^{-s}}$$

and $\zeta(s) \neq 0$ on this half-plane.

***Proof*:**
The factorization is corollary 2.3.2, the Euler product for ζ, obtained by applying the Euler product for multiplicative functions to the totally multiplicative function $\mathbf{1}$. The non-vanishing is proposition 2.3.4: an absolutely convergent infinite product whose factors are all nonzero has a nonzero value, and the local factors $(1-p^{-s})^{-1}$ are reciprocals, hence never zero. Therefore $\zeta(s) = \prod_p (1-p^{-s})^{-1} \neq 0$ for $\mathrm{Re}(s) > 1$, as required.

The Euler product is the reason ζ encodes the primes at all: it expresses the global function as an assembly of one local factor per prime, so that every analytic statement about ζ on $\mathrm{Re}(s) > 1$ is simultaneously a statement about the primes, and conversely. The non-vanishing recalled here is the easy half of the story. It holds throughout the region of absolute convergence precisely because the obstruction to a zero is local, a single vanishing factor, and no factor vanishes; this is the structural argument of proposition 2.3.4, sharper than the earlier proof through the Möbius series in example 2.2.5. The non-vanishing that matters for the distribution of primes, on the boundary line $\mathrm{Re}(s) = 1$, lies outside this region and is inaccessible to the product argument; it is the subject of section 4.4.

A few numerical values anchor the qualitative picture and connect ζ to constants from elsewhere in mathematics.

Example 4.1.3: Special Values of ζ

The two best-known special values are the even-argument evaluations

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

The first is the Basel problem, solved by Euler; both are proved in [13] (the standard derivations use the Fourier expansion of a periodic function or the product formula for sin, neither of which is developed here). Numerically $\zeta(2) \approx 1.6449$ and $\zeta(4) \approx 1.0823$. The monotone decrease of proposition 4.1.1(3) toward the asymptote $\zeta = 1$ is visible in a short table of real values:

s	1.5	2	3	4	6	10
$\zeta(s)$	2.6124	1.6449	1.2021	1.0823	1.0173	1.0010

Each entry exceeds the next, and the values approach 1 as s grows, the contribution beyond the $n = 1$ term shrinking like 2^{-s}. The value $\zeta(3) = 1.2021\ldots$, the constant appearing at $s = 3$, is *Apéry's constant*; Apéry proved it irrational in 1979. The even values $\zeta(2k)$ are all rational multiples of π^{2k}, a pattern that the functional equation of chapter 6 explains in one stroke, whereas the odd values $\zeta(3), \zeta(5), \ldots$ have no such closed form and remain arithmetically obscure.

The contrast between the even and odd values is the first sign that ζ carries arithmetic information no elementary manipulation of the series reveals. The even values fall out of the functional equation because that equation relates $\zeta(2k)$ to $\zeta(1-2k)$, a value the equation forces to be rational; the odd values are paired by the same equation with $\zeta(-2k)$, which the functional equation makes vanish, so no comparable evaluation results. This asymmetry is what makes $\zeta(3)$ and its odd siblings difficult, and it foreshadows the role of the functional equation as the organizing principle of the next two chapters.

The single tool from chapter 1 that the continuation rests on is the partial-sum formula for $\sum_{n \le x} n^{-s}$. For $\mathrm{Re}(s) > 0$ with $s \neq 1$, proposition 1.4.6 gives

$$\sum_{n \le x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\mathrm{Re}(s)})$$

where the constant $\zeta(s)$ appearing here is, for $\mathrm{Re}(s) > 1$, the value of the defining series, and is given on the wider half-plane $\mathrm{Re}(s) > 0$ by the convergent integral

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{t\}}{t^{s+1}}\, dt$$

with $\{t\}$ the fractional part of t. The integral converges absolutely for $\mathrm{Re}(s) > 0$ because $|\{t\}t^{-s-1}| \le t^{-\mathrm{Re}(s)-1}$ and $\int_1^\infty t^{-\mathrm{Re}(s)-1}\,dt < \infty$ there. The right-hand side of this formula is defined and finite for every s with $\mathrm{Re}(s) > 0$ except $s = 1$, where the term $s/(s-1)$ has a simple pole; it therefore furnishes a value for $\zeta(s)$ on all of $\mathrm{Re}(s) > 0$, agreeing with the series where the series converges. This formula is what produces the continuation, and the next section establishes that the function it defines is holomorphic on $\mathrm{Re}(s) > 0 \setminus \{1\}$ with a simple pole of residue 1 at $s = 1$, thereby extending ζ past the line $\mathrm{Re}(s) = 1$ to which the series confined it.

4.2 Meromorphic Continuation to $\mathrm{Re}(s) > 0$

The Dirichlet series $\sum_{n\ge1} n^{-s}$ of definition 2.1.1 converges only on the half-plane $\mathrm{Re}(s) > 1$, and there it defines $\zeta(s)$ as a holomorphic function (proposition 4.1.1). Outside that half-plane the series is silent: it diverges, and the symbol $\zeta(s)$ has, as yet, no meaning. The whole analytic theory of the primes depends on giving it one. The Abel-summation analysis of proposition 1.4.6 already does so. Comparing the partial sums $\sum_{n\le x} n^{-s}$ against the integral $\int_1^x t^{-s}\,dt$ produced the closed form

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}\,dt$$

in which the right-hand side makes sense on the larger half-plane $\mathrm{Re}(s) > 0$, because the integral converges there. The series-defined ζ on $\mathrm{Re}(s) > 1$ and this integral expression agree where both are defined, so the latter extends the former.

This section records that continuation as a theorem and reads off its local structure. The two elementary pieces on the right behave very differently near the line $\mathrm{Re}(s) = 1$: the rational term $s/(s-1)$ carries a single pole at $s = 1$, while the integral term is holomorphic across the whole half-plane and contributes nothing singular. Isolating the pole and computing the first two coefficients of the Laurent expansion of ζ at $s = 1$ is the work that remains, and it is the structural input that the proof of the Prime Number Theorem will draw on.

Theorem 4.2.1: Meromorphic Continuation of ζ to $\mathrm{Re}(s) > 0$

The function $\zeta(s)$, defined for $\mathrm{Re}(s) > 1$ by the series $\sum_{n=1}^\infty n^{-s}$, extends to a meromorphic function on the half-plane $\mathrm{Re}(s) > 0$ whose only singularity there is a simple pole at $s = 1$ with residue 1. Explicitly, for all s with $\mathrm{Re}(s) > 0$ and $s \ne 1$,

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}\,dt \tag{4.1}$$

where $\{t\} = t - \lfloor t \rfloor$ denotes the fractional part.

***Proof*:**
Write $I(s) = \int_1^\infty \{t\}\,t^{-s-1}\,dt$ for the integral on the right of (4.1), so that the asserted identity reads $\zeta(s) = s/(s-1) - sI(s)$. The argument has three stages: the integral defines a holomorphic function on $\mathrm{Re}(s) > 0$; the right-hand side of (4.1) is therefore meromorphic on $\mathrm{Re}(s) > 0$ with a single simple pole at $s = 1$; and it agrees with the series on $\mathrm{Re}(s) > 1$, so it is the continuation. The residue is computed at the end.

Step 1: The integral $I(s)$ *is holomorphic on* $\mathrm{Re}(s) > 0$. The integrand $g(s,t) = \{t\}\, t^{-s-1}$ is, for each fixed $t \geq 1$, an entire function of s, and for each fixed s it is continuous in t. Since $0 \leq \{t\} < 1$, it is dominated by

$$|g(s,t)| = \{t\}\, t^{-\mathrm{Re}(s)-1} \leq t^{-\mathrm{Re}(s)-1}$$

Fix a compact set $K \subset \{\mathrm{Re}(s) > 0\}$ and let $\sigma_0 = \min_{s \in K} \mathrm{Re}(s) > 0$. Then $|g(s,t)| \leq t^{-\sigma_0 - 1}$ for all $s \in K$ and all $t \geq 1$, and $\int_1^\infty t^{-\sigma_0-1}\, dt = 1/\sigma_0 < \infty$. The dominating function $t^{-\sigma_0-1}$ is independent of s and integrable on $[1, \infty)$, so the integral defining $I(s)$ converges absolutely and uniformly for $s \in K$. A locally uniform limit of holomorphic functions is holomorphic, and a parameter integral with holomorphic integrand and a locally uniform integrable bound is holomorphic in the parameter, with differentiation permissible under the integral sign; this is the standard theorem on holomorphy of integrals depending on a parameter ([13]). Applying it on each such K, and noting that the half-plane $\mathrm{Re}(s) > 0$ is exhausted by such compact sets, shows that I is holomorphic on $\mathrm{Re}(s) > 0$.

Step 2: The right-hand side is meromorphic with a simple pole at $s = 1$. On $\mathrm{Re}(s) > 0$ the term $sI(s)$ is holomorphic, being the product of the entire function s with the holomorphic function $I(s)$ of Step 1. The term $s/(s-1)$ is a rational function whose only singularity is a simple pole at $s = 1$: writing $s/(s-1) = 1 + 1/(s-1)$ exhibits the principal part $1/(s-1)$ explicitly. The sum

$$R(s) := \frac{s}{s-1} - sI(s)$$

is therefore meromorphic on $\mathrm{Re}(s) > 0$, holomorphic away from $s = 1$, and has at $s = 1$ a simple pole, since the holomorphic term $sI(s)$ cannot cancel the simple pole of $s/(s-1)$.

Step 3: R *agrees with* ζ *on* $\mathrm{Re}(s) > 1$ *and is the continuation.* By proposition 1.4.6, for every s with $\mathrm{Re}(s) > 1$ the constant term in the asymptotic expansion of $\sum_{n \leq x} n^{-s}$ is exactly $R(s)$, and that constant equals $\lim_{x\to\infty} \sum_{n \leq x} n^{-s} = \sum_{n=1}^\infty n^{-s} = \zeta(s)$. Hence $R(s) = \zeta(s)$ for all s with $\mathrm{Re}(s) > 1$. The function ζ is holomorphic on the connected open set $\mathrm{Re}(s) > 1$, and R is meromorphic on the connected open set $\mathrm{Re}(s) > 0$ with its only pole at $s = 1$; the two agree on the nonempty open subset $\mathrm{Re}(s) > 1$, which has accumulation points in $\mathrm{Re}(s) > 0$. By the identity theorem ([13]), a meromorphic function on a connected open set is determined by its values on any subset with an accumulation point, so R is the unique meromorphic extension of ζ to $\mathrm{Re}(s) > 0$. This establishes (4.1) and shows that the only singularity of the continued ζ on $\mathrm{Re}(s) > 0$ is the simple pole of R at $s = 1$.

Step 4: The residue at $s = 1$ *is* 1. For a simple pole the residue is the limit of $(s-1)$ times the function. Using (4.1) and the holomorphy of $sI(s)$ at $s = 1$,

$$\lim_{s\to 1}(s-1)\zeta(s) = \lim_{s\to 1}\left[(s-1)\cdot\frac{s}{s-1} - (s-1)\, sI(s)\right] = \lim_{s\to 1}\big[s - (s-1)\, sI(s)\big] = 1$$

the second term vanishing in the limit because $sI(s)$ is finite at $s = 1$. Thus the pole is simple with residue 1, completing the proof.

The continuation is what lets the analytic machinery reach the line $\mathrm{Re}(s) = 1$ and the point $s = 1$, where the series itself gives nothing. Every later argument that extracts arithmetic from $\zeta(s)$ near this line, the non-vanishing on $\mathrm{Re}(s) = 1$, the behavior of the logarithmic derivative, and ultimately the asymptotic $\psi(x) \sim x$, presupposes that ζ is defined and analytic there. The single simple pole at $s = 1$ is the one feature of this extended function that survives into the Prime Number Theorem: when Perron's formula expresses the Chebyshev sum $\psi(x)$ as a contour integral of $-\zeta'(s)/\zeta(s) \cdot x^s/s$, the pole of ζ at $s = 1$ becomes a pole of the integrand with residue x, and that residue is the main term $\psi(x) \sim x$ (chapter 5). That the residue of ζ at the pole is exactly 1, rather than some other constant, is what makes the main term x rather than a multiple of it. The pole is, in this precise sense, the analytic source of the leading order of the distribution of primes.

The continuation obtained here reaches only the half-plane $\mathrm{Re}(s) > 0$, governed by the convergence of the integral in (4.1), which fails once $\mathrm{Re}(s) \leq 0$. Extending ζ to all of $\mathbb{C}$ requires a different mechanism, the functional equation relating $\zeta(s)$ to $\zeta(1-s)$, developed in chapter 6. An alternative route to the same half-plane runs through the Dirichlet eta function: the integral representation of example 2.4.6 expresses $(1 - 2^{1-s})\,\Gamma(s)\,\zeta(s)$ by an integral that already converges on $\mathrm{Re}(s) > 0$, and dividing by the elementary factor $(1 - 2^{1-s})\Gamma(s)$ continues ζ across the line $\mathrm{Re}(s) = 1$ as well. The Abel-summation route taken here is the more elementary of the two, requiring no special functions, and it has the advantage of displaying the pole and its residue directly in the formula.

The pole at $s = 1$ is significant enough to isolate as a corollary in the form most often invoked, the value of the limit defining the residue.

Corollary 4.2.2: The Simple Pole at $s = 1$

The continued function $\zeta(s)$ has a simple pole at $s = 1$, and

$$\lim_{s \to 1} (s-1)\,\zeta(s) = 1$$

***Proof*:**
This is the statement of the pole and residue in theorem 4.2.1: the residue of a simple pole at $s = 1$ is by definition $\lim_{s \to 1}(s-1)\zeta(s)$, and theorem 4.2.1 computes this limit to be 1, as required.

Knowing that the pole is simple with residue 1 fixes the leading term $1/(s-1)$ of the Laurent expansion of ζ at $s = 1$. The constant term is the next datum, and it is not arbitrary: it is the Euler–Mascheroni constant γ, the same number that measures the gap between the harmonic sum $\sum_{n \leq x} 1/n$ and $\log x$ (example 1.4.4). The coincidence is not accidental. Both quantities arise from comparing a sum of n^{-s} against its integral, the harmonic sum at $s = 1$ and the zeta values as $s \to 1$, and the constant γ is the common discrepancy. The formula (4.1) makes the connection precise, because the integral $I(s)$ that appears in it specializes at $s = 1$ to the very integral whose value computes γ.

Proposition 4.2.3: The Laurent Expansion at $s = 1$

Near $s = 1$ the continued zeta function has the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$$

where γ is the Euler–Mascheroni constant.

***Proof*:**
Start from the continuation formula (4.1) of theorem 4.2.1, valid for $\mathrm{Re}(s) > 0$ with $s \neq 1$, and split off the principal part of the rational term by writing $s/(s-1) = 1 + 1/(s-1)$. With $I(s) = \int_1^\infty \{t\}\, t^{-s-1}\, dt$ as before, this gives

$$\zeta(s) = \frac{1}{s-1} + 1 - s\, I(s) \tag{4.2}$$

The leading term $1/(s-1)$ is the principal part already identified in corollary 4.2.2; the remaining part $1 - sI(s)$ is holomorphic at $s = 1$ by Step 1 of the proof of theorem 4.2.1, so its value at $s = 1$ is the constant term of the Laurent expansion and its Taylor expansion supplies the higher-order terms. Evaluating at $s = 1$ requires the single number $I(1)$.

The function I is continuous at $s = 1$, again by the holomorphy of Step 1, so $I(1) = \int_1^\infty \{t\}\, t^{-2}\, dt$, the integral obtained by setting $s = 1$ in the integrand. This is exactly the integral whose value was computed in example 1.4.4 in the course of evaluating the Euler–Mascheroni constant, namely

$$\int_1^\infty \frac{\{t\}}{t^2}\, dt = 1 - \gamma$$

Substituting $s = 1$ and $I(1) = 1 - \gamma$ into the holomorphic part $1 - sI(s)$ of (4.2) gives the constant term

$$1 - 1 \cdot I(1) = 1 - (1 - \gamma) = \gamma$$

Since $1 - sI(s)$ is holomorphic at $s = 1$, it equals $\gamma + O(s-1)$ near $s = 1$, and combining with the principal part yields $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$, as we sought to show.

The constant term γ ties the analytic behavior of ζ at its pole to the elementary asymptotics of the harmonic series, and this is the form in which the pole enters several refined estimates: the closed-form evaluation of the Mertens constant in theorem 3.3.3, for instance, isolates γ precisely as the value $\zeta(1+\rho) = 1/\rho + \gamma + o(1)$ read off from this expansion as $\rho \to 0^+$. The higher coefficients of the Laurent expansion are the *Stieltjes constants* γ_k, defined by $\zeta(s) = 1/(s-1) + \sum_{k \geq 0} \frac{(-1)^k}{k!} \gamma_k\, (s-1)^k$ with $\gamma_0 = \gamma$; they admit closed forms as regularized limits of $\sum_{n \leq x} (\log n)^k / n$ but do not enter the leading-order theory of the primes.

4.3 First Arithmetic Consequences

The continuation theorem of the previous section gives $\zeta(s)$ a single simple pole at $s = 1$ with residue 1 (theorem 4.2.1 and corollary 4.2.2), and the Laurent expansion $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$ records its leading behavior there (proposition 4.2.3). This is a statement about ζ as an analytic function; on its own it says nothing about primes. The bridge to arithmetic is the Euler product, which on $\mathrm{Re}(s) > 1$ ties ζ to the primes one factor at a time, and the two operations that read the primes off the product, the logarithmic derivative $-\zeta'/\zeta$ and the logarithm $\log \zeta$. Applying these operations to a function with a known pole transfers that pole into analytic statements about prime-power sums.

This section carries out three such transfers. The logarithmic derivative $-\zeta'/\zeta$ acquires its own simple pole at $s = 1$, again of residue 1, and its Dirichlet series is the von Mangoldt series $\sum_n \Lambda(n) n^{-s}$

(theorem 2.2.6); this is the analytic object Perron's formula integrates to produce $\psi(x)$. Stripping the higher prime powers from $-\zeta'/\zeta$ isolates the prime series $\Phi(s) = \sum_p (\log p)p^{-s}$, which inherits the same pole. And the logarithm $\log\zeta$, expanded over prime powers, yields an analytic reproof that $\sum_p 1/p$ diverges, with the precise rate $\sum_p p^{-s} = \log(1/(s-1)) + O(1)$ as $s \to 1^+$. Each of these is a meromorphic fact about a half-plane; converting any of them into an asymptotic for $\psi(x)$ requires control of these functions on the boundary line $\mathrm{Re}(s) = 1$, supplied in the next section and exploited in chapter 5.

The first transfer is the most direct. The logarithmic derivative of a meromorphic function turns a pole or zero into a simple pole whose residue counts its order, and ζ has exactly one pole near $s = 1$ and (by the continuation) no zero there. The order of that pole is 1, so $-\zeta'/\zeta$ has a simple pole of residue 1.

Proposition 4.3.1: The Pole of the Logarithmic Derivative at $s = 1$

On the half-plane $\mathrm{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ is the von Mangoldt function (definition 1.1.8). Moreover $-\zeta'/\zeta$ continues to a meromorphic function on a neighborhood of $s = 1$, holomorphic there apart from a simple pole at $s = 1$, and that pole has residue 1.

***Proof*:**
The Dirichlet series identity on $\mathrm{Re}(s) > 1$ is theorem 2.2.6: the von Mangoldt series $\sum_n \Lambda(n)n^{-s}$ converges absolutely on $\mathrm{Re}(s) > 1$ and equals $-\zeta'(s)/\zeta(s)$ there. It remains to analyze the behavior near $s = 1$.

By the continuation theorem (theorem 4.2.1) and the Laurent expansion of proposition 4.2.3, on a punctured neighborhood of $s = 1$ the zeta function has the form

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$$

Factor out the principal part: writing

$$\zeta(s) = \frac{1}{s-1}\big(1 + \gamma(s-1) + O((s-1)^2)\big) = \frac{1}{s-1}\, g(s)$$

defines $g(s) = 1 + \gamma(s-1) + O((s-1)^2)$, which is holomorphic at $s = 1$ with $g(1) = 1 \neq 0$. Since g is holomorphic and nonzero at $s = 1$, it is nonzero on some neighborhood U of $s = 1$, and on U the function $\zeta(s) = g(s)/(s-1)$ is meromorphic with its only pole the simple pole at $s = 1$ contributed by the factor $1/(s-1)$; in particular ζ has no zero on U.

On $U \setminus \{1\}$ the logarithmic derivative is computed from the factorization. Taking logarithmic derivatives turns the product $\zeta = (s-1)^{-1} g$ into a sum,

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{d}{ds}\log\big((s-1)^{-1}\big) + \frac{g'(s)}{g(s)} = -\frac{1}{s-1} + \frac{g'(s)}{g(s)}$$

the first term being the logarithmic derivative of $(s-1)^{-1}$. Because g is holomorphic and nonvanishing on U, the quotient g'/g is holomorphic on U; denote it $h(s)$, so that h is holomorphic at $s=1$. Negating,

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} - h(s)$$

on $U \setminus \{1\}$. The right-hand side is meromorphic on U, holomorphic away from $s=1$, and has at $s=1$ a simple pole whose principal part is $1/(s-1)$, since the holomorphic term $-h(s)$ contributes nothing to the principal part. The coefficient of $(s-1)^{-1}$ is 1, so the residue is

$$\operatorname{Res}_{s=1}\left(-\frac{\zeta'(s)}{\zeta(s)}\right) = 1$$

This identifies the continuation of $-\zeta'/\zeta$ near $s=1$ as a simple pole of residue 1, as we sought to show.

The pole of $-\zeta'/\zeta$ at $s=1$ is the single analytic fact on which the leading order of the Prime Number Theorem rests. Perron's formula applied to the coefficients $a_n = \Lambda(n)$ expresses the Chebyshev function as the contour integral

$$\psi(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\left(-\frac{\zeta'(s)}{\zeta(s)}\right)\frac{x^s}{s}\,ds$$

for any $c>1$ (theorem 2.5.2 and example 2.5.4). Shifting the contour to the left past the line $\operatorname{Re}(s)=1$ picks up the residue of the integrand at the pole $s=1$; since $-\zeta'/\zeta$ has residue 1 there, the integrand $-\zeta'(s)/\zeta(s)\cdot x^s/s$ has residue $x^1/1 = x$, and this residue is exactly the main term $\psi(x)\sim x$. The agreement of the two residue-1 computations, the pole of ζ and the pole of $-\zeta'/\zeta$, is what fixes the main term as x with coefficient 1 rather than some other multiple. What the present proposition does not supply is the legitimacy of the contour shift: moving the line of integration from $\operatorname{Re}(s)=c>1$ to a line $\operatorname{Re}(s)<1$ requires that $-\zeta'/\zeta$ be holomorphic on the line $\operatorname{Re}(s)=1$ itself, away from $s=1$, which in turn requires ζ to have no zeros there. That non-vanishing is the missing input, established in section 4.4, and the quantitative contour estimate that turns it into $\psi(x)\sim x$ is the work of chapter 5.

The von Mangoldt series $\sum_n \Lambda(n)n^{-s}$ runs over all prime powers p^k, weighting each by $\log p$. The contribution of the genuine primes, the terms with $k=1$, is a series over primes alone, and isolating it produces an object closer to the prime-counting function than $-\zeta'/\zeta$ is.

Definition 4.3.2: The Prime Series $\Phi(s)$

The *prime series* is

$$\Phi(s) = \sum_p \frac{\log p}{p^s}$$

the sum running over all primes p, defined for $\operatorname{Re}(s)>1$.

The series converges absolutely on $\operatorname{Re}(s)>1$: its terms are a subseries of the von Mangoldt series $\sum_n \Lambda(n)n^{-s}$, which converges absolutely there (theorem 2.2.6), so Φ is holomorphic on $\operatorname{Re}(s)>1$. The function Φ is the Dirichlet generating function of the coefficients $(\log p)$ supported on primes, and its summatory function is the Chebyshev theta function $\vartheta(x) = \sum_{p\le x}\log p$ of definition 3.1.2, just as $-\zeta'/\zeta$ generates $\psi(x) = \sum_{n\le x}\Lambda(n)$. The two Chebyshev functions differ only by the prime-power terms $\psi(x) - \vartheta(x) = \sum_{k\geq 2} \sum_{p^k \leq x} \log p$, and theorem 3.1.5 shows this difference to be of lower order, so $\vartheta(x) \sim x$ and $\psi(x) \sim x$ stand or fall together. The same redundancy appears analytically: Φ and $-\zeta'/\zeta$ differ by a function holomorphic well past the line $\mathrm{Re}(s) = 1$, so they carry the same singular data near $s = 1$.

Example 4.3.3: The Prime Series at $s = 2$

At $s = 2$ the series $\Phi(2) = \sum_p (\log p)p^{-2}$ converges, and its first terms are

$$\frac{\log 2}{4} + \frac{\log 3}{9} + \frac{\log 5}{25} + \frac{\log 7}{49} + \frac{\log 11}{121} + \cdots$$

Numerically the first five terms contribute $0.1733+0.1221+0.0644+0.0397+0.0198 = 0.4193$, and the full sum is $\Phi(2) = 0.4931\ldots$. The companion logarithmic-derivative value is $-\zeta'(2)/\zeta(2) = 0.5700\ldots$, the full von Mangoldt sum $\sum_n \Lambda(n)n^{-2}$; the difference $-\zeta'(2)/\zeta(2) - \Phi(2) = 0.0769\ldots$ collects the prime-power terms $\sum_p \sum_{k\geq 2} (\log p)p^{-2k}$, the largest of which is $(\log 2)(4^{-2} + 4^{-3} + \cdots) = (\log 2)/12 = 0.0578$. The prime-power correction is small even at $s = 2$ and shrinks as $\mathrm{Re}(s)$ grows, the first sign of the general bound that makes Φ and $-\zeta'/\zeta$ interchangeable near the line $\mathrm{Re}(s) = 1$.

The relation between Φ and $-\zeta'/\zeta$ is exact and explicit: their difference is the sum of the prime-power terms with exponent $k \geq 2$, and that sum converges on a half-plane reaching well to the left of $\mathrm{Re}(s) = 1$. The difference is therefore holomorphic where it matters, and the pole of $-\zeta'/\zeta$ at $s = 1$ passes to Φ unchanged.

Proposition 4.3.4: Φ and the Logarithmic Derivative

For $\mathrm{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \Phi(s) + R(s), \qquad R(s) = \sum_p \sum_{k=2}^{\infty} \frac{\log p}{p^{ks}}$$

The remainder $R(s)$ converges absolutely on $\mathrm{Re}(s) > 1/2$ and is holomorphic there. Consequently $\Phi(s) - 1/(s-1)$ extends holomorphically to a neighborhood of $s = 1$: the prime series Φ has a simple pole at $s = 1$ with residue 1.

***Proof*:**

Step 1: The decomposition. By theorem 2.2.6 the logarithmic derivative is the von Mangoldt series, which by definition 1.1.8 sums $\log p$ over all prime powers p^k with $k \geq 1$:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{ks}}$$

on $\mathrm{Re}(s) > 1$, where the rearrangement of the absolutely convergent series into a double sum over (p, k) is the bijection between prime powers and pairs (p, k). Splitting off the $k = 1$ terms, which form $\Phi(s)$ by definition 4.3.2, leaves

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{\log p}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{\log p}{p^{ks}} = \Phi(s) + R(s)$$

which is the asserted identity.

Step 2: R is holomorphic on $\mathrm{Re}(s) > 1/2$. Fix a compact set $K \subset \{\mathrm{Re}(s) > 1/2\}$ and let $\sigma_0 = \min_{s\in K} \mathrm{Re}(s) > 1/2$. For $s \in K$ the modulus of the (p, k)-term is $(\log p)p^{-k\sigma_0}$ or smaller, and summing the geometric inner series over $k \geq 2$,

$$\sum_{k=2}^{\infty} \frac{\log p}{p^{k\sigma_0}} = \frac{\log p}{p^{2\sigma_0}} \cdot \frac{1}{1 - p^{-\sigma_0}} \leq \frac{\log p}{p^{2\sigma_0}} \cdot \frac{1}{1 - 2^{-\sigma_0}}$$

using $p \geq 2$ so that $1-p^{-\sigma_0} \geq 1-2^{-\sigma_0} > 0$. The constant $(1-2^{-\sigma_0})^{-1}$ is finite and independent of p, so it suffices that $\sum_p (\log p)p^{-2\sigma_0}$ converges. Since $2\sigma_0 > 1$, choose $\varepsilon > 0$ with $2\sigma_0 - \varepsilon > 1$; then $\log p \leq C_\varepsilon\, p^\varepsilon$ for a constant C_ε and all p, whence

$$\sum_p \frac{\log p}{p^{2\sigma_0}} \leq C_\varepsilon \sum_p \frac{1}{p^{2\sigma_0-\varepsilon}} \leq C_\varepsilon \sum_{n=1}^{\infty} \frac{1}{n^{2\sigma_0-\varepsilon}} < \infty$$

the last series converging because its exponent exceeds 1. The bound is independent of $s \in K$, so the double series for R converges absolutely and uniformly on K. Each summand $(\log p)p^{-ks}$ is entire in s, and a locally uniform limit of holomorphic functions is holomorphic ([13]); exhausting $\{\mathrm{Re}(s) > 1/2\}$ by compact sets K shows R is holomorphic on $\mathrm{Re}(s) > 1/2$.

Step 3: The pole of Φ. On $\mathrm{Re}(s) > 1$ the identity of Step 1 rearranges to $\Phi(s) = -\zeta'(s)/\zeta(s) - R(s)$. By proposition 4.3.1 the function $-\zeta'/\zeta$ continues to a neighborhood U of $s = 1$ as a meromorphic function whose only singularity is a simple pole of residue 1, with $-\zeta'(s)/\zeta(s) = 1/(s-1) - h(s)$ for some h holomorphic on U. By Step 2, R is holomorphic on the neighborhood $U \cap \{\mathrm{Re}(s) > 1/2\}$ of $s = 1$. Therefore on this neighborhood

$$\Phi(s) - \frac{1}{s-1} = \left(-\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1}\right) - R(s) = -h(s) - R(s)$$

is a difference of two functions holomorphic at $s = 1$, hence holomorphic at $s = 1$. Thus $\Phi(s) = 1/(s-1) + (\text{holomorphic near } s = 1)$, so Φ has a simple pole at $s = 1$ whose principal part is $1/(s-1)$, that is, a simple pole of residue 1, completing the proof.

The prime series Φ is the analytic object that most directly governs the primes themselves, as opposed to the prime powers, and its pole at $s = 1$ is the analytic form of the statement that the primes have density $1/\log x$ in the sense made precise by the Prime Number Theorem. Perron's formula applied to Φ, with coefficients $a_n = \log p$ for $n = p$ prime and $a_n = 0$ otherwise, produces the contour integral for the Chebyshev theta function $\vartheta(x) = \sum_{p\leq x} \log p$, and the residue of $\Phi(s)x^s/s$ at the pole $s = 1$ is again x, giving the main term $\vartheta(x) \sim x$. The decomposition $-\zeta'/\zeta = \Phi + R$ with R holomorphic past the line $\mathrm{Re}(s) = 1$ is the analytic counterpart of the elementary estimate $\psi(x) - \vartheta(x) = O(\sqrt{x}\log x)$: in both pictures the prime powers of exponent $k \geq 2$ contribute a lower-order, harmless correction, and the substance lies entirely with the primes. The non-vanishing of ζ on $\mathrm{Re}(s) = 1$ enters here exactly as it does for $-\zeta'/\zeta$, since Φ differs from it only by the holomorphic R; once that non-vanishing is in hand, Φ and $-\zeta'/\zeta$ may be used interchangeably in the contour argument of chapter 5.

The third transfer works with $\log\zeta$ rather than its derivative, and it reproves, by analytic means, the oldest quantitative fact about the primes. The expansion of $\log\zeta$ over prime powers (proposition 2.3.5) has the prime sum $\sum_p p^{-s}$ as its leading piece, and the pole of ζ forces $\log\zeta$ to blow up as $s \to 1^+$. Since the higher prime-power terms stay bounded, the blow-up must come from the prime sum, and the Laurent expansion of ζ makes the rate explicit.

Proposition 4.3.5: Analytic Divergence of the Prime Sum

As $s \to 1^+$ along the real axis,

$$\sum_p \frac{1}{p^s} = \log \frac{1}{s-1} + O(1)$$

In particular $\sum_p 1/p$ diverges.

***Proof*:**
Step 1: Separate the prime sum from the prime powers in $\log \zeta$. For real $s > 1$, proposition 2.3.5 gives the absolutely convergent expansion

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k\, p^{ks}} = \sum_p \frac{1}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{1}{k\, p^{ks}}$$

where the logarithm is the real logarithm of $\zeta(s) > 1$, and the second sum splits off the prime-power terms with $k \geq 2$. Write $E(s)$ for that second sum.

Step 2: The prime-power tail is bounded. For $s \geq 1$ the bound established in example 2.3.6 applies: dropping the factor $1/k \leq 1$ and summing the geometric inner series,

$$0 \leq E(s) = \sum_p \sum_{k=2}^{\infty} \frac{1}{k\, p^{ks}} \leq \sum_p \sum_{k=2}^{\infty} \frac{1}{p^{ks}} \leq \sum_p \sum_{k=2}^{\infty} \frac{1}{p^k} = \sum_p \frac{p^{-2}}{1-p^{-1}}$$

the middle inequality using $p^{ks} \geq p^k$ for $s \geq 1$. The final sum is bounded by a convergent series independent of s: since $p^{-2}/(1-p^{-1}) = 1/(p^2 - p) = 1/(p(p-1))$, comparison with all integers gives

$$\sum_p \frac{1}{p(p-1)} \leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1$$

the sum telescoping. Hence $0 \leq E(s) \leq 1$ for all $s \geq 1$, so $E(s) = O(1)$ as $s \to 1^+$.

Step 3: The size of $\log \zeta(s)$ *near* $s = 1$. By the Laurent expansion of proposition 4.2.3, near $s = 1$,

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1) = \frac{1}{s-1}\big(1 + \gamma(s-1) + O((s-1)^2)\big)$$

For real s slightly greater than 1 both sides are positive, so taking the real logarithm,

$$\log \zeta(s) = \log \frac{1}{s-1} + \log\big(1 + \gamma(s-1) + O((s-1)^2)\big)$$

As $s \to 1^+$ the argument of the second logarithm tends to 1, so that term tends to $\log 1 = 0$ and is in particular $O(1)$. Therefore

$$\log \zeta(s) = \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

Step 4: Combine. Subtracting the bounded tail of Step 2 from the expansion of Step 1 and inserting Step 3,

$$\sum_p \frac{1}{p^s} = \log \zeta(s) - E(s) = \left(\log \frac{1}{s-1} + O(1)\right) - O(1) = \log \frac{1}{s-1} + O(1)$$

as $s \to 1^+$. The right-hand side tends to $+\infty$ as $s \to 1^+$, since $\log(1/(s-1)) \to +\infty$. If the series $\sum_p 1/p$ converged, then for every $s > 1$ the comparison $\sum_p p^{-s} \le \sum_p p^{-1}$ would bound $\sum_p p^{-s}$ by a finite constant uniformly in s, contradicting the divergence to $+\infty$ just established. Hence $\sum_p 1/p$ diverges, completing the proof.

The divergence of $\sum_p 1/p$ is Euler's theorem, and the rate $\sum_p p^{-s} \sim \log(1/(s-1))$ is its analytic sharpening: the prime sum grows like $\log(1/(s-1))$ as s approaches the pole, the same logarithmic rate by which $\log \zeta(s)$ itself blows up. This is the Dirichlet-series form of the Mertens estimate $\sum_{p \le x} 1/p = \log\log x + M + O(1/\log x)$ of theorem 3.3.3. The two are linked by the heuristic $s - 1 \leftrightarrow 1/\log x$ that runs throughout the subject: setting $s = 1 + 1/\log x$ turns $\log(1/(s-1))$ into $\log\log x$, and the analytic statement becomes the leading term of the Mertens statement. The agreement reflects the same arithmetic seen through two summation methods, the sharp cutoff $\sum_{p \le x}$ and the smooth analytic weighting $\sum_p p^{-s}$; the constant M and the error $O(1/\log x)$ are the finer data that the smooth version compresses into its $O(1)$. The Mertens theorem, proved by elementary partial summation in chapter 3, is the stronger statement, with an explicit constant and error term that the present proof does not reach. What the analytic proof offers in exchange is method: it derives the divergence from the single fact that ζ has a pole at $s = 1$, and the same argument applied to a Dirichlet L-function $L(s, \chi)$, whose non-vanishing at $s = 1$ replaces the pole of ζ, yields the divergence of $\sum_{p \equiv a} 1/p$ over primes in an arithmetic progression, the analytic core of Dirichlet's theorem in chapter 7.

All three results of this section are statements about half-planes and about behavior as $s \to 1^+$ along the real axis. The pole of $-\zeta'/\zeta$, the pole of Φ, and the rate of divergence of $\sum_p p^{-s}$ each isolate the singular structure of an analytic function at the point $s = 1$. None of them yet delivers an asymptotic for $\psi(x)$, $\vartheta(x)$, or $\pi(x)$, because the passage from the singularity to the counting function runs through Perron's formula (theorem 2.5.2 and example 2.5.4), and that passage demands a contour shift across the line $\mathrm{Re}(s) = 1$. The shift is legitimate only where the integrand is holomorphic, and the integrand $-\zeta'(s)/\zeta(s) \cdot x^s/s$ is holomorphic on the line $\mathrm{Re}(s) = 1$ precisely when ζ has no zeros there. Establishing that $\zeta(1 + it) \ne 0$ for all real t, the one piece of analytic input that lies beyond the elementary Euler-product arguments, is the subject of the next section, after which chapter 5 converts the pole at $s = 1$ into the asymptotic $\psi(x) \sim x$.

4.4 Non-Vanishing on the Line $\mathrm{Re}(s) = 1$

The preceding sections fixed the analytic structure of $\zeta(s)$ on the half-plane $\mathrm{Re}(s) > 0$: a single simple pole at $s = 1$ with residue 1 (theorem 4.2.1), and the logarithmic derivative $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n) n^{-s}$ carrying the arithmetic. What remains is the one analytic fact that the elementary Euler-product argument of chapter 2 could not reach. The non-vanishing of $\zeta(s)$ throughout $\mathrm{Re}(s) > 1$ followed from the factorization $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, in which no local factor vanishes (corollary 4.1.2); but that argument lives inside the region of absolute convergence and says nothing on the boundary line $\mathrm{Re}(s) = 1$, where the product no longer converges. The decisive analytic fact behind the Prime Number Theorem is that $\zeta(s)$ has no zeros on this line either: $\zeta(1 + it) \neq 0$ for every real t.

This section proves it by the positivity argument of Mertens and de la Vallée Poussin, organized around the elementary trigonometric inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$. The mechanism is the logarithmic expansion $\log\zeta(s) = \sum_{p,k} 1/(k\,p^{ks})$ of proposition 2.3.5, whose real part on a vertical line is a sum of cosines $\cos(kt\log p)$ weighted by positive coefficients. Feeding the three real parts at heights 0, t, and $2t$ into the inequality produces a single positive combination, and that positivity is incompatible with a zero of ζ on the line: a zero at $1 + it_0$ would force the combination to negative infinity as s approaches the line, a contradiction. The argument is short once the inequality is in hand.

Lemma 4.4.1: The Inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$

For every real number θ,

$$3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \geq 0$$

***Proof*:**
The double-angle identity $\cos 2\theta = 2\cos^2\theta - 1$ gives

$$3 + 4\cos\theta + \cos 2\theta = 3 + 4\cos\theta + (2\cos^2\theta - 1) = 2\cos^2\theta + 4\cos\theta + 2$$

The right-hand side is $2(\cos^2\theta + 2\cos\theta + 1) = 2(\cos\theta + 1)^2$. A real square is nonnegative and the factor 2 is positive, so $2(1 + \cos\theta)^2 \geq 0$ for every real θ, as we sought to show.

The choice of coefficients $3, 4, 1$ is not arbitrary. The argument below combines $\log\zeta$ at three heights, weighting the height 0 contribution by 3, the height t contribution by 4, and the height $2t$ contribution by 1, and the combination is useful only if the resulting trigonometric sum is nonnegative for every angle. Any nonnegative combination $a_0 + a_1\cos\theta + a_2\cos 2\theta$ with $a_0 = a_2$ doubled appropriately would serve in principle, but $(3, 4, 1)$ is the lightest integer choice that both stays nonnegative and gives the height 0 term enough weight to overpower a single zero on the line; the equality case $\theta = \pi$, where the expression vanishes, shows the coefficients cannot be lowered while preserving positivity. The same inequality, applied to Dirichlet L-functions, drives the non-vanishing $L(1, \chi) \neq 0$ behind primes in arithmetic progressions (chapter 7).

The positivity now transfers from angles to the zeta function. The link is the expansion of $\log\zeta$ along a vertical line, whose real part is precisely a weighted sum of cosines.

Theorem 4.4.2: Non-Vanishing of ζ on $\mathrm{Re}(s) = 1$

The Riemann zeta function does not vanish on the line $\mathrm{Re}(s) = 1$ away from the pole: $\zeta(1 + it) \neq 0$ for every real $t \neq 0$.

***Proof*:**
Throughout, write $s = \sigma + it$ with $\sigma > 1$ real, so that s lies in the half-plane of absolute convergence where the Euler product and its logarithm are valid. The argument has two stages: a positivity estimate valid for all $\sigma > 1$, and a contradiction derived from a hypothetical zero on the line.

Step 1: The fundamental inequality $|\zeta(\sigma)|^3\,|\zeta(\sigma+it)|^4\,|\zeta(\sigma+2it)| \geq 1$ *for* $\sigma > 1$*.* By proposition 2.3.5, for $\mathrm{Re}(s) > 1$ the logarithm of ζ is given by the absolutely convergent double series

$$\log\zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k\,p^{ks}}$$

For $s = \sigma+it$ the summand is $k^{-1}p^{-k\sigma}p^{-ikt} = k^{-1}p^{-k\sigma}e^{-ikt\log p}$, whose real part is $k^{-1}p^{-k\sigma}\cos(kt\log p)$ since $\mathrm{Re}(e^{-i\alpha}) = \cos\alpha$. Absolute convergence permits taking the real part term by term, so

$$\mathrm{Re}\log\zeta(\sigma+it) = \sum_p \sum_{k=1}^{\infty} \frac{\cos(kt\log p)}{k\,p^{k\sigma}} \tag{4.3}$$

Now form the combination of (4.3) at the three heights 0, t, and $2t$ with weights $3, 4, 1$. Each term carries the common positive coefficient $c_{p,k} = (k\,p^{k\sigma})^{-1}$, and writing $\theta_{p,k} = kt\log p$ for the angle at height t, the angle at height 0 is 0 and the angle at height $2t$ is $2\theta_{p,k}$. Hence

$$3\mathrm{Re}\log\zeta(\sigma) + 4\mathrm{Re}\log\zeta(\sigma+it) + \mathrm{Re}\log\zeta(\sigma+2it) = \sum_p \sum_{k=1}^{\infty} c_{p,k}\big(3 + 4\cos\theta_{p,k} + \cos 2\theta_{p,k}\big)$$

the rearrangement of the three absolutely convergent series into one being permissible. By lemma 4.4.1 each bracket is nonnegative, and each coefficient $c_{p,k}$ is positive, so every term of the combined series is nonnegative and therefore

$$3\mathrm{Re}\log\zeta(\sigma) + 4\mathrm{Re}\log\zeta(\sigma+it) + \mathrm{Re}\log\zeta(\sigma+2it) \geq 0 \tag{4.4}$$

For any nonzero complex number w and any branch of its logarithm, $\mathrm{Re}\log w = \log|w|$. Each ζ value here is nonzero, lying in the half-plane $\mathrm{Re}(s) > 1$ (corollary 4.1.2), so (4.4) reads $3\log|\zeta(\sigma)| + 4\log|\zeta(\sigma+it)| + \log|\zeta(\sigma+2it)| \geq 0$. Exponentiating, and using that exp is increasing,

$$|\zeta(\sigma)|^3\,|\zeta(\sigma+it)|^4\,|\zeta(\sigma+2it)| \geq 1 \qquad (\sigma > 1) \tag{4.5}$$

This inequality holds for every real t and every $\sigma > 1$, and it is the only consequence of positivity that the argument needs.

Step 2: A zero on the line contradicts (4.5)*.* Suppose, for contradiction, that $\zeta(1+it_0) = 0$ for some real $t_0 \neq 0$. Apply (4.5) with $t = t_0$ and let $\sigma \to 1^+$ along the real axis; the three factors are estimated separately. The estimates compare each factor with a power of $\sigma - 1$, and the exponents are arranged so that the product tends to 0, contradicting the lower bound 1.

The first factor blows up at the controlled rate of the pole. By corollary 4.2.2 the continued ζ has a simple pole at $s = 1$ with $\lim_{\sigma\to1^+}(\sigma-1)\zeta(\sigma) = 1$, so $(\sigma-1)\zeta(\sigma) \to 1$ and in particular $(\sigma-1)|\zeta(\sigma)| \to 1$. Thus there is a constant $A > 0$ and a threshold $\sigma_1 \in (1, 2]$ such that

$$|\zeta(\sigma)| \leq \frac{A}{\sigma - 1} \qquad (1 < \sigma \leq \sigma_1) \tag{4.6}$$

indeed any $A > 1$ works for σ near 1. Cubing, $|\zeta(\sigma)|^3 \leq A^3(\sigma - 1)^{-3}$.

The second factor vanishes at the controlled rate of the assumed zero. The point $1 + it_0$ is not the pole, since $t_0 \neq 0$, so by theorem 4.2.1 the function ζ is holomorphic in a neighborhood of $1 + it_0$. By hypothesis it vanishes there, hence vanishes to some finite order $m \geq 1$: there is a function h holomorphic near $1 + it_0$ with $h(1 + it_0) \neq 0$ and $\zeta(s) = (s - (1 + it_0))^m h(s)$ on that neighborhood (the local factorization of a holomorphic function at a zero, [13]). Restricting to $s = \sigma + it_0$ with $\sigma > 1$, the displacement from the zero is $s - (1 + it_0) = \sigma - 1$, a positive real number, so

$$\zeta(\sigma + it_0) = (\sigma - 1)^m \, h(\sigma + it_0)$$

As $\sigma \to 1^+$ the factor $h(\sigma + it_0) \to h(1 + it_0)$, a finite value, so $|h(\sigma + it_0)|$ is bounded by some constant $B > 0$ for σ in an interval $(1, \sigma_2]$ with $\sigma_2 \in (1, \sigma_1]$. Since $m \geq 1$ and $0 < \sigma - 1 < 1$ on this interval (shrinking σ_2 below 2 if needed), $(\sigma - 1)^m \leq (\sigma - 1)$, and therefore

$$|\zeta(\sigma + it_0)| = (\sigma - 1)^m \, |h(\sigma + it_0)| \leq B(\sigma - 1) \qquad (1 < \sigma \leq \sigma_2) \tag{4.7}$$

Raising to the fourth power, $|\zeta(\sigma + it_0)|^4 \leq B^4(\sigma - 1)^4$.

The third factor stays bounded. The point $1 + 2it_0$ is also distinct from the pole, since $2t_0 \neq 0$, so by theorem 4.2.1 the function ζ is holomorphic at $1 + 2it_0$, hence continuous there. Consequently $\zeta(\sigma + 2it_0) \to \zeta(1 + 2it_0)$ as $\sigma \to 1^+$, a finite limit, and there is a constant $C > 0$ and a threshold $\sigma_3 \in (1, \sigma_2]$ with

$$|\zeta(\sigma + 2it_0)| \leq C \qquad (1 < \sigma \leq \sigma_3) \tag{4.8}$$

No claim about the order of ζ at $1 + 2it_0$ is needed; mere boundedness suffices, and that is guaranteed by continuity whether or not ζ happens to vanish there.

Combining the three estimates (4.6), (4.7), and (4.8), valid simultaneously for $1 < \sigma \leq \sigma_3$, gives

$$|\zeta(\sigma)|^3 \, |\zeta(\sigma + it_0)|^4 \, |\zeta(\sigma + 2it_0)| \leq \frac{A^3}{(\sigma - 1)^3} \cdot B^4(\sigma - 1)^4 \cdot C = A^3 B^4 C \, (\sigma - 1)$$

The right-hand side tends to 0 as $\sigma \to 1^+$. Yet by (4.5) the left-hand side is at least 1 for every $\sigma > 1$. Letting $\sigma \to 1^+$ forces $1 \leq \lim_{\sigma \to 1^+} A^3 B^4 C(\sigma - 1) = 0$, an impossibility. The assumption $\zeta(1 + it_0) = 0$ is therefore untenable, and $\zeta(1 + it_0) \neq 0$ for every real $t_0 \neq 0$, completing the proof.

The arithmetic of the exponents is the heart of the matter. The pole contributes a singularity of order one, so the cube $|\zeta(\sigma)|^3$ grows like $(\sigma - 1)^{-3}$; a zero of order $m \geq 1$ makes the fourth power $|\zeta(\sigma + it_0)|^4$ vanish at least like $(\sigma - 1)^4$; and the bounded third factor contributes nothing. The net exponent $-3 + 4 = +1$ is positive, so the product vanishes, and this is exactly the contradiction. Had the weights been chosen differently, the balance could fail: weights $(1, 2, 1)$, for instance, would give net exponent $-1 + 2 = +1$ as well, but the corresponding trigonometric polynomial $1 + 2\cos\theta + \cos 2\theta = 2\cos^2\theta + 2\cos\theta$ is *not* nonnegative for all θ, so Step 1 would fail. The pair of requirements, nonnegativity of the cosine polynomial and a strictly positive net exponent, is what singles out $(3, 4, 1)$, and the equality $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$ is the smallest integer solution meeting both. A single zero on the line cannot survive this squeeze.

This non-vanishing is the analytic linchpin of the Prime Number Theorem. The proof of $\psi(x) \sim x$ in chapter 5 begins from the Perron representation $\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \big(-\zeta'(s)/\zeta(s)\big) x^s / s \, ds$ of example 2.5.4, taken on a vertical line $\mathrm{Re}(s) = c$ with $c > 1$, and proceeds by pushing the contour leftward onto the line $\mathrm{Re}(s) = 1$. Each zero of ζ on or near that line would be a pole of the integrand $-\zeta'(s)/\zeta(s)$, an obstruction to the contour shift contributing an extra term to $\psi(x)$ of size comparable to the main term. theorem 4.4.2 is precisely the statement that there is no such obstruction on the line itself: the only pole of the integrand encountered on $\mathrm{Re}(s) = 1$ is the one at $s = 1$ coming from the pole of ζ (proposition 4.3.1), whose residue x is the main term $\psi(x) \sim x$. The non-vanishing is what licenses the contour to rest on $\mathrm{Re}(s) = 1$ with no pole obstruction except $s = 1$, and it is in this sense that the asymptotic distribution of primes is encoded in a single line of the complex plane.

4.4.3 Historical Note: Hadamard and de la Vallée Poussin The Prime Number Theorem was conjectured by Gauss and Legendre near the close of the eighteenth century and resisted proof for nearly a hundred years. It was settled independently in 1896 by Jacques Hadamard and Charles-Jean de la Vallée Poussin, each building on Riemann's 1859 program of studying $\pi(x)$ through the complex-analytic properties of $\zeta(s)$. The technical core of both proofs is the non-vanishing of $\zeta(s)$ on the line $\mathrm{Re}(s) = 1$; the positivity argument through the inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$ presented here is the streamlined form later extracted by Mertens and refined by de la Vallée Poussin, and is the version that has become standard. Qualitative non-vanishing on the line suffices to prove $\psi(x) \sim x$, but not to control the error term. De la Vallée Poussin went further, converting the same positivity into a *quantitative* zero-free region of the form $\sigma > 1 - c/\log|t|$ to the left of the line, which yields an explicit error bound in the Prime Number Theorem; these zero-free regions, and the error terms they produce, are developed in chapter 5.

The non-vanishing is recorded once more in the form in which the proof of the Prime Number Theorem invokes it, as a statement about the logarithmic derivative.

Corollary 4.4.4: Regularity of the Logarithmic Derivative on $\mathrm{Re}(s) = 1$

The logarithmic derivative $-\zeta'(s)/\zeta(s)$ is holomorphic on the line $\mathrm{Re}(s) = 1$ except for a simple pole at $s = 1$. Equivalently, $\zeta(s)$ has no zeros on $\mathrm{Re}(s) = 1$.

***Proof*:**
By theorem 4.2.1 the function ζ is holomorphic on a neighborhood of the line $\mathrm{Re}(s) = 1$ except at $s = 1$, where it has a simple pole. The logarithmic derivative $-\zeta'/\zeta$ of a meromorphic function is holomorphic at every point that is neither a zero nor a pole of ζ; at a zero of order m it has a simple pole with residue $-m$, and at a pole of order n it has a simple pole with residue n (the standard local computation for $-\zeta'/\zeta$, [13]). On the line $\mathrm{Re}(s) = 1$ the only pole of ζ is the simple pole at $s = 1$, which contributes a simple pole of $-\zeta'/\zeta$ at $s = 1$ with residue 1 (proposition 4.3.1). It remains to rule out zeros of ζ on the line, the other possible source of poles of $-\zeta'/\zeta$. The point $s = 1$ is the pole, not a zero. For every other point $1 + it$ with $t \neq 0$, theorem 4.4.2 gives $\zeta(1 + it) \neq 0$. Hence ζ has no zeros on $\mathrm{Re}(s) = 1$, so $-\zeta'/\zeta$ has no poles there beyond the simple pole at $s = 1$, and is holomorphic on the rest of the line, as required.

5

The Prime Number Theorem

The four preceding chapters assembled, piece by piece, every ingredient the proof of the Prime Number Theorem requires. chapter 3 reduced the statement $\pi(x) \sim x/\log x$ to the assertion $\psi(x) \sim x$ about the Chebyshev function, and to the equivalent assertion $\vartheta(x) \sim x$ about its prime-only counterpart, through the equivalence of theorem 3.1.5; the same chapter pinned the order of magnitude of these functions, proving $\vartheta(x) \asymp x$ by Chebyshev's elementary count of prime powers (theorem 3.2.3). chapter 4 then supplied the analytic facts that the elementary theory could not reach: the zeta function continues meromorphically across the line $\mathrm{Re}(s) = 1$ with a single simple pole at $s = 1$ of residue 1 (theorem 4.2.1), its logarithmic derivative $-\zeta'(s)/\zeta(s)$ inherits that pole (proposition 4.3.1), and, the one fact lying just beyond the Euler-product arguments, $\zeta(s)$ has no zeros on the line $\mathrm{Re}(s) = 1$ (theorem 4.4.2). The goal of this chapter is to combine these into the asymptotic $\psi(x) \sim x$, equivalently $\vartheta(x) \sim x$ and $\pi(x) \sim x/\log x$, which is the Prime Number Theorem.

The route is the analytic proof of Hadamard and de la Vallée Poussin from 1896, in the streamlined form due to D. J. Newman in 1980 [16] and presented by Zagier [24], which trades the heavy contour estimates of the original arguments for a single Tauberian theorem. The strategy is to package the primes into the Dirichlet series $\Phi(s) = \sum_p (\log p)p^{-s}$, to convert the non-vanishing of ζ on $\mathrm{Re}(s) = 1$ into a holomorphic extension of Φ to the closed half-plane $\mathrm{Re}(s) \geq 1$ except for one pole, and to feed that extension into a Tauberian theorem that forces the integral $\int_1^\infty (\vartheta(x) - x)/x^2\, dx$ to converge; the convergence of this integral, together with the monotonicity of ϑ, yields $\vartheta(x) \sim x$. This first section lays out the plan in full and states the central analytic claim that the next three sections establish. The Laplace-transform machinery and Newman's theorem occupy section 5.2 and section 5.3, the proof itself section 5.4; the final section, section 5.5, sharpens qualitative non-vanishing into a quantitative zero-free region and reads off an error term, the first quantitative refinement of the asymptotic.

5.1 The Analytic Strategy

The Prime Number Theorem is a single asymptotic statement, but its proof draws on two streams of input developed in separate chapters, and the first task is to see precisely how they join. The elementary side, from chapter 3, contributes two facts: that the three counting functions ψ, ϑ, and π rise or fall together, so that proving the asymptotic for any one of them proves it for all three; and that $\vartheta(x)$ is already known to be of the correct order x, bounded above and below by constant multiples of it. These reduce the entire theorem to the one limit $\vartheta(x)/x \to 1$. The analytic side, from chapter 4, contributes the behavior of $\zeta(s)$ and of the prime series $\Phi(s)$ near and on the line $\mathrm{Re}(s) = 1$. This section records the reduction, then states the plan by which the analytic inputs produce the missing limit.

The reduction itself is a restatement of results already proved, collected here so that the rest of the chapter may speak only of ϑ.

Proposition 5.1.1: Reduction of the Prime Number Theorem to $\vartheta(x) \sim x$

The following three asymptotic statements are equivalent:

$$\pi(x) \sim \frac{x}{\log x}, \qquad \vartheta(x) \sim x, \qquad \psi(x) \sim x$$

Moreover $\vartheta(x) = O(x)$, and the ratio $\vartheta(x)/x$ is bounded. Consequently the Prime Number Theorem in any of its three forms is equivalent to the single statement $\vartheta(x) \sim x$, and to prove it it suffices to prove that $\vartheta(x)/x \to 1$.

***Proof*:**
The equivalence of the three asymptotics is theorem 3.1.5, where $\pi(x) \sim x/\log x$, $\vartheta(x) \sim x$, and $\psi(x) \sim x$ are shown to imply one another by Abel summation transferring between the prime-counting function π (definition 3.1.1) and the Chebyshev functions ϑ and ψ (definition 3.1.2). The bound $\vartheta(x) = O(x)$ is the upper half of Chebyshev's estimates (theorem 3.2.3), which produces constants $0 < c_1 < c_2$ with $c_1 x \le \vartheta(x) \le \psi(x) \le c_2 x$ for all large x; the right-hand inequality gives $\vartheta(x) \le c_2 x$, so $\vartheta(x) = O(x)$ and $\vartheta(x)/x \le c_2$ is bounded above, while $\vartheta(x)/x \ge c_1$ bounds it below. Since the three statements are equivalent and $\vartheta(x) \sim x$ is by definition the assertion $\vartheta(x)/x \to 1$, establishing that single limit establishes all three, completing the proof.

The proposition is the hinge of the chapter. It says that the entire content of the Prime Number Theorem is concentrated in the behavior of one function, $\vartheta(x) = \sum_{p \le x} \log p$, and that the work already done has narrowed the target to a single limit: $\vartheta(x)/x$ is known to lie between two positive constants, and what remains is to show that it converges, and that its limit is exactly 1. The Chebyshev bounds do half the job, confining the ratio to a bounded interval; the analytic theory must do the other half, collapsing that interval to the point 1. The boundedness recorded here is not incidental to the proof that follows. It is the hypothesis that makes the Tauberian step legitimate: a Tauberian theorem extracts an asymptotic for $\vartheta(x)$ from the analytic behavior of its generating function only under a one-sided growth condition, and $\vartheta(x) = O(x)$ is exactly that condition.

With the target fixed as $\vartheta(x) \sim x$, the plan of the proof can be stated in three steps. Each step is the subject of one of the next three sections; the purpose here is to show how they fit together and where each draws on the analytic structure of chapter 4.

5.1.2 Key Ideas: The Plan of the Proof The proof of $\vartheta(x) \sim x$ proceeds in three steps.

1. *The prime series extends to the line.* The prime series $\Phi(s) = \sum_p (\log p)\, p^{-s}$ of definition 4.3.2, holomorphic on $\mathrm{Re}(s) > 1$, has a simple pole at $s = 1$ of residue 1 and extends holomorphically to the closed half-plane $\mathrm{Re}(s) \geq 1$ with no singularity other than that pole. This is the analytic heart of the argument, and it is where the non-vanishing of ζ enters: by proposition 4.3.4 the difference $-\zeta'(s)/\zeta(s) - \Phi(s)$ is holomorphic on $\mathrm{Re}(s) > 1/2$, so Φ and $-\zeta'/\zeta$ have the same singularities up to that line, and by corollary 4.4.4 the function $-\zeta'/\zeta$ is holomorphic on all of $\mathrm{Re}(s) = 1$ except for the simple pole at $s = 1$, precisely because ζ has no zeros on the line (theorem 4.4.2).

2. *An integral converges.* The extension of Φ to the line forces the convergence of the integral
$$\int_1^\infty \frac{\vartheta(x) - x}{x^2}\, dx$$
The passage from step (1) to this convergence runs through a Laplace transform: Φ is recast as the Laplace transform of a function built from ϑ (the subject of section 5.2), and the holomorphic extension of that transform to the boundary line is exactly the hypothesis of the Tauberian theorem of Newman established in section 5.3, whose conclusion is the convergence of the integral. This step is carried out in section 5.4.

3. *Convergence forces the asymptotic.* The convergence of $\int_1^\infty (\vartheta(x) - x)/x^2\, dx$, combined with the fact that ϑ is nondecreasing, forces $\vartheta(x) \sim x$. A monotone function whose normalized deviation from x has a convergent integral cannot stay bounded away from x: were $\vartheta(x) \geq \lambda x$ for some $\lambda > 1$ on a sequence of points, or $\vartheta(x) \leq \lambda x$ for some $\lambda < 1$, monotonicity would make the integrand bounded below by a positive constant over intervals of fixed logarithmic length, contradicting convergence. This final step, also in section 5.4, yields $\vartheta(x)/x \to 1$ and hence, by proposition 5.1.1, the Prime Number Theorem.

The three steps trace a path from the complex plane to the real line: a holomorphic extension across $\mathrm{Re}(s) = 1$ in step (1), a statement about a real integral in step (2), and a real-variable asymptotic in step (3). The transitions between them are the substance of the next three sections, and each transition is where a piece of analysis does work that no elementary argument could. What the roadmap makes visible, and what is worth dwelling on before the details, is why the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ is the one input on which everything turns.

The decisive point is step (1), and within it the holomorphic extension of Φ to the closed half-plane. The series $\Phi(s) = \sum_p (\log p) p^{-s}$ converges only on $\mathrm{Re}(s) > 1$; by itself it gives no value on the boundary, and the convergence of the integral in step (2) is a statement about the behavior of ϑ that is sensitive to exactly that boundary. The bridge is the identity $-\zeta'(s)/\zeta(s) = \Phi(s) + R(s)$ of proposition 4.3.4, in which R is holomorphic well past the line, on $\mathrm{Re}(s) > 1/2$. This identity transfers the question about Φ on the line into a question about $-\zeta'/\zeta$ on the line, and the singularities of $-\zeta'/\zeta$ are dictated entirely by the zeros and the pole of ζ. The pole of ζ at $s = 1$ produces the one permitted singularity, the simple pole of Φ of residue 1, which becomes the main term in the asymptotic. Every zero of ζ on the line would produce an additional pole of $-\zeta'/\zeta$, hence of Φ, and would destroy the holomorphic extension that step (1) asserts. A single zero at a point $1 + it_0$ with $t_0 \neq 0$ would be a pole of Φ off the real axis, the Laplace transform of step (2) would fail to extend holomorphically to that point of the boundary, the hypothesis of Newman's theorem would not hold, and the integral would have no reason to converge. The non-vanishing $\zeta(1+it) \neq 0$ for all real $t \neq 0$, proved in theorem 4.4.2, is precisely the statement that no such obstruction exists: the only singularity of Φ on the closed half-plane $\mathrm{Re}(s) \geq 1$ is the pole at $s = 1$, and the extension of step (1) goes through. This is the sense in which the distribution of primes is governed by a single vertical line in the complex plane, and why the analytic input that the elementary theory of chapter 3 could not supply, the absence of zeros on that line, is the input that completes the proof.

5.2 The Laplace Transform

The strategy of section 5.1 reduces the Prime Number Theorem to the single asymptotic $\vartheta(x) \sim x$, and proposition 5.1.1 traces the chain that delivers $\pi(x) \sim x/\log x$ once that asymptotic is in hand. What remains is to prove $\vartheta(x) \sim x$ from the analytic data assembled in chapter 4: the prime series $\Phi(s) = \sum_p (\log p)p^{-s}$ (definition 4.3.2), its simple pole at $s = 1$ with residue 1 (proposition 4.3.4), and the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ (corollary 4.4.4). The bridge between the Dirichlet-series object Φ and the counting function ϑ is an integral transform, the *Laplace transform*, and the route runs through the substitution $x = e^t$ that turns the counting function $\vartheta(x)$ on $[1, \infty)$ into a function of t on $[0, \infty)$.

This section records the Laplace transform and computes the two integrals the proof in section 5.3 and section 5.4 requires. The first expresses the transform of $\vartheta(e^t)$ in terms of $\Phi(s)/s$; the second packages this into an auxiliary function $g(s)$ whose holomorphic extension to the closed half-plane $\mathrm{Re}(s) \geq 0$ is the one analytic input Newman's tauberian theorem will consume.

Definition 5.2.1: The Laplace Transform

Let $f\colon [0, \infty) \to \mathbb{C}$ be locally integrable. The *Laplace transform* of f is the function

$$(\mathcal{L}f)(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

defined for those $s \in \mathbb{C}$ for which the integral converges.

The Laplace transform is the one-sided companion of the Mellin transform of section 2.4, and the two are linked by the same change of variables that linked the Mellin transform to a two-sided Laplace transform in proposition 2.4.3. Under the substitution $t = e^{-x}$, the Mellin transform $(\mathcal{M}F)(s) = \int_0^\infty F(t)t^{s-1}\, dt$ becomes the two-sided Laplace integral $\int_{-\infty}^\infty F(e^{-x})e^{-sx}\, dx$; restricting a function to the half-line and integrating only over $t \geq 0$ gives the one-sided transform of definition 5.2.1. As with both of those transforms, the modulus $|e^{-st}| = e^{-\mathrm{Re}(s)t}$ depends only on $\mathrm{Re}(s)$, so the region of convergence is a half-plane $\mathrm{Re}(s) > \sigma_0$ invariant under vertical translation, the abscissa σ_0 depending on the growth of f.

Example 5.2.2: The Laplace Transform of a Constant

Take $f(t) = 1$ for all $t \geq 0$. The transform is

$$(\mathcal{L}1)(s) = \int_0^\infty e^{-st}\, dt = \left[-\frac{1}{s}\, e^{-st}\right]_{t=0}^{t=\infty} = \frac{1}{s}$$

for $\mathrm{Re}(s) > 0$. The boundary evaluation at $t = \infty$ vanishes because $|e^{-st}| = e^{-\mathrm{Re}(s)t} \to 0$ exactly when $\operatorname{Re}(s) > 0$, and the value at $t = 0$ is $-1/s$; the abscissa of convergence is $\sigma_0 = 0$. The transform $1/s$ is holomorphic on $\operatorname{Re}(s) > 0$ and continues meromorphically to $\mathbb{C}$ with a single simple pole at $s = 0$, the boundary point of the half-plane of convergence. This $1/s$ is the term that will subtract against the pole of $\Phi(s+1)/(s+1)$ at $s = 0$ in proposition 5.2.4.

The function whose transform the proof needs is $\vartheta(e^t)$, the Chebyshev theta function (definition 3.1.2) reparametrized by $t = \log x$. Its transform is, up to the factor $1/s$, the prime series Φ.

Proposition 5.2.3: The Laplace Transform of $\vartheta(e^t)$

For $\operatorname{Re}(s) > 1$,

$$\int_0^\infty \vartheta(e^t)\, e^{-st}\, dt = \int_1^\infty \vartheta(x)\, x^{-s-1}\, dx = \frac{\Phi(s)}{s}$$

where $\Phi(s) = \sum_p (\log p)\, p^{-s}$ is the prime series of definition 4.3.2.

***Proof*:**
Throughout write $\sigma = \operatorname{Re}(s)$ and assume $\sigma > 1$.

Step 1: The change of variables $x = e^t$. On the integral $\int_0^\infty \vartheta(e^t)e^{-st}\, dt$ substitute $x = e^t$, so $t = \log x$, $dt = dx/x$, and $e^{-st} = x^{-s}$. As t runs over $[0, \infty)$ the variable x runs over $[1, \infty)$, and

$$\int_0^\infty \vartheta(e^t)\, e^{-st}\, dt = \int_1^\infty \vartheta(x)\, x^{-s}\, \frac{dx}{x} = \int_1^\infty \vartheta(x)\, x^{-s-1}\, dx$$

The substitution $x = e^t$ is a C^1 diffeomorphism of $[0, \infty)$ onto $[1, \infty)$, so it preserves absolute convergence; convergence of the resulting integral is verified in Step 3. This establishes the first equality.

Step 2: Abel summation against $d\vartheta$. The function ϑ is the summatory function of the coefficients $a_n = \log n$ for n prime and $a_n = 0$ otherwise, so $\vartheta(x) = \sum_{n \le x} a_n$, and its Stieltjes measure $d\vartheta(x)$ places mass $\log p$ at each prime p. Apply Abel summation (theorem 1.4.1) in its Stieltjes form to the integral $\int_1^\infty \vartheta(x)x^{-s-1}\, dx$. Take the test function $f(x) = x^{-s}$, which is continuously differentiable on $[1, \infty)$ with $f'(x) = -s\, x^{-s-1}$, so that $x^{-s-1} = -f'(x)/s$. For a finite upper limit $X \ge 1$, integration by parts for the Riemann–Stieltjes integral gives

$$\int_1^X \vartheta(x)\, x^{-s-1}\, dx = -\frac{1}{s} \int_1^X \vartheta(x)\, f'(x)\, dx = -\frac{1}{s} \left(\Big[\vartheta(x) f(x)\Big]_1^X - \int_1^X f(x)\, d\vartheta(x) \right)$$

the boundary term and the Stieltjes integral arising from $\int u\, dv = [uv] - \int v\, du$ with $u = \vartheta$, $v = f$. Since $\vartheta(x) = 0$ for $x < 2$, the boundary contribution at $x = 1$ vanishes, and $\int_1^X f(x)\, d\vartheta(x) = \sum_{p \le X} (\log p)\, p^{-s}$ because $d\vartheta$ assigns mass $\log p$ to each prime $p \le X$ and the integrand f is evaluated at $x = p$. Therefore

$$\int_1^X \vartheta(x)\, x^{-s-1}\, dx = -\frac{1}{s}\, \vartheta(X)\, X^{-s} + \frac{1}{s} \sum_{p \le X} \frac{\log p}{p^s}$$

Step 3: The boundary term vanishes and the limit is $\Phi(s)/s$. By Chebyshev's estimates (theorem 3.2.3), $\vartheta(x) = O(x)$, so there is a constant C with $\vartheta(X) \le CX$ for all $X \ge 1$. Hence

$$\left|\vartheta(X)\, X^{-s}\right| = \vartheta(X)\, X^{-\sigma} \le C\, X^{1-\sigma}$$

and since $\sigma > 1$ the exponent $1 - \sigma$ is negative, so $X^{1-\sigma} \to 0$ as $X \to \infty$; the boundary term tends to 0. The same bound shows the integral converges absolutely: $|\vartheta(x)\, x^{-s-1}| = \vartheta(x)\, x^{-\sigma-1} \le Cx \cdot x^{-\sigma-1} = Cx^{-\sigma}$, and $\int_1^\infty x^{-\sigma}\, dx < \infty$ because $\sigma > 1$. Letting $X \to \infty$,

$$\int_1^\infty \vartheta(x)\, x^{-s-1}\, dx = \frac{1}{s} \sum_p \frac{\log p}{p^s} = \frac{\Phi(s)}{s}$$

the sum on the right being $\Phi(s)$ by definition 4.3.2, convergent on $\sigma > 1$. Combining with Step 1 gives all three equalities, as we sought to show.

The proposition is the analytic dictionary entry that the whole proof turns on: it converts the prime series Φ, an object built from the primes through a Dirichlet series, into the Laplace transform of the counting function $\vartheta(e^t)$, an object built from the same primes through their summatory behavior. The factor $1/s$ is the signature of summation: passing from the coefficients $\log p$ to their partial sums ϑ divides the generating function by s, exactly as the Laplace transform of a constant in example 5.2.2 produced $1/s$ from the summatory function of the unit mass at the origin. The pole of Φ at $s = 1$ (proposition 4.3.4) therefore becomes a pole of $\Phi(s)/s$ at $s = 1$, with the kernel factor $1/s$ contributing the value 1 there, so the residue is again 1; this is the pole that will furnish the main term $\vartheta(x) \sim x$. The representation is valid only on $\mathrm{Re}(s) > 1$, where both Φ converges and the boundary term decays, and the entire difficulty of the Prime Number Theorem lies in pushing analytic control of this transform onto the line $\mathrm{Re}(s) = 1$.

That push is mediated by a single auxiliary function. Newman's tauberian theorem, the analytic engine of section 5.3, applies to the Laplace transform of a *bounded* function whose transform extends holomorphically across the imaginary axis. The function $\vartheta(e^t)$ is not bounded, but the normalized quantity $\vartheta(e^t)e^{-t}$ is, by Chebyshev's estimates, and subtracting its limiting value 1 produces a function tending to a transform with the required regularity.

Proposition 5.2.4: The Auxiliary Function $g(s)$

Define $f\colon [0, \infty) \to \mathbb{R}$ by

$$f(t) = \vartheta(e^t)\, e^{-t} - 1$$

and let $g(s) = \int_0^\infty f(t)\, e^{-st}\, dt$ be its Laplace transform. Then for $\mathrm{Re}(s) > 0$,

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

***Proof*:**
Write $\sigma = \mathrm{Re}(s)$ and assume $\sigma > 0$. The integrand splits as $f(t)e^{-st} = \vartheta(e^t)e^{-t}e^{-st} - e^{-st} = \vartheta(e^t)e^{-(s+1)t} - e^{-st}$, so over any finite range $[0, T]$,

$$\int_0^T f(t)\, e^{-st}\, dt = \int_0^T \vartheta(e^t)\, e^{-(s+1)t}\, dt - \int_0^T e^{-st}\, dt$$

the splitting being legitimate on a bounded interval where each piece is integrable. The two pieces are handled separately, then the limit $T \to \infty$ is taken.

Step 1: The first integral. Since $\sigma > 0$, the shifted argument satisfies $\mathrm{Re}(s+1) = \sigma + 1 > 1$, so proposition 5.2.3 applies at $s+1$ in place of s:

$$\int_0^\infty \vartheta(e^t)\, e^{-(s+1)t}\, dt = \frac{\Phi(s+1)}{s+1}$$

the integral converging absolutely because $\mathrm{Re}(s+1) > 1$. In particular the partial integrals $\int_0^T \vartheta(e^t)e^{-(s+1)t}\, dt$ converge to $\Phi(s+1)/(s+1)$ as $T \to \infty$.

Step 2: The second integral. By example 5.2.2, for $\sigma > 0$,

$$\int_0^\infty e^{-st}\, dt = \frac{1}{s}$$

so $\int_0^T e^{-st}\, dt \to 1/s$ as $T \to \infty$.

Step 3: Combine. Both pieces converge as $T \to \infty$, so their difference converges and $g(s) = \int_0^\infty f(t)e^{-st}\, dt$ exists for $\sigma > 0$, with

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

completing the proof.

The function g packages exactly the data Newman's theorem will need. Its definition is engineered so that the integrand $f(t) = \vartheta(e^t)e^{-t} - 1$ is bounded: Chebyshev's estimates (theorem 3.2.3) give $\vartheta(x) \le (\log 4)\, x$ for all $x \ge 1$, while $\vartheta(x) \ge 0$ holds trivially, so $0 \le \vartheta(x)/x \le \log 4$ for every $x \ge 1$ and the normalized quantity $\vartheta(e^t)e^{-t} = \vartheta(e^t)/e^t$ stays inside $[0, \log 4]$. Hence $f(t) = \vartheta(e^t)/e^t - 1$ stays inside the bounded range $[-1, \log 4 - 1]$ for all $t \ge 0$; the lower Chebyshev bound $\vartheta(x) \ge c_1 x$, valid for large x, pins $f(t)$ eventually above $c_1 - 1$ as well. Boundedness of f on $[0, \infty)$ is the hypothesis the tauberian theorem of section 5.3 imposes on the function being transformed. The formula $g(s) = \Phi(s+1)/(s+1) - 1/s$ then exhibits g, on $\mathrm{Re}(s) > 0$, as a difference of two functions each carrying a pole at $s = 0$. The shifted prime series $\Phi(s+1)$ has its pole at $s+1 = 1$, that is, at $s = 0$, with residue 1 (proposition 4.3.4), so $\Phi(s+1)/(s+1)$ has a simple pole at $s = 0$ with residue 1; the second term $1/s$ has a simple pole at $s = 0$ with residue 1 as well. The two principal parts cancel: g is holomorphic at $s = 0$, where it would otherwise inherit the singularity of Φ.

This cancellation is the crux. The entire proof of the Prime Number Theorem reduces to showing that g extends holomorphically to the closed half-plane $\mathrm{Re}(s) \ge 0$, and the analytic facts of chapter 4 are precisely what make that extension possible. At $s = 0$ the pole of Φ at the point $s = 1$ is exactly cancelled by the term $1/s$, as computed above; this is where the residue-1 pole of Φ enters, and it is the only point of the line $\mathrm{Re}(s) = 0$ at which $\Phi(s+1)$ is singular. At every other point $s = it$ with $t \ne 0$, the shifted argument $s + 1 = 1 + it$ lies on the line $\mathrm{Re} = 1$ away from the pole, where Φ is holomorphic because ζ has no zeros there (corollary 4.4.4), the non-vanishing of ζ on $\mathrm{Re}(s) = 1$ established in chapter 4. The two inputs together, the pole of Φ at $s = 1$ cancelling $1/s$ and the non-vanishing of ζ on the rest of the line, give g a holomorphic extension to a neighborhood of

$\mathrm{Re}(s) \geq 0$, the hypothesis that drives section 5.3 and section 5.4.

A numerical look at f makes its boundedness, the structural feature that lets the argument proceed, concrete.

Example 5.2.5: g at a Real Point

The integrand $f(t) = \vartheta(e^t)/e^t - 1$ measures the deviation of $\vartheta(x)/x$ from 1 at $x = e^t$. Tabulating $\vartheta(x)/x - 1$ at a range of x, using the values of $\vartheta(x)$ from example 3.1.3 together with further computation, gives

x	$\vartheta(x)/x$	$\vartheta(x)/x - 1$
10	0.5347	-0.4653
100	0.8373	-0.1627
1000	0.9562	-0.0438
10000	0.9896	-0.0104
100000	0.9969	-0.0031

Every entry in the last column lies in the interval $[-1, \log 4 - 1] \approx [-1, 0.386]$ guaranteed by theorem 3.2.3, since $0 \leq \vartheta(x)/x \leq \log 4$ for all $x \geq 1$; the upper constant $\log 4 \approx 1.386$ caps $\vartheta(x)/x$ for every x, and the lower Chebyshev constant $\log 2 \approx 0.693$ confines $\vartheta(x)/x$ from below once x is large, so $\vartheta(x)/x - 1$ cannot escape a fixed range no matter how large x grows. The table also shows the values drifting toward 0, consistent with the conjectured limit $\vartheta(x)/x \to 1$; but the proof does not assume this convergence. It assumes only the boundedness visible in the table, that $f(t)$ stays within fixed bounds for all $t \geq 0$. This boundedness is exactly the hypothesis Newman's theorem (section 5.3) will require of f, and the convergence $\vartheta(x)/x \to 1$ is the conclusion the theorem will deliver, not an input it demands.

5.3 Newman's Tauberian Theorem

The analytic engine of the proof of the Prime Number Theorem is a single theorem of D. J. Newman about Laplace transforms [16], in the streamlined form given by Zagier [24]. The Perron representation of example 2.5.4 expresses the prime-counting data through a contour integral of $-\zeta'/\zeta$, and chapter 4 supplied the analytic input that the integrand has no pole on the line $\mathrm{Re}(s) = 1$ except the one at $s = 1$. What remains is to convert that holomorphy-up-to-the-line into an asymptotic for the counting function, and the conversion is exactly the business of a *Tauberian* theorem: a result that upgrades a statement about the boundary behavior of a transform into a statement about convergence of the function it transforms. Newman's theorem is the cleanest such statement adapted to the situation at hand, and its proof is a short contour argument requiring nothing past Cauchy's integral formula.

The shape of the statement is the following. A bounded function f on $[0, \infty)$ has a Laplace transform $g(s) = \int_0^\infty f(t)e^{-st}\,dt$ that converges and is holomorphic on the open half-plane $\mathrm{Re}(s) > 0$. Convergence of the *improper integral* $\int_0^\infty f(t)\,dt$ is the assertion that g extends continuously to the boundary point $s = 0$ with the limit equal to the integral, and this is a strictly stronger demand than boundedness of f alone. Newman's theorem supplies the missing hypothesis in the form most useful for the zeta function: if g extends not just continuously but *holomorphically* across the imaginary axis, then the improper integral converges and equals $g(0)$. Holomorphy up to the line is the Tauberian side condition, and boundedness of f is the standing hypothesis that the application will draw from Chebyshev's bound.

Theorem 5.3.1: Newman's Analytic Theorem

Let $f\colon [0,\infty) \to \mathbb{C}$ be bounded and integrable (Riemann or Lebesgue) on every finite interval $[0,T]$, and for $\mathrm{Re}(s) > 0$ define

$$g(s) = \int_0^\infty f(t)e^{-st}\,dt$$

The integral converges absolutely and g is holomorphic on $\mathrm{Re}(s) > 0$. Suppose g extends to a holomorphic function on an open set containing the closed half-plane $\mathrm{Re}(s) \geq 0$. Then the improper integral $\int_0^\infty f(t)\,dt$ converges and

$$\int_0^\infty f(t)\,dt = g(0)$$

***Proof*:**
Write $B = \sup_{t\geq 0} |f(t)|$, finite by hypothesis. For $\mathrm{Re}(s) > 0$ the bound $|f(t)e^{-st}| \leq Be^{-\mathrm{Re}(s)t}$ and $\int_0^\infty e^{-\mathrm{Re}(s)t}\,dt = 1/\mathrm{Re}(s) < \infty$ show that the defining integral converges absolutely; that g is holomorphic on $\mathrm{Re}(s) > 0$ is the holomorphy of a parameter integral whose integrand is holomorphic in s and dominated locally uniformly, established in [13]. The substance is the boundary statement, proved by a contour integral.

Step 1: The truncated transforms and the goal. For each $T > 0$ set

$$g_T(s) = \int_0^T f(t)e^{-st}\,dt$$

The integrand is entire in s for each fixed t, and the integral runs over the finite interval $[0,T]$, so g_T is entire (again by holomorphy of a parameter integral, now over a compact interval, [13]). The improper integral $\int_0^\infty f(t)\,dt$ is by definition $\lim_{T\to\infty} \int_0^T f(t)\,dt = \lim_{T\to\infty} g_T(0)$, so the assertion to be proved is exactly

$$g_T(0) \longrightarrow g(0) \qquad (T \to \infty)$$

The difference $g(0) - g_T(0)$ will be written as a contour integral and estimated.

Step 2: A contour and the Cauchy representation. Fix $R > 0$ for the remainder of the argument. By hypothesis g is holomorphic on an open set $\Omega \supseteq \{\mathrm{Re}(s) \geq 0\}$. The intersection of Ω with the disc $\{|s| \leq R\}$ is an open neighborhood of the compact arc $\{|s| = R,\ \mathrm{Re}(s) \geq 0\} \cup \{\mathrm{Re}(s) = 0,\ |s| \leq R\}$, so there is a $\delta = \delta(R) > 0$, which may be taken with $\delta < R$, such that g is holomorphic on a neighborhood of the closed region

$$D = \{\, s \big| |s| \leq R,\ \mathrm{Re}(s) \geq -\delta \,\}$$

Let $C = \partial D$ be its positively oriented boundary, consisting of the major arc $A = \{|s| = R,\ \mathrm{Re}(s) \geq -\delta\}$ of the circle of radius R together with the vertical segment $\{\mathrm{Re}(s) = -\delta,\ |\mathrm{Im}(s)| \leq \sqrt{R^2 - \delta^2}\}$. The point $s = 0$ lies in the interior of D, and g_T is entire, so the function

$$s \longmapsto \big(g(s) - g_T(s)\big)\, e^{sT} \left(\frac{1}{s} + \frac{s}{R^2}\right)$$

is holomorphic on a neighborhood of D except for a simple pole at $s = 0$. The factor $1/s + s/R^2$ has residue 1 at $s = 0$ (the term s/R^2 is regular there), and $e^{sT} = 1$ at $s = 0$, so the residue of the displayed function at $s = 0$ is $g(0) - g_T(0)$. Cauchy's residue theorem ([13]) gives

$$g(0) - g_T(0) = \frac{1}{2\pi i} \int_C \big(g(s) - g_T(s)\big)\, e^{sT} \left(\frac{1}{s} + \frac{s}{R^2}\right) ds \tag{5.1}$$

The plan is to split C into the part in $\mathrm{Re}(s) > 0$ and the part in $\mathrm{Re}(s) \le 0$, and to bound each.

Step 3: The kernel on the circle $|s| = R$. On the circle $|s| = R$ one has $\bar{s} = R^2/s$, hence

$$\frac{1}{s} + \frac{s}{R^2} = \frac{\bar{s}}{R^2} + \frac{s}{R^2} = \frac{s + \bar{s}}{R^2} = \frac{2\mathrm{Re}(s)}{R^2}$$

so that on $|s| = R$,

$$\left| e^{sT} \left(\frac{1}{s} + \frac{s}{R^2}\right) \right| = e^{\mathrm{Re}(s)T}\, \frac{2\,|\mathrm{Re}(s)|}{R^2} \tag{5.2}$$

This identity is the reason the second term s/R^2 was inserted; it is recorded for use on both halves of the contour.

Step 4: The right half $C_+ = C \cap \{\mathrm{Re}(s) > 0\}$. Here C_+ is the right portion $\{|s| = R,\ \mathrm{Re}(s) > 0\}$ of the circle, a semicircle of length πR. For $\mathrm{Re}(s) > 0$ the tail estimate

$$|g(s) - g_T(s)| = \left| \int_T^\infty f(t) e^{-st}\, dt \right| \le B \int_T^\infty e^{-\mathrm{Re}(s)t}\, dt = \frac{B\, e^{-\mathrm{Re}(s)T}}{\mathrm{Re}(s)}$$

holds, where the first equality uses that the integral defining $g(s)$ converges and exceeds that defining $g_T(s)$ by the tail over $[T, \infty)$. Multiplying by the kernel bound (5.2) and using $|\mathrm{Re}(s)| = \mathrm{Re}(s) > 0$,

$$\left| \big(g(s) - g_T(s)\big)\, e^{sT} \left(\frac{1}{s} + \frac{s}{R^2}\right) \right| \le \frac{B\, e^{-\mathrm{Re}(s)T}}{\mathrm{Re}(s)} \cdot e^{\mathrm{Re}(s)T}\, \frac{2\mathrm{Re}(s)}{R^2} = \frac{2B}{R^2}$$

The integrand on C_+ is bounded by $2B/R^2$ pointwise, so by the standard length estimate ([13]) the contribution of C_+ to (5.1) satisfies

$$\left| \frac{1}{2\pi i} \int_{C_+} \big(g(s) - g_T(s)\big)\, e^{sT} \left(\frac{1}{s} + \frac{s}{R^2}\right) ds \right| \le \frac{1}{2\pi} \cdot \frac{2B}{R^2} \cdot \pi R = \frac{B}{R}$$

This bound is uniform in T.

Step 5: The left half, the g_T part. Let $C_- = C \cap \{\mathrm{Re}(s) \le 0\}$ be the remainder of the contour: the left portion of the arc A together with the vertical segment at $\mathrm{Re}(s) = -\delta$. On C_- the integrand of (5.1) is treated by separating g from g_T, which is legitimate because each piece is integrable on the compact contour. Consider first the g_T piece,

$$I_T^- = \frac{1}{2\pi i}\int_{C_-} g_T(s)\, e^{sT}\left(\frac{1}{s}+\frac{s}{R^2}\right) ds$$

Since g_T is entire, the integrand $g_T(s)\, e^{sT}(1/s + s/R^2)$ is holomorphic on the punctured plane $\mathbb{C}\setminus\{0\}$, in particular on the open left half-plane $\{\mathrm{Re}(s) < 0\}$. The contour C_- and the left semicircle

$$C'_- = \{\, |s| = R,\ \mathrm{Re}(s) \le 0\,\}$$

share the two endpoints $\pm iR$, and the closed region they bound lies entirely in $\{\mathrm{Re}(s) \le 0\}$, with the only point of nonholomorphy $s = 0$ on its boundary rather than its interior. Cauchy's theorem ([13]) therefore permits replacing C_- by C'_- in I_T^- without changing its value. On C'_- one has $\mathrm{Re}(s) \le 0$, and for such s the entire transform is bounded by

$$|g_T(s)| = \left|\int_0^T f(t)e^{-st}\, dt\right| \le B\int_0^T e^{-\mathrm{Re}(s)t}\, dt = B\,\frac{e^{-\mathrm{Re}(s)T}-1}{-\mathrm{Re}(s)} \le \frac{B\, e^{-\mathrm{Re}(s)T}}{|\mathrm{Re}(s)|}$$

using $-\mathrm{Re}(s) = |\mathrm{Re}(s)| > 0$ on the open left semicircle and $e^{-\mathrm{Re}(s)T} - 1 \le e^{-\mathrm{Re}(s)T}$. Multiplying by the kernel bound (5.2),

$$\left|g_T(s)\, e^{sT}\left(\frac{1}{s}+\frac{s}{R^2}\right)\right| \le \frac{B\, e^{-\mathrm{Re}(s)T}}{|\mathrm{Re}(s)|}\cdot e^{\mathrm{Re}(s)T}\,\frac{2\,|\mathrm{Re}(s)|}{R^2} = \frac{2B}{R^2}$$

the factors $e^{\pm\mathrm{Re}(s)T}$ and $|\mathrm{Re}(s)|$ cancelling exactly as on the right half. The semicircle C'_- has length πR, so the length estimate gives

$$|I_T^-| \le \frac{1}{2\pi}\cdot\frac{2B}{R^2}\cdot\pi R = \frac{B}{R}$$

again uniformly in T.

Step 6: The left half, the g part. The complementary piece is

$$J_T^- = \frac{1}{2\pi i}\int_{C_-} g(s)\, e^{sT}\left(\frac{1}{s}+\frac{s}{R^2}\right) ds$$

Here R, hence δ and the contour C_-, are fixed, and $T \to \infty$. The function $g(s)\,(1/s + s/R^2)$ is continuous on the compact set C_-: the only candidate for a singularity is $s = 0$, but C_- has $\mathrm{Re}(s) \le 0$ and meets the imaginary axis only at the two points $\pm iR$, where $|s| = R \ne 0$, so $s = 0 \notin C_-$ and the factor $1/s$ is bounded there. Let

$$M = \max_{s\in C_-}\left|g(s)\left(\frac{1}{s}+\frac{s}{R^2}\right)\right| < \infty$$

a constant depending on R and δ but not on T. The remaining factor satisfies $|e^{sT}| = e^{\mathrm{Re}(s)T} \le 1$ on C_-, since $\mathrm{Re}(s) \le 0$, so the integrand of J_T^- is dominated by the constant M on the finite-length contour C_-, uniformly in T. For each fixed $s \in C_-$ with $\mathrm{Re}(s) < 0$ one has $e^{\mathrm{Re}(s)T} \to 0$ as $T \to \infty$; the exceptional points $\pm iR$, where $\mathrm{Re}(s) = 0$, form a set of measure zero on the contour. By the dominated convergence theorem ([14]), applied to the arc-length integral with dominating function M,

$$J_T^- \longrightarrow 0 \qquad (T \to \infty)$$

This is the only one of the three pieces that depends on letting T grow, and it is where holomorphy of g across the line is used: only because g continues to $\mathrm{Re}(s) = -\delta$ can the contour C_- be pushed off the imaginary axis into the region $\mathrm{Re}(s) < 0$ where e^{sT} decays.

Step 7: Assembling the estimate. Splitting (5.1) as $C = C_+ \cup C_-$ and the C_- integral as its g and g_T parts,

$$g(0) - g_T(0) = \underbrace{\frac{1}{2\pi i}\int_{C_+} (\cdots)\,ds}_{\text{Step 4}} \;-\; \underbrace{I_T^-}_{\text{Step 5}} \;+\; \underbrace{J_T^-}_{\text{Step 6}}$$

where the sign on I_T^- records that the g_T contribution over C_- enters (5.1) as $\int_{C_-}(g - g_T) = J_T^- - I_T^-$. Steps 4 and 5 bound the first two terms by B/R each, uniformly in T, and Step 6 sends the third to 0. Hence

$$\limsup_{T\to\infty} |g(0) - g_T(0)| \le \frac{B}{R} + \frac{B}{R} + 0 = \frac{2B}{R}$$

The left-hand side does not depend on R, while the right-hand side is $2B/R$ for every $R > 0$. Letting $R \to \infty$ forces $\limsup_{T\to\infty} |g(0) - g_T(0)| = 0$, so $g_T(0) \to g(0)$. By Step 1 this is the statement that $\int_0^\infty f(t)\,dt$ converges with value $g(0)$, completing the proof.

The hypothesis does double duty, and its two halves play distinct roles. Boundedness of f is what makes the tail estimates of Steps 4 and 5 produce the bound $2B/R^2$ on the circle; without a uniform bound on f the kernel could not absorb the growth of $e^{\mathrm{Re}(s)T}$ on the right, and the argument would collapse. Holomorphy of g across the line is the Tauberian content: it is the hypothesis that lets the contour in Step 6 be displaced into $\{\mathrm{Re}(s) < 0\}$, where the exponential e^{sT} decays and the g-integral vanishes in the limit. Strip away the holomorphy and only boundedness of f remains, which guarantees that g is defined and holomorphic on $\mathrm{Re}(s) > 0$ but says nothing about the boundary; the improper integral $\int_0^\infty f(t)\,dt$ may then diverge by oscillation even though f is bounded, as the example $f(t) = \cos t$ shows, where $g(s) = s/(s^2+1)$ has poles at $s = \pm i$ on the imaginary axis and the holomorphy hypothesis fails.

The name *Tauberian* marks the logical direction of the theorem. An *Abelian* theorem deduces the behavior of a transform from the convergence of the function: if $\int_0^\infty f(t)\,dt$ converges, then $g(s) \to g(0)$ as $s \to 0^+$, a statement of continuity that is comparatively soft. A Tauberian theorem reverses the implication, recovering convergence of $\int_0^\infty f(t)\,dt$ from analytic regularity of g at the boundary, and such reversals are false without a side condition that restrains the function from oscillating, the *Tauberian condition.* In Newman's theorem the side condition is the boundedness of f together with the holomorphic extension of g, and the value $g(0)$ that the holomorphic transform assigns to the boundary point is promoted from a formal or Abel-summed value into the genuine value of the improper integral.

5.3.2 Key Ideas: The Kernel $e^{sT}(1/s + s/R^2)$ The factor $e^{sT}(1/s + s/R^2)$ in (5.1) is engineered so that a single contour integral does three things at once. The term $1/s$ carries a simple pole at $s = 0$ with residue 1; against the holomorphic factor $g(s) - g_T(s)$ it produces the residue $g(0) - g_T(0)$, which is the quantity to be estimated. The factor e^{sT} is the device that lets the truncation $g - g_T$ be controlled: on the right half-plane the tail $g(s) - g_T(s)$ carries a decay $e^{-\mathrm{Re}(s)T}$, and the inserted e^{sT} cancels it, while on the left half-plane the same e^{sT} decays and kills the g-integral as $T \to \infty$. The role of the added term s/R^2 is to tame the kernel on the circle $|s| = R$: by the identity $1/s + s/R^2 = 2\mathrm{Re}(s)/R^2$ valid on that circle, the modulus of the kernel collapses to $e^{\mathrm{Re}(s)T}\,(2|\mathrm{Re}(s)|/R^2)$, in which the prefactor $2|\mathrm{Re}(s)|/R^2$ supplies exactly the factor $\mathrm{Re}(s)$ needed to cancel the $1/\mathrm{Re}(s)$ from the tail bound $Be^{-\mathrm{Re}(s)T}/\mathrm{Re}(s)$, and the residual $e^{\pm\mathrm{Re}(s)T}$ cancels the exponential. The net bound $2B/R^2$ on both semicircles is symmetric in the two halves precisely because the kernel modulus depends only on $|\mathrm{Re}(s)|$. Without the s/R^2 term the bare $1/s$ would contribute $1/R$ on the circle with no $\mathrm{Re}(s)$ factor to absorb the $1/\mathrm{Re}(s)$ from the tail, and the right-half estimate would diverge as s approaches the imaginary axis; the corrective term costs nothing at $s = 0$, where it is regular, so the pole and its residue are untouched.

The application in section 5.4 fits the theorem exactly. The function to which Newman's theorem will be applied is built from the Chebyshev theta function ϑ (definition 3.1.2), arranged so that the boundedness hypothesis becomes the assertion that $\vartheta(x)/x$ is bounded, which is the content of Chebyshev's estimates $c_1 x \le \vartheta(x) \le c_2 x$ in theorem 3.2.3. The holomorphic-extension hypothesis becomes the non-vanishing of ζ on $\mathrm{Re}(s) = 1$ from theorem 4.4.2, in the form that $-\zeta'/\zeta$, and with it the relevant transform, continues across the line save for the controlled pole at $s = 1$ (corollary 4.4.4). Newman's theorem then returns the convergence of an improper integral in ϑ, and a short argument converts that convergence into $\vartheta(x) \sim x$, hence into the Prime Number Theorem through the equivalence of theorem 3.1.5. The theorem proved here is thus the one analytic lever that turns the boundary data on ζ into an asymptotic count of primes.

5.4 Proof of the Prime Number Theorem

The pieces assembled in the three preceding sections now combine into the theorem. The auxiliary function g of section 5.2, defined on $\mathrm{Re}(s) > 0$ as the Laplace transform of $f(t) = \vartheta(e^t)e^{-t} - 1$, was computed in proposition 5.2.4 to be $g(s) = \Phi(s + 1)/(s + 1) - 1/s$. The pole of Φ at $s = 1$ (proposition 4.3.4) cancels against the term $1/s$ at $s = 0$, and the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ (corollary 4.4.4, equivalently theorem 4.4.2) removes every other obstruction on the imaginary axis; together they extend g holomorphically across $\mathrm{Re}(s) = 0$. Newman's theorem (theorem 5.3.1) then converts that extension into the convergence of $\int_0^\infty f(t)\,dt$, which the substitution $x = e^t$ turns into the convergence of $\int_1^\infty (\vartheta(x) - x)/x^2\,dx$.

What remains after that convergence is purely real-variable. The integrand $(\vartheta(x) - x)/x^2$ records the normalized deviation of ϑ from its conjectured asymptotic, and the monotonicity of ϑ forbids that deviation from persisting: a function that climbs only upward cannot keep $\vartheta(x)/x$ away from 1 without forcing the integral to diverge. This section carries out both halves, the analytic convergence and the monotonicity argument, and assembles them into the headline result of the chapter.

Theorem 5.4.1: Convergence of the Key Integral

The improper integral
$$\int_1^\infty \frac{\vartheta(x) - x}{x^2}\, dx$$
converges.

***Proof*:**
The proof applies Newman's theorem (theorem 5.3.1) to the function f of proposition 5.2.4 and then changes variables. The work is in verifying the holomorphic-extension hypothesis, which occupies Steps 2 and 3.

Step 1: The function and its transform. Define $f\colon [0,\infty) \to \mathbb{R}$ by
$$f(t) = \vartheta(e^t)\, e^{-t} - 1$$
The function f is bounded: by Chebyshev's estimates (theorem 3.2.3) one has $0 \le \vartheta(x) \le (\log 4)\, x$ for all $x \ge 1$, so writing $x = e^t$ gives $0 \le \vartheta(e^t)\, e^{-t} = \vartheta(x)/x \le \log 4$, whence $-1 \le f(t) \le \log 4 - 1$ for every $t \ge 0$. The function f is also locally integrable, indeed Riemann integrable on every finite interval $[0, T]$, since $\vartheta(e^t)$ is a nondecreasing step function with finitely many jumps on any bounded interval and e^{-t} is continuous. Both hypotheses on the transformed function in theorem 5.3.1 are thus met. Its Laplace transform is, by proposition 5.2.4, the function
$$g(s) = \int_0^\infty f(t)\, e^{-st}\, dt = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}, \qquad \mathrm{Re}(s) > 0 \tag{5.3}$$
holomorphic on $\mathrm{Re}(s) > 0$. To invoke Newman's theorem it remains to show that g extends holomorphically to an open set containing the closed half-plane $\mathrm{Re}(s) \ge 0$.

Step 2: Holomorphic extension at $s = 0$. Near $s = 0$ the shifted argument $s + 1$ is near 1, where Φ has its only singularity. By proposition 4.3.4 the prime series has a simple pole at $s = 1$ with residue 1, so $\Phi(w) - 1/(w-1)$ is holomorphic in a neighborhood of $w = 1$; setting $w = s + 1$, the function
$$\Phi(s+1) - \frac{1}{s}$$
is holomorphic in a neighborhood of $s = 0$, since $w - 1 = s$. Call this holomorphic function $h_0(s)$, so that $\Phi(s+1) = 1/s + h_0(s)$ near $s = 0$. Substituting into (5.3),
$$g(s) = \frac{1}{s+1}\left(\frac{1}{s} + h_0(s)\right) - \frac{1}{s} = \frac{1}{s(s+1)} - \frac{1}{s} + \frac{h_0(s)}{s+1}$$
The first two terms combine through the partial-fraction identity $1/(s(s+1)) = 1/s - 1/(s+1)$, which gives
$$\frac{1}{s(s+1)} - \frac{1}{s} = \left(\frac{1}{s} - \frac{1}{s+1}\right) - \frac{1}{s} = -\frac{1}{s+1}$$
so the singular term $1/s$ cancels exactly, leaving
$$g(s) = -\frac{1}{s+1} + \frac{h_0(s)}{s+1} = \frac{h_0(s) - 1}{s+1} \tag{5.4}$$

near $s = 0$. The numerator $h_0(s) - 1$ is holomorphic near $s = 0$, and the denominator $s + 1$ is nonzero there (it equals 1 at $s = 0$), so the right-hand side of (5.4) is holomorphic in a neighborhood of $s = 0$. This is the holomorphic extension of g to the origin.

Step 3: Holomorphic extension at $s = i\tau$ with $\tau \neq 0$. Fix a real $\tau \neq 0$ and consider the point $s = i\tau$ on the imaginary axis. The shifted argument is $s + 1 = 1 + i\tau$, a point on the line $\mathrm{Re} = 1$ distinct from the pole $s = 1$. By proposition 4.3.4 the prime series decomposes as

$$\Phi(w) = -\frac{\zeta'(w)}{\zeta(w)} - R(w)$$

where R is holomorphic on $\mathrm{Re}(w) > 1/2$. The point $w = 1 + i\tau$ lies in this half-plane, so R is holomorphic in a neighborhood of it. The logarithmic derivative $-\zeta'/\zeta$ is holomorphic on the line $\mathrm{Re}(w) = 1$ except for the simple pole at $w = 1$, by corollary 4.4.4, which records the non-vanishing $\zeta(1 + i\tau) \neq 0$ for $\tau \neq 0$ of theorem 4.4.2; since $\tau \neq 0$, the point $1 + i\tau$ is not the pole, so $-\zeta'/\zeta$ is holomorphic in a neighborhood of $w = 1 + i\tau$. Therefore Φ, the difference of two functions holomorphic near $w = 1 + i\tau$, is holomorphic in a neighborhood of that point, hence so is $w \mapsto \Phi(w)/w$ because $w = 1 + i\tau \neq 0$. Translating back by $w = s + 1$, the function $\Phi(s+1)/(s+1)$ is holomorphic in a neighborhood of $s = i\tau$. The remaining term $-1/s$ of (5.3) is holomorphic at $s = i\tau$ as well, since $i\tau \neq 0$. Their sum g is thus holomorphic in a neighborhood of every point $i\tau$ with $\tau \neq 0$.

Step 4: Assembling the extension and applying Newman's theorem. The function g is holomorphic on the open half-plane $\mathrm{Re}(s) > 0$ by Step 1. Step 2 gives a neighborhood of $s = 0$ on which g extends holomorphically, and Step 3 gives, for each $\tau \neq 0$, a neighborhood of $i\tau$ on which g extends holomorphically; the extensions agree with g on the overlaps with $\mathrm{Re}(s) > 0$, where they all equal the original transform, so by the identity theorem ([13]) they patch into a single holomorphic function on the union. That union is an open set containing the open half-plane $\mathrm{Re}(s) > 0$ together with every point of the imaginary axis $\mathrm{Re}(s) = 0$, hence an open set containing the closed half-plane $\mathrm{Re}(s) \geq 0$. The hypotheses of theorem 5.3.1 are satisfied: f is bounded and locally integrable, and its transform g extends holomorphically across the line. The theorem concludes that the improper integral $\int_0^\infty f(t)\,dt$ converges (with value $g(0)$).

Step 5: The change of variables $x = e^t$. The convergence just obtained is

$$\int_0^\infty \big(\vartheta(e^t)\,e^{-t} - 1\big)\,dt = \lim_{T\to\infty} \int_0^T \big(\vartheta(e^t)\,e^{-t} - 1\big)\,dt < \infty$$

On each truncation $\int_0^T$ substitute $x = e^t$, so $t = \log x$, $dt = dx/x$, and $e^{-t} = 1/x$; as t runs over $[0, T]$ the variable x runs over $[1, e^T]$. The substitution is a C^1 diffeomorphism of $[0, T]$ onto $[1, e^T]$, so

$$\int_0^T \big(\vartheta(e^t)\,e^{-t} - 1\big)\,dt = \int_1^{e^T} \left(\frac{\vartheta(x)}{x} - 1\right)\frac{dx}{x} = \int_1^{e^T} \frac{\vartheta(x) - x}{x^2}\,dx$$

the last equality combining the two terms over the common denominator x^2. As $T \to \infty$ the upper limit $e^T \to \infty$ through all values, and the left-hand side converges by Step 4, so the right-hand side converges to the same finite limit:

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2}\,dx = \int_0^\infty \big(\vartheta(e^t)\,e^{-t} - 1\big)\,dt < \infty$$

Therefore the improper integral converges, as we sought to show.

The theorem is the analytic payload of the entire chapter delivered in a single real statement. Everything complex-analytic has been spent: the meromorphic continuation of ζ, the pole at $s = 1$, the non-vanishing on the line, Newman's contour argument. What survives is the assertion that $\vartheta(x) - x$, measured against the natural scale x^2 that makes the integrand the logarithmic-mean deviation $(\vartheta(x)/x - 1)/x$, integrates to a finite value. This is already a strong statement about how rarely and by how little $\vartheta(x)$ can stray from x, but it is not yet the asymptotic $\vartheta(x) \sim x$. A finite integral permits the integrand to be large on thin sets, and a priori $\vartheta(x)/x$ could oscillate, dipping below 1 and rising above it infinitely often, so long as the excursions are brief enough that their total weighted contribution stays bounded. Ruling out such oscillation is the role of monotonicity, and it is the one property of ϑ that the analytic machinery never used.

The next result isolates exactly that real-variable principle. It is stated for a general nondecreasing function so that the mechanism, monotonicity against a convergent integral, stands clear of the arithmetic.

Lemma 5.4.2: Monotonicity Forces the Asymptotic

Let $\vartheta\colon [1, \infty) \to [0, \infty)$ be nondecreasing, and suppose the integral

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2}\, dx$$

converges. Then $\vartheta(x) \sim x$ as $x \to \infty$, that is, $\vartheta(x)/x \to 1$.

***Proof*:**

The argument is by contradiction, ruling out in turn that $\vartheta(x)/x$ exceeds 1 along a sequence and that it falls below 1 along a sequence. Both directions exploit the same device: over an interval whose endpoints are in a fixed ratio, monotonicity of ϑ bounds the integrand below (respectively above) by a positive (respectively negative) constant independent of where the interval sits, while the Cauchy criterion for the convergent integral forces the integral over any such far-out interval to be small. The Cauchy criterion is the statement that since $\int_1^\infty (\vartheta(t) - t)/t^2\, dt$ converges, for every $\varepsilon > 0$ there is an X_0 with

$$\left| \int_a^b \frac{\vartheta(t) - t}{t^2}\, dt \right| < \varepsilon \qquad \text{whenever } X_0 \le a \le b \tag{5.5}$$

the integral over $[a, b]$ being the difference of the partial integrals up to b and up to a.

Step 1: The upper excursion cannot occur. Suppose, for contradiction, that there is a constant $\lambda > 1$ and arbitrarily large x with $\vartheta(x) \ge \lambda x$. Fix such an x and consider the interval $[x, \lambda x]$. For t in this interval, $t \le \lambda x$, and because ϑ is nondecreasing and $t \ge x$,

$$\vartheta(t) \ge \vartheta(x) \ge \lambda x$$

so the numerator satisfies $\vartheta(t) - t \ge \lambda x - t \ge 0$ throughout $[x, \lambda x]$, the last inequality because $t \le \lambda x$. Hence

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2}\, dt \;\ge\; \int_x^{\lambda x} \frac{\lambda x - t}{t^2}\, dt$$

In the integral on the right substitute $t = ux$, so $dt = x\,du$, $t^2 = u^2x^2$, and as t runs over $[x, \lambda x]$ the variable u runs over $[1, \lambda]$:

$$\int_x^{\lambda x} \frac{\lambda x - t}{t^2}\,dt = \int_1^{\lambda} \frac{\lambda x - ux}{u^2x^2}\,x\,du = \int_1^{\lambda} \frac{\lambda - u}{u^2}\,du$$

the factors of x cancelling completely. The resulting quantity

$$c_\lambda^+ := \int_1^{\lambda} \frac{\lambda - u}{u^2}\,du$$

is a positive constant depending only on λ, not on x: the integrand $(\lambda - u)/u^2$ is strictly positive for $u \in [1, \lambda)$ and continuous, so its integral over $[1, \lambda]$ is strictly positive. The antiderivative $\int(\lambda - u)/u^2\,du = -\lambda/u - \log u$ evaluates it to $c_\lambda^+ = \lambda - 1 - \log\lambda$, which is positive for every $\lambda > 1$. Thus for every one of the arbitrarily large x with $\vartheta(x) \geq \lambda x$,

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2}\,dt \;\geq\; c_\lambda^+ > 0$$

a fixed positive lower bound. But by the Cauchy criterion (5.5) with $\varepsilon = c_\lambda^+/2$, there is an X_0 beyond which every such integral, over an interval $[x, \lambda x]$ with $x \geq X_0$, has absolute value below $c_\lambda^+/2$. Choosing one of the arbitrarily large x with $x \geq X_0$ produces an integral that is simultaneously $\geq c_\lambda^+$ and $< c_\lambda^+/2$, a contradiction. Hence no such $\lambda > 1$ exists, and $\limsup_{x\to\infty} \vartheta(x)/x \leq 1$.

Step 2: The lower excursion cannot occur. Suppose, for contradiction, that there is a constant λ with $0 < \lambda < 1$ and arbitrarily large x with $\vartheta(x) \leq \lambda x$. Fix such an x and consider now the interval $[\lambda x, x]$, which lies below x. For t in this interval, $t \geq \lambda x$, and because ϑ is nondecreasing and $t \leq x$,

$$\vartheta(t) \leq \vartheta(x) \leq \lambda x$$

so the numerator satisfies $\vartheta(t) - t \leq \lambda x - t \leq 0$ throughout $[\lambda x, x]$, the last inequality because $t \geq \lambda x$. Hence

$$\int_{\lambda x}^{x} \frac{\vartheta(t) - t}{t^2}\,dt \;\leq\; \int_{\lambda x}^{x} \frac{\lambda x - t}{t^2}\,dt$$

Substituting $t = ux$ as before, with u now running over $[\lambda, 1]$,

$$\int_{\lambda x}^{x} \frac{\lambda x - t}{t^2}\,dt = \int_{\lambda}^{1} \frac{\lambda - u}{u^2}\,du$$

again with all factors of x cancelling. The quantity

$$c_\lambda^- := \int_{\lambda}^{1} \frac{\lambda - u}{u^2}\,du$$

is a strictly negative constant depending only on λ: on $(\lambda, 1]$ the integrand $(\lambda - u)/u^2$ is strictly negative and continuous, so $c_\lambda^- < 0$. The same antiderivative $-\lambda/u - \log u$ evaluates it to $c_\lambda^- = 1 - \lambda + \log\lambda$, which is negative for every λ with $0 < \lambda < 1$. Thus for every one of the arbitrarily large x with $\vartheta(x) \leq \lambda x$,

$$\int_{\lambda x}^{x} \frac{\vartheta(t) - t}{t^2}\,dt \;\leq\; c_\lambda^- < 0$$

a fixed negative upper bound. The Cauchy criterion (5.5) with $\varepsilon = |c_\lambda^-|/2$ gives an X_0 beyond which every integral over $[\lambda x, x]$ with $\lambda x \geq X_0$ has absolute value below $|c_\lambda^-|/2$. Selecting an arbitrarily large x with $\lambda x \geq X_0$ yields an integral that is at once $\leq c_\lambda^- = -|c_\lambda^-|$ and greater than $-|c_\lambda^-|/2$, a contradiction. Hence no such $\lambda \in (0,1)$ exists, and $\liminf_{x\to\infty} \vartheta(x)/x \geq 1$.

Steps 1 and 2 give $\limsup_{x\to\infty} \vartheta(x)/x \leq 1 \leq \liminf_{x\to\infty} \vartheta(x)/x$. Since the liminf never exceeds the limsup, both equal 1, so $\lim_{x\to\infty} \vartheta(x)/x = 1$, completing the proof.

The lemma turns a statement about an average into a statement about a limit, and monotonicity is the entire reason the conversion is legitimate. The integrand $(\vartheta(t) - t)/t^2$ has a convergent integral, which controls only the cumulative deviation; the integrand could in principle be large at isolated places. Monotonicity forbids isolation. If ϑ overshoots λx at a single point x, it stays at least λx for the entire interval $[x, \lambda x]$, because it cannot come back down, and the overshoot is therefore spread across a window of multiplicative width λ, contributing a fixed positive amount to the integral. The same window appears in both directions, and its scale invariance, the fact that $c_\lambda^\pm$ depends on λ but not on x, is what lets a single excursion at arbitrarily large x contradict the smallness guaranteed far out by the Cauchy criterion. The choice of weight x^2 in the denominator is exactly what produces this invariance: it makes the substitution $t = ux$ scale the integral to a form free of x, so a deviation at height x costs the same regardless of x. Replace ϑ by a function that is not monotone, and the argument collapses, since a brief spike could be undone immediately and the window of forced deviation would vanish.

With both halves in place, the proof of the Prime Number Theorem is immediate. The function ϑ supplied by arithmetic is nondecreasing, and theorem 5.4.1 furnishes exactly the convergent integral the lemma requires.

Theorem 5.4.3: The Prime Number Theorem

As $x \to \infty$,

$$\vartheta(x) \sim x, \qquad \psi(x) \sim x, \qquad \pi(x) \sim \frac{x}{\log x}$$

where ϑ and ψ are the Chebyshev functions (definition 3.1.2) and π is the prime counting function (definition 3.1.1).

***Proof*:**
The Chebyshev theta function $\vartheta(x) = \sum_{p\leq x} \log p$ is nondecreasing in x, since increasing x can only admit further primes into the sum, each contributing a nonnegative term $\log p$. By theorem 5.4.1 the integral $\int_1^\infty (\vartheta(x) - x)/x^2\, dx$ converges. The two hypotheses of lemma 5.4.2, monotonicity of ϑ and convergence of this integral, are therefore met, and the lemma yields

$$\vartheta(x) \sim x$$

By the equivalence of the three asymptotic forms (theorem 3.1.5), $\vartheta(x) \sim x$ holds if and only if $\psi(x) \sim x$ and if and only if $\pi(x) \sim x/\log x$. Hence all three hold simultaneously, completing the proof.

This is the theorem the chapter was built to prove, and its history measures the distance the argument has travelled. The asymptotic $\pi(x) \sim x/\log x$ was conjectured by Gauss and Legendre from numerical tables near the end of the eighteenth century, refined by Gauss into the sharper guess $\pi(x) \sim \mathrm{Li}(x)$, and proved only in 1896, independently by Jacques Hadamard and Charles-Jean de la Vallée Poussin. The century of delay separates the two kinds of information the statement contains. The *order* of $\pi(x)$, that it grows like $x/\log x$ up to bounded factors, is elementary: Chebyshev obtained it in 1852 by comparing the central binomial coefficient against its prime factorization, and that argument, reproduced in theorem 3.2.3, places $\vartheta(x)/x$ between $\log 2$ and $\log 4$ using nothing past unique factorization and Legendre's formula. What the elementary method cannot reach is the *constant*. Chebyshev's corridor has width a factor of two, and no sharpening of the binomial comparison closes it to the single value 1. The constant is analytic in origin. It is the residue of the pole of ζ at $s = 1$, transmitted through the pole of $-\zeta'/\zeta$ and of Φ at $s = 1$, each of residue exactly 1 (proposition 4.3.4), and it reaches the counting function as the value $g(0)$ that Newman's theorem assigns at the boundary. The non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ is what licenses that boundary value to exist, and it is precisely the input that the elementary theory of chapter 3 could not produce. The Prime Number Theorem is, in this sense, the meeting of two centuries of method: the order from Chebyshev's combinatorics, the constant from Riemann's complex analysis of ζ.

5.4.4 Historical Note: Riemann's Memoir and the Elementary Proof The path proved here is the one Riemann charted in his 1859 memoir, which proposed studying $\pi(x)$ through the analytic continuation of $\zeta(s)$ and the location of its zeros, though Riemann himself proved no asymptotic. Hadamard and de la Vallée Poussin completed the program in 1896 with the non-vanishing of ζ on $\mathrm{Re}(s) = 1$ as the technical core. For half a century afterward it was believed that the analytic input was unavoidable, that no proof could avoid the complex zeros of ζ. This was refuted in 1949 by Atle Selberg and Paul Erdős, who gave an *elementary* proof, in the technical sense of avoiding complex analysis, built on Selberg's symmetry formula $\sum_{p\le x}(\log p)^2 + \sum_{pq\le x}(\log p)(\log q) = 2x\log x + O(x)$. The elementary proof is longer and less transparent than the analytic one and yields a weaker error term; the analytic method through zero-free regions, taken up in section 5.5, remains the route to quantitative refinements. Newman's 1980 contour argument [16], in Zagier's exposition [24], is the streamlining of the analytic proof followed in this chapter.

The theorem is best felt against the numbers that first suggested it, where the slow approach of $\pi(x)$ to $x/\log x$ and the closer fit of the logarithmic integral are both visible.

Example 5.4.5: The Prime Number Theorem Numerically

Write $\mathrm{Li}(x) = \int_2^x dt/\log t$ for the logarithmic integral, the approximation to $\pi(x)$ that emerged inside the proof of theorem 3.1.5. Comparing $\pi(x)$ against the two approximations $x/\log x$ and $\mathrm{Li}(x)$ at three widely spaced values of x gives

x	$\pi(x)$	$\dfrac{x}{\log x}$	$\dfrac{\pi(x)}{x/\log x}$	$\mathrm{Li}(x)$	$\dfrac{\pi(x)}{\mathrm{Li}(x)}$
10^3	168	144.8	1.1605	176.6	0.9515
10^6	78498	72382.4	1.0845	78626.5	0.9984
10^9	50847534	48254942.4	1.0537	50849233.9	0.99997

The fourth column, $\pi(x)/(x/\log x)$, is the ratio whose limit the theorem asserts to be 1. It does tend to 1, but slowly: even at $x = 10^9$ it is still 1.05, off by five percent, and the convergence is governed by the next-order term, which is of relative size $1/\log x$, equal to $1/\log 10^9 \approx 0.048$ at $x = 10^9$, matching the observed gap. The logarithmic integral $\mathrm{Li}(x)$ is the better approximation by a wide margin: the last column $\pi(x)/\operatorname{Li}(x)$ reaches 1 far faster, already agreeing to four decimal places at $x = 10^9$. The reason is that $\operatorname{Li}(x) = x/\log x + x/(\log x)^2 + \cdots$ captures the secondary term $x/(\log x)^2$ that $x/\log x$ omits, and it is exactly this secondary term that accounts for the persistent excess of $\pi(x)$ over $x/\log x$ seen in the fourth column. Both approximations are asymptotically correct, $\pi(x)/(x/\log x) \to 1$ and $\pi(x)/\operatorname{Li}(x) \to 1$, since $\operatorname{Li}(x) \sim x/\log x$; the theorem is the statement that the ratios converge, and the table is the statement that for $x/\log x$ the convergence is unhurried. A consequence read directly from $\pi(x) \sim x/\log x$, made precise in corollary 5.4.6 below, is that the n-th prime p_n satisfies $p_n \sim n \log n$: at $n = 10^4$ the prime is $p_n = 104729$ against $n \log n \approx 92103$, the ratio 1.137 drifting toward 1 at the same logarithmic rate.

The asymptotic for p_n used at the close of the example is the most frequently quoted reformulation of the theorem, and it follows by inverting $\pi(x) \sim x/\log x$. Two further consequences are worth recording in the same breath: the precise sense in which primes thin out, and the inversion itself.

Corollary 5.4.6: Consequences

Let p_n denote the n-th prime. Then as $n \to \infty$,

$$p_n \sim n \log n$$

Moreover the density of primes near x is asymptotically $1/\log x$: the probability that a randomly chosen integer in $[1, x]$ is prime, namely $\pi(x)/x$, satisfies $\pi(x)/x \sim 1/\log x$.

***Proof*:**
The density statement is a restatement of the theorem: dividing $\pi(x) \sim x/\log x$ from theorem 5.4.3 by x gives $\pi(x)/x \sim 1/\log x$, which is the asserted density. The substance is the asymptotic for p_n, proved by inverting the relation $\pi(x) \sim x/\log x$.

Step 1: A logarithmic comparison. Set $x = p_n$ in the theorem. By definition $\pi(p_n) = n$, the count of primes up to the n-th prime, so theorem 5.4.3 gives

$$n = \pi(p_n) \sim \frac{p_n}{\log p_n}, \qquad \text{that is} \qquad \frac{n \log p_n}{p_n} \to 1 \tag{5.6}$$

Taking logarithms of $n \sim p_n/\log p_n$, and using that log of a quantity tending to 1 tends to 0,

$$\log n = \log p_n - \log\log p_n + o(1)$$

Since $p_n \to \infty$, both $\log p_n \to \infty$ and $\log\log p_n \to \infty$, but the second grows far slower; dividing the last display by $\log p_n$,

$$\frac{\log n}{\log p_n} = 1 - \frac{\log\log p_n}{\log p_n} + \frac{o(1)}{\log p_n} \longrightarrow 1$$

because $\log\log p_n/\log p_n \to 0$. Hence $\log n \sim \log p_n$.

Step 2: Inverting. Rewrite (5.6) as $p_n \sim n \log p_n$. Substituting the equivalence $\log p_n \sim \log n$ from Step 1,

$$p_n \sim n \log p_n \sim n \log n$$

where the replacement of $\log p_n$ by $\log n$ is legitimate because the two are asymptotically equal and n is a common factor: $p_n/(n \log n) = (p_n/(n \log p_n)) \cdot (\log p_n/\log n) \to 1 \cdot 1 = 1$. Therefore $p_n \sim n \log n$, as we sought to show.

The corollary translates the theorem into the language of the primes as a sequence. The statement $p_n \sim n \log n$ says that the n-th prime sits near $n \log n$, so the primes, though individually unpredictable, are on average spaced $\log n \approx \log p_n$ apart at height p_n: the gap between consecutive primes near x is, on average, about $\log x$. This is the same fact as the density $1/\log x$ viewed from the dual side, counting the spacing between primes rather than their frequency, and the two are reciprocal as one expects. The average gap $\log x$ grows without bound, so the primes do thin out, but only logarithmically, far more slowly than the squares or the powers of 2; between x and $2x$ there are still about $x/\log x$ primes for every large x. The estimate $p_n \sim n \log n$ is the crudest term of a longer expansion, $p_n = n \log n + n \log \log n - n + o(n)$, whose secondary terms come from the same logarithmic-integral correction that made $\mathrm{Li}(x)$ outperform $x/\log x$ in example 5.4.5; the leading term alone is the direct content of the Prime Number Theorem.

5.5 Error Terms and the Zero-Free Region

The proof of the Prime Number Theorem completed in section 5.4 rests on a single qualitative fact: $\zeta(s)$ has no zeros on the line $\mathrm{Re}(s) = 1$ (theorem 4.4.2). That fact is exactly enough to force $\psi(x) \sim x$, equivalently $\vartheta(x) \sim x$ and $\pi(x) \sim x/\log x$ (theorem 5.4.3), but it is silent on the rate of convergence. Non-vanishing on the line says the ratio $\psi(x)/x$ tends to 1; it does not say how fast, and the Tauberian route of Newman's theorem, which trades all quantitative information for a clean limit, gives no error term at all. To measure the speed one needs a quantitative strengthening: not just that ζ avoids the line, but that it avoids an open region to the *left* of the line, a region whose width near a given height controls how far the contour of integration can be pushed past $\mathrm{Re}(s) = 1$ and hence how small the resulting error can be made.

This section proves the de la Vallée Poussin zero-free region, the first such quantitative result, in full: there is a constant $c > 0$ with $\zeta(s) \neq 0$ throughout $\sigma \geq 1 - c/\log(|t| + 2)$. The argument is the quantitative form of the positivity inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$ that proved qualitative non-vanishing in chapter 4, sharpened by a growth bound for ζ near the line. The width of this region then dictates an error term, $\psi(x) = x + O\big(x \exp(-c\sqrt{\log x})\big)$, whose derivation by contour optimization is stated with a precise reference. The section closes with the Riemann–von Mangoldt explicit formula, the Riemann Hypothesis, and the modern Korobov–Vinogradov region, each placing the elementary error term in the wider setting that the functional equation of the next chapter opens up.

The proof of the zero-free region needs one analytic input beyond the structural facts of chapter 4: a bound on the size of ζ and its derivative just to the left of $\mathrm{Re}(s) = 1$. The continuation formula of proposition 1.4.6 supplies it directly.

Lemma 5.5.1: A Growth Bound for ζ near the Line

There is a constant $A > 0$ such that

$$|\zeta(s)| \leq A \log |t| \qquad \text{and} \qquad |\zeta'(s)| \leq A (\log |t|)^2$$

uniformly for all $s = \sigma + it$ with $\sigma \geq 1 - \frac{1}{\log |t|}$ and $|t| \geq 5$.

***Proof*:**
By the reflection $\zeta(\bar{s}) = \overline{\zeta(s)}$, which holds because ζ is real on the real axis and analytic, both moduli are symmetric under $t \mapsto -t$, so it suffices to treat $t \geq 5$. Write $\sigma = \mathrm{Re}(s)$ throughout. The threshold $t \geq 5$ gives $\log t \geq \log 5 > \frac{3}{2}$, which keeps every point used below inside the half-plane $\mathrm{Re}(s) > 0$ where the continuation formula of proposition 1.4.6 is valid: the lower edge of the region satisfies $1 - 1/\log t \geq 1 - 1/\log 5 > 0$, and the Cauchy circle of Step 3 stays in $\mathrm{Re}(w) > 0$ as well.

Step 1: A truncated continuation formula. proposition 1.4.6 gives, for $\mathrm{Re}(s) > 0$ and $s \neq 1$ and every real $x \geq 1$,

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\sigma})$$

with an absolute implied constant. Solving for $\zeta(s)$,

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} - \frac{x^{1-s}}{1-s} + O(x^{-\sigma}) \tag{5.7}$$

The estimate of the three pieces will be carried out at the choice $x = t$, the truncation height matched to the imaginary part.

Step 2: The bound for $|\zeta(s)|$. The argument is carried out uniformly in a width parameter: for each fixed $\kappa \geq 1$ there is a constant $A_0(\kappa)$ with $|\zeta(s)| \leq A_0(\kappa) \log t$ for all $s = \sigma + it$ with $\sigma \geq 1 - \kappa/\log t$, $\sigma > 0$, and $t \geq 2$. The positivity constraint $\sigma > 0$ is what makes the continuation formula available; in the stated region of the lemma ($\kappa = 1$, $t \geq 5$) it holds automatically since $1 - 1/\log t > 0$ there, while in the $\kappa = 3$ application of Step 3 it is checked directly at each point. Fix $\kappa \geq 1$ and such an s, and take $x = t$ in (5.7); the continuation formula applies since $\mathrm{Re}(s) = \sigma > 0$ and $s \neq 1$. First note that for these σ,

$$t^{1-\sigma} = \exp\big((1-\sigma)\log t\big) \leq \exp\left(\frac{\kappa}{\log t} \cdot \log t\right) = e^{\kappa}$$

since $1 - \sigma \leq \kappa/\log t$; this bound on $t^{1-\sigma}$ uses only the region constraint, not the sign of σ. The three pieces are estimated in turn.

For the Dirichlet sum, write $|n^{-s}| = n^{-\sigma} = n^{-1}\, n^{1-\sigma}$ and bound the factor $n^{1-\sigma}$ for $1 \leq n \leq x$ by $\max(x^{1-\sigma}, 1)$. For $\sigma \leq 1$ the exponent $1 - \sigma$ is nonnegative, so $n^{1-\sigma} \leq x^{1-\sigma}$; for $\sigma > 1$ it is negative, so $n^{1-\sigma} \leq 1$. In both cases $n^{1-\sigma} \leq \max(x^{1-\sigma}, 1) \leq e^{\kappa}$, using $x^{1-\sigma} = t^{1-\sigma} \leq e^{\kappa}$ and

$1 \le e^\kappa$. Hence $n^{-\sigma} \le n^{-1} e^\kappa$ for $n \le x$, and

$$\left|\sum_{n\le x} \frac{1}{n^s}\right| \le e^\kappa \sum_{n \le x} \frac{1}{n} \le e^\kappa\big(1+\log x\big) = e^\kappa\,(1+\log t)$$

the harmonic sum bounded by $1+\log x$ through comparison with $\int_1^x du/u$.

For the rational term, $|x^{1-s}/(1-s)| = x^{1-\sigma}/|1-s|$, and $|1-s| = |(1-\sigma)-it| \ge |t| = t \ge 2$, so

$$\left|\frac{x^{1-s}}{1-s}\right| = \frac{t^{1-\sigma}}{|1-s|} \le \frac{e^\kappa}{t} \le \frac{e^\kappa}{2}$$

For the error term, $x^{-\sigma} = t^{-\sigma} = t^{-1}\,t^{1-\sigma} \le t^{-1}e^\kappa \le e^\kappa/2$. Summing the three contributions,

$$|\zeta(s)| \le e^\kappa(1+\log t) + \frac{e^\kappa}{2} + O\!\left(\frac{e^\kappa}{2}\right) \le A_0(\kappa)\log t$$

for a suitable constant $A_0(\kappa)$ and all $t \ge 2$, since $\log t$ dominates the constants once $t \ge 2$ and $1+\log t \le (1+1/\log 2)\log t$ there. This is the asserted bound on $|\zeta|$, valid on $\sigma \ge 1-\kappa/\log t$ with $\sigma > 0$.

Step 3: The bound for $|\zeta'(s)|$ by a Cauchy estimate. Fix $s = \sigma + it$ with $\sigma \ge 1 - 1/\log t$ and $t \ge 5$, and let Γ be the positively oriented circle of radius $r = 1/(2\log t)$ centered at s. Since $t \ge 5$ gives $\log t \ge \log 5$, the radius satisfies $r \le 1/(2\log 5) < 1$. Each point $w = u + iv$ on Γ satisfies

$$u \ge \sigma - r \ge 1 - \frac{1}{\log t} - \frac{1}{2\log t} = 1 - \frac{3}{2\log t}, \qquad |v-t| \le r < 1$$

so $|v| \ge t-1 \ge 4$ and $|v| \le t+1 \le 2t$, the last because $1 \le t$. The lower bound on u is positive: $\log t \ge \log 5 > \frac{3}{2}$ gives $1 - 3/(2\log t) > 0$, so $\mathrm{Re}(w) = u > 0$ and w lies in the half-plane where ζ is defined and the bound of Step 2 is available. From $|v| \le 2t$ and $t \ge 5$ one gets $\log|v| \le \log(2t) = \log 2 + \log t \le 2\log t$, hence $1/\log t \le 2/\log|v|$, and therefore

$$u \ge 1 - \frac{3}{2\log t} \ge 1 - \frac{3}{\log|v|}$$

so w lies in the region of Step 2 with $\kappa = 3$, imaginary part v satisfying $|v| \ge 4 \ge 2$, and $\mathrm{Re}(w) = u > 0$. Step 2 with $\kappa = 3$ therefore applies to every $w \in \Gamma$, giving $|\zeta(w)| \le A_0(3)\log|v| \le 2A_0(3)\log t$, the last step using $\log|v| \le 2\log t$. Thus $|\zeta(w)| \le A_1 \log t$ for an absolute constant $A_1 = 2A_0(3)$ and every $w \in \Gamma$. The Cauchy integral formula for the derivative ([13]) gives

$$\zeta'(s) = \frac{1}{2\pi i}\int_\Gamma \frac{\zeta(w)}{(w-s)^2}\,dw$$

the contour Γ lying in $\mathrm{Re}(w) > 0$, $w \ne 1$, where ζ is holomorphic. On Γ one has $|w-s| = r$, so $|(w-s)^{-2}| = r^{-2}$, and the length of Γ is $2\pi r$; the standard length estimate yields

$$|\zeta'(s)| \le \frac{1}{2\pi}\cdot\frac{\max_{w\in\Gamma}|\zeta(w)|}{r^2}\cdot 2\pi r = \frac{1}{r}\max_{w\in\Gamma}|\zeta(w)| \le (2\log t)\cdot A_1\log t = 2A_1\,(\log t)^2$$

Taking $A = \max(A_0(1), 2A_1)$ gives both bounds simultaneously on the stated region, completing the proof.

The bound is crude in its dependence on σ, holding uniformly across the thin strip $1 - 1/\log|t| \le \sigma$ with no improvement as σ increases past 1, but its dependence on t is what matters: ζ grows no faster than a power of $\log|t|$ near the line. This polynomial-in-log growth is the quantitative substitute for the boundedness that holds deeper inside the half-plane $\mathrm{Re}(s) > 1$, and it is the only size information the zero-free argument consumes. The choice $x = |t|$ in the proof is the balance point: a larger truncation would shrink the error term $O(x^{-\sigma})$ but inflate the Dirichlet sum, and a smaller one would do the reverse, so matching the truncation height to the imaginary part is what produces the single factor of $\log|t|$ rather than a power of $|t|$. The derivative bound costs one further factor of $\log|t|$, the price of the radius $r \asymp 1/\log|t|$ in the Cauchy estimate, the largest scale of radius on which the growth bound for ζ still applies.

Example 5.5.2: The Growth Bound at $s = 1 + it$

On the line itself, $\sigma = 1$, the lemma reads $|\zeta(1+it)| = O(\log|t|)$ and $|\zeta'(1+it)| = O((\log|t|)^2)$. The first is the familiar statement that ζ grows slowly along the line $\mathrm{Re}(s) = 1$. At the concrete height $t = 100$ the bound gives $|\zeta(1+100i)| \le A\log 100 = A \cdot 4.605\ldots$, while the true value is $|\zeta(1+100i)| \approx 1.63$, comfortably inside the bound for any reasonable constant; the gap reflects the crudeness of the constant A, not of the $\log|t|$ shape. The logarithmic growth is genuine, not an artifact: the partial sums $\sum_{n \le |t|} n^{-1-it}$ that dominate (5.7) have modulus of order $\log|t|$ for typical t, since the phases $n^{-it} = e^{-it\log n}$ fail to produce full cancellation in the harmonic-weighted sum. The same computation with the second derivative, a Cauchy estimate on a circle of radius $\asymp 1/\log|t|$ applied to the bound for ζ', would give $|\zeta''(1+it)| = O((\log|t|)^3)$, each derivative costing one additional logarithm.

The growth bound is the missing quantitative ingredient. The qualitative non-vanishing theorem of chapter 4 ruled out zeros *on* the line by showing that a zero there would drive the positive combination $|\zeta(\sigma)|^3|\zeta(\sigma+it)|^4|\zeta(\sigma+2it)|$ below 1 as $\sigma \to 1^+$. The quantitative version measures how far a zero must sit from the line for the same combination to remain admissible, and the distance it extracts is exactly $c/\log(|t|+2)$.

Theorem 5.5.3: The de la Vallée Poussin Zero-Free Region

There is a constant $c > 0$ such that $\zeta(s) \ne 0$ throughout the region

$$\sigma \ge 1 - \frac{c}{\log(|t|+2)}$$

In particular ζ has no zeros in an open neighborhood of the line $\mathrm{Re}(s) = 1$ whose width at height t shrinks like $1/\log|t|$.

***Proof*:**

The half-plane $\mathrm{Re}(s) > 1$ contains no zeros, by the Euler product (corollary 4.1.2), and a finite portion of the line is handled by continuity, so it suffices to find $c > 0$ such that no zero $\rho = \beta + i\gamma$ has

$$\beta > 1 - \frac{c}{\log(|\gamma|+2)}$$

for $|\gamma|$ large. By the symmetry $\zeta(\bar{s}) = \overline{\zeta(s)}$, zeros come in conjugate pairs, so it is enough to treat $\gamma \ge \gamma_0$ for a fixed $\gamma_0 \ge 7$, the range $0 \le \gamma < \gamma_0$ being a compact piece of the closed half-plane $\mathrm{Re}(s) \ge 1$ on which ζ is nonzero (no zeros on $\mathrm{Re}(s) = 1$ by theorem 4.4.2, none in $\mathrm{Re}(s) > 1$)

and hence bounded away from zero on a strip $\sigma \geq 1 - c'$ by continuity and compactness. Fix a zero $\rho = \beta + i\gamma$ with $\gamma \geq \gamma_0$ and write $L = \log(\gamma + 2)$.

Step 1: The positivity inequality. For real $\sigma > 1$ the von Mangoldt series of proposition 4.3.1 converges absolutely, and taking real parts of $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ at $s = \sigma + i\tau$ gives, since $\Lambda(n) \geq 0$ and $|n^{-i\tau}| = 1$ with $\mathrm{Re}(n^{-i\tau}) = \cos(\tau \log n)$,

$$\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\tau)\right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^\sigma} \cos(\tau \log n)$$

Form the combination with weights $3, 4, 1$ at the heights $\tau = 0, \gamma, 2\gamma$:

$$3\left(-\frac{\zeta'}{\zeta}(\sigma)\right) + 4\,\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) + \mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + 2i\gamma)\right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^\sigma}\big(3 + 4\cos\theta_n + \cos 2\theta_n\big)$$

with $\theta_n = \gamma \log n$. By lemma 4.4.1 every bracket $3 + 4\cos\theta_n + \cos 2\theta_n = 2(1 + \cos\theta_n)^2 \geq 0$, and the coefficients $\Lambda(n)n^{-\sigma}$ are nonnegative, so the entire sum is nonnegative:

$$3\left(-\frac{\zeta'}{\zeta}(\sigma)\right) + 4\,\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) + \mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + 2i\gamma)\right) \geq 0 \tag{5.8}$$

This is the quantitative engine; the three terms are now bounded above individually, the first from the pole and the latter two from the growth bound, and the inequality is read backward to constrain β.

Step 2: The pole term. By proposition 4.3.1 the function $-\zeta'/\zeta$ has near $s = 1$ the form $-\zeta'(s)/\zeta(s) = 1/(s-1) - h(s)$ with h holomorphic in a fixed neighborhood of $s = 1$, hence bounded on the segment $1 < \sigma \leq 2$. Therefore there is a constant A_2 with

$$-\frac{\zeta'}{\zeta}(\sigma) \leq \frac{1}{\sigma - 1} + A_2 \qquad (1 < \sigma \leq 2) \tag{5.9}$$

the inequality being equality up to the bounded term $-h(\sigma)$, which contributes at most a constant.

Step 3: The off-line terms from the growth bound. The two remaining terms are controlled by a standard bound on the logarithmic derivative of a function of moderate growth, applied to ζ. The bound states that there is a constant $A_3 > 0$ such that for $1 < \sigma \leq 2$ and any height τ with $|\tau| \geq \gamma_0$,

$$\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\tau)\right) \leq A_3 \log(|\tau| + 2) - \sum_{\rho'} \mathrm{Re}\frac{1}{\sigma + i\tau - \rho'} \tag{5.10}$$

where the sum runs over the zeros ρ' of ζ with $|\mathrm{Im}\rho' - \tau| \leq 1$, and every term of that sum is nonnegative because $\mathrm{Re}(\sigma + i\tau - \rho') = \sigma - \beta' > 1 - 1 = 0$ for a zero $\rho' = \beta' + i\gamma'$ (which has $\beta' \leq 1$). The inequality (5.10) is the partial-fraction estimate for $-\zeta'/\zeta$ obtained from the Borel–Carathéodory theorem and Jensen's formula applied to ζ on the disc of radius slightly larger than 1 centered at $2 + i\tau$: Borel–Carathéodory converts the upper bound $|\zeta(s)| \leq A \log|t|$ of lemma 5.5.1 on the boundary circle into a bound on ζ'/ζ in the interior, and Jensen's formula identifies the zeros ρ' as the only obstruction, contributing the displayed negative partial-fraction terms; the derivation is standard and is carried out in full in [6, Ch. 12] and [11, Ch. 6], using the complex-analytic tools of [13] (the Borel–Carathéodory theorem and Jensen's formula), with $\log(|\tau|+2)$ entering as the logarithm of the growth bound's M. Two applications of (5.10) are needed.

At height $\tau = 2\gamma$, drop the entire nonnegative sum to obtain the clean upper bound

$$\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma+2i\gamma)\right) \le A_3\log(2\gamma+2) \le A_4 L \tag{5.11}$$

for a constant A_4, using $\log(2\gamma+2) \le 2\log(\gamma+2) = 2L$ for $\gamma \ge 1$.

At height $\tau = \gamma$, the chosen zero $\rho = \beta + i\gamma$ has $\mathrm{Im}\rho = \gamma = \tau$, so $|\mathrm{Im}\rho - \tau| = 0 \le 1$ and ρ appears in the sum of (5.10). Keep only its term and drop the rest, all nonnegative:

$$\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma+i\gamma)\right) \le A_3\log(\gamma+2) - \mathrm{Re}\frac{1}{\sigma+i\gamma-\rho} = A_3L - \mathrm{Re}\frac{1}{\sigma-\beta}$$

since $\sigma + i\gamma - \rho = \sigma + i\gamma - (\beta+i\gamma) = \sigma - \beta$ is real and positive. Thus

$$\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma+i\gamma)\right) \le A_3L - \frac{1}{\sigma-\beta} \tag{5.12}$$

the term $-1/(\sigma-\beta)$ recording the proximity of $\sigma+i\gamma$ to the zero.

Step 4: Combining the bounds. Substitute (5.9), (5.12), and (5.11) into the positivity inequality (5.8):

$$0 \le 3\left(\frac{1}{\sigma-1} + A_2\right) + 4\left(A_3L - \frac{1}{\sigma-\beta}\right) + A_4L$$

Collecting the logarithmic terms into a single constant times L, there is a constant $A_5 = 3A_2/L + 4A_3 + A_4 \le A_6$ (bounded since $L \ge \log(\gamma_0+2) \ge 1$) with

$$\frac{4}{\sigma-\beta} \le \frac{3}{\sigma-1} + A_6L \tag{5.13}$$

valid for every $\sigma \in (1,2]$. This single inequality, true for all such σ, will be made to contradict β being too close to 1 by an appropriate choice of σ.

Step 5: Optimizing σ. Choose

$$\sigma = 1 + \frac{\delta}{L}, \qquad \delta > 0 \text{ a constant to be fixed,}$$

which lies in $(1,2]$ once $\delta \le L$, true for large γ since $L \to \infty$. Then $\sigma - 1 = \delta/L$, so $3/(\sigma-1) = 3L/\delta$, and (5.13) becomes

$$\frac{4}{\sigma-\beta} \le \frac{3L}{\delta} + A_6L = \left(\frac{3}{\delta} + A_6\right)L$$

Inverting,

$$\sigma - \beta \geq \frac{4}{(3/\delta + A_6)L} = \frac{4\delta}{(3 + A_6\delta)L}$$

Now $\beta = \sigma - (\sigma - \beta) \leq \sigma - \dfrac{4\delta}{(3 + A_6\delta)L} = 1 + \dfrac{\delta}{L} - \dfrac{4\delta}{(3 + A_6\delta)L}$, that is,

$$\beta \leq 1 - \frac{1}{L}\left(\frac{4\delta}{3 + A_6\delta} - \delta\right) = 1 - \frac{\delta}{L} \cdot \frac{4 - (3 + A_6\delta)}{3 + A_6\delta} = 1 - \frac{\delta}{L} \cdot \frac{1 - A_6\delta}{3 + A_6\delta}$$

The bracketed factor $(1 - A_6\delta)/(3 + A_6\delta)$ is positive precisely when $\delta < 1/A_6$. Fix any such δ, for instance $\delta = 1/(2A_6)$, giving the positive constant

$$c = \frac{\delta(1 - A_6\delta)}{3 + A_6\delta} = \frac{1}{2A_6} \cdot \frac{1/2}{3 + 1/2} = \frac{1}{14\, A_6} > 0$$

With this δ,

$$\beta \leq 1 - \frac{c}{L} = 1 - \frac{c}{\log(\gamma + 2)}$$

for every zero $\rho = \beta + i\gamma$ with $\gamma \geq \gamma_0$. Shrinking c if necessary to also cover the compact range $0 \leq \gamma < \gamma_0$, where a fixed zero-free strip $\sigma \geq 1 - c'$ exists by the non-vanishing on the line, the bound $\beta \leq 1 - c/\log(|\gamma| + 2)$ holds for all zeros, which is the assertion that $\zeta(s) \neq 0$ for $\sigma > 1 - c/\log(|t| + 2)$, completing the proof.

The proof is the qualitative non-vanishing argument run with a ruler. In chapter 4 the inequality (5.8) was used only to exclude $\beta = 1$: a zero on the line would make $-4/(\sigma - 1)$ blow up faster than $+3/(\sigma - 1)$ as $\sigma \to 1^+$, breaking nonnegativity. Here the same competition between the $3/(\sigma - 1)$ from the pole and the $4/(\sigma - \beta)$ from the zero is resolved quantitatively, and the factor $4 > 3$ is what does the work: because the zero is weighted by 4 and the pole by 3, the zero cannot approach the pole's line faster than a definite rate without the weighted sum turning negative. The growth bound enters only through the constant A_3L in (5.10), a term of size $\log\gamma$; it sets the scale $1/L = 1/\log(\gamma + 2)$ of the forbidden region, since balancing the constant against the pole's $1/(\sigma - 1)$ forces the optimal offset $\sigma - 1 \asymp 1/L$, and the zero is then trapped at distance $\asymp 1/L$ from the line. The shape $c/\log(|t| + 2)$ is thus the direct fingerprint of the logarithmic growth of ζ: a slower-growing ζ would widen the region, and the conjectural bound $|\zeta(1 + it)| = O((\log|t|)^\varepsilon)$ would not, by this argument alone, change the log in the denominator, because the method's loss is in the constant, not the shape.

The width of the zero-free region translates directly into an error term for $\psi(x)$. The mechanism is the truncated Perron formula: the contour can be pushed left to the boundary of the region, and the further left it goes, the smaller the factor x^σ on the shifted line, hence the smaller the error.

Theorem 5.5.4: The Prime Number Theorem with Error Term

There is a constant $c > 0$ such that

$$\psi(x) = x + O\Big(x \exp\big(-c\sqrt{\log x}\big)\Big)$$

and correspondingly, with $\mathrm{Li}(x) = \int_2^x dt/\log t$,

$$\pi(x) = \mathrm{Li}(x) + O\Big(x \exp\big(-c'\sqrt{\log x}\big)\Big)$$

for a constant $c' > 0$.

The full proof is a contour-optimization computation: the truncation height and the depth of the contour shift must be chosen as coupled functions of x, and tracking the resulting error through the curved boundary of the zero-free region is several pages of estimation with no conceptual turning point beyond the choices themselves. The argument is given in full in [6, Ch. 18] and [11, Ch. 6]; only the structure and the origin of the term $\exp(-c\sqrt{\log x})$ are recorded here.

***Proof Key ideas*:**
Apply the truncated Perron formula (theorem 2.5.3) to the von Mangoldt coefficients $a_n = \Lambda(n)$, whose Dirichlet series is $-\zeta'/\zeta$ (example 2.5.4), at a height T to be chosen:

$$\psi(x) = \frac{1}{2\pi i}\int_{c_0 - iT}^{c_0 + iT}\left(-\frac{\zeta'}{\zeta}(s)\right)\frac{x^s}{s}\,ds + R(x,T)$$

with $c_0 = 1 + 1/\log x$ and an explicit error $R(x,T)$ from the truncation. *Step 1: shift the contour.* Move the segment of integration left to a contour hugging the boundary $\sigma = 1 - c/\log(|t|+2)$ of the zero-free region of theorem 5.5.3, closing it off with horizontal segments at $\pm T$. The shift is legitimate because $-\zeta'/\zeta$ is holomorphic in the region swept, save for the single pole at $s = 1$; the residue of $(-\zeta'/\zeta)(s)\,x^s/s$ there is x, by the residue-1 pole of proposition 4.3.1, and this residue is the main term. *Step 2: bound the shifted integrand.* On and near the boundary contour the size of $-\zeta'/\zeta$ is controlled by the growth bound lemma 5.5.1, which gives $-\zeta'/\zeta = O((\log|t|)^2)$ there, while the factor x^s contributes $x^\sigma = x \cdot x^{-c/\log(|t|+2)} = x\exp\big(-c\log x/\log(|t|+2)\big)$, exponentially smaller than x by an amount that worsens as $|t|$ grows. *Step 3: optimize T.* The truncation error $R(x,T)$ decreases as T grows, while the contribution of the shifted contour increases with T through the $\log(|t|+2)$ in the exponent; balancing the two competing errors fixes the optimal height. Setting $\log T \asymp \sqrt{\log x}$ equalizes them, because the dominant saving $\exp(-c\log x/\log T)$ on the contour and the dominant loss from the truncation both become $\exp(-c''\sqrt{\log x})$ at that height. The common value $\exp(-c''\sqrt{\log x})$, times x, is the error term. *Step 4: pass to $\pi(x)$.* The estimate for ψ transfers to ϑ by removing the prime-power contribution (of size $O(\sqrt{x}\log x)$, absorbed into the error), and to $\pi(x)$ by partial summation against $1/\log t$ (theorem 1.4.1), which converts the main term x into $\mathrm{Li}(x) = \int_2^x dt/\log t$ and preserves an error of the same exponential shape with a possibly smaller constant c'. The full execution of these four steps is carried out in [6, Ch. 18] and [11, Ch. 6], completing the proof.

The error term $x\exp(-c\sqrt{\log x})$ is smaller than $x/(\log x)^N$ for every fixed N, since $\exp(-c\sqrt{\log x})$ decays faster than any negative power of $\log x$, yet it is far larger than the power saving $x^{1-\delta}$ that a zero-free region of *fixed* width δ to the left of the line would yield. The square root in the exponent is the exact arithmetic of the optimization: the contour saving is $\exp(-c\log x/\log T)$ and the truncation loss grows with $\log T$, so the two balance when $(\log x)/\log T \asymp \log T$, that is, $\log T \asymp \sqrt{\log x}$, making each error $\exp(-c\sqrt{\log x})$. A wider zero-free region, one whose boundary recedes from the line more slowly in $|t|$, would permit a deeper contour shift at each height and hence a larger exponent; the shape of the region and the shape of the error term are two readings of the same data. The constant c here is not the same as the constant in the zero-free region, but it is determined by it, and any improvement in the width of the region propagates to an improvement in the error.

The error term and the zero-free region are two faces of a single object, made precise by the explicit formula, which expresses the deviation $\psi(x) - x$ as a sum over the zeros of ζ.

5.5.5 Note: The Explicit Formula The deviation of ψ from its main term is given exactly, not just asymptotically, by the *Riemann–von Mangoldt explicit formula*: for $x > 1$ not a prime power,

$$\psi_0(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log\left(1 - \frac{1}{x^2}\right)$$

where $\psi_0(x) = \frac{1}{2}\big(\psi(x^-) + \psi(x^+)\big)$ is the normalized Chebyshev function and the sum runs over the nontrivial zeros ρ of ζ, taken in conjugate pairs and in order of increasing $|\mathrm{Im}\rho|$. The main term x is the residue at the pole $s = 1$; each zero $\rho = \beta + i\gamma$ contributes an oscillation $-x^\rho/\rho$ of size $x^\beta/|\rho|$; the constant $-\log(2\pi)$ is the value at $s = 0$ from the kernel; and the final term collects the trivial zeros at $s = -2, -4, \ldots$. Its proof lies beyond the present chapter: it requires the meromorphic continuation of ζ to all of $\mathbb{C}$ through the functional equation (chapter 6) and the Hadamard product factorization of the entire function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, which expresses ξ as a product over its zeros and thereby converts the contour integral of $-\zeta'/\zeta$ into the sum over ρ. The full derivation is given in [6, Ch. 17]; see also [11]. The formula foreshadows the role of the functional equation in chapter 6: every term but the first is governed by the symmetry $s \leftrightarrow 1 - s$ and the zeros it organizes, and the error term of theorem 5.5.4 is precisely a bound on the zero-sum $\sum_\rho x^\rho/\rho$ obtained by knowing that all $\beta < 1$ stay a definite distance below 1.

The explicit formula reframes every statement about the error term as a statement about the location of the zeros. The size of $\sum_\rho x^\rho/\rho$ is governed by $\sup_\rho \beta = \sup_\rho \mathrm{Re}(\rho)$: the larger the real parts of the zeros, the larger the oscillations x^β, and the worse the error. The de la Vallée Poussin region bounds β below 1 by a margin that narrows with height, and the resulting error $x\exp(-c\sqrt{\log x})$ is what that margin yields. The optimal possible control corresponds to the zeros lying as far left as they can, on the line $\mathrm{Re}(s) = 1/2$, which is the content of the Riemann Hypothesis.

Conjecture 5.5.6: The Riemann Hypothesis

Every nontrivial zero ρ of the Riemann zeta function satisfies $\mathrm{Re}(\rho) = \frac{1}{2}$.

The hypothesis is equivalent to the sharpest possible error term in the Prime Number Theorem. If every nontrivial zero has $\beta = \frac{1}{2}$, then each term x^ρ/ρ in the explicit formula has size $x^{1/2}/|\rho|$, and summing over zeros with the known density produces the bound below; conversely, that bound forces the zeros onto the line, so the equivalence runs both ways.

Proposition 5.5.7: Riemann Hypothesis and the Error Term

The Riemann Hypothesis holds if and only if

$$\psi(x) = x + O\big(\sqrt{x}\,(\log x)^2\big)$$

and this error term is essentially optimal: the deviation $\psi(x) - x$ is not $o(\sqrt{x})$, so the exponent $\frac{1}{2}$ cannot be lowered.

The equivalence is proved in [11, Ch. 13]; the forward direction reads the bound off the explicit formula under $\beta \equiv \frac{1}{2}$, and the reverse recovers $\sup \beta \leq \frac{1}{2}$ from the bound by a contour argument, while the optimality is a consequence of the oscillation forced by the infinitely many zeros on the line. Unconditionally, the best known zero-free region is wider than de la Vallée Poussin's but still far from the half-plane $\mathrm{Re}(s) > \frac{1}{2}$ that the Riemann Hypothesis asserts: the *Korobov–Vinogradov* region states that $\zeta(s) \neq 0$ for

$$\sigma \geq 1 - \frac{c}{(\log |t|)^{2/3}(\log\log |t|)^{1/3}}$$

established by Vinogradov's method for estimating exponential sums and proved in full in [21, Ch. 6]. By the same contour optimization that produced theorem 5.5.4 from de la Vallée Poussin's region, this wider region yields the best unconditional error term $\psi(x) = x + O\big(x\exp(-c(\log x)^{3/5}(\log\log x)^{-1/5})\big)$.

The progression from de la Vallée Poussin to Korobov–Vinogradov to the Riemann Hypothesis is a progression in the width of a zero-free region and, by the explicit formula, in the size of an error term. De la Vallée Poussin's region has width $\asymp 1/\log|t|$ and gives the error $x\exp(-c\sqrt{\log x})$; Korobov–Vinogradov widens the region near the real axis by a power of $\log\log|t|$ and improves the exponent from $\frac{1}{2}$ to $\frac{3}{5}$; the Riemann Hypothesis would push the region all the way to the open half-plane $\mathrm{Re}(s) > \frac{1}{2}$, the widest a zero-free region can be, since zeros are known to lie on $\mathrm{Re}(s) = \frac{1}{2}$ and so the region cannot extend past it, and would replace the exponential saving by the square-root power $x^{1/2}$. Each step trades a better understanding of where the zeros are not for a sharper count of the primes, and the line $\mathrm{Re}(s) = \frac{1}{2}$ marks the limit of what any zero-free region can achieve. The functional equation of the next chapter is what brings that line into view, by exhibiting the symmetry $s \leftrightarrow 1 - s$ under which the critical line $\mathrm{Re}(s) = \frac{1}{2}$ is the axis, and by supplying the continuation and the product factorization that the explicit formula and the Riemann Hypothesis both presuppose.

6

The Functional Equation and the Critical Strip

The previous two chapters extended $\zeta(s)$ only as far as the half-plane $\mathrm{Re}(s) > 0$ and drew the Prime Number Theorem out of its behavior near the line $\mathrm{Re}(s) = 1$: the simple pole at $s = 1$ (theorem 4.2.1) supplied the main term $\psi(x) \sim x$, and the non-vanishing on the line $\mathrm{Re}(s) = 1$ (theorem 4.4.2) supplied the convergence that turned that main term into theorem 5.4.3. None of it touched the region $\mathrm{Re}(s) \leq 0$, where the defining series diverges and the integral of (4.1) fails to converge. This chapter continues $\zeta(s)$ to the entire complex plane and uncovers the symmetry that governs the continuation: the *functional equation*, a single identity relating $\zeta(s)$ to $\zeta(1-s)$. The mechanism is a pairing of two transcendental functions, the gamma function $\Gamma(s)$ and the Jacobi theta function $\theta(t)$, the first supplying the archimedean factor that completes ζ into a symmetric object and the second supplying, through its own transformation law under $t \mapsto 1/t$, the reflection $s \leftrightarrow 1-s$ itself.

The consequences are immediate once the symmetry is in hand. The poles of Γ at the non-positive integers, transferred through the completed zeta function of section 6.3, force ζ to vanish at the negative even integers: these are the *trivial zeros*. The same machinery discharges the foreshadowed evaluations of example 4.1.3, expressing every even value $\zeta(2k)$ as a rational multiple of π^{2k} and giving in particular the value $\zeta(2) = \pi^2/6$ promised there. What the symmetry leaves undetermined is the location of the remaining zeros, the nontrivial ones, all of which lie in the *critical strip* $0 \leq \mathrm{Re}(s) \leq 1$; the assertion that they lie on the central line $\mathrm{Re}(s) = 1/2$ is the Riemann Hypothesis (**??** 5.5.6), and the functional equation is what makes that line the natural object of the conjecture. The development begins with the gamma function, the analytic prerequisite on which the completed zeta function, and hence the entire chapter, rests.

6.1 The Gamma Function

The gamma function appeared in chapter 2 as the Mellin transform of e^{-t} (example 2.4.2) and as the inseparable companion of $\zeta(s)$ in the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t - 1)\,dt$ of theorem 2.4.5. There it was used only on the half-plane $\mathrm{Re}(s) > 0$ where its defining integral converges. The functional equation requires Γ on the whole plane: the completed zeta function $\xi(s)$ of section 6.3 is built from the factor $\Gamma(s/2)$, and the symmetry $s \leftrightarrow 1 - s$ that ξ enjoys can only be read off once Γ is understood as a meromorphic function with its full pole structure. This section develops that structure in full: the meromorphic continuation of Γ to $\mathbb{C}$, its poles and their residues, its absence of zeros, and the two identities (reflection and duplication) that, together with the asymptotic estimate of Stirling, control its size and analytic behavior throughout the plane.

The starting point is the integral that defined Γ in chapter 2, recorded here as the definition for this chapter.

Definition 6.1.1: The Gamma Function

The *gamma function* is defined for $\mathrm{Re}(s) > 0$ by the integral

$$\Gamma(s) = \int_0^\infty e^{-t}\,t^{s-1}\,dt$$

the Mellin transform of e^{-t} evaluated at s (example 2.4.2). The integral converges absolutely on $\mathrm{Re}(s) > 0$ and defines a holomorphic function there.

The convergence and holomorphy were established in example 2.4.2: near $t = 0^+$ the integrand is comparable to $t^{\mathrm{Re}(s)-1}$, integrable precisely when $\mathrm{Re}(s) > 0$, and near $t = \infty$ the factor e^{-t} forces convergence for every s. Holomorphy follows from the standard theorem on integrals depending holomorphically on a parameter, applied on compact subsets of the half-plane ([13]). The half-plane $\mathrm{Re}(s) > 0$ is the natural domain of the integral, but it is not the natural domain of the function: Γ extends past it, and the extension is forced by a single functional relation.

The relation in question is the integration-by-parts identity $\Gamma(s+1) = s\,\Gamma(s)$, the analytic form of the factorial recursion $n! = n \cdot (n-1)!$. Read forward it is a routine identity; read backward it is a continuation device, for it expresses $\Gamma(s)$ in terms of $\Gamma(s+1)$, a value with a larger real part, and so pushes the domain of definition leftward one unit at a time.

Theorem 6.1.2: Meromorphic Continuation of Γ

The gamma function satisfies the functional relation

$$\Gamma(s+1) = s\,\Gamma(s) \tag{6.1}$$

for all s with $\mathrm{Re}(s) > 0$. Through this relation Γ extends to a meromorphic function on all of $\mathbb{C}$, holomorphic everywhere except for simple poles at the non-positive integers $s = 0, -1, -2, \ldots$, with

$$\mathrm{Res}_{s=-n}\Gamma(s) = \frac{(-1)^n}{n!} \qquad (n = 0, 1, 2, \ldots)$$

The continued function has no zeros: $\Gamma(s) \neq 0$ for every $s \in \mathbb{C}$.

***Proof*:**
The proof has three stages: the functional relation on the half-plane of convergence; the continuation it forces, with the location and order of the poles; and the residues, followed by the absence of zeros.

Step 1: The functional relation. Fix s with $\mathrm{Re}(s) > 0$. Integration by parts in $\Gamma(s+1) = \int_0^\infty e^{-t} t^s \, dt$, with $u = t^s$ and $dv = e^{-t}\, dt$, gives $du = s\, t^{s-1}\, dt$ and $v = -e^{-t}$, so that

$$\Gamma(s+1) = \left[-e^{-t} t^s\right]_0^\infty + s \int_0^\infty e^{-t} t^{s-1}\, dt$$

The boundary term vanishes at both ends: as $t \to \infty$ the factor e^{-t} dominates the power t^s, and as $t \to 0^+$ the power $t^s \to 0$ because $\mathrm{Re}(s) > 0$. The remaining integral is $s\,\Gamma(s)$, establishing (6.1).

Step 2: The continuation and the poles. Solving (6.1) for $\Gamma(s)$ gives $\Gamma(s) = \Gamma(s+1)/s$. Iterating the relation N times, valid as an identity between meromorphic functions wherever both sides are defined, expresses

$$\Gamma(s) = \frac{\Gamma(s+N)}{s(s+1)(s+2)\cdots(s+N-1)} \tag{6.2}$$

Fix an integer $N \geq 1$ and consider (6.2) on the half-plane $\mathrm{Re}(s) > -N$. The numerator $\Gamma(s+N)$ is holomorphic there, because $\mathrm{Re}(s+N) > 0$ puts its argument in the domain of definition 6.1.1. The denominator is a polynomial in s vanishing exactly at $s = 0, -1, -2, \ldots, -(N-1)$, each a simple zero. The right-hand side of (6.2) is therefore a meromorphic function on $\mathrm{Re}(s) > -N$, holomorphic except for simple poles at $s = 0, -1, \ldots, -(N-1)$, and on the overlap $\mathrm{Re}(s) > 0$ it agrees with the integral $\Gamma(s)$ by (6.1). Two meromorphic functions on the connected open set $\mathrm{Re}(s) > -N$ agreeing on the nonempty open subset $\mathrm{Re}(s) > 0$ are equal, by the identity theorem ([13]); hence (6.2) furnishes the meromorphic continuation of Γ to $\mathrm{Re}(s) > -N$. The poles introduced for one value of N persist for every larger value, since enlarging N multiplies numerator and denominator by the additional factors $\Gamma(s+N+1)/\Gamma(s+N) = s+N$ in a way consistent with (6.1); letting $N \to \infty$ assembles a single meromorphic function on all of $\mathbb{C}$, holomorphic except for simple poles at every non-positive integer.

Step 3: The residues and the absence of zeros. Fix $n \geq 0$ and compute the residue at $s = -n$ from (6.2) with any $N > n$. Near $s = -n$ the only vanishing factor in the denominator is $(s+n)$, so the residue is the limit

$$\mathrm{Res}_{s=-n}\Gamma(s) = \lim_{s \to -n} (s+n)\,\Gamma(s) = \frac{\Gamma(-n+N)}{(-n)(-n+1)\cdots(-1)\cdot(1)(2)\cdots(N-1-n)}$$

where in the denominator the factor $(s+n)$ has been removed and the remaining factors evaluated at $s = -n$. The product of the negative factors $(-n)(-n+1)\cdots(-1)$ has n terms and equals $(-1)^n n!$; the product of the positive factors $(1)(2)\cdots(N-1-n)$ equals $(N-1-n)!$; and the numerator is $\Gamma(N-n) = (N-n-1)!$ since $N-n$ is a positive integer (example 6.1.5). The positive-factor product cancels the numerator, leaving

$$\mathrm{Res}_{s=-n}\Gamma(s) = \frac{(N-n-1)!}{(-1)^n n!\,(N-1-n)!} = \frac{(-1)^n}{n!}$$

as claimed. For the absence of zeros, suppose $\Gamma(s_0) = 0$ for some $s_0 \in \mathbb{C}$; necessarily s_0 is not a pole. Reflection (theorem 6.1.3) gives $\Gamma(s_0)\Gamma(1-s_0) = \pi/\sin(\pi s_0)$, and the right-hand side is finite and nonzero unless s_0 is an integer. If s_0 is not an integer, the product $\Gamma(s_0)\Gamma(1-s_0)$ is nonzero, so $\Gamma(s_0) \neq 0$, a contradiction. If s_0 is a positive integer, $\Gamma(s_0) = (s_0-1)! \neq 0$ (example 6.1.5); and s_0 cannot be a non-positive integer, those being the poles. Hence Γ has no zeros, completing the proof.

The pole structure recorded here is the feature that the rest of the chapter turns into arithmetic. The completed zeta function of section 6.3 carries the factor $\Gamma(s/2)$, whose argument $s/2$ runs through the poles $0, -1, -2, \ldots$ of Γ exactly when s runs through $0, -2, -4, \ldots$. At each such s the factor $\Gamma(s/2)$ becomes infinite; since the completed function is finite there (away from $s = 0$), the companion factor $\zeta(s)$ must vanish to compensate. The poles of Γ at the non-positive integers are, through this transfer, the source of the trivial zeros of ζ at the negative even integers $s = -2, -4, -6, \ldots$, made precise in corollary 6.4.1. That Γ has no zeros is equally significant: it guarantees that the gamma factor introduces no *spurious* zeros into the completed function, so that every zero of $\xi(s)$ traces back to a zero of $\zeta(s)$ and not to an accident of the archimedean factor.

The recursion (6.1) fixes the values of Γ at integers and half-integers, but it says nothing about the relation between $\Gamma(s)$ and $\Gamma(1-s)$, the pairing that the functional equation will need. That relation is the reflection formula, and it is most cleanly obtained through the Beta integral, the two-variable companion of Γ.

Theorem 6.1.3: The Reflection Formula

For every $s \in \mathbb{C}$ that is not an integer,

$$\Gamma(s)\,\Gamma(1-s) = \frac{\pi}{\sin(\pi s)} \tag{6.3}$$

***Proof*:**
The argument proves (6.3) first on the strip $0 < \mathrm{Re}(s) < 1$ and then extends it to all non-integer s by analytic continuation. The strip is reached through the Beta function, whose relation to Γ is established as the first step.

Step 1: The Beta integral and its evaluation by Γ. For complex x, y with $\mathrm{Re}(x) > 0$ and $\mathrm{Re}(y) > 0$ define the *Beta integral*

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$$

which converges absolutely there: near $t = 0$ the integrand is comparable to $t^{\mathrm{Re}(x)-1}$ and near $t = 1$ to $(1-t)^{\mathrm{Re}(y)-1}$, both integrable. The identity $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ is obtained from the product of the two gamma integrals. By definition 6.1.1,

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^\infty e^{-u-v}\,u^{x-1}v^{y-1}\,du\,dv$$

the product of two absolutely convergent integrals written as a double integral over the first quadrant, the interchange into a single double integral justified by Tonelli's theorem applied to the absolute values $e^{-u-v}u^{\mathrm{Re}(x)-1}v^{\mathrm{Re}(y)-1}$, whose double integral is the finite product $\Gamma(\mathrm{Re}x)\Gamma(\mathrm{Re}y)$ ([14]). Change variables by $u = r\tau$, $v = r(1-\tau)$ with $r \in (0,\infty)$ and $\tau \in (0,1)$; this maps the strip $\{(r,\tau)\}$ bijectively onto the first quadrant, with $u+v=r$ and Jacobian

$$\frac{\partial(u,v)}{\partial(r,\tau)} = \det\begin{pmatrix} \tau & r \\ 1-\tau & -r \end{pmatrix} = -r\tau - r(1-\tau) = -r$$

so $du\,dv = r\,dr\,d\tau$. Substituting,

$$\Gamma(x)\Gamma(y) = \int_0^1\int_0^\infty e^{-r}\,(r\tau)^{x-1}\big(r(1-\tau)\big)^{y-1}\,r\,dr\,d\tau = \int_0^\infty e^{-r}r^{x+y-1}\,dr\int_0^1 \tau^{x-1}(1-\tau)^{y-1}\,d\tau$$

the separation of the double integral into a product again justified by Tonelli on the absolute values. The first factor is $\Gamma(x+y)$ and the second is $B(x,y)$, giving

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \tag{6.4}$$

for $\mathrm{Re}(x), \mathrm{Re}(y) > 0$.

Step 2: Reduction of the reflection product to a single integral. Take $y = 1-x$ with $0 < \mathrm{Re}(x) < 1$, so both $\mathrm{Re}(x) > 0$ and $\mathrm{Re}(1-x) > 0$ hold and (6.4) applies. Since $x+y=1$ and $\Gamma(1)=1$ (example 6.1.5), (6.4) reads

$$\Gamma(x)\,\Gamma(1-x) = B(x,1-x) = \int_0^1 t^{x-1}(1-t)^{-x}\,dt$$

The substitution $u = t/(1-t)$, equivalently $t = u/(1+u)$, carries $t\in(0,1)$ to $u\in(0,\infty)$ with $1-t = 1/(1+u)$ and $dt = du/(1+u)^2$. The integrand transforms as

$$t^{x-1}(1-t)^{-x}\,dt = \left(\frac{u}{1+u}\right)^{x-1}(1+u)^x\cdot\frac{du}{(1+u)^2} = u^{x-1}(1+u)^{-(x-1)+x-2}\,du = \frac{u^{x-1}}{1+u}\,du$$

the powers of $(1+u)$ combining to the single exponent $-(x-1)+x-2=-1$, so that

$$\Gamma(x)\,\Gamma(1-x) = \int_0^\infty \frac{u^{x-1}}{1+u}\,du \tag{6.5}$$

an integral converging for $0<\mathrm{Re}(x)<1$, since the integrand is comparable to $u^{\mathrm{Re}(x)-1}$ near $u=0$ and to $u^{\mathrm{Re}(x)-2}$ near $u=\infty$.

Step 3: Evaluation of the integral and continuation. The integral on the right of (6.5) evaluates to $\pi/\sin(\pi x)$ for $0<\mathrm{Re}(x)<1$. This is a contour integral computed by the residue theorem: integrating $z^{x-1}/(1+z)$ around a keyhole contour about the branch cut along the positive real axis picks up the residue at the simple pole $z=-1$, and the two sides of the cut combine to give $(1-e^{2\pi i(x-1)})\int_0^\infty u^{x-1}/(1+u)\,du = 2\pi i\,e^{\pi i(x-1)}$, which rearranges to $\pi/\sin(\pi x)$. The full keyhole computation is carried out in [13, Residue theorem chapter]; the result is

$$\int_0^\infty \frac{u^{x-1}}{1+u}\,du = \frac{\pi}{\sin(\pi x)} \qquad (0<\mathrm{Re}(x)<1)$$

Combining this with (6.5) gives (6.3) on the strip $0<\mathrm{Re}(s)<1$. Both sides of (6.3) are meromorphic on all of $\mathbb{C}$: the left side $\Gamma(s)\Gamma(1-s)$ by theorem 6.1.2, the right side $\pi/\sin(\pi s)$ as the ratio of an entire function and the entire function $\sin(\pi s)$. They agree on the strip $0<\mathrm{Re}(s)<1$, a set with accumulation points, so by the identity theorem ([13]) they agree wherever both are defined, that is, for every non-integer s. This establishes (6.3), as we sought to show.

The reflection formula is the algebraic heart of the symmetry $s \leftrightarrow 1-s$. It pairs the value of Γ at s with its value at $1-s$ through the elementary factor $\pi/\sin(\pi s)$, and this pairing is exactly the form the gamma factors take in the functional equation of theorem 6.3.3: the completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ inherits its reflection symmetry from the interplay of this formula with the theta transformation of section 6.2. The formula also re-derives the pole structure of theorem 6.1.2 from a different vantage. The right-hand side $\pi/\sin(\pi s)$ has simple poles at every integer; on the negative side $s = -n$ $(n \geq 0)$, the factor $\Gamma(1-s) = \Gamma(1+n) = n!$ is finite and nonzero, so the pole must come from $\Gamma(s)$, confirming that Γ has a simple pole there. On the positive side $s = n \geq 1$, the factor $\Gamma(s) = (n-1)!$ is finite, so the pole of $\pi/\sin(\pi s)$ is absorbed by $\Gamma(1-s)$, which indeed has a pole at $1-n \leq 0$. The reflection formula thus encodes the entire pole pattern of Γ in the zeros of $\sin(\pi s)$. Combined with the residues of theorem 6.1.2, it is the tool that, in section 6.4, converts the poles of $\Gamma(s/2)$ into the trivial zeros of ζ and the values of Γ at half-integers into the special values $\zeta(2k)$.

A second multiplicative identity relates Γ at s and $s+1/2$ to Γ at $2s$. Where reflection pairs a point with its mirror image across $\mathrm{Re}(s) = 1/2$, the duplication formula interpolates between the lattice of integers and the lattice of half-integers, and it is the gamma-function input to the relation between $\zeta(s)$ and the completed function built from $\Gamma(s/2)$.

Theorem 6.1.4: Legendre Duplication and Stirling

The gamma function satisfies the *Legendre duplication formula*

$$\Gamma(s)\,\Gamma\big(s+\tfrac{1}{2}\big) = 2^{1-2s}\sqrt{\pi}\,\Gamma(2s) \tag{6.6}$$

for all $s \in \mathbb{C}$ at which both sides are defined. Moreover Γ obeys the asymptotic estimate of Stirling: for $|\arg s| \leq \pi - \delta$ with $\delta > 0$ fixed, as $|s| \to \infty$,

$$\log\Gamma(s) = \big(s-\tfrac{1}{2}\big)\log s - s + \tfrac{1}{2}\log(2\pi) + O\left(\frac{1}{|s|}\right) \tag{6.7}$$

with the principal branch of the logarithm, the implied constant depending only on δ.

***Proof*:**
The duplication formula is proved in full through the Beta integral; the complex Stirling estimate is treated by key ideas with a precise reference, for the reason stated below.

Step 1: The duplication formula. Assume first $\mathrm{Re}(s) > 0$, so that all Beta integrals below converge; the identity then extends to all admissible s by analytic continuation exactly as in theorem 6.1.3. Apply (6.4) with $x = y = s$:

$$B(s,s) = \frac{\Gamma(s)^2}{\Gamma(2s)} = \int_0^1 \big(t(1-t)\big)^{s-1}\,dt$$

Substitute $t = (1+u)/2$, so $dt = du/2$ and $t(1-t) = (1-u^2)/4$, carrying $t \in (0,1)$ to $u \in (-1,1)$:

$$B(s,s) = \int_{-1}^{1}\left(\frac{1-u^2}{4}\right)^{s-1}\frac{du}{2} = 2^{1-2s}\int_{-1}^{1}(1-u^2)^{s-1}\,du$$

where $\frac{1}{2}\cdot 4^{1-s} = 2^{-1}2^{2-2s} = 2^{1-2s}$. The integrand is even in u, so the integral over $(-1,1)$ is twice the integral over $(0,1)$. In that integral substitute $v = u^2$, $dv = 2u\,du$, hence $du = \frac{1}{2}v^{-1/2}\,dv$:

$$2\int_0^1 (1-u^2)^{s-1}\,du = 2\int_0^1 (1-v)^{s-1}\cdot \tfrac{1}{2}v^{-1/2}\,dv = \int_0^1 v^{-1/2}(1-v)^{s-1}\,dv = B\big(\tfrac{1}{2}, s\big)$$

the last integral being the Beta integral with parameters $\frac{1}{2}$ and s. Assembling the two displays,

$$\frac{\Gamma(s)^2}{\Gamma(2s)} = 2^{1-2s}\,B\big(\tfrac{1}{2}, s\big) = 2^{1-2s}\,\frac{\Gamma\big(\frac{1}{2}\big)\,\Gamma(s)}{\Gamma\big(s+\frac{1}{2}\big)}$$

using (6.4) once more for $B(\frac{1}{2}, s)$. Now $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (example 6.1.5). Multiplying both sides by $\Gamma(s+\frac{1}{2})/\Gamma(s)$ clears the common factor $\Gamma(s)$ and yields

$$\frac{\Gamma(s)\,\Gamma\big(s+\frac{1}{2}\big)}{\Gamma(2s)} = 2^{1-2s}\sqrt{\pi}$$

which is (6.6). Both sides of (6.6) are meromorphic on $\mathbb{C}$, so the identity, proved on $\mathrm{Re}(s) > 0$, holds wherever both sides are defined by the identity theorem ([13]).

Step 2: Stirling's estimate. The real-variable form $\log\Gamma(x+1) = x\log x - x + \frac{1}{2}\log(2\pi x) + O(1/x)$ as $x \to \infty$ is theorem 1.5.7; since $\log\Gamma(x+1) = \log x + \log\Gamma(x)$ and $\frac{1}{2}\log(2\pi x) = \frac{1}{2}\log(2\pi) + \frac{1}{2}\log x$, subtracting $\log x$ recovers (6.7) for real $s = x \to +\infty$, with $(x-\frac{1}{2})\log x = x\log x - \frac{1}{2}\log x$. The extension to the complex sector $|\arg s| \le \pi - \delta$ requires more than the Euler–Maclaurin computation that produced the real case: it requires controlling the error term uniformly as $\arg s$ ranges over the sector, which is most efficiently done either from the Binet integral representation of $\log\Gamma$ or by combining the real-axis asymptotic with the functional relation (6.1) and a Phragmén–Lindelöf argument to propagate the bound off the real axis. The full complex argument is several pages of estimation with no new conceptual content beyond the real case, and is carried out in [13, Stirling/asymptotics chapter]; see also [21, Ch. 4] for the form with explicit error terms in the sector. The estimate (6.7) is as stated, completing the proof.

Stirling's estimate is the quantitative control that the analytic theory of ζ ultimately rests on. The completed function $\xi(s)$ of section 6.3 is entire of order one, and that order is a direct consequence of (6.7): the gamma factor $\Gamma(s/2)$ grows like $\exp(\frac{1}{2}|s|\log|s|)$ in vertical strips, and it is this growth rate, balanced against the polynomial size of ζ in the strip, that determines the density of the zeros and underlies the explicit formula of box 5.5.5. The duplication formula, for its part, is the algebraic identity that allows the factor $\Gamma(s/2)$ appearing in $\xi(s)$ to be related to the factor $\Gamma(s)$ appearing in the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t-1)\,dt$ of theorem 2.4.5; the two normalizations differ by exactly the elementary factor that (6.6) supplies. Together, reflection and duplication exhaust the multiplicative identities of Γ that the functional equation needs.

The abstract development is best anchored by computing Γ at the values where it is most often used: the integers, where it reproduces the factorial, and the half-integers, where it produces the powers of $\sqrt{\pi}$ that appear throughout the theory.

Example 6.1.5: Special Values of Γ

1. *Integer values.* Direct evaluation of definition 6.1.1 at $s = 1$ gives

$$\Gamma(1) = \int_0^\infty e^{-t}\, dt = 1$$

The recursion (6.1) then gives $\Gamma(2) = 1 \cdot \Gamma(1) = 1$, $\Gamma(3) = 2 \cdot \Gamma(2) = 2$, and by induction $\Gamma(n) = (n-1)!$ for every positive integer n: if $\Gamma(n) = (n-1)!$ then $\Gamma(n+1) = n\,\Gamma(n) = n \cdot (n-1)! = n!$. Thus Γ interpolates the factorial, with $\Gamma(n+1) = n!$.

2. *The value at $\frac{1}{2}$.* Setting $s = \frac{1}{2}$ in the reflection formula (6.3) gives $\Gamma(\frac{1}{2})^2 = \Gamma(\frac{1}{2})\Gamma(1-\frac{1}{2}) = \pi/\sin(\pi/2) = \pi$. Since $\Gamma(\frac{1}{2}) = \int_0^\infty e^{-t}t^{-1/2}\, dt > 0$ is a positive real number, taking the positive square root gives

$$\Gamma\big(\tfrac{1}{2}\big) = \sqrt{\pi}$$

The same value follows from the Gaussian integral: the substitution $t = w^2$ in $\Gamma(\frac{1}{2})$ gives $\Gamma(\frac{1}{2}) = \int_0^\infty e^{-w^2} \cdot w^{-1} \cdot 2w\, dw = 2\int_0^\infty e^{-w^2}\, dw = \int_{-\infty}^\infty e^{-w^2}\, dw = \sqrt{\pi}$ ([17]).

3. *Half-integer values.* The recursion (6.1) propagates the value at $\frac{1}{2}$ to all half-integers. From $\Gamma(s+1) = s\,\Gamma(s)$ with $s = \frac{1}{2}, \frac{3}{2}, \ldots,$

$$\Gamma\big(\tfrac{3}{2}\big) = \tfrac{1}{2}\sqrt{\pi}, \qquad \Gamma\big(\tfrac{5}{2}\big) = \tfrac{3}{2} \cdot \tfrac{1}{2}\sqrt{\pi} = \tfrac{3}{4}\sqrt{\pi}$$

and in general, for $n \geq 0$,

$$\Gamma\big(\tfrac{1}{2} + n\big) = \frac{(2n)!}{4^n\, n!}\sqrt{\pi}$$

The closed form is verified by induction: it holds at $n = 0$, and the step from n to $n+1$ multiplies by $\frac{1}{2} + n = (2n+1)/2$, while the right side is multiplied by $\frac{(2n+2)!}{4^{n+1}(n+1)!}\Big/\frac{(2n)!}{4^n n!} = \frac{(2n+1)(2n+2)}{4(n+1)} = \frac{2n+1}{2}$, the two factors agreeing.

4. *Stirling at $s = 10$.* The estimate (6.7) at the integer $s = 10$, where $\log\Gamma(10) = \log 9! = \log(362{,}880) \approx 12.8018$, gives the approximation

$$\big(10 - \tfrac{1}{2}\big)\log 10 - 10 + \tfrac{1}{2}\log(2\pi) \approx 9.5 \cdot 2.302585 - 10 + 0.918939 \approx 12.7935$$

matching $\log\Gamma(10)$ to three significant figures. The discrepancy is $12.8018 - 12.7935 \approx 0.0083$, in agreement with the predicted error $O(1/|s|) = O(1/10)$: the first correction term of the full Stirling expansion (theorem 1.5.7) is $B_2/(2 \cdot 1 \cdot 10) = 1/(12 \cdot 10) \approx 0.00833$, and adding it brings the approximation to 12.8018, recovering $\log\Gamma(10)$ to the displayed precision.

6.2 Theta Functions and Poisson Summation

The functional equation of the zeta function relates $\zeta(s)$ to $\zeta(1-s)$, exchanging the half-plane $\mathrm{Re}(s) > 1$ where the series converges with its mirror image $\mathrm{Re}(s) < 0$. The source of this symmetry is not visible in the Dirichlet series itself; it lies one level down, in a symmetry of the Gaussian. The function $x \mapsto e^{-\pi x^2}$ is its own Fourier transform, and rescaling the variable converts that self-duality into the relation $t \leftrightarrow 1/t$ for the Gaussian $e^{-\pi x^2 t}$. Summing over the integer lattice turns this analytic fact into an identity for the Jacobi theta function, the generating function $\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$, by way of the Poisson summation formula, which equates a sum over $\mathbb{Z}$ to a sum of Fourier-transform values over $\mathbb{Z}$.

This section develops the three ingredients in order: Poisson summation, the Fourier transform of a Gaussian, and the theta function with its transformation law $\theta(1/t) = \sqrt{t}\,\theta(t)$. The theta functional equation is the geometric statement; the functional equation of ζ, obtained in section 6.3 by feeding θ into a Mellin transform, is its arithmetic form. The $t \leftrightarrow 1/t$ symmetry established here becomes, after the change of variables $t \mapsto t^{-1}$ inside the Mellin integral, the $s \leftrightarrow 1-s$ symmetry of the completed zeta function.

The first ingredient is Poisson summation. It is the statement that periodizing a function and reading off its Fourier coefficients recovers the values of the Fourier transform at integer points, so that the total mass of a function over the integer lattice equals the total mass of its transform over the dual lattice. The hypotheses must rule out the two ways such an identity can fail: the sum $\sum_n f(n)$ must converge, and the periodization must have a convergent Fourier series that represents it pointwise. Schwartz decay, or any decay forcing both f and $\hat{f}$ to fall off faster than $|x|^{-1-\delta}$, secures both.

Theorem 6.2.1: Poisson Summation

Let $f\colon \mathbb{R} \to \mathbb{C}$ be a function for which there exist constants $C > 0$ and $\delta > 0$ with

$$|f(x)| \le \frac{C}{(1+|x|)^{1+\delta}}, \qquad |\widehat{f}(\xi)| \le \frac{C}{(1+|\xi|)^{1+\delta}}$$

for all $x, \xi \in \mathbb{R}$, where the Fourier transform is normalized by

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\, e^{-2\pi i x\xi}\, dx$$

(every Schwartz function satisfies these bounds, as does $f(x) = e^{-\pi x^2 t}$ for fixed $t > 0$). Then both series below converge absolutely and

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \widehat{f}(n) \tag{6.8}$$

***Proof*:**
The argument periodizes f, expands the periodization in a Fourier series, and evaluates at the origin. Each step is justified by the stated decay.

Step 1: The periodization is a continuous 1*-periodic function.* Define

$$F(x) = \sum_{n\in\mathbb{Z}} f(x+n)$$

For x in the compact interval $[0,1]$ and $|n| \ge 2$ one has $|x+n| \ge |n| - 1 \ge |n|/2$, so $|f(x+n)| \le C\,(1+|n|/2)^{-1-\delta}$, a bound independent of x and summable over n because $\sum_n (1+|n|/2)^{-1-\delta} < \infty$ for $\delta > 0$. By the Weierstrass M-test the series defining F converges absolutely and uniformly on $[0,1]$, hence its sum F is continuous there; the same bound on every translate $[k, k+1]$ gives continuity on all of $\mathbb{R}$. Replacing x by $x+1$ permutes the summands, so $F(x+1) = F(x)$, and F is 1-periodic.

Step 2: The Fourier coefficients of F are the values $\widehat{f}(m)$. The m-th Fourier coefficient of the continuous periodic function F is

$$c_m = \int_0^1 F(x)\, e^{-2\pi i m x}\, dx = \int_0^1 \sum_{n\in\mathbb{Z}} f(x+n)\, e^{-2\pi i m x}\, dx$$

The uniform convergence from Step 1 permits interchanging the sum and the integral over the bounded interval $[0,1]$. Substituting $y = x+n$ in the n-th term, and using $e^{-2\pi i m x} = e^{-2\pi i m(y-n)} = e^{-2\pi i m y}$ since $e^{2\pi i m n} = 1$ for integers m, n, gives

$$c_m = \sum_{n\in\mathbb{Z}} \int_0^1 f(x+n)\, e^{-2\pi i m x}\, dx = \sum_{n\in\mathbb{Z}} \int_n^{n+1} f(y)\, e^{-2\pi i m y}\, dy = \int_{-\infty}^{\infty} f(y)\, e^{-2\pi i m y}\, dy$$

the consecutive intervals $[n, n+1]$ tiling $\mathbb{R}$, with the reassembly of the integral justified by absolute convergence of $\int_{\mathbb{R}} |f| < \infty$, itself a consequence of the decay hypothesis. The last integral is $\widehat{f}(m)$.

Step 3: The Fourier series represents F pointwise at $x = 0$. By Step 2 the Fourier coefficients satisfy $|c_m| = |\widehat{f}(m)| \le C\,(1+|m|)^{-1-\delta}$, so $\sum_m |c_m| < \infty$ and the Fourier series $\sum_m c_m e^{2\pi i m x}$ converges absolutely and uniformly. A continuous periodic function whose Fourier coefficients are absolutely summable equals the sum of its Fourier series at every point ([17]); the uniform limit of the partial sums is a continuous function with the same Fourier coefficients as F, and a continuous periodic function is determined by its Fourier coefficients. Evaluating the resulting identity $F(x) = \sum_m c_m e^{2\pi i m x}$ at $x = 0$ yields

$$\sum_{n\in\mathbb{Z}} f(n) = F(0) = \sum_{m\in\mathbb{Z}} c_m = \sum_{m\in\mathbb{Z}} \widehat{f}(m)$$

which is (6.8), completing the proof.

Poisson summation is a self-duality of the integer lattice: the lattice $\mathbb{Z}$ is its own dual under the pairing $(x,\xi) \mapsto e^{2\pi i x \xi}$, and the formula says that a function and its Fourier transform deposit the same total mass on $\mathbb{Z}$. The two sides see the function at complementary scales. The left side, $\sum_n f(n)$, samples f at integer arguments and is dominated by the behavior of f near the origin when f is concentrated there; the right side, $\sum_n \widehat{f}(n)$, samples the transform, which is concentrated near the origin precisely when f is spread out. This exchange of "concentrated near 0" for "spread out" is the mechanism that will convert small-t behavior of the theta function into large-t behavior, and it is the analytic content of the functional equation. The decay hypotheses are exactly what is needed to make every interchange in the proof legitimate; for the single function this section applies the formula to, the Gaussian, the hypotheses hold with room to spare, since a Gaussian and its transform decay faster than any power.

The function Poisson summation will be applied to is the Gaussian $e^{-\pi x^2 t}$, and the right side of (6.8) requires its Fourier transform. The defining property of the Gaussian among all functions is that it is an eigenfunction of the Fourier transform: the standard Gaussian $e^{-\pi x^2}$ is fixed, and rescaling the width sends the transform to the reciprocal width, with a compensating factor of $t^{-1/2}$.

Lemma 6.2.2: The Fourier Transform of a Gaussian

For each $t > 0$, the Fourier transform of the Gaussian $g_t(x) = e^{-\pi x^2 t}$, in the normalization $\widehat{g_t}(\xi) = \int_{\mathbb{R}} g_t(x)\, e^{-2\pi i x \xi}\, dx$, is

$$\widehat{g_t}(\xi) = t^{-1/2}\, e^{-\pi \xi^2/t} \tag{6.9}$$

In particular $g_1 = e^{-\pi x^2}$ is its own Fourier transform.

***Proof*:**
The standard Gaussian is treated first, and the general case follows by a change of variables.

Step 1: The transform of $g_1(x) = e^{-\pi x^2}$ is g_1. Set $\varphi(\xi) = \widehat{g_1}(\xi) = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x \xi}\, dx$. The integrand and its ξ-derivative $-2\pi i x\, e^{-\pi x^2} e^{-2\pi i x \xi}$ are dominated in absolute value by the integrable functions $e^{-\pi x^2}$ and $2\pi |x|\, e^{-\pi x^2}$, uniformly for ξ in any bounded set, so differentiation under the integral sign is justified ([14]) and

$$\varphi'(\xi) = \int_{\mathbb{R}} e^{-\pi x^2}\, (-2\pi i x)\, e^{-2\pi i x \xi}\, dx$$

The factor $-2\pi x\, e^{-\pi x^2}$ is the derivative $\frac{d}{dx} e^{-\pi x^2}$, so integration by parts with $u = e^{-2\pi i x \xi}$ and $dv = i\,(\frac{d}{dx} e^{-\pi x^2})\, dx$ gives, the boundary term vanishing because $e^{-\pi x^2} \to 0$ as $x \to \pm\infty$,

$$\varphi'(\xi) = i\Big[e^{-2\pi i x \xi}\, e^{-\pi x^2}\Big]_{-\infty}^{\infty} - i \int_{\mathbb{R}} e^{-\pi x^2}\, (-2\pi i \xi)\, e^{-2\pi i x \xi}\, dx = -2\pi \xi\, \varphi(\xi)$$

This first-order linear differential equation has solution $\varphi(\xi) = \varphi(0)\, e^{-\pi \xi^2}$. The initial value is the Gaussian integral $\varphi(0) = \int_{\mathbb{R}} e^{-\pi x^2}\, dx = 1$, the normalization built into the factor π in the exponent ([14]). Hence $\varphi(\xi) = e^{-\pi \xi^2} = g_1(\xi)$.

Step 2: Rescale to obtain g_t. Fix $t > 0$ and write $g_t(x) = g_1(\sqrt{t}\, x)$. Substituting $u = \sqrt{t}\, x$, so $du = \sqrt{t}\, dx$, in the defining integral gives

$$\widehat{g_t}(\xi) = \int_{\mathbb{R}} g_1(\sqrt{t}\, x)\, e^{-2\pi i x \xi}\, dx = \frac{1}{\sqrt{t}} \int_{\mathbb{R}} g_1(u)\, e^{-2\pi i u (\xi/\sqrt{t})}\, du = \frac{1}{\sqrt{t}}\, \widehat{g_1}\left(\frac{\xi}{\sqrt{t}}\right)$$

By Step 1, $\widehat{g_1}(\xi/\sqrt{t}) = e^{-\pi(\xi/\sqrt{t})^2} = e^{-\pi \xi^2/t}$, so $\widehat{g_t}(\xi) = t^{-1/2}\, e^{-\pi \xi^2/t}$, which is (6.9). Setting $t = 1$ recovers $\widehat{g_1} = g_1$, as we sought to show.

The identity (6.9) packages the entire dependence on t into the reciprocal $1/t$ in the exponent and the prefactor $t^{-1/2}$. A narrow Gaussian (large t, sharply peaked at the origin) transforms to a wide one (small $1/t$, spread out), and conversely; the Fourier transform interchanges the two regimes, with the prefactor $t^{-1/2}$ enforcing the correct normalization of total mass. This is the precise sense in which the Gaussian realizes the uncertainty principle with equality, and it is the single input that makes the theta function transform cleanly: summing (6.9) over the lattice will send t to $1/t$ and produce the factor $\sqrt{t}$ that appears in the theta functional equation.

The theta function is the lattice sum of the Gaussian. Collecting the values $e^{-\pi n^2 t}$ over all integers n produces a function of t that encodes the Gaussian at every integer point at once, and it is the object on which Poisson summation acts.

Definition 6.2.3: The Jacobi Theta Function

The *Jacobi theta function* is the series

$$\theta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad t > 0$$

Separating the term $n = 0$ from the pair $\pm n$ for $n \geq 1$, which contribute equally, gives the equivalent form

$$\theta(t) = 1 + 2\sum_{n=1}^{\infty} e^{-\pi n^2 t} \tag{6.10}$$

For every $t > 0$ the series converges absolutely and rapidly: the n-th term is $e^{-\pi n^2 t}$, and for $n \geq 1$ this is bounded by $e^{-\pi n t} \cdot e^{-\pi(n^2-n)t} \leq (e^{-\pi t})^n$, so the tail is dominated by a convergent geometric series with ratio $e^{-\pi t} < 1$. The convergence is locally uniform on $t > 0$, since on any interval $t \geq t_0 > 0$ the same geometric bound holds with ratio $e^{-\pi t_0} < 1$; hence θ is continuous, indeed real-analytic, on $(0, \infty)$. As $t \to +\infty$ every term with $n \neq 0$ tends to 0, and the geometric bound shows the tail $\sum_{n\geq 1} e^{-\pi n^2 t} \leq e^{-\pi t}/(1 - e^{-\pi t})$ vanishes, so

$$\theta(t) = 1 + 2\sum_{n=1}^{\infty} e^{-\pi n^2 t} \longrightarrow 1 \qquad (t \to +\infty)$$

the $n = 0$ term surviving alone. The behavior as $t \to 0^+$ is the opposite extreme: every term tends to 1, the series loses its geometric decay, and $\theta(t)$ diverges. Quantifying that divergence is exactly what the functional equation does.

The functional equation is the transformation law of θ under $t \mapsto 1/t$. It is the lattice-sum form of the Gaussian self-duality of lemma 6.2.2, obtained by applying Poisson summation to the Gaussian and reading the prefactor $t^{-1/2}$ as the source of the $\sqrt{t}$.

Theorem 6.2.4: The Functional Equation of the Theta Function

For all $t > 0$,

$$\theta\left(\frac{1}{t}\right) = \sqrt{t}\,\theta(t) \tag{6.11}$$

***Proof*:**

Fix $t > 0$ and apply Poisson summation (theorem 6.2.1) to the Gaussian $g_t(x) = e^{-\pi x^2 t}$. The function g_t satisfies the decay hypotheses of theorem 6.2.1: it decays faster than any power of $|x|$, and by lemma 6.2.2 its transform $\widehat{g_t}(\xi) = t^{-1/2} e^{-\pi \xi^2/t}$ does as well, so both required bounds hold (with any $\delta > 0$, for a suitable constant C depending on t). The formula (6.8) therefore applies and gives

$$\sum_{n\in\mathbb{Z}} g_t(n) = \sum_{n\in\mathbb{Z}} \widehat{g_t}(n)$$

The left side is $\sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} = \theta(t)$ by the definition of θ. For the right side, lemma 6.2.2 gives

$\widehat{g}_t(n) = t^{-1/2}\, e^{-\pi n^2/t}$, so

$$\sum_{n\in\mathbb{Z}} \widehat{g}_t(n) = t^{-1/2} \sum_{n\in\mathbb{Z}} e^{-\pi n^2/t} = t^{-1/2}\, \theta\left(\frac{1}{t}\right)$$

the last equality being the definition of θ evaluated at $1/t$. Equating the two sides yields

$$\theta(t) = t^{-1/2}\, \theta\left(\frac{1}{t}\right) \tag{6.12}$$

Multiplying both sides of (6.12) by $t^{1/2} = \sqrt{t}$ isolates $\theta(1/t)$ on one side and gives $\theta(1/t) = \sqrt{t}\,\theta(t)$, which is (6.11), completing the proof.

The transformation law (6.11) is the symmetry that the functional equation of ζ inherits. It relates the value of θ at t to its value at the reciprocal $1/t$, exchanging the regime t large, where $\theta(t) \to 1$, with the regime t small, where $\theta(t)$ blows up. The point $t = 1$ is the fixed point of the involution $t \mapsto 1/t$, and (6.11) reads $\theta(1) = \sqrt{1}\,\theta(1)$ there, a tautology that locates the axis of symmetry. This is the same structural picture that will govern ζ: an involution with a distinguished fixed point, relating the function on one side to its values on the other. The connection is made precise in section 6.3 through the Mellin transform of chapter 2. Integrating $\theta(t) - 1$ against $t^{s/2}$ produces, up to elementary factors, the completed zeta function $\xi(s)$; the change of variables $t \mapsto 1/t$ inside that integral turns the theta symmetry $t \leftrightarrow 1/t$ into the reflection $s \leftrightarrow 1 - s$, because $t^{s/2}\, dt/t$ becomes $t^{(1-s)/2}\, dt/t$ under $t \mapsto 1/t$ once the factor $\sqrt{t}$ from (6.11) is absorbed. The quadratic exponent n^2 in $e^{-\pi n^2 t}$ is what forces the Mellin pairing to be against the half-power $t^{s/2}$ rather than t^s, and this halving is what centers the zeta symmetry at $s = \frac{1}{2}$ rather than at an integer, the analytic origin of the critical line $\mathrm{Re}(s) = \frac{1}{2}$. In this sense the entire functional-equation apparatus of the next section is a translation of the single identity (6.11) into the language of Dirichlet series.

The asymmetry between the two limiting regimes of θ is what the functional equation quantifies, and a numerical table makes both the asymmetry and the symmetry $t \leftrightarrow 1/t$ concrete before they are put to analytic use.

Example 6.2.5: Decay and the Modular Transformation

The functional equation (6.11) converts the elementary limit $\theta(s) \to 1$ as $s \to +\infty$ into the precise blow-up rate of θ at the origin. Setting $u = 1/t$, so that $t \to 0^+$ corresponds to $u \to +\infty$, the relation $\theta(1/t) = \sqrt{t}\,\theta(t)$ reads $\theta(u) = \sqrt{1/u}\,\theta(1/u)$, hence

$$\theta\left(\frac{1}{u}\right) = \sqrt{u}\,\theta(u) = \sqrt{u}\,\big(1 + o(1)\big) \qquad (u \to +\infty)$$

since $\theta(u) \to 1$. Writing $t = 1/u$, this is the asymptotic

$$\theta(t) \sim t^{-1/2} \qquad (t \to 0^+) \tag{6.13}$$

The two ends of the range thus behave oppositely: $\theta(t) - 1$ decays to 0 faster than any power of t as $t \to +\infty$ (geometrically, by the bound of definition 6.2.3), while $\theta(t)$ itself diverges like $t^{-1/2}$ as $t \to 0^+$. This asymmetry, rapid decay at one end against a $t^{-1/2}$ singularity at the other, is precisely what splits the Mellin integral defining $\xi(s)$ into a convergent piece and a pair of elementary polar terms in section 6.3.

A short table of values, computed by summing the rapidly convergent series (6.10) to ten significant figures, exhibits both the symmetry and the divergence. Each pair $(t, 1/t)$ is related by (6.11):

t	$1/4$	$1/2$	1	2	4
$\theta(t)$	2.000013949	1.419495488	1.086434811	1.003734885	1.000006975

The growth as t decreases toward 0 is the $t^{-1/2}$ trend of (6.13): at $t = 1/4$ the value $\theta(1/4) = 2.0000\ldots$ is already close to $t^{-1/2} = 2$, and at $t = 4$ the value $\theta(4) = 1.0000\ldots$ is close to its limit 1. The functional equation is verified directly against these entries. For the pair $t = 4$ and $1/t = 1/4$,

$$\sqrt{4}\,\theta(4) = 2 \times 1.000006975 = 2.000013950, \qquad \theta\big(\tfrac{1}{4}\big) = 2.000013949$$

agreeing to the recorded precision, so $\theta(1/4)\,/\,(\sqrt{4}\,\theta(4)) = 1.0000000000$. The same check on the pair $t = 2$, $1/t = 1/2$ gives $\sqrt{2}\,\theta(2) = 1.41421356 \times 1.003734885 = 1.419495487$ against $\theta(1/2) = 1.419495488$, again a ratio of 1 to ten digits. The fixed point $t = 1$ of the involution $t \mapsto 1/t$ is self-paired, and the entry $\theta(1) = 1.086434811$ satisfies $\sqrt{1}\,\theta(1) = \theta(1)$ trivially; this value equals $\pi^{1/4}/\Gamma(3/4)$ in closed form, the one argument at which θ admits an evaluation in terms of the gamma function.

6.3 The Completed Zeta Function and Its Functional Equation

The two preceding sections assembled the ingredients: the gamma function with its full pole structure (section 6.1), and the Jacobi theta function with its transformation law $\theta(1/t) = \sqrt{t}\,\theta(t)$ under the involution $t \mapsto 1/t$ (section 6.2). Multiplying $\zeta(s)$ by the archimedean gamma factor $\pi^{-s/2}\Gamma(s/2)$ produces a single function, the completed zeta function, that is symmetric under $s \leftrightarrow 1 - s$. This symmetry is Riemann's functional equation. The mechanism that produces it is the Mellin transform of chapter 2: writing the completed function as the Mellin transform of $\theta(t) - 1$ converts the theta symmetry $t \leftrightarrow 1/t$, after the change of variables $t \mapsto 1/t$ inside the integral, into the reflection $s \leftrightarrow 1 - s$.

The functional equation does two things at once. It continues $\zeta(s)$ from the half-plane $\mathrm{Re}(s) > 0$ reached in chapter 4 to the entire complex plane, and it pins the continuation down by a reflection symmetry whose axis is the line $\mathrm{Re}(s) = 1/2$. Every later structural fact about ζ, the trivial zeros of section 6.4, the special values $\zeta(2k)$ promised in example 4.1.3, and the centering of the Riemann Hypothesis on the critical line, is read off from this one identity.

The object that carries the symmetry is ζ multiplied by the gamma factor at the real place.

Definition 6.3.1: The Completed Zeta Function

The *local gamma factor at the real place* is

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)$$

with Γ the gamma function of definition 6.1.1. The *completed zeta function* is the product

$$\Lambda(s) = \pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)\,\zeta(s) = \Gamma_{\mathbb{R}}(s)\,\zeta(s)$$

defined initially on the half-plane $\mathrm{Re}(s) > 1$, where both factors are holomorphic and $\zeta(s)$ is given by its Dirichlet series.

The notation $\Gamma_{\mathbb{R}}$ marks the factor as attached to the real place of $\mathbb{Q}$, the place corresponding to the ordinary absolute value, in contrast to the p-adic factors $(1 - p^{-s})^{-1}$ of the Euler product attached to the finite places (corollary 4.1.2). The completed function $\Lambda(s)$ multiplies the Euler product over the finite places by the one missing factor over the infinite place, and it is this complete product, not ζ alone, that enjoys the clean symmetry. The closely related *Riemann xi function*

$$\xi(s) = \frac{1}{2}\,s(s-1)\,\Lambda(s) = \frac{1}{2}\,s(s-1)\,\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)\,\zeta(s)$$

clears the two poles of Λ (located in theorem 6.3.3) by the elementary factor $\frac{1}{2}s(s-1)$, producing the entire function whose Hadamard product over its zeros drives the explicit formula of box 5.5.5; the factor $s(s-1)$ is itself invariant under $s \mapsto 1-s$, so ξ inherits the symmetry of Λ unchanged.

Example 6.3.2: The Completed Factor at $s = 2$

At $s = 2$ the gamma factor is $\Gamma_{\mathbb{R}}(2) = \pi^{-1}\Gamma(1) = \pi^{-1}$, using $\Gamma(1) = 1$ from example 6.1.5. Since $\zeta(2) = \pi^2/6$ (example 4.1.3), the completed value is

$$\Lambda(2) = \pi^{-1}\cdot\frac{\pi^2}{6} = \frac{\pi}{6}$$

The symmetry proved below forces $\Lambda(-1) = \Lambda(2) = \pi/6$ as well. This is a nontrivial constraint: the reflected point $s = -1$ lies outside the half-plane of definition 6.3.1, where neither $\zeta(-1)$ nor $\Lambda(-1)$ has yet been assigned a value, and the functional equation is precisely what defines them. Reading the constraint backward, $\Lambda(-1) = \Gamma_{\mathbb{R}}(-1)\zeta(-1) = \pi^{1/2}\Gamma(-\frac{1}{2})\zeta(-1)$; with $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$ from the recursion (6.1) and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (example 6.1.5), this reads $\pi^{1/2}\cdot(-2\sqrt{\pi})\cdot\zeta(-1) = -2\pi\,\zeta(-1)$, and equating to $\pi/6$ gives $\zeta(-1) = -1/12$. The functional equation thus turns the elementary datum $\zeta(2) = \pi^2/6$ into the value $\zeta(-1) = -1/12$; this is the computation carried out systematically in section 6.4.

The motivating idea behind the gamma factor is the integral representation of example 2.4.2. The gamma function is the Mellin transform of e^{-t}, and rescaling its argument produces, for each fixed n, the Mellin transform of the Gaussian $e^{-\pi n^2 t}$. Summing over n assembles the gamma factor times $\zeta(s)$ as the Mellin transform of the theta series, and the theta functional equation then delivers the symmetry. This is Riemann's argument, carried out in full below.

Theorem 6.3.3: The Functional Equation of the Riemann Zeta Function

The completed zeta function extends to a meromorphic function on all of $\mathbb{C}$, holomorphic everywhere except for simple poles at $s = 0$ and $s = 1$, and satisfies the functional equation

$$\Lambda(s) = \Lambda(1 - s) \tag{6.14}$$

for all $s \in \mathbb{C}$. Equivalently,

$$\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)\,\zeta(s) = \pi^{-(1-s)/2}\,\Gamma\Big(\frac{1-s}{2}\Big)\,\zeta(1 - s)$$

***Proof*:**
The argument has four stages: the Mellin representation of Λ on $\mathrm{Re}(s) > 1$; the splitting of the Mellin integral at $t = 1$; the application of the theta functional equation to the integral over $(0, 1)$; and the reading-off of the symmetry, the continuation, and the poles from the resulting formula.

Step 1: The Mellin representation of Λ. Fix s with $\mathrm{Re}(s) > 1$. The starting point is the integral identity

$$\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)\,n^{-s} = \int_0^\infty e^{-\pi n^2 t}\,t^{s/2-1}\,dt \tag{6.15}$$

valid for each integer $n \geq 1$. To see it, begin from the definition $\Gamma(s/2) = \int_0^\infty e^{-u}\,u^{s/2-1}\,du$ of definition 6.1.1, which converges since $\mathrm{Re}(s/2) > 0$, and substitute $u = \pi n^2 t$, so $du = \pi n^2\,dt$ and $u^{s/2-1} = (\pi n^2)^{s/2-1} t^{s/2-1}$. This gives

$$\Gamma\Big(\frac{s}{2}\Big) = \int_0^\infty e^{-\pi n^2 t}\,(\pi n^2)^{s/2-1}\,t^{s/2-1}\cdot \pi n^2\,dt = (\pi n^2)^{s/2}\int_0^\infty e^{-\pi n^2 t}\,t^{s/2-1}\,dt$$

and dividing by $(\pi n^2)^{s/2} = \pi^{s/2} n^s$ yields (6.15). Summing (6.15) over $n \geq 1$ and using $\sum_{n\geq 1} n^{-s} = \zeta(s)$ gives, on the left, $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \Lambda(s)$. On the right the sum of the integrals is the integral of the sum:

$$\Lambda(s) = \sum_{n=1}^\infty \int_0^\infty e^{-\pi n^2 t}\,t^{s/2-1}\,dt = \int_0^\infty \left(\sum_{n=1}^\infty e^{-\pi n^2 t}\right) t^{s/2-1}\,dt = \int_0^\infty \psi(t)\,t^{s/2-1}\,dt \tag{6.16}$$

where $\psi(t) = \sum_{n\geq 1} e^{-\pi n^2 t} = \frac{1}{2}(\theta(t) - 1)$ by (6.10). The interchange of sum and integral is justified by Tonelli's theorem ([14]) applied to the nonnegative integrand $e^{-\pi n^2 t}\,t^{\mathrm{Re}(s)/2-1}$: the iterated integral $\sum_n \int_0^\infty e^{-\pi n^2 t} t^{\mathrm{Re}(s)/2-1}\,dt = \pi^{-\mathrm{Re}(s)/2}\Gamma(\mathrm{Re}(s)/2)\sum_n n^{-\mathrm{Re}(s)}$ is finite because $\mathrm{Re}(s) > 1$ makes the last series converge, so Tonelli permits the interchange for the absolute values, and the integrand here is already nonnegative. This establishes the Mellin representation (6.16) of Λ on $\mathrm{Re}(s) > 1$.

Step 2: Splitting at $t = 1$. Break the integral (6.16) at $t = 1$:

$$\Lambda(s) = \int_0^1 \psi(t)\,t^{s/2-1}\,dt + \int_1^\infty \psi(t)\,t^{s/2-1}\,dt \tag{6.17}$$

The second integral converges for every $s \in \mathbb{C}$ and defines an entire function. On $[1,\infty)$ the bound $\psi(t) \le e^{-\pi t}/(1-e^{-\pi t})$ established in section 6.2 shows $\psi(t)$ decays faster than any power of t, so for every fixed s the integrand $\psi(t)t^{s/2-1}$ is dominated by $\psi(t)t^{\mathrm{Re}(s)/2-1}$, integrable on $[1,\infty)$; holomorphy in s follows from differentiation under the integral sign, justified on compact subsets by the same domination ([13]). The first integral, over $(0,1)$, is where the theta functional equation enters, because $\psi(t)$ is not small as $t \to 0^+$.

Step 3: The theta functional equation on $(0,1)$. The functional equation $\theta(1/t) = \sqrt{t}\,\theta(t)$ of theorem 6.2.4, written in terms of $\psi = \frac{1}{2}(\theta-1)$, relates $\psi(t)$ to $\psi(1/t)$. From $\theta(t) = 2\psi(t)+1$ and $\theta(1/t) = 2\psi(1/t)+1$, the relation $\theta(1/t) = \sqrt{t}\,\theta(t)$ reads $2\psi(1/t)+1 = \sqrt{t}\,(2\psi(t)+1)$, and solving for $\psi(t)$ gives

$$\psi(t) = \frac{1}{\sqrt{t}}\,\psi\left(\frac{1}{t}\right) + \frac{1}{2\sqrt{t}} - \frac{1}{2} \tag{6.18}$$

Now substitute $t \mapsto 1/t$ in the first integral of (6.17). With $t = 1/r$, $dt = -r^{-2}\,dr$, the range $t \in (0,1)$ becomes $r \in (1,\infty)$, and $t^{s/2-1} = r^{1-s/2}$, so

$$\int_0^1 \psi(t)\,t^{s/2-1}\,dt = \int_1^\infty \psi\left(\frac{1}{r}\right) r^{1-s/2}\,r^{-2}\,dr = \int_1^\infty \psi\left(\frac{1}{r}\right) r^{-s/2-1}\,dr$$

By (6.18) with t replaced by $1/r$, namely $\psi(1/r) = \sqrt{r}\,\psi(r) + \frac{1}{2}\sqrt{r} - \frac{1}{2}$, the integrand becomes

$$\psi\left(\frac{1}{r}\right) r^{-s/2-1} = \left(\sqrt{r}\,\psi(r) + \frac{\sqrt{r}}{2} - \frac{1}{2}\right) r^{-s/2-1} = \psi(r)\,r^{-(s-1)/2-1} + \frac{1}{2}\,r^{-(s+1)/2} - \frac{1}{2}\,r^{-s/2-1}$$

the first power following from $\frac{1}{2} - s/2 - 1 = -(s-1)/2 - 1$. The two elementary terms integrate explicitly over $[1,\infty)$, convergently when $\mathrm{Re}(s) > 1$:

$$\frac{1}{2}\int_1^\infty r^{-(s+1)/2}\,dr = \frac{1}{2}\cdot\frac{1}{(s+1)/2-1} = \frac{1}{s-1}, \qquad \frac{1}{2}\int_1^\infty r^{-s/2-1}\,dr = \frac{1}{2}\cdot\frac{1}{s/2} = \frac{1}{s}$$

Therefore

$$\int_0^1 \psi(t)\,t^{s/2-1}\,dt = \int_1^\infty \psi(r)\,r^{(1-s)/2-1}\,dr + \frac{1}{s-1} - \frac{1}{s} \tag{6.19}$$

where the surviving integral is written with the exponent $-(s-1)/2-1 = (1-s)/2-1$.

Step 4: Symmetry, continuation, and poles. Substituting (6.19) into (6.17) gives

$$\Lambda(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty \psi(t)\left(t^{s/2-1} + t^{(1-s)/2-1}\right)dt \tag{6.20}$$

derived for $\mathrm{Re}(s) > 1$, where every step was valid. The right-hand side, however, makes sense for every $s \in \mathbb{C}$. The integral $\int_1^\infty \psi(t)(t^{s/2-1} + t^{(1-s)/2-1})\,dt$ converges for all s and is entire, by the rapid decay of ψ on $[1,\infty)$ exactly as in Step 2, applied now to each of the two powers; the remaining terms $1/(s-1) - 1/s$ are meromorphic on $\mathbb{C}$ with simple poles at $s = 1$ and $s = 0$. Hence the right-hand side of (6.20) is a meromorphic function on $\mathbb{C}$, holomorphic except for simple poles at $s = 0$ and $s = 1$, agreeing with $\Lambda(s)$ on $\mathrm{Re}(s) > 1$; by the identity theorem ([13]) it is the meromorphic continuation of Λ to all of $\mathbb{C}$. The residues are $\mathrm{Res}_{s=1}\Lambda = 1$ and $\mathrm{Res}_{s=0}\Lambda = -1$, the coefficients of $1/(s-1)$ and $1/s$.

The functional equation is now read directly off (6.20). The substitution $s \mapsto 1-s$ fixes the right-hand side: it interchanges the two powers $t^{s/2-1}$ and $t^{(1-s)/2-1}$ inside the integral, leaving the integral unchanged, and it sends

$$\frac{1}{s-1} - \frac{1}{s} \longmapsto \frac{1}{(1-s)-1} - \frac{1}{1-s} = \frac{1}{-s} - \frac{1}{1-s} = -\frac{1}{s} + \frac{1}{s-1}$$

which is the same pair of terms. Both the integral and the rational part are therefore invariant under $s \mapsto 1-s$, so $\Lambda(s) = \Lambda(1-s)$ for all s, which is (6.14). The equivalent form follows by writing out $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Lambda(1-s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$ from definition 6.3.1, completing the proof.

The symmetry expressed by (6.14) has its axis at the line $\mathrm{Re}(s) = 1/2$, the fixed-point set of the involution $s \mapsto 1-s$. The completed function takes equal values at any two points reflected across this line, and the line itself is where a point coincides with its mirror image. This is the analytic origin of the central role of $\mathrm{Re}(s) = 1/2$: it is not imposed on the theory but forced by the functional equation, descending from the fixed point $t = 1$ of the theta involution $t \mapsto 1/t$ through the half-power $t^{s/2}$ of the Mellin pairing in (6.16). The quadratic exponent n^2 in $e^{-\pi n^2 t}$ is what makes that pairing a pairing against $t^{s/2}$ rather than t^s, and the resulting halving is what centers the symmetry at $s = 1/2$ rather than at an integer; the Riemann Hypothesis (**??** 5.5.6) places every nontrivial zero on this axis, and (6.14) is what singles the axis out.

The two poles of Λ sit symmetrically about the axis, at $s = 1$ and $s = 0$, and the symmetry exchanges them. The pole at $s = 1$ is the pole of ζ itself (corollary 4.2.2): the gamma factor $\Gamma_{\mathbb{R}}(1) = \pi^{-1/2}\Gamma(1/2) = 1$ is finite and nonzero there, so Λ inherits the simple pole of ζ with the same residue, $\mathrm{Res}_{s=1}\Lambda = \mathrm{Res}_{s=1}\zeta = 1$, consistent with Step 4. The pole at $s = 0$, by contrast, is not a pole of ζ, which is holomorphic at the origin; it is the pole of the gamma factor, since $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ has a simple pole at $s = 0$ inherited from the pole of Γ at 0 (theorem 6.1.2). Under $s \leftrightarrow 1-s$ the analytic pole of ζ at $s = 1$ is matched with the archimedean pole of the gamma factor at $s = 0$: the two poles are reflections of one another, one arithmetic and one transcendental, and the functional equation is what identifies them.

The gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ is the factor at the real place of $\mathbb{Q}$, and its appearance here is the first instance of a general pattern. A completed L-function is the product of the Dirichlet-series Euler factors over the finite places with an archimedean factor over the infinite place, built from $\Gamma_{\mathbb{R}}$ or its complex analogue, and it is the completed product, never the bare Dirichlet series, that satisfies a clean functional equation relating s to $1-s$ (or to $k-s$ in the appropriate normalization). The completed Dirichlet L-functions, taken up in a later chapter of this book, carry exactly such a factor, chosen according to the parity of the character, and the completed Dedekind zeta functions, also treated later, carry one factor $\Gamma_{\mathbb{R}}$ for each real place and one factor $\Gamma_{\mathbb{C}}$ for each complex place of the number field. The single factor $\Gamma_{\mathbb{R}}$ attached to ζ here, the zeta function of $\mathbb{Q}$, is the simplest case of this product-over-places structure, and the functional equation of theorem 6.3.3 is its prototype.

With Λ continued to all of $\mathbb{C}$, the continuation of ζ itself follows by dividing out the gamma factor, and the symmetric functional equation (6.14) converts into the asymmetric form relating $\zeta(s)$ directly to $\zeta(1-s)$.

Corollary 6.3.4: Continuation of ζ to the Whole Plane

The Riemann zeta function continues to a meromorphic function on all of $\mathbb{C}$ via

$$\zeta(s) = \frac{\pi^{s/2}\,\Lambda(s)}{\Gamma(s/2)} \tag{6.21}$$

whose only pole is the simple pole at $s = 1$, with residue 1. In asymmetric form the functional equation reads

$$\zeta(s) = 2^s\,\pi^{s-1}\,\sin\Big(\frac{\pi s}{2}\Big)\,\Gamma(1-s)\,\zeta(1-s) \tag{6.22}$$

for all $s \in \mathbb{C}$.

***Proof*:**
The continuation comes from solving definition 6.3.1 for ζ; the asymmetric form comes from substituting the symmetric functional equation and converting the gamma factors by reflection and duplication.

Step 1: The continuation and its single pole. Solving $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ for ζ gives (6.21). The factor $\pi^{s/2}$ is entire and nowhere zero, and Λ is meromorphic on $\mathbb{C}$ with simple poles only at $s = 0, 1$ by theorem 6.3.3, so the right-hand side of (6.21) is meromorphic on $\mathbb{C}$. Its poles can only occur where Λ has a pole, and its zeros only where $1/\Gamma(s/2)$ has a zero, that is, where $\Gamma(s/2)$ has a pole. At $s = 1$ the factor $\Gamma(s/2) = \Gamma(1/2) = \sqrt{\pi}$ is finite and nonzero, so the simple pole of Λ at $s = 1$ passes to ζ; the residue is $\operatorname{Res}_{s=1}\zeta = \pi^{1/2}\operatorname{Res}_{s=1}\Lambda\,/\,\Gamma(1/2) = \pi^{1/2}\cdot 1\,/\,\sqrt{\pi} = 1$, recovering corollary 4.2.2. At $s = 0$ the factor $1/\Gamma(s/2)$ has a zero, because $\Gamma(s/2)$ has a pole there (theorem 6.1.2); this zero cancels the simple pole of Λ at $s = 0$, so ζ is holomorphic at the origin. No other poles of Λ exist, and $\Gamma(s/2) \neq 0$ everywhere by theorem 6.1.2, so $1/\Gamma(s/2)$ contributes no poles; the only pole of ζ is the simple pole at $s = 1$ with residue 1.

Step 2: Conversion to the asymmetric form. The symmetric equation (6.14) solved for $\zeta(s)$ reads

$$\zeta(s) = \pi^{s/2}\,\pi^{-(1-s)/2}\,\frac{\Gamma\big(\frac{1-s}{2}\big)}{\Gamma\big(\frac{s}{2}\big)}\,\zeta(1-s) = \pi^{\,s-1/2}\,\frac{\Gamma\big(\frac{1-s}{2}\big)}{\Gamma\big(\frac{s}{2}\big)}\,\zeta(1-s) \tag{6.23}$$

using $\frac{s}{2} - \frac{1-s}{2} = s - \frac{1}{2}$. The ratio of gamma values is reduced to elementary functions in two moves. First, the duplication formula (6.6) of theorem 6.1.4 at $z = s/2$, namely $\Gamma(z)\Gamma(z+\frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z)$, gives

$$\Gamma\Big(\frac{s}{2}\Big)\,\Gamma\Big(\frac{s+1}{2}\Big) = 2^{1-s}\sqrt{\pi}\,\Gamma(s) \tag{6.24}$$

Second, the reflection formula (6.3) of theorem 6.1.3 at $z = (1+s)/2$, using $\sin\big(\pi\frac{1+s}{2}\big) = \cos\big(\frac{\pi s}{2}\big)$, gives

$$\Gamma\Big(\frac{1-s}{2}\Big)\,\Gamma\Big(\frac{1+s}{2}\Big) = \frac{\pi}{\sin\big(\pi\frac{1+s}{2}\big)} = \frac{\pi}{\cos\big(\frac{\pi s}{2}\big)} \tag{6.25}$$

since $\Gamma(\frac{1-s}{2}) = \Gamma(1-\frac{1+s}{2})$. Dividing (6.25) by (6.24) cancels the common factor $\Gamma(\frac{1+s}{2}) = \Gamma(\frac{s+1}{2})$ and isolates the required ratio:

$$\frac{\Gamma\big(\frac{1-s}{2}\big)}{\Gamma\big(\frac{s}{2}\big)} = \frac{\pi}{\cos\big(\frac{\pi s}{2}\big)}\cdot\frac{1}{2^{1-s}\sqrt{\pi}\,\Gamma(s)} = \frac{\sqrt{\pi}\,2^{\,s-1}}{\cos\big(\frac{\pi s}{2}\big)\,\Gamma(s)}$$

Substituting this into (6.23) and collecting the powers of π, with $\pi^{s-1/2} \cdot \pi^{1/2} = \pi^s$, gives

$$\zeta(s) = \frac{\pi^s\, 2^{\,s-1}}{\cos\!\left(\frac{\pi s}{2}\right)\Gamma(s)}\,\zeta(1-s) \tag{6.26}$$

It remains to clear $\Gamma(s)$ from the denominator. The reflection formula (6.3) gives $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$, hence $1/\Gamma(s) = \sin(\pi s)\,\Gamma(1-s)/\pi$, and the double-angle identity $\sin(\pi s) = 2\sin\!\left(\frac{\pi s}{2}\right)\cos\!\left(\frac{\pi s}{2}\right)$ turns (6.26) into

$$\zeta(s) = \pi^s\, 2^{\,s-1} \cdot \frac{1}{\cos\!\left(\frac{\pi s}{2}\right)} \cdot \frac{2\sin\!\left(\frac{\pi s}{2}\right)\cos\!\left(\frac{\pi s}{2}\right)\Gamma(1-s)}{\pi}\,\zeta(1-s)$$

The factor $\cos\!\left(\frac{\pi s}{2}\right)$ cancels, and $2^{s-1} \cdot 2 = 2^s$ while $\pi^s/\pi = \pi^{s-1}$, leaving

$$\zeta(s) = 2^s\, \pi^{s-1} \sin\!\left(\frac{\pi s}{2}\right)\Gamma(1-s)\,\zeta(1-s)$$

which is (6.22). The identity was derived from the meromorphic equation (6.14) through identities valid wherever both sides are defined, and both sides of (6.22) are meromorphic on $\mathbb{C}$, so it holds for all s by the identity theorem ([13]), as we sought to show.

The asymmetric form (6.22) is the version of the functional equation most directly applied to values of ζ, since it expresses $\zeta(s)$ for s in the left half-plane in terms of $\zeta(1-s)$ for $1-s$ in the right half-plane, where the Dirichlet series converges. The three elementary factors record what the symmetric form hides inside the gamma factors. The factor $\zeta(1-s)$ supplies the values to be reflected; the factor $\Gamma(1-s)$ supplies the analytic continuation, contributing poles at the positive integers $s = 1, 2, 3, \ldots$ that must be cancelled or accounted for; and the factor $\sin(\pi s/2)$ is the source of the trivial zeros, vanishing at the even integers $s = 0, \pm 2, \pm 4, \ldots$. The interplay of these factors at the negative integers is exactly the computation of section 6.4: at $s = -2k$ the factor $\sin(\pi s/2)$ vanishes while the others stay finite, forcing $\zeta(-2k) = 0$, whereas at $s = 1-2k$ all three elementary factors are finite and nonzero, since $\sin(\pi(1-2k)/2) = \cos(\pi k) = (-1)^k$ and $\Gamma(1-(1-2k)) = \Gamma(2k) = (2k-1)!$, so the equation expresses $\zeta(1-2k)$ as the nonzero rational value that, reflected, yields $\zeta(2k)$.

Example 6.3.5: The Functional Equation at $s = 2$

The asymmetric equation (6.22) at $s = 2$ reproduces the value $\zeta(-1) = -1/12$ from the known value $\zeta(2) = \pi^2/6$, the same computation as in example 6.3.2 carried out through the elementary factors. Setting $s = -1$ in (6.22), so that $1 - s = 2$, gives

$$\zeta(-1) = 2^{-1}\,\pi^{-2} \sin\!\left(-\frac{\pi}{2}\right)\Gamma(2)\,\zeta(2)$$

The factors evaluate as $2^{-1} = \frac{1}{2}$, π^{-2}, $\sin(-\pi/2) = -1$, $\Gamma(2) = 1$ (example 6.1.5), and $\zeta(2) = \pi^2/6$ (example 4.1.3). Multiplying,

$$\zeta(-1) = \frac{1}{2} \cdot \frac{1}{\pi^2} \cdot (-1) \cdot 1 \cdot \frac{\pi^2}{6} = -\frac{1}{12}$$

the powers of π cancelling exactly, as they must for the rational value. The same equation read at an even negative integer shows the trivial zeros directly: at $s = -2$, the factor $\sin(\pi s/2) = \sin(-\pi) = 0$, while $2^{-2}\pi^{-3}$, $\Gamma(3) = 2$, and $\zeta(3)$ are all finite, so $\zeta(-2) = 0$. The single factor $\sin(\pi s/2)$ carries the vanishing.

6.4 Trivial Zeros and Special Values

The functional equation of section 6.3 continued $\zeta(s)$ to the whole plane and tied $\zeta(s)$ to $\zeta(1-s)$; this section reads off what that symmetry forces at the integers. The completed function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ of definition 6.3.1 is holomorphic and finite away from its two poles at $s = 0$ and $s = 1$, while the gamma factor $\Gamma(s/2)$ carries poles of its own at the negative even integers. Where the gamma factor blows up but Λ stays finite, the only remaining factor, ζ, must vanish: these are the trivial zeros of ζ at $s = -2, -4, -6, \ldots$. The same accounting, applied at the negative odd integers and the origin, pins ζ to the Bernoulli numbers of definition 1.5.1, giving the rational values $\zeta(0) = -1/2$ and $\zeta(1-2k) = -B_{2k}/(2k)$.

Reflecting these rational values across the line $\mathrm{Re}(s) = 1/2$ through the functional equation produces the values at the positive even integers, expressing each $\zeta(2k)$ as a rational multiple of π^{2k}. In particular $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$, discharging the evaluations foreshadowed in example 4.1.3: what was cited there to external analysis is now a consequence of the functional equation, derived from the rational value $\zeta(1-2k)$ by reflection. The contrast with the odd values, which the same reflection sends to the vanishing $\zeta(-2k)$ and so leaves without any closed form, is the structural reason for the difficulty noted in chapter 4.

The trivial zeros come first, read directly off the pole structure of the completed function.

Corollary 6.4.1: The Trivial Zeros

For every integer $k \geq 1$,

$$\zeta(-2k) = 0$$

These are the *trivial zeros* of the Riemann zeta function. Together with the points of the critical strip $0 \leq \mathrm{Re}(s) \leq 1$, they account for every zero of ζ: outside the closed strip, the only zeros of ζ are the negative even integers $s = -2, -4, -6, \ldots$.

***Proof*:**
The argument reads the vanishing of $\zeta(-2k)$ off definition 6.3.1 and then locates all zeros outside the strip.

Step 1: Vanishing at the negative even integers. Fix $k \geq 1$ and set $s = -2k$. By theorem 6.3.3, the completed function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is meromorphic on $\mathbb{C}$ with its only poles the simple poles at $s = 0$ and $s = 1$; since $-2k \notin \{0, 1\}$, the value $\Lambda(-2k)$ is finite. Solving definition 6.3.1 for ζ as in (6.21) gives

$$\zeta(-2k) = \frac{\pi^{-k}\,\Lambda(-2k)}{\Gamma(-k)}$$

where $\Gamma(s/2) = \Gamma(-k)$ at $s = -2k$ and $\pi^{s/2} = \pi^{-k}$. The factor π^{-k} is finite, and $\Lambda(-2k)$ is finite by the preceding sentence. The denominator $\Gamma(-k)$, on the other hand, is infinite: Γ has a simple pole at the non-positive integer $-k$ (theorem 6.1.2). Equivalently, $1/\Gamma(s/2)$ has a zero at $s = -2k$, since $1/\Gamma$ is entire and vanishes exactly at the poles of Γ. A finite numerator times the zero $1/\Gamma(-k) = 0$ gives

$$\zeta(-2k) = \pi^{-k}\,\Lambda(-2k) \cdot \frac{1}{\Gamma(-k)} = \pi^{-k}\,\Lambda(-2k) \cdot 0 = 0$$

as claimed. The mechanism is the transfer recorded after theorem 6.1.2: the pole of the archimedean factor $\Gamma(s/2)$ forces the arithmetic factor ζ to vanish, so that the product Λ stays finite.

Step 2: These are the only zeros outside the strip. Suppose s_0 is a zero of ζ with $\mathrm{Re}(s_0) \notin [0,1]$. The Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of corollary 4.1.2 converges on $\mathrm{Re}(s) > 1$ and is a product of nonzero factors, none of which vanishes, so ζ has no zero with $\mathrm{Re}(s) > 1$; hence $\mathrm{Re}(s_0) < 0$. On the half-plane $\mathrm{Re}(s) < 0$ the asymmetric functional equation (6.22) of corollary 6.3.4 applies, reading

$$\zeta(s_0) = 2^{s_0}\,\pi^{s_0-1}\sin\Big(\frac{\pi s_0}{2}\Big)\,\Gamma(1-s_0)\,\zeta(1-s_0)$$

The reflected point has $\mathrm{Re}(1-s_0) > 1$, so $\zeta(1-s_0) \neq 0$ by the Euler-product argument just given. The factor $\Gamma(1-s_0)$ is finite and nonzero: $1-s_0$ has positive real part, away from the poles of Γ, and Γ has no zeros (theorem 6.1.2). The factors 2^{s_0} and π^{s_0-1} are nonzero. The product can therefore vanish only through the factor $\sin(\pi s_0/2)$, which is zero precisely when $s_0/2 \in \mathbb{Z}$, that is, when s_0 is an even integer. Combined with $\mathrm{Re}(s_0) < 0$, this forces $s_0 \in \{-2,-4,-6,\dots\}$. Every zero outside the closed strip is thus a negative even integer, and Step 1 shows each negative even integer is indeed a zero, completing the proof.

The trivial zeros are exactly the zeros the functional equation manufactures from the pole structure of the gamma factor; they carry no arithmetic content and are accounted for completely. What the corollary leaves open is the location of the remaining zeros, those inside the critical strip $0 \le \mathrm{Re}(s) \le 1$. The Euler product rules out zeros with $\mathrm{Re}(s) > 1$, the functional equation reflects this to rule out all zeros with $\mathrm{Re}(s) < 0$ except the trivial ones, and the non-vanishing on the line $\mathrm{Re}(s) = 1$ of theorem 4.4.2 (with its reflection $\mathrm{Re}(s) = 0$) pushes the boundary inward; the zeros that survive, the *nontrivial* zeros, all lie in the open strip $0 < \mathrm{Re}(s) < 1$. The Riemann Hypothesis (**??** 5.5.6) asserts that they lie on the central line $\mathrm{Re}(s) = 1/2$, and it is the trivial zeros having been disposed of here that lets the conjecture be stated about the strip alone.

The same division of Λ by the gamma factor that produced the trivial zeros produces, at the negative *odd* integers and at the origin, finite values rather than zeros, because there the gamma factor $\Gamma(s/2)$ is finite. Those values are rational, and they are exactly the Bernoulli numbers.

Theorem 6.4.2: Values at the Negative Integers

The Riemann zeta function takes the value

$$\zeta(0) = -\frac{1}{2}$$

at the origin, and for every integer $k \ge 1$,

$$\zeta(1-2k) = -\frac{B_{2k}}{2k} \tag{6.27}$$

where B_{2k} is the Bernoulli number of definition 1.5.1. The remaining negative integer values are the trivial zeros $\zeta(-2k) = 0$ of corollary 6.4.1.

***Proof*:**
The value at the origin is computed from the Laurent data of Λ and Γ at $s = 0$; the values at the negative odd integers come from the integral representation of theorem 2.4.5, deformed to a loop contour on which the Bernoulli numbers appear as residues.

Step 1: The value $\zeta(0) = -1/2$. Both $\Lambda(s)$ and $\Gamma(s/2)$ have simple poles at $s = 0$, and their quotient through (6.21) is finite there, so $\zeta(0)$ is computed as a ratio of residues. From (6.21),

$$\zeta(s) = \pi^{s/2}\,\frac{\Lambda(s)}{\Gamma(s/2)}$$

and ζ is holomorphic at $s = 0$ by corollary 6.3.4, so $\zeta(0) = \lim_{s\to 0}\zeta(s)$. The factor $\pi^{s/2} \to 1$ as $s \to 0$. For the quotient $\Lambda(s)/\Gamma(s/2)$, multiply numerator and denominator by s and take limits separately. The numerator has the simple pole $\mathrm{Res}_{s=0}\Lambda = -1$ recorded in the proof of theorem 6.3.3, so $\lim_{s\to 0} s\,\Lambda(s) = -1$. For the denominator, the recursion (6.1) gives $\Gamma(s/2) = \Gamma(s/2+1)/(s/2) = (2/s)\,\Gamma(s/2+1)$, and $\Gamma(s/2+1) \to \Gamma(1) = 1$ as $s \to 0$ (example 6.1.5), so $\lim_{s\to 0} s\,\Gamma(s/2) = 2$. Therefore

$$\zeta(0) = \lim_{s\to 0}\pi^{s/2}\,\frac{s\,\Lambda(s)}{s\,\Gamma(s/2)} = 1\cdot\frac{-1}{2} = -\frac{1}{2}$$

the cancellation of the factor s being legitimate because both one-sided limits exist and the denominator limit is nonzero.

Step 2: A loop representation of ζ. The continuation of corollary 6.3.4 is recast as a contour integral, starting from the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t-1)\,dt$ of theorem 2.4.5, valid for $\mathrm{Re}(s) > 1$. Let H_ε be the *Hankel contour*: the path coming in from $+\infty$ along the upper side of the positive real axis, traversing the circle $|z| = \varepsilon$ once counterclockwise around the origin, and returning to $+\infty$ along the lower side. On H_ε the function $(-z)^{s-1} = e^{(s-1)\log(-z)}$ is defined with the branch of $\log(-z)$ that is real for $z < 0$, equivalently $\arg(-z) \in (-\pi, \pi)$; on the upper edge $\log(-z) = \log|z| - i\pi$ and on the lower edge $\log(-z) = \log|z| + i\pi$. The claim is the *Riemann–Hankel representation*

$$\zeta(s) = -\frac{\Gamma(1-s)}{2\pi i}\int_{H_\varepsilon}\frac{(-z)^{s-1}}{e^z-1}\,dz \tag{6.28}$$

valid for all $s \in \mathbb{C}$ (the integral is independent of $\varepsilon \in (0, 2\pi)$, since the integrand is holomorphic in the punctured disc $0 < |z| < 2\pi$ between two such contours, and defines an entire function of s). To establish (6.28), fix $\mathrm{Re}(s) > 1$ and shrink the radius $\varepsilon \to 0$. On the circle $|z| = \varepsilon$ the integrand is $O(\varepsilon^{\mathrm{Re}(s)-1}\cdot\varepsilon^{-1}) = O(\varepsilon^{\mathrm{Re}(s)-2})$ in modulus, since $1/(e^z-1) = O(1/|z|)$ near the origin, and the circle has length $2\pi\varepsilon$; the contribution is $O(\varepsilon^{\mathrm{Re}(s)-1}) \to 0$ as $\varepsilon \to 0$ because $\mathrm{Re}(s) > 1$. The two straight edges survive. On the upper edge, $z = t$ with t running from ∞ down to 0 and $(-z)^{s-1} = e^{(s-1)(\log t - i\pi)} = t^{s-1}e^{-i\pi(s-1)}$; on the lower edge, $z = t$ with t running from 0 up to ∞ and $(-z)^{s-1} = t^{s-1}e^{i\pi(s-1)}$. The upper edge is traversed from ∞ to 0, contributing the negative of $e^{-i\pi(s-1)}\int_0^\infty t^{s-1}/(e^t-1)\,dt$, while the lower edge contributes $+e^{i\pi(s-1)}\int_0^\infty t^{s-1}/(e^t-1)\,dt$; their sum is

$$\int_{H_\varepsilon}\frac{(-z)^{s-1}}{e^z-1}\,dz \xrightarrow{\varepsilon\to 0} \left(e^{i\pi(s-1)} - e^{-i\pi(s-1)}\right)\int_0^\infty\frac{t^{s-1}}{e^t-1}\,dt = 2i\sin\bigl(\pi(s-1)\bigr)\,\Gamma(s)\zeta(s)$$

using theorem 2.4.5 for the surviving real integral. Now $\sin(\pi(s-1)) = -\sin(\pi s)$, and the reflection formula (6.3) gives $\sin(\pi s)\,\Gamma(s) = \pi/\Gamma(1-s)$, so the limiting value is

$$\int_{H_\varepsilon} \frac{(-z)^{s-1}}{e^z - 1}\,dz = 2i\sin\big(\pi(s-1)\big)\,\Gamma(s)\zeta(s) = -2i\sin(\pi s)\,\Gamma(s)\zeta(s) = -\frac{2\pi i\,\zeta(s)}{\Gamma(1-s)}$$

for $\mathrm{Re}(s) > 1$, which rearranges to (6.28). The integral over H_ε converges for every $s \in \mathbb{C}$, because away from the origin the contour stays a bounded distance from the poles of $1/(e^z-1)$ at $z = 2\pi in$ and the factor $1/(e^z-1)$ decays as $\mathrm{Re}(z) \to +\infty$, so it defines an entire function of s; multiplying by the meromorphic factor $-\Gamma(1-s)/(2\pi i)$ gives a meromorphic function agreeing with ζ on $\mathrm{Re}(s) > 1$, hence equal to the continued ζ on all of $\mathbb{C}$ by the identity theorem ([13]). This is (6.28).

Step 3: Collapse to a residue at a negative odd integer. Set $s = 1-2k$ in (6.28). The exponent $s-1 = -2k$ is an even negative integer, so $(-z)^{s-1} = (-z)^{-2k} = z^{-2k}$ is single-valued, the branch ambiguity disappears, and the two edges of H_ε cancel: their integrands are now identical functions of t traversed in opposite directions. Only the circle $|z| = \varepsilon$ remains, traversed once counterclockwise, so the contour integral equals $2\pi i$ times the residue of the integrand at $z = 0$:

$$\int_{H_\varepsilon} \frac{z^{-2k}}{e^z-1}\,dz = 2\pi i\,\mathrm{Res}_{z=0}\frac{z^{-2k}}{e^z-1}$$

The residue is read from the generating function of definition 1.5.1. Writing $\frac{1}{e^z-1} = \frac{1}{z}\cdot\frac{z}{e^z-1} = \frac{1}{z}\sum_{m=0}^{\infty}\frac{B_m}{m!}z^m$, the integrand is

$$\frac{z^{-2k}}{e^z-1} = z^{-2k-1}\sum_{m=0}^{\infty}\frac{B_m}{m!}z^m = \sum_{m=0}^{\infty}\frac{B_m}{m!}z^{m-2k-1}$$

whose residue, the coefficient of z^{-1}, is the term with $m = 2k$, namely $B_{2k}/(2k)!$. Substituting $s = 1-2k$ into the representation (6.28), with $\Gamma(1-s) = \Gamma(2k) = (2k-1)!$ (example 6.1.5) and the minus sign carried by (6.28),

$$\zeta(1-2k) = -\frac{(2k-1)!}{2\pi i}\cdot 2\pi i\,\frac{B_{2k}}{(2k)!} = -(2k-1)!\,\frac{B_{2k}}{(2k)!} = -\frac{B_{2k}}{2k}$$

the cancellation of $2\pi i$ against $2\pi i$ and of $(2k-1)!$ against $(2k)! = 2k\cdot(2k-1)!$ leaving the rational value, which is (6.27).

Step 4: Consistency check via the functional equation. The value (6.27) is confirmed independently from the asymmetric functional equation (6.22), which at $s = 1-2k$ reads, with $1-s = 2k$,

$$\zeta(1-2k) = 2^{1-2k}\,\pi^{-2k}\,\sin\left(\frac{\pi(1-2k)}{2}\right)\Gamma(2k)\,\zeta(2k)$$

Here $\sin(\pi(1-2k)/2) = \sin(\pi/2-\pi k) = \cos(\pi k) = (-1)^k$, and $\Gamma(2k) = (2k-1)!$. For $k = 1$ this gives $\zeta(-1) = 2^{-1}\pi^{-2}\cdot(-1)\cdot 1\cdot\zeta(2) = -\frac{1}{2}\pi^{-2}\cdot(\pi^2/6) = -1/12$, matching $-B_2/(2\cdot 1) = -(1/6)/2 = -1/12$ from (6.27). The general identity (6.27) is consistent with (6.22) for every k, as theorem 6.4.3 makes precise by inverting this relation, completing the proof.

The values $\zeta(1-2k) = -B_{2k}/(2k)$ are the arithmetic core of the negative-integer behavior. They are rational, and they are the same rational numbers that govern sums of powers and the Euler–Maclaurin expansion of chapter 1, which is the first hint that the zeta function at negative integers carries the combinatorics of the Bernoulli numbers rather than the transcendence of π. The vanishing of the odd-indexed Bernoulli numbers $B_3 = B_5 = \cdots = 0$ (proposition 1.5.2) is consistent with the picture: (6.27) involves only the even-indexed B_{2k}, never the odd ones, so the formula never references a vanishing Bernoulli number, and the trivial zeros at the negative even integers are a separate phenomenon coming from the gamma factor, not from any vanishing of B. The value $\zeta(0) = -1/2$ fits the pattern as the case "$k = 0$" in the sense that $-B_0/0$ is replaced by the residue computation of Step 1, with $B_0 = 1$ entering through $\Gamma(s/2+1) \to 1$.

The functional equation now runs in the opposite direction. Having pinned ζ at the negative odd integers $1-2k$ to the rational values $-B_{2k}/(2k)$, reflecting across $\mathrm{Re}(s) = 1/2$ pins ζ at the positive even integers $2k$, and the reflection introduces exactly the power of π that turns the rational value into the transcendental one.

Theorem 6.4.3: Values at the Positive Even Integers

For every integer $k \geq 1$,

$$\zeta(2k) = (-1)^{k+1} \frac{B_{2k}\,(2\pi)^{2k}}{2\,(2k)!} \tag{6.29}$$

where B_{2k} is the Bernoulli number of definition 1.5.1. In particular $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$, the values foreshadowed in example 4.1.3.

***Proof*:**

The asymmetric functional equation (6.22) of corollary 6.3.4, evaluated at $s = 1-2k$, relates $\zeta(1-2k)$ to $\zeta(2k)$; substituting the value of $\zeta(1-2k)$ from theorem 6.4.2 and solving for $\zeta(2k)$ yields (6.29). The elementary factors in (6.22) are exactly those produced from the gamma factors of the symmetric equation (6.14) by the reflection formula (6.3) of theorem 6.1.3 and the duplication formula (6.6) of theorem 6.1.4, carried out in the proof of corollary 6.3.4; the computation below uses the asymmetric form directly, so those two identities enter through it. The equation (6.22) at $s = 1-2k$, with $1-s = 2k$, reads

$$\zeta(1-2k) = 2^{1-2k}\,\pi^{-2k} \sin\left(\frac{\pi(1-2k)}{2}\right) \Gamma(2k)\,\zeta(2k) \tag{6.30}$$

The trigonometric factor simplifies by the addition formula:

$$\sin\left(\frac{\pi(1-2k)}{2}\right) = \sin\left(\frac{\pi}{2} - \pi k\right) = \cos(\pi k) = (-1)^k$$

and the gamma factor is $\Gamma(2k) = (2k-1)!$ by example 6.1.5, since $2k$ is a positive integer. Substituting these and the value $\zeta(1-2k) = -B_{2k}/(2k)$ of (6.27) into (6.30),

$$-\frac{B_{2k}}{2k} = 2^{1-2k}\,\pi^{-2k}\,(-1)^k\,(2k-1)!\,\zeta(2k)$$

Solving for $\zeta(2k)$ requires dividing by the factor $2^{1-2k}\pi^{-2k}(-1)^k(2k-1)!$, which is nonzero, giving

$$\zeta(2k) = -\frac{B_{2k}}{2k} \cdot \frac{1}{2^{1-2k}\,\pi^{-2k}\,(-1)^k\,(2k-1)!} = -\frac{B_{2k}}{2k} \cdot 2^{2k-1}\,\pi^{2k}\,(-1)^{-k}\,\frac{1}{(2k-1)!}$$

using $1/2^{1-2k} = 2^{2k-1}$, $1/\pi^{-2k} = \pi^{2k}$, and $1/(-1)^k = (-1)^{-k} = (-1)^k$. Collecting the factorials with $2k \cdot (2k-1)! = (2k)!$ and the signs with $-(-1)^k = (-1)^{k+1}$,

$$\zeta(2k) = (-1)^{k+1}\, B_{2k}\, \frac{2^{2k-1}\,\pi^{2k}}{(2k)!}$$

Finally $2^{2k-1}\pi^{2k} = \frac{1}{2}\, 2^{2k}\pi^{2k} = \frac{1}{2}(2\pi)^{2k}$, which gives (6.29).

For the stated special cases, take $k = 1$ and $k = 2$. With $B_2 = 1/6$ (definition 1.5.1),

$$\zeta(2) = (-1)^2\, \frac{B_2\,(2\pi)^2}{2 \cdot 2!} = \frac{(1/6) \cdot 4\pi^2}{4} = \frac{\pi^2}{6}$$

and with $B_4 = -1/30$,

$$\zeta(4) = (-1)^3\, \frac{B_4\,(2\pi)^4}{2 \cdot 4!} = -\frac{(-1/30) \cdot 16\pi^4}{48} = \frac{16\pi^4}{1440} = \frac{\pi^4}{90}$$

recovering the values of example 4.1.3, completing the proof.

The even values are rational multiples of powers of π for a structural reason now visible in the proof: the functional equation ties $\zeta(2k)$ to the rational number $\zeta(1-2k) = -B_{2k}/(2k)$, and the elementary factors 2^{2k}, π^{2k}, and $1/(2k)!$ that mediate the reflection supply exactly one factor π^{2k}, turning the rational value into a rational multiple of π^{2k}. The transcendence of $\zeta(2k)$ is therefore inherited from π alone; no transcendence enters through the arithmetic side, which contributes only the rational B_{2k}. The odd values $\zeta(3), \zeta(5), \ldots$ have no analogous expression precisely because the reflection sends them to the *other* family of negative-integer values: the functional equation pairs $\zeta(2k+1)$ with $\zeta(-2k)$, which is a trivial zero by corollary 6.4.1, so the reflection yields the identity $0 = 0$ rather than an evaluation. This is the exact mechanism behind the difficulty foreshadowed in chapter 4: the even values fall out because their mirror images are nonzero rationals, while the odd values resist because their mirror images vanish, leaving the functional equation silent. Whether $\zeta(2k+1)$ is even irrational is known only for $\zeta(3)$, Apéry's constant of example 4.1.3; the closed-form question is open for every odd argument.

The values at small integers are collected and verified below, showing both families, the rational negative-integer values and the π-power positive even values, side by side.

Example 6.4.4: The Special Values, Computed

The formulas (6.27) and (6.29), together with $\zeta(0) = -1/2$ from theorem 6.4.2, evaluate ζ at the first several integers from the Bernoulli numbers $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$ of definition 1.5.1.

1. *Negative integers.* From $\zeta(1-2k) = -B_{2k}/(2k)$:

$$\zeta(-1) = -\frac{B_2}{2} = -\frac{1/6}{2} = -\frac{1}{12}, \qquad \zeta(-3) = -\frac{B_4}{4} = -\frac{-1/30}{4} = \frac{1}{120}$$

   and $\zeta(-5) = -B_6/6 = -(1/42)/6 = -1/252$. The trivial zeros fill in the even arguments: $\zeta(-2) = \zeta(-4) = \cdots = 0$ by corollary 6.4.1.

2. *Positive even integers.* From $\zeta(2k) = (-1)^{k+1}B_{2k}(2\pi)^{2k}/(2(2k)!)$, the case $k = 3$ with

$B_6 = 1/42$ gives

$$\zeta(6) = (-1)^4 \frac{B_6 (2\pi)^6}{2 \cdot 6!} = \frac{(1/42) \cdot 64\pi^6}{2 \cdot 720} = \frac{64\pi^6}{42 \cdot 1440} = \frac{64\pi^6}{60480} = \frac{\pi^6}{945}$$

alongside $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$ from theorem 6.4.3.

Assembling the two families,

s	0	-1	-3	2	4	6
$\zeta(s)$	$-\frac{1}{2}$	$-\frac{1}{12}$	$\frac{1}{120}$	$\frac{\pi^2}{6}$	$\frac{\pi^4}{90}$	$\frac{\pi^6}{945}$

One even value is checked numerically against the defining series. For $\zeta(6)$, the closed form gives $\pi^6/945 = 961.389\ldots/945 = 1.017343\ldots$, while the partial sum $\sum_{n=1}^{4} n^{-6} = 1 + 1/64 + 1/729 + 1/4096 = 1.017241\ldots$ already agrees to three decimal places, the difference 1.02×10^{-4} matching the size of the tail $\sum_{n \geq 5} n^{-6}$, which is bounded by $\int_4^\infty t^{-6}\, dt = 1/(5 \cdot 4^5) \approx 1.95 \times 10^{-4}$. The notorious value $\zeta(-1) = -1/12$, which the divergent series $1 + 2 + 3 + \cdots$ is sometimes said to "equal" in physics, is here simply the value of the analytically continued ζ at $s = -1$, the rational number $-B_2/2$ produced by the functional equation, with no summation of the divergent series involved.

6.5 The Critical Strip and the Riemann Hypothesis

The functional equation $\Lambda(s) = \Lambda(1-s)$ of theorem 6.3.3 does more than continue ζ to the whole plane; it imposes a reflection symmetry whose axis is the line $\mathrm{Re}(s) = 1/2$. The trivial zeros of section 6.4, lying at the even negative integers, account for every zero in the left half-plane, and the half-plane $\mathrm{Re}(s) \geq 1$ carries none by the non-vanishing on the line (theorem 4.4.2). What remains is a vertical strip of width one, the critical strip, into which the symmetry $s \leftrightarrow 1-s$ folds all the other zeros. These are the nontrivial zeros, the zeros that carry the arithmetic content of ζ through the explicit formula of box 5.5.5.

The functional equation arranges the nontrivial zeros into a configuration symmetric across the critical line, and a second symmetry, complex conjugation, pairs each zero with its reflection across the real axis. The two symmetries together confine the zeros to quadruples unless they fall on the critical line itself. The Riemann Hypothesis (**??** 5.5.6) is the assertion that the quadruples always degenerate, that every nontrivial zero lies on the axis of symmetry. This section fixes the terminology, establishes the symmetries the functional equation forces, states the asymptotic count of the zeros in the strip, and introduces the Hardy Z-function, the real-valued companion of ζ on the critical line whose zeros are exactly the critical-line zeros of ζ.

Definition 6.5.1: The Critical Strip and the Critical Line

The *critical strip* is the closed vertical strip

$$\{s \in \mathbb{C} \big| 0 \leq \mathrm{Re}(s) \leq 1\}$$

The *critical line* is the vertical line $\mathrm{Re}(s) = 1/2$, the axis of symmetry of the functional equation. A *nontrivial zero* of ζ is a zero lying in the critical strip; equivalently (as proposition 6.5.3 shows the strip endpoints carry no zeros) a zero with $0 < \mathrm{Re}(s) < 1$. The nontrivial zeros of ζ are exactly the zeros of the entire function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ of box 5.5.5.

The qualifier "nontrivial" separates these zeros from the trivial zeros at $s = -2, -4, -6, \ldots$ of corollary 6.4.1, which lie outside the strip and are completely understood. The identification of the nontrivial zeros with the zeros of ξ is what makes the strip the right object: the elementary prefactor $\frac{1}{2}s(s-1)$ removes the pole of ζ at $s = 1$ and the pole of the gamma factor at $s = 0$, and the gamma factor $\pi^{-s/2}\Gamma(s/2)$ has no zeros and supplies poles at $s = 0, -2, -4, \ldots$ that cancel the trivial zeros of ζ one for one. What survives in ξ is an entire function whose only zeros are the nontrivial zeros of ζ, none of them at $s = 0$ or $s = 1$. Because ξ is entire and (by Stirling, theorem 6.1.4) of order one, it has infinitely many zeros, all inside the strip; the Hadamard product over them is what converts the explicit formula's contour integral into a sum over ρ.

Example 6.5.2: The First Nontrivial Zero

The lowest nontrivial zero in the upper half-plane is

$$\rho_1 = \tfrac{1}{2} + i\gamma_1, \qquad \gamma_1 = 14.134725\ldots$$

a value with $\mathrm{Re}(\rho_1) = 1/2$ exactly, consistent with the Riemann Hypothesis. It lies in the interior of the critical strip, $0 < \frac{1}{2} < 1$, and not on either boundary line: it is not at $\mathrm{Re}(s) = 1$, in accordance with theorem 4.4.2, and not at $\mathrm{Re}(s) = 0$, where ξ is nonzero. By the conjugate symmetry of proposition 6.5.3 the point $\bar{\rho}_1 = \frac{1}{2} - i\gamma_1$ is also a zero, so the zeros nearest the real axis form the pair $\frac{1}{2} \pm 14.134725\ldots i$. Since this zero sits on the critical line, the reflection $1 - \rho_1 = \frac{1}{2} - i\gamma_1$ coincides with $\bar{\rho}_1$, and the would-be quadruple of proposition 6.5.3 collapses to a conjugate pair; the numerical value $14.134725\ldots$ is computed in [6, Ch. 15].

The two boundary lines of the strip and the region outside it must first be cleared of nontrivial zeros, and the symmetries that organize the zeros inside the strip must be established, before the count of theorem 6.5.4 can be stated. Both follow from the functional equation together with the non-vanishing on $\mathrm{Re}(s) = 1$.

Proposition 6.5.3: Symmetry of the Nontrivial Zeros

The nontrivial zeros of ζ lie in the open strip $0 < \mathrm{Re}(s) < 1$. They are symmetric under the two involutions

$$s \longmapsto 1 - s \qquad \text{and} \qquad s \longmapsto \bar{s}$$

that is, if ρ is a nontrivial zero then so are $1 - \rho$, $\bar{\rho}$, and $1 - \bar{\rho}$. Consequently the nontrivial zeros occur in quadruples $\{\rho,\ 1-\rho,\ \bar{\rho},\ 1-\bar{\rho}\}$, which degenerate to a conjugate pair $\{\rho, \bar{\rho}\}$ precisely when ρ lies on the critical line $\mathrm{Re}(s) = 1/2$.

***Proof*:**
The argument locates the zeros in the open strip, then derives each of the two symmetries, then reads off the quadruple structure.

Step 1: No nontrivial zeros with $\mathrm{Re}(s) \geq 1$. For $\mathrm{Re}(s) > 1$ the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of corollary 4.1.2 converges to a nonzero value, since each factor is nonzero and the product converges absolutely; hence $\zeta(s) \neq 0$ there. On the line $\mathrm{Re}(s) = 1$ itself, $\zeta(1+it) \neq 0$ for every real $t \neq 0$ by theorem 4.4.2, and at $t = 0$ the point $s = 1$ is the pole of ζ (corollary 6.3.4), not a zero. Therefore ζ has no zeros in the closed half-plane $\mathrm{Re}(s) \geq 1$.

Step 2: No zeros with $\mathrm{Re}(s) \leq 0$ *except the trivial ones.* The asymmetric functional equation (6.22) of corollary 6.3.4 reads

$$\zeta(s) = 2^s\,\pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\,\zeta(1-s)$$

Fix s with $\mathrm{Re}(s) < 0$, so that $\mathrm{Re}(1-s) > 1$. By Step 1 the factor $\zeta(1-s)$ is nonzero. The factors 2^s and π^{s-1} are nowhere zero, and $\Gamma(1-s)$ is nonzero everywhere, having no zeros (theorem 6.1.2); since $\mathrm{Re}(1-s) > 1 > 0$, the point $1-s$ is not a pole of Γ either, so $\Gamma(1-s)$ is finite and nonzero. The only factor that can vanish is $\sin(\pi s/2)$, which is zero exactly at the even integers; in the range $\mathrm{Re}(s) < 0$ these are $s = -2, -4, -6, \ldots$, the trivial zeros of corollary 6.4.1. Hence in $\mathrm{Re}(s) < 0$ the function ζ vanishes only at the trivial zeros. On the line $\mathrm{Re}(s) = 0$ the same identity applies with $\mathrm{Re}(1-s) = 1$: the factor $\zeta(1-s) = \zeta(1-it)$ is nonzero for $t \neq 0$ by theorem 4.4.2, the factor $\sin(\pi s/2) = \sin(\pi it/2)$ is nonzero for $t \neq 0$, and $\Gamma(1-s)$ is finite and nonzero on this line; at $s = 0$ the factor $\sin(\pi s/2)$ vanishes, but $\zeta(1-s) = \zeta(1)$ is the pole, and the product is the finite value $\zeta(0) = -1/2$ (theorem 6.4.2), not a zero. Thus $\mathrm{Re}(s) = 0$ carries no zeros at all. Combining with Step 1, every zero other than the trivial zeros lies in the open strip $0 < \mathrm{Re}(s) < 1$, which is the asserted location of the nontrivial zeros.

Step 3: Symmetry under $s \mapsto 1-s$. Let ρ be a nontrivial zero, so $0 < \mathrm{Re}(\rho) < 1$ and $\zeta(\rho) = 0$. The gamma factor $\Gamma_{\mathbb{R}}(\rho) = \pi^{-\rho/2}\Gamma(\rho/2)$ is finite and nonzero at ρ, since $\Gamma(\rho/2)$ has its poles only at $\rho/2 = 0, -1, -2, \ldots$, that is at $\rho = 0, -2, -4, \ldots$, none of which lie in the open strip. Hence the completed function $\Lambda(\rho) = \Gamma_{\mathbb{R}}(\rho)\zeta(\rho) = 0$. By the functional equation (6.14), $\Lambda(1-\rho) = \Lambda(\rho) = 0$. The reflected point $1-\rho$ also lies in the open strip, since $\mathrm{Re}(1-\rho) = 1 - \mathrm{Re}(\rho) \in (0,1)$, so its gamma factor $\Gamma_{\mathbb{R}}(1-\rho)$ is again finite and nonzero by the same reasoning. From $\Lambda(1-\rho) = \Gamma_{\mathbb{R}}(1-\rho)\zeta(1-\rho) = 0$ and $\Gamma_{\mathbb{R}}(1-\rho) \neq 0$ it follows that $\zeta(1-\rho) = 0$. Thus $1-\rho$ is a nontrivial zero.

Step 4: Symmetry under $s \mapsto \bar{s}$. On the half-plane $\mathrm{Re}(s) > 1$ the Dirichlet series gives $\zeta(\bar{s}) = \sum_{n\geq 1} n^{-\bar{s}} = \sum_{n\geq 1} \overline{n^{-s}} = \overline{\zeta(s)}$, the middle equality holding because $n^{-\bar{s}} = \overline{n^{-s}}$ for the positive integer base n. Both $\zeta(\bar{s})$ and $\overline{\zeta(s)}$ are holomorphic on $\mathbb{C} \setminus \{1\}$ (the continuation of corollary 6.3.4, composed with conjugation, is holomorphic), and they agree on the open set $\mathrm{Re}(s) > 1$, so by the identity theorem ([13]) the relation

$$\zeta(\bar{s}) = \overline{\zeta(s)} \tag{6.31}$$

holds for all $s \neq 1$. If ρ is a nontrivial zero then $\zeta(\bar{\rho}) = \overline{\zeta(\rho)} = \bar{0} = 0$, and $\bar{\rho}$ lies in the open strip because $\mathrm{Re}(\bar{\rho}) = \mathrm{Re}(\rho) \in (0,1)$, so $\bar{\rho}$ is a nontrivial zero.

Step 5: The quadruple structure. The two involutions $s \mapsto 1 - s$ and $s \mapsto \bar{s}$ commute, their composite being $s \mapsto 1 - \bar{s}$. Applying them to a nontrivial zero ρ produces the four points ρ, $1 - \rho$, $\bar{\rho}$, and $1 - \bar{\rho}$, each a nontrivial zero by Steps 3 and 4. Writing $\rho = \beta + i\gamma$ with $0 < \beta < 1$, the four points are $\beta + i\gamma$, $(1 - \beta) - i\gamma$, $\beta - i\gamma$, and $(1 - \beta) + i\gamma$, the vertices of a rectangle centered at $\frac{1}{2}$ on the real axis, with sides parallel to the axes. This rectangle is nondegenerate, and the four zeros distinct, exactly when $\beta \neq 1/2$ and $\gamma \neq 0$. Conjugation pairs ρ with $\bar{\rho}$ symmetrically about the real axis, so a zero off the real axis ($\gamma \neq 0$) is never alone; the count along the line $\mathrm{Re}(s) = 1/2$ is the case where the reflection $s \mapsto 1 - s$ adds nothing new, since there $\beta = 1 - \beta$ forces $1 - \rho = \bar{\rho}$ and the quadruple collapses to the conjugate pair $\frac{1}{2} \pm i\gamma$. Thus a nontrivial zero ρ with $\gamma \neq 0$ generates a full quadruple of four distinct zeros unless it lies on the critical line, where it generates only the pair $\{\rho, \bar{\rho}\}$, as we sought to show.

The two symmetries have different sources, and the distinction is worth keeping in view. The conjugate symmetry $\zeta(\bar{s}) = \overline{\zeta(s)}$ of (6.31) is elementary, an instance of the Schwarz reflection principle that holds for any Dirichlet series with real coefficients; it places the zeros symmetrically about the real axis and is the reason the explicit formula of box 5.5.5 groups the zeros into conjugate pairs $\rho, \bar{\rho}$, so that the oscillatory terms $x^{\rho}/\rho + x^{\bar{\rho}}/\bar{\rho}$ combine into a real quantity. The reflection symmetry $s \mapsto 1 - s$ is the substantive one: it is the functional equation, and it places the zeros symmetrically about the critical line. A nontrivial zero off the line at $\beta + i\gamma$ with $\beta > 1/2$ forces a partner at $(1 - \beta) + i\gamma$ with $1 - \beta < 1/2$, on the opposite side of the line and equally far from it. The Riemann Hypothesis is the statement that this never happens, that the reflection $s \mapsto 1 - s$ fixes every nontrivial zero, which is possible only on the fixed-point line $\mathrm{Re}(s) = 1/2$.

The symmetry confines the zeros but does not count them. The distribution of the nontrivial zeros up to a given height is governed by an asymptotic formula, the foundation of every quantitative statement about the zeros and the starting point for the explicit formula's error analysis.

Theorem 6.5.4: The Zero-Counting Formula

Let $N(T)$ denote the number of nontrivial zeros $\rho = \beta + i\gamma$ of ζ with $0 < \gamma \leq T$, counted with multiplicity. Then

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$$

as $T \to \infty$.

The full proof is a careful application of the argument principle to ξ on a rectangle, together with Stirling's estimate for the gamma factor; the computation tracks the change in $\arg \xi$ along three sides of the rectangle and bounds the contribution of the fourth, and its several pages of estimation carry no conceptual turning point beyond the two inputs named. The argument is given in full in [6, Ch. 15] and [21, Ch. 14]; the key ideas and the origin of each term are recorded here.

***Proof Key ideas*:**
Fix a large T avoiding the ordinates of zeros, and let R be the positively oriented boundary of the rectangle with corners $-1, 2, 2 + iT, -1 + iT$, chosen so that all nontrivial zeros with $0 < \gamma \leq T$ lie strictly inside (the strip $0 < \mathrm{Re}(s) < 1$ sits inside $-1 < \mathrm{Re}(s) < 2$) and none lies on R.

Step 1: The argument principle for ξ. The function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire (definition 6.5.1), and the rectangle $R = [-1, 2] \times [0, T]$ lies in the upper half-plane, so its zeros inside R are exactly the nontrivial zeros of ζ with $0 < \gamma \leq T$. The argument principle ([13])

counts these as

$$N(T) = \frac{1}{2\pi}\Delta_R \arg\xi$$

the normalized total change in $\arg\xi$ around R. Two symmetries reduce this contour. The conjugate symmetry $\xi(\bar{s}) = \overline{\xi(s)}$, inherited from (6.31) and the real prefactor, together with the reflection $\xi(s) = \xi(1-s)$ gives $\xi(1-\bar{s}) = \overline{\xi(s)}$, so reflecting R across the line $\mathrm{Re}(s) = 1/2$ (the map $s \mapsto 1-\bar{s}$) conjugates ξ and leaves $|\Delta \arg\xi|$ unchanged; hence the change in $\arg\xi$ over the left half of R equals the change over the right half, and $\Delta_R \arg\xi = 2\,\Delta_{R'} \arg\xi$, where R' is the right portion of the boundary running $\frac{1}{2} \to 2 \to 2+iT \to \frac{1}{2}+iT$. On the bottom segment $\frac{1}{2} \to 2$ the function ξ is real and positive, so $\arg\xi$ is constant there and that segment contributes nothing; the change over R' equals the change along the half-contour L running $2 \to 2+iT \to \frac{1}{2}+iT$. Assembling these reductions,

$$\pi N(T) = \Delta_L \arg\xi$$

the change in $\arg\xi$ along L.

Step 2: The gamma factor gives the main term. Split $\arg\xi = \arg\big(\frac{1}{2}s(s-1)\big)+\arg\big(\pi^{-s/2}\Gamma(s/2)\big)+\arg\zeta(s)$ and track each summand along the half-contour L. The change in $\arg\big(\pi^{-s/2}\Gamma(s/2)\big)$ along L is evaluated by Stirling's estimate (6.7) of theorem 6.1.4: at the endpoint $1/2+iT$ the formula $\log\Gamma(s/2)$ contributes an imaginary part $\mathrm{Im}\log\Gamma(\frac{1}{4}+\frac{iT}{2})$ of size $\frac{T}{2}\log\frac{T}{2}-\frac{T}{2}+O(\log T)$, and the factor $\pi^{-s/2}$ contributes $-\frac{T}{2}\log\pi$. Combining and using $\log\frac{T}{2}-\log\pi = \log\frac{T}{2\pi}$ gives the change $\frac{T}{2}\log\frac{T}{2\pi}-\frac{T}{2}+O(\log T)$ along L; dividing by the factor π from the relation $\pi N(T) = \Delta_L \arg\xi$ of Step 1 produces the main term

$$\frac{T}{2\pi}\log\frac{T}{2\pi}-\frac{T}{2\pi}$$

The elementary factor $\frac{1}{2}s(s-1)$ contributes a bounded change, of order $O(1)$, absorbed into the error.

Step 3: The zeta factor gives the error term. The remaining summand is $\frac{1}{\pi}$ times the change in $\arg\zeta$ along the half-contour L. On the segment $2 \to 2+iT$ this change is bounded, so up to $O(1)$ it equals the customary

$$S(T) = \frac{1}{\pi}\arg\zeta\big(\tfrac{1}{2}+iT\big)$$

the argument taken by continuous variation from $s = 2$, where $\arg\zeta = 0$. The bound $S(T) = O(\log T)$ comes from controlling the number of sign changes of $\mathrm{Re}\zeta$ along the horizontal segment, which is in turn bounded by Jensen's inequality applied to ζ on a disc of fixed radius about $2+iT$, where ζ has size polynomial in T. This $O(\log T)$ is the entire error term. The full execution of all three steps, including the contour avoidance and the Jensen bound on $S(T)$, is carried out in [6, Ch. 15] and [21, Ch. 14], completing the proof.

The formula says the nontrivial zeros have density $\frac{1}{2\pi}\log\frac{T}{2\pi}$ near height T: the average gap between consecutive ordinates γ near T is $2\pi/\log T$, shrinking slowly as one climbs the critical strip. The main term is produced entirely by the archimedean data, the gamma factor and the power of π that complete ζ, and is therefore known unconditionally and exactly; the arithmetic of ζ enters only through the secondary term $S(T)$, whose true size is the subject of much of the finer theory. The Riemann Hypothesis is not needed for the count: $N(T)$ is the number of zeros in the strip regardless of where in the strip they sit, and the formula holds whether or not they lie on the critical line. What the Riemann Hypothesis would add is the assertion that all $N(T)$ of these zeros have $\beta = 1/2$, refining a statement about how many zeros there are into a statement about where they are.

The count motivates a change of viewpoint. To study the zeros on the critical line directly, one restricts ζ to that line and removes the phase that the gamma factor introduces, producing a real-valued function whose sign changes locate the zeros.

Definition 6.5.5: The Hardy Z-Function

The *Riemann–Siegel theta function* (distinct from the Jacobi theta function of section 6.2, despite the shared symbol; the two never appear together) is

$$\theta(t) = \arg\Big(\pi^{-it/2}\,\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big)\Big)$$

the argument taken by the principal branch, continuous in t with $\theta(0) = 0$. The *Hardy Z-function* is

$$Z(t) = e^{i\theta(t)}\,\zeta\big(\tfrac{1}{2} + it\big)$$

defined for real t. It is real-valued, satisfies $|Z(t)| = |\zeta(\frac{1}{2} + it)|$, and its real zeros are exactly the zeros of ζ on the critical line.

***Proof*:**
The three asserted properties are established in turn; the first is the substantive one and rests on the functional equation restricted to the critical line.

Step 1: $Z(t)$ is real-valued. Recall the real gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ of definition 6.3.1 and the completed function $\Lambda(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$. On the critical line $s = \frac{1}{2} + it$,

$$\Gamma_{\mathbb{R}}\big(\tfrac{1}{2} + it\big) = \pi^{-(1/2+it)/2}\,\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big) = \pi^{-1/4}\cdot\pi^{-it/2}\,\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big)$$

The factor $\pi^{-1/4}$ is a positive real number, so $\Gamma_{\mathbb{R}}(\frac{1}{2}+it)$ has the same argument as $\pi^{-it/2}\Gamma(\frac{1}{4}+\frac{it}{2})$, namely $\theta(t)$. Writing $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) = |\Gamma_{\mathbb{R}}(\frac{1}{2} + it)|\,e^{i\theta(t)}$ therefore holds, and consequently

$$Z(t) = e^{i\theta(t)}\zeta\big(\tfrac{1}{2} + it\big) = \frac{\Gamma_{\mathbb{R}}(\frac{1}{2} + it)}{|\Gamma_{\mathbb{R}}(\frac{1}{2} + it)|}\,\zeta\big(\tfrac{1}{2} + it\big) = \frac{\Lambda(\frac{1}{2} + it)}{|\Gamma_{\mathbb{R}}(\frac{1}{2} + it)|} \tag{6.32}$$

using $\Gamma_{\mathbb{R}}(\frac{1}{2}+it)\zeta(\frac{1}{2}+it) = \Lambda(\frac{1}{2}+it)$. The denominator $|\Gamma_{\mathbb{R}}(\frac{1}{2}+it)|$ is a positive real number (the gamma factor has no zeros and no poles on the line, as $\frac{1}{4} + \frac{it}{2}$ is never a pole of Γ). It remains to show the numerator $\Lambda(\frac{1}{2} + it)$ is real. The functional equation (6.14) at $s = \frac{1}{2} + it$ gives $\Lambda(\frac{1}{2} + it) = \Lambda(1 - (\frac{1}{2} + it)) = \Lambda(\frac{1}{2} - it)$. The completed function inherits the Schwarz reflection $\Lambda(\bar{s}) = \overline{\Lambda(s)}$ from (6.31): indeed $\Gamma_{\mathbb{R}}(\bar{s}) = \pi^{-\bar{s}/2}\Gamma(\bar{s}/2) = \overline{\pi^{-s/2}}\,\overline{\Gamma(s/2)} = \overline{\Gamma_{\mathbb{R}}(s)}$, since Γ has real Taylor coefficients and so commutes with conjugation, and multiplying by $\zeta(\bar{s}) = \overline{\zeta(s)}$ yields $\Lambda(\bar{s}) = \overline{\Lambda(s)}$. Applied at $s = \frac{1}{2} + it$, whose conjugate is $\frac{1}{2} - it$, this reads $\Lambda(\frac{1}{2} - it) = \overline{\Lambda(\frac{1}{2} + it)}$. Chaining the two relations,

$$\Lambda\big(\tfrac{1}{2} + it\big) = \Lambda\big(\tfrac{1}{2} - it\big) = \overline{\Lambda\big(\tfrac{1}{2} + it\big)}$$

so $\Lambda(\frac{1}{2} + it)$ equals its own conjugate and is real. By (6.32), $Z(t)$ is a real number divided by a positive real number, hence real.

Step 2: $|Z(t)| = |\zeta(\frac{1}{2} + it)|$. The phase factor has modulus one, $|e^{i\theta(t)}| = 1$, since $\theta(t)$ is real. Therefore $|Z(t)| = |e^{i\theta(t)}|\,|\zeta(\frac{1}{2} + it)| = |\zeta(\frac{1}{2} + it)|$ directly from the definition.

Step 3: The real zeros of Z are the critical-line zeros of ζ. For real t, the factor $e^{i\theta(t)}$ is nonzero, so $Z(t) = 0$ if and only if $\zeta(\frac{1}{2} + it) = 0$. Thus t is a real zero of Z exactly when $\frac{1}{2} + it$ is a zero of ζ on the critical line, with matching multiplicities since $e^{i\theta(t)}$ is a nonvanishing analytic factor. This is the asserted correspondence, completing the proof.

The Z-function trades a complex-valued function on a line for an equivalent real-valued one, and the gain is that real zeros can be detected by sign changes. Whenever $Z(t_1)$ and $Z(t_2)$ have opposite signs, the intermediate value theorem forces a zero of Z, hence a zero of ζ, between them on the critical line; this is the mechanism behind every numerical and theoretical verification that zeros lie on the line, including Hardy's theorem that infinitely many do. The factorization (6.32) shows Z is the completed function Λ on the line, stripped of the positive modulus of its gamma factor, so the zeros of Z, the zeros of Λ on the line, and the critical-line zeros of ζ all coincide. The Riemann–Siegel theta function $\theta(t)$ that supplies the phase grows like $\frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2}$ by the same Stirling estimate that produced the main term of theorem 6.5.4, and the count $N(T)$ is, up to the error $S(T)$, the number of times $\theta(t)$ sweeps through a multiple of π; the zero-counting formula and the Z-function are thus two readings of the same gamma-factor asymptotics.

6.5.6 Note: The Riemann Hypothesis Revisited The Riemann Hypothesis (**??** 5.5.6) was stated in chapter 5 as the assertion that every nontrivial zero has $\mathrm{Re}(s) = 1/2$. The functional equation now exhibits why this line is singled out: by proposition 6.5.3 the nontrivial zeros are symmetric across the critical line under $s \mapsto 1 - s$, and $\mathrm{Re}(s) = 1/2$ is the axis of that symmetry, the fixed-point set of the involution. The Riemann Hypothesis is therefore the statement that every nontrivial zero sits *on* the axis of symmetry, so that the quadruple $\{\rho, 1-\rho, \bar{\rho}, 1-\bar{\rho}\}$ always degenerates to the conjugate pair $\{\rho, \bar{\rho}\}$. The consequence for the distribution of primes is recorded in proposition 5.5.7: the hypothesis is equivalent to the error bound $\psi(x) = x + O(\sqrt{x}\,(\log x)^2)$, the sharpest possible, since the size of the zero-sum $\sum_\rho x^\rho/\rho$ in the explicit formula is governed by $\sup_\rho \mathrm{Re}(\rho)$. With the functional equation in hand, every ingredient of that explicit formula (box 5.5.5) is now established: the entire completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, whose zeros are exactly the nontrivial zeros, and the product factorization over those zeros that converts the contour integral of $-\zeta'/\zeta$ into the sum over ρ.

7

Dirichlet Characters and L-Functions

The six preceding chapters built the analytic theory of the primes around a single function, the Riemann zeta function $\zeta(s) = \sum_{n\geq 1} n^{-s}$. Its Euler product (corollary 2.3.2) encoded unique factorization analytically; its pole at $s = 1$ (theorem 4.2.1) measured the abundance of the primes; its non-vanishing on the line $\mathrm{Re}(s) = 1$ (theorem 4.4.2) converted that abundance into the Prime Number Theorem (theorem 5.4.3); and its functional equation (theorem 6.3.3), assembled from the gamma and theta functions, fixed the symmetry of the whole picture. That theory counts all primes at once. It says nothing about how the primes distribute themselves among the residue classes modulo a fixed integer m, the question of whether the arithmetic progression $a, a + m, a + 2m, \ldots$ with $\gcd(a, m) = 1$ contains infinitely many primes. To reach the residue classes one must split the prime-counting machinery according to congruence, and the device that performs the splitting is the Dirichlet character: a homomorphism of the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$, extended to all of $\mathbb{Z}$ by zero on the integers sharing a factor with m. Twisting the zeta construction by such a character χ produces a Dirichlet L-function $L(s, \chi) = \sum_{n\geq 1} \chi(n)\, n^{-s}$, and the family of these functions, one for each character modulo m, resolves the single count $\sum_n n^{-s}$ into $\varphi(m)$ separate counts, one attached to each residue class.

This chapter is the character-twisted parallel of chapters 4 to 6. Each structural feature of ζ has its analogue for $L(s, \chi)$: an Euler product over the primes, a meromorphic continuation (entire when χ is nonprincipal, since the pole of ζ survives only in the principal twist), and a functional equation relating $L(s, \chi)$ to $L(1 - s, \bar{\chi})$, the reflection now mediated not only by the gamma factor but by a finite exponential sum, the Gauss sum, whose absolute value $\sqrt{m}$ supplies the constant in the symmetry. Above all there is one fact with no analogue in the zeta theory because it is automatic there, where the pole forces $\zeta(s) \to \infty$ as $s \to 1$: the non-vanishing $L(1, \chi) \neq 0$ for every nonprincipal character χ. This single inequality is what prevents the contributions of the nonprincipal characters from cancelling the main term, and it yields Dirichlet's theorem, that every arithmetic progression

$a \bmod m$ with $\gcd(a, m) = 1$ contains infinitely many primes, distributed with density $1/\varphi(m)$ among the primes (chapter 7 closes on this result). The development begins with the characters themselves, the finite group-theoretic objects on which the entire analytic edifice is raised.

7.1 Dirichlet Characters

The structure of the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$ was settled in section 0.4: it is a finite abelian group of order $\varphi(m)$, cyclic precisely when $m \in \{1, 2, 4, p^k, 2p^k\}$ and a product of cyclic factors in general. The arithmetic functions of the previous chapters were defined directly on $\mathbb{Z}$, and the multiplicative ones, such as $n \mapsto n^{-s}$, were exactly the functions respecting the multiplication of $\mathbb{Z}$. To detect congruence classes one needs functions that also respect the multiplication of $(\mathbb{Z}/m\mathbb{Z})^\times$, and the homomorphisms from this finite abelian group to the multiplicative group $\mathbb{C}^\times$ are the natural such objects. Extended from the unit group to all integers by setting them to zero where they would otherwise be undefined, these homomorphisms become arithmetic functions in their own right, the Dirichlet characters. This section constructs them, counts them, records the two facts about them on which everything later depends (that they form a group dual to $(\mathbb{Z}/m\mathbb{Z})^\times$, and that they satisfy orthogonality relations), and isolates the refinements (the conductor and the parity) that the functional equation will require.

Definition 7.1.1: Dirichlet Character

Let $m \in \mathbb{Z}_{>0}$. A *Dirichlet character modulo m* is a function $\chi\colon \mathbb{Z} \to \mathbb{C}$ satisfying the following three conditions:

1. *Complete multiplicativity:* $\chi(ab) = \chi(a)\chi(b)$ for all $a, b \in \mathbb{Z}$;
2. *Periodicity:* $\chi(a + m) = \chi(a)$ for all $a \in \mathbb{Z}$;
3. *Support on the units:* $\chi(a) \neq 0$ if and only if $\gcd(a, m) = 1$.

Equivalently, χ arises from a group homomorphism $\widetilde{\chi}\colon (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$ by setting $\chi(a) = \widetilde{\chi}(a \bmod m)$ when $\gcd(a, m) = 1$ and $\chi(a) = 0$ otherwise. The integer m is the *modulus* of χ.

The three conditions and the homomorphism description are two presentations of the same object, and the equivalence is worth tracing once. A group homomorphism $\widetilde{\chi}\colon (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$ extended by zero is completely multiplicative: if both a and b are coprime to m, then $\chi(ab) = \widetilde{\chi}(\bar{a}\bar{b}) = \widetilde{\chi}(\bar{a})\widetilde{\chi}(\bar{b}) = \chi(a)\chi(b)$, and if either shares a factor with m, then so does ab, so both sides vanish; it is periodic because its value depends only on the residue $a \bmod m$; and it is supported on the units by construction. Conversely, any χ satisfying the three conditions restricts to a function $(\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$ (well defined by periodicity, nonzero by condition (iii)) that is multiplicative by condition (i), hence a group homomorphism. The values of a character are constrained: applying condition (i) to $a^{\varphi(m)}$ for $\gcd(a, m) = 1$ and using Euler's theorem (theorem 0.3.12), $a^{\varphi(m)} \equiv 1 \pmod{m}$, gives $\chi(a)^{\varphi(m)} = \chi(1) = 1$, so every nonzero value of χ is a $\varphi(m)$-th root of unity and in particular has absolute value 1.

The passage from the group homomorphism $\widetilde{\chi}$ to the arithmetic function χ is the move that makes character theory available to analytic number theory: a homomorphism of $(\mathbb{Z}/m\mathbb{Z})^\times$ knows nothing about the integers, but its extension by zero is a completely multiplicative function on $\mathbb{Z}$, the same class of function as $n \mapsto n^{-s}$, and so can be fed into a Dirichlet series. The remainder of this chapter is the study of the series $\sum_n \chi(n)n^{-s}$ so produced. The notational convention, followed throughout, is to write χ for both the character on $\mathbb{Z}$ and the underlying homomorphism on $(\mathbb{Z}/m\mathbb{Z})^\times$, the meaning being fixed by the argument.

The count and the group law for these objects come directly from the duality theory of finite abelian groups, applied to $G = (\mathbb{Z}/m\mathbb{Z})^\times$.

Theorem 7.1.2: Characters as the Dual Group

Fix $m \in \mathbb{Z}_{>0}$ and write $G = (\mathbb{Z}/m\mathbb{Z})^\times$. Restriction to the units, $\chi \mapsto \widetilde{\chi}$, is a bijection between the Dirichlet characters modulo m and the group homomorphisms $G \to \mathbb{C}^\times$, that is, the elements of the dual group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$. Consequently:

1. there are exactly $\varphi(m)$ Dirichlet characters modulo m;
2. under pointwise multiplication $(\chi_1\chi_2)(n) = \chi_1(n)\chi_2(n)$ the Dirichlet characters modulo m form a finite abelian group, with identity the principal character and inverse $\chi^{-1} = \bar{\chi}$ the complex conjugate;
3. this group is isomorphic to G itself; and
4. every character takes values in the $\varphi(m)$-th roots of unity.

***Proof*:**
The bijection in the opening sentence is the equivalence established after definition 7.1.1: extension by zero sends a homomorphism $G \to \mathbb{C}^\times$ to a Dirichlet character, restriction to the units sends a Dirichlet character to such a homomorphism, and the two operations are mutually inverse. Under this bijection pointwise multiplication of characters on $\mathbb{Z}$ corresponds to pointwise multiplication of homomorphisms on G, since the bijection is given by restriction and multiplication is computed pointwise on both sides; the bijection is therefore a group isomorphism between the Dirichlet characters modulo m and the dual group $\widehat{G}$. It remains to import the structure of $\widehat{G}$.

The group G is finite abelian (section 0.4). For a finite abelian group G the dual $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$ is a group under pointwise multiplication, with identity the trivial homomorphism $g \mapsto 1$ and inverse $\widetilde{\chi}^{-1}(g) = \widetilde{\chi}(g)^{-1} = \overline{\widetilde{\chi}(g)}$, the last equality because $\widetilde{\chi}(g)$ is a root of unity (every element of G has finite order, so its image does too), hence has modulus 1; and there is a (non-canonical) isomorphism $\widehat{G} \cong G$, so in particular $|\widehat{G}| = |G|$. These are the standard facts on characters of finite abelian groups; their proof is given in [5], the cyclic case by sending a generator to a chosen root of unity and the general case by the invariant factor decomposition $G \cong \prod_i \mathbb{Z}/d_i\mathbb{Z}$, under which $\widehat{G} \cong \prod_i \widehat{\mathbb{Z}/d_i\mathbb{Z}} \cong \prod_i \mathbb{Z}/d_i\mathbb{Z} \cong G$.

Transporting these facts across the isomorphism establishes the numbered claims. The count $|\widehat{G}| = |G| = \varphi(m)$ gives (1). The group law and the identification of the identity and inverse give (2), the principal character being the extension by zero of the trivial homomorphism (definition 7.1.3 below) and $\bar{\chi}$ the extension of $\overline{\widetilde{\chi}}$, which agrees with χ^{-1} on the units and is zero off them. The isomorphism $\widehat{G} \cong G$ gives (3). Finally (4) restates the value constraint already noted after definition 7.1.1: for $\gcd(n, m) = 1$ the element $\widetilde{\chi}(\bar{n})$ has order dividing $|G| = \varphi(m)$, so $\chi(n)^{\varphi(m)} = 1$, completing the proof.

The theorem fixes the population of characters and, more usefully, gives that population the structure of a group isomorphic to $(\mathbb{Z}/m\mathbb{Z})^\times$. The isomorphism $\widehat{G} \cong G$ is not canonical, depending on a choice of generators, but its mere existence is what is used: it guarantees exactly $\varphi(m)$ characters, no more and no fewer, and this exact count is what makes the orthogonality relations below into clean equalities rather than inequalities. When $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic, say generated by a primitive root g (section 0.4), a character is determined by the single value $\chi(g)$, which may be any $\varphi(m)$-th root of unity, and the $\varphi(m)$ choices of $\chi(g)$ enumerate the $\varphi(m)$ characters; this is the situation for prime moduli and is the source of the concrete tables in example 7.1.7. For non-cyclic moduli a character is specified by its values on a set of generators of the invariant factors, one root of unity of the appropriate order for each factor.

Among the $\varphi(m)$ characters one is distinguished as the identity of the group, and it warrants a name because it plays the role for L-functions that ζ itself plays.

Definition 7.1.3: The Principal Character

The *principal character* modulo m, denoted χ_0, is the Dirichlet character

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n,m) = 1, \\ 0 & \text{if } \gcd(n,m) > 1 \end{cases}$$

the extension by zero of the trivial homomorphism $(\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$.

The principal character is the multiplicative indicator function of the integers coprime to m: it is 1 exactly where a character is allowed to be nonzero, and so records support without recording any finer information about the residue class. Its L-function will turn out to be the Riemann zeta function with the Euler factors at the primes dividing m removed (proposition 7.2.3), which is why χ_0, alone among the characters modulo m, produces a function with a pole. Every other character distinguishes residue classes nontrivially and produces an entire L-function.

The single computational fact that powers the entire theory of primes in progressions is the orthogonality of characters. It is the precise sense in which the $\varphi(m)$ characters modulo m form a complete, mutually independent system of “frequencies” on $(\mathbb{Z}/m\mathbb{Z})^\times$, and it is the device that extracts a single residue class from a sum over all characters.

Theorem 7.1.4: Orthogonality Relations

Let $m \in \mathbb{Z}_{>0}$. The Dirichlet characters modulo m satisfy:

1. *Column orthogonality.* For every integer n with $\gcd(n, m) = 1$,
$$\sum_{\chi \bmod m} \chi(n) = \begin{cases} \varphi(m) & \text{if } n \equiv 1 \pmod{m}, \\ 0 & \text{otherwise} \end{cases}$$
the sum running over all $\varphi(m)$ Dirichlet characters modulo m.

2. *Row orthogonality.* For every Dirichlet character χ modulo m,
$$\sum_{n \bmod m} \chi(n) = \begin{cases} \varphi(m) & \text{if } \chi = \chi_0, \\ 0 & \text{otherwise} \end{cases}$$
the sum running over a complete set of residues n modulo m (equivalently, over the $\varphi(m)$ residues coprime to m, the others contributing 0).

Consequently, for any integers a, n with $\gcd(a, m) = 1$,
$$\frac{1}{\varphi(m)} \sum_{\chi \bmod m} \chi(n)\,\overline{\chi(a)} = \begin{cases} 1 & \text{if } n \equiv a \pmod{m}, \\ 0 & \text{otherwise} \end{cases} \tag{7.1}$$
so that the left-hand side is the indicator function of the residue class a modulo m, supported on integers coprime to m.

***Proof*:**
Both relations are the orthogonality relations for characters of the finite abelian group $G = (\mathbb{Z}/m\mathbb{Z})^\times$, proved in [5]; the task here is to translate them into the language of Dirichlet characters and then to derive (7.1). Throughout, the Dirichlet characters modulo m are identified with the elements of $\widehat{G}$ by theorem 7.1.2, and a residue n coprime to m is identified with the corresponding element $\bar{n} \in G$.

Step 1: Row orthogonality. For a character $\widetilde{\chi} \in \widehat{G}$, the group-character orthogonality of [5] states that $\sum_{g \in G} \widetilde{\chi}(g)$ equals $|G|$ if $\widetilde{\chi}$ is trivial and 0 otherwise. The direct argument is short: if $\widetilde{\chi}$ is trivial the sum is $\sum_g 1 = |G| = \varphi(m)$, while if $\widetilde{\chi}$ is nontrivial there is some $h \in G$ with $\widetilde{\chi}(h) \neq 1$, and since $g \mapsto hg$ permutes G,
$$S := \sum_{g \in G} \widetilde{\chi}(g) = \sum_{g \in G} \widetilde{\chi}(hg) = \widetilde{\chi}(h) \sum_{g \in G} \widetilde{\chi}(g) = \widetilde{\chi}(h)\, S$$
whence $(1 - \widetilde{\chi}(h))S = 0$ and $S = 0$. Translating to $\mathbb{Z}$: summing χ over a complete residue system modulo m, the residues with $\gcd(n, m) > 1$ contribute 0 by support, and the residues with $\gcd(n, m) = 1$ contribute $\sum_{g \in G} \widetilde{\chi}(g)$, which is $\varphi(m)$ for $\chi = \chi_0$ and 0 otherwise. This is (2).

Step 2: Column orthogonality. Fix n coprime to m and put $g = \bar{n} \in G$. The dual statement of [5] is that $\sum_{\widetilde{\chi} \in \widehat{G}} \widetilde{\chi}(g)$ equals $|G|$ if g is the identity and 0 otherwise; this is row orthogonality applied to the second-dual group $\widehat{\widehat{G}}$, under the canonical evaluation isomorphism $G \cong \widehat{\widehat{G}}$ sending g to the character $\widetilde{\chi} \mapsto \widetilde{\chi}(g)$. The direct argument mirrors Step 1: if g is the identity then every $\widetilde{\chi}(g) = 1$ and the sum is $|\widehat{G}| = \varphi(m)$ by theorem 7.1.2; if $g \neq 1$ then, because $\widehat{G}$ separates points of G (a consequence of $\widehat{G} \cong G$, again [5]), there is a character ψ with $\psi(g) \neq 1$, and since $\widetilde{\chi} \mapsto \psi\widetilde{\chi}$ permutes $\widehat{G}$,

$$T := \sum_{\widetilde{\chi}\in\widehat{G}} \widetilde{\chi}(g) = \sum_{\widetilde{\chi}\in\widehat{G}} (\psi\widetilde{\chi})(g) = \psi(g) \sum_{\widetilde{\chi}\in\widehat{G}} \widetilde{\chi}(g) = \psi(g)\, T$$

so $(1 - \psi(g))T = 0$ and $T = 0$. Since $g = \bar{n}$ is the identity of G exactly when $n \equiv 1 \pmod{m}$, this is (1).

Step 3: The indicator identity. Let a be coprime to m, so $\bar{a} \in G$ is invertible, and let n be arbitrary. If $\gcd(n, m) > 1$ then $\chi(n) = 0$ for every χ, so the left-hand side of (7.1) is 0; and n is then not congruent to a modulo m (as $\gcd(a, m) = 1$), so the right-hand side is 0 as well. If $\gcd(n, m) = 1$, then $\overline{\chi(a)} = \chi(a)^{-1} = \chi(a^{-1})$ where a^{-1} is any integer with $aa^{-1} \equiv 1 \pmod{m}$ (it exists since $\gcd(a, m) = 1$), using $|\chi(a)| = 1$ and complete multiplicativity. Hence $\chi(n)\overline{\chi(a)} = \chi(na^{-1})$, and summing over χ and applying column orthogonality (1) to the integer na^{-1},

$$\sum_{\chi \bmod m} \chi(n)\overline{\chi(a)} = \sum_{\chi \bmod m} \chi(na^{-1}) = \begin{cases} \varphi(m) & \text{if } na^{-1} \equiv 1 \pmod{m}, \\ 0 & \text{otherwise} \end{cases}$$

The condition $na^{-1} \equiv 1 \pmod{m}$ is equivalent to $n \equiv a \pmod{m}$. Dividing by $\varphi(m)$ yields (7.1), as we sought to show.

Identity (7.1) is what the entire chapter uses, and the mechanism it provides deserves to be stated plainly. To study primes $p \equiv a \pmod{m}$ one would like to sum some prime-counting quantity over exactly those primes, but the condition $p \equiv a$ is not multiplicative and cannot be inserted directly into a Dirichlet series. Orthogonality removes the obstruction. The right-hand side of (7.1) is precisely the indicator of the congruence $n \equiv a$, and the left-hand side replaces that single non-multiplicative condition by an average of the values $\chi(n)$, each of which *is* completely multiplicative. Inserting the average into a sum over primes and exchanging the order of summation turns one sum over the progression $a \bmod m$ into a weighted combination of $\varphi(m)$ sums over all primes, the χ-th of which is governed by the L-function $L(s, \chi)$ developed in section 7.2. The principal character χ_0 contributes the main term, carrying the pole that forces divergence, while each nonprincipal χ contributes a correction that stays bounded precisely because $L(1, \chi) \neq 0$. This decomposition, indicator-into-characters followed by character-into-L-function, is the strategy of Dirichlet's theorem in section 7.5; the non-multiplicative congruence is dissolved into multiplicative pieces, analyzed one character at a time, and reassembled.

For the functional equation the relevant refinements are not the group structure but two finer invariants of an individual character: whether it belongs to its modulus or is inherited from a smaller one, and how it behaves under negation.

Definition 7.1.5: Primitive Characters and the Conductor

Let χ be a Dirichlet character modulo m, and let d be a positive divisor of m. A Dirichlet character χ^* modulo d *induces* χ if

$$\chi(n) = \chi^*(n) \qquad \text{for all } n \text{ with } \gcd(n, m) = 1$$

The character χ is *induced* by χ^* in this case, and is called *imprimitive* if it is induced by some character to a proper divisor $d < m$; it is *primitive* if no such proper divisor exists. The *conductor* of χ, written $\mathfrak{f}(\chi)$ or $\mathfrak{f}_\chi$, is the smallest positive divisor d of m such that χ is induced by a character modulo d.

The conductor is well defined: the principal character modulo 1 induces the principal character modulo m (both equal 1 on the integers coprime to m, and modulo 1 every integer is coprime), so the set of divisors d for which χ is induced by a character modulo d is nonempty, and being a finite set of positive integers it has a least element. The character inducing χ at level $\mathfrak{f}(\chi)$ is unique and is itself primitive; this and the basic theory of induction (that the inducing character at the conductor is the "true" character, of which χ is a copy with extra Euler factors stripped out) are developed in [6] and recur in section 7.3, where only primitive characters carry a clean functional equation. A character is primitive exactly when it equals its own conductor-level character, $\mathfrak{f}(\chi) = m$. The principal character χ_0 modulo $m > 1$ is imprimitive, with conductor 1, since it is induced by the trivial character modulo 1; the only primitive character of conductor 1 is the trivial character itself.

Definition 7.1.6: Even and Odd Characters

A Dirichlet character χ modulo m is *even* if $\chi(-1) = +1$ and *odd* if $\chi(-1) = -1$. These two cases are exhaustive: complete multiplicativity gives $\chi(-1)^2 = \chi((-1)^2) = \chi(1) = 1$, so $\chi(-1) \in \{+1, -1\}$.

The parity records the behavior of χ under $n \mapsto -n$, since $\chi(-n) = \chi(-1)\chi(n) = \pm\chi(n)$. It is the single binary invariant that decides which gamma factor completes $L(s, \chi)$ into a symmetric function in section 7.3: an even character is completed with the factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ already used for ζ in definition 6.3.1, while an odd character requires the shifted factor $\pi^{-(s+1)/2}\Gamma((s+1)/2)$, the shift by 1 in the argument reflecting the sign change. The principal character is even, since $\chi_0(-1) = 1$ whenever $\gcd(-1, m) = 1$, which always holds. Quadratic characters can be of either parity: the Legendre symbol $\left(\frac{\cdot}{p}\right)$ (definition 0.5.3) is even when $p \equiv 1 \pmod 4$ and odd when $p \equiv 3 \pmod 4$, by the first supplement to quadratic reciprocity (section 0.5) computing $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

The smallest moduli make all of these notions concrete. The next example tabulates every character modulo 4 and modulo 5, identifies the principal one, computes each parity, and locates the conductors, and it isolates the one nonprincipal character modulo 4 that recurs throughout the subject.

Example 7.1.7: Characters Modulo 4 and Modulo 5

Modulo 4. The unit group is $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$, cyclic of order $\varphi(4) = 2$ generated by 3 (since $3^2 = 9 \equiv 1$). By theorem 7.1.2 there are exactly two characters, each determined by its value on the generator 3, which must be a square root of unity, that is, ± 1:

$n \bmod 4$	1	3
$\chi_0(n)$	1	1
$\chi_1(n)$	1	-1

with $\chi_0(n) = \chi_1(n) = 0$ for even n. Here χ_0 is principal. The nonprincipal character χ_1 satisfies $\chi_1(-1) = \chi_1(3) = -1$, so χ_1 is odd; it is primitive, of conductor 4, since the only divisors of 4 are $1, 2, 4$, the only character modulo 1 or 2 is principal (as $\varphi(1) = \varphi(2) = 1$), and a principal character cannot induce the nonprincipal χ_1. Concretely χ_1 is the function

$$\chi_1(n) = \begin{cases} 1 & n \equiv 1 \pmod 4, \\ -1 & n \equiv 3 \pmod 4, \\ 0 & n \text{ even} \end{cases}$$

the unique nonprincipal character modulo 4. Its L-function is

$$L(s, \chi_1) = \sum_{n \geq 1} \frac{\chi_1(n)}{n^s} = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \frac{1}{9^s} - \cdots = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^s} = \beta(s)$$

the *Dirichlet beta function*, the L-function attached to the Gaussian integers $\mathbb{Z}[i]$, whose value $\beta(1) = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \pi/4$ is the Leibniz series. This character and its L-function recur in sections 7.2 and 7.3.

Modulo 5. The unit group is $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$, cyclic of order $\varphi(5) = 4$ generated by the primitive root 2, whose powers run $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 3$ modulo 5 (section 0.4). A character is determined by $\chi(2)$, which must be a fourth root of unity: one of $1, i, -1, -i$. Writing $i = \sqrt{-1}$, the four characters are

$n \bmod 5$	1	2	4	3
$\chi_0(n)$	1	1	1	1
$\psi(n)$	1	i	-1	$-i$
$\psi^2(n)$	1	-1	1	-1
$\psi^3(n)$	1	$-i$	-1	i

where ψ is the character with $\psi(2) = i$, and the columns are ordered by the powers $2^0, 2^1, 2^2, 2^3$ of the generator so that each row reads off the geometric progression $\psi^j(2)^k$. The four characters are $\chi_0 = \psi^0, \psi, \psi^2, \psi^3$, forming a cyclic group of order 4 generated by ψ, isomorphic to $(\mathbb{Z}/5\mathbb{Z})^\times$ as theorem 7.1.2 requires, with $\psi^{-1} = \bar{\psi} = \psi^3$. To read off the parities, note $-1 \equiv 4 \pmod 5$, the column $n = 4$: thus $\chi_0(-1) = 1$ and $\psi^2(-1) = 1$ are even, while $\psi(-1) = -1$ and $\psi^3(-1) = -1$ are odd. The character ψ^2, taking values ± 1 with $\psi^2(2) = -1$, is the quadratic character modulo 5; it is the Legendre symbol $\left(\frac{\cdot}{5}\right)$, recording that $1, 4$ are the quadratic residues and $2, 3$ the non-residues modulo 5, consistent with its evenness and $5 \equiv 1 \pmod 4$. The three nonprincipal characters ψ, ψ^2, ψ^3 are primitive of conductor 5, since the only proper divisor of 5 is 1 and its sole character is principal, so no nonprincipal character modulo 5 is induced from below; the principal character χ_0, by contrast, is imprimitive with conductor 1, induced by the trivial character modulo 1.

7.2 Dirichlet L-Functions

The previous section produced the characters as completely multiplicative arithmetic functions, the same class of function as $n \mapsto n^{-s}$ from which the Euler product of ζ was built. Feeding a character χ modulo m into a Dirichlet series produces the central object of the chapter, the Dirichlet L-function $L(s,\chi) = \sum_{n\geq 1} \chi(n)\, n^{-s}$. It is the character-twisted analogue of the Riemann zeta function: each prime contributes a single Euler factor, now weighted by the value $\chi(p)$, and the resulting product encodes the distribution of the residue class data that χ records. The Euler product, the analytic continuation, and the location of poles all transfer from the zeta theory of chapter 4, with one structural split that determines everything later. The principal character reproduces a lightly modified ζ, complete with its pole at $s = 1$; every other character produces a function holomorphic across the line $\mathrm{Re}(s) = 1$, and the value of that function at $s = 1$, shown nonzero in section 7.4, is the quantity that drives Dirichlet's theorem on primes in arithmetic progressions in section 7.5.

This section defines $L(s,\chi)$, factors it as an Euler product, and continues it past the line $\mathrm{Re}(s) = 1$: trivially for the principal character, by reduction to ζ, and by Abel summation against the bounded character sums for every nonprincipal character. The example at the end evaluates the simplest nonprincipal L-functions to small moduli, including the one whose value at $s = 1$ is the Leibniz series $\pi/4$.

Definition 7.2.1: Dirichlet L-Function

Let χ be a Dirichlet character modulo m (definition 7.1.1). The *Dirichlet L-function* attached to χ is the Dirichlet series

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

a function of the complex variable s. The series converges absolutely on the half-plane $\mathrm{Re}(s) > 1$.

Absolute convergence on $\mathrm{Re}(s) > 1$ is immediate from the bound $|\chi(n)| \leq 1$ of theorem 7.1.2(4): every nonzero value of χ is a root of unity, so $|\chi(n)| \in \{0,1\}$, and therefore

$$\sum_{n=1}^{\infty} \left| \frac{\chi(n)}{n^s} \right| = \sum_{n=1}^{\infty} \frac{|\chi(n)|}{n^\sigma} \leq \sum_{n=1}^{\infty} \frac{1}{n^\sigma}$$

where $\sigma = \mathrm{Re}(s)$, and the comparison series $\sum_n n^{-\sigma}$ converges for $\sigma > 1$. The abscissa of absolute convergence (definition 2.1.3) of $L(s,\chi)$ is thus at most 1. For the principal character χ_0 it equals 1, since $L(s,\chi_0)$ differs from $\zeta(s)$ by finitely many Euler factors (proposition 7.2.3 below) and so inherits the divergence of the harmonic series at $s = 1$; for nonprincipal characters the abscissa of absolute convergence is still 1, but the series converges conditionally on the larger half-plane $\mathrm{Re}(s) > 0$, the content of theorem 7.2.4. On $\mathrm{Re}(s) > 1$ the series defines a holomorphic function, by the general analyticity of Dirichlet series in their half-plane of absolute convergence (theorem 2.1.7).

The same completely multiplicative structure that gave ζ its Euler product gives one to every L-function. This is the first point at which the residue class information carried by χ enters the analytic picture: the factor at a prime p records the value $\chi(p)$, which depends only on $p \bmod m$.

Theorem 7.2.2: The Euler Product

Let χ be a Dirichlet character modulo m. For $\mathrm{Re}(s) > 1$,

$$L(s,\chi) = \prod_p \frac{1}{1-\chi(p)\,p^{-s}}$$

the product running over all primes and converging absolutely. The factor at a prime $p \big| m$ equals 1, since $\chi(p) = 0$ there, so the product is effectively over the primes not dividing m. In particular $L(s,\chi) \neq 0$ for $\mathrm{Re}(s) > 1$.

***Proof*:**
A Dirichlet character is completely multiplicative by condition (i) of definition 7.1.1, and its abscissa of absolute convergence is $\sigma_a(\chi) \le 1$, as recorded after definition 7.2.1. For $\mathrm{Re}(s) > 1 \ge \sigma_a(\chi)$ the Euler product for multiplicative functions (theorem 2.3.1) applies, and since χ is totally multiplicative the local factor at p is the geometric series

$$\sum_{k=0}^{\infty} \frac{\chi(p^k)}{p^{ks}} = \sum_{k=0}^{\infty} \left(\frac{\chi(p)}{p^s}\right)^k = \frac{1}{1-\chi(p)\,p^{-s}}$$

the geometric series converging because $|\chi(p)p^{-s}| = |\chi(p)|\,p^{-\sigma} \le p^{-\sigma} < 1$ for $\sigma > 1$. This gives the stated product, and theorem 2.3.1 guarantees its absolute convergence. For a prime $p \big| m$ the value $\chi(p) = 0$ by condition (iii) of definition 7.1.1, so the factor is $(1-0)^{-1} = 1$ and drops out, leaving the product over $p \nmid m$.

For the non-vanishing, the local factors $(1-\chi(p)p^{-s})^{-1}$ are reciprocals of nonzero complex numbers, hence themselves nonzero: at $p \big| m$ the factor is 1, and at $p \nmid m$ the quantity $1-\chi(p)p^{-s}$ has $|\chi(p)p^{-s}| < 1$, so it cannot vanish. An absolutely convergent infinite product with no vanishing factor has a nonzero value (proposition 2.3.4), so $L(s,\chi) \neq 0$ throughout $\mathrm{Re}(s) > 1$, as we sought to show.

The Euler product is what makes $L(s,\chi)$ an instrument for studying primes rather than a formal series: it converts the global function into an assembly of one factor per prime, with the residue class of p entering through $\chi(p)$. Taking a logarithm of the product, as in proposition 2.3.5, will turn $L(s,\chi)$ into a sum over prime powers weighted by χ, and orthogonality (theorem 7.1.4) will then sieve out a single residue class from the family of such sums over all χ. The non-vanishing on $\mathrm{Re}(s) > 1$ recorded here is the easy half of the non-vanishing story, valid wherever the product converges; the non-vanishing that the theory of primes in progressions actually needs is at the boundary point $s = 1$ itself, where the product no longer converges and a separate argument is required (section 7.4).

The two regimes promised in the chapter introduction now separate. The principal character carries the pole; everything else is holomorphic at $s = 1$. The principal case is settled first, by direct comparison with ζ.

Proposition 7.2.3: The Principal L-Function and ζ

Let χ_0 be the principal character modulo m (definition 7.1.3). For $\operatorname{Re}(s) > 1$,

$$L(s, \chi_0) = \zeta(s) \prod_{p \mid m} \left(1 - p^{-s}\right)$$

Consequently $L(s, \chi_0)$ extends to a meromorphic function on $\operatorname{Re}(s) > 0$ whose only singularity there is a simple pole at $s = 1$, with residue

$$\operatorname{Res}_{s=1} L(s, \chi_0) = \prod_{p \mid m} \left(1 - \frac{1}{p}\right) = \frac{\varphi(m)}{m}$$

***Proof*:**
Step 1: Factoring out the missing Euler factors. On $\operatorname{Re}(s) > 1$ the Euler product of theorem 7.2.2 for χ_0 runs over the primes $p \nmid m$, since $\chi_0(p) = 1$ for $p \nmid m$ and $\chi_0(p) = 0$ for $p \mid m$. The product for ζ (corollary 2.3.2) runs over all primes, with factor $(1 - p^{-s})^{-1}$ at each. Both products converge absolutely for $\operatorname{Re}(s) > 1$, so they may be compared factor by factor and the finite collection of factors at the primes $p \mid m$ separated off:

$$L(s, \chi_0) = \prod_{p \nmid m} \frac{1}{1 - p^{-s}} = \left(\prod_{p} \frac{1}{1 - p^{-s}}\right) \prod_{p \mid m} \left(1 - p^{-s}\right) = \zeta(s) \prod_{p \mid m} \left(1 - p^{-s}\right)$$

the middle equality multiplying and dividing by the finitely many factors at $p \mid m$, which is legitimate because each such factor $(1 - p^{-s})^{-1}$ is finite and nonzero for $\operatorname{Re}(s) > 1$. This is the displayed identity.

Step 2: Continuation and the pole. The finite product $P(s) = \prod_{p \mid m} (1 - p^{-s})$ is an entire function of s, a finite product of entire functions. By theorem 4.2.1 the factor $\zeta(s)$ continues meromorphically to $\operatorname{Re}(s) > 0$ with a single simple pole at $s = 1$ of residue 1 and no other singularity. The product $\zeta(s)P(s)$ therefore continues to a meromorphic function on $\operatorname{Re}(s) > 0$, and since P is entire, its only possible pole is at $s = 1$, inherited from ζ. Whether the pole survives is decided by the value of P at $s = 1$: it survives precisely when $P(1) \neq 0$. Here

$$P(1) = \prod_{p \mid m} \left(1 - \frac{1}{p}\right)$$

a finite product of factors each strictly between 0 and 1 (for every prime $p \geq 2$ one has $0 < 1 - 1/p < 1$), hence $P(1) \neq 0$. The pole of ζ at $s = 1$ thus passes to $L(s, \chi_0)$ as a simple pole, and there is no other singularity on $\operatorname{Re}(s) > 0$.

Step 3: The residue. Since P is holomorphic at $s = 1$ and ζ has a simple pole there,

$$\mathrm{Res}_{s=1}L(s,\chi_0) = \mathrm{Res}_{s=1}\big(\zeta(s)P(s)\big) = P(1)\,\mathrm{Res}_{s=1}\zeta(s) = P(1)\cdot 1 = \prod_{p \mid m}\left(1 - \frac{1}{p}\right)$$

using $\mathrm{Res}_{s=1}\zeta(s) = 1$ from corollary 4.2.2 and the fact that the residue of a product of a holomorphic function with a simple pole is the holomorphic value times the residue. The final product equals $\varphi(m)/m$: the totient is multiplicative with $\varphi(p^k) = p^k(1 - 1/p)$ at prime powers (definition 0.3.5 and the multiplicativity of φ), so writing $m = \prod_{p \mid m} p^{a_p}$ gives

$$\frac{\varphi(m)}{m} = \prod_{p \mid m}\frac{\varphi(p^{a_p})}{p^{a_p}} = \prod_{p \mid m}\left(1 - \frac{1}{p}\right)$$

which is $P(1)$, completing the proof.

The principal L-function is ζ with the Euler factors at the primes dividing m deleted. Removing finitely many nonzero, holomorphic factors changes neither the meromorphic continuation nor the location of the pole, only the residue: the residue 1 of ζ is scaled by $\prod_{p \mid m}(1 - 1/p) = \varphi(m)/m$, the density of the integers coprime to m. This is the analytic form of a counting fact. The pole of $L(s,\chi_0)$ at $s = 1$ measures the abundance of the integers coprime to m, exactly as the pole of ζ measures the abundance of all integers, and the residue $\varphi(m)/m$ is the proportion of residue classes modulo m that are coprime to m. In the decomposition of section 7.5 this pole is the main term: it is the single source of divergence, and the principal character is the only character of modulus m that contributes one. Every nonprincipal character must therefore contribute a bounded correction, and the next theorem is what guarantees boundedness.

For a nonprincipal character the series $\sum_n \chi(n)n^{-s}$ does not converge absolutely at $s = 1$, but it converges conditionally on a much larger half-plane than absolute convergence reaches. The mechanism is cancellation: the partial sums $\sum_{n \le x}\chi(n)$ do not grow, because over each complete period the values of χ sum to zero. Abel summation converts that boundedness into convergence of the series.

Theorem 7.2.4: Continuation for Nonprincipal Characters

Let χ be a nonprincipal Dirichlet character modulo m. Then the series $\sum_{n \ge 1}\chi(n)n^{-s}$ converges for every s with $\mathrm{Re}(s) > 0$, and the convergence is uniform on compact subsets of the half-plane $\mathrm{Re}(s) > 0$. Consequently $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > 0$. In particular $L(s,\chi)$ is holomorphic at $s = 1$; the principal L-function aside, no L-function of modulus m has a pole.

***Proof*:**
Step 1: The character sums are bounded. Write $S(x) = \sum_{n \le x}\chi(n)$ for the partial sums of the coefficients. The claim is that $|S(x)| \le \varphi(m)$ for all $x \ge 0$. The values of χ are periodic modulo m (condition (ii) of definition 7.1.1), so the sum over any block of m consecutive integers is the

sum over one full period,

$$\sum_{n=a+1}^{a+m} \chi(n) = \sum_{n \bmod m} \chi(n) = 0$$

the last equality being row orthogonality (theorem 7.1.4(2)), which gives 0 precisely because $\chi \neq \chi_0$. Hence in $S(x)$ all complete blocks of length m contribute nothing, and only the final incomplete block of fewer than m terms survives. That block has at most m terms, at most $\varphi(m)$ of which are nonzero (the integers coprime to m in a period number exactly $\varphi(m)$, the rest having $\chi(n) = 0$), and each nonzero term has $|\chi(n)| = 1$ by theorem 7.1.2(4). Therefore

$$|S(x)| \leq \varphi(m) \qquad \text{for all } x \geq 0$$

a single bound independent of x.

Step 2: Abel summation. Fix s with $\sigma = \mathrm{Re}(s) > 0$, and apply Abel's summation formula (theorem 1.4.1) with coefficients $a_n = \chi(n)$, so that the summatory function is $A(t) = S(t)$, and with the test function $f(t) = t^{-s}$, which is continuously differentiable on $[1, \infty)$ with $f'(t) = -s\, t^{-s-1}$. For $x \geq 1$,

$$\sum_{n \leq x} \frac{\chi(n)}{n^s} = S(x)\, x^{-s} + s \int_1^x S(t)\, t^{-s-1}\, dt \tag{7.2}$$

the sign on the integral following from $f'(t) = -s\, t^{-s-1}$. Each piece is now controlled by the bound of Step 1. The boundary term satisfies $|S(x)\, x^{-s}| \leq \varphi(m)\, x^{-\sigma} \to 0$ as $x \to \infty$, since $\sigma > 0$. The integrand satisfies $|S(t)\, t^{-s-1}| \leq \varphi(m)\, t^{-\sigma-1}$, and $\int_1^\infty t^{-\sigma-1}\, dt = 1/\sigma < \infty$, so the integral $\int_1^\infty S(t)\, t^{-s-1}\, dt$ converges absolutely. Letting $x \to \infty$ in (7.2), the boundary term vanishes and the partial sums converge:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = s \int_1^\infty S(t)\, t^{-s-1}\, dt \qquad (\mathrm{Re}(s) > 0) \tag{7.3}$$

This establishes convergence of the series for every s with $\mathrm{Re}(s) > 0$. The argument is the boundedness criterion behind corollary 1.4.9: bounded partial sums against a factor decaying to 0 force the series to converge, here for the complex test function t^{-s} rather than the monotone real one of the corollary.

Step 3: Holomorphy. It remains to see that the convergence is locally uniform, so that the limit is holomorphic. Let K be a compact subset of $\{\mathrm{Re}(s) > 0\}$ and set $\sigma_0 = \min_{s \in K} \mathrm{Re}(s) > 0$ and $C = \max_{s \in K} |s| < \infty$. For $s \in K$ the tail of the integral representation (7.3) beyond x is bounded, using Step 1, by

$$\left| s \int_x^\infty S(t)\, t^{-s-1}\, dt \right| \leq C\, \varphi(m) \int_x^\infty t^{-\sigma_0 - 1}\, dt = \frac{C\, \varphi(m)}{\sigma_0}\, x^{-\sigma_0}$$

together with the boundary term $|S(x) x^{-s}| \leq \varphi(m)\, x^{-\sigma_0}$ from (7.2). Both bounds are independent of $s \in K$ and tend to 0 as $x \to \infty$, so the partial sums $\sum_{n \leq x} \chi(n) n^{-s}$ converge to $L(s, \chi)$ uniformly on K. Each partial sum is a finite sum of the entire functions n^{-s}, hence holomorphic, and a locally uniform limit of holomorphic functions is holomorphic ([13]). Therefore $L(s, \chi)$ is holomorphic on $\mathrm{Re}(s) > 0$. In particular it is holomorphic at $s = 1$, with no pole there, completing the proof.

The contrast with the principal case is the structural fact on which the rest of the chapter turns. The principal character recovers a tweaked zeta function, carrying the pole at $s = 1$ that forces divergence and supplies the main term in every count weighted by χ_0. A nonprincipal character, by contrast, yields a function holomorphic across the entire half-plane $\mathrm{Re}(s) > 0$, the pole having been killed by the cancellation of the character sum over each period. The single number that now matters is the *value* $L(1, \chi)$. Because $L(s, \chi)$ is holomorphic at $s = 1$, this value is a finite complex number rather than an infinity, and the decomposition of section 7.5 expresses the count of primes in a fixed residue class as the pole-carrying principal term plus a sum of corrections, one for each nonprincipal χ, the χ-th correction governed by $\log L(s, \chi)$ near $s = 1$. The correction stays bounded, and so fails to cancel the main term, exactly when $L(1, \chi) \neq 0$, so that $\log L(s, \chi)$ does not run off to $-\infty$. That nonvanishing, $L(1, \chi) \neq 0$ for every nonprincipal χ, is the one fact with no counterpart in the zeta theory, where the pole makes the analogous statement automatic; it is established in section 7.4 and converted into Dirichlet's theorem in section 7.5.

The smallest nonprincipal characters give L-functions that can be written out term by term, and one of them evaluates at $s = 1$ to a series whose sum is classical.

Example 7.2.5: L-Functions to Small Moduli

Modulus 4. The unique nonprincipal character modulo 4 is the odd character χ_1 of example 7.1.7, with $\chi_1(n) = 1$ for $n \equiv 1$, $\chi_1(n) = -1$ for $n \equiv 3$, and $\chi_1(n) = 0$ for n even, all modulo 4. Its L-function is

$$L(s, \chi_1) = \sum_{n=1}^{\infty} \frac{\chi_1(n)}{n^s} = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \frac{1}{9^s} - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^s} = \beta(s)$$

the Dirichlet beta function. By theorem 7.2.4 this series converges for every s with $\mathrm{Re}(s) > 0$, so in particular at $s = 1$ it converges to the value

$$L(1, \chi_1) = \beta(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots = \frac{\pi}{4}$$

the Leibniz series. The evaluation $\beta(1) = \pi/4$ follows from the power series $\arctan x = \sum_{k \geq 0} (-1)^k x^{2k+1}/(2k+1)$, valid for $|x| \leq 1$: the series at $x = 1$ is $\sum_{k \geq 0} (-1)^k/(2k+1)$, and since the series converges at the endpoint $x = 1$, Abel's limit theorem identifies its sum with $\lim_{x \to 1^-} \arctan x = \arctan 1 = \pi/4$ ([13]). The partial sums of $\beta(1)$ converge slowly, as the alternating tail bound $|\beta(1) - \sum_{k=0}^{N} (-1)^k/(2k+1)| \leq 1/(2N+3)$ shows; the first several are

N	$\sum_{k=0}^{N} \frac{(-1)^k}{2k+1}$
0	1.000000
1	0.666667
2	0.866667
3	0.723810
4	0.834921
9	0.760460
49	0.780399

drifting toward $\pi/4 = 0.785398\ldots$ at the rate $1/(2N)$ dictated by the alternating bound. That $L(1, \chi_1) = \pi/4 \neq 0$ is the nonvanishing of section 7.4 made explicit in the first nontrivial case, and it is the analytic input that proves there are infinitely many primes in each of the progressions $1, 5, 9, \ldots$ and $3, 7, 11, \ldots$ modulo 4.

Modulus 3. The unit group $(\mathbb{Z}/3\mathbb{Z})^\times = \{1, 2\}$ is cyclic of order $\varphi(3) = 2$, so there are two characters: the principal χ_0 and one nonprincipal character χ with $\chi(1) = 1$, $\chi(2) = \chi(-1) = -1$, and $\chi(n) = 0$ for $3 \mid n$. The character χ is odd, of conductor 3. Its L-function is

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = 1 - \frac{1}{2^s} + \frac{1}{4^s} - \frac{1}{5^s} + \frac{1}{7^s} - \frac{1}{8^s} + \cdots$$

the signs running $+, -$ over the residues $1, 2$ modulo 3 and the multiples of 3 omitted. Again theorem 7.2.4 gives convergence and holomorphy on $\mathrm{Re}(s) > 0$, and the value

$$L(1, \chi) = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots = \frac{\pi}{3\sqrt{3}}$$

is nonzero, the closed form $\pi/(3\sqrt{3})$ being the modulus-3 analogue of the Leibniz series; its nonvanishing yields infinitely many primes in each of the progressions 1 mod 3 and 2 mod 3. The principal character modulo 3, by contrast, gives $L(s, \chi_0) = \zeta(s)(1 - 3^{-s})$ by proposition 7.2.3, with a simple pole at $s = 1$ of residue $1 - 1/3 = 2/3 = \varphi(3)/3$.

7.3 Gauss Sums and the Functional Equation

The functional equation of ζ (theorem 6.3.3) was assembled in chapter 6 from two ingredients: the gamma factor $\Gamma_\mathbb{R}(s) = \pi^{-s/2}\Gamma(s/2)$, which completed the bare Dirichlet series into a symmetric object $\Lambda(s) = \Gamma_\mathbb{R}(s)\zeta(s)$, and the theta transformation $\theta(1/t) = \sqrt{t}\,\theta(t)$ (theorem 6.2.4), which supplied the symmetry $\Lambda(s) = \Lambda(1-s)$ by feeding through a Mellin transform. The L-function of a primitive Dirichlet character satisfies a functional equation of the same shape, relating s to $1-s$, but with one new feature absent from the zeta case: the reflection is no longer an exact equality but carries a constant of modulus 1, the *root number*, and that constant is built from a finite exponential sum attached to the character, the Gauss sum. The presence of the constant is forced by the loss of self-duality. The theta series $\theta(t)$ transforms into itself under $t \mapsto 1/t$ because the trivial character is its own conjugate; a character-twisted theta series transforms into the series of the *conjugate* character, and the Gauss sum is precisely the proportionality factor that mediates the exchange.

This section constructs the Gauss sum, computes its absolute value (the single fact that makes the root number have modulus 1), defines the completed L-function with the gamma factor chosen by the parity of the character, and proves the functional equation by twisting Riemann's theta argument by χ. The proof is given in full for even characters, where the gamma factor is the same $\Gamma_\mathbb{R}$ used for ζ and the twisted theta function is the direct analogue of θ; the odd case requires a theta series weighted by n and is carried through in full along the same path. The closing example carries the entire apparatus through the smallest nontrivial character, modulo 4, where the Gauss sum is $2i$ and the root number is 1.

Definition 7.3.1: The Gauss Sum

Let χ be a Dirichlet character modulo m (definition 7.1.1). The *Gauss sum* attached to χ is the finite exponential sum

$$\tau(\chi) = \sum_{a=1}^{m} \chi(a)\, e^{2\pi i a/m}$$

the sum running over a complete set of residues modulo m. More generally, for an integer n the *twisted Gauss sum* is

$$\tau_n(\chi) = \sum_{a=1}^{m} \chi(a)\, e^{2\pi i n a/m}$$

so that $\tau(\chi) = \tau_1(\chi)$.

The Gauss sum pairs the character χ against the additive character $a \mapsto e^{2\pi i a/m}$ of $\mathbb{Z}/m\mathbb{Z}$, and it is the basic object linking the multiplicative structure that χ records to the additive structure carried by the exponential. Each summand has modulus $|\chi(a)| \leq 1$, so the trivial bound is $|\tau(\chi)| \leq \varphi(m)$ (only the $\varphi(m)$ residues coprime to m contribute, the rest having $\chi(a) = 0$); the content of the next theorem is that for a primitive character this bound is improved to the exact value $|\tau(\chi)| = \sqrt{m}$, a square-root cancellation that is the source of the root number's modulus. The terminology and the definition specialize, when χ is the Legendre symbol $\left(\frac{\cdot}{p}\right)$ (definition 0.5.3) modulo a prime p, to the quadratic Gauss sum $\sum_a \left(\frac{a}{p}\right) e^{2\pi i a/p}$ whose evaluation governs quadratic reciprocity (section 0.5); the present definition is the extension of that classical object to an arbitrary character.

The absolute value of the Gauss sum, and the closely related product $\tau(\chi)\tau(\bar{\chi})$, rest on a single algebraic identity: for a primitive character the twisted sum $\tau_n(\chi)$ is a scalar multiple of $\tau(\chi)$, the scalar being $\bar{\chi}(n)$. This *separability* is the key lemma, and it is the one place where primitivity (definition 7.1.5) is indispensable.

Theorem 7.3.2: Modulus of the Gauss Sum

Let χ be a primitive Dirichlet character modulo m. Then:

1. *Separability.* For every integer n,

$$\tau_n(\chi) = \bar{\chi}(n)\, \tau(\chi)$$

2. *Modulus.* The Gauss sum has absolute value

$$|\tau(\chi)| = \sqrt{m}$$

3. *Conjugate and product.* The Gauss sums of χ and $\bar{\chi}$ satisfy

$$\tau(\bar{\chi}) = \chi(-1)\, \overline{\tau(\chi)}, \qquad \tau(\chi)\, \tau(\bar{\chi}) = \chi(-1)\, m$$

***Proof*:**
The argument establishes separability first, then derives the modulus by summing $|\tau_n(\chi)|^2$ over n in two ways, and finally reads off the conjugate relation.

Step 1: Separability when $\gcd(n,m) = 1$. Suppose first that n is coprime to m, so $\bar{n}$ is a unit modulo m. As a runs over a complete residue system modulo m, so does $c = na$, and multiplicativity gives $\chi(a) = \chi(n^{-1}c) = \chi(n)^{-1}\chi(c) = \bar{\chi}(n)\chi(c)$, the last equality because $|\chi(n)| = 1$ forces $\chi(n)^{-1} = \overline{\chi(n)} = \bar{\chi}(n)$ (theorem 7.1.2). The exponential is unchanged in the new variable, $e^{2\pi ina/m} = e^{2\pi ic/m}$, so

$$\tau_n(\chi) = \sum_{a \bmod m} \chi(a)\, e^{2\pi ina/m} = \sum_{c \bmod m} \bar{\chi}(n)\,\chi(c)\, e^{2\pi ic/m} = \bar{\chi}(n)\,\tau(\chi)$$

This step uses only multiplicativity and the invertibility of n, not primitivity, and it holds for every character.

Step 2: Separability when $\gcd(n,m) > 1$. Here $\bar{\chi}(n) = 0$, so the claim is $\tau_n(\chi) = 0$. This is the assertion that requires primitivity: for an imprimitive character $\tau_n(\chi)$ can be nonzero at integers n sharing a factor with m. Write $d = \gcd(n,m) > 1$ and $m = dm'$, so m' is a *proper* divisor of m, that is, $m' < m$. Write $n = dn_0$; then $\gcd(n_0, m') = 1$, since any common prime factor of n_0 and m' would multiply d to a larger common divisor of n and m. The exponential simplifies to $e^{2\pi ina/m} = e^{2\pi in_0a/m'}$, which depends only on a modulo m'. Grouping the residues a modulo m by their reduction modulo m', so that a is sorted into the class $a \equiv b \pmod{m'}$ indexed by a residue b modulo m',

$$\tau_n(\chi) = \sum_{b \bmod m'} e^{2\pi in_0b/m'} \sum_{\substack{a \bmod m \\ a\equiv b \pmod{m'}}} \chi(a) = \sum_{b \bmod m'} e^{2\pi in_0b/m'}\, T(b), \qquad T(b) := \sum_{\substack{a \bmod m \\ a\equiv b \pmod{m'}}} \chi(a)$$

the inner sum $T(b)$ running over the d residues modulo m in the class of b modulo m'. It suffices to show $T(b) = 0$ for every b.

The primitivity criterion. The vanishing of every $T(b)$ rests on a restatement of primitivity (definition 7.1.5) in terms of the reduction map. The claim is that, for a primitive χ modulo m and any proper divisor $m' < m$, there exists an integer c with $c \equiv 1 \pmod{m'}$, $\gcd(c,m) = 1$, and $\chi(c) \neq 1$. Suppose, for contradiction, that no such c exists, so $\chi(c) = 1$ for every c coprime to m with $c \equiv 1 \pmod{m'}$. This forces χ to be induced from level m', contradicting primitivity, as follows. Define a function χ^* modulo m' by setting $\chi^*(b) = 0$ when $\gcd(b,m') > 1$, and otherwise choosing any integer a with $a \equiv b \pmod{m'}$ and $\gcd(a,m) = 1$ and setting $\chi^*(b) = \chi(a)$. Such an a exists whenever $\gcd(b,m') = 1$: let r be the product of the primes dividing m but not m', so $\gcd(m',r) = 1$, and use the Chinese Remainder Theorem (theorem 0.3.7) on the coprime moduli m' and r to solve $a \equiv b \pmod{m'}$ and $a \equiv 1 \pmod{r}$ simultaneously. Then a is coprime to m: a prime $p \big| m'$ does not divide a because $a \equiv b \pmod{m'}$ with $\gcd(b,m') = 1$, while a prime $p \big| m$ with $p \nmid m'$ divides r, so $a \equiv 1 \pmod{p}$ gives $p \nmid a$; these exhaust the prime divisors of m. The value $\chi^*(b)$ is independent of the choice of a: if a, a' both lie in the class of b modulo m' and are coprime to m, then $c = a'a^{-1}$ satisfies $c \equiv 1 \pmod{m'}$ and $\gcd(c,m) = 1$, so $\chi(c) = 1$ by hypothesis and $\chi(a') = \chi(c)\chi(a) = \chi(a)$. The function χ^* is completely multiplicative: for b, b' both coprime to m', unit lifts a, a' have product aa' a unit lift of bb', so $\chi^*(bb') = \chi(aa') = \chi(a)\chi(a') = \chi^*(b)\chi^*(b')$, while if either of b, b' shares a factor with m' then so does bb' and both sides are 0. Since χ^* agrees with χ on the integers coprime to m, the character χ is induced by χ^* modulo the proper divisor $m' < m$, contradicting the primitivity of χ. This proves the criterion.

Vanishing of the inner sums. Fix b and let c be the integer supplied by the criterion, with $c \equiv 1 \pmod{m'}$, $\gcd(c, m) = 1$, and $\chi(c) \neq 1$. Multiplication by c permutes the residues modulo m (as c is a unit) and preserves residues modulo m' (as $c \equiv 1 \pmod{m'}$), so it permutes the set $\{a \bmod m : a \equiv b \pmod{m'}\}$ over which $T(b)$ is summed. Reindexing that sum by $a \mapsto ca$ and using complete multiplicativity,

$$T(b) = \sum_{\substack{a \bmod m \\ a\equiv b \pmod{m'}}} \chi(a) = \sum_{\substack{a \bmod m \\ a\equiv b \pmod{m'}}} \chi(ca) = \chi(c)\, T(b)$$

whence $(1 - \chi(c))\, T(b) = 0$, and since $\chi(c) \neq 1$ this gives $T(b) = 0$. This is the same averaging device that proved row orthogonality (theorem 7.1.4), applied now to the restriction of χ to the residues fixed modulo m' rather than to the full group. As b was arbitrary, every inner sum vanishes, so $\tau_n(\chi) = 0$ when $\gcd(n, m) > 1$. Combining with Step 1 gives $\tau_n(\chi) = \bar{\chi}(n)\tau(\chi)$ for all n, which is (1); the same vanishing is carried out by the conductor-theoretic route in [6, Ch. 9] and [11, Ch. 9].

Step 3: The modulus. Evaluate the sum

$$S = \sum_{n \bmod m} |\tau_n(\chi)|^2$$

in two ways. On one hand, separability (1) gives $\tau_n(\chi) = \bar{\chi}(n)\tau(\chi)$, so $|\tau_n(\chi)|^2 = |\bar{\chi}(n)|^2\, |\tau(\chi)|^2$, which equals $|\tau(\chi)|^2$ when $\gcd(n, m) = 1$ and 0 otherwise; summing over the $\varphi(m)$ residues coprime to m,

$$S = \varphi(m)\, |\tau(\chi)|^2$$

On the other hand, expanding the square directly,

$$|\tau_n(\chi)|^2 = \tau_n(\chi)\, \overline{\tau_n(\chi)} = \sum_{a \bmod m} \sum_{b \bmod m} \chi(a)\, \bar{\chi}(b)\, e^{2\pi i n(a-b)/m}$$

and summing over a complete residue system n modulo m, the geometric sum $\sum_{n \bmod m} e^{2\pi i n(a-b)/m}$ equals m when $a \equiv b \pmod m$ and 0 otherwise. Only the diagonal $a = b$ survives:

$$S = m \sum_{a \bmod m} \chi(a)\, \bar{\chi}(a) = m \sum_{a \bmod m} |\chi(a)|^2 = m\, \varphi(m)$$

the final equality because $|\chi(a)|^2 = 1$ for the $\varphi(m)$ residues coprime to m and 0 for the rest. Equating the two evaluations, $\varphi(m)\, |\tau(\chi)|^2 = m\, \varphi(m)$, and since $\varphi(m) \neq 0$, this gives $|\tau(\chi)|^2 = m$, hence $|\tau(\chi)| = \sqrt{m}$, which is (2).

Step 4: The conjugate relation and the product. Conjugating the defining sum and using $\overline{e^{2\pi i a/m}} = e^{-2\pi i a/m} = e^{2\pi i(-1)a/m}$,

$$\overline{\tau(\chi)} = \sum_{a \bmod m} \overline{\chi(a)}\, e^{-2\pi i a/m} = \sum_{a \bmod m} \bar{\chi}(a)\, e^{2\pi i(-1)a/m} = \tau_{-1}(\bar{\chi})$$

the last sum being the twisted Gauss sum of $\bar\chi$ at $n=-1$. Since $\bar\chi$ is primitive (its conductor equals that of χ), separability (1) applied to $\bar\chi$ at $n=-1$ gives $\tau_{-1}(\bar\chi) = \overline{\bar\chi}(-1)\,\tau(\bar\chi) = \chi(-1)\,\tau(\bar\chi)$. Therefore $\overline{\tau(\chi)} = \chi(-1)\,\tau(\bar\chi)$, and multiplying both sides by $\chi(-1) \in \{\pm 1\}$ (definition 7.1.6) yields $\tau(\bar\chi) = \chi(-1)\,\overline{\tau(\chi)}$. Substituting this into the product,

$$\tau(\chi)\,\tau(\bar\chi) = \tau(\chi)\cdot\chi(-1)\,\overline{\tau(\chi)} = \chi(-1)\,|\tau(\chi)|^2 = \chi(-1)\,m$$

using (2), which establishes (3), completing the proof.

The square-root cancellation $|\tau(\chi)| = \sqrt{m}$ is the arithmetic fact that controls the entire functional equation. The trivial bound $\varphi(m)$ would make the Gauss sum as large as the number of nonzero terms; the actual size $\sqrt{m}$ is the square root of the number of terms, the hallmark of the values $\chi(a)e^{2\pi i a/m}$ behaving like independent unit vectors whose sum has length $\sqrt{m}$ rather than $\varphi(m)$. This cancellation is what will force the root number to lie on the unit circle: the root number is $\tau(\chi)$ divided by $\sqrt{m}$ and a power of i, so the equality $|\tau(\chi)| = \sqrt{m}$ is exactly the statement that the resulting constant has modulus 1. The product formula $\tau(\chi)\tau(\bar\chi) = \chi(-1)m$ is the algebraic identity that makes the two-sided functional equation consistent: applying the reflection twice, once for χ and once for $\bar\chi$, multiplies the two root numbers, and that product must be 1 for the double reflection to return to the start. Separability, the engine behind both, is the precise sense in which a primitive character "diagonalizes" the additive characters of $\mathbb{Z}/m\mathbb{Z}$, each twist τ_n being the same vector $\tau(\chi)$ scaled by the character value $\bar\chi(n)$.

The completed L-function is now defined by attaching the gamma factor dictated by the parity of χ, exactly as $\Lambda(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ attached $\Gamma_{\mathbb{R}}$ to ζ. The parity enters as a shift of the gamma argument: an even character keeps the factor $\Gamma_{\mathbb{R}}$ of definition 6.3.1, while an odd character shifts the argument by 1, reflecting the sign change $\chi(-n) = -\chi(n)$ that the symmetric theta series cannot see and that an antisymmetric one must carry.

Definition 7.3.3: The Completed Dirichlet L-Function

Let χ be a primitive Dirichlet character modulo m, with parity $a \in \{0,1\}$ defined by

$$a = \begin{cases} 0 & \text{if } \chi \text{ is even } (\chi(-1) = +1), \\ 1 & \text{if } \chi \text{ is odd } (\chi(-1) = -1) \end{cases}$$

(definition 7.1.6). The *completed L-function* of χ is

$$\Lambda(s,\chi) = \left(\frac{m}{\pi}\right)^{(s+a)/2} \Gamma\left(\frac{s+a}{2}\right) L(s,\chi)$$

with $L(s,\chi)$ the Dirichlet L-function of definition 7.2.1 and Γ the gamma function of definition 6.1.1, defined initially on $\mathrm{Re}(s) > 1$ where the series for $L(s,\chi)$ converges absolutely.

The completed L-function multiplies the Dirichlet series, the product over the finite places, by a single archimedean factor at the real place of $\mathbb{Q}$, the factor $(m/\pi)^{(s+a)/2}\Gamma((s+a)/2)$. For the trivial character $\chi = \chi_0$ modulo 1, which is even, the parity is $a = 0$, the conductor is $m = 1$, and the definition reduces to $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \Lambda(s)$, recovering the completed zeta function of definition 6.3.1; the present definition is its character-twisted generalization, with the conductor m entering through the power $(m/\pi)^{(s+a)/2}$ in place of $\pi^{-s/2}$. The shape of the gamma factor, with argument $(s+a)/2$, is the archimedean Euler factor of χ: the even factor $\Gamma((s)/2)$ is the gamma factor of the trivial representation of the real place, and the odd factor $\Gamma((s+1)/2)$ is the gamma factor of the sign representation, the two one-dimensional representations of $\mathbb{R}^\times/\mathbb{R}^\times_{>0}$ distinguished by their behavior at -1. This parity-indexed choice of archimedean factor is the prototype of the local factors at infinity that recur for the Dedekind zeta functions, taken up in a later chapter of this book, and beyond.

The functional equation now follows, by twisting Riemann's argument by χ. The completed L-function is entire (the pole of ζ was tied to the constant term of the theta series, which the character twist annihilates), and it satisfies a reflection $s \leftrightarrow 1-s$ that carries the root number.

Theorem 7.3.4: The Functional Equation of $L(s,\chi)$

Let χ be a primitive Dirichlet character modulo m with parity $a \in \{0,1\}$, and assume $\chi \neq \chi_0$ (equivalently $m > 1$). Then the completed L-function $\Lambda(s,\chi)$ of definition 7.3.3 extends to an entire function of s and satisfies the functional equation

$$\Lambda(s,\chi) = \epsilon(\chi)\,\Lambda(1-s,\bar{\chi}) \tag{7.4}$$

where the *root number* is

$$\epsilon(\chi) = \frac{\tau(\chi)}{i^a\sqrt{m}}$$

a complex number of modulus $|\epsilon(\chi)| = 1$.

***Proof*:**
That $|\epsilon(\chi)| = 1$ is immediate from theorem 7.3.2(2): the numerator $\tau(\chi)$ has modulus $\sqrt{m}$, the denominator $i^a\sqrt{m}$ has modulus $\sqrt{m}$, so the quotient has modulus 1. The substance is the entirety and the reflection (7.4). The even case $a = 0$ is proved in full, twisting the theta argument of theorem 6.3.3; the odd case $a = 1$ runs along the identical path with a weighted theta series, carried out in full next.

Step 1: The twisted theta function (even case). Assume χ even, $a = 0$. Define the *twisted theta function*

$$\theta_\chi(t) = \sum_{n\in\mathbb{Z}} \chi(n)\, e^{-\pi n^2 t/m}, \qquad t > 0$$

the character-weighted analogue of $\theta(t) = \sum_n e^{-\pi n^2 t}$ (definition 6.2.3), with the conductor m scaling the exponent. The series converges absolutely and locally uniformly on $t > 0$ by the same geometric domination as for θ, since $|\chi(n)| \leq 1$. The $n = 0$ term vanishes because $\chi(0) = 0$, and because χ is even, $\chi(-n) = \chi(n)$, so pairing $\pm n$ gives

$$\theta_\chi(t) = 2\sum_{n=1}^{\infty} \chi(n)\, e^{-\pi n^2 t/m} \tag{7.5}$$

The functional equation of this twisted theta is the analogue of theorem 6.2.4, now exchanging χ with its conjugate:

$$\theta_\chi\left(\frac{1}{t}\right) = \frac{\tau(\chi)}{\sqrt{m}}\,\sqrt{t}\,\theta_{\bar{\chi}}(t) \tag{7.6}$$

established in Step 2.

Step 2: Poisson summation gives (7.6). Separability (theorem 7.3.2(1)) for the primitive character $\bar{\chi}$ reads $\sum_{a \bmod m} \bar{\chi}(a)e^{2\pi ina/m} = \chi(n)\,\tau(\bar{\chi})$ for all n (using $\bar{\bar{\chi}} = \chi$), and since $\tau(\bar{\chi}) \neq 0$ by theorem 7.3.2(2), this inverts to the finite Fourier expansion

$$\chi(n) = \frac{1}{\tau(\bar{\chi})} \sum_{a \bmod m} \bar{\chi}(a)\, e^{2\pi ina/m} \tag{7.7}$$

valid for every integer n. Insert (7.7) into the definition of θ_χ and exchange the finite sum over a with the sum over n:

$$\theta_\chi(t) = \frac{1}{\tau(\bar{\chi})} \sum_{a \bmod m} \bar{\chi}(a) \sum_{n\in\mathbb{Z}} e^{2\pi ina/m}\, e^{-\pi n^2 t/m}$$

The inner sum is $\sum_{n\in\mathbb{Z}} F_a(n)$ for the function $F_a(x) = e^{2\pi iax/m}\, e^{-\pi x^2 t/m}$ on $\mathbb{R}$, a Gaussian modulated by the additive character $e^{2\pi iax/m}$. Its Fourier transform, in the normalization of theorem 6.2.1, is computed from lemma 6.2.2: the Gaussian $g(x) = e^{-\pi x^2(t/m)}$ has $\widehat{g}(\xi) = (t/m)^{-1/2}e^{-\pi\xi^2(m/t)} = (m/t)^{1/2}e^{-\pi\xi^2 m/t}$, and modulation by $e^{2\pi iax/m}$ shifts the transform by a/m, so

$$\widehat{F_a}(\xi) = \left(\frac{m}{t}\right)^{1/2} e^{-\pi(\xi-a/m)^2 m/t}$$

Both F_a and $\widehat{F_a}$ decay faster than any power, so theorem 6.2.1 applies and

$$\sum_{n\in\mathbb{Z}} F_a(n) = \sum_{k\in\mathbb{Z}} \widehat{F_a}(k) = \left(\frac{m}{t}\right)^{1/2} \sum_{k\in\mathbb{Z}} e^{-\pi(km-a)^2/(mt)}$$

the exponent rewritten via $(k - a/m)^2(m/t) = (km - a)^2/(mt)$. Substituting back,

$$\theta_\chi(t) = \frac{1}{\tau(\bar{\chi})} \left(\frac{m}{t}\right)^{1/2} \sum_{a \bmod m} \bar{\chi}(a) \sum_{k\in\mathbb{Z}} e^{-\pi(km-a)^2/(mt)}$$

Set $b = km - a$. As a ranges over residues modulo m and k over $\mathbb{Z}$, the integer b ranges over all of $\mathbb{Z}$, with $a \equiv -b \pmod{m}$, so $\bar{\chi}(a) = \bar{\chi}(-b) = \bar{\chi}(-1)\bar{\chi}(b) = \bar{\chi}(b)$ because $\bar{\chi}$ is even (the parity of $\bar{\chi}$ equals that of χ). The exponent is $(km - a)^2/(mt) = b^2/(mt) = b^2(1/t)/m$, so the double sum collapses to a single sum over b:

$$\sum_{a \bmod m} \bar{\chi}(a) \sum_{k\in\mathbb{Z}} e^{-\pi(km-a)^2/(mt)} = \sum_{b\in\mathbb{Z}} \bar{\chi}(b)\, e^{-\pi b^2(1/t)/m} = \theta_{\bar{\chi}}\left(\frac{1}{t}\right)$$

Therefore $\theta_\chi(t) = \tau(\bar{\chi})^{-1}(m/t)^{1/2}\, \theta_{\bar{\chi}}(1/t)$. Replacing t by $1/t$ gives

$$\theta_\chi\left(\frac{1}{t}\right) = \tau(\bar{\chi})^{-1}\,\sqrt{m}\,\sqrt{t}\,\theta_{\bar{\chi}}(t) = \frac{\sqrt{m}}{\tau(\bar{\chi})}\,\sqrt{t}\,\theta_{\bar{\chi}}(t)$$

The prefactor simplifies by the product relation $\tau(\chi)\tau(\bar{\chi}) = \chi(-1)\,m = m$ of theorem 7.3.2(3), valid here because χ is even: solving for $\tau(\bar{\chi}) = m/\tau(\chi)$ gives $\sqrt{m}/\tau(\bar{\chi}) = \sqrt{m}\,\tau(\chi)/m = \tau(\chi)/\sqrt{m}$. This is (7.6).

Step 3: The Mellin transform (even case). For $\mathrm{Re}(s) > 1$, form the Mellin transform of $\frac{1}{2}\theta_\chi$ against $t^{s/2}$. Using (7.5) and the per-term integral computed by the substitution $u = \pi n^2 t/m$ (so $\int_0^\infty e^{-\pi n^2 t/m} t^{s/2-1}\, dt = (m/\pi)^{s/2} n^{-s}\,\Gamma(s/2)$, exactly as in (6.15) with π replaced by π/m),

$$\frac{1}{2}\int_0^\infty \theta_\chi(t)\, t^{s/2-1}\, dt = \sum_{n=1}^\infty \chi(n)\left(\frac{m}{\pi}\right)^{s/2} n^{-s}\,\Gamma\!\left(\frac{s}{2}\right) = \left(\frac{m}{\pi}\right)^{s/2}\Gamma\!\left(\frac{s}{2}\right) L(s,\chi) = \Lambda(s,\chi) \tag{7.8}$$

the interchange of sum and integral justified by Tonelli's theorem ([14]) applied to the absolute values, the iterated integral $\sum_n \int_0^\infty e^{-\pi n^2 t/m} t^{\mathrm{Re}(s)/2-1}\, dt = (m/\pi)^{\mathrm{Re}(s)/2}\Gamma(\mathrm{Re}(s)/2)\sum_n n^{-\mathrm{Re}(s)}$ being finite for $\mathrm{Re}(s) > 1$. This is the analogue of the Mellin representation (6.16) of Λ, with the parity making θ_χ play the role of $\theta - 1$.

Step 4: Symmetry and continuation (even case). Split (7.8) at $t = 1$:

$$2\,\Lambda(s,\chi) = \int_0^1 \theta_\chi(t)\, t^{s/2-1}\, dt + \int_1^\infty \theta_\chi(t)\, t^{s/2-1}\, dt$$

In the first integral substitute $t \mapsto 1/t$, then apply (7.6):

$$\int_0^1 \theta_\chi(t)\, t^{s/2-1}\, dt = \int_1^\infty \theta_\chi\!\left(\frac{1}{t}\right) t^{-s/2-1}\, dt = \frac{\tau(\chi)}{\sqrt{m}}\int_1^\infty \theta_{\bar\chi}(t)\, t^{(1-s)/2-1}\, dt$$

the exponent following from $\frac{1}{2} - \frac{s}{2} - 1 = \frac{1-s}{2} - 1$. Therefore

$$2\,\Lambda(s,\chi) = \int_1^\infty \theta_\chi(t)\, t^{s/2-1}\, dt + \frac{\tau(\chi)}{\sqrt{m}}\int_1^\infty \theta_{\bar\chi}(t)\, t^{(1-s)/2-1}\, dt \tag{7.9}$$

Both integrals are over $[1,\infty)$, where each of θ_χ and $\theta_{\bar\chi}$ decays faster than any power of t (the bound (7.5) gives $|\theta_\chi(t)| \le 2\sum_{n\ge1} e^{-\pi n^2 t/m}$, geometric in t), so both integrals converge for *every* $s \in \mathbb{C}$ and define entire functions, by differentiation under the integral sign exactly as in Step 2 of theorem 6.3.3. The right-hand side of (7.9) is thus entire, and it agrees with $2\Lambda(s,\chi)$ on $\mathrm{Re}(s) > 1$; by the identity theorem ([13]) it is the entire continuation of $\Lambda(s,\chi)$. There are no polar terms because the twist by the nonprincipal character χ removed the constant term of the theta series that produced the poles $1/(s-1) - 1/s$ for ζ; this is the analytic form of the holomorphy of $L(s,\chi)$ at $s = 1$ (theorem 7.2.4).

The same computation applied to $\bar\chi$ with s replaced by $1 - s$ gives

$$2\,\Lambda(1-s,\bar\chi) = \int_1^\infty \theta_{\bar\chi}(t)\, t^{(1-s)/2-1}\, dt + \frac{\tau(\bar\chi)}{\sqrt{m}}\int_1^\infty \theta_\chi(t)\, t^{s/2-1}\, dt$$

Multiply this identity by $\tau(\chi)/\sqrt{m}$. The cross term acquires the factor

$$\frac{\tau(\chi)}{\sqrt{m}}\cdot\frac{\tau(\bar\chi)}{\sqrt{m}} = \frac{\tau(\chi)\tau(\bar\chi)}{m} = \frac{\chi(-1)\, m}{m} = \chi(-1) = 1$$

by theorem 7.3.2(3) and $\chi(-1) = 1$ for even χ. Hence

$$\frac{\tau(\chi)}{\sqrt{m}}\cdot 2\,\Lambda(1-s,\bar\chi) = \frac{\tau(\chi)}{\sqrt{m}}\int_1^\infty \theta_{\bar\chi}(t)\, t^{(1-s)/2-1}\, dt + \int_1^\infty \theta_\chi(t)\, t^{s/2-1}\, dt = 2\,\Lambda(s,\chi)$$

the right-hand side being exactly (7.9). With $a = 0$ the root number is $\epsilon(\chi) = \tau(\chi)/(i^0\sqrt{m}) = \tau(\chi)/\sqrt{m}$, so this reads $\Lambda(s,\chi) = \epsilon(\chi)\,\Lambda(1-s,\bar{\chi})$, which is (7.4) in the even case.

Step 5: The odd case. For χ odd, $a = 1$, the symmetric theta series θ_χ vanishes identically (pairing $\pm n$ cancels, since $\chi(-n) = -\chi(n)$), and the correct object is the weighted theta function

$$\theta_\chi^{(1)}(t) = \sum_{n\in\mathbb{Z}} n\,\chi(n)\,e^{-\pi n^2 t/m}$$

whose factor n restores a nonzero sum for an odd character: the summand $n\chi(n)$ is even in n, since $(-n)\chi(-n) = (-n)(-\chi(n)) = n\chi(n)$, so $\theta_\chi^{(1)}(t) = 2\sum_{n\geq 1} n\,\chi(n)\,e^{-\pi n^2 t/m}$ (the $n = 0$ term vanishing), absolutely and locally uniformly convergent on $t > 0$. The Fourier transform of the weighted Gaussian is obtained from lemma 6.2.2 by differentiation: for $g(x) = e^{-\pi x^2(t/m)}$, with $\widehat{g}(\xi) = (m/t)^{1/2}e^{-\pi\xi^2 m/t}$ as in Step 2, differentiating $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x)e^{-2\pi i x\xi}\,dx$ under the integral gives $\widehat{g}\,'(\xi) = -2\pi i\,\widehat{xg}(\xi)$, hence

$$\widehat{xg}(\xi) = \frac{1}{-2\pi i}\,\widehat{g}\,'(\xi) = -i\left(\frac{m}{t}\right)^{3/2}\xi\,e^{-\pi\xi^2 m/t}$$

using $1/i = -i$ and $\widehat{g}\,'(\xi) = -2\pi(m/t)^{3/2}\xi\,e^{-\pi\xi^2 m/t}$. Modulating by $e^{2\pi i a x/m}$ shifts the transform by a/m, so $F_a^{(1)}(x) = x\,e^{2\pi i a x/m}e^{-\pi x^2 t/m}$ has $\widehat{F_a^{(1)}}(\xi) = -i\,(m/t)^{3/2}(\xi - a/m)\,e^{-\pi(\xi-a/m)^2 m/t}$. Inserting the finite Fourier expansion (7.7) of $\chi(n)$ into $\theta_\chi^{(1)}$, exchanging the finite a-sum with the n-sum, and applying Poisson summation (theorem 6.2.1) to $F_a^{(1)}$ gives

$$\theta_\chi^{(1)}(t) = \frac{1}{\tau(\bar{\chi})}\sum_{a \bmod m}\bar{\chi}(a)\sum_{k\in\mathbb{Z}}\widehat{F_a^{(1)}}(k)$$

Setting $b = km - a$, so that $k - a/m = b/m$ and $(k - a/m)^2 m/t = b^2(1/t)/m$, and using $\bar{\chi}(a) = \bar{\chi}(-b) = \bar{\chi}(-1)\bar{\chi}(b) = -\bar{\chi}(b)$ since $\bar{\chi}$ is odd, the double sum collapses to $-\frac{1}{m}\,\theta_{\bar{\chi}}^{(1)}(1/t)$, whence

$$\theta_\chi^{(1)}(t) = \frac{1}{\tau(\bar{\chi})}\,(-i)\left(\frac{m}{t}\right)^{3/2}\left(-\frac{1}{m}\right)\theta_{\bar{\chi}}^{(1)}\left(\frac{1}{t}\right) = \frac{i\sqrt{m}}{\tau(\bar{\chi})}\,t^{-3/2}\,\theta_{\bar{\chi}}^{(1)}\left(\frac{1}{t}\right)$$

Replacing t by $1/t$ and simplifying the prefactor by the product relation $\tau(\chi)\tau(\bar{\chi}) = \chi(-1)m = -m$ of theorem 7.3.2(3) (so $i\sqrt{m}/\tau(\bar{\chi}) = i\sqrt{m}\,\tau(\chi)/(-m) = -i\,\tau(\chi)/\sqrt{m} = \tau(\chi)/(i\sqrt{m})$, using $-i = 1/i$) produces the transformation law

$$\theta_\chi^{(1)}\left(\frac{1}{t}\right) = \frac{\tau(\chi)}{i\,\sqrt{m}}\,t^{3/2}\,\theta_{\bar{\chi}}^{(1)}(t)$$

the factor $i^{-1} = i^{-a}$ being the source of the i^a in the denominator of $\epsilon(\chi)$, and the power $t^{3/2}$ reflecting the shifted gamma argument $(s+1)/2$.

Step 6: Mellin transform and reflection (odd case). For $\mathrm{Re}(s) > 1$, the Mellin transform of $\frac{1}{2}\theta_\chi^{(1)}$ against $t^{(s+1)/2}$ reproduces the completed L-function. By the per-term integral $\int_0^\infty e^{-\pi n^2 t/m}\,t^{(s+1)/2-1}\,dt = (m/\pi)^{(s+1)/2}n^{-(s+1)}\Gamma((s+1)/2)$ and $n\cdot n^{-(s+1)} = n^{-s}$,

$$\frac{1}{2}\int_0^\infty \theta_\chi^{(1)}(t)\,t^{(s+1)/2-1}\,dt = \left(\frac{m}{\pi}\right)^{(s+1)/2}\Gamma\Big(\frac{s+1}{2}\Big)L(s,\chi) = \Lambda(s,\chi)$$

the interchange justified by Tonelli ([14]) as in Step 3. Splitting the integral at $t = 1$, substituting $t \mapsto 1/t$ in the piece over $(0, 1)$, and applying the transformation law just established (exactly as in Step 4) gives

$$2\,\Lambda(s,\chi) = \int_1^\infty \theta_\chi^{(1)}(t)\, t^{(s+1)/2-1}\, dt + \frac{\tau(\chi)}{i\sqrt{m}} \int_1^\infty \theta_{\bar\chi}^{(1)}(t)\, t^{((1-s)+1)/2-1}\, dt$$

both integrals over $[1, \infty)$ entire by the rapid decay of $\theta_\chi^{(1)}$ and $\theta_{\bar\chi}^{(1)}$, so $\Lambda(s, \chi)$ continues to an entire function (no polar terms: the character twist annihilates the constant term of the theta series that produced the poles of $\Lambda(s)$, consistent with the holomorphy of $L(s, \chi)$ in theorem 7.2.4). The same identity for $\bar\chi$ with s replaced by $1 - s$, multiplied by $\tau(\chi)/(i\sqrt{m})$, has cross-coefficient

$$\frac{\tau(\chi)}{i\sqrt{m}} \cdot \frac{\tau(\bar\chi)}{i\sqrt{m}} = \frac{\tau(\chi)\tau(\bar\chi)}{i^2\, m} = \frac{-m}{-m} = 1$$

by theorem 7.3.2(3) with $\chi(-1) = -1$ and $i^2 = -1$; it therefore reproduces $2\,\Lambda(s, \chi)$, giving $\Lambda(s, \chi) = \epsilon(\chi)\,\Lambda(1 - s, \bar\chi)$ with $\epsilon(\chi) = \tau(\chi)/(i\sqrt{m})$. This is (7.4) in the odd case, completing the proof.

The completed L-function is the product of the finite-place Euler factors, encoded in $L(s, \chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$ (theorem 7.2.2), with the single archimedean gamma factor chosen by the parity of χ, and it is this completed product, never the bare Dirichlet series, that satisfies the clean reflection $s \leftrightarrow 1 - s$. Every structural feature of the functional equation of ζ (theorem 6.3.3) is reproduced: the gamma factor at the real place, the symmetry about $\mathrm{Re}(s) = 1/2$, the entirety once the polar contribution is removed. The one feature with no counterpart in the zeta theory is the root number $\epsilon(\chi)$. For ζ the reflection is an exact equality, $\Lambda(s) = \Lambda(1 - s)$, because ζ is its own "conjugate" and the trivial Gauss sum contributes the constant 1; for a character the reflection sends χ to $\bar\chi$ and inserts the constant $\epsilon(\chi)$, a complex number on the unit circle whose value records the fine arithmetic of χ through the Gauss sum $\tau(\chi)$. This constant of modulus 1 is the prototype of the *local epsilon-factors* that organize the functional equations of all higher L-functions: it is the global epsilon-factor of χ, and in the local theory it factors into a product of local constants, one at each place, the archimedean place contributing the power i^{-a} and the finite places contributing the Gauss sum. The completed Dedekind zeta functions and Hecke L-functions, taken up in a later chapter of this book, carry root numbers of exactly this kind, and the Gauss sum computed here is the first instance of the construction that produces them.

The smallest odd primitive character makes the Gauss sum, the root number, and the resulting symmetry fully explicit, and identifies the completed L-function with a symmetric object.

Example 7.3.5: The Gauss Sum and Functional Equation Modulo 4

Let χ be the nonprincipal character modulo 4 of example 7.1.7, with $\chi(1) = 1$, $\chi(3) = -1$, and $\chi(n) = 0$ for even n. It is odd, so its parity is $a = 1$, and it is real, so $\bar\chi = \chi$ and χ is its own conjugate. Its L-function is the Dirichlet beta function $L(s, \chi) = \beta(s)$ of example 7.2.5.

The Gauss sum. Only the residues $a = 1, 3$ contribute, since $\chi(2) = \chi(4) = 0$:

$$\tau(\chi) = \chi(1)\, e^{2\pi i\cdot 1/4} + \chi(3)\, e^{2\pi i\cdot 3/4} = e^{i\pi/2} - e^{i3\pi/2} = i - (-i) = 2i$$

so $|\tau(\chi)| = 2 = \sqrt{4} = \sqrt{m}$, confirming theorem 7.3.2(2) in this case. The product relation is also visible directly: $\tau(\bar\chi) = \tau(\chi) = 2i$ since χ is real, and $\tau(\chi)\tau(\bar\chi) = (2i)^2 = -4 = \chi(-1) \cdot 4$, consistent with $\chi(-1) = -1$ from theorem 7.3.2(3).

The root number. With $a = 1$,

$$\epsilon(\chi) = \frac{\tau(\chi)}{i^a \sqrt{m}} = \frac{2i}{i^1 \cdot 2} = \frac{2i}{2i} = 1$$

The root number is 1, so the functional equation (7.4) for this character reads

$$\Lambda(s, \chi) = \Lambda(1 - s, \bar{\chi}) = \Lambda(1 - s, \chi)$$

a genuine symmetry, using $\bar{\chi} = \chi$. Written out from definition 7.3.3 with $m = 4$ and $a = 1$,

$$\Lambda(s, \chi) = \left(\frac{4}{\pi}\right)^{(s+1)/2} \Gamma\left(\frac{s+1}{2}\right) \beta(s)$$

and the symmetry $\Lambda(s, \chi) = \Lambda(1 - s, \chi)$ relates $\beta(s)$ to $\beta(1 - s)$ through the gamma factors, with axis $\mathrm{Re}(s) = 1/2$. This is the functional equation of the Dirichlet beta function, which is the nontrivial factor of the Dedekind zeta function of the Gaussian field $\mathbb{Q}(i)$, in the factorization $\zeta_{\mathbb{Q}(i)}(s) = \zeta(s)\,\beta(s)$ cutting out $\mathbb{Q}(i)$ by the quadratic character χ. The vanishing of the root number's phase (its value is exactly $+1$) is the statement that the completed beta function is symmetric across the critical line with no twist, and it holds because χ is a real (quadratic) character: for such a character the Gauss sum is $\tau(\chi) = i^a \sqrt{m}$ (here $\tau(\chi) = 2i = i\sqrt{4}$ with $a = 1$), which forces $\epsilon(\chi) = \tau(\chi)/(i^a \sqrt{m}) = +1$ directly, independent of any arithmetic of the field. The contrast with a character whose root number is not 1 is what the general theorem records: for a non-real character χ the reflection exchanges χ and $\bar{\chi}$ and inserts a root number $\epsilon(\chi) \neq 1$ on the unit circle, and only the modulus $|\epsilon(\chi)| = 1$ is forced in general, the phase carrying the arithmetic of the Gauss sum.

7.4 Non-Vanishing of $L(1, \chi)$

The decomposition of section 7.2 reduced the count of primes in a residue class to a principal term carrying a pole and a sum of corrections, one for each nonprincipal character, the χ-th correction governed by $\log L(s, \chi)$ as $s \to 1^+$. That correction stays bounded, and so fails to swallow the main term, exactly when $L(1, \chi) \neq 0$, so that $\log L(s, \chi)$ does not run to $-\infty$. The single inequality $L(1, \chi) \neq 0$ for every nonprincipal χ is therefore the linchpin of Dirichlet's theorem. It is the L-function analogue of the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ (theorem 4.4.2), the input that converted the pole of ζ into the Prime Number Theorem; here the same kind of boundary non-vanishing converts the pole of $L(s, \chi_0)$ into a count of primes in each progression.

The argument splits according to the nature of χ. A complex character ($\chi \neq \bar{\chi}$) is disposed of by a counting argument on the full product $\prod_\chi L(s, \chi)$: a zero at $s = 1$ would be accompanied by a second zero from the conjugate character, and two zeros cannot be absorbed by the single pole that the product carries. The real (quadratic) characters, those with $\chi = \bar{\chi}$, are the hard case, since such a character is its own conjugate and a zero at $s = 1$ comes with no free partner. The resolution is an argument of Landau on Dirichlet series with non-negative coefficients, applied to $\zeta(s)L(s, \chi)$: a zero at $s = 1$ would force this product to be holomorphic on $\mathrm{Re}(s) > 0$, hence (by Landau) representable by a convergent Dirichlet series there, but its coefficients are large enough at the perfect squares to make the series diverge well to the right of $\mathrm{Re}(s) = 0$. The whole development rests on one product and its coefficients, constructed first.

Proposition 7.4.1: Behaviour of the Product $\prod_\chi L(s,\chi)$ at $s=1$

Fix $m \in \mathbb{Z}_{>0}$ and set

$$\zeta_m(s) = \prod_{\chi \bmod m} L(s,\chi)$$

the product running over all $\varphi(m)$ Dirichlet characters modulo m. For $\operatorname{Re}(s) > 1$:

1. ζ_m has the Euler product

$$\zeta_m(s) = \prod_{p \nmid m} \left(1 - p^{-f_p s}\right)^{-\varphi(m)/f_p}$$

   where $f_p = \operatorname{ord}_m(p)$ is the multiplicative order of p in $(\mathbb{Z}/m\mathbb{Z})^\times$;

2. ζ_m is a Dirichlet series $\sum_{n\geq 1} c_n n^{-s}$ with non-negative real coefficients $c_n \geq 0$ and constant term $c_1 = 1$; and

3. the real function $\sigma \mapsto \zeta_m(\sigma)$ on $(1,\infty)$ satisfies $\zeta_m(\sigma) \to +\infty$ as $\sigma \to (1/\varphi(m))^+$, so the Dirichlet series of (2) has abscissa of convergence at least $1/\varphi(m)$.

***Proof*:**
Step 1: The logarithm of the product. Each factor $L(s,\chi)$ is the Dirichlet series of the totally multiplicative function χ (definition 7.1.1), with absolutely convergent Euler product on $\operatorname{Re}(s) > 1$ (theorem 7.2.2) and $|\chi(p)p^{-s}| < 1$ there. Proposition 2.3.5 applies to each χ and gives, for $\operatorname{Re}(s) > 1$, the absolutely convergent expansion

$$\log L(s,\chi) = \sum_p \sum_{k=1}^{\infty} \frac{\chi(p)^k}{k\,p^{ks}} = \sum_{p\nmid m} \sum_{k=1}^{\infty} \frac{\chi(p^k)}{k\,p^{ks}}$$

the second form using $\chi(p)^k = \chi(p^k)$ by complete multiplicativity and discarding the primes $p \big| m$, where $\chi(p) = 0$. Summing over the $\varphi(m)$ characters and interchanging the order of summation, which is permitted because the triple sum converges absolutely (each inner double sum does, and there are finitely many characters),

$$\log \zeta_m(s) = \sum_{\chi \bmod m} \log L(s,\chi) = \sum_{p\nmid m} \sum_{k=1}^{\infty} \frac{1}{k\,p^{ks}} \sum_{\chi \bmod m} \chi(p^k)$$

where the branch of $\log \zeta_m$ is the sum of the branches of $\log L(s,\chi)$ fixed by proposition 2.3.5.

Step 2: Orthogonality collapses the character sum. For $p \nmid m$ the integer p^k is coprime to m, so column orthogonality (theorem 7.1.4(1)) evaluates the inner sum:

$$\sum_{\chi \bmod m} \chi(p^k) = \begin{cases} \varphi(m) & \text{if } p^k \equiv 1 \pmod{m}, \\ 0 & \text{otherwise} \end{cases}$$

The congruence $p^k \equiv 1 \pmod{m}$ holds precisely when $f_p = \operatorname{ord}_m(p)$ divides k, that is, when

$k = jf_p$ for some $j \geq 1$. Substituting and writing $k = jf_p$,

$$\log \zeta_m(s) = \varphi(m) \sum_{p \nmid m} \sum_{j=1}^{\infty} \frac{1}{j f_p \, p^{jf_p s}} \tag{7.10}$$

a double series with non-negative real coefficients on the real axis $s = \sigma > 1$. Exponentiating term by term, the contribution of a single prime p is

$$\exp\left(\frac{\varphi(m)}{f_p} \sum_{j=1}^{\infty} \frac{(p^{-f_p s})^j}{j} \right) = \exp\left(-\frac{\varphi(m)}{f_p} \log\bigl(1 - p^{-f_p s}\bigr) \right) = \bigl(1 - p^{-f_p s}\bigr)^{-\varphi(m)/f_p}$$

using $\sum_{j \geq 1} z^j/j = -\log(1-z)$ with $z = p^{-f_p s}$, valid since $|p^{-f_p s}| < 1$ for $\mathrm{Re}(s) > 0$. Taking the product over $p \nmid m$ gives the Euler product of (1).

Step 3: Non-negative coefficients and the constant term. The exponent $\varphi(m)/f_p$ is a positive integer, since $f_p = \mathrm{ord}_m(p)$ divides $|(\mathbb{Z}/m\mathbb{Z})^\times| = \varphi(m)$ by Lagrange's theorem. Each factor of the Euler product expands by the generalized binomial series

$$\bigl(1 - p^{-f_p s}\bigr)^{-\varphi(m)/f_p} = \sum_{j=0}^{\infty} \binom{\varphi(m)/f_p + j - 1}{j} p^{-jf_p s}$$

whose coefficients $\binom{\varphi(m)/f_p+j-1}{j}$ are non-negative integers and whose constant term ($j = 0$) is 1. A product of Dirichlet series each with non-negative coefficients and constant term 1 is again a Dirichlet series with non-negative coefficients and constant term 1: the coefficient c_n of n^{-s} in the product is a finite sum of products of the individual non-negative coefficients, and c_1 is the product of the constant terms. This is the absolutely convergent rearrangement of the product into a single Dirichlet series, valid on $\mathrm{Re}(s) > 1$ where every factor converges absolutely. This gives (2).

Step 4: Divergence at $s = 1/\varphi(m)$. Work on the real axis, $s = \sigma > 1/\varphi(m)$, where every term of (7.10) is non-negative. Discarding all terms with $j \geq 2$ leaves the lower bound

$$\log \zeta_m(\sigma) \geq \varphi(m) \sum_{p \nmid m} \frac{1}{f_p \, p^{f_p \sigma}}$$

For each $p \nmid m$ the order f_p divides $\varphi(m)$, so $f_p \leq \varphi(m)$, and at σ approaching $1/\varphi(m)$ from above the exponent satisfies $f_p \sigma \leq \varphi(m)\sigma$. Hence $p^{f_p\sigma} \leq p^{\varphi(m)\sigma}$, and since also $f_p \leq \varphi(m)$,

$$\log \zeta_m(\sigma) \geq \varphi(m) \sum_{p \nmid m} \frac{1}{\varphi(m) \, p^{\varphi(m)\sigma}} = \sum_{p \nmid m} \frac{1}{p^{\varphi(m)\sigma}}$$

As $\sigma \to (1/\varphi(m))^+$ the exponent $\varphi(m)\sigma \to 1^+$, and $\sum_p p^{-\varphi(m)\sigma}$ diverges to $+\infty$ by the divergence of the prime sum (proposition 4.3.5), the finitely many primes $p|m$ omitted from the sum affecting it by a bounded amount. Therefore $\log \zeta_m(\sigma) \to +\infty$, so $\zeta_m(\sigma) \to +\infty$. Because the coefficients c_n are non-negative, the real function $\zeta_m(\sigma)$ on the interval of convergence is the increasing limit of the partial sums, so a finite value of the series at a real point σ_0 forces convergence for all $\sigma \geq \sigma_0$; the blow-up at $1/\varphi(m)$ thus shows the abscissa of convergence is at least $1/\varphi(m)$, which is (3) and completes the proof.

The product $\zeta_m(s)$ assembled here is a prototype of the Dedekind zeta function of the cyclotomic field $\mathbb{Q}(\zeta_m)$, which factors as exactly this product of Dirichlet L-functions over the characters modulo m; that factorization, and the parallel one for quadratic fields, is taken up in chapter 8. The two properties extracted are the only ones the non-vanishing proof uses. First, ζ_m is a Dirichlet series with non-negative coefficients and constant term 1, in particular real and at least 1 at every real σ in its range. Second, that series cannot be summed too far to the left: it already diverges at $s = 1/\varphi(m)$, a point strictly to the right of $\mathrm{Re}(s) = 0$. Non-negativity of coefficients is the hypothesis of Landau's theorem, and the divergence at a positive real point is the contradiction that theorem will be played against. The mechanism is identical to the one behind the non-vanishing of ζ on $\mathrm{Re}(s) = 1$: there the inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$ (lemma 4.4.1) forced a product of zeta values to be large; here the non-negativity of the coefficients of ζ_m plays the same positivity role, now packaged as a statement about an entire Dirichlet series rather than a single trigonometric inequality.

With the product in hand, the non-vanishing itself follows. The proof treats the complex and real characters separately, the real case being the one that requires the full strength of the positivity.

Theorem 7.4.2: Non-Vanishing at $s = 1$

Let χ be a nonprincipal Dirichlet character modulo m. Then

$$L(1, \chi) \neq 0$$

***Proof*:**
By theorem 7.2.4 every nonprincipal $L(s, \chi)$ is holomorphic on $\mathrm{Re}(s) > 0$, so $L(1, \chi)$ is a well-defined finite value and the statement $L(1, \chi) \neq 0$ is meaningful. Two facts about conjugation are used throughout. First, $L(s, \bar{\chi}) = \overline{L(\bar{s}, \chi)}$ for $\mathrm{Re}(s) > 0$, since term by term $\overline{\chi(n)/n^{\bar{s}}} = \bar{\chi}(n)/n^s$ and the series converges (theorem 7.2.4); evaluating at the real point $s = 1$ gives $L(1, \bar{\chi}) = \overline{L(1, \chi)}$, so $L(1, \chi) = 0$ if and only if $L(1, \bar{\chi}) = 0$. Second, $\bar{\chi}$ is again a nonprincipal character modulo m (theorem 7.1.2(2)), so the conjugate factor is one of the factors of ζ_m.

Step 1: Complex characters. Suppose first $\chi \neq \bar{\chi}$, and suppose toward a contradiction that $L(1, \chi) = 0$. Then $L(1, \bar{\chi}) = \overline{L(1, \chi)} = 0$ as well, and χ and $\bar{\chi}$ are two *distinct* nonprincipal characters modulo m, each contributing a factor vanishing at $s = 1$ to the product $\zeta_m(s) = \prod_{\psi \bmod m} L(s, \psi)$. Account for the order of ζ_m at $s = 1$ factor by factor. The principal factor $L(s, \chi_0)$ has a simple pole at $s = 1$ (proposition 7.2.3), contributing order -1. The factors $L(s, \chi)$ and $L(s, \bar{\chi})$ each have a zero at $s = 1$, contributing order at least $+1$ apiece. Every remaining factor is holomorphic at $s = 1$ (theorem 7.2.4), contributing order at least 0. The order of the product at $s = 1$ is the sum of the orders of the factors, hence at least

$$(-1) + (+1) + (+1) + 0 = +1 > 0$$

so ζ_m is holomorphic at $s = 1$ with a zero there, $\zeta_m(1) = 0$. The single pole of ζ_m on $\mathrm{Re}(s) > 0$ is the one from χ_0, now cancelled, so ζ_m is holomorphic on all of $\mathrm{Re}(s) > 0$.

Apply Landau's theorem on Dirichlet series with non-negative coefficients ([6, Ch. 4]): if a Dirichlet series $\sum_n c_n n^{-s}$ has $c_n \geq 0$ for all n and extends to a function holomorphic on the half-plane $\mathrm{Re}(s) > \sigma_0$, then the series itself converges on $\mathrm{Re}(s) > \sigma_0$, so its abscissa of convergence is at most σ_0. The series $\zeta_m = \sum_n c_n n^{-s}$ has non-negative coefficients (proposition 7.4.1(2)) and, under the present assumption, extends holomorphically to $\mathrm{Re}(s) > 0$; Landau's theorem then forces its abscissa of convergence to be at most 0. But proposition 7.4.1(3) shows the abscissa is at least $1/\varphi(m) > 0$, a contradiction. Therefore $L(1,\chi) \neq 0$ for $\chi \neq \bar{\chi}$.

Step 2: Real characters. Suppose now $\chi = \bar{\chi}$, so χ takes only the real values $\{0, \pm 1\}$ (each nonzero value is a root of unity equal to its own conjugate), and suppose toward a contradiction that $L(1,\chi) = 0$. The self-conjugate χ supplies no second zero, so the product ζ_m alone no longer yields the contradiction; instead work with the smaller product

$$F(s) = \zeta(s)\, L(s,\chi)$$

the Riemann zeta function times the single L-function. By theorem 4.2.1 the factor $\zeta(s)$ is holomorphic on $\mathrm{Re}(s) > 0$ except for a simple pole at $s = 1$, and $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > 0$ (theorem 7.2.4). At $s = 1$ the simple pole of ζ, of order -1, meets the assumed zero of $L(s,\chi)$, of order at least $+1$, so the product F is holomorphic at $s = 1$; away from $s = 1$ both factors are holomorphic. Thus F is holomorphic on all of $\mathrm{Re}(s) > 0$.

Step 3: The coefficients of F. For $\mathrm{Re}(s) > 1$ both $\zeta(s) = \sum_n n^{-s}$ and $L(s,\chi) = \sum_n \chi(n) n^{-s}$ converge absolutely, so their product is the Dirichlet series of the convolution: $F(s) = \sum_{n \geq 1} a_n n^{-s}$ with

$$a_n = \sum_{d \mid n} \chi(d)$$

since ζ has all coefficients 1. The arithmetic function $n \mapsto a_n$ is the Dirichlet convolution of χ with the constant function 1, both multiplicative, hence multiplicative. On a prime power p^k it evaluates to

$$a_{p^k} = \sum_{i=0}^{k} \chi(p)^i = 1 + \chi(p) + \chi(p)^2 + \cdots + \chi(p)^k$$

a geometric sum whose ratio $\chi(p) \in \{0, +1, -1\}$ is real. The three cases give

$$a_{p^k} = \begin{cases} k+1 & \text{if } \chi(p) = +1, \\ 1 & \text{if } \chi(p) = 0, \\ 1 & \text{if } \chi(p) = -1 \text{ and } k \text{ even}, \\ 0 & \text{if } \chi(p) = -1 \text{ and } k \text{ odd} \end{cases}$$

In every case $a_{p^k} \geq 0$, and whenever k is even $a_{p^k} \geq 1$. By multiplicativity $a_n = \prod_{p^k \| n} a_{p^k} \geq 0$ for all n, so F is a Dirichlet series with non-negative coefficients. Moreover, if $n = \ell^2$ is a perfect square then every exponent in the factorization of n is even, so every local factor $a_{p^k} \geq 1$ and hence $a_{\ell^2} \geq 1$.

Step 4: Landau's contradiction. The non-negativity of the coefficients a_n together with the holomorphy of F on $\mathrm{Re}(s) > 0$ feeds into Landau's theorem ([6, Ch. 4]) exactly as in Step 1: the abscissa of convergence of $\sum_n a_n n^{-s}$ would be at most 0, so the series would converge for every real $\sigma > 0$. Test this against the perfect-square lower bound. For real $\sigma > 0$, all terms being non-negative,

$$F(\sigma) = \sum_{n=1}^{\infty} \frac{a_n}{n^\sigma} \geq \sum_{\ell=1}^{\infty} \frac{a_{\ell^2}}{(\ell^2)^\sigma} \geq \sum_{\ell=1}^{\infty} \frac{1}{\ell^{2\sigma}}$$

using $a_{\ell^2} \geq 1$ from Step 3. The right-hand series is the harmonic-type series $\sum_\ell \ell^{-2\sigma}$, which diverges for $2\sigma \leq 1$, that is, for $\sigma \leq 1/2$. Hence the Dirichlet series for F diverges at every real $\sigma \leq 1/2$, so its abscissa of convergence is at least $1/2$. This contradicts the value at most 0 forced by Landau's theorem. The assumption $L(1,\chi) = 0$ is therefore untenable, and $L(1,\chi) \neq 0$ for real nonprincipal χ, completing the proof.

The two cases meet the same obstruction from opposite directions. A complex character carries its own undoing: a zero at $s = 1$ duplicates into a zero of the conjugate factor, and the resulting double zero overwhelms the single pole that the product ζ_m is permitted to have, leaving a Dirichlet series with non-negative coefficients that is holomorphic where it has no right to converge. A real character has no conjugate partner, and the substitute is the product $\zeta(s)L(s,\chi)$, whose coefficients $a_n = \sum_{d|n} \chi(d)$ are visibly non-negative and at least 1 on the perfect squares; a zero at $s = 1$ would again make this holomorphic on $\mathrm{Re}(s) > 0$, contradicting the divergence its square terms guarantee. The class number formula of chapter 8 gives an alternative route to the real case, expressing $L(1,\chi)$ for a quadratic character as a positive multiple of the class number of the associated quadratic field, manifestly nonzero; the Landau argument above is the elementary route, requiring only the positivity of a Dirichlet series, and it is the one on which Dirichlet's theorem is built in section 7.5.

This non-vanishing is exactly the input Dirichlet's theorem needs. In the decomposition of the count of primes $p \equiv a \pmod m$, orthogonality (theorem 7.1.4) expresses the count through the weighted sum $\sum_{\chi \bmod m} \bar{\chi}(a) \log L(s,\chi)$ as $s \to 1^+$. The principal term $\bar{\chi}_0(a) \log L(s,\chi_0)$ inherits the pole of $L(s,\chi_0)$ at $s = 1$ (proposition 7.2.3), so $\log L(s,\chi_0) \to +\infty$ and supplies the divergence that forces infinitely many primes. Every nonprincipal term $\bar{\chi}(a) \log L(s,\chi)$ must stay bounded for the divergence to be attributed to the principal term alone, and boundedness is precisely what $L(1,\chi) \neq 0$ secures: since $L(s,\chi)$ is holomorphic and nonzero at $s = 1$, the logarithm $\log L(s,\chi)$ extends continuously to a finite value $\log L(1,\chi)$ there, with no pole and no descent to $-\infty$. The non-vanishing thus quarantines the singularity in the single principal character, so that the count of primes in the progression $a \bmod m$ diverges, and Dirichlet's theorem follows in section 7.5.

The smallest moduli make the non-vanishing concrete, as the values $L(1,\chi)$ were already computed in example 7.2.5; the next example collects them and exhibits the closed forms that display each as a nonzero number.

Example 7.4.3: $L(1,\chi)$ for Small Moduli

The values of $L(1,\chi)$ at the nonprincipal characters of the smallest moduli are evaluated by recognizing each series as a classical sum, and every one is nonzero, in agreement with theorem 7.4.2.

Modulus 4. The unique nonprincipal character is the odd character χ_1 of example 7.1.7, and its L-function at $s = 1$ is the Leibniz series (example 7.2.5):

$$L(1,\chi_1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots = \frac{\pi}{4} \neq 0$$

Modulus 3. The unique nonprincipal character χ modulo 3 is odd, and (example 7.2.5)

$$L(1,\chi) = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots = \frac{\pi}{3\sqrt{3}} \neq 0$$

Both characters here are quadratic and odd, the hard case of theorem 7.4.2, and the table records their values alongside the modulus-5 quadratic character ψ^2 of example 7.1.7, whose value $L(1,\psi^2)$

admits the closed form below:

m	χ	parity	$L(1,\chi)$
3	nonprincipal	odd	$\pi/(3\sqrt{3}) \approx 0.6046$
4	χ_1	odd	$\pi/4 \approx 0.7854$
5	ψ^2	even	$2\log\big((1+\sqrt{5})/2\big)/\sqrt{5} \approx 0.4304$

Every entry is strictly positive, hence nonzero. The closed forms are not coincidences: each is an instance of the analytic class number formula, which evaluates $L(1,\chi)$ for a primitive quadratic character of conductor d in terms of arithmetic invariants of the quadratic field $\mathbb{Q}(\sqrt{\pm d})$. For an odd character the formula produces a rational multiple of π divided by $\sqrt{d}$ (here $\pi/(3\sqrt{3})$ and $\pi/4$, the imaginary fields $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$), while for an even character it produces a logarithm of the fundamental unit (here $\log\big((1+\sqrt{5})/2\big)$ is the logarithm of the fundamental unit of the real field $\mathbb{Q}(\sqrt{5})$). In each case the relevant arithmetic invariant is a positive quantity, which is the structural reason $L(1,\chi) \neq 0$, and chapter 8 derives this formula and with it a second proof that these values cannot vanish.

7.5 Dirichlet's Theorem and Equidistribution

The four preceding sections built every component the proof requires. The characters of section 7.1 resolve a residue class into a sum over frequencies; the L-functions of section 7.2 package the primes of each frequency into an Euler product, with the principal character carrying a pole at $s = 1$ (proposition 7.2.3) and every other character holomorphic there (theorem 7.2.4); and the non-vanishing $L(1,\chi) \neq 0$ of section 7.4 (theorem 7.4.2) guarantees that the holomorphic ones contribute nothing singular. Assembling these three facts proves Dirichlet's theorem: every arithmetic progression $a, a+m, a+2m, \ldots$ with $\gcd(a,m) = 1$ contains infinitely many primes. The mechanism is the one that drove the analytic proof of the infinitude of all primes in chapter 4, run one residue class at a time. There the pole of ζ at $s = 1$ forced $\sum_p p^{-s} \to +\infty$; here orthogonality isolates the single residue class a, the pole of $L(s,\chi_0)$ supplies the same divergence, and the non-vanishing of the other L-functions confines that divergence to the class a rather than letting it leak away.

The same computation does more than count: it weighs. Carried one step further, it shows that the divergence is shared equally among the $\varphi(m)$ admissible residue classes, each contributing the term $\frac{1}{\varphi(m)} \log \frac{1}{s-1}$ to the prime sum as $s \to 1^+$. This is the statement that the primes are equidistributed modulo m, with Dirichlet density $1/\varphi(m)$ in each class coprime to m. The section closes with the smallest moduli made numerical, where the equidistribution is visible in prime-counting data and where the residual fluctuation around equality already exhibits the Chebyshev bias.

The proof is a single calculation: expand $\log L(s,\chi)$ as a sum over primes, weight each character by $\overline{\chi(a)}$, sum over all characters, and let orthogonality collapse the double family of sums onto the one residue class a. Each step is in place from the earlier sections; the work is the bookkeeping that tracks where the pole goes.

Theorem 7.5.1: Dirichlet's Theorem on Primes in Arithmetic Progressions

Let a and m be integers with $m \geq 1$ and $\gcd(a, m) = 1$. Then the arithmetic progression

$$a, \ a+m, \ a+2m, \ \ldots$$

contains infinitely many primes. More precisely, the sum of reciprocals of the primes in the progression diverges:

$$\sum_{\substack{p \\ p \equiv a \pmod m}} \frac{1}{p} = +\infty$$

***Proof*:**
Step 1: The prime sum for one character. Fix a Dirichlet character χ modulo m. For real $s > 1$ the Euler product of theorem 7.2.2 converges absolutely and has no vanishing factor, so $L(s, \chi) \neq 0$ and proposition 2.3.5 gives the absolutely convergent logarithmic expansion

$$\log L(s, \chi) = \sum_p \sum_{k=1}^{\infty} \frac{\chi(p)^k}{k\, p^{ks}} = \sum_p \frac{\chi(p)}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{\chi(p)^k}{k\, p^{ks}}$$

the second equality splitting off the $k = 1$ term from the prime-power tail. Write $R(s, \chi)$ for that tail. The tail is bounded uniformly in χ and in $s \geq 1$: using $|\chi(p)| \leq 1$ and the estimate already made for ζ in example 2.3.6,

$$|R(s, \chi)| \leq \sum_p \sum_{k=2}^{\infty} \frac{|\chi(p)|^k}{k\, p^{ks}} \leq \sum_p \sum_{k=2}^{\infty} \frac{1}{p^k} = \sum_p \frac{1}{p(p-1)} \leq 1$$

the middle inequality dropping $1/k \leq 1$ and using $p^{ks} \geq p^k$ for $s \geq 1$, and the final bound being the telescoping sum of example 2.3.6. Thus $R(s, \chi) = O(1)$ as $s \to 1^+$, with an implied constant independent of χ, and

$$\log L(s, \chi) = \sum_p \frac{\chi(p)}{p^s} + O(1) \qquad (s \to 1^+) \tag{7.11}$$

The series $\sum_p \chi(p) p^{-s}$ is the part of $\log L(s, \chi)$ that sees the primes linearly; everything else is bounded.

Step 2: Weight by $\overline{\chi(a)}$ and sum over characters. Multiply (7.11) by the constant $\overline{\chi(a)}$, which has modulus $|\overline{\chi(a)}| = 1$ since $\gcd(a, m) = 1$, and sum over all $\varphi(m)$ characters χ modulo m. The bounded terms remain bounded, $\sum_\chi \overline{\chi(a)}\, O(1) = O(1)$, there being finitely many characters, so

$$\sum_{\chi \bmod m} \overline{\chi(a)} \log L(s, \chi) = \sum_{\chi \bmod m} \overline{\chi(a)} \sum_p \frac{\chi(p)}{p^s} + O(1)$$

Interchanging the finite character sum with the prime sum on the right (legitimate for $s > 1$, where each $\sum_p \chi(p) p^{-s}$ converges absolutely),

$$\sum_{\chi \bmod m} \overline{\chi(a)} \log L(s, \chi) = \sum_p \frac{1}{p^s} \left(\sum_{\chi \bmod m} \chi(p)\, \overline{\chi(a)} \right) + O(1)$$

The inner character sum is exactly the quantity orthogonality evaluates. By the indicator identity (7.1) of theorem 7.1.4, for $\gcd(a,m)=1$,

$$\sum_{\chi \bmod m} \chi(p)\,\overline{\chi(a)} = \begin{cases} \varphi(m) & \text{if } p \equiv a \pmod{m}, \\ 0 & \text{otherwise} \end{cases}$$

so the prime sum collapses onto the single residue class a:

$$\sum_{\chi \bmod m} \overline{\chi(a)} \log L(s,\chi) = \varphi(m) \sum_{\substack{p \\ p\equiv a \pmod{m}}} \frac{1}{p^s} + O(1) \qquad (s \to 1^+) \tag{7.12}$$

This identity is the heart of the matter: the left-hand side is a sum of $\varphi(m)$ logarithms of L-functions, and the right-hand side is the prime sum over the progression, multiplied by $\varphi(m)$. The behaviour of the left-hand side as $s \to 1^+$ now controls the prime sum.

Step 3: The principal character contributes the pole. Separate the principal character χ_0 from the rest of the left-hand side of (7.12). For χ_0 one has $\overline{\chi_0(a)} = 1$, and by proposition 7.2.3 the function $L(s,\chi_0) = \zeta(s)\prod_{p|m}(1-p^{-s})$ has a simple pole at $s=1$. For real $s \to 1^+$ this gives $L(s,\chi_0) \to +\infty$, and quantitatively the principal term has the same logarithmic blow-up as $\log\zeta$: combining (7.11) for χ_0 with the prime-sum estimate $\sum_p p^{-s} = \log\frac{1}{s-1} + O(1)$ of proposition 4.3.5 and noting $\chi_0(p) = 1$ for $p \nmid m$ while the finitely many primes $p|m$ alter the sum by $O(1)$,

$$\log L(s,\chi_0) = \sum_{p\nmid m} \frac{1}{p^s} + O(1) = \log\frac{1}{s-1} + O(1) \qquad (s \to 1^+) \tag{7.13}$$

so $\log L(s,\chi_0) \to +\infty$.

Step 4: The nonprincipal characters stay bounded. For each nonprincipal χ the function $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > 0$ (theorem 7.2.4) and, decisively, nonzero at $s=1$:

$$L(1,\chi) \neq 0$$

by theorem 7.4.2. A function holomorphic and nonzero at a point has a logarithm holomorphic in a neighbourhood of that point, so $\log L(s,\chi)$ extends continuously to the finite value $\log L(1,\chi)$ as $s \to 1^+$. In particular each nonprincipal term is bounded:

$$\overline{\chi(a)} \log L(s,\chi) = \overline{\chi(a)} \log L(1,\chi) + o(1) = O(1) \qquad (s \to 1^+)$$

This is the step where the non-vanishing earns its place. Were $L(1,\chi) = 0$ for some nonprincipal χ, the term $\log L(s,\chi)$ would tend to $-\infty$ as $s \to 1^+$, and could cancel the pole from χ_0, leaving the prime sum bounded and the argument inconclusive. The content of theorem 7.4.2 is precisely that this cancellation cannot happen.

Step 5: Conclusion. Insert Steps 3 and 4 into (7.12). The left-hand side is the principal term plus the bounded nonprincipal terms,

$$\sum_{\chi \bmod m} \overline{\chi(a)} \log L(s,\chi) = \log\frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

the single unbounded contribution coming from χ_0. Equating with the right-hand side of (7.12) and dividing by $\varphi(m)$,

$$\sum_{\substack{p \\ p\equiv a \pmod m}} \frac{1}{p^s} = \frac{1}{\varphi(m)} \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+) \tag{7.14}$$

As $s \to 1^+$ the right-hand side tends to $+\infty$, since $\log \frac{1}{s-1} \to +\infty$ and $\varphi(m) \geq 1$ is a fixed positive integer. Therefore $\sum_{p\equiv a} p^{-s} \to +\infty$ as $s \to 1^+$. If only finitely many primes lay in the progression, this sum would be a finite sum of terms each bounded by p^{-1}, hence bounded uniformly in $s > 1$, contradicting the divergence. Hence the progression contains infinitely many primes, and letting $s \to 1^+$ in the same inequality (or comparing $\sum_{p\equiv a} p^{-s} \leq \sum_{p\equiv a} p^{-1}$ for $s > 1$) shows $\sum_{p\equiv a} 1/p$ diverges, completing the proof.

The proof is the analytic argument for the infinitude of primes from chapter 4, refracted through the characters. Where the pole of ζ at $s = 1$ forced $\sum_p p^{-s} \to +\infty$ and so produced infinitely many primes (chapter 3, proposition 4.3.5), the pole of $L(s, \chi_0)$ does the same job inside a single residue class. Orthogonality is the instrument that performs the restriction: the weighted sum $\sum_\chi \overline{\chi(a)}\chi(p)$ is the indicator of the class a, so summing $\log L(s, \chi)$ against these weights converts a statement about all primes into a statement about the primes $p \equiv a$. The principal character is the only one with a pole, and it alone supplies the divergence. Every nonprincipal character would, if its L-function vanished at $s = 1$, contribute a competing term $\log L(s, \chi) \to -\infty$ capable of cancelling that pole; the non-vanishing $L(1, \chi) \neq 0$ is exactly the guarantee that no such cancellation occurs. This is the same structural role played by the non-vanishing of ζ on the line $\mathrm{Re}(s) = 1$ in the proof of the Prime Number Theorem (theorem 4.4.2): in both arguments the pole carries the arithmetic, and a boundary non-vanishing result protects the pole from being annihilated, the difference being that for ζ the obstruction sits on the whole line $\mathrm{Re}(s) = 1$, whereas here it is concentrated at the single point $s = 1$ for each twisted function $L(s, \chi)$.

7.5.2 Note: Toward Chebotarev Dirichlet's theorem is the abelian case of a far more general equidistribution law. The residue class of a prime p modulo m is the same data as the Frobenius conjugacy class of p in the Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ of the cyclotomic field, an abelian group, and the L-functions $L(s, \chi)$ are the factors into which the Dedekind zeta function of $\mathbb{Q}(\zeta_m)$ decomposes. For an arbitrary finite Galois extension $K/\mathbb{Q}$ with group G, possibly nonabelian, the Chebotarev density theorem asserts that the Frobenius classes of the unramified primes are equidistributed among the conjugacy classes of G, a given class C being hit with density $|C|/|G|$. Dirichlet's theorem is the case G abelian, where every conjugacy class is a single element. The proof follows the same template: the Dedekind zeta function of K factors through Artin L-functions attached to the irreducible characters of G, and the non-vanishing of these L-functions at $s = 1$ confines the pole to the trivial character. The factorization of the cyclotomic Dedekind zeta function into Dirichlet L-functions, the first instance of this pattern, is taken up in chapter 8.

Equation (7.14) obtained in the proof is sharper than the bare infinitude. The coefficient of the logarithmic singularity is $1/\varphi(m)$, the *same* for every admissible class a, independent of which class was chosen. This uniformity is the analytic expression of equidistribution: the primes do not just populate every progression, they populate the $\varphi(m)$ progressions in equal proportion. Making this precise requires only a definition and a division.

Definition 7.5.3: Dirichlet Density

Let A be a set of primes. The *Dirichlet density* of A is

$$\delta(A) = \lim_{s\to 1^+} \frac{\displaystyle\sum_{p\in A} p^{-s}}{\displaystyle\sum_{p} p^{-s}}$$

when the limit exists, the denominator running over all primes. Since $\sum_p p^{-s} = \log \frac{1}{s-1} + O(1)$ as $s \to 1^+$ (proposition 4.3.5), the limit may equivalently be written

$$\delta(A) = \lim_{s\to 1^+} \frac{\displaystyle\sum_{p\in A} p^{-s}}{\log \frac{1}{s-1}}$$

A set with $\delta(A) > 0$, in particular, is infinite.

Dirichlet density measures the relative weight of a set of primes through the analytic weighting $p \mapsto p^{-s}$ as s descends to the pole, rather than through the sharp cutoff $p \le x$ of the counting function $\pi(x; m, a)$. It is the notion of size native to the L-function method, and it is exactly the quantity the proof of theorem 7.5.1 computed. A related but a priori stronger notion, the *natural density* $\lim_{x\to\infty} \pi(x; m, a)/\pi(x)$, weights every prime up to x equally; the natural density of each progression also equals $1/\varphi(m)$, but that statement requires the Prime Number Theorem for arithmetic progressions and lies beyond the elementary L-function argument used here. When the natural density of a set of primes exists, it equals the Dirichlet density, so the two agree wherever both are defined.

Theorem 7.5.4: Dirichlet Density and Equidistribution

Let a and m be integers with $m \ge 1$ and $\gcd(a, m) = 1$. The set of primes $p \equiv a \pmod{m}$ has Dirichlet density

$$\delta\big(\{p \equiv a \pmod{m}\}\big) = \frac{1}{\varphi(m)}$$

Equivalently, as $s \to 1^+$,

$$\sum_{\substack{p \\ p\equiv a \pmod{m}}} \frac{1}{p^s} = \frac{1}{\varphi(m)} \log \frac{1}{s-1} + O(1)$$

The $\varphi(m)$ residue classes coprime to m thus carry the primes in equal proportion, each with density $1/\varphi(m)$, and together they account for all of the prime sum: the densities sum to $\varphi(m) \cdot \frac{1}{\varphi(m)} = 1$.

***Proof*:**
The asymptotic

$$\sum_{\substack{p \\ p\equiv a \pmod m}} \frac{1}{p^s} = \frac{1}{\varphi(m)} \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

is (7.14), established in the course of proving theorem 7.5.1: the only character contributing a singularity to the weighted sum $\sum_\chi \overline{\chi(a)} \log L(s,\chi)$ is the principal one, whose contribution is $\log \frac{1}{s-1} + O(1)$ by (7.13), and dividing by $\varphi(m)$ distributes this singularity with coefficient $1/\varphi(m)$. Dividing the displayed asymptotic by $\sum_p p^{-s} = \log \frac{1}{s-1} + O(1)$ (proposition 4.3.5),

$$\frac{\sum_{p\equiv a} p^{-s}}{\sum_p p^{-s}} = \frac{\frac{1}{\varphi(m)} \log \frac{1}{s-1} + O(1)}{\log \frac{1}{s-1} + O(1)} = \frac{\frac{1}{\varphi(m)} + O\Big(1/\log \frac{1}{s-1}\Big)}{1 + O\Big(1/\log \frac{1}{s-1}\Big)}$$

the second equality dividing numerator and denominator by $\log \frac{1}{s-1}$, which tends to $+\infty$ as $s \to 1^+$ so that both $O(1)$ terms become $O(1/\log \frac{1}{s-1}) \to 0$. Letting $s \to 1^+$, the right-hand side tends to $\frac{1}{\varphi(m)}/1 = 1/\varphi(m)$. By definition 7.5.3 the Dirichlet density of the progression is therefore $1/\varphi(m)$.

For the final clause, the $\varphi(m)$ residue classes coprime to m partition the primes not dividing m, and the finitely many primes $p \big| m$ contribute $O(1)$ to every sum and so do not affect a density. Summing the densities over the $\varphi(m)$ admissible classes gives $\varphi(m) \cdot \frac{1}{\varphi(m)} = 1$, the density of the set of all primes, as we sought to show.

The coefficient $1/\varphi(m)$ is the analytic measure of the principle that, to the resolution the L-function method can detect, the primes prefer no residue class over another. The pole of $L(s,\chi_0)$ at $s = 1$, with residue $\varphi(m)/m$ (proposition 7.2.3), is the single source of the prime sum's divergence, and orthogonality distributes that one pole evenly across the $\varphi(m)$ admissible classes, giving each the coefficient $1/\varphi(m)$. The residue $\varphi(m)/m$ records that only $\varphi(m)$ of the m residue classes modulo m can contain more than one prime at all, the rest sharing a common factor with m; among those $\varphi(m)$ classes the division is exactly equal. The equidistribution is a statement to leading order: it fixes the coefficient of $\log \frac{1}{s-1}$ but says nothing about the bounded $O(1)$ remainder, and it is in that remainder that the finer arithmetic of the primes modulo m resides, including the systematic bias the next example records.

The smallest moduli put the density $1/\varphi(m)$ within reach of direct computation, and the prime-counting data show both the equidistribution and the size of the fluctuation around it.

Example 7.5.5: Primes in Progressions mod 4 and mod 10
Modulus 4. Since $\varphi(4) = 2$, the admissible classes are 1 and 3, each of Dirichlet density 1/2 by theorem 7.5.4. Every odd prime falls into one of them (the even prime $2 \equiv 2$ is the sole prime in a non-admissible class), and the two classes are the progressions

$$1, 5, 9, 13, 17, \ldots \qquad \text{and} \qquad 3, 7, 11, 19, 23, \ldots$$

Writing $\pi(x; 4, a) = \#\{p \le x : p \equiv a \,(\text{mod } 4)\}$ for the counting function of each class, the values up to several powers of ten are:

x	$\pi(x;4,1)$	$\pi(x;4,3)$	$\pi(x;4,3)-\pi(x;4,1)$
10^2	11	13	2
10^3	80	87	7
10^4	609	619	10
10^5	4783	4808	25
10^6	39175	39322	147

The two counts track each other closely: at $x = 10^6$ the ratio $\pi(x;4,1)/\pi(x;4,3) = 39175/39322 = 0.9963\ldots$ is near 1, and each count is close to half of $\pi(10^6) - 1 = 78497$ (subtracting the prime 2), namely $39175 + 39322 = 78497$, split as 0.4990 and 0.5010 of the total. This is the density $1/2$ visible in finite data.

Modulus 10. Since $\varphi(10) = 4$, the admissible classes are $1, 3, 7, 9$, each of Dirichlet density $1/4$. These are the primes ending in the decimal digits $1, 3, 7, 9$ respectively (the digits $0, 2, 4, 5, 6, 8$ forcing a factor of 2 or 5, so 2 and 5 are the only primes outside these four classes). The counts are:

x	$\pi(x;10,1)$	$\pi(x;10,3)$	$\pi(x;10,7)$	$\pi(x;10,9)$
10^2	5	7	6	5
10^3	40	42	46	38
10^4	306	310	308	303
10^5	2387	2402	2411	2390
10^6	19617	19665	19621	19593

At $x = 10^6$ the four counts lie within 0.4% of their common mean 19624, and each is close to one quarter of $\pi(10^6) - 2 = 78496$ (subtracting 2 and 5), namely 19624, illustrating the density $1/4$ in each of the four classes.

7.5.6 Remark: The Chebyshev Bias The modulus-4 table shows the count $\pi(x;4,3)$ exceeding $\pi(x;4,1)$ at every listed x, with a difference that grows but remains small beside either count. This is the *Chebyshev bias*: primes congruent to 3 modulo 4 tend to lead those congruent to 1, despite both classes having density $1/2$. The bias is a phenomenon of the $O(1)$ remainder term in theorem 7.5.4, not of the leading $\frac{1}{\varphi(m)} \log \frac{1}{s-1}$ term, which is exactly equal for the two classes; the equidistribution theorem is silent about it. The bias does not contradict equidistribution and does not persist as a fixed inequality: the sign of $\pi(x;4,3) - \pi(x;4,1)$ does change for suitable x, the first sign change occurring near $x = 26{,}861$, and under the generalized Riemann hypothesis the set of x for which $\pi(x;4,3) > \pi(x;4,1)$ has logarithmic density approximately 0.9959, close to but not equal to 1. The explanation traces to the squares: the class 1 modulo 4 contains the squares of odd primes ($p^2 \equiv 1$ for p odd), giving that class a slight head start in the prime-power count $\psi(x;4,a)$ that the prime count then echoes. A precise account belongs to the theory of the explicit formula and the distribution of zeros of $L(s,\chi)$; see [11, Ch. 4] for the analytic treatment.

8

The Dedekind Zeta Function and the Class Number Formula

chapters 4 to 7 built the analytic theory of the primes over a single base field, the field $\mathbb{Q}$ of rational numbers. The Riemann zeta function $\zeta(s) = \sum_{n\geq 1} n^{-s}$ packaged the rational primes through its Euler product, its pole at $s = 1$ measured their abundance, and twisting the construction by a Dirichlet character χ produced the family of L-functions $L(s, \chi)$ that resolved the primes among the residue classes modulo a fixed integer. Every object in that theory, the characters included, was attached to $\mathbb{Q}$: the Dirichlet characters are functions on $\mathbb{Z}$, and the splitting they detect is the splitting of integers into congruence classes. This chapter changes the base field. The field $\mathbb{Q}$ is replaced by an arbitrary number field K, a finite extension of $\mathbb{Q}$ of degree $d = [K : \mathbb{Q}] = r_1 + 2r_2$, where r_1 counts the real embeddings $K \hookrightarrow \mathbb{R}$ and r_2 counts the conjugate pairs of complex embeddings $K \hookrightarrow \mathbb{C}$; in place of $\mathbb{Z}$ stands the ring of integers $\mathcal{O}_K$, the integral closure of $\mathbb{Z}$ in K. The role formerly played by the rational primes is now played by the nonzero prime ideals of $\mathcal{O}_K$, and the central question becomes how a rational prime p decomposes among them.

The function that organizes this decomposition is the *Dedekind zeta function*

$$\zeta_K(s) = \sum_{\mathfrak{a}} (N\mathfrak{a})^{-s},$$

the sum running over the nonzero integral ideals $\mathfrak{a} \subseteq \mathcal{O}_K$, where $N\mathfrak{a} = [\mathcal{O}_K : \mathfrak{a}]$ is the absolute ideal norm. It is the correct generalization of $\zeta(s) = \zeta_{\mathbb{Q}}(s)$: its Euler product runs over the prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$, so that the local factor at a rational prime p records, through the primes $\mathfrak{p}$ lying above p and their residue degrees, exactly how p splits in K. Like ζ, the function ζ_K extends meromorphically past its half-plane of convergence and carries a single simple pole at $s = 1$; but where the residue of ζ at $s = 1$ was simply 1, the residue of ζ_K is a single constant

$$\kappa_K = \mathrm{Res}_{s=1}\zeta_K(s) = \frac{2^{r_1}(2\pi)^{r_2}\, h_K R_K}{w_K\sqrt{|d_K|}}$$

assembled at once from all the basic arithmetic invariants of K: the class number $h_K = |\mathrm{Cl}\,(K)|$, the regulator R_K, the discriminant d_K, and the number w_K of roots of unity in K. This identity, the *analytic class number formula*, is the central result of the chapter (section 8.3). The five sections develop it in order: section 8.1 defines ζ_K, establishes its convergence and Euler product, and works out the Gaussian field as a first example; section 8.2 carries out the geometry-of-numbers count of ideals of bounded norm that produces the pole; section 8.3 assembles the count into the residue κ_K and so proves the class number formula; section 8.4 factors ζ_K for quadratic and cyclotomic K into Dirichlet L-functions, recovering the nonvanishing $L(1,\chi) \neq 0$ as a corollary; and section 8.5 establishes the functional equation of ζ_K and the conductor-discriminant relation. Throughout the chapter, K denotes a number field of degree $d = r_1 + 2r_2$.

8.0.1 Remark: Algebraic Prerequisites From this chapter onward the book treats algebraic number theory as a co-requisite, drawing on it freely and citing [2] for the results it uses. The facts invoked are the structural basis of the theory of $\mathcal{O}_K$: that $\mathcal{O}_K$ is a Dedekind domain, in which every nonzero ideal factors uniquely into prime ideals; the multiplicativity of the absolute ideal norm, $N(\mathfrak{a}\mathfrak{b}) = N\mathfrak{a} \cdot N\mathfrak{b}$ and $N((\alpha)) = |N_{K/\mathbb{Q}}(\alpha)|$; the finiteness of the ideal class group $\mathrm{Cl}\,(K)$; Minkowski's geometry of numbers, including the canonical embedding $\iota\colon K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^d$ and its covolume computation together with Minkowski's convex-body bound; Dirichlet's unit theorem $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r_1+r_2-1}$ and the regulator R_K; and the Dedekind–Kummer theorem governing the splitting of a rational prime p in $\mathcal{O}_K$. None of these results is reproved here. They are stated where used and referenced to [2], which is the companion volume on the algebraic side.

8.1 The Dedekind Zeta Function

The Riemann zeta function admits two equivalent presentations: as the Dirichlet series $\sum_{n\geq 1} n^{-s}$ summed over positive integers, and as the Euler product $\prod_p (1-p^{-s})^{-1}$ assembled from one factor per rational prime. The first encodes counting, the second encodes the multiplicative structure of $\mathbb{Z}$, and the equality of the two is the analytic form of unique factorization. The Dedekind zeta function of a number field K is built on the same two presentations, with the integers replaced by the nonzero integral ideals of $\mathcal{O}_K$ and the rational primes replaced by the nonzero prime ideals. The equivalence of the two presentations rests, exactly as over $\mathbb{Q}$, on unique factorization, which for $\mathcal{O}_K$ is the unique factorization of ideals into prime ideals supplied by the Dedekind property ([2, unique factorization of ideals]).

Definition 8.1.1: The Dedekind Zeta Function

Let K be a number field with ring of integers $\mathcal{O}_K$. The *Dedekind zeta function* of K is the series

$$\zeta_K(s) = \sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-s},$$

the sum running over the nonzero integral ideals $\mathfrak{a} \subseteq \mathcal{O}_K$, where $N\mathfrak{a} = [\mathcal{O}_K : \mathfrak{a}]$ is the absolute norm of $\mathfrak{a}$. Equivalently, grouping the ideals by their norm,

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{a_K(n)}{n^s}, \qquad a_K(n) = \#\{\mathfrak{a} \subseteq \mathcal{O}_K \mid N\mathfrak{a} = n\},$$

so that ζ_K is the Dirichlet series with coefficients the ideal-counting function a_K.

The two forms agree term by term: each nonzero integral ideal $\mathfrak{a}$ contributes $(N\mathfrak{a})^{-s}$ to the first sum, and collecting the ideals of a fixed norm n contributes $a_K(n)\, n^{-s}$ to the second. The coefficients $a_K(n)$ are finite for every n, since an ideal of norm n contains the integer n (as $\mathfrak{a} \supseteq n\mathcal{O}_K$ whenever $N\mathfrak{a} = n$, because $[\mathcal{O}_K : \mathfrak{a}] = n$ kills the quotient) and there are only finitely many ideals containing the fixed nonzero ideal $n\mathcal{O}_K$; the second form is therefore a genuine Dirichlet series with finite coefficients.

The simplest case recovers the function with which the analytic theory began.

8.1.2 Remark: The Rational Case Recovers ζ For $K = \mathbb{Q}$ one has $\mathcal{O}_K = \mathbb{Z}$, and the nonzero integral ideals of $\mathbb{Z}$ are exactly the principal ideals $(n) = n\mathbb{Z}$ for $n \geq 1$, each with norm $N((n)) = [\mathbb{Z} : n\mathbb{Z}] = n$. Distinct positive integers give distinct ideals, so $a_{\mathbb{Q}}(n) = \#\{(m) \mid m \geq 1,\ N((m)) = n\} = 1$ for every n, and the definition collapses to

$$\zeta_{\mathbb{Q}}(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s),$$

the Riemann zeta function of chapter 4. The Dedekind zeta function is thus a strict generalization of ζ, reducing to it over the base field $\mathbb{Q}$.

The first analytic task is to locate the half-plane on which the defining series makes sense. The comparison is with a power of the Riemann zeta function, and the exponent is the degree d of the field.

Proposition 8.1.3: Absolute Convergence and Holomorphy on $\mathrm{Re}(s) > 1$

The series defining $\zeta_K(s)$ converges absolutely on the half-plane $\mathrm{Re}(s) > 1$ and defines a holomorphic function there. More precisely, for real $\sigma > 1$ the comparison

$$\sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-\sigma} \leq \zeta(\sigma)^d$$

holds, where $d = [K : \mathbb{Q}]$.

***Proof*:**
Step 1: The coefficient function a_K is multiplicative. Let $m, n \geq 1$ be coprime. An ideal $\mathfrak{a}$ with $N\mathfrak{a} = mn$ factors uniquely into prime powers, $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}}$, by the Dedekind property ([2, unique factorization of ideals]), and by multiplicativity of the absolute norm $N\mathfrak{a} = \prod_{\mathfrak{p}} (N\mathfrak{p})^{v_{\mathfrak{p}}}$ ([2, multiplicativity of the ideal norm]). Since $N\mathfrak{p} = p^{f_{\mathfrak{p}}}$ is a power of the rational prime p below $\mathfrak{p}$, grouping the prime ideals according to the rational prime they lie over partitions $\mathfrak{a}$ as a product $\mathfrak{a} = \mathfrak{b}\mathfrak{c}$ with $N\mathfrak{b} = m$ and $N\mathfrak{c} = n$, where $\mathfrak{b}$ collects the prime powers above the primes dividing m and $\mathfrak{c}$ those above the primes dividing n; because $\gcd(m, n) = 1$, no rational prime divides both, so this splitting is well defined and unique. The map $\mathfrak{a} \mapsto (\mathfrak{b}, \mathfrak{c})$ is therefore a bijection between the ideals of norm mn and the pairs of ideals of norms m and n, which gives $a_K(mn) = a_K(m)\, a_K(n)$. Thus a_K is multiplicative.

Step 2: The comparison with $\zeta(\sigma)^d$. Fix a real number $\sigma > 1$. Every term $(N\mathfrak{a})^{-\sigma}$ is positive, so the sum may be rearranged freely; grouping the ideals by their factorization into prime powers and using $N\mathfrak{a} = \prod_{\mathfrak{p}} (N\mathfrak{p})^{v_{\mathfrak{p}}}$ gives, by unique factorization of ideals,

$$\sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-\sigma} = \prod_{\mathfrak{p}} \sum_{k=0}^{\infty} (N\mathfrak{p})^{-k\sigma} = \prod_{\mathfrak{p}} \big(1 - (N\mathfrak{p})^{-\sigma}\big)^{-1},$$

each geometric series converging because $N\mathfrak{p} \geq 2$ forces $(N\mathfrak{p})^{-\sigma} < 1$. Group the prime ideals by the rational prime p below them:

$$\prod_{\mathfrak{p}} \big(1 - (N\mathfrak{p})^{-\sigma}\big)^{-1} = \prod_{p} \prod_{\mathfrak{p} | p} \big(1 - (N\mathfrak{p})^{-\sigma}\big)^{-1}$$

For a fixed rational prime p, the primes $\mathfrak{p}|p$ satisfy $\sum_{\mathfrak{p}|p} e_{\mathfrak{p}} f_{\mathfrak{p}} = d$ ([2, the degree identity $\sum_{\mathfrak{p}|p} e_{\mathfrak{p}} f_{\mathfrak{p}} = d$]), where $e_{\mathfrak{p}}$ and $f_{\mathfrak{p}}$ are the ramification index and residue degree; since each $e_{\mathfrak{p}}, f_{\mathfrak{p}} \geq 1$, there are at most d primes above p. Moreover $N\mathfrak{p} = p^{f_{\mathfrak{p}}} \geq p$, so $(N\mathfrak{p})^{-\sigma} \leq p^{-\sigma}$ and hence

$$\big(1 - (N\mathfrak{p})^{-\sigma}\big)^{-1} \leq \big(1 - p^{-\sigma}\big)^{-1}$$

The inner product over the at most d primes $\mathfrak{p}|p$ is therefore bounded by $(1 - p^{-\sigma})^{-d}$, and consequently

$$\sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-\sigma} = \prod_{p} \prod_{\mathfrak{p} | p} \big(1 - (N\mathfrak{p})^{-\sigma}\big)^{-1} \leq \prod_{p} \big(1 - p^{-\sigma}\big)^{-d} = \Big(\prod_{p} \big(1 - p^{-\sigma}\big)^{-1}\Big)^{d} = \zeta(\sigma)^d,$$

the last equality being the Euler product for ζ (corollary 2.3.2). Since $\zeta(\sigma) < \infty$ for $\sigma > 1$, the displayed bound is finite.

Step 3: Absolute convergence and holomorphy on $\mathrm{Re}(s) > 1$. For any s with $\sigma = \mathrm{Re}(s) > 1$ and any ideal $\mathfrak{a}$ one has $|(N\mathfrak{a})^{-s}| = (N\mathfrak{a})^{-\sigma}$, so the bound of Step 2 gives $\sum_{\mathfrak{a}} |(N\mathfrak{a})^{-s}| = \sum_{\mathfrak{a}} (N\mathfrak{a})^{-\sigma} \leq \zeta(\sigma)^d < \infty$, establishing absolute convergence on $\mathrm{Re}(s) > 1$. In the Dirichlet-series form $\zeta_K(s) = \sum_{n \geq 1} a_K(n) n^{-s}$ this is absolute convergence of a Dirichlet series at every point of $\mathrm{Re}(s) > 1$. On any compact subset of the half-plane $\mathrm{Re}(s) > 1$ there is a real $\sigma_0 > 1$ with $\mathrm{Re}(s) \geq \sigma_0$ throughout, and then $|a_K(n) n^{-s}| \leq a_K(n) n^{-\sigma_0}$ with $\sum_n a_K(n) n^{-\sigma_0} \leq \zeta(\sigma_0)^d < \infty$

independent of s; the Weierstrass M-test gives uniform convergence on the compact set. Each partial sum is holomorphic, so the locally uniform limit $\zeta_K(s)$ is holomorphic on $\mathrm{Re}(s) > 1$ ([13]; this is the convergence theory of Dirichlet series developed in chapter 2).

The product over prime ideals that appeared inside the convergence proof is the Euler product of ζ_K, and it deserves to be stated in its own right: it is the analytic form of unique factorization of ideals, the exact analogue for K of the Euler product of ζ (corollary 2.3.2).

Theorem 8.1.4: Euler Product

For $\mathrm{Re}(s) > 1$ the Dedekind zeta function factors as an absolutely convergent product over the nonzero prime ideals of $\mathcal{O}_K$,

$$\zeta_K(s) = \prod_{\mathfrak{p}} \left(1 - (N\mathfrak{p})^{-s}\right)^{-1},$$

the product taken over all nonzero prime ideals $\mathfrak{p} \subseteq \mathcal{O}_K$.

***Proof*:**
Step 1: Each local factor is a norm-weighted geometric series. Fix s with $\sigma = \mathrm{Re}(s) > 1$. For a nonzero prime ideal $\mathfrak{p}$ one has $|(N\mathfrak{p})^{-s}| = (N\mathfrak{p})^{-\sigma} \le 2^{-\sigma} < 1$, so the geometric series sums to

$$\left(1 - (N\mathfrak{p})^{-s}\right)^{-1} = \sum_{k=0}^{\infty} (N\mathfrak{p})^{-ks} = \sum_{k=0}^{\infty} (N\mathfrak{p}^{k})^{-s},$$

the second equality using multiplicativity of the norm, $N\mathfrak{p}^{k} = (N\mathfrak{p})^{k}$ ([2, multiplicativity of the ideal norm]). The local factor is thus the sum of $(N\mathfrak{q})^{-s}$ over the powers $\mathfrak{q} = \mathfrak{p}^{k}$, $k \ge 0$, of the single prime $\mathfrak{p}$.

Step 2: Absolute convergence of the product. An infinite product $\prod_{\mathfrak{p}}(1 - (N\mathfrak{p})^{-s})^{-1}$ converges absolutely precisely when $\sum_{\mathfrak{p}} |(N\mathfrak{p})^{-s}| < \infty$. Grouping primes by the rational prime below them and using $N\mathfrak{p} \ge p$ together with the fact that at most d primes lie over each p (as in Step 2 of proposition 8.1.3),

$$\sum_{\mathfrak{p}} (N\mathfrak{p})^{-\sigma} = \sum_{p} \sum_{\mathfrak{p} | p} (N\mathfrak{p})^{-\sigma} \le \sum_{p} d\, p^{-\sigma} = d \sum_{p} p^{-\sigma} < \infty$$

for $\sigma > 1$, since $\sum_p p^{-\sigma} \le \zeta(\sigma) < \infty$. The product therefore converges absolutely.

Step 3: The product equals the sum. Because the product converges absolutely, its factors may be multiplied out and the resulting terms summed in any order. Multiplying the geometric series of Step 1 over all $\mathfrak{p}$, a generic term in the expansion is a finite product $\prod_{\mathfrak{p}}(N\mathfrak{p}^{k_{\mathfrak{p}}})^{-s}$ with $k_{\mathfrak{p}} \ge 0$ and all but finitely many $k_{\mathfrak{p}}$ equal to zero; by multiplicativity of the norm this equals $(N\mathfrak{a})^{-s}$ for the ideal $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$. Unique factorization of ideals ([2, unique factorization in Dedekind domains]) asserts that the assignment $(k_{\mathfrak{p}})_{\mathfrak{p}} \mapsto \mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$ is a bijection from the finitely-supported exponent sequences onto the nonzero integral ideals. Hence every nonzero ideal $\mathfrak{a}$ arises exactly once, and the expansion of the product is precisely

$$\prod_{\mathfrak{p}} \left(1 - (N\mathfrak{p})^{-s}\right)^{-1} = \sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-s} = \zeta_K(s),$$

the absolute convergence established in proposition 8.1.3 justifying the rearrangement into the single sum over ideals. (Equivalently, a_K is multiplicative by Step 1 of proposition 8.1.3, and the identity is the Euler product for a multiplicative Dirichlet coefficient, theorem 2.3.1, applied to the series $\sum_n a_K(n)n^{-s}$ with local factor $\sum_k a_K(p^k)p^{-ks}$ regrouped over the primes above p.)

As with the Riemann zeta function, the Euler product immediately yields nonvanishing throughout the region of absolute convergence: a convergent product of nonzero factors cannot vanish.

Corollary 8.1.5: Nonvanishing on $\mathrm{Re}(s) > 1$

$\zeta_K(s) \neq 0$ for every s with $\mathrm{Re}(s) > 1$.

***Proof*:**
By theorem 8.1.4 the value $\zeta_K(s)$ is an absolutely convergent product over the prime ideals $\mathfrak{p}$, each factor $(1 - (N\mathfrak{p})^{-s})^{-1}$ being the reciprocal of the nonzero number $1 - (N\mathfrak{p})^{-s}$ (nonzero because $|(N\mathfrak{p})^{-s}| = (N\mathfrak{p})^{-\sigma} < 1$ for $\sigma > 1$). An absolutely convergent infinite product all of whose factors are nonzero has a nonzero value (proposition 2.3.4). Hence $\zeta_K(s) \neq 0$.

The convergence and nonvanishing established here mirror exactly the situation for ζ on $\mathrm{Re}(s) > 1$, and they are the easy part of the theory: both rest only on the Euler product, which lives inside the region of absolute convergence. The substantial analytic content of the chapter, the continuation of ζ_K across the line $\mathrm{Re}(s) = 1$ and the determination of its residue there, requires the geometric counting of ideals taken up in section 8.2. Before turning to it, the smallest nontrivial field illustrates how the Euler product encodes splitting, and it previews the factorization that section 8.4 establishes in general.

Example 8.1.6: The Gaussian Field $\mathbb{Q}(i)$
Take $K = \mathbb{Q}(i)$, the field of Gaussian rationals, with ring of integers $\mathcal{O}_K = \mathbb{Z}[i]$, the ring of Gaussian integers. The splitting of a rational prime p in $\mathbb{Z}[i]$ is governed by its residue modulo 4 ([2, splitting of primes in $\mathbb{Q}(i)$]):

1. if $p \equiv 1 \pmod 4$, then p *splits*, $p\mathbb{Z}[i] = \mathfrak{p}\bar{\mathfrak{p}}$ with $\mathfrak{p} \neq \bar{\mathfrak{p}}$ two distinct primes each of norm $N\mathfrak{p} = N\bar{\mathfrak{p}} = p$;
2. if $p \equiv 3 \pmod 4$, then p is *inert*, $p\mathbb{Z}[i]$ remains prime, a single prime of norm $N(p\mathbb{Z}[i]) = p^2$;
3. if $p = 2$, then p *ramifies*, $2\mathbb{Z}[i] = (1+i)^2$ (up to a unit), a single prime $(1+i)$ of norm $N((1+i)) = 2$.

The Euler factor of $\zeta_{\mathbb{Q}(i)}$ at p is the product of $(1 - (N\mathfrak{p})^{-s})^{-1}$ over the primes $\mathfrak{p} \big| p$, and the three cases give

$$p \equiv 1\colon\; \left(1 - p^{-s}\right)^{-2}, \qquad p \equiv 3\colon\; \left(1 - p^{-2s}\right)^{-1}, \qquad p = 2\colon\; \left(1 - 2^{-s}\right)^{-1}$$

Let χ_{-4} denote the nontrivial Dirichlet character modulo 4 (the character written χ_1 in chapter 7), the character with $\chi_{-4}(p) = 1$ for $p \equiv 1 \pmod 4$, $\chi_{-4}(p) = -1$ for $p \equiv 3 \pmod 4$, and $\chi_{-4}(2) = 0$. Each Euler factor above factors as $(1 - p^{-s})^{-1}(1 - \chi_{-4}(p)\, p^{-s})^{-1}$:

- for $p \equiv 1 \pmod 4$, $\chi_{-4}(p) = 1$ and $(1 - p^{-s})^{-1}(1 - p^{-s})^{-1} = (1 - p^{-s})^{-2}$;
- for $p \equiv 3 \pmod 4$, $\chi_{-4}(p) = -1$ and $(1 - p^{-s})^{-1}(1 + p^{-s})^{-1} = (1 - p^{-2s})^{-1}$;
- for $p = 2$, $\chi_{-4}(2) = 0$ and $(1 - 2^{-s})^{-1}(1 - 0)^{-1} = (1 - 2^{-s})^{-1}$.

In every case the Euler factor of $\zeta_{\mathbb{Q}(i)}$ at p equals the product of the p-th Euler factors of $\zeta(s)$ and of $L(s, \chi_{-4})$. Taking the product over all primes and invoking the Euler products of ζ (corollary 2.3.2) and of $L(s, \chi_{-4})$ (theorem 7.2.2), valid on $\mathrm{Re}(s) > 1$, this factor-by-factor identity gives

$$\zeta_{\mathbb{Q}(i)}(s) = \zeta(s)\, L(s, \chi_{-4}), \qquad \mathrm{Re}(s) > 1$$

The decomposition exhibits the residue class data of $\mathbb{Z}[i]$ (the splitting type of p) as the value $\chi_{-4}(p)$ of a Dirichlet character, the prototype of the factorization of ζ_K for quadratic fields established in section 8.4. It also connects the analytic invariants of the two threads: the pole of $\zeta_{\mathbb{Q}(i)}$ at $s = 1$ is the pole of ζ, and its residue is $\mathrm{Res}_{s=1}\zeta(s) \cdot L(1, \chi_{-4}) = L(1, \chi_{-4})$, the finite nonzero value $L(1, \chi_{-4}) = \pi/4$ computed in chapter 7. The class number formula of section 8.3 will recover this same residue from the invariants of $\mathbb{Q}(i)$ directly.

8.2 Ideal Counting and Lattice Point Geometry

The Euler product of theorem 8.1.4 presents $\zeta_K(s)$ as a Dirichlet series $\sum_{\mathfrak{a}} (N\mathfrak{a})^{-s} = \sum_{n \geq 1} a_K(n)\, n^{-s}$ whose coefficients $a_K(n) = \#\{\mathfrak{a} : N\mathfrak{a} = n\}$ count integral ideals of given norm. Whether this series has a pole at $s = 1$, and what its residue there is, is decided entirely by the average size of $a_K(n)$, that is by the growth of the summatory function

$$A_K(x) = \sum_{n \leq x} a_K(n) = \#\{\mathfrak{a} \subseteq \mathcal{O}_K \text{ integral} : N\mathfrak{a} \leq x\}$$

The principal object of this section is the asymptotic law governing $A_K(x)$, whose leading coefficient is exactly the constant κ_K of the notation contract, an expression assembling the class number, the regulator, the discriminant, the number of roots of unity, and the signature of K into a single number. The pole of ζ_K at $s = 1$ and its residue (extracted analytically in section 8.3) are then immediate consequences of this counting estimate, and with them the whole of the analytic class number formula.

The estimate is geometric in nature. An integral ideal of bounded norm is, after passing to a fixed representative of its class, a principal ideal generated by an algebraic integer of bounded field norm, and such generators are exactly the lattice points of the Minkowski embedding lying inside a dilate of a fixed region. Counting them is a problem in the geometry of numbers: the number of lattice points of a full lattice Λ inside a large homothetic copy tS of a fixed body S is asymptotically $\mathrm{vol}(S)/\mathrm{vol}(\mathbb{R}^d/\Lambda)$ times t^d, with an error controlled by the surface area of S. The same lattice-point principle underlies the elementary hyperbola counting of chapter 1, now carried out in dimension $d = [K : \mathbb{Q}]$ against the norm-form region rather than under a hyperbola in the plane.

The argument is organized into five steps. *Step 1* reduces the count to a single ideal class. *Step 2* converts counting in a class to counting principal ideals in a fixed ideal, modulo the action of the unit group. *Step 3* realizes that count geometrically through the Minkowski embedding. *Step 4* is the lattice-point counting estimate itself. *Step 5* evaluates the volume of the relevant region as $2^{r_1}\pi^{r_2}R_K$, the appearance of the regulator being the heart of the matter. The algebraic prerequisites (that $\mathcal{O}_K$ is a Dedekind domain with finite class group, the multiplicativity of the norm, the covolume of an ideal under the Minkowski embedding, and Dirichlet's unit theorem) are imported from [2].

Theorem 8.2.1: Asymptotics of the Ideal-Counting Function

Let K be a number field of degree $d = [K:\mathbb{Q}] = r_1 + 2r_2$, with r_1 real and r_2 conjugate pairs of complex embeddings. Then, as $x \to \infty$,

$$A_K(x) = \#\{\mathfrak{a} \subseteq \mathcal{O}_K \text{ integral} : N\mathfrak{a} \le x\} = \kappa_K\, x + O\big(x^{1-1/d}\big)$$

where

$$\kappa_K = \frac{2^{r_1}(2\pi)^{r_2}\, h_K R_K}{w_K\sqrt{|d_K|}}$$

with h_K the class number, R_K the regulator, d_K the discriminant, and w_K the number of roots of unity in K. The implied constant depends only on K.

The proof divides the count over the ideal class group and reduces, by the proposition stated immediately after, to a single class; that proposition carries the entire geometric content. The deduction of the theorem from the proposition is the bookkeeping of *Step 1*.

***Proof*:**
Step 1 (reduction to one ideal class). The set of nonzero integral ideals of $\mathcal{O}_K$ is partitioned by the ideal class group $\mathrm{Cl}(K)$, which is finite of order $h_K = |\mathrm{Cl}(K)|$ ([2]). Since the norm $N\mathfrak{a}$ depends only on $\mathfrak{a}$,

$$A_K(x) = \sum_{C\in \mathrm{Cl}(K)} \#\{\mathfrak{a} \in C : N\mathfrak{a} \le x\} \tag{8.1}$$

a sum of h_K terms. By proposition 8.2.2 below, each term equals

$$\#\{\mathfrak{a} \in C : N\mathfrak{a} \le x\} = \frac{2^{r_1}(2\pi)^{r_2}\, R_K}{w_K\sqrt{|d_K|}}\, x + O\big(x^{1-1/d}\big)$$

with the same main term for every class C. Summing over the h_K classes multiplies the main term by h_K, and the sum of h_K error terms (the number of summands being the constant h_K) is again $O(x^{1-1/d})$. Hence

$$A_K(x) = h_K \cdot \frac{2^{r_1}(2\pi)^{r_2}\, R_K}{w_K\sqrt{|d_K|}}\, x + O\big(x^{1-1/d}\big) = \frac{2^{r_1}(2\pi)^{r_2}\, h_K R_K}{w_K\sqrt{|d_K|}}\, x + O\big(x^{1-1/d}\big)$$

which is the asserted asymptotic with leading coefficient κ_K.

The whole of the work is therefore the per-class count, which is the next proposition. Its proof, occupying Steps 2 through 5, is the geometric core of the chapter.

Proposition 8.2.2: Counting in a Fixed Ideal Class

For every ideal class $C \in \mathrm{Cl}\,(K)$, the number of integral ideals in C of norm at most x satisfies, as $x \to \infty$,

$$\#\{\mathfrak{a} \in C : N\mathfrak{a} \leq x\} = \frac{2^{r_1}(2\pi)^{r_2}\,R_K}{w_K\sqrt{|d_K|}}\,x + O\big(x^{1-1/d}\big)$$

with implied constant depending only on K (and not on C).

The proof relies on one external input from the geometry of numbers, the lattice-point estimate stated and proved next, after which Steps 2 through 5 assemble the proposition.

Lemma 8.2.3: Lipschitz Lattice-Point Principle

Let $\Lambda \subset \mathbb{R}^d$ be a full lattice and let $S \subset \mathbb{R}^d$ be a bounded set whose boundary ∂S is contained in the union of the images of finitely many Lipschitz maps $[0,1]^{d-1} \to \mathbb{R}^d$. Then, as $t \to \infty$,

$$\#(\Lambda \cap tS) = \frac{\mathrm{vol}(S)}{\mathrm{vol}(\mathbb{R}^d/\Lambda)}\,t^d + O\big(t^{d-1}\big)$$

the implied constant depending only on Λ and on the Lipschitz constants and number of the parametrizing maps.

***Proof*:**
This is the Lipschitz lattice-point estimate of the geometry of numbers; a complete proof is given in [10, Ch. VI, §2]. The mechanism is the following. Tile $\mathbb{R}^d$ by the translates of a fundamental parallelepiped P of Λ, of volume $\mathrm{vol}(P) = \mathrm{vol}(\mathbb{R}^d/\Lambda)$, and to each lattice point $\lambda \in \Lambda$ attach the tile $\lambda + P$. Counting tiles whose corner λ lies in tS approximates the volume: the union of these tiles differs from tS only by tiles meeting the boundary $\partial(tS) = t\,\partial S$, so

$$\big|\,\#(\Lambda \cap tS)\cdot \mathrm{vol}(P) - \mathrm{vol}(tS)\,\big| \leq \mathrm{vol}(P)\cdot \#\{\lambda \in \Lambda : (\lambda + P) \cap t\,\partial S \neq \varnothing\}$$

A tile meeting $t\,\partial S$ lies within bounded distance (the diameter of P) of that set. The hypothesis on ∂S bounds the number of such tiles: the image of $[0,1]^{d-1}$ under a Lipschitz map with constant L, scaled by t, can be covered by $O\big((Lt)^{d-1}\big)$ balls of unit radius, each meeting $O(1)$ tiles, so each of the finitely many boundary pieces accounts for $O(t^{d-1})$ tiles. Summing over the finitely many pieces, the boundary contributes $O(t^{d-1})$ tiles. Since $\mathrm{vol}(tS) = \mathrm{vol}(S)\,t^d$ by homogeneity of Lebesgue measure, dividing through by $\mathrm{vol}(P)$ gives $\#(\Lambda \cap tS) = \mathrm{vol}(S)\,t^d/\mathrm{vol}(\mathbb{R}^d/\Lambda) + O(t^{d-1})$, as asserted.

***Proof of proposition 8.2.2*:**
Fix the class C throughout.

Step 2 (from a class to principal ideals in a fixed ideal, modulo units). Choose, once and for all, an integral ideal $\mathfrak{b}$ lying in the inverse class C^{-1}; such an ideal exists because every class contains integral ideals ([2]). The strategy is to clear the obstruction to principality by multiplying through by $\mathfrak{b}$. The map

$$\mathfrak{a} \longmapsto \mathfrak{a}\mathfrak{b} \tag{8.2}$$

sends an integral ideal $\mathfrak{a} \in C$ to the integral ideal $\mathfrak{ab}$, whose class is $C \cdot C^{-1} = 1$, hence $\mathfrak{ab}$ is *principal*, say $\mathfrak{ab} = (\alpha)$. The generator α is nonzero, and since $\mathfrak{ab} \subseteq \mathfrak{b}$ one has $\alpha \in \mathfrak{b}$. Conversely, if $(\alpha) \subseteq \mathfrak{b}$ is a nonzero principal ideal with $\alpha \in \mathfrak{b}$, then $\mathfrak{b} \big| (\alpha)$ in the sense of Dedekind divisibility, so $(\alpha) = \mathfrak{ab}$ for a unique integral ideal $\mathfrak{a} = (\alpha)\mathfrak{b}^{-1}$, whose class is C. Thus (8.2) is a bijection

$$\{\mathfrak{a} \in C : \mathfrak{a} \text{ integral}\} \xrightarrow{\sim} \{(\alpha) : 0 \neq \alpha \in \mathfrak{b}\}/\sim \tag{8.3}$$

between integral ideals in C and nonzero principal ideals generated by elements of $\mathfrak{b}$, where on the right one counts principal *ideals*, that is, generators up to units: $(\alpha) = (\alpha')$ if and only if $\alpha' = u\alpha$ for some unit $u \in \mathcal{O}_K^\times$.

The bijection respects norms. By multiplicativity of the absolute norm and the identity $N((\alpha)) = |N_{K/\mathbb{Q}}(\alpha)|$ ([2]),

$$|N_{K/\mathbb{Q}}(\alpha)| = N((\alpha)) = N(\mathfrak{ab}) = N\mathfrak{a} \cdot N\mathfrak{b} \tag{8.4}$$

so the constraint $N\mathfrak{a} \leq x$ transcribes to $|N_{K/\mathbb{Q}}(\alpha)| \leq x\, N\mathfrak{b}$. Therefore

$$\#\{\mathfrak{a} \in C : N\mathfrak{a} \leq x\} = \#\big\{\alpha \in \mathfrak{b} \setminus \{0\} : |N_{K/\mathbb{Q}}(\alpha)| \leq x\, N\mathfrak{b}\big\}/\sim \tag{8.5}$$

the quotient again being by the multiplicative action of $\mathcal{O}_K^\times$. The problem is now to count, up to units, the nonzero elements of $\mathfrak{b}$ whose field norm is bounded by $x\, N\mathfrak{b}$.

Step 3 (geometry of numbers). Counting elements of $\mathfrak{b}$ is a lattice problem under the Minkowski embedding. Let $\sigma_1, \dots, \sigma_{r_1}$ be the real embeddings of K and $\sigma_{r_1+1}, \dots, \sigma_{r_1+r_2}$ one from each conjugate pair of complex embeddings, and write

$$\iota : K \hookrightarrow V := \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^d, \qquad \iota(\alpha) = \big(\sigma_1(\alpha), \dots, \sigma_{r_1}(\alpha); \sigma_{r_1+1}(\alpha), \dots, \sigma_{r_1+r_2}(\alpha)\big)$$

for the canonical (Minkowski) embedding. Under ι the ideal $\mathfrak{b}$ maps to a full lattice $\iota(\mathfrak{b}) \subset V \cong \mathbb{R}^d$ of covolume

$$\mathrm{vol}\big(\mathbb{R}^d/\iota(\mathfrak{b})\big) = 2^{-r_2}\sqrt{|d_K|}\; N\mathfrak{b} \tag{8.6}$$

where $\mathbb{R}^d$ carries Lebesgue measure transported through the identification $\mathbb{C} \cong \mathbb{R}^2$ by $z \mapsto (\mathrm{Re} z, \mathrm{Im} z)$ ([2]; the factor 2^{-r_2} records that identification). On V the norm form

$$|N(v)| = \prod_{j=1}^{r_1} |x_j| \cdot \prod_{k=1}^{r_2} |z_k|^2, \qquad v = (x_1, \dots, x_{r_1}; z_1, \dots, z_{r_2})$$

is the continuous extension of $\alpha \mapsto |N_{K/\mathbb{Q}}(\alpha)| = \prod_j |\sigma_j(\alpha)| \cdot \prod_k |\sigma_{r_1+k}(\alpha)|^2$ to all of V, and it is *homogeneous of degree d*: for $t > 0$,

$$|N(tv)| = t^{r_1} \cdot (t^2)^{r_2}\, |N(v)| = t^{\,r_1+2r_2}|N(v)| = t^d\, |N(v)| \tag{8.7}$$

since each real coordinate scales linearly and each complex coordinate contributes its squared modulus.

It remains to handle the quotient by units. By Dirichlet's unit theorem the unit group factors as $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r_1+r_2-1}$, where μ_K is the finite cyclic group of roots of unity, of order w_K, and the free part has rank $r := r_1 + r_2 - 1$ ([2]). The torsion μ_K acts on $\iota(\mathfrak{b}) \setminus \{0\}$ freely, since a root of unity fixing a nonzero α would equal 1; it partitions the nonzero elements into orbits of size exactly w_K. The free part acts through the logarithmic embedding, and one selects a fundamental domain for it on the norm-one hypersurface. Write

$$V^\times := \{v \in V : |N(v)| \neq 0\}$$

for the locus where no coordinate vanishes. There is a measurable *cone* $D \subseteq V^\times$ (closed under positive real scaling, $tD = D$ for $t > 0$) which is a fundamental domain for the action of the free units $\mathcal{O}_K^\times/\mu_K \cong \mathbb{Z}^r$ on $V^\times$, so that each orbit of the free units meets D in exactly one point; the construction of D and its cone property are recorded in [10, Ch. VI, §3] and [15, Ch. VII, §5]. Setting

$$S := \{v \in D : 0 < |N(v)| \leq 1\} \tag{8.8}$$

the cone property together with the homogeneity (8.7) gives, for every $Y > 0$, the homothety relation

$$\{v \in D : 0 < |N(v)| \leq Y\} = Y^{1/d}\, S \tag{8.9}$$

indeed $v \in D$ with $|N(v)| \leq Y$ if and only if $w := Y^{-1/d}v \in D$ (as D is a cone) with $|N(w)| = Y^{-1}|N(v)| \leq 1$, that is $w \in S$. The element-count of Step 2, taken up to the full unit group, is now a lattice-point count in such a region divided by w_K for the torsion: selecting the unique representative of each free-unit orbit inside D and one of the w_K torsion representatives,

$$\#\{\mathfrak{a} \in C : N\mathfrak{a} \leq x\} = \frac{1}{w_K}\,\#\Big(\iota(\mathfrak{b}) \cap (x\,N\mathfrak{b})^{1/d} S\Big) \tag{8.10}$$

where (8.9) with $Y = x\,N\mathfrak{b}$ rewrites $\{v \in D : 0 < |N(v)| \leq x\,N\mathfrak{b}\}$ as $(x\,N\mathfrak{b})^{1/d}S$ and (8.5) supplies the norm bound. The region S is bounded, and its boundary is the union of finitely many images of Lipschitz maps from $[0,1]^{d-1}$, by the explicit description of D through the logarithmic coordinates of [10, Ch. VI, §3]; this regularity is exactly the hypothesis of lemma 8.2.3. (In the rank-zero cases $r = 0$ the cone D is all of $V^\times$ and S is visibly a disk or an interval, as computed in Step 5; no unit theorem is needed there.)

Step 4 (lattice-point counting). Apply lemma 8.2.3 with the lattice $\Lambda = \iota(\mathfrak{b})$, the region S of (8.8), and the scaling parameter $t = (x\,N\mathfrak{b})^{1/d}$, so that $t^d = x\,N\mathfrak{b}$ and $t^{d-1} = (x\,N\mathfrak{b})^{(d-1)/d} = O(x^{1-1/d})$ as $x \to \infty$ (with $N\mathfrak{b}$ a constant depending only on C). Using the covolume (8.6),

$$\#\Big(\iota(\mathfrak{b}) \cap (x\,N\mathfrak{b})^{1/d} S\Big) = \frac{\mathrm{vol}(S)}{2^{-r_2}\sqrt{|d_K|}\;N\mathfrak{b}}\,x\,N\mathfrak{b} + O\big(x^{1-1/d}\big) = \frac{\mathrm{vol}(S)}{2^{-r_2}\sqrt{|d_K|}}\,x + O\big(x^{1-1/d}\big) \tag{8.11}$$

the factor $N\mathfrak{b}$ cancelling between numerator and covolume. This cancellation is the structural reason the count is independent of the choice of $\mathfrak{b}$, and (since Step 5 shows $\mathrm{vol}(S)$ to depend only on K) independent of the class C. Dividing by w_K as in (8.10),

$$\#\{\mathfrak{a} \in C : N\mathfrak{a} \leq x\} = \frac{\mathrm{vol}(S)}{w_K\,2^{-r_2}\sqrt{|d_K|}}\,x + O\big(x^{1-1/d}\big) \tag{8.12}$$

Everything is now explicit except $\mathrm{vol}(S)$, the volume of the norm-bounded fundamental region.

Step 5 (the volume constant). The claim that finishes the proof is

$$\mathrm{vol}(S) = 2^{r_1}\pi^{r_2}\,R_K \tag{8.13}$$

The two rank-zero cases, where $r = r_1 + r_2 - 1 = 0$ and the regulator is by convention $R_K = 1$ (the empty determinant), are elementary and are done in full; they are also the cases of immediate use in section 8.3.

Case $\mathbb{Q}$ ($r_1 = 1$, $r_2 = 0$, $d = 1$). Here $V = \mathbb{R}$, the unit group is $\{\pm 1\}$ with $w_K = 2$ and free rank 0, so $D = \mathbb{R}^\times$ and the norm form is $|N(x)| = |x|$. The region is

$$S = \{x \in \mathbb{R} : 0 < |x| \leq 1\} = [-1, 0) \cup (0, 1]$$

of length $\mathrm{vol}(S) = 2 = 2^1 = 2^{r_1}$, in agreement with (8.13) since $\pi^0 R_{\mathbb{Q}} = 1$. As a check, (8.12) then gives $\#\{\mathfrak{a} \subseteq \mathbb{Z} : N\mathfrak{a} \leq x\} = \frac{2}{2 \cdot 1 \cdot 1}\, x + O(1) = x + O(1)$, the count of positive integers at most x, as it must be.

Case imaginary quadratic ($r_1 = 0$, $r_2 = 1$, $d = 2$). Here $V = \mathbb{C}$, the free unit rank is 0, and $w_K \in \{2, 4, 6\}$ is the number of roots of unity. The cone D is all of $\mathbb{C}^\times$ and the norm form is $|N(z)| = |z|^2$, so

$$S = \{z \in \mathbb{C} : 0 < |z|^2 \leq 1\} = \{z \in \mathbb{C} : 0 < |z| \leq 1\}$$

the closed unit disk with the origin removed. Its area is that of the unit disk,

$$\mathrm{vol}(S) = \pi \cdot 1^2 = \pi = \pi^1 = \pi^{r_2}$$

in agreement with (8.13) since $2^0 R_K = 1$. This is the geometric heart of the imaginary quadratic class number formula: the per-class count (8.12) becomes $\frac{\pi}{w_K\, 2^{-1}\sqrt{|d_K|}}\, x = \frac{2\pi}{w_K \sqrt{|d_K|}}\, x$, recovering the coefficient $2^{r_1}(2\pi)^{r_2} R_K / (w_K \sqrt{|d_K|})$ with $r_1 = 0$, $r_2 = 1$, $R_K = 1$.

General case. When $r = r_1 + r_2 - 1 \geq 1$ the volume (8.13) is computed by a change of variables to "norm-and-logarithm" coordinates adapted to the unit action, in which the integral over S separates into an angular part contributing the powers of 2 and π, a radial part of length 1 (from the norm constraint $0 < |N(v)| \leq 1$), and a part over the free-unit directions whose volume is by definition the covolume of the logarithmic unit lattice, namely the regulator R_K. In outline: pass each real coordinate to $(\operatorname{sign} x_j,\ |x_j|)$ and each complex coordinate to polar form $(\theta_k,\ \rho_k)$ with $\rho_k = |z_k|$; the sign choices and angles θ_k contribute the factor $2^{r_1} \cdot (2\pi)^{r_2}$ on integration, after which the radial variables $(|x_j|, \rho_k)$ are transformed to $(\log |N(v)|;\ \ell_1, \ldots, \ell_r)$, the ℓ_i being the coordinates of the logarithmic embedding along the unit directions. The fundamental domain D becomes a unit box in the ℓ-coordinates of volume equal to the covolume of the log-unit lattice, that is R_K; the norm constraint $0 < |N(v)| \leq 1$ cuts out a single unit of length in the $\log |N(v)|$ direction; and the Jacobian of the substitution is constant, arranged so that a factor 2^{r_2} is absorbed against the $\rho_k\, d\rho_k\, d\theta_k$ area element of each complex place, leaving $2^{r_1}\pi^{r_2}$ in place of $2^{r_1}(2\pi)^{r_2}$. The complete Jacobian computation, including the precise normalization of the logarithmic coordinates and the verification that the resulting box has volume R_K, is carried out in [10, Ch. VI, §3] and, in the form adapted to the present residue, in [15, Ch. VII, §5]. The outcome is (8.13), consistent with both rank-zero cases above (where $R_K = 1$ and the box degenerates to a point).

Conclusion. Insert (8.13) into (8.12) and simplify the constant. The factor 2^{-r_2} in the denominator inverts to 2^{r_2} in the numerator, so

$$\frac{\mathrm{vol}(S)}{w_K\, 2^{-r_2}\sqrt{|d_K|}} = \frac{2^{r_1}\pi^{r_2} R_K}{w_K\, 2^{-r_2}\sqrt{|d_K|}} = \frac{2^{r_1}\, 2^{r_2}\pi^{r_2} R_K}{w_K \sqrt{|d_K|}} = \frac{2^{r_1}(2\pi)^{r_2} R_K}{w_K \sqrt{|d_K|}} \tag{8.14}$$

the last equality from $2^{r_2}\pi^{r_2} = (2\pi)^{r_2}$. This is precisely the coefficient asserted in proposition 8.2.2. The leading coefficient depends only on the invariants of K, not on the class C; the implied constant in the error term, which a priori depends on the chosen ideal $\mathfrak{b}$ (and hence on C), is made uniform across the classes by fixing one representative $\mathfrak{b}$ per class and taking the largest constant among the h_K choices.

The estimate just proved is the entire arithmetic input to the analytic class number formula. Two features deserve emphasis. First, the leading coefficient κ_K is genuinely a product of *all* the basic invariants of K at once: the signature (r_1, r_2) through the powers of 2 and 2π, the class number h_K from the number of summands in (8.1), the regulator R_K from the volume (8.13), the discriminant through the covolume (8.6), and the roots of unity w_K from the torsion of the unit group. No single step produces the formula; it assembles from the geometry of every step together. Second, the error exponent $1 - 1/d$ is dictated by the codimension-one boundary of the norm region: the surface of tS scales as t^{d-1}, and with $t = (x\,N\mathfrak{b})^{1/d}$ this is $x^{(d-1)/d} = x^{1-1/d}$. For $K = \mathbb{Q}$ the exponent degenerates to 0, recovering the trivial $A_{\mathbb{Q}}(x) = x + O(1)$; for higher degree the boundary error is a genuine $o(x)$ but no better at this level of generality.

A comparison with the rational case of chapter 1 situates the argument. There the counting was of lattice points (a, b) under the hyperbola $ab \le x$, a two-dimensional region with main term $x \log x$ for the divisor sum; here the region is the d-dimensional norm body tS, its volume linear in x (after the d-th-root scaling) because the norm form is homogeneous of degree d, matching the ambient dimension exactly. The boundary-error principle is identical in both: a region with rectifiable boundary contains a number of lattice points equal to its volume divided by the covolume, up to a surface-order error. What is new over $\mathbb{Q}$ is the quotient by units, which forces the passage to a fundamental domain and is the precise point at which the regulator enters. In the next section this counting asymptotic is converted, by a direct Dirichlet-series argument, into the pole of ζ_K at $s = 1$ with residue κ_K, completing the analytic class number formula.

8.3 The Analytic Class Number Formula

The two preceding sections prepared the two halves of a single statement. section 8.1 introduced the Dedekind zeta function $\zeta_K(s) = \sum_{\mathfrak{a}} N\mathfrak{a}^{-s} = \sum_{n\ge1} a_K(n)\, n^{-s}$ as the Dirichlet series whose coefficients $a_K(n) = \#\{\mathfrak{a} : N\mathfrak{a} = n\}$ count integral ideals by norm, and established its Euler product over the prime ideals of $\mathcal{O}_K$ (theorem 8.1.4). section 8.2 fixed the average size of those coefficients: the summatory function $A_K(x) = \sum_{n\le x} a_K(n)$, the number of integral ideals of norm at most x, satisfies the asymptotic $A_K(x) = \kappa_K\, x + O\big(x^{1-1/d}\big)$, where κ_K is the geometric constant assembled from the basic invariants of K (theorem 8.2.1). What remains is to read the analytic structure of ζ_K off this counting estimate, and to identify the constant κ_K in arithmetic terms.

The passage from the counting estimate to the analytic structure is purely formal, and it reproduces, for an arbitrary number field, the mechanism by which the meromorphic continuation of the Riemann zeta function was extracted from the elementary asymptotics of $\sum_{n\le x} n^{-s}$ in theorem 4.2.1. A Dirichlet series with non-negative coefficients whose partial sums grow linearly, $A_K(x) = \kappa_K x + O(x^{1-\delta})$, is governed near $s = 1$ entirely by the leading coefficient κ_K: the series converges on $\mathrm{Re}(s) > 1$, continues across the line $\mathrm{Re}(s) = 1$, and acquires there a single simple pole whose residue is exactly κ_K. The error term, of size $O(x^{1-\delta})$, contributes only a holomorphic correction on the wider half-plane $\mathrm{Re}(s) > 1 - \delta$. Isolating this principle as a lemma, applicable to any series of this shape, and then feeding it the specific estimate of theorem 8.2.1, yields the analytic class number formula in two strokes.

Lemma 8.3.1: From Counting to a Pole

Let $(a(n))_{n\geq 1}$ be a sequence of non-negative real numbers, and suppose its summatory function $A(x) = \sum_{n\leq x} a(n)$ satisfies

$$A(x) = \kappa\, x + O\big(x^{1-\delta}\big)$$

for some constant κ and some $\delta \in (0,1]$. Then the Dirichlet series $f(s) = \sum_{n\geq 1} a(n)\, n^{-s}$ converges absolutely on the half-plane $\mathrm{Re}(s) > 1$ and extends to a meromorphic function on the half-plane $\mathrm{Re}(s) > 1-\delta$ whose only singularity there is a simple pole at $s = 1$ with residue κ. Explicitly, writing $A(x) = \kappa x + E(x)$ with $E(x) = O(x^{1-\delta})$, for all s with $\mathrm{Re}(s) > 1-\delta$ and $s \neq 1$,

$$f(s) = \kappa + \frac{\kappa}{s-1} + s\int_1^{\infty} E(x)\, x^{-s-1}\, dx \tag{8.15}$$

***Proof*:**
Since $A(x) = \kappa x + O(x^{1-\delta})$ with $\delta \leq 1$, the partial sums $A(x)$ are $O(x)$, so $a(n) = A(n) - A(n-1) = O(n)$; hence $|a(n)|n^{-\sigma} = O(n^{1-\sigma})$, and $\sum_n a(n)n^{-s}$ converges absolutely for $\mathrm{Re}(s) = \sigma > 2$. This crude bound is superseded below, where the integral representation will give convergence on the whole half-plane $\mathrm{Re}(s) > 1$; the present remark only secures a region on which f is defined as a series, so that the identity theorem may later be invoked. The argument has four stages: an integral representation of f valid for $\mathrm{Re}(s) > 1$; the splitting of that integral into a polar term and an error integral; the holomorphy of the error integral on $\mathrm{Re}(s) > 1-\delta$; and the identification of the pole and its residue.

Step 1: The integral representation $f(s) = s\int_1^\infty A(x)\, x^{-s-1}\, dx$ for $\mathrm{Re}(s) > 1$. Fix s with $\sigma = \mathrm{Re}(s) > 1$. For each $n \geq 1$ the elementary identity

$$n^{-s} = s\int_n^{\infty} x^{-s-1}\, dx$$

holds, since $\sigma > 0$ makes the integral convergent and the fundamental theorem of calculus evaluates it to $\big[-x^{-s}\big]_n^\infty = n^{-s}$. Multiplying by $a(n) \geq 0$ and summing over n,

$$f(s) = \sum_{n\geq 1} a(n)\, n^{-s} = \sum_{n\geq 1} a(n)\, s\int_n^{\infty} x^{-s-1}\, dx = s\int_1^{\infty}\Big(\sum_{n\leq x} a(n)\Big)x^{-s-1}\, dx = s\int_1^{\infty} A(x)\, x^{-s-1}\, dx$$

The interchange of summation and integration is justified by Tonelli's theorem ([14]): the summand-integrand $a(n)\,|x^{-s-1}| = a(n)\, x^{-\sigma-1}$ is non-negative, and the resulting iterated integral s replaced by σ gives $\sigma\int_1^\infty A(x)x^{-\sigma-1}\, dx$, which is finite because $A(x) = O(x)$ forces the integrand to be $O(x^{-\sigma})$ with $\sigma > 1$. The collapse of the double sum to $\sum_{n\leq x} a(n) = A(x)$ uses that the condition $n \leq x$ is exactly the range on which the inner integral $\int_n^\infty$ contributes to the outer integral $\int_1^\infty(\,\cdot\,)\, dx$ at the point x. This is partial summation in integral form, the same device that produced the representation of ζ in theorem 4.2.1.

Step 2: Splitting off the polar term. Substitute $A(x) = \kappa x + E(x)$ into the representation of Step 1. The two pieces integrate separately for $\mathrm{Re}(s) > 1$, where both converge absolutely. The polar piece is

$$s\int_1^\infty \kappa x\cdot x^{-s-1}\,dx = \kappa s\int_1^\infty x^{-s}\,dx = \kappa s\cdot\frac{1}{s-1} = \kappa + \frac{\kappa}{s-1}$$

the integral $\int_1^\infty x^{-s}\,dx = 1/(s-1)$ being valid for $\mathrm{Re}(s) > 1$, and the last equality using $s/(s-1) = 1 + 1/(s-1)$. The error piece is $s\int_1^\infty E(x)\,x^{-s-1}\,dx$. Combining,

$$f(s) = \kappa + \frac{\kappa}{s-1} + s\int_1^\infty E(x)\,x^{-s-1}\,dx \qquad (\mathrm{Re}(s) > 1) \tag{8.16}$$

which is (8.15) on the region $\mathrm{Re}(s) > 1$ where it was derived.

Step 3: The error integral is holomorphic on $\mathrm{Re}(s) > 1-\delta$. Let $J(s) = \int_1^\infty E(x)\,x^{-s-1}\,dx$ denote the error integral. By hypothesis there is a constant $C > 0$ with $|E(x)| \le C\,x^{1-\delta}$ for all $x \ge 1$, so the integrand is dominated by

$$\left|E(x)\,x^{-s-1}\right| = |E(x)|\,x^{-\sigma-1} \le C\,x^{1-\delta}\,x^{-\sigma-1} = C\,x^{-\sigma-\delta}$$

where $\sigma = \mathrm{Re}(s)$. Fix a compact set $\mathcal{K} \subset \{\mathrm{Re}(s) > 1-\delta\}$ and put $\sigma_0 = \min_{s\in\mathcal{K}}\mathrm{Re}(s)$, so that $\sigma_0 > 1-\delta$, equivalently $\sigma_0 + \delta > 1$. Then for every $s \in \mathcal{K}$ and every $x \ge 1$ the bound $|E(x)\,x^{-s-1}| \le C\,x^{-\sigma_0-\delta}$ holds, and

$$\int_1^\infty C\,x^{-\sigma_0-\delta}\,dx = \frac{C}{\sigma_0+\delta-1} < \infty$$

because $\sigma_0 + \delta - 1 > 0$. The dominating function $C\,x^{-\sigma_0-\delta}$ is independent of s and integrable on $[1,\infty)$, so $J(s)$ converges absolutely and uniformly for $s \in \mathcal{K}$. For each fixed $x \ge 1$ the integrand $E(x)\,x^{-s-1}$ is an entire function of s, and a parameter integral with holomorphic integrand and a locally uniform integrable bound is holomorphic in the parameter, by the standard theorem on holomorphy of integrals depending on a parameter ([13]). Since the half-plane $\mathrm{Re}(s) > 1-\delta$ is exhausted by such compact sets $\mathcal{K}$, the function J is holomorphic on all of $\mathrm{Re}(s) > 1-\delta$. Consequently $s\,J(s)$ is holomorphic there as well, being the product of the entire function s with J.

Step 4: Continuation, the pole, and the residue. The right-hand side of (8.15),

$$R(s) := \kappa + \frac{\kappa}{s-1} + s\,J(s)$$

is, by Step 3, meromorphic on $\mathrm{Re}(s) > 1-\delta$: the term κ is constant, $s\,J(s)$ is holomorphic, and $\kappa/(s-1)$ is rational with a single simple pole at $s = 1$. By (8.16) the functions R and f agree on $\mathrm{Re}(s) > 1$, and there f is holomorphic (the representation of Step 1 shows f extends holomorphically to $\mathrm{Re}(s) > 1$, where the series, the integral, and hence f all converge, refining the crude region $\mathrm{Re}(s) > 2$ of absolute convergence noted at the outset). The set $\{\mathrm{Re}(s) > 1\}$ is a nonempty open subset of the connected open set $\{\mathrm{Re}(s) > 1-\delta\}$ and has accumulation points there, so by the identity theorem ([13]) R is the unique meromorphic extension of f to $\mathrm{Re}(s) > 1-\delta$. Its only singularity is the simple pole of $\kappa/(s-1)$ at $s = 1$, the holomorphic term $s\,J(s)$ being unable to cancel it. For the residue, $s\,J(s)$ is holomorphic, hence finite, at $s = 1$, so

$$\mathrm{Res}_{s=1} f = \lim_{s\to1}(s-1)\,f(s) = \lim_{s\to1}\Big[(s-1)\kappa + \kappa + (s-1)\,s\,J(s)\Big] = \kappa$$

the first and third terms vanishing in the limit. Thus the pole is simple with residue κ, and (8.15) holds throughout $\mathrm{Re}(s) > 1-\delta$ with $s \neq 1$.

The lemma is stated for an abstract sequence so that the only field-specific input is the counting estimate. In particular, the non-negativity of the coefficients was used in Step 1 only to license Tonelli's theorem in the interchange; for a general (signed) Dirichlet series the same conclusion holds, by a routine absolute-convergence argument in place of Tonelli, but the present formulation is exactly adapted to the ideal-counting coefficients $a_K(n) \geq 0$. With the lemma in hand, the analytic class number formula is immediate.

Theorem 8.3.2: The Analytic Class Number Formula

Let K be a number field of degree $d = [K : \mathbb{Q}] = r_1 + 2r_2$, with r_1 real embeddings and r_2 conjugate pairs of complex embeddings. The Dedekind zeta function $\zeta_K(s)$ extends to a meromorphic function on the half-plane $\mathrm{Re}(s) > 1 - 1/d$ whose only singularity there is a simple pole at $s = 1$. Its residue at this pole is

$$\mathrm{Res}_{s=1}\zeta_K(s) = \lim_{s\to 1}(s-1)\,\zeta_K(s) = \frac{2^{r_1}(2\pi)^{r_2}\,h_K\,R_K}{w_K\,\sqrt{|d_K|}} = \kappa_K \tag{8.17}$$

where $h_K = |\mathrm{Cl}\,(K)|$ is the class number, $R_K = \mathrm{Reg}(K)$ the regulator, w_K the number of roots of unity in K, and d_K the discriminant.

***Proof*:**
The coefficients $a_K(n) = \#\{\mathfrak{a} : N\mathfrak{a} = n\}$ are non-negative integers, and by theorem 8.2.1 their summatory function satisfies $A_K(x) = \kappa_K\,x + O\big(x^{1-1/d}\big)$, with the constant κ_K given by the right-hand side of (8.17). This is precisely the hypothesis of lemma 8.3.1 with $a(n) = a_K(n)$, $\kappa = \kappa_K$, and $\delta = 1/d \in (0, 1]$. The lemma then asserts that $\zeta_K(s) = \sum_n a_K(n)n^{-s}$ extends meromorphically to $\mathrm{Re}(s) > 1 - 1/d$ with a single simple pole at $s = 1$ of residue κ_K, which is the claim.

The residue (8.17) is the central identity of the chapter, and it deserves to be read for what it asserts. The left-hand side is a purely analytic quantity, the strength of the pole of a meromorphic function built from the Euler product over prime ideals; by theorem 8.1.4 it encodes, factor by factor, the splitting behavior of every rational prime in K. The right-hand side is a single number assembled from the coarsest arithmetic invariants of K all at once: the class number h_K, which measures the failure of $\mathcal{O}_K$ to be a principal ideal domain; the regulator R_K, which measures the volume of the lattice of units; the order w_K of the group of roots of unity; and the discriminant d_K, which measures the ramification. That a single analytic limit should equal a product of these otherwise unrelated quantities is the substance of the formula, and it is the prototype of a pervasive phenomenon: the leading behavior of an L-function at a distinguished point is a regulator of arithmetic data. The conjectures of Birch and Swinnerton-Dyer, and more generally the Beilinson and Bloch–Kato conjectures, are the modern descendants of this single residue computation.

The factor κ_K specializes, at $K = \mathbb{Q}$, to the residue of the Riemann zeta function, and the formula is consistent with the special L-values computed analytically in chapter 7. The following examples verify each arithmetic instance.

Example 8.3.3: Four Residues from Their Invariants
Each computation evaluates the right-hand side of (8.17) from the invariants of K, taken from [2]. Throughout, the regulator of a field with unit rank $r_1 + r_2 - 1 = 0$ is the empty product $R_K = 1$,

and for a real quadratic field the regulator is $R_K = \log \varepsilon$, where $\varepsilon > 1$ is the fundamental unit.

The rational field $K = \mathbb{Q}$. Here $r_1 = 1$, $r_2 = 0$, $h_K = 1$, $R_K = 1$ (unit rank 0), $w_K = 2$ (the roots of unity are ± 1), and $d_K = 1$, so $|d_K| = 1$. The formula gives

$$\kappa_{\mathbb{Q}} = \frac{2^1(2\pi)^0 \cdot 1 \cdot 1}{2 \cdot \sqrt{1}} = \frac{2}{2} = 1$$

and $\zeta_{\mathbb{Q}} = \zeta$ is the Riemann zeta function, so this recovers the simple pole of residue 1 established in corollary 4.2.2. The class number formula contains the residue of ζ as its degenerate case.

The Gaussian field $K = \mathbb{Q}(i)$. The ring of integers is $\mathcal{O}_K = \mathbb{Z}[i]$, with $r_1 = 0$, $r_2 = 1$, and discriminant $d_K = -4$, so $|d_K| = 4$. The unit group is $\{\pm 1, \pm i\}$, the full group of fourth roots of unity, so $w_K = 4$ and the unit rank $r_1 + r_2 - 1 = 0$ gives $R_K = 1$. The class number is $h_K = 1$, since $\mathbb{Z}[i]$ is a principal ideal domain. Hence

$$\kappa_{\mathbb{Q}(i)} = \frac{2^0(2\pi)^1 \cdot 1 \cdot 1}{4 \cdot \sqrt{4}} = \frac{2\pi}{4 \cdot 2} = \frac{2\pi}{8} = \frac{\pi}{4}$$

This value admits an independent check. The Dedekind zeta function factors as $\zeta_{\mathbb{Q}(i)}(s) = \zeta(s)\, L(s, \chi_{-4})$, the product of the Riemann zeta function with the Dirichlet L-function of the nontrivial character χ_{-4} modulo 4 (the factorization is proved in section 8.4). Taking residues at $s = 1$, where $L(s, \chi_{-4})$ is holomorphic, gives

$$\operatorname{Res}_{s=1} \zeta_{\mathbb{Q}(i)}(s) = \Big(\operatorname{Res}_{s=1} \zeta(s)\Big) \cdot L(1, \chi_{-4}) = 1 \cdot L(1, \chi_{-4})$$

and $L(1, \chi_{-4}) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \pi/4$ is the Leibniz series evaluated in chapter 7. The two routes agree: $\kappa_{\mathbb{Q}(i)} = \pi/4$.

The field $K = \mathbb{Q}(\sqrt{-5})$. The ring of integers is $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, with $r_1 = 0$, $r_2 = 1$. Since $-5 \equiv 3 \pmod 4$, the discriminant is $d_K = 4 \cdot (-5) = -20$, so $|d_K| = 20$ and $\sqrt{|d_K|} = 2\sqrt{5}$. The only units are ± 1, so $w_K = 2$ and the unit rank 0 gives $R_K = 1$. The class number is $h_K = 2$: the ideal $(2, 1 + \sqrt{-5})$ is non-principal, and $\mathrm{Cl}\,(K) \cong \mathbb{Z}/2\mathbb{Z}$. Hence

$$\kappa_{\mathbb{Q}(\sqrt{-5})} = \frac{2^0(2\pi)^1 \cdot 2 \cdot 1}{2 \cdot 2\sqrt{5}} = \frac{4\pi}{4\sqrt{5}} = \frac{\pi}{\sqrt{5}}$$

This is the first field in the list with $h_K > 1$, and the class number $h_K = 2$ enters the residue directly as a factor of 2. The failure of unique factorization in $\mathbb{Z}[\sqrt{-5}]$, the historical origin of ideal theory, is visible analytically as a doubling of the residue relative to the principal case.

The real quadratic field $K = \mathbb{Q}(\sqrt{2})$. Here $r_1 = 2$, $r_2 = 0$, and the unit rank is $r_1 + r_2 - 1 = 1$, so the regulator is nontrivial and records the fundamental unit. Since $2 \equiv 2 \pmod 4$, the discriminant is $d_K = 4 \cdot 2 = 8$, so $|d_K| = 8$ and $\sqrt{|d_K|} = 2\sqrt{2}$. The roots of unity are ± 1, so $w_K = 2$, and the class number is $h_K = 1$. The fundamental unit is $\varepsilon = 1 + \sqrt{2}$, of norm $N_{K/\mathbb{Q}}(1 + \sqrt{2}) = 1 - 2 = -1$, so $R_K = \log(1 + \sqrt{2})$. Hence

$$\kappa_{\mathbb{Q}(\sqrt{2})} = \frac{2^2(2\pi)^0 \cdot 1 \cdot \log(1 + \sqrt{2})}{2 \cdot 2\sqrt{2}} = \frac{4 \log(1 + \sqrt{2})}{4\sqrt{2}} = \frac{\log(1 + \sqrt{2})}{\sqrt{2}}$$

In contrast to the three preceding fields, where the regulator was the empty product 1, here the residue is transcendental in a new way: it is a logarithm of an algebraic number, the fundamental unit. The regulator factor R_K is the contribution that distinguishes the real case, where there are infinitely many units, from the imaginary case, where there are only finitely many.

The two imaginary-quadratic computations follow a general pattern. The factor $2^{r_1}(2\pi)^{r_2}$ in the residue is, for an imaginary quadratic field, equal to 2π, and after division by $w_K\sqrt{|d_K|}$ it produces exactly a rational multiple of $\pi/\sqrt{|d_K|}$, namely the special value $L(1,\chi_{d_K})$ of the quadratic Dirichlet L-function attached to the field. This is the structural reason behind the closed forms tabulated in chapter 7: the value $L(1,\chi)$ of a primitive quadratic character was computed there by analytic means (Gauss sums and the Fourier expansion of the sawtooth function), and the class number formula now reproduces the same numbers from pure arithmetic, the class number, the units, and the discriminant of the associated quadratic field. The two derivations are independent, and their agreement is a theorem rather than a coincidence.

8.3.4 Remark: The Numerical Coincidence Made a Theorem The agreement observed for $\mathbb{Q}(i)$, that the arithmetically computed residue $\pi/4$ coincides with the analytically computed Leibniz value $L(1,\chi_{-4})$, is an instance of a general identity. For any imaginary quadratic field $K=\mathbb{Q}(\sqrt{D})$ of fundamental discriminant $D=d_K<0$, the residue formula (8.17) reads

$$\frac{2\pi\, h_K}{w_K\sqrt{|D|}} = \mathrm{Res}_{s=1}\zeta_K(s) = L(1,\chi_D)$$

where the last equality uses the factorization $\zeta_K(s)=\zeta(s)\,L(s,\chi_D)$ together with $\mathrm{Res}_{s=1}\zeta(s)=1$ from corollary 4.2.2. Equating the outer terms gives Dirichlet's class number formula in its classical form, $L(1,\chi_D)=2\pi h_K/(w_K\sqrt{|D|})$, which expresses an analytic L-value as a positive rational multiple of $\pi/\sqrt{|D|}$ with the class number as numerator. The positivity of h_K is, in this light, the structural reason $L(1,\chi_D)\neq 0$, an arithmetic counterpart to the analytic non-vanishing proved in theorem 7.4.2. The factorization $\zeta_K=\zeta\cdot L(\cdot\,,\chi_D)$ that underlies the whole bridge is established, for both signs of the discriminant and with the real-quadratic regulator in place of the $\pi/\sqrt{|D|}$ factor, in section 8.4; that section turns the numerical agreement recorded here into the general theorem.

8.4 Factoring ζ_K and the Algebraic Non-Vanishing of $L(1,\chi)$

The class number formula of section 8.3 extracts a single transcendental constant, the residue κ_K, from the pole of ζ_K at $s=1$, and reads off from it all the basic invariants of K at once. For a general number field that residue is the end of the story: ζ_K is an irreducible analytic object, not built from anything simpler. For the abelian fields, however, ζ_K is not irreducible at all. When $K/\mathbb{Q}$ is Galois with abelian Galois group, the Euler product of ζ_K regroups, prime by prime, into a product of Dirichlet L-functions, one factor for each character of the Galois group. This section carries out the regrouping in the two cases where it can be made completely explicit by hand: the quadratic fields, where ζ_K splits as $\zeta(s)L(s,\chi_D)$ for a single quadratic character, and the cyclotomic fields, where ζ_K splits as the product of $L(s,\chi)$ over all the Dirichlet characters of conductor dividing the cyclotomic level.

The factorization is not a formal coincidence. Reading it from left to right, it expresses the Dedekind zeta function, with its known pole, as a product of L-functions; reading it from right to left, it expresses each L-function as a ratio of zeta functions whose poles and residues are already in hand. The latter direction is the more consequential. For a quadratic field the identity $L(s,\chi_D)=\zeta_K(s)/\zeta(s)$ exhibits $L(s,\chi_D)$ as a quotient of two functions each with a simple pole of known residue at $s=1$, so the quotient is holomorphic and nonzero there, and its value is forced to be the positive number κ_K.

This is a second, purely algebraic, proof that $L(1, \chi) \neq 0$ for a real primitive character, the subtle case of theorem 7.4.2, where the conjugate-character argument of chapter 7 collapses and the proof had to fall back on Landau's theorem applied to the auxiliary product $\zeta(s)L(s, \chi)$. The geometry of numbers, packaged into the residue κ_K, replaces the positivity argument entirely, and along the way produces Dirichlet's original closed formulas for $L(1, \chi_D)$ in terms of the class number and the fundamental unit.

The quadratic case rests on the splitting law for primes in a quadratic field, which the algebraic prerequisites record in the form of the Kronecker symbol.

Theorem 8.4.1: Quadratic Factorization

Let $K = \mathbb{Q}(\sqrt{m})$ be a quadratic field with fundamental discriminant $D = d_K$, and let $\chi_D = \left(\frac{D}{\cdot}\right)$ be the associated Kronecker symbol, a real primitive Dirichlet character modulo $|D|$ ([2]). Then for $\mathrm{Re}(s) > 1$,

$$\zeta_K(s) = \zeta(s)\, L(s, \chi_D) \tag{8.18}$$

***Proof*:**
Both sides of (8.18) are given on $\mathrm{Re}(s) > 1$ by absolutely convergent Euler products: the left by the product of ζ_K over the prime ideals of $\mathcal{O}_K$ (theorem 8.1.4), the right by the product of ζ over the rational primes (corollary 2.3.2) times the product of $L(s, \chi_D)$ over the rational primes (theorem 7.2.2). Every nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ contains a unique rational prime p (namely the positive generator of $\mathfrak{p} \cap \mathbb{Z}$, a prime because $\mathcal{O}_K/\mathfrak{p}$ is a field of characteristic p), so the prime ideals of $\mathcal{O}_K$ are partitioned into the finite blocks $\{\mathfrak{p} : \mathfrak{p} \big| p\mathcal{O}_K\}$ indexed by the rational primes p. Because the Euler product of ζ_K converges absolutely on $\mathrm{Re}(s) > 1$, its factors may be reordered and regrouped freely, and grouping the prime ideals of $\mathcal{O}_K$ according to the rational prime p they lie above turns the product into a product over rational primes,

$$\zeta_K(s) = \prod_{p} \prod_{\mathfrak{p} \big| p} \left(1 - N\mathfrak{p}^{-s}\right)^{-1}$$

the inner product running over the finitely many prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$ dividing $p\mathcal{O}_K$. In a quadratic field there are at most two such primes, and the norm of each is $N\mathfrak{p} = p^f$ with f the residue degree, since $\mathcal{O}_K/\mathfrak{p}$ is the finite field of $N\mathfrak{p}$ elements containing $\mathbb{F}_p$ with degree f ([2]). The two products on the right of (8.18) likewise factor over p, with the p-th factor

$$\left(1 - p^{-s}\right)^{-1}\left(1 - \chi_D(p)p^{-s}\right)^{-1}$$

It therefore suffices to prove, for each rational prime p, the local identity

$$\prod_{\mathfrak{p} \big| p} \left(1 - N\mathfrak{p}^{-s}\right)^{-1} = \left(1 - p^{-s}\right)^{-1}\left(1 - \chi_D(p)p^{-s}\right)^{-1} \tag{8.19}$$

since taking the product of (8.19) over all p then yields (8.18), the rearrangement of the doubly indexed product being legitimate by absolute convergence. The factorization of $p\mathcal{O}_K$ in a quadratic field falls into exactly three types, governed by the value $\chi_D(p) \in \{+1, -1, 0\}$ through the quadratic splitting law (the Dedekind–Kummer theorem specialized to a quadratic field, equivalently the law of quadratic reciprocity, [2] and section 0.5). Each type is checked in turn.

Step 1: p splits, $\chi_D(p) = +1$. Here $p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$ with $\mathfrak{p}_1 \neq \mathfrak{p}_2$ two distinct primes, each of residue degree 1, so $N\mathfrak{p}_1 = N\mathfrak{p}_2 = p$. The left side of (8.19) is

$$\left(1 - p^{-s}\right)^{-1}\left(1 - p^{-s}\right)^{-1} = \left(1 - p^{-s}\right)^{-2}$$

The right side is $\left(1 - p^{-s}\right)^{-1}\left(1 - (+1)p^{-s}\right)^{-1} = \left(1 - p^{-s}\right)^{-2}$. The two agree.

Step 2: p is inert, $\chi_D(p) = -1$. Here $p\mathcal{O}_K = \mathfrak{p}$ is prime, of residue degree 2, so $N\mathfrak{p} = p^2$. The left side of (8.19) is the single factor

$$\left(1 - p^{-2s}\right)^{-1} = \left(1 - p^{-s}\right)^{-1}\left(1 + p^{-s}\right)^{-1}$$

using the factorization $1 - X^2 = (1 - X)(1 + X)$ with $X = p^{-s}$. The right side is $\left(1 - p^{-s}\right)^{-1}\big(1 - (-1)p^{-s}\big)^{-1} = \left(1 - p^{-s}\right)^{-1}\left(1 + p^{-s}\right)^{-1}$. The two agree.

Step 3: p ramifies, $\chi_D(p) = 0$. Ramification occurs precisely at the primes dividing the discriminant, $p \big| D$, which is exactly where $\chi_D(p) = 0$ (the Kronecker symbol vanishes on the primes dividing its modulus $|D|$). Here $p\mathcal{O}_K = \mathfrak{p}^2$ with $\mathfrak{p}$ of residue degree 1, so $N\mathfrak{p} = p$, and the left side of (8.19) is the single factor

$$\left(1 - p^{-s}\right)^{-1}$$

The right side is $\left(1 - p^{-s}\right)^{-1}\left(1 - 0 \cdot p^{-s}\right)^{-1} = \left(1 - p^{-s}\right)^{-1} \cdot 1 = \left(1 - p^{-s}\right)^{-1}$. The two agree.

In all three cases the local identity (8.19) holds, and these cases are exhaustive. Taking the product over all rational primes p gives (8.18) on $\mathrm{Re}(s) > 1$.

The three cases of the proof line up the three factorization types of a rational prime in K with the three values of the quadratic character: a split prime ($\chi_D(p) = +1$) contributes two factors of norm p, an inert prime ($\chi_D(p) = -1$) one factor of norm p^2, and a ramified prime ($\chi_D(p) = 0$) one factor of norm p, and in each case the resulting Euler factor of ζ_K is exactly the product of the Euler factor of ζ with the Euler factor of $L(s, \chi_D)$. The identity (8.18) is the analytic incarnation of the splitting law: the entire arithmetic of how rational primes decompose in K is encoded in the single equation $\zeta_K = \zeta \cdot L(\,\cdot\,, \chi_D)$, with the character χ_D reading off the decomposition type from the residue of p modulo $|D|$.

Comparing Dirichlet-series coefficients in (8.18) records the splitting law one more way, now in terms of the ideal-counting function of section 8.2. On $\mathrm{Re}(s) > 1$ the left side is $\sum_{n \geq 1} a_K(n) n^{-s}$ with $a_K(n) = \#\{\mathfrak{a} : N\mathfrak{a} = n\}$ the number of integral ideals of norm n (definition 8.1.1), while the right side is the Dirichlet product of $\zeta(s) = \sum_n n^{-s}$ with $L(s, \chi_D) = \sum_n \chi_D(n) n^{-s}$, whose n-th coefficient is the convolution $\sum_{d \mid n} \chi_D(d)$. Equating coefficients gives, for every $n \geq 1$,

$$a_K(n) = \sum_{d \mid n} \chi_D(d) \tag{8.20}$$

a closed formula for the number of ideals of norm n in a quadratic field as a divisor sum of the Kronecker symbol. This is the exact quadratic instance of the counting that section 8.2 performed by the geometry of numbers: the multiplicative function $n \mapsto a_K(n)$, which there was summed over $n \leq x$ to extract the residue κ_K, is here identified term by term with $\sum_{d|n} \chi_D(d)$, and its average size is governed by $L(1, \chi_D)$. In particular $a_K(p) = 1 + \chi_D(p)$ equals 2, 0, or 1 according as p splits, is inert, or ramifies, recovering the prime-by-prime splitting law from the coefficient identity.

The factorization is consistent with the pole structure already established for both sides, and the consistency is worth recording, since it is the mechanism behind the non-vanishing theorem to come. On $\mathrm{Re}(s) > 1$ the right side of (8.18) is the product of ζ, which continues to a function with a simple pole at $s = 1$ (theorem 4.2.1), and $L(s, \chi_D)$, which continues to a function holomorphic at $s = 1$ (theorem 7.2.4, χ_D being nonprincipal). The product thus has at $s = 1$ a pole of order at most one, and exactly one provided $L(1, \chi_D) \neq 0$. The left side, ζ_K, was shown in section 8.3 to have a simple pole at $s = 1$, of residue $\kappa_K \neq 0$. Equating the two descriptions, the simple pole of ζ_K must come from the simple pole of the ζ factor, and the holomorphic factor $L(s, \chi_D)$ cannot vanish at $s = 1$, for a zero there would cancel the pole and leave ζ_K holomorphic at $s = 1$, contradicting the pole of residue κ_K. This is exactly the non-vanishing argument, made quantitative in the next theorem.

The factorization, combined with the analytic facts already established for ζ and ζ_K, yields the non-vanishing of $L(1, \chi_D)$ as an immediate consequence, with no positivity argument at all.

Theorem 8.4.2: Algebraic Non-Vanishing of $L(1, \chi_D)$

For every fundamental discriminant D, the value $L(1, \chi_D)$ is finite, nonzero, and positive; in fact

$$L(1, \chi_D) = \kappa_K = \frac{2^{r_1} (2\pi)^{r_2} \, h_K R_K}{w_K \sqrt{|d_K|}} \tag{8.21}$$

where $K = \mathbb{Q}(\sqrt{m})$ is the quadratic field of discriminant $D = d_K$ and $\kappa_K = \mathrm{Res}_{s=1} \zeta_K(s)$.

***Proof*:**
By theorem 8.4.1, on the half-plane $\mathrm{Re}(s) > 1$ one has $\zeta_K(s) = \zeta(s) L(s, \chi_D)$, and since $\zeta(s) \neq 0$ there (corollary 4.1.2), the factorization may be solved for $L(s, \chi_D)$:

$$L(s, \chi_D) = \frac{\zeta_K(s)}{\zeta(s)} \qquad (\mathrm{Re}(s) > 1) \tag{8.22}$$

Each side of (8.22) continues meromorphically to a neighbourhood of $s = 1$: the numerator ζ_K by theorem 8.3.2, where it is shown to have a simple pole at $s = 1$ with residue $\mathrm{Res}_{s=1} \zeta_K = \kappa_K$, and the denominator ζ by theorem 4.2.1, with a simple pole at $s = 1$ of residue $\mathrm{Res}_{s=1} \zeta = 1$ (corollary 4.2.2). Write the Laurent expansions at $s = 1$,

$$\zeta_K(s) = \frac{\kappa_K}{s-1} + O(1), \qquad \zeta(s) = \frac{1}{s-1} + O(1)$$

Since χ_D is a nonprincipal Dirichlet character (it is primitive of conductor $|D| > 1$, hence nonprincipal), the L-function $L(s, \chi_D)$ is holomorphic on $\mathrm{Re}(s) > 0$ (theorem 7.2.4); in particular it is holomorphic at $s = 1$, so the quotient on the right of (8.22) must be holomorphic at $s = 1$ as well. Computing the limit of that quotient as $s \to 1$, the two simple poles cancel:

$$L(1, \chi_D) = \lim_{s \to 1} \frac{\zeta_K(s)}{\zeta(s)} = \lim_{s \to 1} \frac{\kappa_K/(s-1) + O(1)}{1/(s-1) + O(1)} = \lim_{s \to 1} \frac{\kappa_K + O(s-1)}{1 + O(s-1)} = \frac{\kappa_K}{1} = \kappa_K$$

where the second equality multiplies numerator and denominator by $s-1$. This is (8.21). The residue κ_K is strictly positive, being a product and quotient of the strictly positive quantities 2^{r_1}, $(2\pi)^{r_2}$, $h_K \geq 1$, $R_K > 0$, $w_K \geq 1$, and $\sqrt{|d_K|} > 0$, so $L(1,\chi_D) = \kappa_K > 0$. In particular $L(1,\chi_D) \neq 0$.

This is a second proof, independent of the one in chapter 7, that $L(1,\chi) \neq 0$ for a real primitive Dirichlet character χ. Every real primitive Dirichlet character is the Kronecker symbol χ_D of a unique fundamental discriminant D ([2]), so the present theorem 8.4.2 covers exactly the real-character case of theorem 7.4.2, which was the subtle case there: the case in which the conjugate-character argument of chapter 7 collapses and the proof had to fall back on Landau's theorem applied to $\zeta(s)L(s,\chi)$. The two proofs locate the non-vanishing in different places. The earlier proof derives it from the positivity of the coefficients of a Dirichlet series, an analytic input; the present one derives it from the strict positivity of the geometric constant κ_K, an input from the geometry of numbers, by exhibiting $L(1,\chi_D)$ as the ratio of the residue of ζ_K to the residue of ζ at $s=1$. The complex-character case of theorem 7.4.2 is not reached this way, and remains the province of the conjugate-pair argument; only the real characters arise as χ_D.

Specializing (8.21) to the two signatures of a quadratic field recovers Dirichlet's original class number formulas. The signature is dictated by the sign of D: a negative discriminant gives an imaginary quadratic field, with no real embeddings and one pair of complex embeddings, while a positive discriminant gives a real quadratic field, with two real embeddings and no complex ones.

Corollary 8.4.3: Imaginary Quadratic Class Number Formula

Let $D < 0$ be a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$ the corresponding imaginary quadratic field. Then

$$L(1,\chi_D) = \frac{2\pi\, h_K}{w_K\sqrt{|D|}} \tag{8.23}$$

where h_K is the class number and w_K the number of roots of unity in K.

***Proof*:**
An imaginary quadratic field has signature $r_1 = 0$, $r_2 = 1$, since the two embeddings $K \hookrightarrow \mathbb{C}$ are a single conjugate pair of complex embeddings. The unit rank is $r_1 + r_2 - 1 = 0$, so $\mathcal{O}_K^\times$ is finite (Dirichlet's unit theorem, [2]) and the regulator of a rank-zero unit lattice is $R_K = 1$ by the convention of section 8.3. Substituting $r_1 = 0$, $r_2 = 1$, $R_K = 1$, and $|d_K| = |D|$ into (8.21),

$$L(1,\chi_D) = \frac{2^0(2\pi)^1\, h_K \cdot 1}{w_K\sqrt{|D|}} = \frac{2\pi\, h_K}{w_K\sqrt{|D|}}$$

which is (8.23).

Corollary 8.4.4: Real Quadratic Class Number Formula

Let $D > 0$ be a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$ the corresponding real quadratic field. Then

$$L(1, \chi_D) = \frac{2\, h_K \log \varepsilon_K}{\sqrt{D}} \tag{8.24}$$

where h_K is the class number and $\varepsilon_K > 1$ is the fundamental unit of K.

***Proof*:**
A real quadratic field has signature $r_1 = 2$, $r_2 = 0$, since both embeddings $K \hookrightarrow \mathbb{R}$ are real. The unit rank is $r_1 + r_2 - 1 = 1$, so $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}$ with $\mu_K = \{\pm 1\}$ the only roots of unity in a real field, whence $w_K = 2$; the unit lattice has rank 1 and is generated modulo torsion by the fundamental unit $\varepsilon_K > 1$, so the regulator is the single entry $R_K = \log \varepsilon_K$ ([2] and section 8.3). Substituting $r_1 = 2$, $r_2 = 0$, $R_K = \log \varepsilon_K$, $w_K = 2$, and $|d_K| = D$ into (8.21),

$$L(1, \chi_D) = \frac{2^2 (2\pi)^0\, h_K \log \varepsilon_K}{2\sqrt{D}} = \frac{4\, h_K \log \varepsilon_K}{2\sqrt{D}} = \frac{2\, h_K \log \varepsilon_K}{\sqrt{D}}$$

which is (8.24).

The two formulas exhibit the dichotomy between imaginary and real quadratic fields in its sharpest form. For $D < 0$ the value $L(1, \chi_D)$ is a rational multiple of $\pi/\sqrt{|D|}$, the transcendence coming entirely from the archimedean factor 2π of the complex place, and the arithmetic content is the class number h_K (the root-of-unity count w_K equals 2 except for the two fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$, where it is 4 and 6). For $D > 0$ the value is instead a rational multiple of $\log \varepsilon_K/\sqrt{D}$, the transcendence now residing in the regulator $\log \varepsilon_K$, and the arithmetic content is the product $h_K \log \varepsilon_K$ of the class number with the regulator. The same residue κ_K produces both, the signature deciding which invariants survive.

The imaginary formula (8.23) matches the closed forms computed by hand in chapter 7 for the smallest moduli. Taking $D = -4$, the field is $K = \mathbb{Q}(i)$, with class number $h_K = 1$ and $w_K = 4$ roots of unity (the four powers of i), so

$$L(1, \chi_{-4}) = \frac{2\pi \cdot 1}{4 \cdot \sqrt{4}} = \frac{2\pi}{4 \cdot 2} = \frac{\pi}{4}$$

recovering the value of the Leibniz series $1 - \frac{1}{3} + \frac{1}{5} - \cdots$ found in chapter 7, the L-function of the unique nonprincipal character modulo 4. Taking $D = -20$, the field is $K = \mathbb{Q}(\sqrt{-5})$, the standard example of a field with non-trivial class group: here $h_K = 2$ and $w_K = 2$, and $|D| = 20$, so

$$L(1, \chi_{-20}) = \frac{2\pi \cdot 2}{2 \cdot \sqrt{20}} = \frac{2\pi}{\sqrt{20}} = \frac{2\pi}{2\sqrt{5}} = \frac{\pi}{\sqrt{5}}$$

The factor of 2 over the naive guess $\pi/\sqrt{20}$ is exactly the class number $h_K = 2$, the analytic manifestation of the failure of unique factorization in $\mathbb{Z}[\sqrt{-5}]$. Two further values exhibit the exceptional root-of-unity counts and the real case. For $D = -3$ the field is $K = \mathbb{Q}(\sqrt{-3})$, which contains the sixth roots of unity, so $w_K = 6$ and $h_K = 1$, giving

$$L(1, \chi_{-3}) = \frac{2\pi \cdot 1}{6\sqrt{3}} = \frac{\pi}{3\sqrt{3}}$$

the closed form found for the unique nonprincipal character modulo 3 in chapter 7. For $D = 5$ the field is the real quadratic $K = \mathbb{Q}(\sqrt{5})$, with $h_K = 1$ and fundamental unit $\varepsilon_K = (1+\sqrt{5})/2$ the golden ratio, so by (8.24)

$$L(1,\chi_5) = \frac{2 \cdot 1 \cdot \log\big((1+\sqrt{5})/2\big)}{\sqrt{5}}$$

again matching the closed form recorded for the quadratic character modulo 5 in chapter 7, where the value appeared as a logarithm of the fundamental unit with no explanation; the explanation is the real-quadratic class number formula. The class number formula thus turns a transcendental special value of an L-function into a count of ideal classes (and, for $D > 0$, a regulator), and conversely computes the special value once the class number and fundamental unit, both finite and algorithmically computable from D, are known.

The quadratic factorization is the degree-two case of a general principle: the Dedekind zeta function of an abelian extension of $\mathbb{Q}$ factors as a product of Dirichlet L-functions indexed by the characters of the Galois group. The cyclotomic fields realize the principle in its cleanest form, and they are exactly the prototype promised in chapter 7, where the product $\prod_\chi L(s,\chi)$ over the characters modulo m was studied as a stand-in for the Dedekind zeta function of $\mathbb{Q}(\zeta_m)$.

Proposition 8.4.5: Cyclotomic Factorization

Let $m \geq 1$ and let $K = \mathbb{Q}(\zeta_m)$ be the m-th cyclotomic field, with Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$. Then for $\mathrm{Re}(s) > 1$,

$$\zeta_K(s) = \prod_\chi L(s,\chi^*) \tag{8.25}$$

the product running over the Dirichlet characters χ modulo m (equivalently over the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$), where χ^* denotes the primitive character of conductor dividing m that induces χ.

***Proof*:**
Both sides of (8.25) are absolutely convergent Euler products on $\mathrm{Re}(s) > 1$, the left over the prime ideals of $\mathcal{O}_K$ (theorem 8.1.4) and the right over the rational primes for each of the $\varphi(m)$ characters (theorem 7.2.2). Grouping the prime ideals of $\mathcal{O}_K$ over the rational prime p below them, it suffices to prove for each rational prime p the local identity

$$\prod_{\mathfrak{p} | p} \big(1 - N\mathfrak{p}^{-s}\big)^{-1} = \prod_\chi \big(1 - \chi^*(p)p^{-s}\big)^{-1} \tag{8.26}$$

the product over χ ranging over the characters modulo m, and then take the product over all p. The proof of the unramified case is given in full; the ramified case is reduced to a precise citation.

Step 1: The splitting of an unramified prime. Let $p \nmid m$. Such a prime is unramified in $\mathbb{Q}(\zeta_m)$, the discriminant of $\mathbb{Q}(\zeta_m)$ being divisible only by the primes dividing m ([2]), so the ramification index is $e = 1$. The splitting is governed by the order of the Frobenius. Under the isomorphism $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$, which sends an automorphism σ to the residue a with $\sigma(\zeta_m) = \zeta_m^a$, the Frobenius automorphism at p (well defined up to conjugacy, but the group is abelian, so it is a single element) corresponds to the residue of p itself, since Frob_p acts as the p-power map modulo any prime above p. The decomposition group at p is the cyclic group generated by Frob_p, of order equal to the order of p in $(\mathbb{Z}/m\mathbb{Z})^\times$, and this order is the common residue degree of the primes above p. Thus the residue degree of every prime $\mathfrak{p}$ above p is

$$f = \operatorname{ord}_m(p)$$

the multiplicative order of p in $(\mathbb{Z}/m\mathbb{Z})^\times$, and the number of primes above p is

$$g = \frac{\varphi(m)}{f}$$

since $efg = \varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$ with ramification index $e = 1$ ([2]). Each of the g primes $\mathfrak{p}$ above p has $N\mathfrak{p} = p^f$, so the left side of (8.26) is

$$\prod_{\mathfrak{p}|p} \left(1 - N\mathfrak{p}^{-s}\right)^{-1} = \left(1 - p^{-fs}\right)^{-g} \tag{8.27}$$

a single factor raised to the power g.

Step 2: The character side, via the values $\chi(p)$. For $p \nmid m$ every character χ modulo m has $\chi^*(p) = \chi(p)$, since the inducing character χ^* and χ agree on the integers coprime to m and p is one of them; the conductor bookkeeping affects only the primes dividing m. The right side of (8.26) is therefore $\prod_\chi (1 - \chi(p)p^{-s})^{-1}$, and the task is to evaluate the polynomial

$$\prod_\chi \left(1 - \chi(p)X\right), \qquad X = p^{-s}$$

Consider the evaluation homomorphism

$$\mathrm{ev}_p \colon \widehat{(\mathbb{Z}/m\mathbb{Z})^\times} \longrightarrow \mathbb{C}^\times, \qquad \chi \longmapsto \chi(p)$$

from the character group to the multiplicative group of $\mathbb{C}$. Its image is a finite subgroup of $\mathbb{C}^\times$, hence a group of roots of unity; the image is exactly the group μ_f of f-th roots of unity, because $\chi(p)$ is an f-th root of unity for every χ (as $\chi(p)^f = \chi(p^f) = \chi(1) = 1$, using $p^f \equiv 1 \pmod m$ from $f = \operatorname{ord}_m(p)$), and conversely every f-th root of unity is attained (the cyclic group $\langle p \rangle \subseteq (\mathbb{Z}/m\mathbb{Z})^\times$ generated by p has order f, and the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$ restrict onto all f characters of $\langle p \rangle$, which take p to each f-th root of unity). The kernel of ev_p has order

$$|\ker \mathrm{ev}_p| = \frac{|\widehat{(\mathbb{Z}/m\mathbb{Z})^\times}|}{|\mu_f|} = \frac{\varphi(m)}{f} = g$$

using $|\widehat{(\mathbb{Z}/m\mathbb{Z})^\times}| = \varphi(m)$ (theorem 7.1.2). By the orbit count for the surjection ev_p, every fibre has the same size g, so as χ ranges over all $\varphi(m)$ characters the value $\chi(p)$ runs through each f-th root of unity exactly g times.

Step 3: The Euler-factor identity for $p \nmid m$. Grouping the $\varphi(m)$ linear factors $1 - \chi(p)X$ by the common value $\chi(p) = \omega$, each f-th root of unity ω occurring g times,

$$\prod_\chi (1 - \chi(p)X) = \prod_{\omega \in \mu_f} (1 - \omega X)^g = \Big(\prod_{\omega \in \mu_f} (1 - \omega X)\Big)^g = (1 - X^f)^g$$

the last equality being the factorization of $1 - X^f = \prod_{\omega\in\mu_f}(1-\omega X)$ into linear factors over the f-th roots of unity (the roots of $1 - X^f$ are exactly the inverses ω^{-1}, which range over μ_f as ω does). Taking reciprocals and substituting $X = p^{-s}$,

$$\prod_{\chi}\left(1-\chi(p)p^{-s}\right)^{-1} = \left(1-p^{-fs}\right)^{-g} \tag{8.28}$$

Comparing (8.28) with (8.27), both sides of the local identity (8.26) equal $\left(1-p^{-fs}\right)^{-g}$, so (8.26) holds for every unramified prime $p \nmid m$.

Step 4: The ramified primes and the conductor bookkeeping. For the finitely many primes $p|m$, the prime p ramifies in $\mathbb{Q}(\zeta_m)$, the equality $\chi^*(p) = \chi(p)$ can fail (an imprimitive χ has $\chi(p) = 0$ while its primitive inducing character χ^*, of conductor prime to p, may have $\chi^*(p) \neq 0$), and the matching of the local Euler factor of ζ_K at p with the product $\prod_\chi(1-\chi^*(p)p^{-s})^{-1}$ requires tracking the conductors of the inducing characters together with the ramification of p. This local computation, and the verification that the primitive characters χ^* supplying the factors of (8.25) are exactly the primitive Dirichlet characters of conductor dividing m, are carried out in full in [23, Ch. 3]; the same factorization is established from the conductor-discriminant formula in [15, Ch. VII, (10.16)]. Granting the ramified local identity from these sources and combining it with the unramified identity (8.26) proved in Steps 1–3, the product over all rational primes p yields (8.25) on $\mathrm{Re}(s) > 1$.

A small case makes the unramified mechanism concrete. Take $m = 5$, so $K = \mathbb{Q}(\zeta_5)$ has degree $\varphi(5) = 4$ and Galois group $(\mathbb{Z}/5\mathbb{Z})^\times$, cyclic of order 4 generated by 2; the four Dirichlet characters modulo 5 are $\chi_0, \psi, \psi^2, \psi^3$ in the notation of chapter 7, all primitive except the principal χ_0 (which has conductor 1), and (8.25) reads $\zeta_K(s) = \zeta(s)L(s,\psi)L(s,\psi^2)L(s,\psi^3)$. Consider three unramified primes. For $p = 11 \equiv 1 \pmod 5$ the order is $f = \mathrm{ord}_5(11) = 1$, so $g = 4$ and 11 splits into four primes of norm 11; on the character side every $\chi(11) = \chi(1) = 1$, so $\prod_\chi(1-\chi(11)X) = (1-X)^4$, matching $(1-11^{-s})^{-4}$. For $p = 2$, a generator, the order is $f = \mathrm{ord}_5(2) = 4$, so $g = 1$ and 2 is inert with a single prime of norm $2^4 = 16$; the values $\chi(2)$ run once through each fourth root of unity $\{1, i, -1, -i\}$, so $\prod_\chi(1-\chi(2)X) = \prod_{\omega^4=1}(1-\omega X) = 1 - X^4$, matching $(1-2^{-4s})^{-1}$. For $p = 19 \equiv 4 \pmod 5$ the order is $f = \mathrm{ord}_5(19) = \mathrm{ord}_5(4) = 2$ (since $4^2 = 16 \equiv 1$), so $g = 2$ and 19 splits into two primes of norm 19^2; the values $\chi(19)$ run twice through the square roots of unity $\{1, -1\}$, so $\prod_\chi(1-\chi(19)X) = ((1-X)(1+X))^2 = (1-X^2)^2$, matching $(1-19^{-2s})^{-2}$. In each case $efg = 4 = [\mathbb{Q}(\zeta_5):\mathbb{Q}]$ and the local factors agree, as Steps 1–3 require.

The decomposition is again consistent with the pole at $s = 1$, and now the consistency runs the other way, from the non-vanishing to the residue. Among the $\varphi(m)$ factors of (8.25) exactly one comes from the trivial character: the primitive character inducing χ_0 modulo m is the trivial character modulo 1, whose L-function is ζ itself, contributing the lone simple pole at $s = 1$. Every other factor $L(s,\chi^*)$ is attached to a nontrivial primitive character, hence is holomorphic on $\mathrm{Re}(s) > 0$ (theorem 7.2.4) and, by the non-vanishing $L(1,\chi^*) \neq 0$ (theorem 7.4.2), is also nonzero at $s = 1$. The product (8.25) therefore has a simple pole at $s = 1$, and taking residues,

$$\kappa_K = \mathrm{Res}_{s=1}\zeta_K(s) = \Big(\mathrm{Res}_{s=1}\zeta(s)\Big)\prod_{\chi^*\neq 1} L(1,\chi^*) = \prod_{\chi^*\neq 1} L(1,\chi^*)$$

the residue of ζ being 1 (corollary 4.2.2). For the cyclotomic field the class number formula thus equates the geometric residue $\kappa_K = 2^{r_1}(2\pi)^{r_2}h_K R_K/(w_K\sqrt{|d_K|})$ with a product of special values

$\prod_{\chi^*\neq 1} L(1,\chi^*)$ of Dirichlet L-functions; this is the analytic identity from which Dirichlet's class number formulas for cyclotomic fields, and the decomposition of the regulator into a determinant of logarithms of cyclotomic units, are derived in [23, Ch. 3 and Ch. 4].

The cyclotomic factorization (8.25) fulfills the forward reference of chapter 7, where the product $\prod_{\chi \bmod m} L(s,\chi)$ was introduced as the cyclotomic prototype and used, through the non-negativity of its coefficients, to drive the non-vanishing of $L(1,\chi)$. That product is, up to the conductor bookkeeping for the imprimitive characters at the ramified primes, the Dedekind zeta function $\zeta_{\mathbb{Q}(\zeta_m)}$, and the unramified Euler-factor identity proved above, $\prod_\chi(1-\chi(p)X) = (1-X^f)^g$ with $f = \operatorname{ord}_m(p)$ and $g = \varphi(m)/f$, is precisely the identity that gave the prototype its Euler product $\prod_{p\nmid m}(1-p^{-fs})^{-g}$ in chapter 7. The factorization places that prototype in its proper context: it is the Galois-theoretic decomposition of an abelian zeta function, the first instance of a pattern that, for a general abelian extension $K/\mathbb{Q}$, writes ζ_K as the product of the Dirichlet L-functions attached to the characters of $\mathrm{Gal}(K/\mathbb{Q})$ (viewed as Dirichlet characters via the Artin reciprocity map of [12]), and that, for a nonabelian extension, becomes the factorization of ζ_K into Artin L-functions. In every such factorization the pole of ζ_K at $s=1$, with its residue κ_K, is carried by the single trivial character, whose L-function is ζ, while every nontrivial factor is holomorphic and nonzero at $s=1$; the non-vanishing of those factors is what confines the pole to one place and, read as the residue identity above, expresses the arithmetic constant κ_K as a product of the boundary values $L(1,\chi)$. This is the structural reason the non-vanishing of $L(1,\chi)$ and the analytic class number formula are two faces of the same fact, and it is the entry point, pursued for nonabelian G through the Chebotarev density theorem and Artin L-functions, to the general analytic theory of L-functions of number fields.

8.5 The Functional Equation for $\zeta_K(s)$

The analytic class number formula of section 8.3 extracted the residue of ζ_K at its single pole $s=1$; it left untouched the behavior of ζ_K to the left of the line $\mathrm{Re}(s)=1$, where the defining series and Euler product of section 8.1 no longer converge. For $K=\mathbb{Q}$ the corresponding gap was closed in chapter 6 by the functional equation, which continued ζ to the whole plane and exhibited the reflection $s \leftrightarrow 1-s$ as the governing symmetry. The same symmetry holds for every number field. Completing ζ_K by the archimedean factors attached to the real and complex places of K, and by a power of $|d_K|$ playing the role of the conductor, produces a function Λ_K invariant under $s \mapsto 1-s$. This is *Hecke's functional equation*, the subject of the present section.

The general theorem is stated and its proof, a lattice-theoretic generalization of Riemann's theta argument, is deferred to the standard references by a precise citation. For abelian K the functional equation is recovered in full from the material already developed: the factorization $\zeta_K = \prod_\chi L(s,\chi)$ of section 8.4 reduces it to the functional equations of the Dirichlet L-functions established in chapter 7, and the reduction turns the bookkeeping of archimedean factors and conductors into two clean numerical matchings: the Legendre duplication formula pairs the gamma factors, and the conductor-discriminant formula multiplies the conductors to $|d_K|$.

Definition 8.5.1: The Completed Dedekind Zeta Function

Let K be a number field of degree $d = r_1 + 2r_2$, with r_1 real places and r_2 complex places, discriminant d_K, and Dedekind zeta function ζ_K of definition 8.1.1. Write the *archimedean factors*

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right), \qquad \Gamma_{\mathbb{C}}(s) = 2\,(2\pi)^{-s}\,\Gamma(s)$$

with Γ the gamma function of definition 6.1.1. The *completed Dedekind zeta function* of K is

$$\Lambda_K(s) = |d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\,\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s) \tag{8.29}$$

defined initially on the half-plane $\mathrm{Re}(s) > 1$, where $\zeta_K(s)$ is given by its absolutely convergent Dirichlet series and each archimedean factor is holomorphic.

The factor $\Gamma_{\mathbb{R}}(s)$ is attached to each real place of K and is identical to the local factor at the real place of $\mathbb{Q}$ from definition 6.3.1; the factor $\Gamma_{\mathbb{C}}(s)$ is attached to each complex place, that is, to each conjugate pair of complex embeddings, and is the local factor at a complex place. The two are linked by the Legendre duplication formula: as verified in (8.35) below,

$$\Gamma_{\mathbb{R}}(s)\,\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s) \tag{8.30}$$

so that a complex place behaves, archimedean-factor-wise, as one real place at s together with one real place at $s+1$. The power $|d_K|^{s/2}$ is the conductor-type factor: it generalizes the trivial conductor 1 of $\mathbb{Q}$, whose discriminant is $d_{\mathbb{Q}} = 1$, and its presence is what makes the completed function symmetric rather than only meromorphic.

8.5.2 Remark: Reduction to the Riemann case For $K = \mathbb{Q}$ one has $r_1 = 1$, $r_2 = 0$, $d_K = 1$, and $\zeta_K = \zeta$ (definition 8.1.1). The definition (8.29) then reads

$$\Lambda_{\mathbb{Q}}(s) = 1^{s/2}\,\Gamma_{\mathbb{R}}(s)^1\,\Gamma_{\mathbb{C}}(s)^0\,\zeta(s) = \pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\zeta(s) = \Lambda(s)$$

the completed zeta function of definition 6.3.1. Thus Λ_K is the number-field generalization of Λ, with the single real factor of $\mathbb{Q}$ replaced by r_1 real and r_2 complex factors and the trivial conductor replaced by $|d_K|^{1/2}$.

The functional equation for Λ_K is the following theorem of Hecke. Its proof generalizes the four-stage Riemann argument of theorem 6.3.3 from the lattice $\mathbb{Z} \subset \mathbb{R}$ to the Minkowski lattice $\iota(\mathcal{O}_K) \subset \mathbb{R}^d$, and is given in full in the references cited; the statement is recorded here, and the abelian case is proved independently in theorem 8.5.5.

Theorem 8.5.3: Hecke's Functional Equation

Let K be a number field and Λ_K its completed Dedekind zeta function (definition 8.5.1). Then Λ_K extends to a meromorphic function on all of $\mathbb{C}$, holomorphic everywhere except for simple poles at $s = 0$ and $s = 1$, and satisfies the functional equation

$$\Lambda_K(s) = \Lambda_K(1 - s) \tag{8.31}$$

for all $s \in \mathbb{C}$. Equivalently, ζ_K continues to a meromorphic function on $\mathbb{C}$, holomorphic except for a simple pole at $s = 1$, with $\zeta_K(s)$ recovered from $\Lambda_K(s)$ by dividing out the (nowhere-vanishing) factor $|d_K|^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}$.

***Proof*:**
The full proof is Hecke's, generalizing the theta-and-Mellin argument of theorem 6.3.3. One forms the theta series $\Theta(t) = \sum_{x\in\iota(\mathcal{O}_K)} e^{-\pi t\langle x,x\rangle}$ of the Minkowski lattice $\iota(\mathcal{O}_K) \subset \mathbb{R}^d$ (more precisely, a sum over the lattice translates within each ideal class, weighted to build ζ_K), obtains its transformation law under $t \mapsto 1/t$ from the Poisson summation formula (theorem 6.2.1) applied over the lattice, the covolume $\mathrm{vol}(\mathbb{R}^d/\iota(\mathcal{O}_K)) = 2^{-r_2}\sqrt{|d_K|}$ from [2] entering as the multiplicative constant in the dual-lattice term, and then takes a Mellin transform in t to recover $\Lambda_K(s)$. The self-duality of the theta transformation law produces the symmetry $s \leftrightarrow 1 - s$, and the constant term of the theta series produces the two simple poles, exactly as the constant term of ψ produced the poles of Λ in theorem 6.3.3. The complete argument, including the unit-group integration that handles the $r_1 + r_2 - 1$ regulator directions of [2], is carried out in [15, Ch. VII, §5–§6] and in [10, Ch. XIII–XIV]. The idelic reformulation, in which the functional equation becomes the Poisson summation formula on the adeles and the archimedean factors become local zeta integrals (Tate's thesis), is deferred to [12].

The completed Dedekind zeta function is symmetric with no phase: the constant relating $\Lambda_K(s)$ and $\Lambda_K(1 - s)$ is $+1$. In the language of theorem 7.3.4, the *root number* of ζ_K is $+1$, reflecting that ζ_K is self-dual (the trivial Hecke character is its own conjugate). This is the number-field counterpart of the bare reflection $\Lambda(s) = \Lambda(1 - s)$ for $\mathbb{Q}$, in contrast to the twisted reflection $\Lambda(s, \chi) = \epsilon(\chi)\Lambda(1 - s, \bar{\chi})$ carrying a unit-modulus phase $\epsilon(\chi)$ for a nonprincipal Dirichlet character.

The conductor factor $|d_K|^{s/2}$ admits, for abelian K, an arithmetic decomposition matching the factorization of ζ_K into Dirichlet L-functions. This is the conductor-discriminant formula, which the next theorem records and verifies on the two families of section 8.4, deferring the general proof.

Theorem 8.5.4: Conductor-Discriminant Formula

Let $K/\mathbb{Q}$ be a finite abelian extension, and let X be the associated group of primitive Dirichlet characters, so that

$$\zeta_K(s) = \prod_{\chi \in X} L(s, \chi) \tag{8.32}$$

as in theorem 8.4.1 (quadratic case) and proposition 8.4.5 (cyclotomic case). Then the absolute discriminant of K is the product of the conductors of these characters:

$$|d_K| = \prod_{\chi \in X} \mathfrak{f}_\chi \tag{8.33}$$

where $\mathfrak{f}_\chi$ is the conductor of χ (definition 7.1.5).

***Proof*:**
The identity (8.33) is verified below on the quadratic and cyclotomic families, which are the cases used in this chapter. The general statement, valid for every finite abelian $K/\mathbb{Q}$, is a consequence of the behavior of the discriminant under the conductor-ramification correspondence of class field theory; the complete proof is [15, Ch. VII, (11.9)], and the idelic account is in [12].

Verification, quadratic case. Let $K = \mathbb{Q}(\sqrt{D})$ for a fundamental discriminant D, so $d_K = D$ and $|d_K| = |D|$. By theorem 8.4.1 the character group is $X = \{\chi_0, \chi_D\}$, where χ_0 is the principal character (conductor $\mathfrak{f}_{\chi_0} = 1$, induced from level 1, by definition 7.1.5) and χ_D is the Kronecker symbol of D, a real primitive character modulo $|D|$ (so $\mathfrak{f}_{\chi_D} = |D|$, primitivity being recorded in section 8.4 from [2]). The product of conductors is

$$\prod_{\chi \in X} \mathfrak{f}_\chi = \mathfrak{f}_{\chi_0}\, \mathfrak{f}_{\chi_D} = 1 \cdot |D| = |D| = |d_K|$$

confirming (8.33).

Verification, cyclotomic case. Let $K = \mathbb{Q}(\zeta_p)$ for an odd prime p, of degree $\varphi(p) = p - 1$. By proposition 8.4.5 the character group X consists of the $p-1$ Dirichlet characters modulo p. Of these, the principal character χ_0 has conductor 1. Each of the remaining $p-2$ nonprincipal characters χ modulo p has conductor p: the only divisors of the prime p are 1 and p, the sole character modulo 1 is principal and cannot induce a nonprincipal χ, so no nonprincipal character modulo p is induced from a proper divisor, hence $\mathfrak{f}_\chi = p$ for each. The product of conductors is therefore

$$\prod_{\chi \in X} \mathfrak{f}_\chi = 1 \cdot \underbrace{p \cdots p}_{p-2 \text{ factors}} = p^{p-2}$$

On the other side, the discriminant of $\mathbb{Q}(\zeta_p)$ is $d_K = (-1)^{(p-1)/2}\, p^{p-2}$, so $|d_K| = p^{p-2}$ (the cyclotomic discriminant from [2]; see also [23, Ch. 2]). The two agree, confirming (8.33) in the cyclotomic case. For instance $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ has $|d_K| = 3^1 = 3$, and $\mathbb{Q}(\zeta_5)$ has $|d_K| = 5^3 = 125$.

For abelian K the functional equation of theorem 8.5.3 need not invoke Hecke's general theorem. The factorization (8.32) expresses ζ_K through Dirichlet L-functions, each of which already carries the functional equation of theorem 7.3.4, and assembling those reflections reconstitutes the reflection for Λ_K. The assembly rests on three matchings, which the next theorem makes precise and verifies: the gamma factors pair by the duplication formula (8.30), the character parities count the real and complex places, and the conductors multiply to $|d_K|$ by theorem 8.5.4.

Theorem 8.5.5: Abelian Consistency of the Functional Equation

Let $K/\mathbb{Q}$ be a finite abelian extension with associated group X of primitive Dirichlet characters, so that $\zeta_K(s) = \prod_{\chi \in X} L(s, \chi)$. Then the functional equation

$$\Lambda_K(s) = \Lambda_K(1-s)$$

holds, and follows from the functional equations of the Dirichlet L-functions $L(s, \chi)$, $\chi \in X$ (theorem 7.3.4), together with the conductor-discriminant formula (theorem 8.5.4), without recourse to Hecke's general theorem.

***Proof*:**
The proof matches the completed object Λ_K against the product of the completed L-functions $\Lambda(s, \chi)$ of definition 7.3.3, then transports the reflection of each $\Lambda(s, \chi)$ through that product. The argument runs in four steps: the gamma-factor matching by duplication; the place-count matching of parities to r_1 and r_2; the conductor matching by theorem 8.5.4; and the assembly of the reflection with root number $+1$.

Step 1: The duplication identity. The completed L-function of a primitive character χ of conductor $m = \mathfrak{f}_\chi$ and parity $a \in \{0, 1\}$ is, by definition 7.3.3,

$$\Lambda(s, \chi) = \left(\frac{m}{\pi}\right)^{(s+a)/2} \Gamma\left(\frac{s+a}{2}\right) L(s, \chi) \tag{8.34}$$

For an even character ($a = 0$) the archimedean part of (8.34) is $(m/\pi)^{s/2}\Gamma(s/2) = m^{s/2}\,\Gamma_{\mathbb{R}}(s)$, and for an odd character ($a = 1$) it is $(m/\pi)^{(s+1)/2}\Gamma((s+1)/2) = m^{(s+1)/2}\,\Gamma_{\mathbb{R}}(s+1)$, in the notation of definition 8.5.1. The two real gamma factors combine into a complex gamma factor through the Legendre duplication formula (6.6). Indeed, applying (6.6) at argument $s/2$ gives $\Gamma(s/2)\Gamma((s+1)/2) = 2^{1-s}\sqrt{\pi}\,\Gamma(s)$, whence

$$\Gamma_{\mathbb{R}}(s)\,\Gamma_{\mathbb{R}}(s+1) = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\,\pi^{-(s+1)/2}\Gamma\left(\frac{s+1}{2}\right) = \pi^{-s-1/2}\cdot 2^{1-s}\sqrt{\pi}\,\Gamma(s) = 2^{1-s}\pi^{-s}\Gamma(s) = 2\,(2\pi)^{-s}\Gamma(s) = \Gamma_{\mathbb{C}}(s) \tag{8.35}$$

which is the identity (8.30): one even plus one odd real gamma factor assemble into exactly one complex gamma factor.

Step 2: Parities count the archimedean places. The group X is closed under complex conjugation $\chi \mapsto \bar{\chi}$, since $K/\mathbb{Q}$ is Galois and X is the full group of primitive Dirichlet characters cut out by K, isomorphic to $\mathrm{Gal}(K/\mathbb{Q})$ by the abelian reciprocity underlying theorem 8.4.1 and proposition 8.4.5; conjugation preserves the conductor, and $\bar{\chi}(-1) = \overline{\chi(-1)} = \chi(-1)$ (the value lies in $\{\pm 1\}$), so it preserves parity as well. The count of even and odd characters is fixed by a structural dichotomy special to abelian fields. Complex conjugation is an element σ_∞ of the abelian group $\mathrm{Gal}(K/\mathbb{Q})$, and it corresponds, under the identification of X with $\mathrm{Gal}(K/\mathbb{Q})$, to the residue class of -1, so that $\chi(\sigma_\infty) = \chi(-1)$ for every $\chi \in X$. Because σ_∞ is central (the group is abelian), it acts in the same way on every archimedean embedding of K: either $\sigma_\infty = 1$, and K is totally real, with $r_2 = 0$ and $r_1 = [K : \mathbb{Q}]$; or σ_∞ has order 2, and K is totally imaginary, with $r_1 = 0$ and $[K : \mathbb{Q}] = 2r_2$. (No abelian K has both $r_1 > 0$ and $r_2 > 0$, since a central involution cannot fix one embedding while moving another.) The even characters are those trivial on σ_∞, hence the characters of the quotient $\mathrm{Gal}(K/\mathbb{Q})/\langle\sigma_\infty\rangle = \mathrm{Gal}(K^+/\mathbb{Q})$, where K^+ is the maximal totally real subfield (the fixed field of σ_∞); thus the number of even characters is $[K^+ : \mathbb{Q}]$. In the totally real case $K^+ = K$, giving $[K^+ : \mathbb{Q}] = r_1 = r_1 + r_2$ even characters and no odd characters; in the totally imaginary case $[K^+ : \mathbb{Q}] = [K : \mathbb{Q}]/2 = r_2 = r_1 + r_2$ even characters, the other r_2 characters being odd. Either way,

$$\#\{\chi \in X : \chi(-1) = +1\} = r_1 + r_2, \qquad \#\{\chi \in X : \chi(-1) = -1\} = r_2 \tag{8.36}$$

the even characters numbering $r_1 + r_2$ and the odd characters numbering r_2; this is the place decomposition recorded for abelian K in [15, Ch. VII, §5]. The two cases of the chapter make (8.36) explicit: for real quadratic $K = \mathbb{Q}(\sqrt{D})$, $D > 0$, both χ_0 and χ_D are even (as $\chi_D(-1) = +1$ for $D > 0$), giving $r_1 + r_2 = 2$ even and $r_2 = 0$ odd, consistent with $r_1 = 2$, $r_2 = 0$; for imaginary quadratic $K = \mathbb{Q}(\sqrt{D})$, $D < 0$, χ_0 is even and χ_D is odd (as $\chi_D(-1) = -1$ for $D < 0$), giving $r_1 + r_2 = 1$ even and $r_2 = 1$ odd, consistent with $r_1 = 0$, $r_2 = 1$.

Step 3: The product of completed L-functions. Form the product of (8.34) over all $\chi \in X$, separating its archimedean part from the L-functions. Using $(m/\pi)^{s/2}\Gamma(s/2) = m^{s/2}\Gamma_\mathbb{R}(s)$ for an even character and $(m/\pi)^{(s+1)/2}\Gamma((s+1)/2) = m^{(s+1)/2}\Gamma_\mathbb{R}(s+1)$ for an odd one (Step 1), and writing $m = \mathfrak{f}_\chi$, the archimedean parts multiply to

$$\prod_{\chi\in X}\left(\frac{\mathfrak{f}_\chi}{\pi}\right)^{(s+a_\chi)/2}\Gamma\!\left(\frac{s+a_\chi}{2}\right) = \left(\prod_{\chi \text{ even}} \mathfrak{f}_\chi^{s/2}\right)\left(\prod_{\chi \text{ odd}} \mathfrak{f}_\chi^{(s+1)/2}\right)\Gamma_\mathbb{R}(s)^{\#\{\text{even}\}}\,\Gamma_\mathbb{R}(s+1)^{\#\{\text{odd}\}} \tag{8.37}$$

By (8.36) there are $r_1 + r_2$ even characters and r_2 odd characters, so the gamma part of (8.37) is $\Gamma_\mathbb{R}(s)^{r_1+r_2}\,\Gamma_\mathbb{R}(s+1)^{r_2}$. Splitting off r_2 of the $\Gamma_\mathbb{R}(s)$ factors and pairing each with a $\Gamma_\mathbb{R}(s+1)$ through the duplication identity (8.35) converts r_2 of the products $\Gamma_\mathbb{R}(s)\Gamma_\mathbb{R}(s+1)$ into $\Gamma_\mathbb{C}(s)$, leaving r_1 copies of $\Gamma_\mathbb{R}(s)$:

$$\Gamma_\mathbb{R}(s)^{r_1+r_2}\,\Gamma_\mathbb{R}(s+1)^{r_2} = \Gamma_\mathbb{R}(s)^{r_1}\left(\Gamma_\mathbb{R}(s)\,\Gamma_\mathbb{R}(s+1)\right)^{r_2} = \Gamma_\mathbb{R}(s)^{r_1}\,\Gamma_\mathbb{C}(s)^{r_2} \tag{8.38}$$

This is the gamma matching: each of the r_2 complex places of K arises as one even gamma factor paired with one odd gamma factor, exactly as $\Gamma_\mathbb{R}(s)\Gamma_\mathbb{R}(s+1) = \Gamma_\mathbb{C}(s)$ predicts, while the r_1 remaining even factors are the r_1 real places. The conductor part of (8.37) separates the exponent $s/2$ shared by all characters from the extra $1/2$ carried by each odd character:

$$\left(\prod_{\chi \text{ even}} \mathfrak{f}_\chi^{s/2}\right)\left(\prod_{\chi \text{ odd}} \mathfrak{f}_\chi^{(s+1)/2}\right) = \left(\prod_{\chi\in X} \mathfrak{f}_\chi\right)^{s/2} \cdot \prod_{\chi \text{ odd}} \mathfrak{f}_\chi^{1/2} = |d_K|^{s/2} \cdot c_K \tag{8.39}$$

where the conductor product $\prod_{\chi\in X}\mathfrak{f}_\chi = |d_K|$ is the conductor-discriminant formula (8.33), supplying the factor $|d_K|^{s/2}$, and

$$c_K = \prod_{\chi \text{ odd}} \mathfrak{f}_\chi^{1/2} > 0$$

is a positive constant independent of s. Combining the gamma matching (8.38), the conductor matching (8.39), and the factorization $\prod_{\chi\in X} L(s,\chi)=\zeta_K(s)$ of (8.32), the product of completed L-functions is

$$\prod_{\chi\in X}\Lambda(s,\chi)=c_K\,|d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\,\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s)=c_K\,\Lambda_K(s) \tag{8.40}$$

by the definition (8.29) of Λ_K: the product of completed L-functions equals Λ_K up to the positive constant c_K. As a check, for imaginary quadratic $K=\mathbb{Q}(\sqrt{D})$ the sole odd character is χ_D of conductor $|D|$, so $c_K=\sqrt{|D|}$, while for real quadratic K there are no odd characters and $c_K=1$.

Step 4: Reflection and root number. Each factor on the left of (8.40) satisfies a functional equation. For the principal character $\chi_0\in X$, of conductor 1, the completed function is $\Lambda(s,\chi_0)=\Lambda(s)$ (definition 7.3.3) and the reflection is $\Lambda(s)=\Lambda(1-s)$ of theorem 6.3.3. For each nonprincipal $\chi\in X$, theorem 7.3.4 gives

$$\Lambda(s,\chi)=\epsilon(\chi)\,\Lambda(1-s,\bar\chi),\qquad \epsilon(\chi)=\frac{\tau(\chi)}{i^{a_\chi}\sqrt{\mathfrak{f}_\chi}},\quad |\epsilon(\chi)|=1 \tag{8.41}$$

Multiply (8.41) over all nonprincipal $\chi\in X$ and append the relation for χ_0. Because X is closed under conjugation (Step 2), the assignment $\chi\mapsto\bar\chi$ permutes X, so on the right

$$\prod_{\chi\in X}\Lambda(1-s,\bar\chi)=\prod_{\chi\in X}\Lambda(1-s,\chi)$$

the product being only reindexed. Therefore, writing $W=\prod_{\chi\in X}\epsilon(\chi)$ (with $\epsilon(\chi_0)=1$),

$$\prod_{\chi\in X}\Lambda(s,\chi)=W\prod_{\chi\in X}\Lambda(1-s,\chi) \tag{8.42}$$

It remains to show $W=+1$. Partition X into characters fixed by conjugation and genuinely conjugate pairs. The principal character χ_0 is fixed by conjugation and has $\epsilon(\chi_0)=1$, established above from the reflection $\Lambda(s)=\Lambda(1-s)$. A *nonprincipal* character fixed by conjugation, $\chi=\bar\chi$, is real, hence quadratic, and for such a character the root number is $\epsilon(\chi)=+1$. This is the sign theorem for the quadratic Gauss sum: for a real primitive character χ of parity a_χ the Gauss sum is exactly $\tau(\chi)=i^{a_\chi}\sqrt{\mathfrak{f}_\chi}$ (that is $\sqrt{\mathfrak{f}_\chi}$ for χ even and $i\sqrt{\mathfrak{f}_\chi}$ for χ odd), so $\epsilon(\chi)=\tau(\chi)/(i^{a_\chi}\sqrt{\mathfrak{f}_\chi})=+1$. The modulus $|\tau(\chi)|=\sqrt{\mathfrak{f}_\chi}$ and the relation $\tau(\bar\chi)=\chi(-1)\overline{\tau(\chi)}$ of theorem 7.3.2 reduce this to determining a single sign (they give $\tau(\chi)\in\{\pm\sqrt{\mathfrak{f}_\chi}\}$ for χ even and $\tau(\chi)\in\{\pm i\sqrt{\mathfrak{f}_\chi}\}$ for χ odd); that the sign is $+$, the content of Gauss's theorem on the sign of the quadratic Gauss sum, is the case $\tau(\chi)=i^{a_\chi}\sqrt{\mathfrak{f}_\chi}$ recorded in section 7.3 and proved in full in [6, Ch. 2] and [11, Ch. 9]. For a genuinely complex character, $\chi\neq\bar\chi$, the pair $\{\chi,\bar\chi\}$ contributes $\epsilon(\chi)\,\epsilon(\bar\chi)$, and this product is $+1$ by the reflection alone, with no sign theorem needed: applying (8.41) to $\bar\chi$ at the point $1-s$ gives $\Lambda(1-s,\bar\chi)=\epsilon(\bar\chi)\,\Lambda(s,\chi)$ (using $\bar{\bar\chi}=\chi$), and substituting this into $\Lambda(s,\chi)=\epsilon(\chi)\,\Lambda(1-s,\bar\chi)$ yields $\Lambda(s,\chi)=\epsilon(\chi)\epsilon(\bar\chi)\,\Lambda(s,\chi)$; since $\Lambda(s,\chi)$ is not identically zero (its L-factor is nonvanishing on $\mathrm{Re}(s)>1$ by the Euler product theorem 7.2.2), $\epsilon(\chi)\epsilon(\bar\chi)=1$. Thus every conjugate pair contributes $+1$ and every real character contributes $+1$, so the total is

$$W=\prod_{\chi\in X}\epsilon(\chi)=\prod_{\chi\text{ real}}(+1)\cdot\prod_{\{\chi,\bar\chi\}}\epsilon(\chi)\epsilon(\bar\chi)=+1$$

Inserting $W = +1$ into (8.42) and then dividing the resulting identity $\prod_\chi \Lambda(s,\chi) = \prod_\chi \Lambda(1-s,\chi)$ by the positive constant c_K of (8.40) (which appears on both sides, since (8.40) holds with s replaced by $1-s$ as well) gives

$$\Lambda_K(s) = \frac{1}{c_K}\prod_{\chi\in X}\Lambda(s,\chi) = \frac{1}{c_K}\prod_{\chi\in X}\Lambda(1-s,\chi) = \Lambda_K(1-s)$$

This is the functional equation (8.31) for abelian K, established from theorem 7.3.4 and theorem 8.5.4 alone. The root number of ζ_K is the product $W = +1$ of the root numbers of its L-function factors, recovering the self-duality recorded after theorem 8.5.3.

The abelian computation exhibits, in miniature, how the invariants of K distribute among the Dirichlet L-functions: the signature (r_1, r_2) is read off from the parities of the characters in X by (8.36), the discriminant $|d_K|$ from their conductors by theorem 8.5.4, and the trivial root number from the pairing of conjugate characters and the reality of the quadratic ones. The same three pieces of data, assembled differently, are what the analytic class number formula of theorem 8.3.2 packages into the residue κ_K; the functional equation and the class number formula are the two halves of the analytic theory of ζ_K, the former controlling its symmetry and the latter its single pole.

8.5.6 Remark: One object, three guises, and the road onward The Dedekind zeta function realizes the central objects of this book as special cases of a single construction. For $K = \mathbb{Q}$ it is the Riemann zeta function ζ of chapter 4, with its functional equation (6.14). For abelian K it factors, by (8.32), into the Dirichlet L-functions $L(s,\chi)$ of chapter 7, and its functional equation (8.31) is the product of theirs, as theorem 8.5.5 showed. For general K it is a new object, governed by Hecke's functional equation (8.31), whose proof passes through the geometry of the Minkowski lattice rather than through any factorization. The non-abelian generalization, in which ζ_K for a Galois extension $K/\mathbb{Q}$ with group G factors through the irreducible representations of G into Artin L-functions and the splitting of primes is governed by the Chebotarev density theorem (the non-abelian counterpart of Dirichlet's theorem of section 7.5), continues the arc begun in chapter 7; it is taken up in the successors to that chapter and, in its idelic and automorphic forms, in [12]. The pattern is uniform: an L-function packages the splitting of primes, its pole or its value at $s = 1$ encodes arithmetic invariants, and its functional equation fixes the symmetry of its zeros.

9

Hecke L-Functions

The analytic theory built so far has moved along two axes. Along the first, chapter 7 twisted the Riemann zeta function by a Dirichlet character χ, a character of the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$, and the resulting L-functions $L(s,\chi)$ resolved the rational primes among the residue classes modulo m. Along the second, chapter 8 held the character trivial but changed the base field, replacing $\mathbb{Q}$ by a number field K and the rational primes by the prime ideals of $\mathcal{O}_K$; the Dedekind zeta function $\zeta_K(s) = \sum_{\mathfrak{a}} (N\mathfrak{a})^{-s}$ carried the arithmetic of K, its pole encoding the basic invariants through the class number formula. The two constructions are the orthogonal extremes of a single one. The group $(\mathbb{Z}/m\mathbb{Z})^\times$ is, as chapter 8 hinted and the present chapter makes precise, the *ray class group* of $\mathbb{Q}$ for the modulus $m\infty$; the Dirichlet characters are its characters; and the trivial character on the ray class group of an arbitrary K returns the Dedekind zeta function. This chapter constructs the common generalization: the twist of ζ_K by a character of a ray class group of K, the *Hecke L-function*.

The ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$ refines the ideal class group Cl_K of chapter 8 by imposing congruence and sign conditions selected by a modulus $\mathfrak{m}$. Its characters, the Hecke (ray class) characters, are simultaneously the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$ when $K = \mathbb{Q}$ (chapter 7) and the trivial character when $\mathfrak{m} = 1$ (chapter 8), so the Hecke L-functions specialize to the Dirichlet L-functions on one end and to the Dedekind zeta function on the other, with every intermediate case a genuine new object. The chapter follows the now-familiar template. The present section constructs the ray class group, proves its finiteness, and defines the Hecke characters and their conductors; section 9.2 attaches to each Hecke character its L-series, with its Euler product over the prime ideals of K; section 9.3 completes the L-function with archimedean gamma factors, one for each infinite place; section 9.4 states Hecke's meromorphic continuation and functional equation; and section 9.5 proves the non-vanishing at $s = 1$ and deduces the generalization of Dirichlet's theorem to primes in ray classes of K. The functional equation, whose proof rests on a theta-series argument over the Minkowski lattice that lies on the algebraic side of the subject, is the one result stated with its proof deferred to a precise citation; everything else is established in full.

9.0.1 Remark: Algebraic and Local Prerequisites This chapter retains algebraic number theory as a co-requisite, exactly as in chapter 8, and adds local field theory alongside it. From [2] it draws the structure of $\mathcal{O}_K$ as a Dedekind domain, the group of fractional ideals and the finiteness of the ideal class group Cl_K, the structure of the finite ring $\mathcal{O}_K/\mathfrak{m}_0$ and its unit group $(\mathcal{O}_K/\mathfrak{m}_0)^\times$, the multiplicativity of the absolute ideal norm, the ramification index and residue degree of a prime, and the splitting of primes through the Dedekind–Kummer theorem. From [4], the new co-requisite, it draws the places and completions K_v of a number field, the distinction between the real, complex, and finite places, the local fields and their unramified extensions, and the Haar measure used in the analytic estimates of later sections. The infinite places of K, which played only a passive role in chapter 8 (entering the class number formula through the counts r_1, r_2), become active data here: the infinite part of a modulus is a set of real places, and the archimedean factors of the completed Hecke L-function are indexed by all the infinite places. None of the results imported from these two companions is reproved; each is stated where used and cited to its source.

9.1 Ray Class Groups and Hecke Characters

The ideal class group $\mathrm{Cl}_K = I_K/P_K$ of chapter 8, the quotient of the group I_K of nonzero fractional ideals by the subgroup P_K of principal fractional ideals, measures the failure of $\mathcal{O}_K$ to be a principal ideal domain. It is the coarsest of a family of class groups attached to K. The finer members of the family are obtained by shrinking the subgroup P_K: instead of dividing by all principal ideals, one divides only by those principal ideals (α) whose generator α satisfies prescribed congruence conditions at a finite set of primes and prescribed sign conditions at a set of real places. The data selecting these conditions is a modulus, and the resulting quotient is a ray class group. For $K = \mathbb{Q}$, where the ideal class group is trivial because $\mathbb{Z}$ is a principal ideal domain, this refinement is the entire content: the ray class group for the modulus $m\infty$ recovers $(\mathbb{Z}/m\mathbb{Z})^\times$, the group on which the Dirichlet characters of chapter 7 were built. The general construction is the natural home for both threads.

Definition 9.1.1: Modulus

Let K be a number field. A *modulus* of K is a formal product

$$\mathfrak{m} = \mathfrak{m}_0\, \mathfrak{m}_\infty,$$

in which the *finite part* $\mathfrak{m}_0$ is a nonzero integral ideal of $\mathcal{O}_K$ and the *infinite part* $\mathfrak{m}_\infty$ is a subset of the real places of K, that is, a subset of the r_1 embeddings $\sigma\colon K \hookrightarrow \mathbb{R}$. A real place σ is said to *divide* $\mathfrak{m}$, written $\sigma \big| \mathfrak{m}$, if $\sigma \in \mathfrak{m}_\infty$, and a prime ideal $\mathfrak{p}$ divides $\mathfrak{m}$ if $\mathfrak{p} \big| \mathfrak{m}_0$. The integer $|\mathfrak{m}_\infty|$ is the number of real places in the infinite part.

The notation is multiplicative by convention: writing $\mathfrak{m}_\infty$ as a product $\prod_{\sigma\in\mathfrak{m}_\infty} \sigma$ of distinct real places lets a modulus be manipulated like an ideal, with divisibility, greatest common divisors, and least common multiples computed place by place. The infinite part records only real places; complex places carry no sign and impose no condition, so they never enter a modulus. The two extreme cases frame the constructions to come: $\mathfrak{m} = (1)$, the unit ideal with empty infinite part, imposes no conditions at all, and $\mathfrak{m} = (m)\infty$ over $K = \mathbb{Q}$, with $\mathfrak{m}_0 = (m)$ and $\mathfrak{m}_\infty$ the single real place of $\mathbb{Q}$, is the modulus underlying the Dirichlet characters of modulus m.

The conditions selected by a modulus carve out a subgroup of the fractional ideals coprime to its finite part.

Definition 9.1.2: Ray Class Group

Let $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ be a modulus of K. Let

$$I^{\mathfrak{m}} = \{\mathfrak{a} \in I_K \mid \mathfrak{a} \text{ coprime to } \mathfrak{m}_0\}$$

be the group of nonzero fractional ideals of K coprime to $\mathfrak{m}_0$, that is, those whose factorization into prime ideals involves no prime dividing $\mathfrak{m}_0$. The *ray* modulo $\mathfrak{m}$ is the subgroup

$$P_1^{\mathfrak{m}} = \left\{(\alpha) \in I^{\mathfrak{m}} \;\middle|\; \alpha \in K^\times,\ \alpha \equiv 1 \pmod{\mathfrak{m}_0},\ \sigma(\alpha) > 0 \text{ for all } \sigma \in \mathfrak{m}_\infty\right\}$$

of principal fractional ideals generated by an element α that is congruent to 1 modulo $\mathfrak{m}_0$ and positive at every real place dividing $\mathfrak{m}$. The *ray class group* of K modulo $\mathfrak{m}$ is the quotient

$$\operatorname{Cl}_{\mathfrak{m}}(K) = I^{\mathfrak{m}}/P_1^{\mathfrak{m}},$$

a group whose order, once finiteness is established, is written $h_{\mathfrak{m}}$.

Two points of the definition require the conventions of [2] to be unambiguous. The congruence $\alpha \equiv 1 \pmod{\mathfrak{m}_0}$ for an element $\alpha \in K^\times$ that need not lie in $\mathcal{O}_K$ is interpreted multiplicatively: it means $v_{\mathfrak{p}}(\alpha - 1) \geq v_{\mathfrak{p}}(\mathfrak{m}_0)$ for every prime $\mathfrak{p} \mid \mathfrak{m}_0$, where $v_{\mathfrak{p}}$ is the $\mathfrak{p}$-adic valuation. For $\alpha \in \mathcal{O}_K$ coprime to $\mathfrak{m}_0$ this is the ordinary congruence in the ring $\mathcal{O}_K/\mathfrak{m}_0$. That $P_1^{\mathfrak{m}}$ is a subgroup of $I^{\mathfrak{m}}$ follows because the defining conditions are multiplicative: if α, β both satisfy them, then $\alpha\beta \equiv 1 \pmod{\mathfrak{m}_0}$ and $\sigma(\alpha\beta) = \sigma(\alpha)\sigma(\beta) > 0$ for $\sigma \in \mathfrak{m}_\infty$, and $\alpha^{-1} \equiv 1 \pmod{\mathfrak{m}_0}$ with $\sigma(\alpha^{-1}) > 0$ as well, so the conditions are preserved under multiplication and inversion. The quotient $\operatorname{Cl}_{\mathfrak{m}}(K)$ is therefore a well-defined abelian group, and when no ambiguity arises the base field is suppressed and it is written $\operatorname{Cl}_{\mathfrak{m}}$.

The principal ideals coprime to $\mathfrak{m}_0$ that are *not* subject to the ray conditions form an intermediate subgroup, and naming it makes the comparison with the ideal class group transparent. Write $P^{\mathfrak{m}} = P_K \cap I^{\mathfrak{m}}$ for the group of all principal fractional ideals coprime to $\mathfrak{m}_0$, so that

$$P_1^{\mathfrak{m}} \subseteq P^{\mathfrak{m}} \subseteq I^{\mathfrak{m}}$$

The largest quotient $I^{\mathfrak{m}}/P^{\mathfrak{m}}$ is the ideal class group Cl_K (every ideal class contains a representative coprime to $\mathfrak{m}_0$, so restricting to $I^{\mathfrak{m}}$ loses no class), and the ray class group $\operatorname{Cl}_{\mathfrak{m}} = I^{\mathfrak{m}}/P_1^{\mathfrak{m}}$ surjects onto it. The kernel of that surjection is $P^{\mathfrak{m}}/P_1^{\mathfrak{m}}$, the principal classes modulo the ray, and the computation of this kernel in terms of residues and signs is the heart of the finiteness theorem.

Theorem 9.1.3: Finiteness of the Ray Class Group

Let $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ be a modulus of K. The ray class group $\mathrm{Cl}_{\mathfrak{m}}$ is finite. Writing $\mathcal{O}_{K,\mathfrak{m},1}^\times = \{u \in \mathcal{O}_K^\times \mid u \equiv 1 \pmod{\mathfrak{m}_0},\ \sigma(u) > 0 \text{ for all } \sigma \in \mathfrak{m}_\infty\}$ for the group of units satisfying the ray conditions, there is an exact sequence of abelian groups

$$\mathcal{O}_K^\times \xrightarrow{\theta} (\mathcal{O}_K/\mathfrak{m}_0)^\times \times \{\pm 1\}^{\mathfrak{m}_\infty} \xrightarrow{\psi} \mathrm{Cl}_{\mathfrak{m}} \xrightarrow{\pi} \mathrm{Cl}_K \longrightarrow 1, \tag{9.1}$$

in which θ sends a unit to its residue class and its vector of signs at the places of $\mathfrak{m}_\infty$, the map ψ sends a residue-and-sign datum to the ray class of any principal ideal realizing it, and π forgets the ray conditions. Consequently the order $h_{\mathfrak{m}} = |\mathrm{Cl}_{\mathfrak{m}}|$ is finite and given by

$$h_{\mathfrak{m}} = \frac{h_K\, \varphi(\mathfrak{m}_0)\, 2^{|\mathfrak{m}_\infty|}}{[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]}, \tag{9.2}$$

where $h_K = |\mathrm{Cl}_K|$ is the class number and $\varphi(\mathfrak{m}_0) = |(\mathcal{O}_K/\mathfrak{m}_0)^\times|$ is the ideal-theoretic Euler function.

***Proof*:**
The proof constructs the three maps, verifies exactness at each interior term, and then reads the order formula off the exact sequence. Throughout, the group in the middle is abbreviated

$$M = (\mathcal{O}_K/\mathfrak{m}_0)^\times \times \{\pm 1\}^{\mathfrak{m}_\infty},$$

a finite group of order $\varphi(\mathfrak{m}_0)\, 2^{|\mathfrak{m}_\infty|}$, the first factor finite because $\mathcal{O}_K/\mathfrak{m}_0$ is a finite ring ([2, the finiteness of the residue ring $\mathcal{O}_K/\mathfrak{m}_0$]) and the second a finite product of copies of $\{\pm 1\}$ indexed by the real places dividing $\mathfrak{m}$.

Step 1: The surjection $\pi\colon \mathrm{Cl}_{\mathfrak{m}} \to \mathrm{Cl}_K$ and exactness at Cl_K. The inclusion $I^{\mathfrak{m}} \hookrightarrow I_K$ followed by the quotient $I_K \to \mathrm{Cl}_K = I_K/P_K$ sends $P_1^{\mathfrak{m}}$ into P_K, since a ray ideal is in particular principal, and so descends to a homomorphism $\pi\colon \mathrm{Cl}_{\mathfrak{m}} \to \mathrm{Cl}_K$. It is surjective: every ideal class in Cl_K has a representative coprime to $\mathfrak{m}_0$, because given any fractional ideal $\mathfrak{a}$ one may multiply by a principal ideal (β) to clear the primes dividing $\mathfrak{m}_0$ from its factorization (the approximation furnished by the Chinese Remainder Theorem in $\mathcal{O}_K$ produces such a β, [2, approximation for fractional ideals]), and the modified ideal lies in $I^{\mathfrak{m}}$ and represents the same class. Thus π is onto, which is exactness at Cl_K.

Step 2: The kernel of π is $P^{\mathfrak{m}}/P_1^{\mathfrak{m}}$. An element of $\mathrm{Cl}_{\mathfrak{m}}$ is a class $[\mathfrak{a}]$ with $\mathfrak{a} \in I^{\mathfrak{m}}$, and $\pi[\mathfrak{a}]$ is trivial in Cl_K precisely when $\mathfrak{a} \in P_K$, that is, when $\mathfrak{a} \in P_K \cap I^{\mathfrak{m}} = P^{\mathfrak{m}}$. Hence

$$\ker \pi = P^{\mathfrak{m}}/P_1^{\mathfrak{m}},$$

the image in $\mathrm{Cl}_{\mathfrak{m}} = I^{\mathfrak{m}}/P_1^{\mathfrak{m}}$ of the principal ideals coprime to $\mathfrak{m}_0$.

Step 3: The map $\psi\colon M \to \mathrm{Cl}_{\mathfrak{m}}$ and its image. Given a datum $(\bar{a}, \epsilon) \in M$, with $\bar{a} \in (\mathcal{O}_K/\mathfrak{m}_0)^\times$ and $\epsilon = (\epsilon_\sigma)_{\sigma \in \mathfrak{m}_\infty} \in \{\pm 1\}^{\mathfrak{m}_\infty}$, choose an element $\alpha \in \mathcal{O}_K$ coprime to $\mathfrak{m}_0$ with residue $\alpha \equiv a \pmod{\mathfrak{m}_0}$ and with $\operatorname{sign} \sigma(\alpha) = \epsilon_\sigma$ for every $\sigma \in \mathfrak{m}_\infty$. Such an α exists by simultaneous approximation: the congruence condition fixes α modulo $\mathfrak{m}_0$, and because the real places $\sigma \in \mathfrak{m}_\infty$

are independent of the finite primes, the sign at each can be prescribed freely by adding a suitable element of $\mathfrak{m}_0$ that is large and of the correct sign at σ while remaining small at the other archimedean places ([2, weak approximation across the places of K], or the strong approximation furnished by [4]). Set

$$\psi(\bar{a}, \epsilon) = [(\alpha)] \in \mathrm{Cl}_{\mathfrak{m}}$$

This is well defined: if α' is another element with the same residue and the same signs, then $\beta = \alpha'/\alpha \in K^\times$ satisfies $\beta \equiv 1 \pmod{\mathfrak{m}_0}$ (the residues agree and $\bar{a}$ is a unit modulo $\mathfrak{m}_0$) and $\sigma(\beta) = \sigma(\alpha')/\sigma(\alpha) > 0$ for $\sigma \in \mathfrak{m}_\infty$ (equal signs make the ratio positive), so $(\beta) \in P_1^{\mathfrak{m}}$ and $(\alpha') = (\beta)(\alpha)$ lies in the same ray class as (α). The assignment is a homomorphism, since the product of two representatives represents the product datum, and its image is exactly $P^{\mathfrak{m}}/P_1^{\mathfrak{m}}$: every principal class in $\mathrm{Cl}_{\mathfrak{m}}$ is $[(\alpha)]$ for some $\alpha \in \mathcal{O}_K$ coprime to $\mathfrak{m}_0$ (clearing denominators against $\mathfrak{m}_0$ as in Step 1), and $[(\alpha)] = \psi(\bar{\alpha}, \operatorname{sign}\sigma(\alpha))$. Comparing with Step 2,

$$\operatorname{im}\psi = P^{\mathfrak{m}}/P_1^{\mathfrak{m}} = \ker\pi,$$

which is exactness at $\mathrm{Cl}_{\mathfrak{m}}$.

Step 4: The map $\theta\colon \mathcal{O}_K^\times \to M$ and exactness at M. Define $\theta(u) = (\bar{u}, (\operatorname{sign}\sigma(u))_{\sigma\in\mathfrak{m}_\infty})$, the residue class of the unit u together with its signs at the real places of $\mathfrak{m}$; this is a homomorphism because residues and signs are multiplicative. Exactness at M asserts $\ker\psi = \operatorname{im}\theta$. For the inclusion $\operatorname{im}\theta \subseteq \ker\psi$: if $u \in \mathcal{O}_K^\times$, then $\psi(\theta(u)) = [(u)] = [\mathcal{O}_K]$ is the trivial class, because $(u) = \mathcal{O}_K$ for a unit, so $\theta(u) \in \ker\psi$. For the reverse inclusion $\ker\psi \subseteq \operatorname{im}\theta$: suppose $(\bar{a}, \epsilon) \in \ker\psi$, so that for a representative α as in Step 3 the ideal (α) lies in $P_1^{\mathfrak{m}}$. By the definition of the ray, $(\alpha) \in P_1^{\mathfrak{m}}$ means $(\alpha) = (\gamma)$ for some $\gamma \in K^\times$ with $\gamma \equiv 1 \pmod{\mathfrak{m}_0}$ and $\sigma(\gamma) > 0$ for $\sigma \in \mathfrak{m}_\infty$. Then $u = \alpha/\gamma$ generates the unit ideal $(\alpha)(\gamma)^{-1} = \mathcal{O}_K$, hence $u \in \mathcal{O}_K^\times$, and modulo $\mathfrak{m}_0$ one has $\bar{u} = \bar{\alpha}\cdot\bar{\gamma}^{-1} = \bar{\alpha} = \bar{a}$ since $\bar{\gamma} = 1$, while at each $\sigma \in \mathfrak{m}_\infty$ the sign is $\operatorname{sign}\sigma(u) = \operatorname{sign}\sigma(\alpha)\cdot\operatorname{sign}\sigma(\gamma) = \epsilon_\sigma\cdot(+1) = \epsilon_\sigma$. Therefore $\theta(u) = (\bar{a}, \epsilon)$, so $(\bar{a}, \epsilon) \in \operatorname{im}\theta$. This proves $\ker\psi = \operatorname{im}\theta$.

Step 5: Finiteness. The sequence (9.1) is now exact at M, at $\mathrm{Cl}_{\mathfrak{m}}$, and at Cl_K. The group Cl_K is finite of order h_K ([2, finiteness of the ideal class group]), and M is finite of order $\varphi(\mathfrak{m}_0)\,2^{|\mathfrak{m}_\infty|}$. By exactness at $\mathrm{Cl}_{\mathfrak{m}}$ and at Cl_K, the surjection π realizes Cl_K as the quotient $\mathrm{Cl}_{\mathfrak{m}}/\ker\pi$, and $\ker\pi = \operatorname{im}\psi$ is a quotient of the finite group M, hence finite. Thus $\mathrm{Cl}_{\mathfrak{m}}$ is an extension of the finite group Cl_K by the finite group $\operatorname{im}\psi$, so it is finite, of order $h_{\mathfrak{m}} = |\operatorname{im}\psi|\cdot h_K$.

Step 6: The order formula. Counting along the exact sequence, exactness at $\mathrm{Cl}_{\mathfrak{m}}$ gives $|\operatorname{im}\psi| = |\ker\pi|$ and the first isomorphism theorem applied to π gives $|\mathrm{Cl}_{\mathfrak{m}}| = |\ker\pi|\cdot|\operatorname{im}\pi| = |\operatorname{im}\psi|\cdot h_K$. Applying the first isomorphism theorem to ψ, whose image is $\ker\pi$ and whose kernel is $\operatorname{im}\theta$ by exactness at M,

$$|\operatorname{im}\psi| = \frac{|M|}{|\ker\psi|} = \frac{|M|}{|\operatorname{im}\theta|}$$

The image of θ is the quotient of $\mathcal{O}_K^\times$ by its kernel, and $\ker\theta$ is exactly the group of units with trivial residue and all signs positive, namely $\mathcal{O}_{K,\mathfrak{m},1}^\times$; hence $|\operatorname{im}\theta| = [\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]$. Combining,

$$h_{\mathfrak{m}} = |\operatorname{im}\psi|\cdot h_K = \frac{|M|\,h_K}{[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]} = \frac{h_K\,\varphi(\mathfrak{m}_0)\,2^{|\mathfrak{m}_\infty|}}{[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]},$$

which is (9.2). The index $[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]$ is finite because it divides the order of the finite group M.

The index $[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times]$ measures how many of the residue-and-sign data in M are already realized by units, and it is the only term in (9.2) that is not a direct count of residues, signs, or ideal classes. For $K = \mathbb{Q}$ this index, the class number $h_\mathbb{Q} = 1$, and the structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ combine to recover the setting of chapter 7 exactly.

9.1.4 Remark: The Rational Case Recovers $(\mathbb{Z}/m\mathbb{Z})^\times$ Take $K = \mathbb{Q}$, so $\mathcal{O}_K = \mathbb{Z}$, $h_K = 1$, and $\mathcal{O}_K^\times = \{\pm 1\}$. There is a single real place, the inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$. For the modulus $\mathfrak{m} = (m)\infty$ with $\mathfrak{m}_0 = (m)$ and $\mathfrak{m}_\infty$ this one real place, the middle group of (9.1) is $M = (\mathbb{Z}/m\mathbb{Z})^\times \times \{\pm 1\}$, of order $\varphi(m) \cdot 2$. The unit group $\{\pm 1\}$ maps in by $\theta(\pm 1) = (\overline{\pm 1}, \operatorname{sign}(\pm 1)) = (\pm 1 \bmod m, \pm 1)$, and this map is injective (the unit -1 has sign -1, so $\theta(-1) \neq \theta(1)$ as long as the sign coordinate is present), giving index $[\mathcal{O}_K^\times : \mathcal{O}_{K,\mathfrak{m},1}^\times] = 2$. The order formula (9.2) returns

$$h_\mathfrak{m} = \frac{1 \cdot \varphi(m) \cdot 2}{2} = \varphi(m),$$

and the exact sequence collapses, with Cl_K trivial, to an isomorphism $\operatorname{Cl}_{(m)\infty}(\mathbb{Q}) \cong M/\operatorname{im}\theta \cong (\mathbb{Z}/m\mathbb{Z})^\times$. Concretely, every fractional ideal of $\mathbb{Q}$ coprime to (m) is (a/b) for unique coprime positive integers a, b with $\gcd(ab, m) = 1$, and the ray $P_1^{(m)\infty}$ consists of those (a/b) with $a/b \equiv 1 \pmod m$ and $a/b > 0$; sending (a/b) to $ab^{-1} \bmod m$ identifies $\operatorname{Cl}_{(m)\infty}(\mathbb{Q})$ with $(\mathbb{Z}/m\mathbb{Z})^\times$. This is the *narrow ray class group* of $\mathbb{Q}$, the source of the Dirichlet characters of chapter 7. Dropping the infinite place, the modulus $\mathfrak{m} = (m)$ gives instead $M = (\mathbb{Z}/m\mathbb{Z})^\times$, on which $\theta(\pm 1) = \overline{\pm 1}$; for $m > 2$ the residues $\overline{1}$ and $\overline{-1}$ are distinct, so $\operatorname{im}\theta$ has order 2 and $h_{(m)} = \varphi(m)/2$, the *wide* ray class group, the quotient of $(\mathbb{Z}/m\mathbb{Z})^\times$ by $\{\pm 1\}$. The infinite place is exactly what separates the residues $\pm a$ that the sign condition keeps distinct, and its presence in the modulus $(m)\infty$ is the reason the Dirichlet theory of chapter 7 works with the full group $(\mathbb{Z}/m\mathbb{Z})^\times$ rather than its quotient by $\{\pm 1\}$.

With the ray class group in hand, its characters are defined exactly as the Dirichlet characters were defined as the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$ in chapter 7. A character of the finite abelian group $\operatorname{Cl}_\mathfrak{m}$ is extended to all of $I^\mathfrak{m}$ through the quotient map, and then to all fractional ideals by zero on those not coprime to $\mathfrak{m}_0$, producing a multiplicative function on ideals that is the input to the Hecke L-series of section 9.2.

Definition 9.1.5: Hecke Character and Conductor

Let $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ be a modulus of K. A *Hecke (ray class) character* modulo $\mathfrak{m}$ is a group homomorphism

$$\chi\colon \operatorname{Cl}_{\mathfrak{m}}(K) \longrightarrow \mathbb{C}^\times$$

Through the quotient $I^{\mathfrak{m}} \to \operatorname{Cl}_{\mathfrak{m}}$ it is regarded as a homomorphism on the fractional ideals coprime to $\mathfrak{m}_0$, that is, $\chi(\mathfrak{a}) = \chi([\mathfrak{a}])$ for $\mathfrak{a} \in I^{\mathfrak{m}}$, and it is extended to *all* nonzero fractional ideals by

$$\chi(\mathfrak{a}) = 0 \qquad \text{whenever } \mathfrak{a} \text{ is not coprime to } \mathfrak{m}_0$$

A modulus $\mathfrak{m}'$ with $\mathfrak{m}'|\mathfrak{m}$ (that is, $\mathfrak{m}'_0|\mathfrak{m}_0$ and $\mathfrak{m}'_\infty \subseteq \mathfrak{m}_\infty$) is said to *induce* χ if there is a Hecke character χ' modulo $\mathfrak{m}'$ with $\chi(\mathfrak{a}) = \chi'(\mathfrak{a})$ for every $\mathfrak{a} \in I^{\mathfrak{m}}$; equivalently, χ factors through the natural surjection $\operatorname{Cl}_{\mathfrak{m}} \to \operatorname{Cl}_{\mathfrak{m}'}$. The *conductor* of χ, written $\mathfrak{f}_\chi$, is the smallest modulus dividing $\mathfrak{m}$ that induces χ, and χ is *primitive* if $\mathfrak{f}_\chi = \mathfrak{m}$.

Three remarks fix the relationship of this definition to its two predecessors and to the wider theory. First, because $\operatorname{Cl}_{\mathfrak{m}}$ is finite (theorem 9.1.3), every value $\chi([\mathfrak{a}])$ on an ideal coprime to $\mathfrak{m}_0$ is a root of unity, of order dividing $h_{\mathfrak{m}}$; in particular $|\chi(\mathfrak{a})| = 1$ on $I^{\mathfrak{m}}$ and $\chi(\mathfrak{a}) = 0$ otherwise, so χ is a bounded multiplicative function on ideals. The extension by zero makes χ *completely multiplicative* on the nonzero integral ideals, $\chi(\mathfrak{a}\mathfrak{b}) = \chi(\mathfrak{a})\chi(\mathfrak{b})$ for integral $\mathfrak{a}, \mathfrak{b}$: if both are coprime to $\mathfrak{m}_0$ this is multiplicativity of the character on $\operatorname{Cl}_{\mathfrak{m}}$, and if either is not, then neither is the integral product $\mathfrak{a}\mathfrak{b}$, so both sides vanish. (On fractional ideals the relation can fail, the valuations at primes dividing $\mathfrak{m}_0$ cancelling, as with (p) and $(1/p)$ over $\mathbb{Q}$ for $\mathfrak{m}_0 = (p)$; the Dirichlet series and Euler product of section 9.2 run over integral ideals, where it holds.)

Second, the conductor is well defined. The set of moduli inducing χ is closed under the greatest common divisor of moduli (computed place by place on the finite parts and by intersection on the infinite parts), since if χ factors through both $\operatorname{Cl}_{\mathfrak{m}'}$ and $\operatorname{Cl}_{\mathfrak{m}''}$ it factors through $\operatorname{Cl}_{\gcd(\mathfrak{m}',\mathfrak{m}'')}$; the existence and uniqueness of a smallest such modulus follow exactly as the conductor of a Dirichlet character was shown to exist in definition 7.1.5, the place-by-place greatest common divisor playing the role of the greatest common divisor of integers there.

The two extreme cases are precisely the characters of the earlier chapters.

9.1.6 Remark: Dirichlet Characters and the Trivial Twist For $K = \mathbb{Q}$ and $\mathfrak{m} = (m)\infty$ the isomorphism $\operatorname{Cl}_{(m)\infty}(\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ of box 9.1.4 identifies the Hecke characters modulo $(m)\infty$ with the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$, that is, with the Dirichlet characters modulo m of definition 7.1.1. Under this identification the extension of a Hecke character by zero on ideals not coprime to (m) corresponds to the extension of a Dirichlet character by zero on integers not coprime to m, the conductor of the Hecke character corresponds to the conductor of the Dirichlet character (definition 7.1.5), and primitivity corresponds to primitivity; the Hecke theory of this chapter is, over $\mathbb{Q}$ with this modulus, literally the Dirichlet theory of chapter 7. At the opposite extreme, for any K and $\mathfrak{m} = (1)$ the ray class group is the ordinary (wide) ideal class group $\operatorname{Cl}_{(1)} = \operatorname{Cl}_K$ (no sign conditions and no congruences), and the *trivial* character $\chi = 1$ assigns 1 to every nonzero ideal of $\mathcal{O}_K$. Its L-series, defined in section 9.2, is $\sum_{\mathfrak{a}}(N\mathfrak{a})^{-s} = \zeta_K(s)$, the Dedekind zeta function of definition 8.1.1. The Hecke L-functions thus interpolate between the Dirichlet L-functions and the Dedekind zeta function, recovering each as a boundary case.

Third, the ray class characters defined above are the finite-order Hecke characters, and they do not exhaust Hecke's original notion. A general *Hecke character* (or *Grossencharakter*) modulo $\mathfrak{m}$ is a homomorphism on $I^{\mathfrak{m}}$ whose restriction to the ray $P_1^{\mathfrak{m}}$ need not be trivial but is instead prescribed by an archimedean *infinity-type*: a continuous character of the connected component of the idele class group, given on principal ideals $(\alpha) \in P_1^{\mathfrak{m}}$ by a formula $\chi((\alpha)) = \prod_{v|\infty} \sigma_v(\alpha)^{a_v} |\sigma_v(\alpha)|^{it_v}$ in terms of exponents attached to the infinite places. The ray class characters are exactly those with trivial infinity-type, the ones factoring through the finite quotient $\mathrm{Cl}_{\mathfrak{m}}$; they are also called the characters *of finite order*, since their values are roots of unity. The general Hecke characters, including those of infinite order, are the ones for which Hecke first proved analytic continuation and a functional equation, and their natural formulation is idelic, as characters of the idele class group $\mathbb{A}_K^\times / K^\times$.

9.1.7 Remark: The Idelic Reformulation and the Infinity-Type The cleanest formulation of Hecke characters, due to Tate, replaces the modulus and the ray class group by the *idele class group* $C_K = \mathbb{A}_K^\times / K^\times$, the quotient of the group of ideles of K by the diagonally embedded $K^\times$. A Hecke character in this language is a continuous homomorphism $\chi \colon C_K \to \mathbb{C}^\times$, and the ray class characters of definition 9.1.5 are exactly those that are trivial on a suitable open subgroup (the principal units at the primes dividing $\mathfrak{m}_0$ and the positive reals at the places of $\mathfrak{m}_\infty$), hence factor through a finite quotient $C_K \to \mathrm{Cl}_{\mathfrak{m}}$. The infinity-type is the restriction of χ to the archimedean component of C_K, and it is trivial precisely for the finite-order characters treated in this chapter. This idelic picture, together with the local-global decomposition of the L-function and the harmonic analysis on $\mathbb{A}_K$ that yields the functional equation, is the content of Tate's thesis; it is developed in [15, Ch. VII, §6] and in [1], and is taken up in the companion [12]. The present chapter works with the finite-order characters throughout, and notes at section 9.3 where the infinity-type, were it present, would enter the archimedean factors of the completed L-function.

9.2 Hecke L-Functions: Definition and Euler Product

The ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$ of section 9.1 is a finite abelian group built from the fractional ideals coprime to the modulus $\mathfrak{m}$, and a Hecke character χ modulo $\mathfrak{m}$ is a character of this group, transported to ideals and extended by zero off the ideals coprime to $\mathfrak{m}_0$ (definition 9.1.5). The construction of the present section is the one that has organized every thread of the book: the Dirichlet series $\sum_{\mathfrak{a}} \chi(\mathfrak{a})(N\mathfrak{a})^{-s}$ attached to this character, and its factorization into an Euler product over the prime ideals of $\mathcal{O}_K$. The series converges on the same half-plane $\mathrm{Re}(s) > 1$ as the Dedekind zeta function, for the same reason and by the same comparison, and the Euler product follows from unique factorization of ideals exactly as for ζ_K, the only new ingredient being the complete multiplicativity of χ on the ideals coprime to $\mathfrak{m}_0$, which is built into its definition as a character of $\mathrm{Cl}_{\mathfrak{m}}(K)$.

The resulting function is the common generalization toward which the two threads of Part II have been converging. Its two extreme cases recover the objects already constructed: the trivial character returns the Dedekind zeta function of chapter 8, up to the finitely many Euler factors at the primes dividing $\mathfrak{m}_0$, while for the base field $K = \mathbb{Q}$ a Hecke character is a Dirichlet character and the Hecke L-function is the Dirichlet L-function of chapter 7. The two functions whose analytic theory occupied the previous five chapters are thus the endpoints of a single family, and the deeper analytic facts about that family (the continuation across $\mathrm{Re}(s) = 1$ and the functional equation) are developed in sections 9.3 and 9.4. This section establishes the elementary part of the theory, the part that lives inside the region of absolute convergence and rests on nothing beyond the Euler product.

Definition 9.2.1: Hecke L-Function

Let χ be a Hecke character modulo $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ of the number field K (definition 9.1.5). The *Hecke L-function* attached to χ is the Dirichlet series

$$L(s,\chi) = \sum_{\mathfrak{a}\neq 0} \frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s},$$

the sum running over the nonzero integral ideals $\mathfrak{a} \subseteq \mathcal{O}_K$, where $N\mathfrak{a} = [\mathcal{O}_K : \mathfrak{a}]$ is the absolute norm and $\chi(\mathfrak{a})$ is the value of the character, with the convention $\chi(\mathfrak{a}) = 0$ whenever $\gcd(\mathfrak{a}, \mathfrak{m}_0) \neq 1$. The terms indexed by ideals not coprime to $\mathfrak{m}_0$ therefore vanish, and the sum is effectively over the integral ideals coprime to $\mathfrak{m}_0$.

The definition specializes the Dedekind zeta function of definition 8.1.1 by inserting the character value $\chi(\mathfrak{a})$ as the coefficient of $(N\mathfrak{a})^{-s}$: for the trivial character of the trivial modulus, every coprimality condition is vacuous and $\chi(\mathfrak{a}) = 1$ for all $\mathfrak{a}$, so the series is ζ_K exactly. The role of the modulus is to restrict the sum to the ideals coprime to $\mathfrak{m}_0$ (the others contributing zero) and to fix, through the ray class group, the precise sense in which χ depends on the residue and sign data of an ideal. The two presentations of ζ_K as a sum over ideals and as a Dirichlet series in a single integer variable both transfer: grouping the ideals by their norm gives $L(s,\chi) = \sum_{n\geq 1} c_\chi(n)\, n^{-s}$ with $c_\chi(n) = \sum_{N\mathfrak{a}=n} \chi(\mathfrak{a})$, a Dirichlet series whose coefficients are bounded in absolute value by the ideal-counting function $a_K(n)$ of definition 8.1.1, since $|c_\chi(n)| \leq \sum_{N\mathfrak{a}=n} |\chi(\mathfrak{a})| \leq \sum_{N\mathfrak{a}=n} 1 = a_K(n)$. This last bound is the whole of the convergence theory.

Proposition 9.2.2: Absolute Convergence on $\mathrm{Re}(s) > 1$

The series defining $L(s,\chi)$ converges absolutely on the half-plane $\mathrm{Re}(s) > 1$ and defines a holomorphic function there. More precisely, for real $\sigma > 1$ the termwise comparison

$$\sum_{\mathfrak{a}\neq 0} \left| \frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s} \right| \leq \sum_{\mathfrak{a}\neq 0} (N\mathfrak{a})^{-\sigma} = \zeta_K(\sigma) \leq \zeta(\sigma)^d$$

holds, where $\sigma = \mathrm{Re}(s)$ and $d = [K : \mathbb{Q}]$.

***Proof*:**
The values of a Hecke character are roots of unity or zero: χ is a character of the finite group $\mathrm{Cl}_{\mathfrak{m}}(K)$ on ideals coprime to $\mathfrak{m}_0$, so $\chi(\mathfrak{a})$ is a root of unity there and $|\chi(\mathfrak{a})| = 1$, while $\chi(\mathfrak{a}) = 0$ otherwise (definition 9.1.5). Hence $|\chi(\mathfrak{a})| \leq 1$ for every nonzero integral ideal $\mathfrak{a}$. For any s with $\sigma = \mathrm{Re}(s) > 1$ and any ideal $\mathfrak{a}$ one has $|(N\mathfrak{a})^{-s}| = (N\mathfrak{a})^{-\sigma}$, so each term is bounded by

$$\left| \frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s} \right| = \frac{|\chi(\mathfrak{a})|}{(N\mathfrak{a})^\sigma} \leq \frac{1}{(N\mathfrak{a})^\sigma}$$

Summing over all nonzero integral ideals and applying the convergence of the Dedekind zeta series (proposition 8.1.3), which gives $\sum_{\mathfrak{a}} (N\mathfrak{a})^{-\sigma} = \zeta_K(\sigma) \leq \zeta(\sigma)^d < \infty$ for $\sigma > 1$, yields the displayed comparison and establishes absolute convergence on $\mathrm{Re}(s) > 1$.

For holomorphy, work on a compact subset of the half-plane $\mathrm{Re}(s) > 1$, on which there is a real $\sigma_0 > 1$ with $\mathrm{Re}(s) \geq \sigma_0$ throughout. In the Dirichlet-series form $L(s,\chi) = \sum_{n\geq 1} c_\chi(n)\, n^{-s}$

the coefficients satisfy $|c_\chi(n)| \le a_K(n)$, so $|c_\chi(n)n^{-s}| \le a_K(n)\, n^{-\sigma_0}$ with $\sum_n a_K(n)\, n^{-\sigma_0} = \zeta_K(\sigma_0) < \infty$ independent of s; the Weierstrass M-test gives uniform convergence on the compact set. Each partial sum is holomorphic, so the locally uniform limit $L(s,\chi)$ is holomorphic on $\mathrm{Re}(s) > 1$ (the convergence theory of Dirichlet series, theorem 2.1.7).

The Euler product is the analytic form of unique factorization of ideals, transferred from ζ_K to $L(s,\chi)$ by the one property of χ that distinguishes it from an arbitrary bounded ideal-indexed coefficient: it is completely multiplicative on the ideals coprime to $\mathfrak{m}_0$, because it factors through the abelian group $\mathrm{Cl}_{\mathfrak{m}}(K)$, on which multiplication of ideal classes is the group law.

Theorem 9.2.3: Euler Product

For $\mathrm{Re}(s) > 1$ the Hecke L-function factors as an absolutely convergent product over the nonzero prime ideals of $\mathcal{O}_K$ coprime to $\mathfrak{m}_0$,

$$L(s,\chi) = \prod_{\mathfrak{p}\nmid\mathfrak{m}_0} \left(1 - \chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}\right)^{-1} \tag{9.3}$$

the product running over the nonzero prime ideals $\mathfrak{p} \subseteq \mathcal{O}_K$ not dividing $\mathfrak{m}_0$. The primes $\mathfrak{p}\big|\mathfrak{m}_0$ have $\chi(\mathfrak{p}) = 0$ and contribute the trivial factor 1, so they may be included with no change. In particular $L(s,\chi) \neq 0$ for $\mathrm{Re}(s) > 1$.

***Proof*:**
Step 1: Complete multiplicativity of χ on ideals coprime to $\mathfrak{m}_0$. Let $\mathfrak{a}, \mathfrak{b}$ be nonzero integral ideals coprime to $\mathfrak{m}_0$. Then $\mathfrak{ab}$ is coprime to $\mathfrak{m}_0$, and in the ray class group $\mathrm{Cl}_{\mathfrak{m}}(K) = I^{\mathfrak{m}}/P_1^{\mathfrak{m}}$ the class of $\mathfrak{ab}$ is the product of the classes of $\mathfrak{a}$ and $\mathfrak{b}$, the group law on $\mathrm{Cl}_{\mathfrak{m}}(K)$ being induced by multiplication of ideals (definition 9.1.2). Since χ is a group homomorphism $\mathrm{Cl}_{\mathfrak{m}}(K) \to \mathbb{C}^\times$ pulled back to ideals (definition 9.1.5), it carries this product of classes to the product of values, so $\chi(\mathfrak{ab}) = \chi(\mathfrak{a})\,\chi(\mathfrak{b})$. If instead one of $\mathfrak{a}, \mathfrak{b}$ is not coprime to $\mathfrak{m}_0$, then neither is $\mathfrak{ab}$, so both sides of $\chi(\mathfrak{ab}) = \chi(\mathfrak{a})\chi(\mathfrak{b})$ vanish by the extension-by-zero convention. Thus $\chi(\mathfrak{ab}) = \chi(\mathfrak{a})\,\chi(\mathfrak{b})$ for all nonzero integral ideals $\mathfrak{a}, \mathfrak{b}$, with $\chi(\mathcal{O}_K) = 1$; that is, χ is completely multiplicative on integral ideals. In particular $\chi(\mathfrak{p}^k) = \chi(\mathfrak{p})^k$ for every prime $\mathfrak{p}$ and every $k \ge 0$.

Step 2: Each local factor is the character-weighted geometric series. Fix s with $\sigma = \mathrm{Re}(s) > 1$, and let $\mathfrak{p}$ be a nonzero prime ideal. Then $|\chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}| = |\chi(\mathfrak{p})|\,(N\mathfrak{p})^{-\sigma} \le (N\mathfrak{p})^{-\sigma} < 1$, since $|\chi(\mathfrak{p})| \le 1$ and $N\mathfrak{p} \ge 2$, so the geometric series in $\chi(\mathfrak{p})(N\mathfrak{p})^{-s}$ converges and, using $\chi(\mathfrak{p}^k) = \chi(\mathfrak{p})^k$ from Step 1 together with $N\mathfrak{p}^k = (N\mathfrak{p})^k$,

$$\left(1 - \chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}\right)^{-1} = \sum_{k=0}^{\infty} \left(\chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}\right)^k = \sum_{k=0}^{\infty} \frac{\chi(\mathfrak{p}^k)}{(N\mathfrak{p}^k)^s}$$

The local factor at $\mathfrak{p}$ is thus the sum of $\chi(\mathfrak{q})(N\mathfrak{q})^{-s}$ over the powers $\mathfrak{q} = \mathfrak{p}^k$, $k \ge 0$, of the single prime $\mathfrak{p}$. For $\mathfrak{p}\big|\mathfrak{m}_0$ one has $\chi(\mathfrak{p}) = 0$, so the factor degenerates to $(1-0)^{-1} = 1$, and including or omitting it leaves the product unchanged.

Step 3: Absolute convergence of the product. An infinite product $\prod_{\mathfrak{p}}(1-\chi(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1}$ converges absolutely precisely when $\sum_{\mathfrak{p}} |\chi(\mathfrak{p})(N\mathfrak{p})^{-s}| < \infty$. Using $|\chi(\mathfrak{p})| \le 1$,

$$\sum_{\mathfrak{p}} |\chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}| \le \sum_{\mathfrak{p}} (N\mathfrak{p})^{-\sigma} < \infty \qquad (\sigma > 1)$$

the finiteness of $\sum_{\mathfrak{p}}(N\mathfrak{p})^{-\sigma}$ on $\mathrm{Re}(s) > 1$ being established in the proof of the Euler product for ζ_K (Step 2 of theorem 8.1.4, by grouping primes over the rational prime below them). The product therefore converges absolutely.

Step 4: The product equals the sum. Because the product converges absolutely, its factors may be multiplied out and the resulting terms summed in any order. Multiplying the geometric series of Step 2 over all $\mathfrak{p}$, a generic term in the expansion is a finite product $\prod_{\mathfrak{p}} \chi(\mathfrak{p}^{k_{\mathfrak{p}}})(N\mathfrak{p}^{k_{\mathfrak{p}}})^{-s}$ with $k_{\mathfrak{p}} \ge 0$ and all but finitely many $k_{\mathfrak{p}}$ equal to zero. By complete multiplicativity of χ (Step 1) and of the norm, this term equals $\chi(\mathfrak{a})(N\mathfrak{a})^{-s}$ for the ideal $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$. Unique factorization of ideals ([2, unique factorization of ideals]) asserts that the assignment $(k_{\mathfrak{p}})_{\mathfrak{p}} \mapsto \mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$ is a bijection from the finitely-supported exponent sequences onto the nonzero integral ideals, so every nonzero ideal $\mathfrak{a}$ arises exactly once and the expansion of the product is precisely

$$\prod_{\mathfrak{p}} \left(1 - \chi(\mathfrak{p})\,(N\mathfrak{p})^{-s}\right)^{-1} = \sum_{\mathfrak{a} \ne 0} \frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s} = L(s, \chi)$$

the absolute convergence of proposition 9.2.2 justifying the rearrangement into the single sum over ideals. This is the rearrangement of theorem 8.1.4 carried out with the extra factor $\chi(\mathfrak{a})$, legitimate because χ is completely multiplicative; equivalently, it is the Euler product for the completely multiplicative ideal coefficient χ. Discarding the trivial factors at $\mathfrak{p}\big|\mathfrak{m}_0$ noted in Step 2 leaves the product over $\mathfrak{p} \nmid \mathfrak{m}_0$ of (9.3).

Step 5: Nonvanishing. Each factor $(1 - \chi(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1}$ is the reciprocal of the nonzero number $1 - \chi(\mathfrak{p})(N\mathfrak{p})^{-s}$, nonzero because $|\chi(\mathfrak{p})(N\mathfrak{p})^{-s}| < 1$ for $\sigma > 1$ (and the factor is 1 for $\mathfrak{p}\big|\mathfrak{m}_0$). An absolutely convergent infinite product all of whose factors are nonzero has a nonzero value (proposition 2.3.4), so $L(s, \chi) \ne 0$ throughout $\mathrm{Re}(s) > 1$.

The convergence and nonvanishing recorded here are the exact counterparts, for the Hecke L-function, of the corresponding facts for ζ (corollary 2.3.2, proposition 2.3.4), for a Dirichlet L-function (theorem 7.2.2), and for ζ_K (theorem 8.1.4, corollary 8.1.5). They are the easy half of the analytic theory, valid throughout the region of absolute convergence and resting only on the Euler product; the nonvanishing that the arithmetic applications require is at the boundary $s = 1$, where the product no longer converges and a separate argument is needed (section 9.5). Before turning to the continuation, the two extreme members of the family are identified with the objects already in hand, confirming that the Hecke L-function is the common generalization advertised at the outset.

Proposition 9.2.4: The Two Extreme Cases

Let χ be a Hecke character modulo $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ of K.

1. *(Trivial character.)* If $\chi = \chi_0$ is the trivial character modulo $\mathfrak{m}$ (the character with $\chi_0(\mathfrak{a}) = 1$ for every ideal $\mathfrak{a}$ coprime to $\mathfrak{m}_0$ and $\chi_0(\mathfrak{a}) = 0$ otherwise), then for $\mathrm{Re}(s) > 1$

$$L(s, \chi_0) = \zeta_K(s) \prod_{\mathfrak{p} \mid \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-s}\right) \tag{9.4}$$

the Dedekind zeta function with the Euler factors at the primes dividing $\mathfrak{m}_0$ removed. Consequently $L(s, \chi_0)$ continues to a meromorphic function on a neighbourhood of $s = 1$ whose only singularity there is a simple pole at $s = 1$, inherited from the simple pole of ζ_K (theorem 8.3.2), with residue

$$\mathrm{Res}_{s=1} L(s, \chi_0) = \kappa_K \prod_{\mathfrak{p} \mid \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-1}\right)$$

where $\kappa_K = \mathrm{Res}_{s=1}\zeta_K(s)$ is the constant of the analytic class number formula.

2. *(Base field $\mathbb{Q}$.)* If $K = \mathbb{Q}$ and χ is identified with a Dirichlet character modulo m (definition 7.1.1), then $L(s, \chi)$ is the Dirichlet L-function of definition 7.2.1: for $\mathrm{Re}(s) > 1$,

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \chi(p)\, p^{-s}\right)^{-1} \tag{9.5}$$

the Hecke and Dirichlet definitions agreeing term by term and Euler factor by Euler factor.

***Proof*:**
(1) The trivial character. By theorem 9.2.3 the Euler product of $L(s, \chi_0)$ runs over the primes $\mathfrak{p} \nmid \mathfrak{m}_0$, with $\chi_0(\mathfrak{p}) = 1$ there, so the factor at such $\mathfrak{p}$ is $(1 - (N\mathfrak{p})^{-s})^{-1}$; the factors at $\mathfrak{p} \mid \mathfrak{m}_0$ are 1. The Euler product of ζ_K (theorem 8.1.4) runs over *all* nonzero prime ideals, with the same factor $(1 - (N\mathfrak{p})^{-s})^{-1}$ at each. Both products converge absolutely for $\mathrm{Re}(s) > 1$ (proposition 9.2.2, proposition 8.1.3), so they may be compared factor by factor and the finite collection of factors at the primes $\mathfrak{p} \mid \mathfrak{m}_0$ separated off:

$$L(s, \chi_0) = \prod_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{1 - (N\mathfrak{p})^{-s}} = \left(\prod_{\mathfrak{p}} \frac{1}{1 - (N\mathfrak{p})^{-s}} \right) \prod_{\mathfrak{p} \mid \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-s}\right) = \zeta_K(s) \prod_{\mathfrak{p} \mid \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-s}\right)$$

the middle equality multiplying and dividing by the finitely many factors at $\mathfrak{p} \mid \mathfrak{m}_0$, legitimate because each such factor $(1 - (N\mathfrak{p})^{-s})^{-1}$ is finite and nonzero for $\mathrm{Re}(s) > 1$. This is (9.4). The finite product $P(s) = \prod_{\mathfrak{p} \mid \mathfrak{m}_0} (1 - (N\mathfrak{p})^{-s})$ is entire, and ζ_K continues to a neighbourhood of $s = 1$ with a single simple pole there of residue κ_K (theorem 8.3.2); hence $L(s, \chi_0) = \zeta_K(s)P(s)$ continues to that neighbourhood, with at most a simple pole at $s = 1$. The pole survives iff

$P(1) \neq 0$, and

$$P(1) = \prod_{\mathfrak{p} | \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-1}\right)$$

is a finite product of factors each strictly between 0 and 1, since $N\mathfrak{p} \geq 2$ gives $0 < 1-(N\mathfrak{p})^{-1} < 1$; thus $P(1) \neq 0$ and the simple pole is retained. Because P is holomorphic at $s = 1$, the residue of the product is the residue of ζ_K times the holomorphic value $P(1)$,

$$\mathrm{Res}_{s=1} L(s, \chi_0) = P(1)\, \mathrm{Res}_{s=1} \zeta_K(s) = \kappa_K \prod_{\mathfrak{p} | \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-1}\right)$$

as claimed.

(2) The base field $\mathbb{Q}$. For $K = \mathbb{Q}$ the ring of integers is $\mathbb{Z}$, and the nonzero integral ideals are the principal ideals (n) for $n \geq 1$, each with norm $N((n)) = [\mathbb{Z} : n\mathbb{Z}] = n$ and distinct n giving distinct ideals (definition 8.1.1). Take the modulus $\mathfrak{m} = (m)\infty$, with finite part $\mathfrak{m}_0 = (m)$ and infinite part the single real place of $\mathbb{Q}$, so that $\mathrm{Cl}_{(m)\infty}(\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ and the Hecke characters modulo $(m)\infty$ are exactly the Dirichlet characters modulo m (box 9.1.4, definition 7.1.1); the finite modulus (m) alone gives only the wide quotient by $\{\pm 1\}$, so the real place is required to recover the full group. Under this identification the value on the ideal (n), $n \geq 1$, is $\chi((n)) = \chi(n)$: the isomorphism sends $(n) \mapsto n \bmod m$, so for $\gcd(n, m) = 1$ the value of χ on (n) is the Dirichlet value $\chi(n)$, while for $\gcd(n, m) \neq 1$ both $\chi((n)) = 0$ (extension by zero on ideals not coprime to $\mathfrak{m}_0$) and $\chi(n) = 0$ (condition (iii) of definition 7.1.1) hold. The Hecke sum of definition 9.2.1 therefore reads

$$L(s, \chi) = \sum_{(n) \neq 0} \frac{\chi((n))}{(N(n))^s} = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

which is exactly the defining series of the Dirichlet L-function (definition 7.2.1). For the Euler product, the nonzero prime ideals of $\mathbb{Z}$ are the principal ideals (p) at the rational primes p, with $N((p)) = p$ and $(p) | (m)$ iff $p | m$; the Hecke product (9.3) over $\mathfrak{p} \nmid \mathfrak{m}_0$ thus becomes the product over $p \nmid m$ of $(1 - \chi(p)\, p^{-s})^{-1}$, with trivial factors at $p | m$ where $\chi(p) = 0$. This is precisely the Dirichlet Euler product of theorem 7.2.2,

$$L(s, \chi) = \prod_p \left(1 - \chi(p)\, p^{-s}\right)^{-1}$$

the factor at $p | m$ being 1 in both presentations. The two definitions agree on $\mathrm{Re}(s) > 1$, establishing (9.5).

The two extreme cases delimit the family. At one end stands the Dedekind zeta function: the trivial Hecke character of the trivial modulus has $\mathfrak{m}_0 = \mathcal{O}_K$, the product in (9.4) is empty, and $L(s, \chi_0) = \zeta_K(s)$ on the nose, carrying the simple pole at $s = 1$ whose residue is the analytic class number formula of theorem 8.3.2. A nontrivial modulus with trivial character only deletes finitely many Euler factors, an entire nonzero adjustment at $s = 1$, so the pole and its order persist and the residue is scaled by the explicit local factor $\prod_{\mathfrak{p} | \mathfrak{m}_0} (1 - (N\mathfrak{p})^{-1})$. At the other end stand the Dirichlet L-functions: over $K = \mathbb{Q}$, with the ray class group reducing to $(\mathbb{Z}/m\mathbb{Z})^\times$, the Hecke L-function is the Dirichlet L-function of chapter 7, and the entire apparatus of that chapter (the pole of the principal

L-function, the holomorphy and nonvanishing of the others) is the $K = \mathbb{Q}$ case of the general Hecke theory. Every Hecke L-function shares the dichotomy these two cases exhibit: a single pole at $s = 1$ when χ is trivial, holomorphy at $s = 1$ when χ is nontrivial. Establishing that dichotomy in general (the continuation across $\mathrm{Re}(s) = 1$ together with the location of the pole) is the analytic heart of the chapter, taken up after the archimedean factors are assembled in section 9.3.

9.3 Local Factors and the Completed Hecke L-Function

The Euler product of theorem 9.2.3 assembles the Hecke L-function from one factor per finite prime $\mathfrak{p}$ of K, the local factor $(1 - \chi(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1}$. As with the Riemann zeta function in chapter 6, the Dirichlet L-functions in chapter 7, and the Dedekind zeta function in chapter 8, this finite product is only half of the analytic object. The functional equation taken up in section 9.4 relates $L(s, \chi)$ to $L(1 - s, \bar{\chi})$, and that symmetry becomes visible only after the product over finite primes is multiplied by a finite collection of *archimedean factors*, one for each infinite place of K, and by a conductor-type power that records the arithmetic conductor of χ together with the discriminant of K. The present section carries out this completion for a Hecke character of finite order, the case in which the archimedean factors are exactly the gamma factors $\Gamma_{\mathbb{R}}$ and $\Gamma_{\mathbb{C}}$ already met in definition 8.5.1, shifted at the real places by the sign data of χ.

The guiding principle is that each place of K, finite or infinite, contributes one local factor, and the completed L-function is the product of all of them. At a finite place $\mathfrak{p}$ the local factor is the Euler factor of theorem 9.2.3, governed by the value $\chi(\mathfrak{p})$. At an infinite place v the local factor is a gamma factor, governed by the component of χ at v, that is, by the behavior of χ under the local units at v. For a finite-order character this local component is recorded by a single bit at each real place, the sign $a_v \in \{0, 1\}$ introduced below, generalizing the parity $a \in \{0, 1\}$ that completed the Dirichlet L-functions in definition 7.3.3; the complex places carry no such bit and contribute the uniform factor $\Gamma_{\mathbb{C}}(s)$. The systematic place-by-place formalism that underlies this bookkeeping, in the language of completions K_v, is that of [4, places and completions].

The sign data is the number-field analogue of the parity of a Dirichlet character. Recall from definition 9.1.1 that the modulus $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ has an infinite part $\mathfrak{m}_\infty$, a set of real places of K, and that the ray $P_1^{\mathfrak{m}}$ requires its generators to be totally positive at the places of $\mathfrak{m}_\infty$. A Hecke character χ modulo $\mathfrak{m}$ (definition 9.1.5) therefore carries, at each real place $v \in \mathfrak{m}_\infty$, a sign: the value of χ on a principal ideal whose generator is negative at v but positive at every other place of $\mathfrak{m}_\infty$ and congruent to 1 modulo $\mathfrak{m}_0$.

Definition 9.3.1: The Sign of χ at a Real Place

Let χ be a Hecke character modulo $\mathfrak{m}$ of finite order, with conductor $\mathfrak{f}_\chi$ of finite part $\mathfrak{f}_0$ and infinite part $\mathfrak{f}_\infty$ (definition 9.1.5). For each real place v of K define the *sign* $a_v \in \{0, 1\}$ of χ at v by

$$a_v = \begin{cases} 1 & \text{if } v \in \mathfrak{f}_\infty, \\ 0 & \text{if } v \notin \mathfrak{f}_\infty \end{cases}$$

that is, $a_v = 1$ precisely when v is a real place at which χ is ramified, equivalently when χ is nontrivial (odd) on the local sign at v, and $a_v = 0$ when χ is unramified (even) at v, in particular when $v \notin \mathfrak{m}_\infty$ or χ is trivial there.

The sign a_v is the local component of χ at the real place v, read off as the value of χ on the local units $K_v^\times = \mathbb{R}^\times$ modulo the totally positive units. The two values $a_v \in \{0,1\}$ correspond to the two characters of the group of connected components $\mathbb{R}^\times/\mathbb{R}^\times_{>0} = \{\pm 1\}$: the trivial character (even, $a_v = 0$) and the sign character (odd, $a_v = 1$), exactly the two one-dimensional representations of the real place that distinguished the even and odd Dirichlet characters in definition 7.1.6. At a complex place there is no analogous bit, because $K_v^\times = \mathbb{C}^\times$ is connected and a finite-order character trivial on a neighborhood of the identity is trivial on the positive reals; the complex local component of a finite-order Hecke character is therefore captured entirely by the uniform factor $\Gamma_{\mathbb{C}}(s)$ recorded below.

With the sign data in hand, the archimedean factors and the completed L-function are defined exactly as for the Dedekind zeta function in definition 8.5.1, with each real factor shifted by the sign a_v.

Definition 9.3.2: The Completed Hecke L-Function

Let K be a number field with r_1 real places and r_2 complex places, discriminant d_K, and let χ be a Hecke character of K of finite order with conductor $\mathfrak{f}_\chi$, signs $a_v \in \{0,1\}$ at the real places v (definition 9.3.1), and Hecke L-function $L(s,\chi)$ of definition 9.2.1. With the archimedean factors

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right), \qquad \Gamma_{\mathbb{C}}(s) = 2\,(2\pi)^{-s}\,\Gamma(s)$$

of definition 8.5.1, write the *archimedean local factor at v* as

$$\Gamma_v(s,\chi) = \begin{cases} \Gamma_{\mathbb{R}}(s+a_v) & \text{if } v \text{ is real,} \\ \Gamma_{\mathbb{C}}(s) & \text{if } v \text{ is complex} \end{cases}$$

The *completed Hecke L-function* of χ is

$$\Lambda(s,\chi) = \left(|d_K|\,N\mathfrak{f}_\chi\right)^{s/2} \prod_{v|\infty} \Gamma_v(s,\chi)\,L(s,\chi) = \left(|d_K|\,N\mathfrak{f}_\chi\right)^{s/2}\left(\prod_{v \text{ real}} \Gamma_{\mathbb{R}}(s+a_v)\right)\left(\prod_{v \text{ complex}} \Gamma_{\mathbb{C}}(s)\right)L(s,\chi) \tag{9.6}$$

where $N\mathfrak{f}_\chi = N\mathfrak{f}_0$ is the absolute norm of the finite part of the conductor, defined initially on the half-plane $\mathrm{Re}(s) > 1$, where $L(s,\chi)$ is given by its absolutely convergent Dirichlet series (proposition 9.2.2) and each archimedean factor is holomorphic.

The completed function multiplies the Dirichlet series, the product over the finite places, by one archimedean factor per infinite place and by the conductor-type power $(|d_K|\,N\mathfrak{f}_\chi)^{s/2}$. The power generalizes the two conductor factors met separately in the two extreme cases: the power $(m/\pi)^{(s+a)/2}$ of the Dirichlet completion in definition 7.3.3, whose conductor is the Dirichlet modulus m, and the power $|d_K|^{s/2}$ of the Dedekind completion in definition 8.5.1, whose conductor is the discriminant alone. In the Hecke setting both contributions appear, and they multiply: the discriminant $|d_K|$ records the ramification of $K/\mathbb{Q}$, common to every χ, while the conductor norm $N\mathfrak{f}_\chi$ records the ramification of χ itself, the finite places where χ is nontrivial. For χ trivial, $\mathfrak{f}_\chi = (1)$ and $N\mathfrak{f}_\chi = 1$, so only $|d_K|$ survives; for $K = \mathbb{Q}$, $|d_K| = 1$, so only $N\mathfrak{f}_\chi$ survives. The shape of each gamma factor is the archimedean Euler factor of χ at the corresponding place: a real place contributes $\Gamma_{\mathbb{R}}(s+a_v)$, the gamma factor of the even or odd character of $\mathbb{R}^\times/\mathbb{R}^\times_{>0}$ according to a_v, and a complex place contributes $\Gamma_{\mathbb{C}}(s)$, the gamma factor attached to the connected group $\mathbb{C}^\times$.

Two reductions confirm that definition 9.3.2 is the common generalization of the two completions already constructed. The first specializes the base field to $\mathbb{Q}$, recovering the completed Dirichlet L-function; the second specializes the character to the trivial one, recovering the completed Dedekind zeta function.

9.3.3 Remark: Reduction to the Dirichlet and Dedekind completions *Reduction to $K = \mathbb{Q}$ (completed Dirichlet L-function).* For $K = \mathbb{Q}$ one has $r_1 = 1$, $r_2 = 0$, and $|d_{\mathbb{Q}}| = 1$. A Hecke character modulo $\mathfrak{m}$ of $\mathbb{Q}$ is, under the correspondence of definition 9.1.5, a primitive Dirichlet character modulo the conductor $\mathfrak{f} = N\mathfrak{f}_\chi$ (the finite part of $\mathfrak{f}_\chi$, an ideal $(\mathfrak{f}) \subset \mathbb{Z}$ identified with the positive integer $\mathfrak{f}$), and its sign at the unique real place is the parity $a_v = a \in \{0, 1\}$ of definition 7.1.6. The single archimedean factor is $\Gamma_{\mathbb{R}}(s + a) = \pi^{-(s+a)/2}\Gamma((s + a)/2)$, so (9.6) reads

$$\Lambda(s, \chi) = (1 \cdot \mathfrak{f})^{s/2}\, \pi^{-(s+a)/2}\, \Gamma\left(\frac{s + a}{2}\right) L(s, \chi)$$

To match definition 7.3.3 exactly, write $\mathfrak{f}^{s/2} = \mathfrak{f}^{(s+a)/2}\,\mathfrak{f}^{-a/2}$; the factor $\mathfrak{f}^{-a/2}$ is a constant absorbed into the normalization of the root number (its analogue is the constant relating the two standard normalizations of the completed Dirichlet L-function), and the remaining $\mathfrak{f}^{(s+a)/2}\pi^{-(s+a)/2} = (\mathfrak{f}/\pi)^{(s+a)/2}$ together with $\Gamma((s + a)/2)$ is precisely the factor $(\mathfrak{f}/\pi)^{(s+a)/2}\Gamma((s + a)/2)$ of definition 7.3.3. Thus $\Lambda(s, \chi)$ reduces, up to the conventional constant, to the completed Dirichlet L-function of chapter 7.
Reduction to χ trivial (completed Dedekind zeta function). For the trivial Hecke character $\chi = \chi_0$ (the principal character of the ray class group), $L(s, \chi_0) = \zeta_K(s)$ by proposition 9.2.4, the conductor is $\mathfrak{f}_{\chi_0} = (1)$ with $N\mathfrak{f}_{\chi_0} = 1$, and χ_0 is even at every real place, so $a_v = 0$ for all v. The archimedean product then has r_1 factors $\Gamma_{\mathbb{R}}(s+0) = \Gamma_{\mathbb{R}}(s)$ and r_2 factors $\Gamma_{\mathbb{C}}(s)$, and (9.6) reads

$$\Lambda(s, \chi_0) = (|d_K| \cdot 1)^{s/2}\, \Gamma_{\mathbb{R}}(s)^{r_1}\, \Gamma_{\mathbb{C}}(s)^{r_2}\, \zeta_K(s) = |d_K|^{s/2}\, \Gamma_{\mathbb{R}}(s)^{r_1}\, \Gamma_{\mathbb{C}}(s)^{r_2}\, \zeta_K(s) = \Lambda_K(s)$$

the completed Dedekind zeta function of definition 8.5.1. Thus $\Lambda(s, \chi)$ is the common refinement of the two earlier completions: the single real factor and conductor $\mathfrak{f}$ of the Dirichlet case, and the r_1 real and r_2 complex factors with conductor $|d_K|$ of the Dedekind case, merge into r_1 real factors shifted by the signs a_v, r_2 complex factors, and the combined conductor $|d_K|\, N\mathfrak{f}_\chi$.

The duplication identity $\Gamma_{\mathbb{R}}(s)\,\Gamma_{\mathbb{R}}(s + 1) = \Gamma_{\mathbb{C}}(s)$ of (8.30), proved in theorem 8.5.5, continues to govern the archimedean factors here: a complex place contributes the single factor $\Gamma_{\mathbb{C}}(s)$, which equals the product of one even real factor $\Gamma_{\mathbb{R}}(s)$ and one odd real factor $\Gamma_{\mathbb{R}}(s + 1)$. This is the analytic form of the fact that a complex place of K corresponds to a conjugate pair of complex embeddings, and that the local field $\mathbb{C}$ contains $\mathbb{R}$ with index 2. The signs a_v at the real places, the absence of signs at the complex places, and the conductor power $(|d_K|\, N\mathfrak{f}_\chi)^{s/2}$ are exactly the data needed to render $\Lambda(s, \chi)$ symmetric under $s \mapsto 1 - s$, as the functional equation of theorem 9.4.2 will record.

A remark on the count of real signs places the parity bookkeeping of theorem 8.5.5 in the present framework. For an abelian extension $K/\mathbb{Q}$ with $\zeta_K = \prod_\chi L(s, \chi)$ a product of Dirichlet L-functions, the number of odd characters in the family equals r_2, the number of complex places, by the signature identity (8.36). In the Hecke setting that count is internalized: a single Hecke character carries its own collection of signs $\{a_v\}$ across the real places of K, and the completed function pairs each real factor with the appropriate shift, with no need to distribute the parities across a family. The Hecke completion thus unifies the per-character parity of the Dirichlet theory and the per-place signature of the Dedekind theory into a single per-place sign attached to one character.

The restriction to finite-order characters is what makes the archimedean factors the elementary gamma factors $\Gamma_{\mathbb{R}}(s+a_v)$ and $\Gamma_{\mathbb{C}}(s)$. A general Hecke character (a *Grössencharakter*) need not have finite order: its restriction to the connected component of the idele class group can be a nontrivial continuous character, encoded by an *infinity-type*, a tuple of exponents attached to the infinite places. The infinity-type modifies the archimedean factors, shifting the gamma argument by a complex amount at the real places and replacing $\Gamma_{\mathbb{C}}(s)$ by a factor of type $\Gamma_{\mathbb{C}}(s+|k_v|/2)$ at a complex place v with infinity-type exponent k_v. The construction of these factors, and the local functional equations that they satisfy, belong to the local theory of the next remark.

9.3.4 Remark: The general infinity-type and Tate's local theory The completion constructed above is the finite-order special case of a uniform local-global formalism. For a general Hecke character χ with a nontrivial infinity-type, the local factor at a complex place v with infinity-type exponent $k_v \in \mathbb{Z}$ is of the form $\Gamma_{\mathbb{C}}(s+|k_v|/2)$, and the factor at a real place is shifted accordingly; the completed L-function is again the product of these archimedean factors with the conductor power and the finite Euler product, and it satisfies a functional equation $s \leftrightarrow 1-s$ with a root number assembled from local constants. The conceptual source of this formalism is Tate's thesis: each place v of K contributes a *local zeta integral*, a Mellin-type integral over the local field $K_v^\times$ against a local character and a Schwartz function, and each local zeta integral satisfies a *local functional equation* relating its value at s to its value at $1-s$ through a local epsilon factor; the global functional equation is the product of these local ones, and the archimedean gamma factors $\Gamma_{\mathbb{R}}(s+a_v)$ and $\Gamma_{\mathbb{C}}(s)$ are precisely the local zeta integrals at the infinite places. This place-by-place language rests on the completions K_v and their Haar measures of [4, places and completions]. The systematic development of the local factors and the local functional equations, in the language of the completions K_v, is given for the Hecke completed L-function in [15, Ch. VII, §8]; Tate's local zeta integrals and the idelic reformulation of Hecke characters are deferred in full to [10, Part XIV], to Tate's thesis in [1], and to the companion treatment in [12].

9.4 Analytic Continuation and the Functional Equation

The non-vanishing of section 9.5 and the generalized Dirichlet theorem it supports both presuppose that the Hecke L-function exists as a function on the whole plane, not on the half-plane alone $\mathrm{Re}(s) > 1$ where its Dirichlet series and Euler product (theorem 9.2.3) converge. For the Dedekind zeta function, the case χ trivial, this continuation and the reflection $s \leftrightarrow 1-s$ governing it were recorded in theorem 8.5.3; for the base field $K = \mathbb{Q}$ and a Dirichlet character they were proved in full in theorem 7.3.4 by twisting Riemann's theta argument. The present section states the common generalization. The completed Hecke L-function $\Lambda(s,\chi)$ of definition 9.3.2, formed by attaching to $L(s,\chi)$ the archimedean factors of the real and complex places of K together with a conductor-type power, continues meromorphically and satisfies a reflection exchanging χ with its conjugate $\bar{\chi}$, carrying a phase, the global root number, of absolute value 1. The trivial-character case returns the bare reflection of ζ_K, and the case $K = \mathbb{Q}$ returns the Gauss-sum root number of theorem 7.3.4; the general statement subsumes both.

The full proof is Hecke's, and it generalizes the four-stage theta-and-Mellin argument of theorem 6.3.3 and theorem 7.3.4 in the same way that theorem 8.5.3 generalized it for the trivial character: from the lattice $\mathbb{Z} \subset \mathbb{R}$ to the Minkowski lattice of $\mathcal{O}_K$, now with the lattice sum weighted by the character χ. The statement is recorded here and the proof is deferred to the standard references by precise citation, the trivial-character case of theorem 8.5.3 serving as the model and the abelian case being established independently at the end of the section.

Theorem 9.4.1: Analytic Continuation of the Hecke L-function

Let K be a number field, $\mathfrak{m}$ a modulus, and χ a Hecke character modulo $\mathfrak{m}$, with completed Hecke L-function $\Lambda(s,\chi)$ of definition 9.3.2. Then $\Lambda(s,\chi)$ extends to a meromorphic function on all of $\mathbb{C}$, and:

1. if χ is nontrivial (that is, not the principal character of the ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$, equivalently χ does not arise from the trivial character of $\mathrm{Cl}(K)$ by the conductor reduction of definition 9.1.5), then $\Lambda(s,\chi)$ is entire;
2. if χ is trivial, then $\Lambda(s,\chi) = \Lambda_K(s)$ is the completed Dedekind zeta function of definition 8.5.1, holomorphic on $\mathbb{C}$ except for simple poles at $s=0$ and $s=1$, as in theorem 8.5.3.

Equivalently, the Hecke L-function $L(s,\chi)$ itself continues to a meromorphic function on $\mathbb{C}$, recovered from $\Lambda(s,\chi)$ by dividing out the nowhere-vanishing archimedean and conductor factors of definition 9.3.2; it is entire when χ is nontrivial, and for χ trivial it is ζ_K, holomorphic except for a simple pole at $s=1$.

The mechanism is the one already seen twice in this book, transported to the number field K and twisted by χ. To each ideal class in the ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$ one attaches a theta series, a sum over the lattice $\iota(\mathfrak{a}) \subset \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^d$ of the Gaussian $e^{-\pi t\langle x,x\rangle}$, $\mathfrak{a}$ an integral ideal in the class, the lattice carrying the Minkowski embedding of [2]. Summing these theta series against the character values χ, with the appropriate weighting at the archimedean places (a homogeneous harmonic polynomial encoding the infinity-type of χ, so that the real places contribute even or odd Gaussians and the complex places contribute the rotational characters), assembles a global theta function $\Theta_\chi(t)$ whose Mellin transform in t reproduces $\Lambda(s,\chi)$, exactly as the Mellin transform of θ_χ produced $\Lambda(s,\chi)$ for $K=\mathbb{Q}$ in (7.8) and the Mellin transform of ψ produced $\Lambda(s)$ in theorem 6.3.3. The transformation law of Θ_χ under $t \mapsto 1/t$ comes from the Poisson summation formula (theorem 6.2.1) applied over the lattice $\iota(\mathfrak{a})$, the covolume $\mathrm{vol}(\mathbb{R}^d/\iota(\mathfrak{a})) = 2^{-r_2}\sqrt{|d_K|}\,N\mathfrak{a}$ of [2] entering as the constant in the dual-lattice term and the Gauss sum of χ entering through the Fourier transform of the character weight, just as the Gauss sum $\tau(\chi)$ entered the twisted theta law (7.6) over $\mathbb{Z}$. Splitting the Mellin integral at $t=1$ and applying the theta transformation law to the tail near 0 produces simultaneously the meromorphic continuation and the reflection $s \leftrightarrow 1-s$; for χ nontrivial the character sum kills the constant term of the theta series, so the integral is entire and no poles appear, while for χ trivial the surviving constant term contributes the two simple poles of Λ_K, precisely as the constant term of ψ contributed the poles of Λ in theorem 6.3.3. The complete argument, including the integration over the unit group that accounts for the r_1+r_2-1 regulator directions of [2] and the harmonic-polynomial weights handling the infinity-type, is carried out in [15, Ch. VII, §8] and in [10, Part XIII–XIV]. The idelic proof, in which χ becomes a Hecke character of the idele class group, the theta argument becomes the Poisson summation formula on the adeles, and the archimedean and finite factors alike become local zeta integrals (Tate's thesis), is given in [1] and deferred, in the form aligned with the present notation, to [12].

The reflection that the same argument produces is the functional equation. As for Dirichlet L-

functions, the symmetry exchanges χ with $\bar{\chi}$ and carries a constant of absolute value 1, the global root number; what is new in the number-field setting is that this root number factors as a product over the places of K of local root numbers, one local epsilon factor for each place, finite or archimedean. The finite local factors are built from the Gauss sums of the components of χ at the primes dividing the conductor, generalizing the single Gauss sum $\tau(\chi)$ of definition 7.3.1; the archimedean local factors are the unit-modulus constants i^{-a} familiar from theorem 7.3.4, one for each real and complex place according to the infinity-type. The global root number is their product, and its absolute value is 1 because each local factor has absolute value 1.

Theorem 9.4.2: Functional Equation of the Hecke L-function

Let χ be a *primitive* Hecke character of a number field K, of conductor $\mathfrak{f}_\chi$ (definition 9.1.5) and modulus $\mathfrak{m} = \mathfrak{f}_\chi$, with completed Hecke L-function $\Lambda(s, \chi)$ of definition 9.3.2. Then $\Lambda(s, \chi)$ satisfies the functional equation

$$\Lambda(s, \chi) = W(\chi)\, \Lambda(1 - s, \bar{\chi}) \tag{9.7}$$

for all $s \in \mathbb{C}$, where the *global root number* (or *global epsilon factor*) $W(\chi)$ is a complex constant, independent of s, of absolute value

$$|W(\chi)| = 1 \tag{9.8}$$

and factors as a product

$$W(\chi) = \prod_v W_v(\chi_v) \tag{9.9}$$

over the places v of K of *local root numbers* (local epsilon factors) $W_v(\chi_v)$, one for each archimedean and finite place, each of absolute value 1 and equal to 1 for all but the finitely many places dividing $\mathfrak{f}_\chi$ and the archimedean places. Moreover:

1. if χ is trivial, then $W(\chi) = +1$ and (9.7) is Hecke's functional equation $\Lambda_K(s) = \Lambda_K(1-s)$ of theorem 8.5.3;
2. if $K = \mathbb{Q}$, then χ is a Dirichlet character, $\Lambda(s, \chi)$ is the completed Dirichlet L-function of definition 7.3.3, and (9.7) is the functional equation of theorem 7.3.4, with the global root number equal to the Gauss-sum root number

$$W(\chi) = \epsilon(\chi) = \frac{\tau(\chi)}{i^a \sqrt{\mathfrak{f}_\chi}}, \qquad a \in \{0, 1\} \text{ the parity of } \chi \tag{9.10}$$

A general (imprimitive) Hecke character is induced by a unique primitive character χ^* at its conductor, with $L(s, \chi) = L(s, \chi^*) \prod_{\mathfrak{p} | \mathfrak{m}_0,\, \mathfrak{p} \nmid \mathfrak{f}_\chi} (1 - \chi^*(\mathfrak{p})(N\mathfrak{p})^{-s})$ differing from $L(s, \chi^*)$ by finitely many Euler factors (definition 9.1.5); these break the $s \leftrightarrow 1 - s$ symmetry, so the clean reflection (9.7) is the one carried by the primitive $\Lambda(s, \chi^*)$, which is why primitivity is assumed. The construction of the local root numbers $W_v(\chi_v)$ and the proof that their product has absolute value 1 are the local-to-global content of Tate's thesis and lie beyond the methods of this book; they are carried out in full in [15, Ch. VII, §8] and in [1], the latter constructing each W_v as the constant in the local functional equation of a local zeta integral and proving $|W_v| = 1$ place by place. The decomposition (9.9) is recorded here because it is the structural feature that the abelian comparison below makes visible: the single global root number of ζ_L for an abelian extension L/K is the product of the root numbers of the

Hecke L-functions into which ζ_L factors, and each of those is itself a product of local epsilon factors. The two listed special cases anchor (9.7) against the results already established. The trivial-character case is the self-dual reflection of theorem 8.5.3, where $\bar{\chi} = \chi$ and the root number is $+1$; the case $K = \mathbb{Q}$ is theorem 7.3.4, where the global root number collapses to a single Gauss sum because $\mathbb{Q}$ has one finite local factor per ramified prime and one archimedean factor at the real place, and the product (9.9) reduces to $\epsilon(\chi) = \tau(\chi)/(i^a\sqrt{\mathfrak{f}_\chi})$.

For an abelian extension the functional equation (9.7) of the individual Hecke L-functions need not be taken on faith from the general theorem: it reproduces, by the same assembly that theorem 8.5.5 carried out over $\mathbb{Q}$, the functional equation of the Dedekind zeta function of the extension. The setting is the number-field analogue of the abelian factorization of section 8.4, with the base field $\mathbb{Q}$ replaced by K and Dirichlet characters replaced by Hecke characters of K.

Theorem 9.4.3: Abelian Consistency over a Number Field

Let L/K be a finite abelian extension of number fields, and let X be the (finite) group of Hecke characters of K that cut out L, in the sense that

$$\zeta_L(s) = \prod_{\chi \in X} L(s, \chi) \tag{9.11}$$

the factorization of ζ_L into Hecke L-functions of K provided by the Artin reciprocity map ([12]; for $K = \mathbb{Q}$ this is the cyclotomic and quadratic factorization of section 8.4). Then Hecke's functional equation for ζ_L,

$$\Lambda_L(s) = \Lambda_L(1-s) \tag{9.12}$$

of theorem 8.5.3 applied to the field L, follows from the functional equations (9.7) of the Hecke L-functions $L(s,\chi)$, $\chi \in X$, together with the conductor-discriminant formula expressing the relative discriminant of L/K as the product of the conductors $\mathfrak{f}_\chi$; in particular the global root number of ζ_L is the product $\prod_{\chi \in X} W(\chi) = +1$.

***Proof*:**
The argument is the one of theorem 8.5.5, transported from the base field $\mathbb{Q}$ to the base field K: the completed object Λ_L is matched against the product of the completed Hecke L-functions $\Lambda(s,\chi)$, and the reflection of each factor is transported through the product. Only the two ingredients that change with the base field are recorded; the combinatorial assembly is identical.

Step 1: Matching the completed functions. By definition 9.3.2 each $\Lambda(s,\chi)$ is $L(s,\chi)$ multiplied by the archimedean factors attached to the infinity-type of χ and by a conductor power $(N\mathfrak{f}_\chi\,|d_K|)^{s/2}$ built from the absolute norm of $\mathfrak{f}_\chi$ and the discriminant of the base field K. Forming the product over $\chi \in X$ and using (9.11), the L-factors multiply to ζ_L, the archimedean factors multiply to the archimedean factors of Λ_L (the $r_1(L)$ real and $r_2(L)$ complex places of L being distributed among the infinity-types of the characters in X, by the place-counting that for $K = \mathbb{Q}$ was the parity count (8.36)), and the conductor powers multiply to $|d_L|^{s/2}$ by the relation

$$|d_L| = |d_K|^{[L:K]}\, N_{K/\mathbb{Q}}\Big(\prod_{\chi \in X} \mathfrak{f}_\chi\Big) \tag{9.13}$$

the conductor-discriminant formula for the relative extension L/K (the tower formula $d_L =$

$d_K^{[L:K]} N_{K/\mathbb{Q}}(\mathfrak{d}_{L/K})$ of [2] combined with the class-field-theoretic identity $\mathfrak{d}_{L/K} = \prod_{\chi \in X} \mathfrak{f}_\chi$ for the relative discriminant of [12]; for $K = \mathbb{Q}$ it is theorem 8.5.4). In the normalization of definition 9.3.2 the conductor enters with the clean exponent $s/2$ and the archimedean factors are the bare $\Gamma_\mathbb{R}(s + a_v)$ and $\Gamma_\mathbb{C}(s)$, so no fixed prefactor survives (unlike the $\mathbb{Q}$-normalization of (8.40), whose $(s+a)/2$ conductor exponent left the constant c_K): the conductor powers multiply exactly to $|d_L|^{s/2}$ and the gamma factors exactly to those of Λ_L. Hence

$$\prod_{\chi \in X} \Lambda(s, \chi) = \Lambda_L(s) \tag{9.14}$$

Step 2: Transporting the reflection. The group X is closed under conjugation $\chi \mapsto \bar{\chi}$, since L/K is Galois and $X \cong \mathrm{Gal}(L/K)$, and conjugation preserves the conductor and the infinity-type. Applying the functional equation (9.7) to each factor and reindexing the right-hand side by $\chi \mapsto \bar{\chi}$ (a permutation of X) gives, with $W = \prod_{\chi \in X} W(\chi)$,

$$\prod_{\chi \in X} \Lambda(s, \chi) = W \prod_{\chi \in X} \Lambda(1 - s, \chi) \tag{9.15}$$

exactly as (8.42) over $\mathbb{Q}$. That $W = +1$ follows by the partition of X into conjugation-fixed characters and genuinely conjugate pairs used in theorem 8.5.5: the trivial character contributes $W(\chi_0) = +1$ by part (1) of theorem 9.4.2; each genuinely conjugate pair $\{\chi, \bar{\chi}\}$ contributes $W(\chi)W(\bar{\chi}) = +1$ by the reflection alone, since applying (9.7) to $\bar{\chi}$ at $1 - s$ and substituting yields $W(\chi)W(\bar{\chi}) = 1$ because $\Lambda(s, \chi)$ is not identically zero; and each conjugation-fixed nontrivial character $\chi = \bar{\chi}$ is real-valued, hence the character of a one-dimensional orthogonal representation, whose root number is $W(\chi) = +1$ by the theorem of Fröhlich and Queyrut on the functional equation of real characters ([7]); the finite-order hypothesis is exactly what makes χ orthogonal and is essential here, since a self-dual character carrying a nontrivial infinity-type can have root number -1. Therefore $W = +1$, and combining (9.15) with (9.14) gives $\Lambda_L(s) = W\,\Lambda_L(1 - s) = \Lambda_L(1 - s)$, the functional equation (9.12), with global root number $\prod_{\chi \in X} W(\chi) = +1$.

The consistency theorem is the number-field counterpart of the abelian computation of theorem 8.5.5 (the case $K = \mathbb{Q}$, where the characters in X are Dirichlet characters and ζ_L is the cyclotomic or quadratic zeta function of section 8.4). It exhibits the global root number of ζ_L, namely $+1$, as the product of the root numbers $W(\chi)$ of the Hecke L-functions that compose it, each of which is in turn the product of local epsilon factors of (9.9). The self-duality of ζ_L is thus assembled from the unit-modulus phases of its Hecke factors, the genuinely complex characters cancelling in conjugate pairs and the real ones contributing $+1$ individually, precisely as for ζ_K over $\mathbb{Q}$. What the general theorem of Hecke adds, and what the abelian assembly cannot reach, is the functional equation (9.7) for a Hecke character χ that does not arise from any abelian factorization, where the root number $W(\chi)$ is a genuine product of local epsilon factors with no expression as a single Gauss sum.

9.4.4 Remark: The completed Hecke L-function as the unifying object The functional equation (9.7) closes the analytic theory of the Hecke L-function in the same way the functional equations of chapters 6 to 8 closed the theories of ζ, $L(s,\chi)$, and ζ_K, and it contains all three as special cases. Setting χ trivial returns Hecke's functional equation (8.31) for ζ_K, and within that the case $K = \mathbb{Q}$ returns the reflection (6.14) of the Riemann zeta function; setting $K = \mathbb{Q}$ with χ a Dirichlet character returns the functional equation (7.4) of chapter 7, root number and all. The completed Hecke L-function $\Lambda(s,\chi)$ is the common generalization toward which Part II has been building: a single object whose Euler product packages the splitting of primes in K weighted by a ray class character, whose pole (present only for the trivial character) carries the arithmetic of section 9.5, and whose functional equation (9.7) fixes the symmetry of its zeros with a root number that is a product of local data over the places of K. The non-abelian extension, in which χ is replaced by an irreducible representation of a Galois group and the Hecke L-function by an Artin L-function whose continuation is reduced to the present theorem through Brauer induction, is the subject of chapter 10; the idelic and automorphic reformulation, in which $\Lambda(s,\chi)$ becomes the zeta integral of a Hecke character of the idele class group and (9.7) becomes Poisson summation on the adeles, is the subject of [12] and, beyond it, of the Langlands program.

9.5 Non-Vanishing at $s = 1$ and Arithmetic Applications

The four preceding sections built the Hecke L-function $L(s,\chi)$ attached to a Hecke (ray class) character χ modulo $\mathfrak{m}$, established its Euler product (theorem 9.2.3), and continued it to the whole plane with a functional equation (theorem 9.4.2). One inequality remains before the arithmetic payoff can be collected: the non-vanishing $L(1,\chi) \neq 0$ for every nontrivial χ. This is the exact analogue, over a general number field K, of the non-vanishing $L(1,\chi) \neq 0$ for nonprincipal Dirichlet characters proved in theorem 7.4.2, and it plays the same role: it is the single fact, absent from the theory of ζ_K itself (where the pole makes the corresponding statement automatic), that converts the pole of ζ_K at $s = 1$ into an equidistribution law for the prime ideals of K among the classes of the ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$.

The strategy reproduces the cyclotomic prototype of chapter 7 verbatim, with the residue classes modulo m replaced by the ray classes modulo $\mathfrak{m}$ and the Riemann zeta function replaced by the Dedekind zeta function. One forms the product of $L(s,\chi)$ over all $h_{\mathfrak{m}}$ characters of $\mathrm{Cl}_{\mathfrak{m}}(K)$; orthogonality of those characters renders the product a Dirichlet series in $N\mathfrak{a}$ with non-negative coefficients, while the trivial character contributes the pole of ζ_K at $s = 1$ (theorem 8.3.2). A zero of any single factor at $s = 1$ would cancel that pole and leave a non-negative-coefficient series holomorphic where it cannot converge, contradicting Landau's theorem (proposition 2.1.5). The argument runs exactly as for ζ_m in theorem 7.4.2; only the bookkeeping of ideals in place of integers differs.

The product over the ray class characters is the object on which the whole argument turns. Its two relevant properties, non-negative coefficients and divergence at a real point to the right of the abscissa 0, are recorded first.

Proposition 9.5.1: The Product $\prod_\chi L(s,\chi)$ over Ray Class Characters

Fix a modulus $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$ of K (definition 9.1.1) and set

$$\zeta_\mathfrak{m}(s) = \prod_{\chi \bmod \mathfrak{m}} L(s,\chi)$$

the product running over all $h_\mathfrak{m}$ Hecke characters of the ray class group $\mathrm{Cl}_\mathfrak{m}(K)$ (definition 9.1.2, theorem 9.1.3). For $\mathrm{Re}(s) > 1$:

1. $\zeta_\mathfrak{m}$ has the Euler product

$$\zeta_\mathfrak{m}(s) = \prod_{\mathfrak{p}\nmid\mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-f_\mathfrak{p} s}\right)^{-h_\mathfrak{m}/f_\mathfrak{p}}$$

   where $f_\mathfrak{p}$ is the order of the class of $\mathfrak{p}$ in $\mathrm{Cl}_\mathfrak{m}(K)$, the product running over the prime ideals of K coprime to $\mathfrak{m}_0$;

2. $\zeta_\mathfrak{m}$ is a Dirichlet series $\sum_\mathfrak{a} c(\mathfrak{a})\,(N\mathfrak{a})^{-s} = \sum_{n\geq 1} c_n n^{-s}$ in the ideal norm, with non-negative real coefficients $c_n \geq 0$ and constant term $c_1 = 1$; and

3. the real function $\sigma \mapsto \zeta_\mathfrak{m}(\sigma)$ on $(1,\infty)$ satisfies $\zeta_\mathfrak{m}(\sigma) \to +\infty$ as $\sigma \to (1/h_\mathfrak{m})^+$, so the Dirichlet series of (2) has abscissa of convergence at least $1/h_\mathfrak{m}$.

***Proof*:**
Step 1: The logarithm of the product. Each factor $L(s,\chi)$ has the absolutely convergent Euler product $\prod_\mathfrak{p}(1-\chi(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1}$ on $\mathrm{Re}(s) > 1$ (theorem 9.2.3), with $|\chi(\mathfrak{p})(N\mathfrak{p})^{-s}| < 1$ there since $N\mathfrak{p} \geq 2$. Taking the principal logarithm factor by factor, exactly as in proposition 2.3.5,

$$\log L(s,\chi) = \sum_\mathfrak{p}\sum_{k=1}^\infty \frac{\chi(\mathfrak{p})^k}{k\,(N\mathfrak{p})^{ks}} = \sum_{\mathfrak{p}\nmid\mathfrak{m}_0}\sum_{k=1}^\infty \frac{\chi(\mathfrak{p}^k)}{k\,(N\mathfrak{p})^{ks}}$$

the second form using $\chi(\mathfrak{p})^k = \chi(\mathfrak{p}^k)$ (complete multiplicativity of χ on ideals, definition 9.1.5) and discarding the primes $\mathfrak{p}\big|\mathfrak{m}_0$, where $\chi(\mathfrak{p}) = 0$ by the convention extending χ to all ideals. Summing over the $h_\mathfrak{m}$ characters and interchanging the order of summation, legitimate because the triple sum converges absolutely and there are finitely many characters,

$$\log\zeta_\mathfrak{m}(s) = \sum_{\chi \bmod \mathfrak{m}} \log L(s,\chi) = \sum_{\mathfrak{p}\nmid\mathfrak{m}_0}\sum_{k=1}^\infty \frac{1}{k\,(N\mathfrak{p})^{ks}} \sum_{\chi \bmod \mathfrak{m}} \chi(\mathfrak{p}^k)$$

the branch of $\log\zeta_\mathfrak{m}$ being the sum of the branches fixed by proposition 2.3.5.

Step 2: Orthogonality collapses the character sum. For $\mathfrak{p} \nmid \mathfrak{m}_0$ the ideal $\mathfrak{p}^k$ is coprime to $\mathfrak{m}_0$, hence defines a class in $\mathrm{Cl}_\mathfrak{m}(K)$, and the characters χ mod $\mathfrak{m}$ are exactly the characters of the finite abelian group $\mathrm{Cl}_\mathfrak{m}(K)$ (definition 9.1.5). Column orthogonality of the characters of a finite abelian group (theorem 7.1.4(1), applied to $\mathrm{Cl}_\mathfrak{m}(K)$ in place of $(\mathbb{Z}/m\mathbb{Z})^\times$) evaluates the inner sum:

$$\sum_{\chi \bmod \mathfrak{m}} \chi(\mathfrak{p}^k) = \begin{cases} h_\mathfrak{m} & \text{if } \mathfrak{p}^k \in P_1^\mathfrak{m}, \\ 0 & \text{otherwise} \end{cases}$$

where $P_1^\mathfrak{m}$ is the ray of principal ideals trivial in $\mathrm{Cl}_\mathfrak{m}(K)$. The class of $\mathfrak{p}^k$ is trivial precisely when the order $f_\mathfrak{p}$ of the class of $\mathfrak{p}$ divides k, that is, when $k = jf_\mathfrak{p}$ for some $j \geq 1$. Substituting $k = jf_\mathfrak{p}$,

$$\log \zeta_\mathfrak{m}(s) = h_\mathfrak{m} \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \sum_{j=1}^{\infty} \frac{1}{jf_\mathfrak{p}\,(N\mathfrak{p})^{jf_\mathfrak{p}s}} \tag{9.16}$$

a double series with non-negative real coefficients on the real axis $s = \sigma > 0$. Exponentiating the contribution of a single prime $\mathfrak{p}$,

$$\exp\left(\frac{h_\mathfrak{m}}{f_\mathfrak{p}} \sum_{j=1}^{\infty} \frac{\left((N\mathfrak{p})^{-f_\mathfrak{p}s}\right)^j}{j}\right) = \left(1 - (N\mathfrak{p})^{-f_\mathfrak{p}s}\right)^{-h_\mathfrak{m}/f_\mathfrak{p}}$$

by $\sum_{j\geq 1} z^j/j = -\log(1-z)$ with $z = (N\mathfrak{p})^{-f_\mathfrak{p}s}$, valid since $|z| < 1$ for $\mathrm{Re}(s) > 0$. Taking the product over $\mathfrak{p} \nmid \mathfrak{m}_0$ gives the Euler product of (1).

Step 3: Non-negative coefficients and the constant term. The exponent $h_\mathfrak{m}/f_\mathfrak{p}$ is a positive integer, since $f_\mathfrak{p}$ is the order of an element of the group $\mathrm{Cl}_\mathfrak{m}(K)$ of order $h_\mathfrak{m}$, and so divides $h_\mathfrak{m}$ by Lagrange's theorem. Each factor of the Euler product expands by the generalized binomial series

$$\left(1 - (N\mathfrak{p})^{-f_\mathfrak{p}s}\right)^{-h_\mathfrak{m}/f_\mathfrak{p}} = \sum_{j=0}^{\infty} \binom{h_\mathfrak{m}/f_\mathfrak{p} + j - 1}{j} (N\mathfrak{p})^{-jf_\mathfrak{p}s}$$

with non-negative integer coefficients and constant term ($j = 0$) equal to 1. Each factor is thus a Dirichlet series in the variable $N\mathfrak{p}$ with non-negative coefficients; grouping the resulting prime-power contributions by the value of the norm $N\mathfrak{a}$ (the norm is completely multiplicative, $N(\mathfrak{a}\mathfrak{b}) = N\mathfrak{a}\cdot N\mathfrak{b}$, [2]) expresses $\zeta_\mathfrak{m}(s) = \sum_\mathfrak{a} c(\mathfrak{a})(N\mathfrak{a})^{-s}$, and collecting ideals of equal norm gives the ordinary Dirichlet series $\sum_n c_n n^{-s}$ with $c_n = \sum_{N\mathfrak{a}=n} c(\mathfrak{a}) \geq 0$ a finite sum of non-negative terms. The constant term is $c_1 = c(\mathcal{O}_K) = 1$, the product of the constant terms of the factors. This rearrangement is the absolutely convergent regrouping of the product on $\mathrm{Re}(s) > 1$, where every factor converges absolutely, and gives (2).

Step 4: Divergence at $s = 1/h_\mathfrak{m}$. Work on the real axis, $s = \sigma > 1/h_\mathfrak{m}$, where every term of (9.16) is non-negative. Discarding the terms with $j \geq 2$,

$$\log \zeta_\mathfrak{m}(\sigma) \geq h_\mathfrak{m} \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{f_\mathfrak{p}\,(N\mathfrak{p})^{f_\mathfrak{p}\sigma}}$$

For each $\mathfrak{p} \nmid \mathfrak{m}_0$ the order $f_\mathfrak{p}$ divides $h_\mathfrak{m}$, so $f_\mathfrak{p} \leq h_\mathfrak{m}$, whence $(N\mathfrak{p})^{f_\mathfrak{p}\sigma} \leq (N\mathfrak{p})^{h_\mathfrak{m}\sigma}$ and, using also $f_\mathfrak{p} \leq h_\mathfrak{m}$,

$$\log \zeta_\mathfrak{m}(\sigma) \geq h_\mathfrak{m} \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{h_\mathfrak{m}\,(N\mathfrak{p})^{h_\mathfrak{m}\sigma}} = \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{(N\mathfrak{p})^{h_\mathfrak{m}\sigma}}$$

As $\sigma \to (1/h_\mathfrak{m})^+$ the exponent $h_\mathfrak{m}\sigma \to 1^+$, and the prime-ideal sum $\sum_\mathfrak{p} (N\mathfrak{p})^{-h_\mathfrak{m}\sigma}$ diverges to $+\infty$: this is the logarithmic divergence of the Dedekind prime sum recorded in (9.18) below (equivalently, $\log \zeta_K(h_{\mathfrak{m}}\sigma) \to +\infty$ as $h_{\mathfrak{m}}\sigma \to 1^+$ by the pole of ζ_K, theorem 8.3.2), the finitely many primes $\mathfrak{p} \big| \mathfrak{m}_0$ omitted altering the sum by a bounded amount. Hence $\log \zeta_{\mathfrak{m}}(\sigma) \to +\infty$, so $\zeta_{\mathfrak{m}}(\sigma) \to +\infty$. Because the coefficients c_n are non-negative, a finite value of the series at a real point σ_0 would force convergence for all $\sigma \geq \sigma_0$; the blow-up at $1/h_{\mathfrak{m}}$ thus shows the abscissa of convergence is at least $1/h_{\mathfrak{m}}$, which is (3).

The product $\zeta_{\mathfrak{m}}(s)$ is the precise analogue of the cyclotomic product $\zeta_m(s)$ of proposition 7.4.1, and by class field theory it is itself a Dedekind zeta function: $\zeta_{\mathfrak{m}} = \zeta_{H_{\mathfrak{m}}}$ is the Dedekind zeta function of the ray class field $H_{\mathfrak{m}}/K$, the abelian extension whose Galois group is identified with $\mathrm{Cl}_{\mathfrak{m}}(K)$ by the Artin map [15, Ch. VI], just as $\zeta_m = \zeta_{\mathbb{Q}(\zeta_m)}$ for the cyclotomic field. This identification is not needed for the non-vanishing proof, which uses only the two extracted properties: $\zeta_{\mathfrak{m}}$ is a Dirichlet series with non-negative coefficients, and it diverges at the real point $s = 1/h_{\mathfrak{m}} > 0$. Non-negativity is the hypothesis of Landau's theorem, and the divergence at a positive real point is the contradiction it is played against.

Theorem 9.5.2: Non-Vanishing at $s = 1$

Let χ be a nontrivial Hecke character modulo $\mathfrak{m}$. Then

$$L(1, \chi) \neq 0$$

***Proof*:**
By theorem 9.4.1 a nontrivial $L(s, \chi)$ is entire, so $L(1, \chi)$ is a well-defined finite value and the statement is meaningful. Two facts about conjugation are used. First, $L(s, \bar{\chi}) = \overline{L(\bar{s}, \chi)}$ on $\mathrm{Re}(s) > 1$, since term by term $\overline{\chi(\mathfrak{a})(N\mathfrak{a})^{-\bar{s}}} = \bar{\chi}(\mathfrak{a})(N\mathfrak{a})^{-s}$, and this identity continues to the whole plane; evaluating at the real point $s = 1$ gives $L(1, \bar{\chi}) = \overline{L(1, \chi)}$, so $L(1, \chi) = 0$ if and only if $L(1, \bar{\chi}) = 0$. Second, $\bar{\chi}$ is again a nontrivial character modulo $\mathfrak{m}$, so its L-function is one of the factors of $\zeta_{\mathfrak{m}}$. The trivial character χ_0 contributes the factor $L(s, \chi_0)$, which by proposition 9.2.4 differs from ζ_K only by the finitely many Euler factors at the primes $\mathfrak{p} \big| \mathfrak{m}_0$,

$$L(s, \chi_0) = \zeta_K(s) \prod_{\mathfrak{p} \big| \mathfrak{m}_0} \left(1 - (N\mathfrak{p})^{-s}\right) \tag{9.17}$$

and so, the deleted factors being holomorphic and nonzero at $s = 1$, inherits from ζ_K a simple pole at $s = 1$ (theorem 8.3.2). This is the only pole among the factors of $\zeta_{\mathfrak{m}}$, every nontrivial factor being entire.

Step 1: Complex characters. Suppose first $\chi \neq \bar{\chi}$, and suppose toward a contradiction $L(1, \chi) = 0$. Then $L(1, \bar{\chi}) = \overline{L(1, \chi)} = 0$ also, and $\chi, \bar{\chi}$ are two *distinct* nontrivial characters modulo $\mathfrak{m}$, each contributing a factor of $\zeta_{\mathfrak{m}}$ vanishing at $s = 1$. Count the order of $\zeta_{\mathfrak{m}}$ at $s = 1$ factor by factor: the trivial factor $L(s, \chi_0)$ contributes order -1 by its simple pole (9.17); the factors $L(s, \chi)$ and $L(s, \bar{\chi})$ contribute order at least $+1$ apiece; every remaining factor is entire, contributing order at least 0. The order of the product at $s = 1$ is the sum of the orders of the factors, hence at least

$$(-1) + (+1) + (+1) + 0 = +1 > 0$$

so $\zeta_{\mathfrak{m}}$ is holomorphic at $s = 1$ with a zero there. The single pole of $\zeta_{\mathfrak{m}}$ being cancelled, $\zeta_{\mathfrak{m}}$ is entire.

Apply Landau's theorem on Dirichlet series with non-negative coefficients (proposition 2.1.5; [6, Ch. 4]): a Dirichlet series with non-negative coefficients that extends holomorphically to a half-plane $\mathrm{Re}(s) > \sigma_0$ has abscissa of convergence at most σ_0. The series $\zeta_{\mathfrak{m}} = \sum_n c_n n^{-s}$ has non-negative coefficients (proposition 9.5.1(2)) and, under the present assumption, is entire, so its abscissa of convergence would be at most 0. But proposition 9.5.1(3) shows the abscissa is at least $1/h_{\mathfrak{m}} > 0$, a contradiction. Therefore $L(1, \chi) \neq 0$ for $\chi \neq \bar{\chi}$.

Step 2: Real characters. Suppose now $\chi = \bar{\chi}$, so χ takes only the values $\{0, \pm 1\}$ (each nonzero value is a root of unity equal to its conjugate, hence ± 1), and χ is a character of order 2 of $\mathrm{Cl}_{\mathfrak{m}}(K)$. Suppose toward a contradiction $L(1, \chi) = 0$. The self-conjugate χ supplies no second zero, so $\zeta_{\mathfrak{m}}$ alone no longer forces the contradiction; instead work with the smaller product

$$F(s) = \zeta_K(s)\, L(s, \chi)$$

the Dedekind zeta function of K times the single L-function. By theorem 8.3.2 the factor ζ_K is holomorphic on $\mathrm{Re}(s) > 0$ apart from a simple pole at $s = 1$, and $L(s, \chi)$ is entire (theorem 9.4.1). At $s = 1$ the simple pole of ζ_K, of order -1, meets the assumed zero of $L(s, \chi)$, of order at least $+1$, so F is holomorphic at $s = 1$; away from $s = 1$ both factors are holomorphic. Thus F is holomorphic on all of $\mathrm{Re}(s) > 0$.

Step 3: The coefficients of F. For $\mathrm{Re}(s) > 1$ both $\zeta_K(s) = \sum_{\mathfrak{a}} (N\mathfrak{a})^{-s}$ and $L(s, \chi) = \sum_{\mathfrak{a}} \chi(\mathfrak{a})(N\mathfrak{a})^{-s}$ converge absolutely, so their product is the Dirichlet series of the ideal-theoretic convolution: $F(s) = \sum_{\mathfrak{a}} b(\mathfrak{a})(N\mathfrak{a})^{-s}$ with

$$b(\mathfrak{a}) = \sum_{\mathfrak{d} \mid \mathfrak{a}} \chi(\mathfrak{d})$$

the sum over integral divisors $\mathfrak{d}$ of $\mathfrak{a}$, since ζ_K has all ideal coefficients equal to 1. The arithmetic function $\mathfrak{a} \mapsto b(\mathfrak{a})$ is the convolution of the completely multiplicative χ with the constant 1, hence multiplicative on ideals (unique factorization into prime ideals, [2]); on a prime-power ideal $\mathfrak{p}^k$,

$$b(\mathfrak{p}^k) = \sum_{i=0}^{k} \chi(\mathfrak{p})^i = 1 + \chi(\mathfrak{p}) + \cdots + \chi(\mathfrak{p})^k$$

a geometric sum with real ratio $\chi(\mathfrak{p}) \in \{0, +1, -1\}$. The three cases give

$$b(\mathfrak{p}^k) = \begin{cases} k+1 & \text{if } \chi(\mathfrak{p}) = +1, \\ 1 & \text{if } \chi(\mathfrak{p}) = 0, \\ 1 & \text{if } \chi(\mathfrak{p}) = -1 \text{ and } k \text{ even}, \\ 0 & \text{if } \chi(\mathfrak{p}) = -1 \text{ and } k \text{ odd} \end{cases}$$

In every case $b(\mathfrak{p}^k) \geq 0$, and $b(\mathfrak{p}^k) \geq 1$ whenever k is even. By multiplicativity $b(\mathfrak{a}) = \prod_{\mathfrak{p}^k \| \mathfrak{a}} b(\mathfrak{p}^k) \geq 0$ for all $\mathfrak{a}$, so collecting ideals of equal norm, $F(s) = \sum_n b_n n^{-s}$ with $b_n = \sum_{N\mathfrak{a}=n} b(\mathfrak{a}) \geq 0$. Moreover, if $\mathfrak{a} = \mathfrak{c}^2$ is a square ideal then every exponent in its factorization is even, so every local factor $b(\mathfrak{p}^k) \geq 1$ and hence $b(\mathfrak{c}^2) \geq 1$; in particular b_n receives a contribution at least 1 whenever $n = (N\mathfrak{c})^2$ is the norm of a square ideal, and the norm-one ideal $\mathcal{O}_K$ already gives $b_1 \geq 1$.

Step 4: Landau's contradiction. The non-negativity of the coefficients b_n together with the holomorphy of F on $\mathrm{Re}(s) > 0$ feeds into Landau's theorem (proposition 2.1.5; [6, Ch. 4]) as in Step 1: the abscissa of convergence of $\sum_n b_n n^{-s}$ would be at most 0, so the series would converge for every real $\sigma > 0$. This is contradicted by a square-ideal lower bound. Restricting the non-negative sum to square ideals $\mathfrak{a} = \mathfrak{c}^2$ and using $b(\mathfrak{c}^2) \geq 1$ together with $N(\mathfrak{c}^2) = (N\mathfrak{c})^2$, for real $\sigma > 0$,

$$F(\sigma) = \sum_{\mathfrak{a}} \frac{b(\mathfrak{a})}{(N\mathfrak{a})^{\sigma}} \geq \sum_{\mathfrak{c}} \frac{b(\mathfrak{c}^2)}{(N\mathfrak{c})^{2\sigma}} \geq \sum_{\mathfrak{c}} \frac{1}{(N\mathfrak{c})^{2\sigma}} = \zeta_K(2\sigma)$$

the sum running over all integral ideals $\mathfrak{c}$. The Dedekind zeta function $\zeta_K(2\sigma)$ diverges as $2\sigma \to 1^+$, by its pole at $s = 1$ (theorem 8.3.2); equivalently, its defining series $\sum_{\mathfrak{c}} (N\mathfrak{c})^{-2\sigma}$ diverges for $2\sigma \leq 1$, that is for $\sigma \leq 1/2$ (proposition 9.2.2, the convergence holding only for $\mathrm{Re} > 1$). Hence the series for F diverges at every real $\sigma \leq 1/2$, so its abscissa of convergence is at least $1/2$, contradicting the value at most 0 forced by Landau's theorem. The assumption $L(1, \chi) = 0$ is therefore untenable, and $L(1, \chi) \neq 0$ for the order-two character χ, completing the proof.

The two cases meet the obstruction from opposite directions, exactly as in theorem 7.4.2. A complex character carries its own undoing: a zero at $s = 1$ duplicates into a zero of the conjugate factor, and the double zero overwhelms the single pole of $\zeta_{\mathfrak{m}}$, leaving a non-negative-coefficient series holomorphic where it cannot converge. A real (order-two) character has no conjugate partner, and the substitute is the product $\zeta_K(s)L(s, \chi)$, whose coefficients $b(\mathfrak{a}) = \sum_{\mathfrak{d} | \mathfrak{a}} \chi(\mathfrak{d})$ are visibly non-negative and at least 1 on the square ideals; a zero at $s = 1$ would again make this holomorphic on $\mathrm{Re}(s) > 0$, contradicting the divergence its square terms guarantee. A second, purely arithmetic route to the order-two case is available, paralleling the algebraic non-vanishing of theorem 8.4.2: the product $\zeta_K(s)L(s, \chi)$ is the Dedekind zeta function of the quadratic extension of K cut out by χ through the Artin map, so $L(s, \chi) = \zeta_{K'}(s)/\zeta_K(s)$ is a ratio of two functions each with a simple pole of known positive residue at $s = 1$ (theorem 8.3.2), forcing $L(1, \chi)$ to equal the positive quotient $\kappa_{K'}/\kappa_K \neq 0$; the precise generalized class number formula underlying this is [15, Ch. VII, §5]. The Landau argument above is the elementary route, requiring only the positivity of a Dirichlet series, and it is the one on which the equidistribution theorem below is built.

The non-vanishing in hand, the distribution of the prime ideals of K among the ray classes follows by the argument that proved Dirichlet's theorem and its density refinement (theorem 7.5.1, theorem 7.5.4), now with the characters of $\mathrm{Cl}_{\mathfrak{m}}(K)$ in the role of the Dirichlet characters and ζ_K in the role of ζ. The notion of density appropriate to prime ideals is the direct analogue of definition 7.5.3.

Definition 9.5.3: Dirichlet Density of a Set of Prime Ideals

Let S be a set of prime ideals of K. The *Dirichlet density* of S is

$$\delta(S) = \lim_{s \to 1^+} \frac{\displaystyle\sum_{\mathfrak{p} \in S} (N\mathfrak{p})^{-s}}{\displaystyle\sum_{\mathfrak{p}} (N\mathfrak{p})^{-s}}$$

when the limit exists, the denominator running over all prime ideals of K.

The denominator has the same logarithmic singularity as for $K = \mathbb{Q}$. Taking the logarithm of the

Euler product of ζ_K (theorem 8.1.4) and isolating the linear term, exactly as in (7.11), the prime-power tail $\sum_{\mathfrak{p}} \sum_{k\geq 2} k^{-1}(N\mathfrak{p})^{-ks}$ is bounded on $s \geq 1$ by $\sum_{\mathfrak{p}}(N\mathfrak{p})^{-2}\big(1-(N\mathfrak{p})^{-1}\big)^{-1} \leq \sum_{\mathfrak{a}}(N\mathfrak{a})^{-2} = \zeta_K(2) < \infty$, so

$$\sum_{\mathfrak{p}} \frac{1}{(N\mathfrak{p})^s} = \log \zeta_K(s) + O(1) = \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+) \tag{9.18}$$

the second equality from the simple pole of ζ_K at $s = 1$ (theorem 8.3.2): near $s = 1$, $\zeta_K(s) = \kappa_K/(s-1) + O(1)$ with $\kappa_K > 0$, so $\log \zeta_K(s) = \log \frac{1}{s-1} + \log \kappa_K + o(1)$. The denominator in definition 9.5.3 therefore tends to $+\infty$ like $\log \frac{1}{s-1}$, and the density may equivalently be written with $\log \frac{1}{s-1}$ in the denominator. For $K = \mathbb{Q}$ the prime ideals are the rational primes, $N(p) = p$, and definition 9.5.3 reduces to definition 7.5.3.

Theorem 9.5.4: Primes in Ray Classes

Let $\mathfrak{m}$ be a modulus of K and $C \in \mathrm{Cl}_{\mathfrak{m}}(K)$ a fixed ray class. The set of prime ideals $\mathfrak{p}$ of K, coprime to $\mathfrak{m}_0$, whose class in $\mathrm{Cl}_{\mathfrak{m}}(K)$ is C has Dirichlet density

$$\delta\big(\{\mathfrak{p} : [\mathfrak{p}] = C\}\big) = \frac{1}{h_{\mathfrak{m}}}$$

Equivalently, as $s \to 1^+$,

$$\sum_{\substack{\mathfrak{p}\nmid\mathfrak{m}_0 \\ [\mathfrak{p}]=C}} \frac{1}{(N\mathfrak{p})^s} = \frac{1}{h_{\mathfrak{m}}} \log \frac{1}{s-1} + O(1)$$

The $h_{\mathfrak{m}}$ ray classes thus carry the prime ideals of K in equal proportion, each with density $1/h_{\mathfrak{m}}$; in particular each class contains infinitely many prime ideals.

***Proof*:**
Step 1: The prime sum for one character. Fix a Hecke character χ modulo $\mathfrak{m}$. For real $s > 1$ the Euler product (theorem 9.2.3) has no vanishing factor, so $L(s,\chi) \neq 0$, and proposition 2.3.5 gives the absolutely convergent expansion

$$\log L(s,\chi) = \sum_{\mathfrak{p}} \frac{\chi(\mathfrak{p})}{(N\mathfrak{p})^s} + \sum_{\mathfrak{p}} \sum_{k=2}^{\infty} \frac{\chi(\mathfrak{p})^k}{k\,(N\mathfrak{p})^{ks}}$$

The prime-power tail is bounded uniformly in χ and in $s \geq 1$: using $|\chi(\mathfrak{p})| \leq 1$ and the bound of (9.18),

$$\left|\sum_{\mathfrak{p}} \sum_{k=2}^{\infty} \frac{\chi(\mathfrak{p})^k}{k\,(N\mathfrak{p})^{ks}}\right| \leq \sum_{\mathfrak{p}} \sum_{k=2}^{\infty} \frac{1}{(N\mathfrak{p})^k} = \sum_{\mathfrak{p}} \frac{1}{N\mathfrak{p}\,(N\mathfrak{p}-1)} \leq \zeta_K(2) < \infty$$

so the tail is $O(1)$ as $s \to 1^+$, with implied constant independent of χ, and

$$\log L(s,\chi) = \sum_{\mathfrak{p}} \frac{\chi(\mathfrak{p})}{(N\mathfrak{p})^s} + O(1) \qquad (s \to 1^+) \tag{9.19}$$

the primes $\mathfrak{p}|\mathfrak{m}_0$ contributing nothing to the linear term, since $\chi(\mathfrak{p}) = 0$ there.

Step 2: Weight by $\overline{\chi(C)}$ and sum over characters. Multiply (9.19) by $\overline{\chi(C)}$, of modulus 1, and sum over all $h_\mathfrak{m}$ characters. The bounded terms remain bounded, and interchanging the finite character sum with the prime-ideal sum (legitimate for $s > 1$),

$$\sum_{\chi \bmod \mathfrak{m}} \overline{\chi(C)} \log L(s,\chi) = \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{(N\mathfrak{p})^s} \left(\sum_{\chi \bmod \mathfrak{m}} \chi(\mathfrak{p})\, \overline{\chi(C)} \right) + O(1)$$

By row orthogonality of the characters of $\mathrm{Cl}_\mathfrak{m}(K)$ (theorem 7.1.4; the indicator identity (7.1), applied to the finite abelian group $\mathrm{Cl}_\mathfrak{m}(K)$), the inner sum is the indicator of the class C:

$$\sum_{\chi \bmod \mathfrak{m}} \chi(\mathfrak{p})\, \overline{\chi(C)} = \begin{cases} h_\mathfrak{m} & \text{if } [\mathfrak{p}] = C, \\ 0 & \text{otherwise} \end{cases}$$

so the prime sum collapses onto the single ray class C:

$$\sum_{\chi \bmod \mathfrak{m}} \overline{\chi(C)} \log L(s,\chi) = h_\mathfrak{m} \sum_{\substack{\mathfrak{p} \nmid \mathfrak{m}_0 \\ [\mathfrak{p}] = C}} \frac{1}{(N\mathfrak{p})^s} + O(1) \qquad (s \to 1^+) \tag{9.20}$$

Step 3: The trivial character contributes the pole. Separate the trivial character χ_0, for which $\overline{\chi_0(C)} = 1$. By (9.19) for χ_0, together with $\chi_0(\mathfrak{p}) = 1$ for $\mathfrak{p} \nmid \mathfrak{m}_0$ and the Dedekind prime sum (9.18) (the finitely many primes $\mathfrak{p} \big| \mathfrak{m}_0$ altering the sum by $O(1)$),

$$\log L(s,\chi_0) = \sum_{\mathfrak{p} \nmid \mathfrak{m}_0} \frac{1}{(N\mathfrak{p})^s} + O(1) = \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+) \tag{9.21}$$

so $\log L(s,\chi_0) \to +\infty$, consistent with the pole of $L(s,\chi_0)$ at $s = 1$ recorded in (9.17).

Step 4: The nontrivial characters stay bounded. For each nontrivial χ the function $L(s,\chi)$ is entire (theorem 9.4.1) and, decisively, nonzero at $s = 1$ by theorem 9.5.2. A function holomorphic and nonzero at a point has a logarithm holomorphic near that point, so $\log L(s,\chi)$ extends continuously to the finite value $\log L(1,\chi)$ as $s \to 1^+$, and each nontrivial term is bounded:

$$\overline{\chi(C)} \log L(s,\chi) = O(1) \qquad (s \to 1^+)$$

This is where the non-vanishing earns its place: were $L(1,\chi) = 0$ for some nontrivial χ, the term $\log L(s,\chi)$ would tend to $-\infty$ and could cancel the pole from χ_0, leaving the prime sum bounded and the conclusion false. Theorem 9.5.2 forbids this.

Step 5: Conclusion. Insert Steps 3 and 4 into the left of (9.20): the principal term contributes $\log \frac{1}{s-1}$, every other term is $O(1)$, so

$$\sum_{\chi \bmod \mathfrak{m}} \overline{\chi(C)} \log L(s,\chi) = \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

Equating with the right of (9.20) and dividing by $h_{\mathfrak{m}}$,

$$\sum_{\substack{\mathfrak{p}\nmid\mathfrak{m}_0 \\ [\mathfrak{p}]=C}} \frac{1}{(N\mathfrak{p})^s} = \frac{1}{h_{\mathfrak{m}}} \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+) \tag{9.22}$$

Dividing (9.22) by the Dedekind prime sum (9.18) and letting $s \to 1^+$, with both $O(1)$ terms becoming $O\big(1/\log \frac{1}{s-1}\big) \to 0$ after division by $\log \frac{1}{s-1}$, gives

$$\delta\big(\{\mathfrak{p} : [\mathfrak{p}] = C\}\big) = \lim_{s\to 1^+} \frac{\frac{1}{h_{\mathfrak{m}}} \log \frac{1}{s-1} + O(1)}{\log \frac{1}{s-1} + O(1)} = \frac{1}{h_{\mathfrak{m}}}$$

the density of definition 9.5.3. The right side of (9.22) tending to $+\infty$, the class C contains infinitely many prime ideals. Summing the densities over the $h_{\mathfrak{m}}$ classes, which partition the prime ideals coprime to $\mathfrak{m}_0$ (and the finitely many $\mathfrak{p}\big|\mathfrak{m}_0$ affect no density), gives $h_{\mathfrak{m}} \cdot \frac{1}{h_{\mathfrak{m}}} = 1$, the density of all prime ideals.

The coefficient $1/h_{\mathfrak{m}}$ is the analytic measure of the principle that, to the resolution the L-function method detects, the prime ideals of K prefer no ray class over another. The pole of ζ_K at $s = 1$, with residue $\kappa_K > 0$ (theorem 8.3.2), is the single source of divergence in the prime-ideal sum, and orthogonality distributes that one pole evenly across the $h_{\mathfrak{m}}$ ray classes. For $K = \mathbb{Q}$ and the modulus $\mathfrak{m} = (m) \cdot \{\infty\}$ given by an integer m and the real place, the ray class group $\mathrm{Cl}_{\mathfrak{m}}(\mathbb{Q})$ is $(\mathbb{Z}/m\mathbb{Z})^\times$ (the totally positive principal ideals (α) with $\alpha \equiv 1 \pmod{m}$ being exactly those with a positive generator $\equiv 1 \pmod{m}$), the Hecke characters are the Dirichlet characters modulo m, and theorem 9.5.4 reduces to the equidistribution $\delta = 1/\varphi(m)$ of theorem 7.5.4. The theorem thus extends Dirichlet's theorem from $\mathbb{Q}$ to every number field, and from congruence classes to ray classes.

The leading-order density does not exhaust the arithmetic content of the pole. As for $K = \mathbb{Q}$, the residue κ_K of ζ_K admits a closed evaluation, the analytic class number formula (theorem 8.3.2), and the same evaluation can be carried out for the ray class data, expressing the leading behaviour of the relevant zeta functions in terms of the ray class number $h_{\mathfrak{m}}$, the regulator, and the discriminant of K. The precise generalized analytic class number formula, giving the leading term of $\zeta_{\mathfrak{m}}$ (equivalently of $\zeta_{H_{\mathfrak{m}}}$) at $s = 1$ in terms of these invariants, is developed in [15, Ch. VII, §5] and [10, Part XIII]; it is the source of the closed forms for $L(1, \chi)$ generalizing the formulas of chapter 8.

9.5.5 Remark: The abelian theory complete, and the road to Chebotarev With theorem 9.5.2 and theorem 9.5.4 the analytic theory of L-functions over a number field is, in its abelian form, complete. The Hecke L-function $L(s, \chi)$ attached to a ray class character unifies the two threads of Part II: for $K = \mathbb{Q}$ it is the Dirichlet L-function $L(s, \chi)$ of chapter 7, and for the trivial character it is the Dedekind zeta function ζ_K of chapter 8 (proposition 9.2.4); its Euler product encodes the splitting of primes, its pole or its value at $s = 1$ encodes the arithmetic invariants of K, and its functional equation (theorem 9.4.2) fixes the symmetry of its zeros. The equidistribution of prime ideals among the ray classes of $\mathrm{Cl}_{\mathfrak{m}}(K)$ is the common generalization of Dirichlet's theorem and the infinitude of primes, valid over every number field. What remains is the passage from abelian to non-abelian. The ray class group $\mathrm{Cl}_{\mathfrak{m}}(K)$ is, by class field theory, the Galois group of the abelian ray class field $H_{\mathfrak{m}}/K$ ([12]), and the equidistribution of prime ideals among its classes is the equidistribution of Frobenius elements in an *abelian* Galois group. For a general finite Galois extension L/K with group G, possibly nonabelian, the same question, how the Frobenius conjugacy classes of the unramified primes of K distribute among the conjugacy classes of G, is answered by the Chebotarev density theorem, the non-abelian generalization of theorem 9.5.4 and theorem 7.5.4. Its proof attaches an L-function not to a character of a ray class group but to each representation of G, the Artin L-function, factors ζ_L through these, and confines the pole to the trivial representation by a non-vanishing argument of the present kind. This is the subject of chapter 10.

10

Artin L-Functions and the Chebotarev Density Theorem

The L-functions of chapter 9 were attached to characters, that is, to one-dimensional representations. A Hecke character is a homomorphism of a ray class group to $\mathbb{C}^\times$, and by the reciprocity law of class field theory the ray class group is itself a Galois group: the Hecke characters modulo $\mathfrak{m}$ are the characters of an abelian Galois group $\mathrm{Gal}(L/K)$, and the splitting of a prime $\mathfrak{p}$ in L is recorded by the single Galois element to which $\mathfrak{p}$ corresponds. This is the entire reach of the abelian theory. The moment L/K is a Galois extension with non-abelian group $G = \mathrm{Gal}(L/K)$, the splitting datum of an unramified prime is no longer a single element of G but a conjugacy class, the class of its Frobenius automorphisms, and a character of G in the one-dimensional sense sees only the abelianization $G/[G,G]$. To capture the full group one must replace characters by representations $\rho\colon G \to \mathrm{GL}(V)$ of arbitrary dimension, and the L-function attached to ρ, formed from the characteristic polynomials of the Frobenius classes acting on V, is the Artin L-function. It is the object that completes the arc of Part II: the Riemann zeta function (chapter 4), the Dirichlet L-functions (chapter 7), the Dedekind zeta function (chapter 8), and the Hecke L-functions (chapter 9) are all Artin L-functions of one-dimensional or trivial representations, and the general construction subsumes them.

The arithmetic payoff is the distribution of the Frobenius classes. Dirichlet's theorem, in its density form (theorem 7.5.4), asserts that the rational primes are equidistributed among the residue classes modulo m, which is the statement that the Frobenius elements in the abelian group $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ are equidistributed among its elements. The Chebotarev density theorem, the summit of this chapter and the non-abelian generalization of Dirichlet's theorem, asserts that for any Galois extension L/K the Frobenius classes $\mathrm{Frob}_\mathfrak{p}$ are equidistributed among the conjugacy classes of G, a class C being attained with Dirichlet density $|C|/|G|$. Its proof runs along the template established for Dirichlet's theorem and refined for the Hecke L-functions: factor the Dedekind zeta function of L into Artin L-functions over the irreducible representations of G, locate the single pole at the trivial representation, and reduce the equidistribution to the non-vanishing of the remaining L-functions at $s = 1$. The present section installs the group-theoretic foundation: it makes precise the passage from a prime to its Frobenius conjugacy class and records the dictionary between that class and the splitting type of the prime.

10.0.1 Remark: Representation-Theoretic and Class-Field-Theoretic Prerequisites This chapter takes the representation theory of finite groups as a co-requisite, replacing the harmonic analysis of finite abelian groups that underlay the Dirichlet and Hecke theories. From [3] it draws the complex representations of a finite group and their characters, the decomposition of a representation into irreducibles and the orthogonality relations for irreducible characters, induced representations and Frobenius reciprocity, and the regular representation and its decomposition; these enter the formalism of section 10.2 and the induction theorem of section 10.3. The algebraic number theory of [2] remains in force, now in its relative form: the decomposition group, inertia group, and Frobenius automorphism of a prime in a Galois extension L/K, together with the transitivity of the Galois action on the primes above a fixed prime of K, are the input to the present section, and the splitting of primes through the Dedekind–Kummer theorem supplies the dictionary at its close. Local data, the completions K_v and the unramified local extensions, are imported from [4] where the local factors require them. The deepest external input is class-field-theoretic: that a one-dimensional Artin L-function coincides with a Hecke L-function, the bridge by which the analytic results of chapter 9 are transported to the non-abelian setting, rests on Artin reciprocity; the identity of the Artin and Hecke L-functions of a one-dimensional representation is established in [15, Ch. VII, §10] and developed idelically in the companion [12], and is invoked, never reproved, at the points in sections 10.3 and 10.4 where it is needed.

10.1 Frobenius Elements and Conjugacy Classes

The splitting of a prime in a Galois extension is governed by a distinguished automorphism, the Frobenius, attached to each prime of the upper field lying above it. For a single prime $\mathfrak{P}$ of L the Frobenius is a well-defined element of the Galois group; but the Artin L-function is indexed by the primes $\mathfrak{p}$ of the base field K, and to a prime of K there correspond several primes of L, each with its own Frobenius. The first task is to show that these elements, as $\mathfrak{P}$ ranges over the primes above a fixed $\mathfrak{p}$, fill out exactly one conjugacy class of the Galois group, so that the splitting of $\mathfrak{p}$ is recorded by a class rather than an element. This conjugacy class is the fundamental datum of the chapter.

Throughout, L/K is a finite Galois extension of number fields with Galois group $G = \mathrm{Gal}(L/K)$, and $\mathfrak{p}$ denotes a nonzero prime ideal of $\mathcal{O}_K$. The rings of integers are $\mathcal{O}_K$ and $\mathcal{O}_L$, and a prime $\mathfrak{P}$ of $\mathcal{O}_L$ lies above $\mathfrak{p}$, written $\mathfrak{P}|\mathfrak{p}$, when $\mathfrak{P} \cap \mathcal{O}_K = \mathfrak{p}$, equivalently when $\mathfrak{P}$ occurs in the factorization $\mathfrak{p}\mathcal{O}_L = \prod_i \mathfrak{P}_i^{e_i}$. The residue fields are $\kappa(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$ and $\kappa(\mathfrak{P}) = \mathcal{O}_L/\mathfrak{P}$, both finite, and the absolute norm of $\mathfrak{p}$ is $N\mathfrak{p} = |\kappa(\mathfrak{p})| = [\mathcal{O}_K : \mathfrak{p}]$.

The decomposition and inertia groups isolate, inside G, the part of the Galois action that fixes a chosen prime of L and the part that acts trivially on its residue field. They are recalled here in the form established in [2]; the present account adds nothing to their construction and uses them only to define the Frobenius class.

For a prime $\mathfrak{P}|\mathfrak{p}$ of L, the *decomposition group* of $\mathfrak{P}$ over K is the stabilizer of $\mathfrak{P}$ in G,

$$D_{\mathfrak{P}} = \{\sigma \in G \mid \sigma\mathfrak{P} = \mathfrak{P}\}$$

the subgroup of automorphisms carrying $\mathfrak{P}$ to itself ([2, decomposition group]). An element of $D_{\mathfrak{P}}$

preserves $\mathfrak{P}$, hence acts on the residue field $\kappa(\mathfrak{P})$, and because it fixes K pointwise it acts as an automorphism of $\kappa(\mathfrak{P})$ over $\kappa(\mathfrak{p})$. The resulting reduction homomorphism

$$D_{\mathfrak{P}} \longrightarrow \mathrm{Gal}\big(\kappa(\mathfrak{P})/\kappa(\mathfrak{p})\big)$$

is surjective, and its kernel is the *inertia group*

$$I_{\mathfrak{P}} = \{\sigma \in D_{\mathfrak{P}} \mid \sigma(x) \equiv x \pmod{\mathfrak{P}} \text{ for all } x \in \mathcal{O}_L\}$$

the automorphisms in $D_{\mathfrak{P}}$ inducing the identity on $\kappa(\mathfrak{P})$ ([2, inertia group]). The prime $\mathfrak{p}$ is *unramified* in L when $I_{\mathfrak{P}}$ is trivial for one, equivalently every, prime $\mathfrak{P}$ above it, which happens for all but the finitely many primes dividing the relative discriminant of L/K.

The extension $\kappa(\mathfrak{P})/\kappa(\mathfrak{p})$ is an extension of finite fields, hence cyclic, with Galois group generated by the field-theoretic Frobenius $x \mapsto x^{N\mathfrak{p}}$, the $N\mathfrak{p}$-power map, where $N\mathfrak{p} = |\kappa(\mathfrak{p})|$. When $\mathfrak{p}$ is unramified the reduction map $D_{\mathfrak{P}} \to \mathrm{Gal}(\kappa(\mathfrak{P})/\kappa(\mathfrak{p}))$ is an isomorphism, so a unique element of $D_{\mathfrak{P}}$ maps to this canonical generator.

For $\mathfrak{p}$ unramified in L and $\mathfrak{P}\big|\mathfrak{p}$, the *Frobenius automorphism* $\mathrm{Frob}_{\mathfrak{P}} \in D_{\mathfrak{P}} \subseteq G$ is the unique element of the decomposition group satisfying

$$\mathrm{Frob}_{\mathfrak{P}}(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}} \qquad \text{for all } x \in \mathcal{O}_L \tag{10.1}$$

([2, Frobenius automorphism]). The congruence (10.1) characterizes $\mathrm{Frob}_{\mathfrak{P}}$ uniquely among the elements of $D_{\mathfrak{P}}$, because an automorphism of $\kappa(\mathfrak{P})$ over $\kappa(\mathfrak{p})$ is determined by its effect on a generator of the cyclic group $\kappa(\mathfrak{P})^\times$ and the $N\mathfrak{p}$-power map is a single such automorphism; the lift to $D_{\mathfrak{P}}$ is unique because the reduction map is injective in the unramified case. The order of $\mathrm{Frob}_{\mathfrak{P}}$ equals the order of the $N\mathfrak{p}$-power map on $\kappa(\mathfrak{P})$, which is the residue degree $f = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})]$, and the decomposition group $D_{\mathfrak{P}} = \langle \mathrm{Frob}_{\mathfrak{P}} \rangle$ is the cyclic group it generates.

The Frobenius is attached to a prime $\mathfrak{P}$ of L, but a single prime $\mathfrak{p}$ of K has several primes above it, permuted transitively by G. The behaviour of the Frobenius under this permutation is the key to passing from a prime of L to a datum depending only on $\mathfrak{p}$.

Proposition 10.1.1: The Frobenius Conjugacy Class

Let L/K be a finite Galois extension with group G, and let $\mathfrak{p}$ be a prime of K unramified in L. For every $\sigma \in G$ and every prime $\mathfrak{P}\big|\mathfrak{p}$ of L,

$$\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma\, \mathrm{Frob}_{\mathfrak{P}}\, \sigma^{-1} \tag{10.2}$$

As $\mathfrak{P}$ ranges over the primes of L above $\mathfrak{p}$, the Frobenius automorphisms $\mathrm{Frob}_{\mathfrak{P}}$ form a single conjugacy class of G.

***Proof*:**
The proof has two parts: the conjugation formula (10.2), and the transitivity of G on the primes above $\mathfrak{p}$, which together identify the set of Frobenius elements with one full conjugacy class.

Step 1: The conjugation formula. Fix $\sigma \in G$ and a prime $\mathfrak{P}\big|\mathfrak{p}$. First, $\sigma\mathfrak{P}$ is again a prime of L above $\mathfrak{p}$: it is a prime ideal because σ is a ring automorphism of $\mathcal{O}_L$, and $\sigma\mathfrak{P} \cap \mathcal{O}_K = \sigma(\mathfrak{P} \cap \mathcal{O}_K) = \sigma\mathfrak{p} = \mathfrak{p}$ since σ fixes K, so $\sigma\mathfrak{P}\big|\mathfrak{p}$. As $\mathfrak{p}$ is unramified, $\mathrm{Frob}_{\sigma\mathfrak{P}}$ is defined, and it is the unique element of $D_{\sigma\mathfrak{P}}$ inducing the $N\mathfrak{p}$-power map on $\kappa(\sigma\mathfrak{P})$. It suffices to verify that the element $\sigma\,\mathrm{Frob}_{\mathfrak{P}}\,\sigma^{-1}$ lies in $D_{\sigma\mathfrak{P}}$ and satisfies the defining congruence (10.1) for $\sigma\mathfrak{P}$; uniqueness then forces the two to coincide.

That $\sigma\,\mathrm{Frob}_{\mathfrak{P}}\,\sigma^{-1}$ stabilizes $\sigma\mathfrak{P}$ is a direct computation. Writing $\tau = \mathrm{Frob}_{\mathfrak{P}}$, which satisfies $\tau\mathfrak{P} = \mathfrak{P}$ because $\tau \in D_{\mathfrak{P}}$,

$$(\sigma\tau\sigma^{-1})(\sigma\mathfrak{P}) = \sigma\tau\mathfrak{P} = \sigma\mathfrak{P}$$

so $\sigma\tau\sigma^{-1} \in D_{\sigma\mathfrak{P}}$. For the congruence, let $y \in \mathcal{O}_L$ be arbitrary and write $y = \sigma(x)$ with $x = \sigma^{-1}(y) \in \mathcal{O}_L$. Applying τ to the congruence and then σ, and using that σ is a ring homomorphism carrying $\mathfrak{P}$ onto $\sigma\mathfrak{P}$, gives

$$(\sigma\tau\sigma^{-1})(y) = \sigma\big(\tau(x)\big) \equiv \sigma\big(x^{N\mathfrak{p}}\big) = \sigma(x)^{N\mathfrak{p}} = y^{N\mathfrak{p}} \pmod{\sigma\mathfrak{P}}$$

where the middle congruence is the image under σ of $\tau(x) \equiv x^{N\mathfrak{p}} \pmod{\mathfrak{P}}$, valid because $a \equiv b \pmod{\mathfrak{P}}$ implies $\sigma(a) \equiv \sigma(b) \pmod{\sigma\mathfrak{P}}$ for the automorphism σ. Thus $\sigma\tau\sigma^{-1} \in D_{\sigma\mathfrak{P}}$ induces the $N\mathfrak{p}$-power map on $\kappa(\sigma\mathfrak{P})$, and by the uniqueness characterizing the Frobenius of $\sigma\mathfrak{P}$,

$$\mathrm{Frob}_{\sigma\mathfrak{P}} = \sigma\tau\sigma^{-1} = \sigma\,\mathrm{Frob}_{\mathfrak{P}}\,\sigma^{-1}$$

which is (10.2).

Step 2: Transitivity and the conjugacy class. The Galois group G acts transitively on the set $\{\mathfrak{P} : \mathfrak{P}|\mathfrak{p}\}$ of primes of L above $\mathfrak{p}$ ([2, transitivity of the Galois action on primes above a fixed prime]). Fix one prime $\mathfrak{P}_0|\mathfrak{p}$ and set $\tau_0 = \mathrm{Frob}_{\mathfrak{P}_0}$. By transitivity every prime above $\mathfrak{p}$ is $\sigma\mathfrak{P}_0$ for some $\sigma \in G$, and by Step 1 its Frobenius is $\mathrm{Frob}_{\sigma\mathfrak{P}_0} = \sigma\tau_0\sigma^{-1}$. Hence the set of Frobenius elements over $\mathfrak{p}$ is

$$\big\{\mathrm{Frob}_{\mathfrak{P}} \;\big|\; \mathfrak{P}|\mathfrak{p}\big\} = \big\{\sigma\tau_0\sigma^{-1} \;\big|\; \sigma \in G\big\}$$

which is exactly the conjugacy class of τ_0 in G. Every conjugate $\sigma\tau_0\sigma^{-1}$ arises, since $\sigma\mathfrak{P}_0$ is a prime above $\mathfrak{p}$ for every σ, and only conjugates arise, by the displayed equality; so the set of Frobenius elements is precisely one conjugacy class.

The proposition licenses a definition depending on the prime $\mathfrak{p}$ of the base field alone.

Definition 10.1.2: Frobenius Class

Let L/K be a finite Galois extension with group G, and let $\mathfrak{p}$ be a prime of K unramified in L. The *Frobenius class* of $\mathfrak{p}$ in L/K, written $\mathrm{Frob}_{\mathfrak{p}}$, is the conjugacy class of G consisting of the Frobenius automorphisms $\mathrm{Frob}_{\mathfrak{P}}$ for the primes $\mathfrak{P}|\mathfrak{p}$ of L. When the meaning is unambiguous the same symbol $\mathrm{Frob}_{\mathfrak{p}}$ is used for any chosen representative $\mathrm{Frob}_{\mathfrak{P}} \in G$ of the class.

The notation deliberately echoes that of the abelian theory. There the Frobenius class is a single element, and $\mathrm{Frob}_{\mathfrak{p}}$ is the Artin symbol attached to $\mathfrak{p}$; in general it is genuinely a class, and any statement made with $\mathrm{Frob}_{\mathfrak{p}}$ as an element must be invariant under conjugation for the choice of representative to be immaterial. The two properties on which the chapter rests are exactly the two anchors just established: the congruence (10.1) characterizing each $\mathrm{Frob}_{\mathfrak{P}}$, and the conjugation formula (10.2) making $\mathrm{Frob}_{\mathfrak{p}}$ a well-defined class.

The splitting type of $\mathfrak{p}$ in L is read off entirely from the Frobenius class. The following dictionary records the correspondence; it is the bridge between the group theory of G and the arithmetic of prime factorizations, and it is what makes the Artin L-function of section 10.2 a generating function for splitting data.

For $\mathfrak{p}$ unramified in L, write $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1 \cdots \mathfrak{P}_g$ for its factorization into distinct primes of L (the exponents are 1 precisely because $\mathfrak{p}$ is unramified), and let f be the common residue degree $[\kappa(\mathfrak{P}_i) : \kappa(\mathfrak{p})]$, the same for every i since G permutes the $\mathfrak{P}_i$ transitively and preserves residue degrees. The fundamental identity for a Galois extension reads $efg = [L : K]$ with $e = 1$ here, so $fg = |G|$. The Frobenius class determines f and g as follows.

Proposition 10.1.3: The Splitting Dictionary

Let L/K be a finite Galois extension with group G of order $n = [L : K]$, and let $\mathfrak{p}$ be a prime of K unramified in L. Let f be the residue degree and g the number of primes of L above $\mathfrak{p}$. Then:

1. f equals the order of any element of the Frobenius class $\mathrm{Frob}_\mathfrak{p}$ in G;
2. $g = n/f$;
3. $\mathfrak{p}$ splits completely in L (that is, $g = n$, equivalently $f = 1$) if and only if $\mathrm{Frob}_\mathfrak{p}$ is the trivial class $\{1\}$;
4. $\mathfrak{p}$ remains inert (that is, $g = 1$, equivalently $f = n$) only if G is cyclic and $\mathrm{Frob}_\mathfrak{p}$ generates it.

***Proof*:**
Item (1) is the computation of the order of the Frobenius recorded above: for a prime $\mathfrak{P}|\mathfrak{p}$, the element $\mathrm{Frob}_\mathfrak{P}$ generates the decomposition group $D_\mathfrak{P}$ and has order equal to that of the $N\mathfrak{p}$-power map on the finite field $\kappa(\mathfrak{P})$, namely the residue degree $f = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})]$. Conjugate elements have equal order, so every element of the class $\mathrm{Frob}_\mathfrak{p}$ has order f, independent of the representative. Item (2) is the fundamental identity $efg = n$ with $e = 1$, the unramified case, giving $fg = n$ and hence $g = n/f$. For item (3), $\mathfrak{p}$ splits completely exactly when $g = n$, equivalently $f = 1$ by item (2), equivalently the Frobenius elements have order 1 by item (1); an element of order 1 is the identity, so the class $\mathrm{Frob}_\mathfrak{p}$ is $\{1\}$, and conversely. For item (4), $\mathfrak{p}$ is inert exactly when $g = 1$, equivalently $f = n$, so a Frobenius element has order $n = |G|$ and thus generates a cyclic subgroup of order $|G|$, forcing $G = \langle \mathrm{Frob}_\mathfrak{P} \rangle$ to be cyclic; the converse direction is immediate since a generator has order $|G|$, giving $f = |G|$ and $g = 1$.

Two specializations connect the dictionary to the earlier chapters and frame the question that the rest of the chapter answers. When G is abelian, every conjugacy class is a single element, so the Frobenius class $\mathrm{Frob}_\mathfrak{p}$ collapses to a single automorphism $\mathrm{Frob}_\mathfrak{p} \in G$, the Artin symbol of $\mathfrak{p}$. This is the setting of chapter 9: for $L = \mathbb{Q}(\zeta_m)$ over $K = \mathbb{Q}$, the group $G \cong (\mathbb{Z}/m\mathbb{Z})^\times$ is abelian, the Frobenius of an unramified prime $p \nmid m$ is the residue class of p (the element $a \mapsto \zeta_m^a$ with $a \equiv p$), and the dictionary returns the splitting of p in the cyclotomic field as the order of p modulo m, exactly the computation underlying the factorization of the cyclotomic Dedekind zeta function in proposition 8.4.5 and the proof of Dirichlet's theorem in section 7.5. The residue class of p modulo m and the Frobenius class of p in $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ are the same datum, and Dirichlet's theorem is the assertion that this datum is equidistributed.

When G is non-abelian the Frobenius class is genuinely a conjugacy class, and the splitting type of $\mathfrak{p}$ records the cycle structure of the class but not the class itself: two primes with conjugate but distinct decomposition-group generators have the same residue degree and the same number of factors, yet may lie in different conjugacy classes and so be distinguished by the higher-dimensional representations of G through their Artin L-functions. The central question of the chapter, posed now and answered by the Chebotarev density theorem in section 10.4, is how the classes $\mathrm{Frob}_{\mathfrak{p}}$ distribute among the conjugacy classes of G as $\mathfrak{p}$ ranges over the unramified primes of K. Chebotarev's theorem asserts that they equidistribute, each class C occurring with Dirichlet density $|C|/|G|$, and so generalizes Dirichlet's theorem from the abelian case (where the classes are singletons and the density $1/|G|$ is the density of primes in a fixed residue class) to all Galois extensions. The route to that theorem passes through the Artin L-functions constructed in the next section.

10.2 Artin L-Functions and Artin Formalism

The Frobenius conjugacy class of section 10.1 records, for each prime $\mathfrak{p}$ of K unramified in the Galois extension L/K, an element of $G = \mathrm{Gal}(L/K)$ well defined up to conjugacy (definition 10.1.2, proposition 10.1.1). A complex representation $\rho\colon G \to \mathrm{GL}(V)$ converts this group-theoretic datum into a linear one: the operator $\rho(\mathrm{Frob}_{\mathfrak{p}})$, a conjugacy class of automorphisms of V, whose characteristic polynomial is a conjugacy invariant and so is attached unambiguously to $\mathfrak{p}$. Assembling one such local factor per prime produces the Artin L-function $L(s, \rho, L/K)$, the object that carries the splitting of primes in L/K refracted through the representation ρ. For the trivial representation it is the Dedekind zeta function ζ_K of chapter 8; for a one-dimensional ρ, which factors through the abelianization of G, it is a Hecke L-function of chapter 9 by the reciprocity law of class field theory; and the general ρ, attached to a possibly nonabelian G, is the genuinely new object of the present chapter.

The arithmetic content of the construction is concentrated in three formal properties, collectively the *Artin formalism*: the L-function is additive in ρ, sends the trivial representation to ζ_K, and is invariant under induction of representations from a subgroup. These three properties are forced by the local definition and require no analytic input; together they reduce the study of every Artin L-function, in principle, to the one-dimensional case, and they yield at once the central structural fact of the theory, the factorization of the Dedekind zeta function ζ_L of the top field into a product of Artin L-functions indexed by the irreducible representations of G. This section establishes the formalism and the factorization, working throughout inside the half-plane $\mathrm{Re}(s) > 1$ of absolute convergence; the continuation of $L(s, \rho, L/K)$ to the rest of the plane, which rests on the formalism together with Brauer's induction theorem, is taken up in section 10.3.

Definition 10.2.1: Artin L-Function

Let L/K be a finite Galois extension of number fields with group $G = \mathrm{Gal}(L/K)$, and let $\rho\colon G \to \mathrm{GL}(V)$ be a finite-dimensional complex representation, V a complex vector space of dimension $\dim V = \dim \rho$. For each nonzero prime $\mathfrak{p}$ of K, choose a prime $\mathfrak{P}$ of L above $\mathfrak{p}$, with decomposition group $D_{\mathfrak{P}}$, inertia group $I_{\mathfrak{P}} \subseteq D_{\mathfrak{P}}$, and let

$$V^{I_{\mathfrak{P}}} = \{\, v \in V : \rho(\tau)v = v \text{ for all } \tau \in I_{\mathfrak{P}} \,\}$$

be the subspace of inertia invariants. The quotient $D_{\mathfrak{P}}/I_{\mathfrak{P}}$ is cyclic, generated by the Frobenius $\mathrm{Frob}_{\mathfrak{P}}$ (definition 10.1.2), and $\rho(\mathrm{Frob}_{\mathfrak{P}})$ acts on $V^{I_{\mathfrak{P}}}$. The *local factor* of the Artin L-function at $\mathfrak{p}$ is

$$L_{\mathfrak{p}}(s,\rho) = \det\bigl(1 - \rho(\mathrm{Frob}_{\mathfrak{P}})\,(N\mathfrak{p})^{-s}\bigm| V^{I_{\mathfrak{P}}}\bigr)^{-1} \tag{10.3}$$

the determinant taken on the inertia-invariant subspace $V^{I_{\mathfrak{P}}}$, and the *Artin L-function* of ρ for the extension L/K is the product over all nonzero primes of K,

$$L(s,\rho,L/K) = \prod_{\mathfrak{p}} \det\bigl(1 - \rho(\mathrm{Frob}_{\mathfrak{p}})\,(N\mathfrak{p})^{-s}\bigm| V^{I_{\mathfrak{P}}}\bigr)^{-1} \tag{10.4}$$

At a prime $\mathfrak{p}$ unramified in L one has $I_{\mathfrak{P}} = 1$, so $V^{I_{\mathfrak{P}}} = V$ and the local factor is the full determinant $\det(1 - \rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s})^{-1}$ on V.

The local factor (10.3) is independent of the two choices implicit in its definition, the prime $\mathfrak{P}|\mathfrak{p}$ and the Frobenius element within its conjugacy class. The verification is the linear-algebra counterpart of the conjugacy invariance of the Frobenius class.

Proposition 10.2.2: Well-Definedness of the Local Factor

The local factor $L_{\mathfrak{p}}(s,\rho)$ of (10.3) does not depend on the choice of prime $\mathfrak{P}$ of L above $\mathfrak{p}$ nor, at an unramified prime, on the choice of Frobenius element in its conjugacy class. Equivalently, the characteristic polynomial $\det(1 - t\,\rho(\mathrm{Frob}_{\mathfrak{P}})|V^{I_{\mathfrak{P}}})$ depends only on $\mathfrak{p}$.

***Proof*:**
Any two primes of L above $\mathfrak{p}$ are conjugate under G: there is $g \in G$ with $\mathfrak{P}' = g\mathfrak{P}$ ([2, transitivity of the Galois action on primes above $\mathfrak{p}$]). Conjugation by g carries the decomposition and inertia groups of $\mathfrak{P}$ to those of $\mathfrak{P}'$, namely $D_{\mathfrak{P}'} = gD_{\mathfrak{P}}g^{-1}$ and $I_{\mathfrak{P}'} = gI_{\mathfrak{P}}g^{-1}$, and carries a Frobenius lift at $\mathfrak{P}$ to one at $\mathfrak{P}'$, so that $\mathrm{Frob}_{\mathfrak{P}'} = g\,\mathrm{Frob}_{\mathfrak{P}}\,g^{-1}$ as elements of $D_{\mathfrak{P}'}/I_{\mathfrak{P}'}$ (proposition 10.1.1). The operator $\rho(g)\colon V \to V$ then restricts to a linear isomorphism $V^{I_{\mathfrak{P}}} \xrightarrow{\sim} V^{I_{\mathfrak{P}'}}$: if $\rho(\tau)v = v$ for all $\tau \in I_{\mathfrak{P}}$, then for $\tau' = g\tau g^{-1} \in I_{\mathfrak{P}'}$ one has $\rho(\tau')\rho(g)v = \rho(g)\rho(\tau)v = \rho(g)v$, so $\rho(g)$ maps $V^{I_{\mathfrak{P}}}$ into $V^{I_{\mathfrak{P}'}}$, and the same argument applied to g^{-1} gives the inverse. Under this isomorphism the action of $\rho(\mathrm{Frob}_{\mathfrak{P}})$ on $V^{I_{\mathfrak{P}}}$ is intertwined with that of $\rho(\mathrm{Frob}_{\mathfrak{P}'}) = \rho(g)\rho(\mathrm{Frob}_{\mathfrak{P}})\rho(g)^{-1}$ on $V^{I_{\mathfrak{P}'}}$, since $\rho(\mathrm{Frob}_{\mathfrak{P}'})\,\rho(g) = \rho(g)\,\rho(\mathrm{Frob}_{\mathfrak{P}})$ on $V^{I_{\mathfrak{P}}}$. Two operators conjugate by an isomorphism of the spaces on which they act have the same characteristic polynomial; hence

$$\det\bigl(1 - t\,\rho(\mathrm{Frob}_{\mathfrak{P}'})\bigm|V^{I_{\mathfrak{P}'}}\bigr) = \det\bigl(1 - t\,\rho(\mathrm{Frob}_{\mathfrak{P}})\bigm|V^{I_{\mathfrak{P}}}\bigr)$$

and setting $t = (N\mathfrak{p})^{-s}$ shows the local factor is independent of $\mathfrak{P}$. At an unramified prime $I_{\mathfrak{P}} = 1$, so $\mathrm{Frob}_{\mathfrak{P}}$ is a genuine element of $D_{\mathfrak{P}}$; replacing it by another representative of the

Frobenius class over $\mathfrak{p}$ conjugates $\rho(\mathrm{Frob}_{\mathfrak{p}})$ within $\mathrm{GL}(V)$ and again leaves the characteristic polynomial unchanged. Thus $L_{\mathfrak{p}}(s,\rho)$ depends only on $\mathfrak{p}$.

Proposition 10.2.3: Convergence

The product (10.4) defining $L(s,\rho,L/K)$ converges absolutely on the half-plane $\mathrm{Re}(s) > 1$ and defines a holomorphic and nowhere-vanishing function there. More precisely, for real $\sigma = \mathrm{Re}(s) > 1$,

$$\prod_{\mathfrak{p}} \left| L_{\mathfrak{p}}(s,\rho) \right| \le \prod_{\mathfrak{p}} \left(1 - (N\mathfrak{p})^{-\sigma}\right)^{-\dim V} = \zeta_K(\sigma)^{\dim V} < \infty \tag{10.5}$$

***Proof*:**
Step 1: The eigenvalues of $\rho(\mathrm{Frob}_{\mathfrak{P}})$ are roots of unity. The group G is finite, so every element $g \in G$ has finite order, say $g^n = 1$ with $n = |G|$. Then $\rho(g)^n = \rho(g^n) = \rho(1) = \mathrm{id}_V$, so the operator $\rho(g)$ satisfies $X^n - 1 = 0$. Its minimal polynomial divides $X^n - 1$, which has distinct roots (the n-th roots of unity), so $\rho(g)$ is diagonalizable and each of its eigenvalues is an n-th root of unity, of absolute value 1. Applying this to $g = \mathrm{Frob}_{\mathfrak{P}}$ acting on V, and hence to its restriction to the invariant subspace $V^{I_{\mathfrak{P}}}$, the eigenvalues $\alpha_1, \dots, \alpha_r$ of $\rho(\mathrm{Frob}_{\mathfrak{P}})$ on $V^{I_{\mathfrak{P}}}$ (where $r = \dim V^{I_{\mathfrak{P}}} \le \dim V$) all satisfy $|\alpha_j| = 1$.

Step 2: The local factor is dominated by a power of the zeta factor. Writing the determinant in (10.3) through the eigenvalues of $\rho(\mathrm{Frob}_{\mathfrak{P}})$ on $V^{I_{\mathfrak{P}}}$,

$$L_{\mathfrak{p}}(s,\rho) = \prod_{j=1}^{r} \left(1 - \alpha_j\,(N\mathfrak{p})^{-s}\right)^{-1}$$

Each factor has, for $\sigma = \mathrm{Re}(s) > 1$, the bound $|1 - \alpha_j (N\mathfrak{p})^{-s}| \ge 1 - |\alpha_j|\,(N\mathfrak{p})^{-\sigma} = 1 - (N\mathfrak{p})^{-\sigma} > 0$, using $|\alpha_j| = 1$ and $N\mathfrak{p} \ge 2$. Hence

$$\left| L_{\mathfrak{p}}(s,\rho) \right| = \prod_{j=1}^{r} \left| 1 - \alpha_j (N\mathfrak{p})^{-s} \right|^{-1} \le \left(1 - (N\mathfrak{p})^{-\sigma}\right)^{-r} \le \left(1 - (N\mathfrak{p})^{-\sigma}\right)^{-\dim V}$$

the last inequality because $r \le \dim V$ and $0 < 1 - (N\mathfrak{p})^{-\sigma} < 1$ makes the exponent increase the value. In particular each local factor is finite and nonzero.

Step 3: The bounding product is $\zeta_K(\sigma)^{\dim V}$. Taking the product of the Step 2 bound over all primes $\mathfrak{p}$ and raising the Euler product of the Dedekind zeta function to the power $\dim V$,

$$\prod_{\mathfrak{p}} \left| L_{\mathfrak{p}}(s,\rho) \right| \le \prod_{\mathfrak{p}} \left(1 - (N\mathfrak{p})^{-\sigma}\right)^{-\dim V} = \left(\prod_{\mathfrak{p}} \left(1 - (N\mathfrak{p})^{-\sigma}\right)^{-1} \right)^{\dim V} = \zeta_K(\sigma)^{\dim V}$$

the middle equality being the Euler product of ζ_K (theorem 8.1.4) at the real point σ, which converges for $\sigma > 1$ (proposition 8.1.3). Thus $\prod_{\mathfrak{p}} |L_{\mathfrak{p}}(s,\rho)|$ is finite, which is (10.5).

Step 4: Absolute convergence, holomorphy, nonvanishing. An infinite product $\prod_{\mathfrak{p}} L_{\mathfrak{p}}(s,\rho)$ of factors $L_{\mathfrak{p}} = \prod_j (1-\alpha_j(N\mathfrak{p})^{-s})^{-1}$ converges absolutely on $\mathrm{Re}(s) > 1$ precisely when $\sum_{\mathfrak{p}} \sum_j |\alpha_j(N\mathfrak{p})^{-s}| < \infty$; and using $|\alpha_j| = 1$ and $r \le \dim V$ this is bounded by $\dim V \cdot \sum_{\mathfrak{p}} (N\mathfrak{p})^{-\sigma}$, finite on $\mathrm{Re}(s) > 1$ by the convergence of the zeta sum (proposition 8.1.3). The product therefore converges absolutely, locally uniformly on $\mathrm{Re}(s) > 1$, where the bound of Step 3 is uniform on each half-plane $\mathrm{Re}(s) \ge \sigma_0 > 1$; the locally uniform limit of the holomorphic partial products is holomorphic. Each factor $L_{\mathfrak{p}}(s,\rho)$ is nonzero by Step 2 (it is the reciprocal of $\prod_j (1-\alpha_j(N\mathfrak{p})^{-s})$, a product of nonzero numbers), and an absolutely convergent infinite product all of whose factors are nonzero has a nonzero value (proposition 2.3.4). Hence $L(s,\rho,L/K) \neq 0$ throughout $\mathrm{Re}(s) > 1$.

The convergence on $\mathrm{Re}(s) > 1$ generalizes the convergence of ζ_K (proposition 8.1.3, the case $\rho = \mathbf{1}$, $\dim V = 1$) and of the Hecke L-functions (proposition 9.2.2, the case $\dim V = 1$ with L/K abelian): in every instance the eigenvalues of Frobenius have absolute value 1, and the local factor is bounded by a power of the local factor of ζ_K. The three formal properties of the construction are likewise visible already at the level of the local factors, and it is to these that the discussion now turns.

Theorem 10.2.4: Artin Formalism

The Artin L-function enjoys the following three properties, all as identities of holomorphic functions on $\mathrm{Re}(s) > 1$.

1. *(Additivity.)* For representations ρ_1, ρ_2 of $G = \mathrm{Gal}(L/K)$,

$$L(s,\rho_1 \oplus \rho_2, L/K) = L(s,\rho_1,L/K)\, L(s,\rho_2,L/K) \tag{10.6}$$

2. *(Triviality.)* For the trivial representation $\mathbf{1}_G$ of G,

$$L(s,\mathbf{1}_G,L/K) = \zeta_K(s) \tag{10.7}$$

3. *(Inductivity.)* For a subgroup $H \le G$ with fixed field L^H, and a representation σ of H,

$$L(s,\mathrm{Ind}_H^G \sigma, L/K) = L(s,\sigma,L/L^H) \tag{10.8}$$

   the right side being the Artin L-function of σ for the Galois extension L/L^H with group $\mathrm{Gal}(L/L^H) = H$.

***Proof*:**
(i) Additivity. Fix a prime $\mathfrak{p}$ of K and a prime $\mathfrak{P}|\mathfrak{p}$ of L, with inertia $I = I_{\mathfrak{P}}$ and Frobenius $\Phi = \mathrm{Frob}_{\mathfrak{P}}$. For the direct sum $\rho = \rho_1 \oplus \rho_2$ on $V = V_1 \oplus V_2$, the inertia invariants split as $V^I = V_1^I \oplus V_2^I$, since a vector (v_1, v_2) is fixed by all of I iff each v_i is, and Φ acts block-diagonally, preserving the splitting. The determinant of a block-diagonal operator is the product of the determinants of its blocks, so

$$\det\bigl(1-\rho(\Phi)(N\mathfrak{p})^{-s}\big|V^I\bigr) = \det\bigl(1-\rho_1(\Phi)(N\mathfrak{p})^{-s}\big|V_1^I\bigr)\, \det\bigl(1-\rho_2(\Phi)(N\mathfrak{p})^{-s}\big|V_2^I\bigr)$$

Taking reciprocals gives $L_{\mathfrak{p}}(s,\rho_1 \oplus \rho_2) = L_{\mathfrak{p}}(s,\rho_1)\, L_{\mathfrak{p}}(s,\rho_2)$, and multiplying over all $\mathfrak{p}$ yields (10.6); the rearrangement of the product of products into the product of the two products is justified by the absolute convergence of all three products on $\mathrm{Re}(s) > 1$ (proposition 10.2.3).

(ii) Triviality. For the trivial representation $\mathbf{1}_G$ on $V = \mathbb{C}$, every $\rho(g)$ is the identity scalar 1, so $V^{I_{\mathfrak{P}}} = \mathbb{C}$ at every prime (the inertia acts trivially) and $\rho(\mathrm{Frob}_{\mathfrak{P}}) = 1$. The local factor is therefore

$$L_{\mathfrak{p}}(s, \mathbf{1}_G) = \det\bigl(1 - 1 \cdot (N\mathfrak{p})^{-s} \big| \mathbb{C}\bigr)^{-1} = \bigl(1 - (N\mathfrak{p})^{-s}\bigr)^{-1}$$

the local factor of ζ_K at $\mathfrak{p}$. Taking the product over all primes $\mathfrak{p}$ of K and invoking the Euler product of the Dedekind zeta function (theorem 8.1.4) gives $L(s, \mathbf{1}_G, L/K) = \prod_{\mathfrak{p}}(1 - (N\mathfrak{p})^{-s})^{-1} = \zeta_K(s)$, which is (10.7). The trivial-representation Artin L-function is thus the Dedekind zeta function of the base field, independent of L.

(iii) Inductivity. Write $E = L^H$ for the fixed field of H, an intermediate field $K \subseteq E \subseteq L$ with $\mathrm{Gal}(L/E) = H$. It suffices to prove the local identity at each prime $\mathfrak{p}$ of K: that the local factor of $\mathrm{Ind}_H^G \sigma$ for L/K at $\mathfrak{p}$ equals the product of the local factors of σ for L/E over the primes $\mathfrak{q}$ of E lying above $\mathfrak{p}$,

$$\det\bigl(1 - \mathrm{Ind}_H^G \sigma\,(\mathrm{Frob}_{\mathfrak{P}})\,(N\mathfrak{p})^{-s} \big| (\mathrm{Ind}_H^G W)^{I_{\mathfrak{P}}}\bigr)^{-1} = \prod_{\mathfrak{q} | \mathfrak{p}} \det\bigl(1 - \sigma(\mathrm{Frob}_{\mathfrak{Q}})\,(N\mathfrak{q})^{-s} \big| W^{I_{\mathfrak{Q}}}\bigr)^{-1} \quad (10.9)$$

where σ acts on W, the left side is formed at the chosen prime $\mathfrak{P}|\mathfrak{p}$ of L, and on the right $\mathfrak{Q}$ denotes a prime of L above $\mathfrak{q}$ (the inner local factor being independent of that choice by proposition 10.2.2 applied to L/E). Multiplying (10.9) over all $\mathfrak{p}$ then reproduces (10.8), because each prime $\mathfrak{q}$ of E lies above exactly one $\mathfrak{p} = \mathfrak{q} \cap E$ of K, so the double product on the right runs over every prime of E exactly once and is, by definition, the Artin L-function $L(s, \sigma, L/E)$.

The local identity (10.9) is the analytic form of the orbit decomposition of the induced representation. The primes $\mathfrak{q}$ of E above $\mathfrak{p}$ correspond to the orbits of the decomposition group $D = D_{\mathfrak{P}}$ acting on the coset space $H\backslash G$, equivalently to the double cosets $H\backslash G/D$; for each $\mathfrak{q}$ the local data ($N\mathfrak{q}$, the inertia $I_{\mathfrak{Q}}$, the Frobenius $\mathrm{Frob}_{\mathfrak{Q}}$) are read off from the corresponding double coset, the residue degree $f(\mathfrak{q}|\mathfrak{p})$ being the length of the $\langle \mathrm{Frob}_{\mathfrak{P}} \rangle$-orbit. The restriction-induction bookkeeping that converts the determinant of $\mathrm{Ind}_H^G \sigma(\mathrm{Frob}_{\mathfrak{P}})$ on the inertia invariants into the product over these orbits is a purely representation-theoretic computation: it is the Frobenius formula for the character of an induced representation ([3, induced representations]), applied to the cyclic group $D/I_{\mathfrak{P}}$ generated by $\mathrm{Frob}_{\mathfrak{P}}$ together with the compatible behaviour of inertia invariants under induction. The complete orbit-by-orbit identification of the two sides of (10.9) is carried out in [15, Ch. VII, (10.4)] and [19, Ch. 7]; it gives (10.9), hence (10.8).

The three properties of theorem 10.2.4 characterize the Artin L-function among the constructions attaching a Dirichlet series to a Galois representation, and they have an immediate and decisive consequence. Applied to the trivial subgroup $H = 1$, inductivity expresses the Dedekind zeta function of the top field L as an Artin L-function for L/K, and the decomposition of the regular representation into irreducibles then resolves ζ_L into a product over the irreducible representations of G.

Theorem 10.2.5: Factorization of the Dedekind Zeta Function

Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$, and let $\widehat{G}$ denote a set of representatives for the isomorphism classes of irreducible complex representations of G. Then, as an identity of holomorphic functions on $\mathrm{Re}(s) > 1$,

$$\zeta_L(s) = \prod_{\rho \in \widehat{G}} L(s, \rho, L/K)^{\dim \rho} \tag{10.10}$$

the product over the irreducible representations ρ of G, each appearing to the power of its dimension. The trivial representation contributes the single factor $L(s, \mathbf{1}_G, L/K) = \zeta_K(s)$, so that

$$\frac{\zeta_L(s)}{\zeta_K(s)} = \prod_{\rho \neq \mathbf{1}_G} L(s, \rho, L/K)^{\dim \rho} \tag{10.11}$$

the product over the nontrivial irreducible representations, is holomorphic on $\mathrm{Re}(s) > 1$.

***Proof*:**
Step 1: ζ_L is the Artin L-function of the induced trivial representation. Apply inductivity (10.8) of theorem 10.2.4 with the trivial subgroup $H = 1 \leq G$, whose fixed field is $L^H = L$ and whose only representation is the trivial one-dimensional $\mathbf{1}_1$. The right side is $L(s, \mathbf{1}_1, L/L)$, the Artin L-function of the trivial representation for the trivial extension L/L, which by triviality (10.7) (with the base field now L) is $\zeta_L(s)$. The left side is $L(s, \mathrm{Ind}_1^G \mathbf{1}_1, L/K)$. Hence

$$\zeta_L(s) = L\big(s, \mathrm{Ind}_1^G \mathbf{1}_1, L/K\big) \tag{10.12}$$

the Artin L-function attached to the induced representation $\mathrm{Ind}_1^G \mathbf{1}_1$, which is the regular representation of G.

Step 2: The regular representation decomposes by dimension. The representation $\mathrm{Ind}_1^G \mathbf{1}_1$ of G induced from the trivial representation of the trivial subgroup is the regular representation $\mathbb{C}[G]$ of G ([3, induced representations]): inducing the trivial character of the trivial subgroup gives the permutation representation on $G/1 = G$, which is $\mathbb{C}[G]$. The decomposition of the regular representation into irreducibles is

$$\mathrm{Ind}_1^G \mathbf{1}_1 \;=\; \mathbb{C}[G] \;\cong\; \bigoplus_{\rho \in \widehat{G}} \rho^{\oplus \dim \rho} \tag{10.13}$$

each irreducible ρ occurring with multiplicity equal to its dimension $\dim \rho$ ([3, regular representation]). This is the representation-theoretic identity underlying $|G| = \sum_\rho (\dim \rho)^2$, obtained by comparing dimensions on the two sides of (10.13).

Step 3: Additivity converts the decomposition into the product. Insert (10.13) into (10.12) and apply additivity (10.6) of theorem 10.2.4, which turns a direct sum of representations into a product of L-functions and a $\dim \rho$-fold repeated summand into a $\dim \rho$-th power:

$$\zeta_L(s) = L\left(s, \bigoplus_{\rho \in \widehat{G}} \rho^{\oplus \dim \rho}, L/K\right) = \prod_{\rho \in \widehat{G}} L(s, \rho, L/K)^{\dim \rho}$$

which is (10.10). Both sides are holomorphic on $\mathrm{Re}(s) > 1$: the left by the convergence of ζ_L (proposition 8.1.3 for the field L), the right by proposition 10.2.3 for each factor, the finite product converging absolutely there.

Step 4: Splitting off the trivial factor. The trivial representation $\mathbf{1}_G \in \widehat{G}$ is one-dimensional and contributes the factor $L(s, \mathbf{1}_G, L/K)^1 = \zeta_K(s)$ by triviality (10.7). Separating this factor from (10.10) gives $\zeta_L(s) = \zeta_K(s) \prod_{\rho \neq \mathbf{1}_G} L(s, \rho, L/K)^{\dim \rho}$, and dividing by $\zeta_K(s)$ (nonzero on $\mathrm{Re}(s) > 1$ by corollary 8.1.5) yields (10.11). Each factor on the right is holomorphic on $\mathrm{Re}(s) > 1$ by proposition 10.2.3, so the finite product is holomorphic there.

The factorization (10.10) is the structural summit of the abelian and non-abelian theory alike. For G abelian every irreducible representation is one-dimensional, $\widehat{G}$ is the character group, and (10.10) reads $\zeta_L = \prod_\chi L(s, \chi, L/K)$ with each one-dimensional factor a Hecke L-function by class field theory ([12, a one-dimensional Artin L-function equals a Hecke L-function]); this recovers, for $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_m)$, the factorization of the cyclotomic Dedekind zeta function into Dirichlet L-functions established at the end of chapter 8 (proposition 8.4.5), and more generally the abelian factorization $\prod_{\chi \in X} L(s, \chi) = \zeta_L$ of theorem 8.5.5. For G nonabelian the higher-dimensional representations contribute genuinely new factors, and (10.11) exhibits the quotient ζ_L/ζ_K as a product of Artin L-functions of nontrivial representations, holomorphic on $\mathrm{Re}(s) > 1$. The Aramata–Brauer theorem, that this quotient is in fact *entire*, and the deeper conjecture of Artin, that each individual $L(s, \rho, L/K)$ with ρ nontrivial irreducible is entire, are the subject of section 10.3; both rest on extending the factorization (10.10) from a product of holomorphic functions on $\mathrm{Re}(s) > 1$ to a relation governing the continuation of each factor across the line $\mathrm{Re}(s) = 1$.

10.3 Meromorphic Continuation via Brauer Induction

The Artin L-function $L(s, \rho, L/K)$ of definition 10.2.1 is, like every L-function in this book, defined by a product over the primes of the base field that converges only on the half-plane $\mathrm{Re}(s) > 1$ (proposition 10.2.3). For the abelian objects of the preceding chapters the passage beyond this half-plane was effected directly: the Riemann zeta function, the Dirichlet L-functions, the Dedekind zeta function, and the Hecke L-functions were each continued by a theta-and-Mellin argument that produced a meromorphic function on all of $\mathbb{C}$ together with a functional equation. No such direct argument is available for a general Artin L-function, because the representation ρ of the Galois group $G = \mathrm{Gal}(L/K)$ need not be one-dimensional, and only the one-dimensional case is, by class field theory, an L-function of the Hecke type whose continuation is already in hand. The continuation of the general Artin L-function is obtained instead by a purely group-theoretic device: a theorem of Brauer expresses the character of any representation of a finite group as an integer combination of characters induced from one-dimensional characters of subgroups, and the formal properties of the Artin L-function (theorem 10.2.4) convert this character identity into a factorization of $L(s, \rho)$ as a finite product of integer powers of one-dimensional Artin L-functions. Each factor is a Hecke L-function, meromorphic on $\mathbb{C}$ by theorem 9.4.1, and a finite product of integer powers of meromorphic functions is meromorphic.

The argument is decisive for meromorphy but silent on holomorphy. The integer exponents supplied by Brauer's theorem may be negative, so the factorization presents $L(s, \rho)$ as a quotient of holomorphic functions, and the poles of the denominator are not, on the face of it, cancelled. Whether they in fact cancel, so that $L(s, \rho)$ is entire whenever ρ is irreducible and nontrivial, is the content of Artin's conjecture, one of the central open problems of the subject. This section states Brauer's theorem and deduces the meromorphic continuation, then records Artin's conjecture and the cases in which it is known, closing with the one unconditional holomorphy statement that the method does deliver: the Aramata–Brauer theorem, which guarantees that the quotient ζ_L/ζ_K is entire, so that no Artin L-function contributes a pole surviving in ζ_L beyond the simple pole already carried by ζ_K.

The single input from the representation theory of finite groups that the continuation requires is the following theorem of Brauer. Its proof is a substantial argument in character theory, resting on the notion of an elementary subgroup (a direct product of a p-group and a cyclic group), and is given in full in the cited sources; only the statement is needed here. The induced character $\mathrm{Ind}_H^G \psi$ of a character ψ of a subgroup $H \leq G$ is recalled from [3, induced representations], where it is constructed together with Frobenius reciprocity; on a conjugacy class it is computed by the standard induction formula averaging ψ over the cosets meeting the class.

Theorem 10.3.1: Brauer's Induction Theorem

Let G be a finite group and let χ be the character of any finite-dimensional complex representation of G. Then there are subgroups $H_i \leq G$, one-dimensional characters ψ_i of H_i, and integers $n_i \in \mathbb{Z}$, such that

$$\chi = \sum_i n_i \,\mathrm{Ind}_{H_i}^G \psi_i \tag{10.14}$$

as a class function on G. The subgroups H_i may be taken to be *elementary*: each H_i is a direct product of a cyclic group and a p-group for some prime p. Equivalently, the virtual characters induced from one-dimensional characters of elementary subgroups generate, over $\mathbb{Z}$, the full character ring $R(G)$ of G.

***Proof*:**
The statement is recorded for use; the proof is not reproduced here. Brauer's theorem is proved in full in [19, Ch. 10], where the reduction to elementary subgroups is carried out through the arithmetic of the character ring and the key congruence properties of induced characters; the application of the theorem to the continuation of L-functions, in the form used below, is the subject of [15, Ch. VII, §10]. The weaker statement of Artin, that χ is a *rational* combination of characters induced from *cyclic* subgroups, is more elementary but insufficient for the present purpose, since clearing denominators would leave a nontrivial power of $L(s,\rho)$ rather than $L(s,\rho)$ itself; Brauer's refinement to integer coefficients and one-dimensional inducing characters is what yields the first power of $L(s,\rho)$ as an honest finite product. The contrast between the two induction theorems, and the necessity of the integral version, is discussed in [19, Ch. 10, §2].

The force of the theorem, for the analytic theory, lies in the two italicized words of its conclusion. That the inducing characters ψ_i are *one-dimensional* places each induced piece within reach of class field theory, which identifies a one-dimensional Artin L-function with a Hecke L-function. That the coefficients n_i are *integers* makes the corresponding factorization of $L(s,\rho)$ a product of integer powers, a meromorphic function, rather than a product of fractional powers, which would introduce branch cuts and destroy single-valuedness. The price paid for integrality is the loss of any sign control on the n_i: Brauer's construction provides no guarantee that the exponents are non-negative, and this is precisely the gap between the meromorphy that the theorem delivers and the holomorphy that Artin's conjecture asserts.

The continuation now follows by transporting (10.14) through the Artin formalism. The two formal properties of the Artin L-function that effect the transport were established in theorem 10.2.4: additivity in the representation, so that the L-function of a direct sum is the product of the L-functions of the summands, and inductivity, so that the L-function of an induced representation $\mathrm{Ind}_H^G W$ over L/K equals the L-function of W itself, formed over the intermediate extension L/L^H with its own Galois group $H = \mathrm{Gal}(L/L^H)$. These two properties are exactly what is needed to convert a $\mathbb{Z}$-linear identity among characters into a multiplicative identity among L-functions.

Theorem 10.3.2: Meromorphic Continuation of Artin L-Functions

Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$, and let $\rho\colon G \to \mathrm{GL}(V)$ be a finite-dimensional complex representation. Then the Artin L-function $L(s, \rho, L/K)$ extends from the half-plane $\mathrm{Re}(s) > 1$ to a meromorphic function on all of $\mathbb{C}$.

***Proof*:**
Step 1: Brauer factorization of the character. Let $\chi_\rho = \mathrm{tr}\rho$ be the character of ρ. By Brauer's induction theorem (theorem 10.3.1) applied to the finite group G there are subgroups $H_i \le G$, one-dimensional characters ψ_i of H_i, and integers $n_i \in \mathbb{Z}$ with

$$\chi_\rho = \sum_i n_i \, \mathrm{Ind}_{H_i}^G \psi_i$$

a finite sum of class functions on G.

Step 2: Transport to a factorization of the L-function. An Artin L-function depends on the representation only through its character, and is additive and inductive in the sense of theorem 10.2.4. Additivity converts the sum over i into a product, and the integer coefficient n_i into the n_i-th power of the corresponding factor; here a negative n_i is read as a positive power of the reciprocal, legitimate because each factor is, by Step 3, a nonzero meromorphic function. Inductivity identifies the factor attached to $\mathrm{Ind}_{H_i}^G \psi_i$, an L-function formed over L/K, with the L-function of ψ_i itself formed over the intermediate field L^{H_i}, whose Galois group over L^{H_i} is $H_i = \mathrm{Gal}(L/L^{H_i})$. Combining the two properties,

$$L(s, \rho, L/K) = \prod_i L\big(s, \psi_i, L/L^{H_i}\big)^{n_i} \tag{10.15}$$

a finite product, valid as an identity of holomorphic functions on $\mathrm{Re}(s) > 1$, where every factor is given by its convergent Euler product (proposition 10.2.3) and is in particular nonzero.

Step 3: Each factor is a meromorphic Hecke L-function. Fix an index i and set $F = L^{H_i}$, so that L/F is Galois with group H_i and ψ_i is a one-dimensional character of $H_i = \mathrm{Gal}(L/F)$. The Artin L-function $L(s, \psi_i, L/F)$ is therefore the L-function of a one-dimensional representation of $\mathrm{Gal}(L/F)$, formed over the base field F. By class field theory, a one-dimensional Artin L-function over a number field equals a Hecke L-function over that field: the Artin reciprocity map identifies the character ψ_i of $\mathrm{Gal}(L/F)$ with a ray class character of F, under which the Frobenius at an unramified prime corresponds to the class of that prime in the ray class group, and the two Euler products agree factor by factor; this identification is one-dimensional Artin reciprocity, established in [15, Ch. VII, §10] and [12]. Hence there is a Hecke character η_i of F with

$$L(s, \psi_i, L/F) = L(s, \eta_i)$$

the Hecke L-function of definition 9.2.1 over the base field F. By Hecke's continuation theorem (theorem 9.4.1) every Hecke L-function extends to a meromorphic function on all of $\mathbb{C}$, holomorphic if η_i is nontrivial and with a single simple pole at $s = 1$ if η_i is trivial. Thus each factor $L(s, \psi_i, L/L^{H_i})$ is meromorphic on $\mathbb{C}$.

Step 4: A finite product of integer powers of meromorphic functions is meromorphic. Each factor on the right of (10.15) is meromorphic on $\mathbb{C}$ by Step 3 and is not identically zero, since on $\mathrm{Re}(s) > 1$ it is given by a convergent nonzero Euler product. The reciprocal of a meromorphic function that is not identically zero is again meromorphic (its poles are the zeros of the original, and conversely), so each factor raised to the integer power n_i, of either sign, is meromorphic on $\mathbb{C}$. A finite product of meromorphic functions is meromorphic. Therefore the right-hand side of (10.15) is meromorphic on $\mathbb{C}$, and it agrees with $L(s, \rho, L/K)$ on the half-plane $\mathrm{Re}(s) > 1$. By the identity theorem the meromorphic function on the right is the unique meromorphic continuation of $L(s, \rho, L/K)$ to $\mathbb{C}$, which is therefore meromorphic on $\mathbb{C}$.

The continuation just established is unconditional, but it comes at the cost of holomorphy. The factorization (10.15) expresses $L(s, \rho)$ as a ratio: the factors with $n_i > 0$ contribute to the numerator, those with $n_i < 0$ to the denominator. The poles of $L(s, \rho)$ can therefore arise from two sources, the poles of the numerator factors and the zeros of the denominator factors, and Brauer's theorem controls neither. Among the Hecke L-functions $L(s, \eta_i)$ only those with η_i trivial have a pole, a single simple pole at $s = 1$; the nontrivial ones are entire. So the only poles that (10.15) can manufacture lie at $s = 1$ (from trivial-η_i factors with $n_i > 0$) and, more seriously, at the zeros of the factors carrying negative exponents. Whether these apparent singularities cancel is exactly what the method leaves open.

The expectation, borne out in every case where it can be tested, is that for an irreducible nontrivial representation the apparent poles cancel completely and the Artin L-function is entire. This is Artin's conjecture.

Definition 10.3.3: Artin's Conjecture

Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$, and let ρ be an irreducible complex representation of G that is nontrivial (not the trivial one-dimensional representation). *Artin's conjecture* asserts that the Artin L-function $L(s, \rho, L/K)$ is entire: it extends to a holomorphic function on all of $\mathbb{C}$, with no poles.

The conjecture is consistent with, and sharpens, the meromorphic continuation of theorem 10.3.2. Brauer's argument establishes only meromorphy, and for a structural reason that the conjecture must overcome: the integers n_i of (10.14) need not be non-negative, so the factorization (10.15) presents $L(s, \rho)$ with genuine denominator factors, and a priori these admit poles of $L(s, \rho)$ at the zeros of those factors. Artin's conjecture asserts that, for ρ irreducible and nontrivial, all such apparent poles cancel. The trivial representation is excluded because $L(s, \mathbf{1}, L/K) = \zeta_K(s)$ has the simple pole of the Dedekind zeta function at $s = 1$; for any other irreducible ρ the conjecture predicts no pole anywhere.

Several families of cases are known unconditionally, in each instance by exhibiting enough non-negativity to defeat the apparent poles.

10.3.4 Remark: Known cases of Artin's conjecture The conjecture is a theorem in the following situations. *One-dimensional ρ:* when ρ is itself one-dimensional, $L(s, \rho, L/K)$ is directly a Hecke L-function by the class field theory step of theorem 10.3.2, and a nontrivial Hecke L-function is entire by theorem 9.4.1; no induction is needed and the conjecture holds. *Monomial representations:* if ρ is induced from a one-dimensional character of a subgroup (a *monomial* representation), then $L(s, \rho)$ is, by inductivity, a single Hecke L-function over the corresponding intermediate field, again entire when ρ is nontrivial; in particular the conjecture holds for every irreducible representation of a group all of whose irreducibles are monomial, such as a supersolvable group, where the factorization (10.15) reduces to a single factor with positive exponent. *Two-dimensional ρ:* certain irreducible two-dimensional representations are now known to be entire through the theory of modular and automorphic forms, the Artin L-function being matched with the L-function of a modular form, which is entire; this is the route by which the conjecture was established for a wide class of two-dimensional Galois representations. The general case remains open. These known cases, and the automorphic methods behind the two-dimensional one, are surveyed in [15, Ch. VII, §10–§12].

Although Artin's conjecture for an individual irreducible ρ is open, the one combination of Artin L-functions that matters most for the arithmetic of L/K, namely the quotient ζ_L/ζ_K, can be shown to be entire unconditionally. The starting point is the factorization of the Dedekind zeta function of the top field through the irreducible representations of the Galois group, established in theorem 10.2.5: with $\widehat{G}$ a set of representatives for the irreducible representations of $G = \mathrm{Gal}(L/K)$,

$$\zeta_L(s) = \prod_{\rho \in \widehat{G}} L(s, \rho, L/K)^{\dim \rho} \tag{10.16}$$

the trivial representation contributing the factor $L(s, \mathbf{1}) = \zeta_K(s)$ to the first power. Separating that factor,

$$\frac{\zeta_L(s)}{\zeta_K(s)} = \prod_{\rho \neq \mathbf{1}} L(s, \rho, L/K)^{\dim \rho} \tag{10.17}$$

the product running over the nontrivial irreducible representations. The Aramata–Brauer theorem asserts that this quotient is entire.

Theorem 10.3.5: Aramata–Brauer

Let L/K be a finite Galois extension. Then the quotient $\zeta_L(s)/\zeta_K(s)$ extends to an entire function on $\mathbb{C}$. Equivalently, the product $\prod_{\rho \neq \mathbf{1}} L(s, \rho, L/K)^{\dim \rho}$ over the nontrivial irreducible representations of $\mathrm{Gal}(L/K)$ is holomorphic on all of $\mathbb{C}$, so that every pole of ζ_L is accounted for by the simple pole of ζ_K at $s = 1$, and ζ_L has no pole that survives in the quotient.

***Proof*:**
The point is that the *regular* representation, and more precisely the augmentation representation ρ_{aug} whose character is $\chi_{\mathrm{reg}} - \mathbf{1}$ and whose L-function is exactly the quotient ζ_L/ζ_K, is, as a virtual character, a *non-negative rational* combination of characters induced from *nontrivial* one-dimensional characters of cyclic subgroups, $\chi_{\mathrm{reg}} - \mathbf{1} = \sum_i \lambda_i \mathrm{Ind}_{C_i}^G \psi_i$ with $\lambda_i \in \mathbb{Q}_{\geq 0}$ and each $\psi_i \neq \mathbf{1}$. This positivity, the content beyond Brauer's theorem, is proved through the theory of Heilbronn characters in [19, Ch. 10] and in [15, Ch. VII, §10]; the coefficients are in general only rational, not integers (for $G = A_4$, for instance, the decomposition forces a coefficient $\frac{1}{2}$),

so the argument passes through a power. Choosing a positive integer N with $N\lambda_i \in \mathbb{Z}_{\geq 0}$ for every i (a common denominator of the finitely many λ_i) and transporting the scaled identity through the additivity and inductivity of theorem 10.2.4 give

$$\left(\frac{\zeta_L(s)}{\zeta_K(s)}\right)^N = \prod_i L\big(s, \psi_i, L/L^{C_i}\big)^{N\lambda_i}, \qquad N\lambda_i \in \mathbb{Z}_{\geq 0}$$

a finite product of *non-negative* integer powers of one-dimensional Artin L-functions, each inducing character ψ_i nontrivial. By one-dimensional reciprocity ([15, Ch. VII, §10]; [12]) each $L(s, \psi_i, L/L^{C_i})$ is a nontrivial Hecke L-function, entire by theorem 9.4.1; a finite product of entire functions raised to non-negative integer powers is entire, so $(\zeta_L/\zeta_K)^N$ is entire. Now ζ_L/ζ_K is meromorphic on $\mathbb{C}$, being the Artin L-function $L(s, \chi_{\text{reg}} - \mathbf{1})$ of theorem 10.3.2, and a meromorphic function whose N-th power is entire is itself entire, since a pole of order $m > 0$ would give the N-th power a pole of order $Nm > 0$. Therefore ζ_L/ζ_K extends to an entire function on $\mathbb{C}$, the assertion.

The Aramata–Brauer theorem is what can be proved unconditionally toward Artin's conjecture. The conjecture would assert that each individual factor $L(s, \rho)^{\dim \rho}$ in (10.17) is entire; Aramata–Brauer asserts only that their product is, which is weaker but suffices for the one statement the arithmetic of L/K most needs. It guarantees that ζ_L and ζ_K have the same pole, a simple pole at $s = 1$, and that no Artin L-function $L(s, \rho)$ with $\rho \neq \mathbf{1}$ drags a pole into ζ_L beyond it. This single fact, that ζ_L/ζ_K is holomorphic and nonzero at $s = 1$ once non-vanishing is added, is the input that the Chebotarev density theorem of section 10.4 will require: the equidistribution of Frobenius classes is read off from the behavior of the Artin L-functions at the edge $s = 1$, and Aramata–Brauer is what licenses the passage from the factorization (10.16) to a statement about the poles of its factors there. The meromorphic continuation of this section, refined at $s = 1$ by Aramata–Brauer and completed by the non-vanishing taken up next, is the analytic foundation on which the non-abelian density theorem rests.

10.4 The Chebotarev Density Theorem

The analytic machinery of the preceding three sections was built for a single purpose, the one this section now collects. section 10.2 attached to each representation ρ of $G = \mathrm{Gal}(L/K)$ an Artin L-function $L(s, \rho, L/K)$ and recorded its two structural identities, the Euler product over the primes of K and the factorization $\zeta_L(s) = \prod_\rho L(s, \rho)^{\dim \rho}$ of the Dedekind zeta function of the top field into Artin L-functions indexed by the irreducible representations of the Galois group (theorem 10.2.5). section 10.3 supplied, by way of Brauer's induction theorem, the meromorphic continuation of every $L(s, \rho)$ to a neighborhood of the line $\mathrm{Re}(s) = 1$ (theorem 10.3.2), together with the Aramata–Brauer theorem that the quotient ζ_L/ζ_K is entire (theorem 10.3.5). What remains is to extract from these the arithmetic conclusion: the distribution of Frobenius classes among the primes of K.

The pattern is the one established for Dirichlet's theorem in chapter 7. There the indicator of a residue class was expanded over the characters of $(\mathbb{Z}/m\mathbb{Z})^\times$, each character contributed the logarithm of a Dirichlet L-function to the prime sum, and only the principal character carried a singularity at $s = 1$, the pole of $\zeta = L(s, \chi_0)$; the non-vanishing of $L(1, \chi) \neq 0$ for $\chi \neq \chi_0$ silenced every other term, leaving the density $1/\varphi(m)$. The non-abelian theorem replaces residue classes by conjugacy classes, Dirichlet characters by irreducible characters of G, Dirichlet L-functions by Artin L-functions, and the non-vanishing $L(1,\chi) \neq 0$ by its exact analogue for Artin L-functions. That analogue is the first order of business; the density theorem follows from it by the same orthogonality computation.

Theorem 10.4.1: Non-Vanishing of Artin L at $s = 1$

Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$, and let ρ be an irreducible representation of G with $\rho \neq \mathbf{1}$. Then $L(s, \rho, L/K)$ is holomorphic and non-zero at $s = 1$:

$$L(1, \rho, L/K) \neq 0$$

with neither a pole nor a zero there.

***Proof*:**
Write $L(s,\rho)$ for $L(s,\rho,L/K)$. The order of vanishing at $s = 1$ is computed directly for each ρ, by reducing through Brauer's induction theorem to one-dimensional L-functions, which class field theory identifies with Hecke L-functions whose behavior at $s = 1$ is known from chapter 9.

Step 1: The order is additive and inductive. For a virtual character χ of G write $\mathrm{ord}(\chi) = \mathrm{ord}_{s=1} L(s,\chi,L/K)$, the order of vanishing at $s = 1$ (negative for a pole), well defined because each $L(s,\chi)$ is meromorphic in a neighborhood of $s = 1$ by theorem 10.3.2. The map ord is additive, since $L(s,\chi_1+\chi_2) = L(s,\chi_1)\,L(s,\chi_2)$ makes the order of a product the sum of the orders (theorem 10.2.4), and inductive, since $L(s, \mathrm{Ind}_H^G \psi, L/K) = L(s,\psi,L/L^H)$ (theorem 10.2.4); it therefore extends to a homomorphism on the ring of virtual characters of G.

Step 2: The order of a one-dimensional L-function. Let $H \leq G$ and let ψ be a one-dimensional character of H. By one-dimensional reciprocity ([15, Ch. VII, §10]; [12]) the Artin L-function $L(s,\psi,L/L^H)$ is the Hecke L-function over $F = L^H$ of a ray class character of F; by Hecke's continuation (theorem 9.4.1) and the Hecke non-vanishing at $s = 1$ (theorem 9.5.2) it is holomorphic and non-zero at $s = 1$ when ψ is nontrivial, and equals $\zeta_F(s)$ with a simple pole at $s = 1$ when ψ is trivial. Hence

$$\mathrm{ord}_{s=1} L(s,\psi,L/L^H) = -\langle \psi, \mathbf{1}_H\rangle = \begin{cases} -1, & \psi = \mathbf{1}_H \\ 0, & \psi \neq \mathbf{1}_H \end{cases} \tag{10.18}$$

the Hecke non-vanishing being precisely what makes the nontrivial order equal to 0 rather than only non-negative.

Step 3: Brauer induction computes the order. By Brauer's induction theorem (theorem 10.3.1) write the character of ρ as $\chi_\rho = \sum_i n_i \,\mathrm{Ind}_{H_i}^G \psi_i$ with $n_i \in \mathbb{Z}$ and each ψ_i a one-dimensional character of a subgroup $H_i \leq G$. Applying the homomorphism ord of Step 1, then the one-dimensional orders (10.18), then Frobenius reciprocity $\langle \mathrm{Ind}_{H_i}^G \psi_i, \mathbf{1}_G\rangle = \langle \psi_i, \mathbf{1}_{H_i}\rangle$ ([3, Frobenius reciprocity]),

$$\mathrm{ord}_{s=1} L(s,\rho) = \sum_i n_i \,\mathrm{ord}_{s=1} L(s,\psi_i,L/L^{H_i}) = -\sum_i n_i \langle \psi_i, \mathbf{1}_{H_i}\rangle = -\sum_i n_i \langle \mathrm{Ind}_{H_i}^G \psi_i, \mathbf{1}_G\rangle = -\langle \chi_\rho, \mathbf{1}_G\rangle \tag{10.19}$$

the final equality by linearity of $\langle\,\cdot\,,\mathbf{1}_G\rangle$ applied to the Brauer decomposition of χ_ρ.

Step 4: Conclusion. For ρ irreducible and nontrivial, $\langle \chi_\rho, \mathbf{1}_G \rangle = 0$, the trivial representation not being a constituent of an irreducible $\rho \neq \mathbf{1}$; so (10.19) gives $\operatorname{ord}_{s=1} L(s,\rho) = 0$. An order of vanishing equal to 0 at $s = 1$ means precisely that the function is holomorphic and non-zero there, that is $L(1,\rho) \neq 0$, which is the assertion. As a consistency check on theorem 10.3.5: weighting (10.19) by $\dim \rho'$ and summing over the nontrivial irreducibles returns $\operatorname{ord}_{s=1}(\zeta_L/\zeta_K) = -\sum_{\rho' \neq \mathbf{1}} (\dim \rho') \langle \chi_{\rho'}, \mathbf{1}_G \rangle = 0$, recovering the holomorphy and non-vanishing of ζ_L/ζ_K at $s = 1$. The standard treatment is [15, Ch. VII, §13].

The proof isolates the precise role of each ingredient. Brauer's induction theorem reduces the order of $L(s,\rho)$ at $s = 1$ to a combination of orders of one-dimensional L-functions; class field theory identifies each of those with a Hecke L-function; and the Hecke non-vanishing of theorem 9.5.2, which keeps each nontrivial Hecke L-function non-zero at $s = 1$, is what pins each such order to 0 rather than only bounding it below. Frobenius reciprocity then assembles the orders into $\operatorname{ord}_{s=1} L(s,\rho) = -\langle \chi_\rho, \mathbf{1}_G \rangle$, which vanishes exactly when ρ contains no trivial constituent. The argument is the exact non-abelian counterpart of the Dirichlet non-vanishing $L(1,\chi) \neq 0$ of theorem 7.4.2, with the Hecke non-vanishing of chapter 9 as the irreducible analytic input.

Lemma 10.4.2: The Logarithmic Frobenius Sum

Let ρ be a representation of $G = \mathrm{Gal}(L/K)$ with character $\chi_\rho = \operatorname{tr}\rho$. As $s \to 1^+$ along the real axis,

$$\sum_{\mathfrak{p}} \chi_\rho(\mathrm{Frob}_{\mathfrak{p}})\,(N\mathfrak{p})^{-s} = \log L(s,\rho,L/K) + O(1) \tag{10.20}$$

the sum running over the primes $\mathfrak{p}$ of K that are unramified in L, and $\mathrm{Frob}_{\mathfrak{p}}$ denoting any element of the Frobenius conjugacy class (so that $\chi_\rho(\mathrm{Frob}_{\mathfrak{p}})$, a class function evaluated on a conjugacy class, is well defined). The branch of the logarithm is the one furnished by the Euler product.

***Proof*:**
Fix s real with $s > 1$. By the Euler product of definition 10.2.1 the Artin L-function factors as $L(s,\rho) = \prod_{\mathfrak{p}} \det\bigl(1 - \rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s} \big| V^{I_{\mathfrak{P}}}\bigr)^{-1}$, the product converging absolutely on $\mathrm{Re}(s) > 1$ (proposition 10.2.3). Taking the logarithm of an absolutely convergent product of nonvanishing factors term by term, and expanding each local logarithm by the identity $-\log\det(1 - A) = \sum_{k\geq 1} \operatorname{tr}(A^k)/k$, valid for a matrix A with $\|A\| < 1$ (the eigenvalues of $\rho(\mathrm{Frob}_{\mathfrak{p}})$ are roots of unity, so $A = \rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s}$ has $\|A\| \leq (N\mathfrak{p})^{-s} < 1$), gives the absolutely convergent double sum

$$\log L(s,\rho) = \sum_{\mathfrak{p}} \sum_{k\geq 1} \frac{1}{k} \operatorname{tr}\bigl(\rho(\mathrm{Frob}_{\mathfrak{p}})^k \big| V^{I_{\mathfrak{P}}}\bigr)\,(N\mathfrak{p})^{-ks} \tag{10.21}$$

in exact parallel with the logarithmic expansion of a Dirichlet Euler product in proposition 2.3.5. For a prime $\mathfrak{p}$ unramified in L the inertia group $I_{\mathfrak{P}}$ is trivial, so $V^{I_{\mathfrak{P}}} = V$ and the trace is taken on all of V; there $\operatorname{tr}(\rho(\mathrm{Frob}_{\mathfrak{p}})^k) = \chi_\rho(\mathrm{Frob}_{\mathfrak{p}}^k)$, and the $k = 1$ term is $\chi_\rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s}$. The claim is that everything in (10.21) beyond these $k = 1$ unramified terms is bounded as $s \to 1^+$.

The higher prime powers ($k \geq 2$). The trace of a unitary matrix on a space of dimension $\dim\rho$ is bounded by $\dim\rho$ in absolute value, so $|\operatorname{tr}(\rho(\mathrm{Frob}_{\mathfrak{p}})^k | V^{I_{\mathfrak{P}}})| \leq \dim\rho$ for every $\mathfrak{p}$ and k. Hence the tail over $k \geq 2$ is dominated, for $s \geq 1$, by

$$\sum_{\mathfrak{p}} \sum_{k \geq 2} \frac{\dim \rho}{k} (N\mathfrak{p})^{-ks} \leq \dim \rho \sum_{\mathfrak{p}} \sum_{k \geq 2} (N\mathfrak{p})^{-k} = \dim \rho \sum_{\mathfrak{p}} \frac{(N\mathfrak{p})^{-2}}{1-(N\mathfrak{p})^{-1}} \leq 2 \dim \rho \sum_{\mathfrak{p}} (N\mathfrak{p})^{-2}$$

using $1-(N\mathfrak{p})^{-1} \geq 1/2$. The final sum $\sum_{\mathfrak{p}}(N\mathfrak{p})^{-2}$ converges (it is dominated by $\sum_{\mathfrak{p}}(N\mathfrak{p})^{-s}$ for $s = 2 > 1$, convergent by proposition 10.2.3 applied to the trivial representation, equivalently by the convergence of $\zeta_K(2)$), so the entire $k \geq 2$ contribution is $O(1)$ uniformly for $s \geq 1$.

The ramified primes. Only finitely many primes $\mathfrak{p}$ of K ramify in L, namely those dividing the relative discriminant ([2, ramification and the discriminant]). For each such $\mathfrak{p}$ the $k = 1$ term $\frac{1}{1}\mathrm{tr}(\rho(\mathrm{Frob}_{\mathfrak{p}})|V^{I_{\mathfrak{P}}})(N\mathfrak{p})^{-s}$ has absolute value at most $(\dim \rho)(N\mathfrak{p})^{-s} \leq \dim \rho$ for $s \geq 1$, and a finite sum of such bounded terms is $O(1)$. (Here $\mathrm{Frob}_{\mathfrak{p}}$ at a ramified prime is taken as the Frobenius of the quotient $D_{\mathfrak{P}}/I_{\mathfrak{P}}$ acting on $V^{I_{\mathfrak{P}}}$, as in definition 10.2.1; only the boundedness of the term is needed.)

Subtracting the two $O(1)$ contributions, the higher prime powers and the finitely many ramified $k = 1$ terms, from (10.21) leaves exactly the sum over unramified $\mathfrak{p}$ of the $k = 1$ terms,

$$\log L(s, \rho) = \sum_{\mathfrak{p} \text{ unram.}} \chi_\rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s} + O(1) \qquad (s \to 1^+)$$

which is (10.20) after transposing the $O(1)$, since $\chi_\rho(\mathrm{Frob}_{\mathfrak{p}}^1) = \chi_\rho(\mathrm{Frob}_{\mathfrak{p}})$. That $\chi_\rho(\mathrm{Frob}_{\mathfrak{p}})$ depends only on the conjugacy class of $\mathrm{Frob}_{\mathfrak{p}}$, and so is unambiguous despite $\mathrm{Frob}_{\mathfrak{p}}$ being defined only up to conjugacy (proposition 10.1.1), is the defining property of a character as a class function.

The expansion (10.20) is the bridge between the analytic and arithmetic sides, exactly as the corresponding expansion was for Dirichlet L-functions in chapter 7. Its left-hand side is a sum over primes weighted by Frobenius data; its right-hand side is the logarithm of an L-function whose behavior at $s = 1$ is now known precisely, holomorphic and non-zero for ρ irreducible nontrivial by theorem 10.4.1, and a simple pole for $\rho = \mathbf{1}$ since $L(s, \mathbf{1}) = \zeta_K(s)$. The order of the pole of $L(s, \rho)$ at $s = 1$ therefore reads

$$\log L(s, \rho) = m_\rho \log \frac{1}{s-1} + O(1) \quad (s \to 1^+), \qquad m_\rho = -\mathrm{ord}_{\,s=1} L(s, \rho) = \begin{cases} 1, & \rho = \mathbf{1} \\ 0, & \rho \neq \mathbf{1} \text{ irreducible} \end{cases} \tag{10.22}$$

where m_ρ is the order of the pole: for $\rho = \mathbf{1}$ the simple pole of ζ_K gives $\log \zeta_K(s) = \log \frac{1}{s-1} + O(1)$ as $s \to 1^+$ (the residue being a nonzero constant, $\log[(s-1)\zeta_K(s)]$ stays bounded), while for irreducible $\rho \neq \mathbf{1}$ the non-vanishing makes $\log L(s, \rho)$ bounded near $s = 1$, so $m_\rho = 0$.

The non-vanishing theorem and the logarithmic expansion combine, by the orthogonality of irreducible characters, into the central result of the chapter. The Dirichlet density of a set of primes of K is defined exactly as in the rational case (definition 9.5.3): for a set S of primes of K,

$$\delta(S) = \lim_{s \to 1^+} \frac{\sum_{\mathfrak{p} \in S}(N\mathfrak{p})^{-s}}{\sum_{\mathfrak{p}}(N\mathfrak{p})^{-s}} = \lim_{s \to 1^+} \frac{\sum_{\mathfrak{p} \in S}(N\mathfrak{p})^{-s}}{\log \frac{1}{s-1}}$$

when the limit exists, the second form using $\sum_{\mathfrak{p}}(N\mathfrak{p})^{-s} = \log \frac{1}{s-1} + O(1)$ as $s \to 1^+$ (eq. (9.18)), itself a transcription of the simple pole of ζ_K.

Theorem 10.4.3: Chebotarev Density Theorem

Let L/K be a finite Galois extension with group $G = \mathrm{Gal}(L/K)$, and let $C \subseteq G$ be a conjugacy class. The set of primes $\mathfrak{p}$ of K, unramified in L, whose Frobenius class $\mathrm{Frob}_{\mathfrak{p}}$ equals C has Dirichlet density

$$\delta\big(\{\mathfrak{p} : \mathrm{Frob}_{\mathfrak{p}} = C\}\big) = \frac{|C|}{|G|}$$

Equivalently, as $s \to 1^+$,

$$\sum_{\substack{\mathfrak{p}\text{ unram.}\\ \mathrm{Frob}_{\mathfrak{p}}=C}} (N\mathfrak{p})^{-s} = \frac{|C|}{|G|}\log\frac{1}{s-1} + O(1) \tag{10.23}$$

In particular, every conjugacy class is the Frobenius of infinitely many primes; the primes split among the Frobenius classes in proportion to the sizes $|C|$ of those classes.

***Proof*:**

Let $\widehat{G}$ be a set of representatives for the irreducible representations of G, with characters $\chi_\rho = \mathrm{tr}\rho$. The proof expands the indicator of the class C over irreducible characters, applies the logarithmic Frobenius sum to each, and uses the non-vanishing theorem to discard every term but one.

Step 1: Expanding the indicator of a conjugacy class. Let $\mathbf{1}_C : G \to \{0,1\}$ be the indicator function of the class C, equal to 1 on C and 0 elsewhere. As C is a union of conjugacy classes (a single one), $\mathbf{1}_C$ is a class function, hence a $\mathbb{C}$-linear combination of the irreducible characters. Column orthogonality of the irreducible characters ([3, character orthogonality]) evaluates the coefficients explicitly: for $g \in G$,

$$\mathbf{1}_C(g) = \frac{|C|}{|G|}\sum_{\rho\in\widehat{G}} \overline{\chi_\rho(C)}\,\chi_\rho(g) \tag{10.24}$$

where $\chi_\rho(C)$ denotes the common value of χ_ρ on the class C. Indeed, the right-hand side is a class function of g; evaluating it on a class C' and using column orthogonality $\sum_\rho \overline{\chi_\rho(C)}\,\chi_\rho(C') = \delta_{C,C'}\,|G|/|C|$ (the orthogonality of the columns of the character table, indexed by conjugacy classes) gives $\frac{|C|}{|G|}\cdot\delta_{C,C'}\,|G|/|C| = \delta_{C,C'}$, which is $\mathbf{1}_C(C')$. As two class functions agreeing on every class, the two sides of (10.24) are equal.

Step 2: From the indicator to a sum over primes. Apply (10.24) with $g = \mathrm{Frob}_{\mathfrak{p}}$ for each prime $\mathfrak{p}$ of K unramified in L. The condition $\mathrm{Frob}_{\mathfrak{p}} = C$ is exactly $\mathbf{1}_C(\mathrm{Frob}_{\mathfrak{p}}) = 1$, so weighting by $(N\mathfrak{p})^{-s}$ and summing over the unramified primes,

$$\sum_{\substack{\mathfrak{p}\text{ unram.}\\ \mathrm{Frob}_{\mathfrak{p}}=C}} (N\mathfrak{p})^{-s} = \sum_{\mathfrak{p}\text{ unram.}} \mathbf{1}_C(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s} = \frac{|C|}{|G|}\sum_{\rho\in\widehat{G}}\overline{\chi_\rho(C)}\sum_{\mathfrak{p}\text{ unram.}}\chi_\rho(\mathrm{Frob}_{\mathfrak{p}})(N\mathfrak{p})^{-s} \tag{10.25}$$

the interchange of the finite sum over $\rho \in \widehat{G}$ with the sum over primes being legitimate for $s > 1$, where every series converges absolutely. The inner prime sum is precisely the left-hand side of the logarithmic Frobenius sum (10.20).

Step 3: Inserting the analytic input. By lemma 10.4.2, for each $\rho \in \widehat{G}$,

$$\sum_{\mathfrak{p}\text{ unram.}} \chi_\rho(\mathrm{Frob}_\mathfrak{p})(N\mathfrak{p})^{-s} = \log L(s,\rho) + O(1) = m_\rho \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

by (10.22), where $m_\rho = 1$ for the trivial representation $\rho = \mathbf{1}$ (the pole of ζ_K) and $m_\rho = 0$ for every irreducible $\rho \neq \mathbf{1}$ (holomorphic and non-zero at $s = 1$ by theorem 10.4.1). Substituting into (10.25),

$$\sum_{\substack{\mathfrak{p}\text{ unram.}\\ \mathrm{Frob}_\mathfrak{p}=C}} (N\mathfrak{p})^{-s} = \frac{|C|}{|G|} \sum_{\rho\in\widehat{G}} \overline{\chi_\rho(C)} \Big(m_\rho \log \frac{1}{s-1} + O(1) \Big) \tag{10.26}$$

In the sum over ρ the coefficient of $\log \frac{1}{s-1}$ is $\sum_\rho \overline{\chi_\rho(C)}\, m_\rho$, and since m_ρ vanishes for every $\rho \neq \mathbf{1}$, only the trivial representation survives in that coefficient. For the trivial representation $\chi_\mathbf{1} \equiv 1$, so $\overline{\chi_\mathbf{1}(C)} = 1$ and $m_\mathbf{1} = 1$; the surviving coefficient is therefore 1. The finitely many bounded terms $\overline{\chi_\rho(C)}\, O(1)$ combine into a single $O(1)$ (the number of representations $|\widehat{G}|$ and the character values $\chi_\rho(C)$ are constants independent of s). Hence

$$\sum_{\substack{\mathfrak{p}\text{ unram.}\\ \mathrm{Frob}_\mathfrak{p}=C}} (N\mathfrak{p})^{-s} = \frac{|C|}{|G|} \log \frac{1}{s-1} + O(1) \qquad (s \to 1^+)$$

which is (10.23).

Step 4: Reading off the density. Dividing (10.23) by $\sum_\mathfrak{p} (N\mathfrak{p})^{-s} = \log \frac{1}{s-1} + O(1)$ (eq. (9.18)),

$$\frac{\sum_{\mathrm{Frob}_\mathfrak{p}=C} (N\mathfrak{p})^{-s}}{\sum_\mathfrak{p} (N\mathfrak{p})^{-s}} = \frac{\frac{|C|}{|G|} \log \frac{1}{s-1} + O(1)}{\log \frac{1}{s-1} + O(1)} = \frac{\frac{|C|}{|G|} + O\Big(1/\log \frac{1}{s-1}\Big)}{1 + O\Big(1/\log \frac{1}{s-1}\Big)}$$

the last equality dividing numerator and denominator by $\log \frac{1}{s-1}$, which tends to $+\infty$ as $s \to 1^+$ so that both $O(1)$ remainders become $O(1/\log \frac{1}{s-1}) \to 0$. Letting $s \to 1^+$, the quotient tends to $\frac{|C|}{|G|}/1 = |C|/|G|$. By the definition of Dirichlet density (definition 9.5.3) the set $\{\mathfrak{p} : \mathrm{Frob}_\mathfrak{p} = C\}$ has density $|C|/|G|$. Since $|C|/|G| > 0$, the set is infinite, and the final clause follows because the conjugacy classes of G partition the unramified primes by their Frobenius and the densities $\sum_C |C|/|G| = |G|/|G| = 1$ exhaust the whole. A complete account, including the effective and unconditional refinements, is in [15, Ch. VII, §13].

Two features of the statement deserve comment. First, the density attached to a class C is $|C|/|G|$, not $1/|G|$: a prime is more likely to have Frobenius in a large conjugacy class than in a small one, simply because there are more group elements to realize. Counting individual *elements* $g \in G$ rather than classes, the proportion of unramified primes $\mathfrak{p}$ for which the Frobenius class contains a fixed g is $|C_g|/|G|$ where C_g is the class of g; the probability that a uniformly random element of G is conjugate to g is the same $|C_g|/|G|$, so the heuristic "$\mathrm{Frob}_\mathfrak{p}$ is equidistributed in G with respect to the uniform measure, read up to conjugacy" is exactly correct. Second, the theorem is genuinely non-abelian: when G is abelian every conjugacy class is a singleton, $|C| = 1$, and the density of each is $1/|G|$, recovering equidistribution of Frobenius elements one element at a time.

10.4.4 Remark: Chebotarev contains Dirichlet The Chebotarev theorem specializes to Dirichlet's theorem on primes in arithmetic progressions (theorem 7.5.4) by the choice $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_m)$, the m-th cyclotomic field. There $G = \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ is abelian ([2, the cyclotomic Galois group]), so conjugacy classes are singletons $C = \{a\}$ indexed by residues $a \in (\mathbb{Z}/m\mathbb{Z})^\times$, each of size $|C| = 1$ and $|G| = \varphi(m)$. The isomorphism sends the Frobenius Frob_p of a prime $p \nmid m$ (which is unramified in $\mathbb{Q}(\zeta_m)$) to the residue class $p \bmod m$, because Frob_p is the automorphism $\zeta_m \mapsto \zeta_m^p$. Thus the condition $\mathrm{Frob}_p = \{a\}$ is the congruence $p \equiv a \pmod{m}$, and theorem 10.4.3 gives those primes density $|C|/|G| = 1/\varphi(m)$, which is precisely theorem 7.5.4. The two arguments are the same: the orthogonality of irreducible characters of G is the orthogonality of Dirichlet characters modulo m under the identification $G \cong (\mathbb{Z}/m\mathbb{Z})^\times$, and the Artin L-functions $L(s, \rho, \mathbb{Q}(\zeta_m)/\mathbb{Q})$ are the Dirichlet L-functions $L(s, \chi)$ under the same identification (the one-dimensional case of the equality of Artin and Hecke L-functions, [12]). The non-abelian theorem is the abelian one with the character theory of a non-commutative group in place of the dual of $(\mathbb{Z}/m\mathbb{Z})^\times$.

10.5 Applications and Outlook

The Chebotarev density theorem (theorem 10.4.3) is what this chapter rests on, and like every density theorem in the book it is useful only when applied to concrete fields. This closing section does three things. It first specializes Chebotarev to the abelian case $L = \mathbb{Q}(\zeta_m)$ and watches Dirichlet's theorem (theorem 7.5.4) reappear as the abelian case of the general statement, completing what chapter 7 began. It then works the smallest genuinely non-abelian example in full, the S_3 extension $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)/\mathbb{Q}$, where the three conjugacy classes split the rational primes into three families of densities $1/6$, $1/2$, $1/3$, each family characterized by an elementary congruence condition on $x^3 - 2$. It closes by placing Artin's conjecture (definition 10.3.3) inside the Langlands program and by recording the single pattern that organizes the entire Part: chapters 4 and 7 to 10 are five instances of one construction.

The abelian case comes first, and it recovers Dirichlet's theorem. Take $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_m)$, the m-th cyclotomic field. The extension $L/\mathbb{Q}$ is Galois, and the Galois group is canonically the unit group modulo m,

$$\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \;\cong\; (\mathbb{Z}/m\mathbb{Z})^\times, \qquad \sigma_a(\zeta_m) = \zeta_m^a \tag{10.27}$$

the automorphism σ_a raising ζ_m to the a-th power corresponding to the residue a ([2]; recorded in proposition 8.4.5). The group $G = (\mathbb{Z}/m\mathbb{Z})^\times$ is abelian, so every conjugacy class is a single element, and the discriminant of $\mathbb{Q}(\zeta_m)$ is divisible only by the primes dividing m, so a rational prime $p \nmid m$ is unramified in L. For such a prime the Frobenius class of definition 10.1.2 is the single element

$$\mathrm{Frob}_p = \sigma_p = (p \bmod m) \in (\mathbb{Z}/m\mathbb{Z})^\times \tag{10.28}$$

since Frob_p acts as the p-power map on the residue field of any prime above p, and under (10.27) the p-power map is exactly σ_p (proposition 10.1.1, computed for this field in the proof of proposition 8.4.5).

Fix a residue class $a \in (\mathbb{Z}/m\mathbb{Z})^\times$. The set of unramified primes with prescribed Frobenius $\mathrm{Frob}_p = \sigma_a$ is, by (10.28), exactly the set of primes $p \equiv a \pmod{m}$. Chebotarev's theorem applied to the abelian extension $L/\mathbb{Q}$ assigns this set the density $|C|/|G|$, where $C = \{\sigma_a\}$ is the conjugacy class; here $|C| = 1$ and $|G| = \varphi(m)$, so

$$\delta(\{p : \mathrm{Frob}_p = \sigma_a\}) = \frac{|C|}{|G|} = \frac{1}{\varphi(m)} \tag{10.29}$$

This is precisely Dirichlet's density statement (theorem 7.5.4): the primes $p \equiv a \pmod m$ have Dirichlet density $1/\varphi(m)$, and in particular there are infinitely many of them, so every arithmetic progression a mod m with $\gcd(a, m) = 1$ contains infinitely many primes. The non-abelian theorem of section 10.4 thus contains the theorem of section 7.5 as its simplest case, the case in which the Galois group is commutative and the Frobenius class degenerates to a single residue.

The recovery is more than a formal match of densities; the two proofs are the same proof read at two levels of generality. Dirichlet's argument resolved the indicator of the class a mod m into Dirichlet characters and attached to each character an L-function, the non-vanishing of $L(1, \chi)$ confining the divergence to the principal character. Chebotarev's argument, specialized to $\mathbb{Q}(\zeta_m)$, resolves the indicator of the Frobenius class into the irreducible characters of $G = (\mathbb{Z}/m\mathbb{Z})^\times$, which are one-dimensional because G is abelian, and attaches to each the Artin L-function $L(s, \chi, L/\mathbb{Q})$. By the reciprocity used in theorem 10.3.2, each one-dimensional Artin L-function is a Hecke L-function, and over $\mathbb{Q}$ a Hecke L-function of $(\mathbb{Z}/m\mathbb{Z})^\times$ is a Dirichlet L-function (proposition 9.2.4); the non-vanishing at $s = 1$ that confines the divergence is theorem 9.5.2, which over $\mathbb{Q}$ is theorem 7.4.2. The factorization of $\zeta_{\mathbb{Q}(\zeta_m)}$ into Dirichlet L-functions recorded in proposition 8.4.5 is the abelian instance of the factorization $\zeta_L = \prod_\rho L(s, \rho, L/\mathbb{Q})^{\dim \rho}$ of theorem 10.2.5, with the irreducible representations ρ all of dimension one.

The first extension whose Galois group is non-abelian is the splitting field of $x^3 - 2$ over $\mathbb{Q}$, and it exhibits every feature of the non-abelian theory in a setting small enough to compute by hand. It is the worked example of the chapter.

Example 10.5.1: The S_3 Extension

Let $L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$ be the splitting field of $x^3 - 2$ over $\mathbb{Q}$. The three roots of $x^3 - 2$ are $\sqrt[3]{2}$, $\zeta_3 \sqrt[3]{2}$, and $\zeta_3^2 \sqrt[3]{2}$, and L is obtained from $\mathbb{Q}$ by adjoining all three, equivalently by adjoining the real root $\sqrt[3]{2}$ and the primitive cube root of unity $\zeta_3 = e^{2\pi i/3}$. The degree is $[L : \mathbb{Q}] = 6$: the cubic $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, giving $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, and ζ_3 has degree 2 over the totally real field $\mathbb{Q}(\sqrt[3]{2})$, which contains no nontrivial root of unity beyond ± 1. The Galois group $G = \mathrm{Gal}(L/\mathbb{Q})$ acts faithfully on the three roots and has order 6, so

$$G \cong S_3 \tag{10.30}$$

the full symmetric group on the three roots. Inside L sit two intermediate fields that organize the analysis: the non-Galois cubic $\mathbb{Q}(\sqrt[3]{2})$, fixed by a transposition, and the quadratic field $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, fixed by the alternating subgroup A_3 and itself Galois over $\mathbb{Q}$ with group $S_3/A_3 \cong \{\pm 1\}$.

The conjugacy classes. The symmetric group S_3 has three conjugacy classes, sorted by cycle type:

class	cycle type	size
$C_1 = \{e\}$	identity	1
$C_2 = \{(12), (13), (23)\}$	transpositions	3
$C_3 = \{(123), (132)\}$	3-cycles	2

The sizes are $1, 3, 2$, summing to $|G| = 6$. By the Chebotarev density theorem (theorem 10.4.3), the rational primes unramified in L (all but $p = 2$ and $p = 3$, the primes dividing $\mathrm{disc}(x^3 - 2) =$

$-108 = -2^2 3^3$) distribute their Frobenius classes in proportion to these sizes:

$$\delta\big(\mathrm{Frob}_p = C_1\big) = \frac{1}{6}, \qquad \delta\big(\mathrm{Frob}_p = C_2\big) = \frac{3}{6} = \frac{1}{2}, \qquad \delta\big(\mathrm{Frob}_p = C_3\big) = \frac{2}{6} = \frac{1}{3} \tag{10.31}$$

The three densities sum to $\frac{1}{6} + \frac{1}{2} + \frac{1}{3} = 1$, accounting (up to the density-zero set of ramified primes) for all rational primes.

Reading the Frobenius off $x^3 - 2 \bmod p$. For an unramified prime $p \notin \{2, 3\}$ the Frobenius class Frob_p acts as a permutation of the three roots of $x^3 - 2$, and its cycle type is exactly the factorization type of $x^3 - 2$ modulo p. This is the Dedekind–Kummer dictionary ([2]): the degrees of the irreducible factors of $x^3 - 2 \bmod p$ are the lengths of the cycles of Frob_p acting on the roots, equivalently the residue degrees of the primes of $\mathbb{Q}(\sqrt[3]{2})$ above p. The three classes correspond to the three factorization patterns of a cubic:

- $\mathrm{Frob}_p = C_1$ (identity, density $\frac{1}{6}$): $x^3 - 2$ has *three* roots modulo p, splitting as a product of three distinct linear factors. Equivalently 2 is a cube modulo p *and* $p \equiv 1 \pmod 3$.
- $\mathrm{Frob}_p = C_2$ (a transposition, density $\frac{1}{2}$): $x^3 - 2$ has exactly *one* root modulo p, factoring as a linear factor times an irreducible quadratic. This is the case $p \equiv 2 \pmod 3$.
- $\mathrm{Frob}_p = C_3$ (a 3-cycle, density $\frac{1}{3}$): $x^3 - 2$ has *no* root modulo p and is irreducible of degree 3. Equivalently $p \equiv 1 \pmod 3$ but 2 is *not* a cube modulo p.

A prime p splits completely in L exactly when $\mathrm{Frob}_p = C_1$, the trivial class, which by the list above happens for density $1/6$ of the primes, those with $p \equiv 1 \pmod 3$ and 2 a cubic residue modulo p; this is the density predicted by the splitting-completely case of Chebotarev, $|C_1|/|G| = 1/6$, and matches the elementary count, the density $\frac{1}{2}$ of primes $p \equiv 1 \pmod 3$ times the conditional density $\frac{1}{3}$ that 2 is a cube among them, $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.

Splitting in the two subfields. The same Frobenius governs the splitting of p in the intermediate fields $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt{-3})$, and cross-checking the two recovers the densities. In the quadratic field $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, the prime p splits or is inert according to the quadratic character: it splits iff $p \equiv 1 \pmod 3$ and is inert iff $p \equiv 2 \pmod 3$, each of density $\frac{1}{2}$ by theorem 7.5.4. Under the quotient $S_3 \twoheadrightarrow S_3/A_3 \cong \{\pm 1\}$, the class C_2 of transpositions maps to the nontrivial element (inert in $\mathbb{Q}(\sqrt{-3})$, $p \equiv 2$) while C_1 and C_3 map to the identity (split in $\mathbb{Q}(\sqrt{-3})$, $p \equiv 1$). The density of C_2 is thus the density $\frac{1}{2}$ of inert primes, consistent with (10.31), and the densities of C_1 and C_3 must partition the split half: indeed $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$. Among the primes $p \equiv 1 \pmod 3$, those for which 2 is a cubic residue (class C_1, p splits completely in $\mathbb{Q}(\sqrt[3]{2})$) and those for which it is not (class C_3, p has a single degree-3 prime in $\mathbb{Q}(\sqrt[3]{2})$, equivalently one unramified prime of residue degree 3) occur in the ratio $1 : 2$, reflecting that the cubic-residue condition cuts the index-3 subgroup of cubes out of $(\mathbb{Z}/p\mathbb{Z})^\times$. In the non-Galois cubic $\mathbb{Q}(\sqrt[3]{2})$ the splitting types are: p splits into three primes (class C_1), one prime of degree 1 and one of degree 2 (class C_2), or one prime of degree 3 (class C_3), exactly the factorization types of $x^3 - 2 \bmod p$.

The Artin factorization. The group S_3 has three irreducible representations: the trivial representation $\mathbf{1}$, the sign representation ϵ (the quadratic character through S_3/A_3), and the standard two-dimensional representation ρ_2. The factorization of theorem 10.2.5 reads

$$\zeta_L(s) = \zeta(s)\, L(s, \epsilon, L/\mathbb{Q})\, L(s, \rho_2, L/\mathbb{Q})^2 \tag{10.32}$$

with ζ the Riemann zeta function (trivial representation, chapter 4), $L(s,\epsilon,L/\mathbb{Q}) = L(s,\chi_{-3})$ the Dirichlet L-function of the quadratic character of $\mathbb{Q}(\sqrt{-3})$ (a one-dimensional Artin L-function, hence a Hecke L-function by the reciprocity of theorem 10.3.2), and $L(s,\rho_2,L/\mathbb{Q})$ the two-dimensional Artin L-function, which equals the Dedekind zeta of the cubic field divided by ζ:

$$L(s,\rho_2,L/\mathbb{Q}) = \frac{\zeta_{\mathbb{Q}(\sqrt[3]{2})}(s)}{\zeta(s)} \tag{10.33}$$

The right side is holomorphic and nonzero at $s = 1$, the simple pole of $\zeta_{\mathbb{Q}(\sqrt[3]{2})}$ cancelling that of ζ, which verifies Artin's conjecture (definition 10.3.3) for ρ_2 in this case: the two-dimensional Artin L-function is entire. Identity (10.33) also exhibits ρ_2 as the representation induced from a nontrivial character of $A_3 \cong \mathbb{Z}/3\mathbb{Z}$, the route by which the induction formalism of theorem 10.2.4 produces it; the factor $L(s,\rho_2,L/\mathbb{Q})^2$ in (10.32) carries the exponent $\dim\rho_2 = 2$.

The S_3 example is the smallest case in which the Frobenius class is genuinely a class rather than an element, and it shows the mechanism by which a non-abelian Galois group encodes arithmetic that no single congruence modulo a fixed integer can capture: the condition $\mathrm{Frob}_p = C_3$, that 2 be a cubic non-residue among the primes $p \equiv 1 \pmod 3$, is not a union of residue classes modulo any m, because the field $\mathbb{Q}(\sqrt[3]{2})$ is not contained in any cyclotomic field. The non-abelian density theorem reaches splitting conditions that lie strictly beyond the abelian theory of chapter 7. Larger groups exhibit the same phenomenon with richer combinatorics: for a quartic field with Galois group A_4 the four conjugacy classes (sizes $1,3,4,4$, namely the identity, the three double transpositions, and two classes of 3-cycles) split the primes into families of densities $\frac{1}{12}, \frac{3}{12}, \frac{4}{12}, \frac{4}{12}$, and for a quintic with group A_5, where every nontrivial representation is at least three-dimensional, Artin's conjecture for the irreducible constituents was for a long time known only in special cases; these lie beyond the hand computations recorded here, and the general non-abelian densities are read off the character table of G exactly as in (10.31).

10.5.2 Remark: Toward Langlands Artin's conjecture (definition 10.3.3), that the Artin L-function $L(s,\rho,L/K)$ of every nontrivial irreducible representation ρ extends to an entire function, is open in general; the meromorphic continuation of theorem 10.3.2, obtained from Brauer's induction theorem (theorem 10.3.1), produces only a function holomorphic except for possible poles, and removing those poles is the substance of the conjecture. The Langlands program supplies the framework in which the conjecture is expected to hold. Its central prediction is a reciprocity: to each n-dimensional representation ρ of the Galois group there should correspond an *automorphic representation* of GL_n over K, in such a way that the Artin L-function $L(s,\rho,L/K)$ coincides with the automorphic L-function of its partner. For automorphic L-functions, holomorphy and the functional equation are theorems rather than conjectures, established by the analytic theory of automorphic forms; so the existence of the automorphic partner would force Artin's conjecture for ρ. The abelian prototype of this reciprocity is already in hand and has been used throughout the chapter: the statement that a one-dimensional Artin L-function equals a Hecke L-function, invoked in the continuation of theorem 10.3.2 and in (10.32), is the case $n=1$, where the automorphic representation of GL_1 is a Hecke character (chapter 9). The case $n=2$ for representations with solvable image was the original setting of the Langlands–Tunnell theorem, an early and decisive confirmation. The theory of automorphic forms that gives these statements their meaning lies outside this book; the idelic reformulation that opens the door to it, and the abelian reciprocity that anchors it, are the subject of [12], and the higher-dimensional theory belongs to the forthcoming companions of the series.

It remains to record the pattern of the Part. The five chapters chapters 4 and 7 to 10 are five realizations of a single construction, and the arc of the Part is the steady widening of that construction's domain. In each case an L-function is attached to an arithmetic situation, and three of its analytic features carry three pieces of arithmetic, always in the same correspondence.

The first feature is the *Euler product*: the L-function factors over primes, and each local factor records how a prime sits in the situation at hand. For ζ the factor $(1-p^{-s})^{-1}$ records only that p is a prime (corollary 2.3.2); for a Dirichlet L-function the factor $(1-\chi(p)p^{-s})^{-1}$ records the residue of p modulo m (theorem 7.2.2); for the Dedekind zeta the factors over the primes $\mathfrak{p}|p$ record the splitting of p in K (theorem 8.1.4); for a Hecke L-function the factor records the ray class of $\mathfrak{p}$ (theorem 9.2.3); and for an Artin L-function the factor $\det(1-\rho(\mathrm{Frob}_\mathfrak{p})N\mathfrak{p}^{-s}\big|V^{I_\mathfrak{P}})^{-1}$ records the Frobenius conjugacy class of $\mathfrak{p}$ through the representation ρ (definition 10.2.1). In every case the L-function is the generating function that packages the splitting of primes.

The second feature is the *behavior at* $s=1$, which carries an arithmetic invariant. The pole of ζ at $s=1$ measures the abundance of the integers (corollary 4.2.2); the residue of ζ_K there is the analytic class number formula, assembling the class number, the regulator, and the discriminant (theorem 8.3.2); the value $L(1,\chi)$ of a Dirichlet or Hecke L-function is a transcendental constant tied, for quadratic characters, to the class number of the associated field (theorem 8.4.2 and corollary 8.4.3); and the order of vanishing or the value of an Artin L-function at the edge encodes the rank data of the situation. The behavior at $s=1$ is the channel through which the analytic object returns arithmetic.

The third feature is *non-vanishing on the line* $\mathrm{Re}(s)=1$, which converts the behavior at $s=1$ into a statement about the distribution of primes. The non-vanishing of ζ on the line yields the Prime Number Theorem (theorems 4.4.2 and 5.4.3); the non-vanishing $L(1,\chi)\neq 0$ yields Dirichlet's theorem and the density $1/\varphi(m)$ (theorems 7.4.2 and 7.5.4); the non-vanishing of Hecke L-functions yields the equidistribution of primes among the ray classes (theorems 9.5.2 and 9.5.4); and the non-vanishing of Artin L-functions on the line, secured through the Hecke L-functions to which Brauer induction reduces them (theorem 10.4.1), yields the Chebotarev density theorem and the densities $|C|/|G|$ (theorem 10.4.3). Each density theorem of the Part is one non-vanishing statement converted into an equidistribution.

Read in this light the Part tells one story. Beginning with the single function ζ counting all primes at once, it twists by a character to separate the primes by residue (chapter 7), passes to a number field to separate them by splitting (chapter 8), twists *that* by a ray class character to recover and generalize both (chapter 9), and finally attaches an L-function to each representation of a non-abelian Galois group, separating the primes by their Frobenius class and culminating in the most general density theorem the abelian-twisted methods can prove (chapter 10). The functional equation runs in parallel throughout, fixing the symmetry of the zeros and so the precise shape of the error terms in every one of these distributions. What remains beyond the reach of these chapters, the holomorphy of the higher-dimensional Artin L-functions and the automorphic reciprocity that would supply it, is exactly the threshold of the modern theory; the idelic and class-field-theoretic language that crosses it is taken up in [12].

Part III

Further Methods in Analytic Number Theory

The L-function method of Part II reached, in the Chebotarev density theorem, the most general distribution law that the analytic theory of a single family of L-functions can prove; it reached at the same time the limits of that method, for the holomorphy of the higher Artin L-functions, and behind it the whole analytic mystery of the zeros, lies beyond it. This Part develops two techniques that stand outside the L-function framework and reach problems it cannot. Both replace the global analysis of a function of a complex variable by direct, finite, arithmetic estimation.

Chapter 11 studies character and exponential sums: finite sums of oscillating terms whose size measures cancellation directly, with no continuation and no contour integration. Chapter 12 develops the sieve, the combinatorial descendant of Eratosthenes' procedure, which counts primes and almost-primes by inclusion and exclusion and reaches the twin primes and the gaps between primes where the L-functions are silent. The two techniques are complementary: the first quantifies the equidistribution that the L-functions predict, the second counts the sifted sequences the L-functions cannot resolve. Each, in the end, meets the analytic theory again at its unproved core, the character sums in the subconvexity problem for L-functions and the sieve at the parity barrier that only the zeros can cross.

The prerequisites revert to those of Part I. The number fields, ray class groups, and Galois representations of Part II recede, surviving only in occasional examples; what carries over is the elementary Fourier analysis over $\mathbb{Z}/q\mathbb{Z}$ and, from chapter 7, the Dirichlet characters and the Gauss sum, which reappears in the very first estimate of the next chapter. The development begins with the character sums.

11

Character Sums and Exponential Sums

The quantities studied in this chapter are finite sums of oscillating terms: a Dirichlet character $\chi(n)$ or a complex exponential $e^{2\pi i f(n)}$ summed over a range of integers. A sum of N such unit terms has trivial modulus at most N, and the whole question is how much smaller it is: the terms cancel, and the sum is small, exactly when the phases are well distributed. Measuring that cancellation is the subject of the chapter, and the estimates obtained feed directly back into the distribution of primes and of arithmetic sequences. The device that makes the first of them possible is the Gauss sum of chapter 7: it converts a sum of a multiplicative character into a sum of additive exponentials whose cancellation can be read off a geometric series, and it is through this conversion that the arithmetic of a character sum becomes visible.

The chapter proceeds from the Pólya–Vinogradov inequality for complete-modulus character sums (section 11.1), through the deeper Burgess bound that alone reaches sums shorter than $\sqrt{q}$ (section 11.2), to the general theory of exponential sums and the differencing technique of Weyl and van der Corput (section 11.3); the Kloosterman sums of section 11.4 are the archetype of a complete exponential sum whose square-root cancellation lies deepest; and the applications of section 11.5 return the estimates to equidistribution, to the divisor and circle problems, and to the threshold of the circle method.

11.0.1 Note: Results Imported from Earlier Chapters The results this chapter and the next draw on from earlier material are few and specific. From chapter 7 they use the characters of $(\mathbb{Z}/q\mathbb{Z})^\times$, the orthogonality relations (theorem 7.1.4), the Gauss sum $\tau(\chi)$ and its modulus $|\tau(\chi)| = \sqrt{q}$ for primitive χ (theorem 7.3.2), and the Legendre symbol (definition 0.5.3); everything else is elementary Fourier analysis over $\mathbb{Z}/q\mathbb{Z}$. The number-field apparatus of Part II is not used, though a few examples refer back to it. The one external input drawn from outside the series is the Weil bound for complete character and exponential sums in one variable, a consequence of the Riemann hypothesis for curves over finite fields; it is stated where first needed (theorem 11.2.1) and its proof, which belongs to algebraic geometry, is cited rather than reproduced.

11.0.2 Convention: Additive Exponential and the Nearest-Integer Norm Throughout this chapter and the next, $e(t)$ denotes the additive character

$$e(t) = e^{2\pi i t}, \qquad t \in \mathbb{R}$$

so that $e(t)$ has period 1, $e(s+t) = e(s)e(t)$, and $|e(t)| = 1$. For an integer a and a modulus q the value $e(a/q)$ is a q-th root of unity, and the Gauss sum of chapter 7 is $\tau(\chi) = \sum_{a \bmod q} \chi(a)\, e(a/q)$ in this notation. The symbol $\|t\|$ denotes the distance from t to the nearest integer,

$$\|t\| = \min_{n\in\mathbb{Z}} |t-n| \in [0, \tfrac{1}{2}]$$

which measures how close $e(t)$ is to 1: the elementary bound $|e(t) - 1| = 2|\sin \pi t| \geq 4\|t\|$ is used repeatedly, as is $|\sin \pi t| = \sin \pi\|t\| \geq 2\|t\|$.

11.1 Character Sums and the Pólya–Vinogradov Inequality

The starting question of the chapter is the size of an incomplete character sum. Fix a Dirichlet character χ modulo q and consider

$$S_\chi(M,N) = \sum_{M<n\leq M+N} \chi(n),$$

the sum of χ over an interval of N consecutive integers. Two bounds are immediate. The triangle inequality gives $|S_\chi(M,N)| \leq N$, since each term has modulus at most 1; and when χ is nonprincipal, summing over a complete period of length q gives 0 by row orthogonality (theorem 7.1.4), so an interval of any length contributes at most a single incomplete period, whence $|S_\chi(M,N)| \leq q$ as well. Combining them, $|S_\chi(M,N)| \leq \min(N,q)$. For intervals much longer than q this is already strong, but the interesting regime is N comparable to or smaller than q, where $\min(N,q)$ carries no cancellation at all: it is the trivial bound, the estimate one writes down before doing any work. The Pólya–Vinogradov inequality is the first genuine improvement, and it shows that a nonprincipal character sum never grows beyond roughly $\sqrt{q}$, whatever the length of the interval.

The mechanism is the one that the Gauss sum was built to provide. A character sum is multiplicative data, insensitive to the additive structure of the interval $M < n \leq M+N$; the exponential sum $\sum_n e(an/q)$, by contrast, is a geometric series whose size is controlled by how far a/q sits from an integer. The Gauss sum interchanges the two, expanding $\chi(n)$ into a combination of the additive characters $n \mapsto e(an/q)$, and it is through this expansion that the cancellation in $S_\chi(M, N)$ becomes visible.

Theorem 11.1.1: Pólya–Vinogradov Inequality

Let χ be a primitive Dirichlet character modulo $q > 1$. Then for all integers M and all $N \geq 1$,

$$\left| \sum_{M<n\leq M+N} \chi(n) \right| \leq \sqrt{q}\,\bigl(1 + \log q\bigr).$$

In particular $|S_\chi(M, N)| \ll \sqrt{q}\,\log q$, with an implied constant that is absolute.

***Proof*:**
The proof expands χ into additive characters by the Gauss sum, bounds each resulting geometric series, and sums the bounds against a harmonic series.

Step 1: The additive expansion. Since χ is primitive modulo q, so is its conjugate $\bar{\chi}$, and the separability of the Gauss sum (theorem 7.3.2) gives, for every integer n,

$$\sum_{a \bmod q} \bar{\chi}(a)\, e\!\left(\frac{an}{q}\right) = \chi(n)\,\tau(\bar{\chi}).$$

The same theorem gives $|\tau(\bar{\chi})| = \sqrt{q}$, so $\tau(\bar{\chi}) \neq 0$, and dividing yields the finite Fourier expansion

$$\chi(n) = \frac{1}{\tau(\bar{\chi})} \sum_{a \bmod q} \bar{\chi}(a)\, e\!\left(\frac{an}{q}\right), \tag{11.1}$$

valid for all n (both sides vanish when $\gcd(n, q) > 1$). This is the identity that trades the multiplicative character χ for the additive characters $n \mapsto e(an/q)$.

Step 2: Summation over the interval. Insert (11.1) into $S_\chi(M, N)$ and exchange the two finite sums:

$$S_\chi(M, N) = \frac{1}{\tau(\bar{\chi})} \sum_{a \bmod q} \bar{\chi}(a) \sum_{M<n\leq M+N} e\!\left(\frac{an}{q}\right).$$

The term $a \equiv 0 \pmod{q}$ contributes nothing, since $\bar{\chi}(0) = 0$ for $q > 1$. For each a with $1 \leq a \leq q-1$ the inner sum is a geometric series with ratio $e(a/q) \neq 1$, so

$$\left| \sum_{M<n\leq M+N} e\!\left(\frac{an}{q}\right) \right| = \left| \frac{e(a(M+N)/q) - e(aM/q)}{e(a/q) - 1} \right| \leq \frac{2}{|e(a/q) - 1|} = \frac{1}{|\sin(\pi a/q)|},$$

using $|e(a/q) - 1| = 2|\sin(\pi a/q)|$ and bounding the numerator by 2.

Step 3: Assembling the bound. Taking $|\bar{\chi}(a)| \leq 1$ and $|\tau(\bar{\chi})|^{-1} = q^{-1/2}$,

$$|S_\chi(M, N)| \leq \frac{1}{\sqrt{q}} \sum_{a=1}^{q-1} \frac{1}{|\sin(\pi a/q)|} = \frac{1}{\sqrt{q}} \sum_{a=1}^{q-1} \frac{1}{\sin(\pi \|a/q\|)}.$$

Since $\sin(\pi t) \geq 2t$ for $0 \leq t \leq \frac{1}{2}$ (concavity of sin on $[0, \pi/2]$), and $\|a/q\| = \min(a, q-a)/q$, each term is at most $q/\bigl(2\min(a, q-a)\bigr)$. As a runs over $1, \ldots, q-1$ the value $\min(a, q-a)$ takes each integer k in the range $1 \leq k \leq \lfloor q/2 \rfloor$ at most twice, so

$$\sum_{a=1}^{q-1} \frac{1}{\sin(\pi a/q)} \leq \frac{q}{2} \sum_{a=1}^{q-1} \frac{1}{\min(a, q-a)} \leq \frac{q}{2} \cdot 2 \sum_{k=1}^{\lfloor q/2 \rfloor} \frac{1}{k} \leq q\bigl(1 + \log\lfloor q/2 \rfloor\bigr) \leq q\,(1 + \log q).$$

Dividing by $\sqrt{q}$ gives $|S_\chi(M, N)| \leq \sqrt{q}\,(1 + \log q)$, as we sought to show.

The inequality is the assertion that a primitive character exhibits square-root cancellation over any interval, up to a logarithm: no matter how the interval is placed or how long it is, the sum $S_\chi(M, N)$ cannot exceed $\sqrt{q}$ by more than a factor of $\log q$. Set against the trivial bound $\min(N, q)$, the improvement is decisive precisely when N is larger than about $\sqrt{q}\,\log q$: for such intervals $\sqrt{q}\,\log q \ll N$, so the character sum is smaller than its number of terms, and cancellation is proven. Below that threshold, for intervals shorter than $\sqrt{q}$, the Pólya–Vinogradov bound exceeds the trivial bound N and says nothing; reaching those short intervals is exactly the achievement of the Burgess bound in the next section.

11.1.2 Remark: The Sharp Constant and the Role of Primitivity The constant obtained above is not optimal. A more careful estimate of the cosecant sum $\sum_{a=1}^{q-1} \sin(\pi a/q)^{-1}$, which is asymptotic to $\frac{2}{\pi}\,q \log q$, replaces the leading factor 1 by $\frac{2}{\pi}$, giving $|S_\chi(M, N)| \leq \bigl(\frac{2}{\pi} + o(1)\bigr)\sqrt{q}\,\log q$; this is the form in which the inequality is usually quoted, and the constant $\frac{2}{\pi}$ is known to be essentially best possible for the pointwise bound. The dependence on primitivity is intrinsic to the method: (11.1) and the evaluation $|\tau(\bar{\chi})| = \sqrt{q}$ both require χ primitive. For an imprimitive χ modulo q induced by a primitive character χ^* modulo the conductor $q^* \big| q$, one applies the theorem to χ^* and accounts for the finitely many primes dividing q but not q^*; the resulting bound is governed by $\sqrt{q^*}\,\log q^*$, the conductor rather than the modulus setting the scale. Under the generalized Riemann hypothesis the logarithm can be lowered to $\log\log q$, a theorem of Montgomery and Vaughan; unconditionally the $\log q$ has never been removed, and doing so even for quadratic characters is a well-known open problem.

The first application is the classical one that motivated the inequality: locating the least quadratic non-residue modulo a prime. If p is an odd prime, the Legendre symbol $\chi = \left(\frac{\cdot}{p}\right)$ is a primitive character modulo p (definition 0.5.3), taking the value $+1$ on quadratic residues and -1 on non-residues. The least non-residue is the first place the partial sums of χ fail to keep pace with the count of integers, and Pólya–Vinogradov pins its location.

Corollary 11.1.3: Least Quadratic Non-residue

Let p be an odd prime and let $n_2(p)$ denote the least positive quadratic non-residue modulo p. Then

$$n_2(p) \ll \sqrt{p}\,\log p.$$

***Proof*:**
Write $\chi = \left(\frac{\cdot}{p}\right)$, a primitive character modulo p. Suppose every integer n with $1 \le n \le N$ is a quadratic residue or a multiple of p; for $N < p$ this means $\chi(n) = 1$ for all $1 \le n \le N$, so $S_\chi(0, N) = \sum_{n \le N} \chi(n) = N$. The Pólya–Vinogradov inequality (theorem 11.1.1) bounds the same sum by $\sqrt{p}\,(1 + \log p)$, forcing $N \le \sqrt{p}\,(1 + \log p)$. Hence some $n \le \sqrt{p}\,(1 + \log p) + 1$ is a non-residue, and $n_2(p) \le \sqrt{p}\,(1 + \log p) + 1 \ll \sqrt{p}\,\log p$, completing the proof.

The estimate $n_2(p) \ll \sqrt{p}\,\log p$ says that a non-residue appears early: among the first $\sqrt{p}\,\log p$ integers, a vanishing fraction of the interval $[1, p]$, the residues cannot maintain their monopoly. The truth is believed to be far smaller, $n_2(p) \ll_\varepsilon p^\varepsilon$ for every $\varepsilon > 0$, and even $n_2(p) \ll (\log p)^{1+\varepsilon}$ under the generalized Riemann hypothesis. The unconditional exponent $\frac{1}{2}$ of Pólya–Vinogradov was first lowered by Vinogradov's own conjecture and then by the Burgess bound to $\frac{1}{4} + \varepsilon$; this improvement is the subject of section 11.2, and it is the reason the short-interval range matters.

11.2 The Burgess Bound

The Pólya–Vinogradov inequality controls a character sum $S_\chi(M, N)$ only when the interval is longer than about $\sqrt{q}$; for $N < \sqrt{q}$ its bound $\sqrt{q}\,\log q$ exceeds the trivial N and proves nothing. Yet the short-interval range is where the arithmetic questions live. The least non-residue, the distribution of residues in a short block, the size of the smallest primitive root: each asks about χ on an interval far shorter than q, and each is invisible to Pólya–Vinogradov. The Burgess bound is the theorem that breaks the $\sqrt{q}$ barrier. It proves cancellation in $S_\chi(M, N)$ for intervals as short as $N \ge q^{1/4+\varepsilon}$, and to this day it remains, for a general modulus, the only unconditional bound that reaches below $\sqrt{q}$.

The method rests on an input of an entirely different depth from the elementary Gauss-sum manipulation of the previous section. Where Pólya–Vinogradov needed only the modulus $|\tau(\chi)| = \sqrt{q}$, Burgess needs square-root cancellation in *complete* sums $\sum_{x \bmod p} \chi(f(x))$ of a character against a polynomial argument. That cancellation is the Riemann hypothesis for the curve $y^d = f(x)$ over the finite field, and it is the one result in this chapter imported from outside the analytic theory.

Theorem 11.2.1: Weil Bound for Character Sums

Let p be a prime, let χ be a Dirichlet character modulo p of exact order $d > 1$, and let $f \in \mathbb{F}_p[x]$ be a polynomial that is not a constant multiple of the d-th power of a polynomial in $\mathbb{F}_p[x]$. Let m be the number of distinct roots of f in $\overline{\mathbb{F}_p}$. Then

$$\left| \sum_{x \bmod p} \chi\big(f(x)\big) \right| \le (m-1)\sqrt{p},$$

where $\chi(f(x))$ is interpreted as 0 when $p \big| f(x)$. In particular, for f of degree k not a d-th power up to scalars, the complete sum is $O_k(\sqrt{p})$.

***Proof Key ideas*:**
The bound is a theorem of Weil; the full proof lies outside the analytic scope of this book, and only the source of the square root is indicated. The sum $\sum_x \chi(f(x))$ counts, with weights given by the character χ, the points on the superelliptic curve $C\colon y^d = f(x)$ over $\mathbb{F}_p$: grouping the x with $f(x)$ in a fixed coset of the d-th powers in $\mathbb{F}_p^\times$ and using the orthogonality of the characters of $\mathbb{F}_p^\times/(\mathbb{F}_p^\times)^d$, the number of affine points of C over $\mathbb{F}_p$ is $\sum_x \sum_{\chi^d=\chi_0} \chi(f(x)) = p + \sum_{\chi\neq\chi_0} \sum_x \chi(f(x))$. The Riemann hypothesis for the curve C, proved by Weil, states that $|\#C(\mathbb{F}_p) - (p+1)| \leq 2g\sqrt{p}$, where g is the genus of C; the hypothesis that f is not a d-th power up to a scalar is what guarantees C is geometrically irreducible, so that the estimate applies. Extracting the single character χ from the sum and bounding the genus by $g \leq (m-1)(d-1)/2$ yields the stated inequality, in the sharpened form with $m-1$ due to the point-counting refinements of Weil and, in this explicit shape, of [18, Ch. 2]. The self-contained account of the Riemann hypothesis for curves is given there and in the algebraic-geometry literature; the genus-1 case is Hasse's theorem on elliptic curves, the special case $\#E(\mathbb{F}_p) = p + 1 - a_p$ with $|a_p| \leq 2\sqrt{p}$ developed in the series' volume on elliptic curves (`NateEllipticCurves`).

The Weil bound is the exact analogue, for a complete sum over $\mathbb{F}_p$, of the square-root cancellation that Pólya–Vinogradov delivered for an incomplete sum: a sum of p unit terms whose phases are governed by a nonconstant polynomial is no larger than $O(\sqrt{p})$. Burgess's insight was that incomplete sums over short intervals can be reduced, by an averaging argument, to a controlled number of these complete sums, and the square-root cancellation then propagates back to the short interval.

Theorem 11.2.2: Burgess Bound

Let χ be a primitive Dirichlet character modulo a prime p. For every integer $r \geq 1$ and every $\varepsilon > 0$, and for all integers M and $N \geq 1$,

$$\left| \sum_{M<n\leq M+N} \chi(n) \right| \ll_{r,\varepsilon} N^{1-1/r}\, p^{(r+1)/(4r^2)+\varepsilon},$$

the implied constant depending only on r and ε. Consequently, for any fixed $\delta > 0$ the sum is $o(N)$ as $p \to \infty$ whenever $N \geq p^{1/4+\delta}$. The same bound holds for a primitive character modulo a general q when $r \in \{1,2,3\}$, and for all $r \geq 1$ when q is cube-free.

***Proof Key ideas*:**
The full proof is long and technical, several pages of moment estimates, and its central input, square-root cancellation in complete sums, is theorem 11.2.1, already stated in full; only the structure of the reduction is given here, in accordance with the convention on deferred proofs. Fix $r \geq 1$.

Step 1: Shifting. The sum $S = \sum_{M<n\leq M+N} \chi(n)$ is nearly unchanged when the interval is translated by a small amount. For integers a, b with $1 \leq a \leq A$ and $1 \leq b \leq B$, where A, B are parameters to be chosen with $AB \leq N$, the shifted sum $\sum_n \chi(n+ab)$ differs from S by at most $2AB$ terms at the two ends. Averaging over all such a, b and using complete multiplicativity $\chi(n+ab) = \chi(a)\chi(a^{-1}n+b)$ for $p \nmid a$ recasts S, up to an error $O(AB)$, as an average of the sums $\sum_b \chi(m+b)$ over the residues $m = a^{-1}n$.

Step 2: Hölder and the passage to a complete sum. Applying Hölder's inequality with exponent $2r$ to the average over m converts the problem into bounding the $2r$-th moment

$$\sum_{m \bmod p} \left| \sum_{b=1}^{B} \chi(m+b) \right|^{2r}.$$

Expanding the $2r$-th power and interchanging the order of summation expresses this moment as a sum, over $2r$-tuples $(b_1, \ldots, b_{2r})$, of the complete character sums $\sum_{m \bmod p} \chi\big((m+b_1)\cdots(m+b_r)\big)\,\overline{\chi}\big((m+b_{r+1})\cdots(m+b_{2r})\big)$.

Step 3: The Weil bound. Each such complete sum is $\sum_m \chi(g(m))$ for the rational-function-turned-polynomial argument determined by the b_i. Unless the numerator and denominator polynomials coincide, so that the argument is a perfect power and the sum is a main term of size p, the argument is not a d-th power up to scalars, and theorem 11.2.1 bounds the sum by $O_r(\sqrt{p})$. The number of degenerate tuples producing a main term is $O_r(B^r)$, and the remaining $O(B^{2r})$ tuples each contribute $O_r(\sqrt{p})$. Thus the moment is $O_r(B^r p + B^{2r}\sqrt{p})$.

Step 4: Optimization. Substituting the moment bound back through the Hölder and shifting steps and choosing A and B to balance the resulting terms (with A a power of p near $p^{1/(2r)}$ and B near N/A) produces the stated exponent $(r+1)/(4r^2)$ on p; the ε absorbs the divisor-type factors $p^{o(1)}$ counting the degenerate tuples and the number of representations. The restriction to $r \le 3$ for general moduli, and its removal for cube-free q, arises because for composite q the complete sums must be estimated modulo prime power factors, where the clean Weil bound is available only under a cube-free hypothesis. The full computation is carried out in [9, Ch. 12] and in Burgess's original papers; the argument above is its skeleton.

The Burgess bound is nontrivial exactly in the range the Pólya–Vinogradov inequality could not reach. Taking r large, the exponent $(r+1)/(4r^2)$ tends to 0 and the factor $N^{1-1/r}$ tends to N, so for any $\delta > 0$ one may fix r large enough that the bound is $\ll N^{1-1/r}p^{1/(4r)+\varepsilon} = o(N)$ once $N \ge p^{1/4+\delta}$. This is the first cancellation in character sums over intervals of length a small power of q, and no unconditional improvement on the exponent $\frac{1}{4}$ is known for a general modulus. The generalized Riemann hypothesis would give cancellation for $N \ge q^{\varepsilon}$, so the gap between $q^{1/4}$ and q^{ε} measures the distance between the Weil bound over $\mathbb{F}_p$ and the still-unproven Riemann hypothesis for the Dirichlet L-functions themselves.

The gain over Pólya–Vinogradov translates immediately into a sharper location of the least non-residue.

Corollary 11.2.3: Burgess's Bound on the Least Non-residue

Let p be an odd prime and $n_2(p)$ the least positive quadratic non-residue modulo p. Then for every $\varepsilon > 0$,

$$n_2(p) \ll_\varepsilon p^{1/(4\sqrt{e})+\varepsilon}.$$

The exponent $1/(4\sqrt{e}) = 0.1516\ldots$ improves the Pólya–Vinogradov exponent $\frac{1}{2}$.

***Proof Key ideas*:**
The reduction combines the Burgess cancellation with the multiplicative closure of the residues, and the full optimization is carried out in Burgess's paper; the mechanism is as follows. Suppose $n_2(p) = N$, so that $\chi(n) = \left(\frac{n}{p}\right) = 1$ for all $1 \le n < N$, where χ is the Legendre symbol. Every integer whose prime factors are all less than N is then a product of residues, hence a residue, so χ takes the value 1 on all N-smooth integers in $[1, x]$. Take $x = p^{1/4+\varepsilon}$, the shortest range in which the Burgess bound (theorem 11.2.2) still forces $\sum_{n \le x} \chi(n) = o(x)$, so that the residues cannot occupy a positive proportion of $[1, x]$. The count of N-smooth integers up to x is, by a standard estimate on smooth numbers, a proportion $\rho(u)$ of the interval with $u = \frac{\log x}{\log N}$ and ρ the Dickman function, and $\rho(u) = u^{-u(1+o(1))}$. Balancing this smooth-number density against the vanishing forced by Burgess, the two are compatible only when u is bounded, and optimizing the resulting exponent produces $N \ll_\varepsilon p^{1/(4\sqrt{e})+\varepsilon}$. The constant $\frac{1}{4\sqrt{e}}$ is exactly the outcome of this balance and is the best exponent known; the details are in [9, §12.4], completing the argument.

Burgess's exponent $\frac{1}{4\sqrt{e}}$ has stood since 1957 as the best unconditional bound on the least non-residue, an instance of a problem where the truth is believed to be $(\log p)^{1+o(1)}$ yet the proven bound remains a fixed power of p. The same short-interval cancellation underlies the best known bounds on the least primitive root modulo p and on the Vinogradov conjecture for the least non-residue in arithmetic progressions; each is an application of theorem 11.2.2 in place of the trivial or Pólya–Vinogradov estimate, and each stalls at the same $q^{1/4}$ barrier that the Weil bound imposes.

11.3 Exponential Sums and Weyl Differencing

The character sums of the previous two sections measure cancellation in a *multiplicative* oscillation, the values $\chi(n)$ of a Dirichlet character. This section takes up the additive counterpart: sums

$$\sum_n e\big(f(n)\big), \qquad e(t) = e^{2\pi i t},$$

in which the phase is a real-valued function f of the summation variable, most often a polynomial. Such exponential sums are the analytic expression of *equidistribution*. A sequence of real numbers spreads out uniformly modulo 1 exactly when the associated exponential sums are small, and the technique for proving them small, differencing, is the additive analogue of the reduction to complete sums that powered the Burgess bound. The section develops the equivalence between equidistribution and exponential-sum cancellation, then the differencing method of Weyl that establishes the cancellation for polynomial phases, and finally van der Corput's refinement, which packages differencing into a reusable inequality.

Definition 11.3.1: Uniform Distribution Modulo One

A sequence $(x_n)_{n\geq 1}$ of real numbers is *uniformly distributed modulo* 1 (or *equidistributed*) if for every subinterval $[a,b) \subseteq [0,1)$,

$$\lim_{N\to\infty} \frac{1}{N} \#\{\, n \leq N : \{x_n\} \in [a,b) \,\} = b - a,$$

where $\{x\} = x - \lfloor x \rfloor$ is the fractional part. Equivalently, the proportion of the first N terms whose fractional parts fall in any interval tends to the length of that interval.

The definition asks that the fractional parts $\{x_n\}$ populate $[0,1)$ evenly in the long run. Testing it directly against every interval is unwieldy; the content of Weyl's criterion is that a single family of test functions, the additive characters $x \mapsto e(hx)$, suffices, converting a statement about counting points in intervals into a statement about the smallness of exponential sums.

Theorem 11.3.2: Weyl's Criterion

A sequence $(x_n)_{n\geq 1}$ of real numbers is uniformly distributed modulo 1 if and only if for every integer $h \neq 0$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} e(hx_n) = 0.$$

***Proof*:**
Write $S_N(f) = \frac{1}{N}\sum_{n\leq N} f(\{x_n\})$ for a 1-periodic function f, and note that equidistribution is precisely the statement $S_N(\mathbf{1}_{[a,b)}) \to b - a = \int_0^1 \mathbf{1}_{[a,b)}$ for every interval.

Step 1: Equidistribution implies the exponential averages vanish. Suppose (x_n) is equidistributed. Then $S_N(f) \to \int_0^1 f$ first for indicator functions of intervals, hence for finite linear combinations of them (step functions), and then for every Riemann-integrable f by squeezing between step functions above and below whose integrals differ by less than any ε. The function $x \mapsto e(hx)$ is continuous, hence Riemann-integrable, so $S_N(e(h\,\cdot)) \to \int_0^1 e(hx)\,dx$, which is 0 for every integer $h \neq 0$. This is the stated limit.

Step 2: The exponential averages vanishing imply equidistribution. Suppose $\frac{1}{N}\sum_{n\leq N} e(hx_n) \to 0$ for all $h \neq 0$. For $h = 0$ the average is identically $1 = \int_0^1 1$. By linearity $S_N(P) \to \int_0^1 P$ for every trigonometric polynomial $P(x) = \sum_{|h|\leq H} c_h e(hx)$. Let f be any continuous 1-periodic function and $\varepsilon > 0$. By the trigonometric approximation theorem ([17]; the Fejér or Weierstrass theorem) there is a trigonometric polynomial P with $\sup_x |f(x) - P(x)| < \varepsilon$. Then

$$\left|S_N(f) - \int_0^1 f\right| \leq |S_N(f) - S_N(P)| + \left|S_N(P) - \int_0^1 P\right| + \left|\int_0^1 (P - f)\right| < 2\varepsilon + \left|S_N(P) - \int_0^1 P\right|,$$

and the middle term tends to 0, so $\limsup_N |S_N(f) - \int_0^1 f| \leq 2\varepsilon$; as ε was arbitrary, $S_N(f) \to \int_0^1 f$ for every continuous f. Finally, given an interval $[a,b)$ and $\varepsilon > 0$, sandwich $\mathbf{1}_{[a,b)}$ between continuous periodic functions $g_- \leq \mathbf{1}_{[a,b)} \leq g_+$ with $\int_0^1 (g_+ - g_-) < \varepsilon$; then $\limsup_N S_N(\mathbf{1}_{[a,b)}) \leq$

$\int g_+ < (b-a)+\varepsilon$ and $\liminf_N S_N(\mathbf{1}_{[a,b)}) \geq \int g_- > (b-a)-\varepsilon$, so $S_N(\mathbf{1}_{[a,b)}) \to b-a$. This is equidistribution, completing the proof.

Weyl's criterion reduces every equidistribution question to the estimation of the exponential sums $\sum_{n\leq N} e(hx_n)$: the sequence spreads out uniformly exactly when these sums are $o(N)$ for each fixed nonzero frequency h. The simplest case is already decisive and sets the pattern for everything that follows.

Example 11.3.3: The Sequence $n\alpha$

Let α be irrational. The sequence $x_n = n\alpha$ is uniformly distributed modulo 1. Indeed, for a fixed integer $h \neq 0$ the number $h\alpha$ is irrational, so $e(h\alpha) \neq 1$, and the geometric sum

$$\left|\sum_{n=1}^{N} e(h\alpha n)\right| = \left|\frac{e(h\alpha(N+1)) - e(h\alpha)}{e(h\alpha)-1}\right| \leq \frac{2}{|e(h\alpha)-1|} = \frac{1}{|\sin(\pi h\alpha)|}$$

is bounded by a constant independent of N. Dividing by N gives $\frac{1}{N}\sum_{n\leq N} e(h\alpha n) \to 0$, and Weyl's criterion (theorem 11.3.2) yields equidistribution. When $\alpha = p/q$ is rational the conclusion fails: the fractional parts $\{n\alpha\}$ cycle through the finite set $\{0, 1/q, \ldots, (q-1)/q\}$ and never populate the intervals between. Irrationality is exactly the dividing line, and the rate of equidistribution is governed by how well α can be approximated by rationals, through the size of $|\sin(\pi h\alpha)|^{-1}$.

For a polynomial phase of higher degree the geometric-series evaluation is unavailable, and the sum $\sum_n e(f(n))$ has no closed form. Weyl's device for handling it is to square the sum and reindex, replacing f by its first differences $f(n+h)-f(n)$, a polynomial of one lower degree. Iterating reduces any polynomial phase to the linear case already understood.

Lemma 11.3.4: Weyl Differencing Inequality

Let $f\colon \mathbb{Z} \to \mathbb{R}$ be any function and let M, N be integers with $N \geq 1$. Then

$$\left|\sum_{M<n\leq M+N} e\big(f(n)\big)\right|^2 \leq N + 2\sum_{h=1}^{N-1}\left|\sum_{M<n\leq M+N-h} e\big(f(n+h)-f(n)\big)\right|.$$

***Proof*:**

Write $T = \sum_{M<n\leq M+N} e(f(n))$. Expanding the square,

$$|T|^2 = \sum_{M<n\leq M+N}\ \sum_{M<m\leq M+N} e\big(f(n)-f(m)\big).$$

Set $h = n-m$, which ranges over $-(N-1) \leq h \leq N-1$; for each such h the pairs (n,m) with $n-m=h$ are those with $M < m \leq M+N$ and $M < m+h \leq M+N$, that is, m in an interval of the stated form. The diagonal $h=0$ contributes $\sum_{M<n\leq M+N} 1 = N$. The terms with h and $-h$ are complex conjugates, since interchanging n and m negates $f(n)-f(m)$, so they sum to twice the real part of the $h>0$ term. Therefore

$$|T|^2 = N + 2\,\mathrm{Re}\sum_{h=1}^{N-1}\ \sum_{M<m\leq M+N-h} e\big(f(m+h)-f(m)\big) \leq N + 2\sum_{h=1}^{N-1}\left|\sum_{M<m\leq M+N-h} e\big(f(m+h)-f(m)\big)\right|,$$

the last step by the triangle inequality and $\mathrm{Re} z \le |z|$, which is the claim, as we sought to show.

The inequality trades one sum with phase f of degree k for a family of inner sums whose phases $f(n+h) - f(n)$ have degree $k-1$ in n: differencing lowers the degree at the cost of an average over the shift h. For a quadratic phase one application already reaches the linear sums of example 11.3.3, and this is the mechanism behind Weyl's equidistribution theorem.

Theorem 11.3.5: Weyl's Equidistribution Theorem

Let $p(x) = \alpha_k x^k + \alpha_{k-1} x^{k-1} + \cdots + \alpha_1 x + \alpha_0$ be a real polynomial for which at least one of the coefficients $\alpha_1, \ldots, \alpha_k$ (a coefficient other than the constant term) is irrational. Then the sequence $\big(p(n)\big)_{n\ge1}$ is uniformly distributed modulo 1. In particular, for every irrational α and every $k \ge 1$ the sequence (αn^k) is equidistributed.

***Proof*:**
By Weyl's criterion (theorem 11.3.2) it suffices to show $\frac{1}{N}\sum_{n\le N} e(h\,p(n)) \to 0$ for each fixed integer $h \ne 0$; replacing p by $h\,p$ (which still has an irrational coefficient among $\alpha_1, \ldots, \alpha_k$), it suffices to treat $h = 1$. The proof is by induction on the degree k of the highest term with irrational coefficient.

Base case $k = 1$. Here $p(n) = \alpha_1 n + \alpha_0$ with α_1 irrational, and $\sum_{n\le N} e(p(n)) = e(\alpha_0)\sum_{n\le N} e(\alpha_1 n)$ is bounded by $|\sin(\pi\alpha_1)|^{-1}$ as in example 11.3.3, so its average tends to 0.

Inductive step. Suppose the claim holds for phases whose top irrational coefficient sits in degree $< k$, and let p have irrational leading coefficient α_k in degree $k \ge 2$. Assume first α_k is the leading coefficient. Weyl differencing (lemma 11.3.4) applied to $f = p$ gives

$$\left|\sum_{n\le N} e(p(n))\right|^2 \le N + 2\sum_{h=1}^{N-1}\left|\sum_{n\le N-h} e\big(p(n+h) - p(n)\big)\right|.$$

For each fixed $h \ge 1$ the difference $p_h(n) = p(n+h) - p(n)$ is a polynomial in n of degree $k-1$ whose leading coefficient is $k\alpha_k h$, which is irrational. By the induction hypothesis the inner sum is $o(N)$ as $N \to \infty$; more quantitatively, writing $D_h(N) = |\sum_{n\le N-h} e(p_h(n))|$, the induction gives $D_h(N)/N \to 0$ for each fixed h. Dividing the differencing inequality by N^2,

$$\frac{1}{N^2}\left|\sum_{n\le N} e(p(n))\right|^2 \le \frac{1}{N} + \frac{2}{N}\cdot\frac{1}{N}\sum_{h=1}^{N-1} D_h(N).$$

The averaged quantity $\frac{1}{N}\sum_{h<N}\frac{D_h(N)}{N}$ is an average of terms each bounded by 1 and each tending to 0 for fixed h; by dominated convergence for the normalized counting measure on $\{1, \ldots, N-1\}$ (equivalently, splitting the sum at $h \le \eta N$ and $h > \eta N$ and using the uniform bound $D_h(N) \le N$), this average tends to 0. Hence the left side tends to 0, so $\frac{1}{N}|\sum_{n\le N} e(p(n))| \to 0$. If the top irrational coefficient α_j lies in degree $j < k$, apply the same differencing $k - j$ times, or simply note that the rational-coefficient terms of higher degree contribute phases that are periodic in n with a fixed period and can be separated out; the argument reduces to the case just treated. This completes the induction and the proof.

Weyl's theorem is the prototype of the differencing method and one of the founding results of analytic number theory: it says that a polynomial sequence with a single irrational coefficient cannot avoid any region of the circle, no matter how the other coefficients are chosen. The quantitative form, obtained by carrying the bounds through the induction rather than passing to the limit, yields the *Weyl sum estimate*: for p of degree k with leading coefficient having a rational approximation $|\alpha_k - a/q| \le q^{-2}$, $\gcd(a,q)=1$, one has $|\sum_{n\le N} e(p(n))| \ll N^{1+\varepsilon}(q^{-1}+N^{-1}+qN^{-k})^{1/2^{k-1}}$. This estimate is what the minor-arc analysis uses in the circle method (section 11.5), where the phases are polynomials and q is the denominator of the rational approximation dictating which arc one is on. The exponent $2^{-(k-1)}$, weakening rapidly with the degree, is the characteristic loss of Weyl differencing and the reason the method is superseded, for large k, by Vinogradov's mean value theorem.

Differencing can be organized into a single inequality that iterates cleanly, due to van der Corput. It is the form of the method most convenient for applications where the phase is not a polynomial but a smooth function, such as the divisor and lattice-point problems of section 11.5.

Theorem 11.3.6: van der Corput's Inequality

Let $z_1,\dots,z_N$ be complex numbers, and set $z_n=0$ for $n\le 0$ and $n>N$. Then for every integer H with $1\le H\le N$,

$$\left|\sum_{n=1}^{N} z_n\right|^2 \le \frac{N+H-1}{H}\sum_{|h|<H}\left(1-\frac{|h|}{H}\right)\sum_n z_{n+h}\,\overline{z_n}.$$

In particular, if $|z_n|\le 1$, then

$$\left|\sum_{n=1}^{N} z_n\right|^2 \le \frac{N+H}{H}\left(N+2\sum_{h=1}^{H-1}\left(1-\frac{h}{H}\right)\left|\sum_n z_{n+h}\,\overline{z_n}\right|\right).$$

***Proof*:**

Consider the identity $H\sum_n z_n = \sum_n\sum_{r=0}^{H-1} z_{n+r}$, valid because as n ranges over $\mathbb{Z}$ and r over $\{0,\dots,H-1\}$, each z_m is counted exactly H times (the padding $z_n=0$ outside $[1,N]$ makes the shifted sums legitimate). The inner sum $\sum_{r=0}^{H-1} z_{n+r}$ is supported on the $N+H-1$ values of n with $2-H\le n\le N$. By the Cauchy–Schwarz inequality applied to this support,

$$H^2\left|\sum_n z_n\right|^2 = \left|\sum_n\sum_{r=0}^{H-1} z_{n+r}\right|^2 \le (N+H-1)\sum_n\left|\sum_{r=0}^{H-1} z_{n+r}\right|^2.$$

Expand the squared modulus: $\left|\sum_r z_{n+r}\right|^2 = \sum_{r,s=0}^{H-1} z_{n+r}\overline{z_{n+s}}$, and sum over n. Setting $h=r-s$ and summing over n first, the number of pairs (r,s) in $\{0,\dots,H-1\}^2$ with $r-s=h$ is $H-|h|$, so

$$\sum_n\left|\sum_{r=0}^{H-1} z_{n+r}\right|^2 = \sum_{|h|<H}(H-|h|)\sum_n z_{n+h}\,\overline{z_n}.$$

Dividing by H^2 and writing $H-|h| = H(1-|h|/H)$ gives the first inequality. For the second, isolate the $h=0$ term $\sum_n |z_n|^2\le N$, pair $\pm h$ into 2Re, bound by absolute values, and use $(N+H-1)/H\le(N+H)/H$, completing the proof.

Van der Corput's inequality is Weyl differencing made self-contained and iterable: the parameter H is free, and choosing it optimally balances the diagonal contribution N/H against the off-diagonal correlations $\sum_n z_{n+h}\overline{z_n}$. Repeated application, the *van der Corput A-process*, lowers the degree of a polynomial phase one step at a time exactly as Weyl differencing does, while the complementary *B-process*, based on the Poisson summation formula and the method of stationary phase, converts a long smooth exponential sum into a shorter dual sum. Alternating the two processes is the classical route to bounds on $\sum_{n\le N} e(f(n))$ for smooth f, and it produces the exponent-pair estimates that yield the subconvexity bounds for ζ on the critical line and the error terms in the divisor and circle problems taken up in section 11.5.

11.4 Kloosterman Sums

The character sums of sections 11.1 and 11.2 were sums of a multiplicative character, and the exponential sums of section 11.3 were sums of an additive character against a polynomial phase. The Kloosterman sum combines the two structures of a finite field in a single complete sum: the phase is additive, but its argument mixes an element of $\mathbb{F}_p^\times$ with its multiplicative inverse. This coupling of addition and inversion makes the Kloosterman sum the archetype of a complete exponential sum whose square-root cancellation is the hardest to establish, inaccessible to elementary differencing and reachable only through the Riemann hypothesis for curves. Kloosterman sums arose in the circle method applied to quadratic forms in four variables and have since become ubiquitous: they are the Fourier coefficients that appear when Poincaré series are unfolded, and their cancellation controls the equidistribution of everything from the angles of Gauss sums to the arithmetic of modular forms.

Definition 11.4.1: Kloosterman Sum

Let $q \ge 1$ be an integer and $a, b \in \mathbb{Z}$. The *Kloosterman sum* is

$$S(a,b;q) = \sum_{\substack{x \bmod q \\ \gcd(x,q)=1}} e\left(\frac{ax+b\bar{x}}{q}\right),$$

where $\bar{x}$ denotes the multiplicative inverse of x modulo q, so that $x\bar{x} \equiv 1 \pmod q$, and the sum runs over the $\varphi(q)$ residues x coprime to q.

The summand pairs each unit x with its inverse $\bar{x}$, weighting the pair by the additive character at $ax + b\bar{x}$. When $b = 0$ the sum degenerates to the Ramanujan sum $\sum_{\gcd(x,q)=1} e(ax/q)$, and when $a = b = 0$ it is simply $\varphi(q)$; the interesting case is a, b both nonzero modulo q, where the interplay of x and $\bar{x}$ prevents any elementary evaluation. The first task is to record the symmetries, which reduce the study of $S(a,b;q)$ to the case of a prime modulus and a single parameter.

Proposition 11.4.2: Elementary Properties of Kloosterman Sums

Let $q \geq 1$ and $a, b, c \in \mathbb{Z}$ with $\gcd(c, q) = 1$. Then:

1. *Reality and symmetry:* $S(a, b; q) = S(b, a; q)$, and $S(a, b; q)$ is a real number.
2. *Homogeneity:* $S(ca, \bar{c}\, b; q) = S(a, b; q)$, where $\bar{c}$ is the inverse of c modulo q; consequently $S(a, b; q) = S(1, ab; q)$ whenever $\gcd(a, q) = 1$, so the sum depends on a, b only through the product ab modulo q in that case.
3. *Multiplicativity:* if $q = q_1 q_2$ with $\gcd(q_1, q_2) = 1$, then
$$S(a, b; q_1 q_2) = S(a\bar{q_2}^{\,2}, b; q_1)\, S(a\bar{q_1}^{\,2}, b; q_2),$$
where $\bar{q_2}$ is the inverse of q_2 modulo q_1 and $\bar{q_1}$ the inverse of q_1 modulo q_2.

***Proof*:**

1. The substitution $x \mapsto \bar{x}$ permutes the units modulo q and exchanges x with $\bar{x}$, sending $ax + b\bar{x}$ to $a\bar{x} + bx$; this gives $S(a, b; q) = S(b, a; q)$. For reality, take complex conjugates and then replace x by $-x$, which permutes the units: $\overline{S(a, b; q)} = \sum_x e\big(-(ax + b\bar{x})/q\big) = \sum_x e\big((a(-x) + b\,\overline{(-x)})/q\big) = S(a, b; q)$, using $\overline{(-x)} = -\bar{x}$ so that $-(ax + b\bar{x})$ becomes $a(-x) + b\,\overline{(-x)}$ under the substitution. Hence $S(a, b; q)$ equals its own conjugate and is real.
2. In $S(ca, \bar{c}b; q) = \sum_x e((cax + \bar{c}b\bar{x})/q)$ substitute $x \mapsto \bar{c}x$, a permutation of the units; then $cax \mapsto ca\bar{c}x = ax$ and $\bar{c}b\bar{x} \mapsto \bar{c}b\,\overline{\bar{c}x} = \bar{c}bc\bar{x} = b\bar{x}$, so the sum is unchanged and equals $S(a, b; q)$. Taking $c = \bar{a}$ (permissible when $\gcd(a, q) = 1$), so that $\bar{c} = a$, gives $S(a, b; q) = S(\bar{a}\, a, ab; q) = S(1, ab; q)$.
3. By the Chinese remainder theorem the map $x \mapsto (x_1, x_2) = (x \bmod q_1, x \bmod q_2)$ is a bijection of units, with inverse $x \equiv q_2\bar{q_2}\, x_1 + q_1\bar{q_1}\, x_2 \pmod{q_1 q_2}$, where $\bar{q_2}q_2 \equiv 1 \pmod{q_1}$ and $\bar{q_1}q_1 \equiv 1 \pmod{q_2}$. Substituting this and the corresponding expression for $\bar{x}$ into the phase, the exponent $(ax+b\bar{x})/(q_1q_2)$ separates modulo 1 into $(a\bar{q_2}\, x_1 + b\bar{q_2}\, \bar{x_1})/q_1 + (a\bar{q_1}\, x_2 + b\bar{q_1}\, \bar{x_2})/q_2$, so the sum factors as $S(a\bar{q_2}, b\bar{q_2}; q_1)\, S(a\bar{q_1}, b\bar{q_1}; q_2)$; applying the homogeneity of part (2) with $c = q_2$ to the first factor and $c = q_1$ to the second brings each into the stated form $S(a\bar{q_2}^{\,2}, b; q_1)\, S(a\bar{q_1}^{\,2}, b; q_2)$. This is the twisted Chinese remainder decomposition.

These are the properties as claimed, completing the proof.

Multiplicativity reduces the evaluation of $S(a, b; q)$ to prime-power moduli, and homogeneity reduces the prime case to the one-parameter family $S(1, n; p)$. For prime powers p^k with $k \geq 2$ the sum can be evaluated in closed form by completing the square in the variable x, the finite-field analogue of stationary phase, and the resulting expressions are the Salié sums, of exact size $2p^{k/2}$ when they do not vanish. The hard case is the prime modulus $q = p$, where no closed form exists and only the size can be estimated. The correct order of magnitude is already visible from the second moment.

Proposition 11.4.3: Second Moment of Kloosterman Sums

Let p be a prime. Then

$$\sum_{a=0}^{p-1} S(a,1;p)^2 = p(p-1),$$

and consequently $\sum_{a=1}^{p-1} S(a,1;p)^2 = p^2 - p - 1$. In particular the average of $S(a,1;p)^2$ over $a \not\equiv 0$ is $p - 1 + O(1/p)$, so the typical size of $S(a,1;p)$ is $\sqrt{p}$.

***Proof*:**
Expand the square and interchange summation:

$$\sum_{a=0}^{p-1} S(a,1;p)^2 = \sum_{a=0}^{p-1} \sum_{\substack{x,y \bmod p \\ \gcd(xy,p)=1}} e\left(\frac{a(x+y) + (\bar{x}+\bar{y})}{p}\right) = \sum_{x,y} e\left(\frac{\bar{x}+\bar{y}}{p}\right) \sum_{a=0}^{p-1} e\left(\frac{a(x+y)}{p}\right).$$

The inner sum over a is p when $x + y \equiv 0 \pmod{p}$ and 0 otherwise. Thus only the pairs with $y \equiv -x$ survive, contributing

$$p \sum_{\substack{x \bmod p \\ \gcd(x,p)=1}} e\left(\frac{\bar{x} + \overline{(-x)}}{p}\right) = p \sum_{x} e\left(\frac{\bar{x} - \bar{x}}{p}\right) = p \sum_{x} 1 = p(p-1),$$

using $\overline{(-x)} = -\bar{x}$, so that $\bar{x} + \overline{(-x)} = 0$. This is the first identity. For the second, the term $a = 0$ is $S(0,1;p)^2$, and $S(0,1;p) = \sum_{x \neq 0} e(\bar{x}/p) = \sum_{u \neq 0} e(u/p) = -1$ (as $\bar{x}$ ranges over all nonzero residues), so $S(0,1;p)^2 = 1$; subtracting gives $\sum_{a=1}^{p-1} S(a,1;p)^2 = p(p-1) - 1 = p^2 - p - 1$, as we sought to show.

The second moment fixes the mean square of $S(a,1;p)$ at essentially p, so $\sqrt{p}$ is the root-mean-square size and no individual bound better than $\sqrt{p^2 - p - 1} < p$ follows from it alone. That every Kloosterman sum is in fact of size at most $2\sqrt{p}$, matching the average pointwise, is the theorem of Weil, and it is as hard to prove as the Weil bound for character sums in section 11.2: both are instances of the Riemann hypothesis for a curve over $\mathbb{F}_p$.

Theorem 11.4.4: Weil's Bound for Kloosterman Sums

Let p be a prime and $a, b \in \mathbb{Z}$ with $p \nmid \gcd(a,b)$, that is, p does not divide both a and b. Then

$$|S(a,b;p)| \leq 2\sqrt{p}.$$

More generally, for any modulus $q \geq 1$ and a, b with $\gcd(a,b,q) = 1$,

$$|S(a,b;q)| \leq \tau(q) \sqrt{\gcd(a,b,q)} \sqrt{q} = O_\varepsilon\left(q^{1/2+\varepsilon}\right),$$

where $\tau(q)$ is the number of divisors of q.

***Proof Key ideas*:**
The prime case is a theorem of Weil, and its proof, like that of theorem 11.2.1, belongs to the theory of curves over finite fields; only the source of the bound is indicated, in keeping with the convention on deferred proofs. Attached to the Kloosterman sum is the Kloosterman sheaf, a rank-2 local system on $\mathbf{G}_m/\mathbb{F}_p$ whose Frobenius trace at the point $t = ab$ is exactly $-S(a, b; p)$. The eigenvalues of Frobenius are the reciprocal roots of the local L-factor, which by the Riemann hypothesis for the associated curve (Weil's theorem) are two complex numbers α, β of absolute value $\sqrt{p}$ each, with $\alpha\beta = p$; hence

$$S(a, b; p) = -(\alpha + \beta), \qquad |\alpha| = |\beta| = \sqrt{p}, \qquad |S(a, b; p)| \leq |\alpha| + |\beta| = 2\sqrt{p}.$$

The angle θ defined by $S(a, b; p) = 2\sqrt{p}\cos\theta$ is the Kloosterman angle, and its distribution as ab varies is governed by the Sato–Tate measure. The general modulus follows from the prime case by the multiplicativity of proposition 11.4.2 and the elementary evaluation of the prime-power (Salié) factors, with the divisor factor $\tau(q)$ recording the number of prime-power components; the details are standard and are carried out in [9, Ch. 11]. The bound $2\sqrt{p}$ is best possible in the sense that the angles θ equidistribute for Sato–Tate, so $S(a, b; p)$ comes arbitrarily close to $\pm 2\sqrt{p}$ infinitely often.

Weil's bound is the exact square-root cancellation predicted by the second moment: a complete Kloosterman sum of $p - 1$ unit terms is no larger than $2\sqrt{p}$, the coupling of x and $\bar{x}$ notwithstanding. Its significance for analytic number theory is that Kloosterman sums are the arithmetic residue left by the circle method and by the spectral theory of automorphic forms, and the saving of $\sqrt{p}$ over the trivial bound p is what converts otherwise divergent main terms into convergent error terms. Before Weil, Kloosterman's own elementary argument through a higher moment gave only $|S(a, b; p)| \ll p^{3/4}$, already enough for his application to quaternary quadratic forms but short of the truth; the passage from $p^{3/4}$ to $2\sqrt{p}$ is precisely the passage from elementary counting to the Riemann hypothesis for curves.

11.4.5 Foreshadowing: Sums of Kloosterman Sums and the Spectral Theory
Weil's bound is optimal for a single Kloosterman sum, but many applications require cancellation in *sums* of Kloosterman sums, $\sum_{c \leq C} S(a, b; c)/c$, where the individual square-root savings are not enough. The Linnik and Selberg conjecture predicts near-total cancellation in such sums, of the form $O_\varepsilon(C^\varepsilon)$ after the $\sqrt{c}$ normalization. The tool that delivers it is not the algebraic geometry of a single curve but the spectral theory of the modular surface: the Kuznetsov trace formula expresses a weighted sum of Kloosterman sums in terms of the eigenvalues of the hyperbolic Laplacian acting on automorphic forms, and the size of the sum is controlled by the location of those eigenvalues. This is the point at which the analytic number theory of this book meets the theory of automorphic forms, which the series takes up in its later volumes; the Kloosterman sum is the bridge between the two.

11.5 Applications

The estimates of the preceding sections were proven for their own sake, but their purpose is to control arithmetic. This closing section collects three applications that show the character and exponential sum bounds at work: the quantitative theory of equidistribution, where an exponential-sum bound becomes an explicit rate of convergence through the Erdős–Turán inequality; the classical lattice-point problems, where van der Corput's method sharpens the elementary error terms of chapter 1; and the circle method, the framework in which Weyl sums and Kloosterman sums together produce asymptotic formulas for the number of representations of an integer. The three illustrate the two faces of the subject, cancellation as equidistribution and cancellation as an error term, and the circle method points ahead to the sieve methods of chapter 12, the complementary technique for the same additive problems.

The equidistribution of section 11.3 was qualitative: Weyl's criterion decides whether $\frac{1}{N}\sum e(hx_n) \to 0$, but not how fast. The rate is measured by the discrepancy, and the Erdős–Turán inequality converts a bound on the Weyl sums directly into a bound on it.

Definition 11.5.1: Discrepancy

The *discrepancy* of the first N terms of a sequence (x_n) of real numbers is

$$D_N = \sup_{0\le a<b\le 1} \left| \frac{1}{N} \#\big\{n \le N : \{x_n\} \in [a,b)\big\} - (b-a) \right|.$$

The sequence is uniformly distributed modulo 1 if and only if $D_N \to 0$, and D_N quantifies the largest deviation of the empirical distribution of $\{x_n\}$ from the uniform one.

Theorem 11.5.2: Erdős–Turán Inequality

There is an absolute constant C such that for every sequence (x_n) and all integers $N \ge 1$ and $H \ge 1$,

$$D_N \le C\left(\frac{1}{H} + \sum_{h=1}^{H} \frac{1}{h} \left| \frac{1}{N} \sum_{n=1}^{N} e(hx_n) \right| \right).$$

***Proof*:**
The proof sandwiches the indicator of an interval between trigonometric polynomials of degree H and evaluates the resulting exponential sums.

Step 1: Selberg's majorants. For an interval $I = [a,b) \subseteq [0,1)$ and an integer $H \ge 1$, Selberg's extremal construction ([11, §1.2]; the Beurling–Selberg majorant of [17]) provides 1-periodic trigonometric polynomials $S^{\pm}(x) = \sum_{|h|\le H} \widehat{S^{\pm}}(h)\, e(hx)$, of degree at most H, with

$$S^{-}(x) \le \mathbf{1}_I(x) \le S^{+}(x) \quad \text{for all } x \in \mathbb{R}/\mathbb{Z},$$

and Fourier coefficients satisfying $\widehat{S^{\pm}}(0) = (b-a) \pm \frac{1}{H+1}$ and $\big|\widehat{S^{\pm}}(h)\big| \le \frac{1}{H+1} + \frac{1}{\pi|h|}$ for $1 \le |h| \le H$. Only these two properties are used; the construction of the polynomials, a self-contained extremal problem in Fourier analysis, is the one ingredient cited rather than reproduced.

Step 2: Counting through the majorant. Write $c_h = \frac{1}{N}\sum_{n\le N} e(hx_n)$, so $|c_h| \le 1$ and $c_{-h} = \overline{c_h}$;

for integer h, $e(h\{x_n\}) = e(hx_n)$. From $\mathbf{1}_I(\{x_n\}) \leq S^+(\{x_n\})$,

$$\frac{1}{N}\#\{n \leq N : \{x_n\} \in I\} \leq \frac{1}{N}\sum_{n\leq N} S^+(\{x_n\}) = \sum_{|h|\leq H} \widehat{S^+}(h)\, c_h = (b-a)+\frac{1}{H+1}+\sum_{1\leq|h|\leq H} \widehat{S^+}(h)\, c_h.$$

Subtracting $b-a$ and bounding the tail, and using that for $1 \leq h \leq H$ one has $\frac{1}{H+1} \leq \frac{1}{h}$, so $\big|\widehat{S^+}(\pm h)\big| \leq \frac{1}{H+1} + \frac{1}{\pi h} \leq \frac{1}{h}\big(1+\frac{1}{\pi}\big)$, together with $|c_{-h}| = |c_h|$,

$$\frac{1}{N}\#\{n \leq N : \{x_n\} \in I\} - (b-a) \leq \frac{1}{H+1} + \sum_{1\leq|h|\leq H} \big|\widehat{S^+}(h)\big|\,|c_h| \leq \frac{1}{H+1} + 2\Big(1+\frac{1}{\pi}\Big)\sum_{h=1}^{H}\frac{1}{h}|c_h|.$$

Step 3: Both bounds and the supremum. The identical computation with the minorant $S^- \leq \mathbf{1}_I$ yields the matching lower bound, so

$$\left|\frac{1}{N}\#\{n \leq N : \{x_n\} \in I\} - (b-a)\right| \leq \frac{1}{H+1} + 2\Big(1+\frac{1}{\pi}\Big)\sum_{h=1}^{H}\frac{1}{h}|c_h|.$$

The right-hand side is independent of the interval I, since the coefficient bounds of Step 1 are uniform. Taking the supremum over $I = [a,b) \subseteq [0,1)$ and using $\frac{1}{H+1} \leq \frac{1}{H}$,

$$D_N \leq \frac{1}{H} + 2\Big(1+\frac{1}{\pi}\Big)\sum_{h=1}^{H}\frac{1}{h}\left|\frac{1}{N}\sum_{n=1}^{N} e(hx_n)\right|,$$

which is the stated inequality with the absolute constant $C = 2(1+1/\pi)$, completing the proof.

The inequality is the quantitative form of Weyl's criterion: a bound on the exponential sums $\sum_{n\leq N} e(hx_n)$ for $h \leq H$ yields a bound on the discrepancy, with H optimized against the quality of the exponential-sum estimate. Applied to a polynomial sequence, it turns the Weyl sum bound of theorem 11.3.5 into an explicit rate.

Example 11.5.3: Discrepancy of $(n\alpha)$ and $(n^2\alpha)$

For $x_n = n\alpha$ with α irrational, the geometric sum of example 11.3.3 gives $|\sum_{n\leq N} e(hn\alpha)| \leq 1/(2\|h\alpha\|)$. Inserting this into the Erdős–Turán inequality, when α has bounded partial quotients (for instance α a quadratic irrational, where $\|h\alpha\| \gg 1/h$), the sum over $h \leq H$ is $\ll \frac{1}{N}\sum_{h\leq H} 1/(h\|h\alpha\|) \ll (\log H)^2/N$, and choosing $H = N$ yields

$$D_N \ll \frac{(\log N)^2}{N}.$$

The fractional parts $\{n\alpha\}$ thus fill $[0,1)$ with discrepancy nearly as small as the trivial lower bound $D_N \gg 1/N$ permits. For the quadratic sequence $x_n = n^2\alpha$ the Weyl sum bound after one differencing gives $|\sum_{n\leq N} e(hn^2\alpha)| \ll N^{1/2+\varepsilon}$ for α of finite type, and the Erdős–Turán inequality yields $D_N \ll N^{-1/2+\varepsilon}$: the squares $\{n^2\alpha\}$ equidistribute, but more slowly than the linear sequence, the loss being the exponent $\frac{1}{2}$ characteristic of a single Weyl differencing step.

A parallel application, beyond the reach of the elementary methods but proven by the same exponential-sum philosophy, is Vinogradov's theorem that $(\{\alpha p\})_{p \text{ prime}}$ is uniformly distributed modulo 1 for every irrational α: the primes, weighted by the von Mangoldt function and analyzed through the exponential sums $\sum_{n\le N}\Lambda(n)e(h\alpha n)$, spread out over the circle exactly as the integers do. The bound on $\sum_n \Lambda(n)e(\alpha n)$ is Vinogradov's estimate, and it is the prototype for the minor-arc analysis in the circle method below.

The second application returns to the lattice-point problems of chapter 1. The divisor sum $\sum_{n\le x} d(n) = x\log x + (2\gamma-1)x + \Delta(x)$ was evaluated in theorem 1.3.3 with the elementary error $\Delta(x) = O(\sqrt{x})$ from the hyperbola method. That bound is a counting estimate, blind to the oscillation of the error term; exponential sums see the oscillation and exploit it.

Theorem 11.5.4: The Divisor Problem via van der Corput

The error term in the divisor sum satisfies

$$\Delta(x) = \sum_{n\le x} d(n) - x\log x - (2\gamma-1)x = O\big(x^{1/3}\log x\big).$$

***Proof Key ideas*:**
The improvement over the elementary $O(\sqrt{x})$ comes from writing $\Delta(x)$ as an exponential sum and applying van der Corput's method; the full computation is a several-page application of the A- and B-processes, and only its structure is indicated. The hyperbola method expresses $\sum_{n\le x} d(n)$ as $2\sum_{n\le\sqrt{x}}\lfloor x/n\rfloor - \lfloor\sqrt{x}\rfloor^2$, and replacing each floor $\lfloor x/n\rfloor$ by $x/n - \frac{1}{2} - \psi(x/n)$, where $\psi(t) = \{t\} - \frac{1}{2}$ is the sawtooth function, isolates $\Delta(x)$ as a sum $-2\sum_{n\le\sqrt{x}}\psi(x/n)$ up to lower-order terms. The Fourier expansion of ψ turns this into a family of exponential sums $\sum_n e(hx/n)$ with the smooth phase $f(n) = hx/n$; van der Corput's inequality (theorem 11.3.6), applied through one B-process (Poisson summation, converting the sum to a dual sum of length $\asymp x/n^2$) followed by the trivial bound, estimates each such sum. Balancing the resulting terms over the ranges of n and h produces the exponent $\frac{1}{3}$; the endpoint bookkeeping contributes the factor $\log x$. The complete argument is in [9, Ch. 2] and, in classical form, in [21, Ch. 12], completing the proof.

The exponent $\frac{1}{3}$ was Voronoi's, and it is the first term in a long history of refinements: successive elaborations of the exponent-pair method lower it toward the conjectured $\frac{1}{4}+\varepsilon$, the current record $\frac{131}{416}$ of Huxley already recorded in chapter 1. The identical method, applied to $\sum_{n\le x} r(n)$ with $r(n)$ the number of representations of n as a sum of two squares, gives the same $O(x^{1/3})$ bound for the error in the Gauss circle problem, the two lattice-point problems being analytically parallel: in both, the error term is an exponential sum with a smooth phase, and van der Corput's method is the instrument that extracts its cancellation.

The final application is the circle method, the framework in which the exponential sums of this chapter address additive problems directly. The setting is the representation of an integer N in a prescribed additive form, and the method reads off the count of representations from the behavior of a generating exponential sum near rational points.

11.5.5 The Circle Method for Waring's Problem Fix $k \geq 2$ and consider the number $r_{s,k}(N)$ of representations of N as a sum of s positive k-th powers, $N = m_1^k + \cdots + m_s^k$. With the Weyl generating sum

$$f(\theta) = \sum_{m \leq N^{1/k}} e(m^k\theta),$$

one has $f(\theta)^s = \sum_n r_{s,k}(n)\, e(n\theta)$, and by orthogonality of the additive characters,

$$r_{s,k}(N) = \int_0^1 f(\theta)^s\, e(-N\theta)\, d\theta.$$

The integral is dissected into *major arcs*, short intervals around rationals a/q with small denominator q, and *minor arcs*, the remainder. On the major arcs $f(\theta)$ is close to a main term computed from Gauss sums, and the contribution is the expected asymptotic $\mathfrak{S}(N)\,\Gamma(1+1/k)^s\,\Gamma(s/k)^{-1}\,N^{s/k-1}$, where the *singular series* $\mathfrak{S}(N) = \sum_{q\geq 1}\sum_{\gcd(a,q)=1}\big(q^{-1}\sum_{m \bmod q} e(am^k/q)\big)^s e(-aN/q)$ encodes the local densities of solutions modulo every prime. On the minor arcs $f(\theta)$ exhibits the cancellation of section 11.3: Weyl's inequality bounds $|f(\theta)|$ by $N^{1/k}$ times a small power saving whenever θ is poorly approximable, and the minor-arc integral is negligible once s is large enough for the power saving to overcome the length. The conclusion is the Hardy–Littlewood asymptotic formula: for $s \geq s_0(k)$ and $\mathfrak{S}(N) \gg 1$,

$$r_{s,k}(N) \sim \mathfrak{S}(N)\,\frac{\Gamma(1+1/k)^s}{\Gamma(s/k)}\,N^{s/k-1},$$

so N is a sum of s positive k-th powers for all large N with the local obstructions accounted for by $\mathfrak{S}(N)$. The Kloosterman sums of section 11.4 enter in Kloosterman's refinement of the method, which lowers the required number of variables $s_0(k)$ by extracting additional cancellation from the minor arcs through the average over the numerators a; this is the origin of the term "Kloosterman sum".

The circle method is the analytic engine for additive problems: Waring's problem, Goldbach's problem, and the representation of integers by quadratic forms all fall to it, and in each case the answer is a main term given by a singular series of local densities and an error term controlled by the exponential-sum estimates of this chapter. Its reach is limited by the minor arcs, precisely where the Weyl and Kloosterman bounds are applied, and improving those bounds is the same problem that has recurred throughout the chapter, from the least non-residue to the divisor problem: how much cancellation can be proven in a sum of arithmetic phases. The sieve methods of chapter 12 attack the same additive and multiplicative problems from the opposite direction, replacing the harmonic analysis of the circle by the combinatorics of inclusion and exclusion, and the two methods together frame the modern analytic theory of the additive behavior of the primes.

12

Sieve Methods

The methods of the previous chapter measured cancellation in sums of oscillating terms. Sieve methods take up a different question, older than the analytic theory and combinatorial at its root: how many integers in a given set survive after those divisible by a collection of small primes are struck out? The sieve of Eratosthenes answers this exactly through inclusion and exclusion, but the exact answer is unusable, its error term swamping its main term as soon as the primes are numerous. The achievement of twentieth-century sieve theory was to trade exactness for control, replacing the alternating Möbius sum by cleverly truncated or weighted approximations whose error terms stay small. The resulting inequalities do not identify the primes exactly, but they bound the count of primes, twin primes, and other sifted sequences from above and below, and they reach problems the L-function methods cannot touch.

Where an L-function encodes the primes through the analytic continuation of an infinite product, a sieve works directly with the finite combinatorics of divisibility, and it asks only for elementary information: how many elements of the set are divisible by each squarefree d. This makes sieves robust and widely applicable, but it also imposes an intrinsic limitation, the parity problem, which prevents an unaided sieve from ever proving that a sifted sequence contains a prime rather than a product of two primes. The chapter develops the main sieves in order of power: the exact but unusable Eratosthenes–Legendre sieve (section 12.1), Brun's combinatorial truncation that first made the sieve quantitative (section 12.2), Selberg's optimized upper-bound sieve (section 12.3), and the large sieve with its arithmetic consequence, the Bombieri–Vinogradov theorem (section 12.4); the closing section (section 12.5) surveys the modern developments, from the parity barrier to the bounded-gaps theorems of the last decade. Only the material of Part I is needed, together with the Mertens estimates of chapter 3 and, in the large sieve, the additive characters $e(t)$ of chapter 11.

12.1 The Eratosthenes–Legendre Sieve

Every sieve begins with the same data: a finite set of integers to be sifted, a set of primes to sift by, and a cutoff below which the sifting primes are drawn. The object of study is the count of survivors.

Definition 12.1.1: Sifting Function

Let $\mathcal{A}$ be a finite sequence of integers, let $\mathcal{P}$ be a set of primes, and let $z \geq 2$ be a real number. Write

$$P(z) = \prod_{\substack{p \in \mathcal{P} \\ p < z}} p$$

for the product of the sifting primes below z. The *sifting function* is

$$S(\mathcal{A}, \mathcal{P}, z) = \#\{\, a \in \mathcal{A} : \gcd(a, P(z)) = 1 \,\},$$

the number of elements of $\mathcal{A}$ divisible by no prime $p \in \mathcal{P}$ with $p < z$. For a squarefree $d \big| P(z)$, write $A_d = \#\{a \in \mathcal{A} : d \big| a\}$ for the number of elements divisible by d, so that $A_1 = |\mathcal{A}|$.

The survivors counted by $S(\mathcal{A}, \mathcal{P}, z)$ are the elements of $\mathcal{A}$ that have been sifted clean of all small prime factors. When $\mathcal{A} = \{n : n \leq x\}$ and $\mathcal{P}$ is the set of all primes, an element survives the sieve at level $z = \sqrt{x}$ exactly when it is 1 or a prime exceeding $\sqrt{x}$, so $S(\mathcal{A}, \mathcal{P}, \sqrt{x}) = \pi(x) - \pi(\sqrt{x}) + 1$; the sieve of Eratosthenes is the statement that striking out the multiples of each prime up to $\sqrt{x}$ leaves precisely the primes in $(\sqrt{x}, x]$. Counting the survivors is a matter of inclusion and exclusion, and the Möbius function performs it.

Proposition 12.1.2: Legendre's Identity

With the notation of definition 12.1.1,

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{d \big| P(z)} \mu(d)\, A_d,$$

the sum running over all squarefree divisors d of $P(z)$.

***Proof*:**
The Möbius function detects the coprimality condition through the identity $\sum_{d \big| n} \mu(d) = \mathbf{1}_{n=1}$ (proposition 1.2.6), applied to $n = \gcd(a, P(z))$. For each $a \in \mathcal{A}$,

$$\sum_{d \big| \gcd(a, P(z))} \mu(d) = \begin{cases} 1 & \text{if } \gcd(a, P(z)) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

so this inner sum is the indicator that a survives the sieve. Summing over all $a \in \mathcal{A}$ and interchanging the order of summation,

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{a \in \mathcal{A}} \sum_{d \big| \gcd(a, P(z))} \mu(d) = \sum_{d \big| P(z)} \mu(d)\, \#\{a \in \mathcal{A} : d \big| a\} = \sum_{d \big| P(z)} \mu(d)\, A_d,$$

where the interchange is justified because $d \mid \gcd(a, P(z))$ is equivalent to $d \mid P(z)$ and $d \mid a$, completing the proof.

Legendre's identity is exact, and for that reason it is the natural starting point; the difficulty it exposes is the reason every later sieve exists. To use the identity one needs an approximation for each A_d. In the typical application the sifting set is regular, meaning that the proportion of its elements divisible by d is close to a multiplicative function $g(d)$:

$$A_d = g(d)\, X + r_d,$$

where $X = |\mathcal{A}|$ is the expected main density, g is multiplicative with $0 \le g(p) < 1$, and r_d is a remainder. Substituting into Legendre's identity splits the sifting function into a main term and an accumulated error,

$$S(\mathcal{A}, \mathcal{P}, z) = X \sum_{d \mid P(z)} \mu(d) g(d) + \sum_{d \mid P(z)} \mu(d)\, r_d = X\, V(z) + R, \qquad V(z) = \prod_{\substack{p \in \mathcal{P} \\ p < z}} \bigl(1 - g(p)\bigr), \tag{12.1}$$

the main term factoring as an Euler product by the multiplicativity of g and μ. The main term $X\, V(z)$ is exactly what a heuristic of independence predicts: each prime p removes a proportion $g(p)$ of the survivors, and the removals behave as if independent. The entire content of a sieve lies in the remainder $R = \sum_{d \mid P(z)} \mu(d)\, r_d$, and here the Eratosthenes–Legendre sieve fails.

Example 12.1.3: The Sieve of Eratosthenes and Its Failure

Take $\mathcal{A} = \{n : n \le x\}$ and $\mathcal{P}$ the set of all primes, so $X = x$, $g(d) = 1/d$, and $A_d = \lfloor x/d \rfloor = x/d + r_d$ with $|r_d| \le 1$. Legendre's identity gives

$$S(\mathcal{A}, \mathcal{P}, z) = x \prod_{p < z} \Bigl(1 - \frac{1}{p}\Bigr) + R, \qquad |R| \le \sum_{d \mid P(z)} 1 = 2^{\pi(z)},$$

since $P(z)$ has $\pi(z)$ prime factors and hence $2^{\pi(z)}$ squarefree divisors, each contributing a remainder of modulus at most 1. By Mertens' third theorem (theorem 3.3.5) the main term is $x \prod_{p<z}(1 - 1/p) = (e^{-\gamma} + o(1))\, x/\log z$. The remainder $2^{\pi(z)}$ is controlled only for z extremely small: to keep it below x one must take $z \le (\log x)$ roughly, and then the main term is $\asymp x/\log\log z$, far larger than the true count of primes. Choosing $z = \log x$ yields only the weak bound $\pi(x) \le S(\mathcal{A}, \mathcal{P}, z) + z \ll x/\log\log x$, which does not even recover Chebyshev's $\pi(x) \ll x/\log x$ (theorem 3.2.3). The exact identity is defeated by the exponential number of divisors: each contributes a harmless $O(1)$, but there are $2^{\pi(z)}$ of them, and their accumulated error destroys the estimate for any z large enough to be interesting.

The failure is instructive. The main term $X\, V(z)$ is exactly right, and if the remainders r_d could be summed with their signs, enormous cancellation would occur; but the trivial bound $|R| \le \sum_{d \mid P(z)} |r_d|$ discards all of it, and the number of terms is too large. Two escapes present themselves, and they organize the rest of the chapter. Brun's idea is to truncate the Möbius sum, keeping only divisors d with few prime factors, so that the number of remainder terms is a manageable power of x rather than $2^{\pi(z)}$; the truncation is arranged, through the Bonferroni inequalities, to yield one-sided bounds rather than an exact identity. Selberg's idea is to replace the Möbius coefficients $\mu(d)$ by optimally chosen weights supported on small d, again bounding rather than computing. Both sacrifice the equality of Legendre's identity for an inequality whose error is controllable, and both, unlike the exact sieve, produce theorems.

12.1.4 Foreshadowing: The Parity Problem There is a limit no elaboration of these ideas can cross. A sieve sees an integer only through the divisibilities $d \mid a$ it is fed, and this data cannot distinguish an integer with an even number of prime factors from one with an odd number: the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ (definition 1.1.6) is, in a precise sense made explicit by Selberg, invisible to the sieve. Consequently a sieve alone can never prove that a sifted sequence contains a number with exactly one prime factor, a prime, as opposed to one with exactly two, a product of two primes; the two cases are indistinguishable to it. This is the *parity problem*, and it is why the sieve results below stop at "almost primes" with a bounded number of factors, and why proving the twin prime conjecture, or that there are infinitely many primes p with $p+2$ prime, requires an input from beyond sieve theory. The precise statement is taken up in section 12.5.

12.2 Brun's Combinatorial Sieve

Brun's insight was that the exact Möbius sum of Legendre's identity need not be evaluated in full. Truncating it, keeping only the divisors d with at most a bounded number of prime factors, both reduces the number of remainder terms to a controllable size and, when the truncation level is chosen with the right parity, converts the exact identity into a one-sided inequality. The combinatorial fact underlying the truncation is a sign property of partial sums of the Möbius function.

Lemma 12.2.1: Sign of the Truncated Möbius Sum

Let $n > 1$ be a squarefree integer with $\omega(n) = r$ distinct prime factors, and let $m \geq 0$ be an integer. Then

$$\sum_{\substack{d \mid n \\ \omega(d) \leq m}} \mu(d) = (-1)^m \binom{r-1}{m},$$

with the convention $\binom{r-1}{m} = 0$ for $m > r-1$. Consequently the truncated sum is nonnegative when m is even and nonpositive when m is odd.

***Proof*:**
Group the divisors by their number of prime factors. Since n is squarefree with r prime factors, the number of divisors $d \mid n$ with $\omega(d) = j$ is $\binom{r}{j}$, and each such d has $\mu(d) = (-1)^j$, so $\sum_{d \mid n,\ \omega(d)=j} \mu(d) = (-1)^j \binom{r}{j}$. Therefore

$$\sum_{\substack{d \mid n \\ \omega(d) \leq m}} \mu(d) = \sum_{j=0}^{m} (-1)^j \binom{r}{j}.$$

The partial alternating sum of binomial coefficients telescopes: using Pascal's rule $\binom{r}{j} = \binom{r-1}{j} + \binom{r-1}{j-1}$, the sum $\sum_{j=0}^{m}(-1)^j\left(\binom{r-1}{j} + \binom{r-1}{j-1}\right)$ collapses, all interior terms cancelling in pairs, to the single surviving term $(-1)^m\binom{r-1}{m}$. Since $\binom{r-1}{m} \geq 0$, the truncated sum has the sign of $(-1)^m$, completing the proof.

The lemma converts the truncation into an inequality. When m is even the truncated Möbius sum is at least the indicator of $\{n = 1\}$ (equal to it at $n = 1$, where the sum is 1, and at least $0 = \mathbf{1}_{n>1}$ elsewhere); when m is odd it is at most that indicator. Summing over the sifting set gives Brun's pure sieve.

Theorem 12.2.2: Brun's Pure Sieve Inequalities

With the notation of definition 12.1.1, for every even integer $m \geq 0$,

$$S(\mathcal{A}, \mathcal{P}, z) \leq \sum_{\substack{d \mid P(z) \\ \omega(d) \leq m}} \mu(d)\, A_d,$$

and for every odd integer $m \geq 1$, the same expression is a lower bound for $S(\mathcal{A}, \mathcal{P}, z)$. In both cases the number of divisors appearing is $\sum_{j \leq m} \binom{\pi_{\mathcal{P}}(z)}{j}$, where $\pi_{\mathcal{P}}(z)$ is the number of primes of $\mathcal{P}$ below z.

***Proof*:**
For each $a \in \mathcal{A}$ apply lemma 12.2.1 to $n = \gcd(a, P(z))$. When $\gcd(a, P(z)) = 1$ the truncated sum is 1 for every $m \geq 0$ (only $d = 1$ contributes). When $\gcd(a, P(z)) > 1$, the truncated sum equals $(-1)^m\binom{r-1}{m}$ with $r = \omega(\gcd(a, P(z))) \geq 1$, which is ≥ 0 for m even and ≤ 0 for m odd. Thus for m even,

$$\mathbf{1}_{\gcd(a,P(z))=1} \leq \sum_{\substack{d \mid \gcd(a,P(z)) \\ \omega(d) \leq m}} \mu(d),$$

and summing over $a \in \mathcal{A}$ and interchanging summation as in proposition 12.1.2 gives the upper bound; the odd case reverses the inequality. The count of divisors is the number of squarefree $d \mid P(z)$ with at most m prime factors, namely $\sum_{j \leq m} \binom{\pi_{\mathcal{P}}(z)}{j}$, as claimed.

Brun's inequalities are the exact identity with its tail discarded, and the discarding is arranged to lose only in one direction. The gain is decisive: whereas the full Legendre sieve carried $2^{\pi(z)}$ remainder terms, the truncated sieve at level m carries only $\sum_{j \leq m} \binom{\pi(z)}{j} \leq \pi(z)^m$, a fixed power of $\pi(z)$. Choosing m growing slowly with x balances the truncation error, the difference between the truncated sum and the true main term, against the accumulated remainder, and the balance is what makes the sieve quantitative. Carried out for the twin primes, it produces the first proof that they are sparse.

Theorem 12.2.3: Brun's Theorem on Twin Primes

Let $\pi_2(x)$ denote the number of primes $p \le x$ for which $p+2$ is also prime. Then

$$\pi_2(x) \ll \frac{x\,(\log\log x)^2}{(\log x)^2}.$$

***Proof Key ideas*:**
The estimate is obtained by applying Brun's pure sieve to a set built from the twin-prime condition, and the full optimization of the truncation level is a technical computation carried out in [9, §6]; the structure is as follows. Sift the sequence $\mathcal{A} = \{n(n+2) : n \le x\}$ by the odd primes: an integer $n \le x$ with both n and $n+2$ prime (and $n > 2$) contributes a product $n(n+2)$ divisible by no odd prime below z once z exceeds $\sqrt{x}$-scale thresholds, so $\pi_2(x)$ is bounded by a sifting function of $\mathcal{A}$ plus a contribution from small primes. For an odd prime p, the congruence $n(n+2) \equiv 0 \pmod{p}$ has two solutions $n \equiv 0, -2 \pmod{p}$, so the local density is $g(p) = 2/p$, and the main term is

$$V(z) = \prod_{2<p<z}\left(1 - \frac{2}{p}\right) \asymp \frac{1}{(\log z)^2},$$

the square arising because two residue classes are removed at each prime, in contrast to the single class $1/p$ of the Eratosthenes sieve. Applying theorem 12.2.2 with truncation level $m \asymp \log\log x$ and cutoff $\log z \asymp \log x/\log\log x$ makes the accumulated remainder $\sum_{\omega(d)\le m} |r_d|$, with $|r_d| \le 2^{\omega(d)}$, smaller than the main term, and the resulting bound is $\pi_2(x) \ll x\,V(z)(\log\log x)^{O(1)} \ll x(\log\log x)^2/(\log x)^2$. The factor $(\log\log x)^2$ is the loss from the truncation and the sub-optimal cutoff forced by keeping the remainder under control; the sharper combinatorial sieves of sections 12.3 and 12.5 remove it, but the order $x/(\log x)^2$ is already the truth. This completes the argument.

The bound $\pi_2(x) \ll x/(\log x)^2$ is of the conjectured order of magnitude: the Hardy–Littlewood prime k-tuple conjecture predicts $\pi_2(x) \sim 2C_2\,x/(\log x)^2$, where the *twin prime constant*

$$C_2 = \prod_{p>2}\left(1 - \frac{1}{(p-1)^2}\right) = 0.6601\ldots$$

corrects the naive density for the correlation between the events "n prime" and "$n+2$ prime". Brun's theorem does not prove this asymptotic, and cannot, but its upper bound of the correct order has an immediate and famous consequence: the twin primes are sparse enough that the sum of their reciprocals converges.

Corollary 12.2.4: Convergence of the Sum over Twin Primes

The sum

$$B = \sum_{p:p+2 \text{ prime}} \left(\frac{1}{p} + \frac{1}{p+2}\right)$$

over all primes p such that $p+2$ is prime converges. Its value $B = 1.902\ldots$ is Brun's constant.

***Proof*:**
It suffices to show $\sum_{p:p+2 \text{ prime}} 1/p < \infty$, since $1/(p+2) < 1/p$. Let $p_1 < p_2 < \cdots$ enumerate the primes p with $p+2$ prime, and recall $\pi_2(t) = \#\{p \leq t : p+2 \text{ prime}\}$. By partial summation (theorem 1.4.1),

$$\sum_{p \leq x,\ p+2 \text{ prime}} \frac{1}{p} = \frac{\pi_2(x)}{x} + \int_2^x \frac{\pi_2(t)}{t^2}\, dt.$$

Brun's theorem (theorem 12.2.3) gives $\pi_2(t) \ll t(\log\log t)^2/(\log t)^2$, so the boundary term tends to 0 and the integrand is $\ll (\log\log t)^2/\big(t(\log t)^2\big)$. The integral $\int_2^\infty (\log\log t)^2/\big(t(\log t)^2\big)\, dt$ converges, as the substitution $u = \log t$ turns it into $\int (\log u)^2/u^2\, du$, which converges at infinity. Hence the partial sums are bounded and increasing, so the series converges, completing the proof.

Brun's constant is one of the few places in the subject where a divergent-looking sum over primes is shown to converge, and the contrast with Euler's theorem is exact: $\sum_p 1/p$ diverges (theorem 3.3.3) because the primes have density $1/\log x$, while $\sum_{p:p+2 \text{ prime}} 1/p$ converges because the twin primes have the smaller density $1/(\log x)^2$. The convergence does not decide whether there are infinitely many twin primes, a series can converge whether its terms are finite or infinite in number, and settling the twin prime conjecture lies beyond the combinatorial sieve for the reason foreshadowed in box 12.1.4. What the sieve delivers is the correct order of magnitude for the count and, through it, the arithmetic constant B.

12.3 The Selberg Sieve

Brun obtained a one-sided inequality by truncating the Möbius function at a fixed number of prime factors. Selberg's sieve reaches the same kind of upper bound by a different and more flexible device: it replaces the Möbius coefficients with a family of real weights left free, then chooses them to minimize the resulting bound. The optimization is a problem in quadratic forms, and its solution gives the sharpest elementary upper-bound sieve, with a clean main term expressed through a single arithmetic sum.

The starting point is an inequality valid for any weights whatsoever, exploiting only that a square is nonnegative.

Lemma 12.3.1: The Λ^2 Inequality

Let $(\lambda_d)_{d \mid P(z)}$ be any real numbers with $\lambda_1 = 1$. Then for every $a \in \mathcal{A}$,

$$\mathbf{1}_{\gcd(a,P(z))=1} \le \Big(\sum_{d \mid \gcd(a,P(z))} \lambda_d \Big)^2,$$

and consequently

$$S(\mathcal{A},\mathcal{P},z) \le \sum_{a\in\mathcal{A}} \Big(\sum_{d \mid \gcd(a,P(z))} \lambda_d \Big)^2 = \sum_{d,e \mid P(z)} \lambda_d \lambda_e \, A_{[d,e]},$$

where $[d,e]$ is the least common multiple and $A_{[d,e]} = \#\{a \in \mathcal{A} : [d,e] \mid a\}$.

***Proof*:**
If $\gcd(a,P(z)) = 1$, the only divisor $d \mid \gcd(a,P(z))$ is $d = 1$, so the inner sum equals $\lambda_1 = 1$ and its square is 1, matching the indicator. If $\gcd(a,P(z)) > 1$, the indicator is 0 while the square is nonnegative, so the inequality holds. This is the pointwise bound. Summing over $a \in \mathcal{A}$,

$$S(\mathcal{A},\mathcal{P},z) \le \sum_{a\in\mathcal{A}} \Big(\sum_{d \mid \gcd(a,P(z))} \lambda_d \Big)^2 = \sum_{a\in\mathcal{A}} \sum_{d,e \mid \gcd(a,P(z))} \lambda_d\lambda_e = \sum_{d,e \mid P(z)} \lambda_d\lambda_e \sum_{\substack{a\in\mathcal{A} \\ [d,e] \mid a}} 1,$$

where the last step interchanges summation and uses that $d \mid a$ and $e \mid a$ together are equivalent to $[d,e] \mid a$. The inner sum is $A_{[d,e]}$, completing the proof.

The inequality holds for every choice of weights, so one is free to choose them to make the right-hand side as small as possible. Restricting the support to $d \le \sqrt{D}$ (so that $[d,e] \le D$ for all terms) and inserting the regularity hypothesis $A_{[d,e]} = g([d,e])\,X + r_{[d,e]}$ splits the bound into a quadratic form in the weights plus a remainder, and minimizing the quadratic form is the content of the method.

Theorem 12.3.2: The Selberg Upper Bound

Let g be multiplicative with $0 < g(p) < 1$ for all $p \in \mathcal{P}$, and suppose $A_d = g(d)X + r_d$ for squarefree $d \big| P(z)$. Let $D \geq 1$ and set $\xi = \sqrt{D}$. Define the multiplicative function h on squarefree integers by

$$h(p) = \frac{g(p)}{1 - g(p)}, \qquad h(d) = \prod_{p \mid d} h(p),$$

and put $G(\xi) = \sum_{d<\xi,\ d \mid P(z)} h(d)$. Then

$$S(\mathcal{A}, \mathcal{P}, z) \leq \frac{X}{G(\xi)} + \sum_{\substack{d,e<\xi \\ d,e \mid P(z)}} \left| r_{[d,e]} \right|.$$

***Proof*:**
Insert $A_{[d,e]} = g([d,e])X + r_{[d,e]}$ into lemma 12.3.1, restricting the weights to λ_d supported on squarefree $d < \xi$ dividing $P(z)$:

$$S(\mathcal{A}, \mathcal{P}, z) \leq X\,Q(\lambda) + R, \qquad Q(\lambda) = \sum_{d,e<\xi} \lambda_d \lambda_e\, g([d,e]), \qquad R = \sum_{d,e<\xi} \lambda_d \lambda_e\, r_{[d,e]}.$$

The proof minimizes the quadratic form $Q(\lambda)$ subject to $\lambda_1 = 1$ by diagonalizing it, then reads off the remainder bound.

Step 1: An identity for $1/g$. Since g is multiplicative with $0 < g(p) < 1$, the function h with $h(p) = g(p)/(1 - g(p))$ satisfies $1/h(p) = 1/g(p) - 1$. Let f be the multiplicative function with $f(p) = 1/h(p)$, that is, $f(m) = 1/h(m)$ for squarefree m. For squarefree k,

$$\sum_{m \mid k} f(m) = \prod_{p \mid k} (1 + f(p)) = \prod_{p \mid k} \frac{1}{g(p)} = \frac{1}{g(k)},$$

the middle equality because $1 + f(p) = 1 + (1/g(p) - 1) = 1/g(p)$. Hence for squarefree d, e, applying this with $k = (d,e)$ and using $g([d,e]) = g(d)g(e)/g((d,e))$ (from $[d,e](d,e) = de$ and multiplicativity),

$$g([d,e]) = g(d)g(e) \sum_{m \mid (d,e)} \frac{1}{h(m)}. \tag{12.2}$$

Step 2: Diagonalization. Substitute (12.2) into $Q(\lambda)$ and exchange the order of summation, collecting the terms by m:

$$Q(\lambda) = \sum_{d,e<\xi} \lambda_d \lambda_e\, g(d)g(e) \sum_{m \mid (d,e)} \frac{1}{h(m)} = \sum_{m<\xi} \frac{1}{h(m)} \Bigg(\sum_{\substack{d<\xi \\ m \mid d}} \lambda_d\, g(d) \Bigg)^2 = \sum_{m<\xi} \frac{y_m^2}{h(m)},$$

where $y_m = \sum_{d<\xi,\ m \mid d} \lambda_d\, g(d)$ and all sums run over squarefree divisors of $P(z)$. The form is diagonal in the y_m.

Step 3: The constraint. The relation $y_m = \sum_{m|d} \lambda_d g(d)$ expresses (y_m) as the divisor-sum transform of $(\lambda_d g(d))$ over multiples; its Möbius inversion is $\lambda_d g(d) = \sum_{d|m} \mu(m/d)\, y_m$. At $d = 1$, where $g(1) = 1$, this reads

$$\lambda_1 = \sum_{m<\xi} \mu(m)\, y_m,$$

so the normalization $\lambda_1 = 1$ is exactly the linear constraint $\sum_{m<\xi} \mu(m)\, y_m = 1$. The transform $(\lambda_d) \leftrightarrow (y_m)$ is triangular and invertible, so minimizing over admissible (λ_d) is the same as minimizing over (y_m) subject to this constraint.

Step 4: Minimization. The relevant m are squarefree divisors of $P(z)$ below ξ, so $\mu(m)^2 = 1$. By the Cauchy–Schwarz inequality,

$$1 = \Big(\sum_{m<\xi} \mu(m)\, y_m\Big)^2 = \Big(\sum_{m<\xi} \frac{y_m}{\sqrt{h(m)}} \cdot \mu(m)\sqrt{h(m)}\Big)^2 \le \Big(\sum_{m<\xi} \frac{y_m^2}{h(m)}\Big)\Big(\sum_{m<\xi} \mu(m)^2\, h(m)\Big) = Q(\lambda)\, G(\xi),$$

since $\sum_{m<\xi} \mu(m)^2 h(m) = \sum_{m<\xi,\ m|P(z)} h(m) = G(\xi)$. Thus $Q(\lambda) \ge 1/G(\xi)$, with equality when $y_m = \mu(m)\, h(m)/G(\xi)$; these values satisfy the constraint ($\sum_m \mu(m) y_m = G(\xi)^{-1} \sum_m h(m) = 1$) and, being finitely supported, correspond through Step 3 to admissible weights with $\lambda_1 = 1$. Hence $\min_{\lambda_1=1} Q(\lambda) = 1/G(\xi)$.

Step 5: The remainder. For the optimal weights one has $|\lambda_d| \le 1$ for every d, a standard estimate on the Selberg weights ([9, §6.5], [11, Lemma in §3.4]); it follows from the closed form for λ_d obtained by inverting Step 3 at the optimum. Therefore $R = \sum_{d,e<\xi} \lambda_d \lambda_e r_{[d,e]} \le \sum_{d,e<\xi} |r_{[d,e]}|$, and combining with $Q(\lambda) = 1/G(\xi)$ gives $S(\mathcal{A}, \mathcal{P}, z) \le X/G(\xi) + \sum_{d,e<\xi} |r_{[d,e]}|$, completing the proof.

The Selberg bound is both sharper and more usable than Brun's. Its main term $X/G(\xi)$ improves on the naive $X\, V(z)$ because $G(\xi) = \sum_{d<\xi} h(d) \ge \sum_{d<\xi} \mu^2(d) \prod_{p|d} g(p) = 1/V(z)$-type sums are larger than the reciprocal of the product, capturing more than the independence heuristic; and the remainder involves only $d, e < \xi = \sqrt{D}$, so D can be taken as large as the remainder $\sum_{[d,e]<D} |r_{[d,e]}|$ permits, typically D a small power of X. The most consequential application is to primes in arithmetic progressions, where the Selberg sieve yields an unconditional upper bound of the right order, the Brun–Titchmarsh theorem.

Theorem 12.3.3: Brun–Titchmarsh

Let $\pi(x; q, a) = \#\{p \le x : p \equiv a \pmod q\}$ for $\gcd(a, q) = 1$. Then, uniformly for $1 \le q < x$ with $x/q \to \infty$,

$$\pi(x; q, a) \le \big(2 + o(1)\big)\, \frac{x}{\varphi(q)\, \log(x/q)}.$$

The fully uniform form, $\pi(x; q, a) \le 2x/\big(\varphi(q) \log(x/q)\big)$ for all $1 \le q < x$ with no error factor, is a theorem of Montgomery and Vaughan.

***Proof*:**
The proof applies the Selberg upper bound (theorem 12.3.2) to the progression, computes the local densities, and estimates the sieve sum $G(\xi)$ from below.

Step 1: Reduction to a sifting problem. Let $z \geq 2$, let $\mathcal{P}$ be the primes not dividing q, and set $\mathcal{A} = \{n \leq x : n \equiv a \pmod q\}$. A prime $p \leq x$ with $p \equiv a \pmod q$ and $p \geq z$ has no prime factor below z, so it survives the sieve; the primes $p < z$ number at most z. Hence

$$\pi(x; q, a) \leq S(\mathcal{A}, \mathcal{P}, z) + z.$$

Step 2: Local densities. The set has $X = |\mathcal{A}| = x/q + O(1)$. For squarefree d with $\gcd(d, q) = 1$, the elements of $\mathcal{A}$ divisible by d are the $n \leq x$ with $n \equiv a \pmod q$ and $n \equiv 0 \pmod d$; by the Chinese remainder theorem these lie in a single class modulo qd, so $A_d = x/(qd) + O(1) = g(d)X + r_d$ with

$$g(d) = \frac{1}{d}, \qquad |r_d| \leq 1.$$

Thus $g(p) = 1/p$, and $h(p) = g(p)/(1 - g(p)) = 1/(p - 1)$, so for squarefree d coprime to q, $h(d) = \prod_{p|d}(p - 1)^{-1} = 1/\varphi(d)$.

Step 3: The Selberg bound. With level $D = \xi^2$, theorem 12.3.2 gives

$$S(\mathcal{A}, \mathcal{P}, z) \leq \frac{X}{G(\xi)} + \sum_{\substack{d,e<\xi \\ \gcd(de,q)=1}} |r_{[d,e]}|, \qquad G(\xi) = \sum_{\substack{d<\xi \\ \gcd(d,q)=1}} \frac{\mu^2(d)}{\varphi(d)}.$$

Since $|r_n| \leq 1$ and each $n = [d, e] < \xi^2$ arises from at most $3^{\omega(n)}$ pairs (d, e), the remainder is at most $\sum_{n<\xi^2} 3^{\omega(n)} \ll \xi^2(\log \xi)^2$.

Step 4: Lower bound for $G(\xi)$. For squarefree d, expanding each factor as a geometric series,

$$\frac{\mu^2(d)}{\varphi(d)} = \prod_{p|d} \frac{1}{p - 1} = \prod_{p|d} \sum_{k\geq 1} \frac{1}{p^k} = \sum_{\substack{m\geq 1 \\ \mathrm{rad}(m)=d}} \frac{1}{m},$$

where $\mathrm{rad}(m)$ is the product of the distinct primes dividing m. Summing over squarefree $d < \xi$ coprime to q, and noting that every $m < \xi$ with $\gcd(m, q) = 1$ has $\mathrm{rad}(m) < \xi$ coprime to q,

$$G(\xi) = \sum_{\substack{d<\xi \\ \gcd(d,q)=1}} \sum_{\mathrm{rad}(m)=d} \frac{1}{m} \geq \sum_{\substack{m<\xi \\ \gcd(m,q)=1}} \frac{1}{m} = \frac{\varphi(q)}{q}\big(\log \xi + O(1)\big),$$

the last equality the standard estimate for the coprime harmonic sum (obtained from $\sum_{m\leq y} 1/m = \log y + O(1)$ by Möbius removal of the primes dividing q, using $\sum_{d|q} \mu(d)/d = \varphi(q)/q$).

Step 5: Conclusion. Choose $\xi = \sqrt{x/q}\,/\,(\log(x/q))^2$, so that $\log \xi = \frac{1}{2}\log(x/q) + O(\log\log(x/q))$ and the remainder $\ll \xi^2(\log \xi)^2 = (x/q)/(\log(x/q))^2 \cdot (1 + o(1))$ is $o(X/G(\xi))$, while $z \leq \xi = o\big(x/(\varphi(q)\log(x/q))\big)$. Therefore, using Step 4,

$$\pi(x; q, a) \leq \frac{X}{G(\xi)} + o\left(\frac{x}{\varphi(q)\log(x/q)}\right) \leq \frac{x/q}{\frac{\varphi(q)}{q} \cdot \frac{1}{2}\log(x/q)}(1 + o(1)) = \big(2 + o(1)\big)\frac{x}{\varphi(q)\log(x/q)},$$

the factor 2 coming from $\log\xi = \frac{1}{2}\log(x/q)$. This is the claim; the removal of the $o(1)$ to give the fully uniform constant 2 is the refinement of Montgomery and Vaughan, [11, §3.4], completing the proof.

The constant 2 in Brun–Titchmarsh is larger by a factor of 2 than the truth: for q fixed and $x \to \infty$ the prime number theorem for arithmetic progressions gives $\pi(x;q,a) \sim x/\big(\varphi(q)\log x\big)$, half the Brun–Titchmarsh bound. The factor of 2 is the price of the parity problem and of uniformity in q, and it is significant: reducing it below 2 for the full range $q < x$ would have consequences for the zeros of L-functions, and the constant 2 is in this sense a barrier of the same nature as the $q^{1/4}$ of the Burgess bound. Applied to the twin-prime sequence $\mathcal{A} = \{n(n+2) : n \le x\}$ with $g(p) = 2/p$, the same Selberg bound sharpens Brun's theorem to

$$\pi_2(x) \le \big(8 + o(1)\big)\, C_2\, \frac{x}{(\log x)^2},$$

removing the spurious $(\log\log x)^2$ of theorem 12.2.3 and matching the conjectured order $2C_2\, x/(\log x)^2$ up to the factor 4 that the parity problem leaves in place.

12.4 The Large Sieve Inequality

The sieves developed so far are *small* sieves: at each prime a bounded number of residue classes is removed, one for the Eratosthenes sieve, two for the twin-prime sieve. The large sieve addresses the opposite regime, in which many residue classes, up to a positive proportion of all of them, are struck out at each prime. Its natural formulation is analytic rather than combinatorial: it is an inequality bounding the mean square of an exponential sum sampled at well-spaced points, and its arithmetic force comes from specializing those points to the Farey fractions. The large sieve is the tool behind the strongest unconditional results on primes in arithmetic progressions, and it is the bridge from the sieve methods of this chapter back to the exponential sums of the last.

Theorem 12.4.1: The Large Sieve Inequality

Let M, N be integers with $N \ge 1$, and let $\alpha_1, \dots, \alpha_R \in \mathbb{R}$ be δ-spaced modulo 1, meaning $\|\alpha_r - \alpha_s\| \ge \delta$ for all $r \ne s$, where $0 < \delta \le \frac{1}{2}$. For arbitrary complex numbers $(a_n)_{M<n\le M+N}$ set

$$S(\alpha) = \sum_{M<n\le M+N} a_n\, e(n\alpha), \qquad e(t) = e^{2\pi i t}$$

in the notation of box 11.0.2. Then

$$\sum_{r=1}^{R} |S(\alpha_r)|^2 \le \big(N - 1 + \delta^{-1}\big) \sum_{M<n\le M+N} |a_n|^2.$$

***Proof*:**
The proof establishes the inequality with the explicit constant $\pi N + \delta^{-1}$ in place of $N - 1 + \delta^{-1}$; the sharp constant stated above, due independently to Selberg and to Montgomery and Vaughan, is obtained by replacing the elementary Gallagher lemma below with an extremal-majorant argument, [9, §7.4–7.5]. The engine is a mean-value inequality of Gallagher.

Step 1: Gallagher's lemma. Let $F\colon \mathbb{R}/\mathbb{Z} \to \mathbb{C}$ be continuously differentiable, let $x_0 \in \mathbb{R}/\mathbb{Z}$, and let $J = [x_0 - \delta/2, x_0 + \delta/2]$. For every $t \in J$, the fundamental theorem of calculus gives $|F(x_0)|^2 = |F(t)|^2 + \int_t^{x_0} (|F|^2)'(u)\,du$. Averaging over $t \in J$ (dividing by $|J| = \delta$),

$$|F(x_0)|^2 = \frac{1}{\delta}\int_J |F(t)|^2\,dt + \frac{1}{\delta}\int_J \left(\int_t^{x_0} (|F|^2)'(u)\,du\right) dt.$$

Interchanging the order of integration in the double integral, the inner factor becomes $\int_J (|F|^2)'(u)\,w(u)\,du$ with weight $w(u) = \frac{1}{2}\big(\delta - 2|u - x_0|\big) \cdot \mathrm{sgn}(x_0 - u)$ of size $|w(u)| = \frac{\delta}{2} - |u - x_0| \le \frac{\delta}{2}$. Hence, using $|(|F|^2)'| = |2\mathrm{Re}(\overline{F}F')| \le 2|F||F'|$,

$$|F(x_0)|^2 \le \frac{1}{\delta}\int_J |F|^2\,dt + \frac{1}{\delta}\cdot\frac{\delta}{2}\int_J 2|F||F'|\,dt = \frac{1}{\delta}\int_J |F|^2\,dt + \int_J |F|\,|F'|\,dt.$$

Step 2: Summation over the spaced points. Because the α_r are δ-spaced modulo 1, the arcs $J_r = [\alpha_r - \delta/2, \alpha_r + \delta/2]$ are pairwise disjoint in $\mathbb{R}/\mathbb{Z}$. Choose $c = M + \frac{N+1}{2}$ and set

$$F(t) = e(-ct)\,S(t) = \sum_{M<n\le M+N} a_n\,e\big((n-c)t\big),$$

so that $|F(\alpha_r)| = |S(\alpha_r)|$. Since $n - n'$ is an integer, $|F(t)|^2$, $|F'(t)|^2$, and $\overline{F}F'$ are all 1-periodic, so F descends to $\mathbb{R}/\mathbb{Z}$ and Step 1 applies. Summing Step 1 over r with the disjoint arcs J_r,

$$\sum_{r=1}^{R} |S(\alpha_r)|^2 = \sum_r |F(\alpha_r)|^2 \le \frac{1}{\delta}\int_0^1 |F|^2\,dt + \int_0^1 |F|\,|F'|\,dt.$$

Step 3: The two integrals. By orthogonality of $t \mapsto e((n-n')t)$ over $\mathbb{R}/\mathbb{Z}$ (the exponent $n - n'$ being an integer, the parameter c cancelling),

$$\int_0^1 |F|^2\,dt = \sum_{M<n\le M+N} |a_n|^2 =: A, \qquad \int_0^1 |F'|^2\,dt = 4\pi^2\sum_n (n-c)^2|a_n|^2 \le 4\pi^2\Big(\frac{N-1}{2}\Big)^2 A,$$

the last bound because $|n - c| \le (N-1)/2$ for $M < n \le M + N$. By the Cauchy–Schwarz inequality $\int_0^1 |F||F'| \le (\int |F|^2)^{1/2}(\int |F'|^2)^{1/2} \le A^{1/2}\cdot \pi(N-1)A^{1/2} = \pi(N-1)A$. Substituting,

$$\sum_{r=1}^{R} |S(\alpha_r)|^2 \le \big(\delta^{-1} + \pi(N-1)\big)A \le \big(\pi N + \delta^{-1}\big)\sum_{M<n\le M+N} |a_n|^2,$$

which is the inequality with the explicit constant, completing the proof.

The inequality says that an exponential sum of length N, whose trivial mean-square size over a full period is $N\sum|a_n|^2$, cannot be large simultaneously at many well-separated frequencies: sampling at R points spaced by δ costs only the factor $N - 1 + \delta^{-1}$, not $R \cdot N$. The arithmetic content appears when the sampling points are the Farey fractions, which are well-spaced automatically.

Corollary 12.4.2: The Arithmetic Large Sieve

For arbitrary complex $(a_n)_{M<n\le M+N}$ and any $Q \ge 1$,

$$\sum_{q\le Q} \sum_{\substack{a=1 \\ \gcd(a,q)=1}}^{q} \left|S\Big(\frac{a}{q}\Big)\right|^2 \le (N+Q^2) \sum_{M<n\le M+N} |a_n|^2.$$

Consequently, if $\mathcal{B} \subseteq \{M+1,\dots,M+N\}$ is a set of integers that, for each prime $p \le Q$, omits at least $\omega(p)$ residue classes modulo p, then

$$|\mathcal{B}| \le \frac{N+Q^2}{L(Q)}, \qquad L(Q) = \sum_{q\le Q} \mu^2(q) \prod_{p|q} \frac{\omega(p)}{p-\omega(p)}.$$

***Proof Key ideas*:**
The Farey fractions a/q with $q \le Q$ and $\gcd(a,q)=1$ are Q^{-2}-spaced modulo 1: for distinct fractions $a/q \ne a'/q'$ with $q,q' \le Q$, $\|a/q-a'/q'\| \ge |aq'-a'q|/(qq') \ge 1/(qq') \ge 1/Q^2$, since the numerator is a nonzero integer. Applying theorem 12.4.1 with $\delta = Q^{-2}$ and $N-1+Q^2 \le N+Q^2$ gives the first inequality. For the second, take $a_n = \mathbf{1}_{\mathcal{B}}(n)$, so $\sum_n |a_n|^2 = |\mathcal{B}|$ and $S(0) = |\mathcal{B}|$; a standard lower bound expresses the left-hand side of the first inequality, restricted to the arithmetic information, as $|\mathcal{B}|^2\, L(Q)$, using that $\mathcal{B}$ avoids $\omega(p)$ classes modulo each p to bound the character sums from below. Combining $|\mathcal{B}|^2 L(Q) \le (N+Q^2)|\mathcal{B}|$ and dividing by $|\mathcal{B}|$ yields the stated bound; the multiplicative rearrangement producing $L(Q)$ is carried out in [9, §7.4], completing the argument.

The arithmetic large sieve bounds the size of any set that avoids many residue classes: the more classes omitted, the larger $L(Q)$ and the smaller the set must be. Its most important application is not to a single set but to primes in progressions on average, where it supplies the input that makes the Bombieri–Vinogradov theorem possible. That theorem is the great unconditional achievement of the large sieve: it asserts that the primes are distributed among the residue classes modulo q as evenly as the generalized Riemann hypothesis would predict, not for every q but on average over q up to nearly $\sqrt{x}$.

Theorem 12.4.3: Bombieri–Vinogradov

For every $A > 0$ there is a $B = B(A) > 0$ such that, with $Q = x^{1/2}(\log x)^{-B}$,

$$\sum_{q\le Q} \max_{\gcd(a,q)=1} \left|\psi(x;q,a) - \frac{x}{\varphi(q)}\right| \ll_A \frac{x}{(\log x)^A},$$

where $\psi(x;q,a) = \sum_{n\le x,\ n\equiv a\,(q)} \Lambda(n)$.

***Proof Key ideas*:**
The theorem is a far-reaching application of the large sieve, and its full proof occupies a chapter of any analytic number theory text; the mechanism is indicated. The error $\psi(x;q,a) - x/\varphi(q)$ is

expressed through Dirichlet characters modulo q by orthogonality (theorem 7.1.4): the deviation is $\frac{1}{\varphi(q)}\sum_{\chi\neq\chi_0}\bar\chi(a)\,\psi(x,\chi)$ with $\psi(x,\chi)=\sum_{n\le x}\Lambda(n)\chi(n)$, so the problem becomes bounding $\sum_{q\le Q}\frac{1}{\varphi(q)}\sum_\chi^*|\psi(x,\chi)|$ summed over primitive characters. The sum $\psi(x,\chi)$ is decomposed by a combinatorial identity for the von Mangoldt function, Vaughan's identity, into bilinear forms $\sum_m\sum_n\alpha_m\beta_n\chi(mn)$; the large sieve inequality (theorem 12.4.1), applied to these bilinear forms summed over q and the primitive characters modulo q, provides exactly the mean-square cancellation needed, the modulus $Q^2=x/(\log x)^{2B}$ matching the length x so that the factor $N+Q^2$ stays of order x. Balancing the pieces produces the saving $(\log x)^{-A}$ with $B(A)$ growing linearly in A. The complete proof is in [6, Ch. 28] and [9, §17.4], completing the sketch.

Bombieri–Vinogradov is often described as the generalized Riemann hypothesis on average, and the description is precise once read correctly. The generalized Riemann hypothesis for the Dirichlet L-functions modulo q gives $\psi(x;q,a)-x/\varphi(q)\ll\sqrt{x}(\log x)^2$ for each individual q, an error of size $x^{1/2+o(1)}$ per modulus. The Bombieri–Vinogradov bound $x/(\log x)^A$, spread over the roughly $\sqrt{x}$ moduli $q\le Q$, amounts to a mean error of $x^{1/2+o(1)}$ per modulus, exactly the quality the generalized Riemann hypothesis would supply term by term; the theorem secures that average strength unconditionally, foregoing only the per-modulus control that would require the hypothesis itself. For most applications the average suffices, and Bombieri–Vinogradov substitutes for the generalized Riemann hypothesis in problems ranging from the Titchmarsh divisor problem to the bounded gaps between primes taken up next. The limit $Q=x^{1/2-o(1)}$ is the natural barrier of the method; extending it to $Q=x^\theta$ with $\theta>1/2$, the Elliott–Halberstam conjecture, remains open and is the hypothesis under which the strongest bounded-gaps results are conditional.

12.5 Modern Directions

The sieves of the preceding sections share a common ceiling. Each produces almost-primes, integers with a bounded number of prime factors, but none produces primes: Brun's method shows the twin-prime sequence is sparse, Selberg's sharpens the count, the large sieve controls progressions on average, yet no combination of them proves that n and $n+2$ are simultaneously prime infinitely often. This section explains the obstruction precisely, surveys the refinements that press against it, and records the recent theorems that circumvent it by importing external input. No proofs are given; the results lie beyond the scope of this book, and the aim is to map the frontier the earlier sections approach.

12.5.1 The Parity Problem The obstruction foreshadowed in box 12.1.4 admits a precise statement, due to Selberg. A sieve estimates a sifted count using only the quantities $A_d = \#\{a \in \mathcal{A} : d|a\}$ for squarefree d up to some level, and any bound it produces must hold for *every* sequence $\mathcal{A}$ consistent with the given A_d. Selberg exhibited two sequences with the same sieve data up to the relevant level, one supported on integers with an even number of prime factors and one on integers with an odd number, whose true sifted counts differ by a constant factor. Because a sieve cannot tell them apart, its upper and lower bounds must straddle both, and the ratio between them cannot be brought below a fixed constant: an unaided sieve loses a factor of 2 precisely because it cannot see the sign of the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ (definition 1.1.6). This is why Brun–Titchmarsh carries the constant 2, why the twin-prime upper bound of section 12.3 exceeds the conjecture by a factor of 4, and why proving that a sifted sequence contains a genuine prime requires information about $\lambda(n)$, equivalently about the zeros of L-functions, that no sieve supplies on its own.

The response to the parity problem was to quantify how much a sieve can achieve and to identify the exact point at which it stalls. The relevant parameter is the dimension of the sieve.

Definition 12.5.2: Sieve Dimension

A sifting problem with local densities $g(p)$ has *dimension* $\kappa \geq 0$ if

$$\sum_{w \leq p < z} g(p) \log p = \kappa \log \frac{z}{w} + O(1)$$

for all $2 \leq w \leq z$, so that $g(p) \approx \kappa/p$ on average. The Eratosthenes sieve, removing one class per prime, has dimension $\kappa = 1$; the twin-prime sieve, removing two, has dimension $\kappa = 2$.

The dimension governs the best result a sieve of a given type can prove. The Rosser–Iwaniec sieve, the optimal combinatorial sieve, attains for each dimension κ a pair of continuous functions F and f describing the sharpest possible upper and lower main terms, and the crucial datum is the *sifting limit* $\beta(\kappa)$: the threshold on the sifting level below which the lower-bound sieve becomes trivial. For dimension $\kappa = 1$ the lower-bound sieve is effective for sifting parameters $s > \beta(1) = 2$, so the linear sieve can find integers with no prime factor below $x^{1/2}$, that is, primes or products of two primes of comparable size; for dimension $\kappa = 2$ the limit is larger and the twin-prime problem sits just beyond it. The Rosser–Iwaniec theory is what makes the following result attainable.

Theorem 12.5.3: Chen's Theorem

There are infinitely many primes p such that $p + 2$ is either prime or a product of two primes. Equivalently, in the notation P_2 for an integer with at most two prime factors, there are infinitely many primes p with $p + 2 = P_2$. Correspondingly, every sufficiently large even integer is the sum of a prime and a P_2.

Chen's theorem, proved in 1973 by a weighted lower-bound sieve of dimension 2 combined with a switching trick and the Bombieri–Vinogradov theorem (theorem 12.4.3) in place of the unavailable generalized Riemann hypothesis, is the closest approach to the twin-prime and Goldbach conjectures that the sieve alone permits. It stops one factor short of a prime, exactly where the parity problem predicts, and no purely sieve-theoretic argument has crossed that gap. The proof is beyond the scope of this book; the full account is in [9, Ch. 25].

The decisive progress on the primes themselves came not from defeating the parity problem but from a different sieve construction that sidesteps it for a weaker conclusion, the existence of small gaps between primes. Rather than asking whether n and $n+2$ are simultaneously prime, one asks only whether some pair among a cluster $n+h_1, \dots, n+h_k$ is simultaneously prime, a question about which the parity obstruction is silent.

Theorem 12.5.4: Bounded Gaps Between Primes

Let p_n denote the n-th prime. Then

$$\liminf_{n\to\infty} (p_{n+1} - p_n) \leq 246.$$

More generally, for every m there are infinitely many intervals of bounded length containing at least m primes; under the Elliott–Halberstam conjecture the gap bound improves to $\liminf(p_{n+1} - p_n) \leq 12$.

The theorem is the culmination of a decade of work. The method of Goldston, Pintz, and Yıldırım in 2005 introduced a sieve weight tuned to detect primes in admissible tuples and reduced the bounded-gaps problem to the level of distribution of the primes, showing that any level beyond 1/2, an inch past Bombieri–Vinogradov, would force bounded gaps. In 2013 Zhang established just enough of that extension, unconditionally, to prove $\liminf(p_{n+1} - p_n) < 7 \times 10^7$, the first finite bound. Within months Maynard and, independently, Tao replaced the one-dimensional GPY weight by a multidimensional sieve that detects several primes in a tuple at once and dispenses with the extension beyond Bombieri–Vinogradov altogether, lowering the bound to 600 and, after collaborative optimization, to 246. The multidimensional sieve is a direct descendant of Selberg's Λ^2 method (section 12.3): the weights are again squares of divisor sums, now indexed by tuples, and again chosen to optimize a quadratic form. The parity problem is not defeated, only evaded: the theorems produce primes in clusters without ever certifying which coordinate is prime.

A parallel line, also sieve-theoretic in origin, concerns primes in arithmetic progressions of their own. The Green–Tao theorem of 2008 states that the primes contain arbitrarily long arithmetic progressions, proved by a transference principle that imports the primes into the framework of Szemerédi's theorem through a sieve-majorant for the primes; it too circumvents the parity problem, by asking a question, the existence of progressions, to which parity poses no obstruction.

These developments close the chapter and the analytic core of the book at its present frontier. The sieve began as Eratosthenes' mechanical procedure for tabulating primes, became through Brun and Selberg a quantitative instrument matching the true order of magnitude of the twin primes and the gaps between primes, and has become in the last decade, joined to the distribution results that the large sieve and Bombieri–Vinogradov provide, the method that proved primes cluster. What it has not done, and provably cannot do alone, is cross the parity barrier to the primes themselves; that crossing awaits the analytic information carried by the zeros of L-functions, the objects with which this book began. The idelic and automorphic methods that promise to control those zeros in the generality the outstanding conjectures require are the subject of the companion volumes of the series, to which [12] is the gateway.

Where to Go From Here

This book has followed the analytic theory of the primes from the elementary counting of Part I to the Chebotarev density theorem and the sieve at the parity barrier. It ends, as the subject itself does, at two frontiers. The L-function method of Part II reaches the zeros of ζ and the L-functions and can go no further without resolving them; the sieve of chapter 12 reaches the almost-primes and cannot cross to the primes without the same analytic information. Every direction beyond this book passes through one of these frontiers, and this closing chapter maps the main routes and the volumes of the series and the standard literature that travel them.

The first direction stays inside the theory of the zeta and Dirichlet L-functions and asks what this book deferred: the location of the zeros. The functional equation of chapter 6 confined them to the critical strip and the Riemann hypothesis places them all on the central line, but the unconditional theory works with what can be proved, zero-free regions and zero-density estimates, and converts them through the explicit formula into sharp error terms for $\pi(x)$ and for primes in progressions. The value-distribution of ζ on the critical line, the moments of $|\zeta(\frac{1}{2}+it)|$, and the pair correlation of the zeros open onto the connection with random matrix theory. These topics are the natural sequel to Parts II, and the standard references are Titchmarsh's *The Theory of the Riemann Zeta-Function* [21], Davenport's *Multiplicative Number Theory* [6], and, for the modern analytic apparatus in full, Iwaniec and Kowalski's *Analytic Number Theory* [9] and Montgomery and Vaughan's *Multiplicative Number Theory I* [11].

The second direction takes up the algebraic thread of chapters 8 to 10. The Hecke L-functions were built there by hand from ray class characters, and their functional equation was stated rather than proved; the idelic reformulation, in which a Hecke character is a character of the idele class group and the functional equation follows from Fourier analysis on the adeles, is the subject of Tate's thesis and the proper home of the whole construction. With it comes class field theory, the Artin reciprocity law that identifies one-dimensional Artin L-functions with Hecke L-functions, invoked but not proved in chapter 10. The series continues this thread in its volume on class field theory, [12]; the classical references are Neukirch's *Algebraic Number Theory* [15], Lang's *Algebraic Number Theory* [10], and the Cassels and Fröhlich volume [1], which contains Tate's thesis itself.

The third direction is the one toward which the end of Part II pointed: the automorphic generalization. Artin's conjecture on the holomorphy of the higher L-functions, left open in chapter 10, is expected to follow from the Langlands program, in which every Galois representation is matched with an automorphic representation whose L-function is holomorphic by the analytic theory of automorphic forms. This is where the analytic and algebraic threads of the book rejoin, and it is a subject of its own, taken up in the series' volumes on automorphic representations and the local Langlands correspondence. The standard entry points are Bump's *Automorphic Forms and Representations*, Gelbart's *Automorphic Forms on Adele Groups*, and, for the classical side nearest to this book, Iwaniec's *Topics in Classical Automorphic Forms*.

The fourth direction develops the methods of Part III to their modern form. The sieve of chapter 12 was presented in its classical shape; the Rosser–Iwaniec theory, the parity-breaking of Friedlander and Iwaniec, and the multidimensional sieve behind the bounded-gaps theorems carry it much further, and the circle method whose outline closed chapter 11 is a subject in itself for Waring's and Goldbach's problems. The comprehensive reference for the sieve is Friedlander and Iwaniec's *Opera de Cribro*, with Halberstam and Richert's *Sieve Methods* for the classical theory; for the circle method, Vaughan's *The Hardy–Littlewood Method* is standard, and Iwaniec and Kowalski's *Analytic Number Theory* [9] again covers both from a unified analytic viewpoint.

These four directions share a single horizon. The zeros of the L-functions govern the error terms of the first, the reciprocity laws of the second and the automorphic correspondence of the third would supply the missing holomorphy, and the parity barrier of the fourth is crossed only by information about those same zeros. The Riemann hypothesis and its generalizations stand behind all of them, and the reader who has followed this book to its end now has the analytic foundation on which the modern attempts to reach them are built.

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